

The Handbook of

ALM in Banking

Managing New Challenges
for Interest Rates, Liquidity
and the Balance Sheet

2nd Edition

EDITED BY ANDREAS BOHN
AND MARIE ELKENBRACHT-HUIZING



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Edited by Andreas Bohn and
Marije Elkenbracht-Huizing



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Introduction

Asset and liability management (ALM) in banking is the function that manages three aspects of the overall balance sheet: capital, interest rate risk and liquidity risk. It focuses predominantly on the structural and strategic elements of these different aspects. ALM looks at the risks embedded in the structure of a balance sheet and aims to find a balance between risk and return. While there are no specific regulations for asset and liability management, several limits and regulations, for, eg, interest rate risk in the banking book (IRRBB), liquidity, funding and capital have to be considered.

This book addresses some of the key features of assets and liabilities in banking and provides practical guidance to manage their respective challenges. We asked industry experts from leading institutions to provide their insight on these challenges. Since the first edition of this book was published in 2014, a stream of new guidelines, principles, regulations and technical standards have been published, including in the “Finalisation of the Basel III Standards”, which was published in December 2017. In particular, elevated requirements for the definition of risk appetite and limits, more guidance on stress testing, and a general strengthening of the risk and control framework as well as elevated reporting requirements have emerged. This made us decide to update the book. It is structured in four parts. [Part I](#) provides a general overview; [Part II](#) focuses on the management of interest rate risk; [Part III](#) covers relevant aspects from a liquidity risk perspective, while [Part IV](#) looks at issues related the steering of capital and balance sheet holistically.

In [Chapter 1](#) Marc Farag, Damian Harland and Dan Nixon provide a general framework for thinking about bank capital and liquidity in the context of the traditional banking model and give an overview of their regulation. Starting from the top, banks and regulators have paid much attention to the need for a sound governance and risk appetite process. In [Chapter 2](#) Koos Timmermans and Wessel Douma share with us their views on governance:

how can banks make sure that their risk appetite and ALM remains up to date? Furthermore, they describe the main sensitivities a typical bank needs to manage and how an enterprise-wide risk appetite framework can be designed to do this. They include a key component of the risk appetite framework: evaluating the outcome of scenario and stress events.

In [Part II](#) we go further into the details of interest rate risk. The regulation and management of this risk has evolved since the first edition, recognising that due to differences in banks and complexity this risk is best captured by a Pillar II approach. In [Chapter 3](#) Roberto Virreira Zijderveld gives us an overview of developments in regulation and guides us through the 12 revised IRRBB principles issued by the Basel Committee on Banking Supervision (BCBS) in 2016. A key change is the requirement for banks to set up robust stress testing of assumptions and model validation controls in order to manage the dependency on the ubiquitous behavioural assumptions in the models used for IRRBB.

Models and their assumptions determine the outcome of metrics used for IRRBB. This is further discussed in the next chapters. First, Giovanni Gentili and Nicola Santini go into the metrics used to calculate value and earnings risks in [Chapter 4](#). In [Chapters 5–8](#) various approaches to model non-maturing deposits are included. With customers having the right to withdraw or add funds, and the bank having the right to change the interest rate, it is a challenge to correctly incorporate these liabilities in risk measures. The chosen methodology can have a significant influence on the outcomes. In [Chapter 5](#) George Soulellis describes approaches to modelling the general run-off profile of deposits from a statistical perspective, while an approach to modelling and risk-managing deposits in an environment of stochastic interest rates and credit spreads is presented by Andreas Bohn in [Chapter 6](#). In [Chapter 7](#) Marije Elkenbracht-Huizing and Bert-Jan Nauta introduce the concept of a fair margin – the increase in customer coupon that makes the value of non-maturing deposits zero – and develop a value-based hedging strategy to stabilise this margin. In [Chapter 8](#) Florentina Paraschiv and Michael Schürle present strategies for optimising the risk–return profile for non-maturing deposits by dynamic replication. On the asset side of the balance sheet, one important product that needs a behavioural model is mortgages. Customers mostly have a right to prepay their mortgages early, leading to prepayment risk. In [Chapter 9](#) Dick Boswinkel explains the various mortgage products, how prepayments can be modelled and the

important drivers to look into. Furthermore, he shows how prepayment risk influences the risk metrics and gives some insight into how to cope with this risk in balance-sheet management. Managing the interest rate risk of a bank in an environment of very low interest rates has become a challenge for most banks; Thomas Becker, Raphael Bulut and Steve Uschmann elaborate on ALM strategies in a low interest rate environment in [Chapter 10](#).

Credit risk also needs to be catered for. Credit spread risk in the banking book was mentioned for the first time in the Basel principles (BCBS 368), which states that banks need to monitor and assess it. In [Chapter 11](#) Raquel Bujalance and Oliver Burnage discuss the difficulties to be overcome and choices that need to be made when incorporating default risk into the metrics. Another important concept in interest rate risk management is hedge accounting. This was introduced to eliminate valuation asymmetries resulting from the different accounting treatment of a hedge and a hedged item, eg, a loan hedged by a swap. In [Chapter 12](#) Bernhard Wondrak explains hedge accounting and how the rules change for International Financial Reporting Standard 9.

[Part III](#) is devoted to liquidity risk. In [Chapter 13](#) Patrick de Neef takes us through regulatory developments since 2014. Next, in [Chapter 14](#) Lennart Gerlagh and Marc Otto describe various aspects of liquidity risk management with a specific focus on determination of the behavioural maturity calendar. This calendar is an important input in stress testing, risk appetite metrics and the basis for funding plans. It is dependent on behavioural models, of which the drivers are also discussed. A core component of the modern ALM function is management of the liquidity buffer or liquidity reserve. Interest rate risk and liquidity risk as well as credit risk and capital consumption have to be reflected. In [Chapter 15](#) Christian Buschmann outlines strategies for the management of the assets covered.

Secured funding has increased in importance since the 2007–9 global financial crisis. In [Chapter 16](#) Federico Galizia and Giovanni Gentili explain the particularities of short- and long-term secured funding instruments and share their thoughts on the collateral needed for these forms of funding and the potential impact this has on unsecured lenders. While secured financing is generally beneficial from a respective debt holder perspective, concerns have been raised about the potential risks to encumbrance for holders of other bank debt. Daniela Migliasso shades light on encumbrance, and

analyses the interaction with the liquidity coverage ratio and net stable funding ratio, describing in [Chapter 17](#) how it can be seen as an opportunity.

After diving into interest rate risk and liquidity risk, in [Part IV](#) we revert to the balance sheet as a whole and capital management in particular. In [Chapter 18](#) Ralf Leiber describes the different types of capital, explains the various capital requirements, discusses potential differences between countries and gives insight into the management of capital. During the global financial crisis it became clear that stress testing exercises executed before the crisis had not sufficiently captured the risks, particularly liquidity and funding risks. This led to the development of regulatory stress testing for the sector as a whole. In [Chapter 19](#) Bernhard Kronfellner, Stephan Süß and Volker Vonhoff take us through the history of stress testing, comparing both US and European regulatory stress tests and giving recommendations on how to best organise and execute stress testing. Reverse stress testing has also increased in importance; with reverse stress testing a bank derives potential scenarios that render the business model unviable. In [Chapter 20](#) Michael Eichhorn and Philippe Mangold discuss a six-step approach to execute reverse stress testing in a structured manner.

Following the financial crisis many new regulations have been introduced for trading books, in particular to contain risks from derivatives. An integrated management of credit risk, liquidity risk as well as capital and term funding requirements has become necessary and needs to be reflected in the ALM framework. Massimo Baldi, Francesco Fede and Andrea Pampolini provide an overview on the integrated management of value adjustments in [Chapter 21](#). Derivatives also pose a challenge to determining the funding strategy, given the dependency of their value on market rates. In [Chapter 22](#) Rene Reinbacher presents an efficient funding strategy for these instruments. The funding strategy is an important input to determine funds transfer pricing (FTP). This topic is further discussed in [Chapter 23](#), in which Robert Schäfer, Pascal Vogt and Peter Neu give an overview of methods to reflect the costs and benefits of funds in a bank's transfer pricing scheme. Here, many previously discussed topics come together: interest and liquidity characteristics based on behavioural models are key inputs in determining the FTP. Widowitz *et al* describe how to use this tool for the effective steering of a bank and share best practices.

The multiple regulatory requirements and constraints limit the shape of the overall balance sheet and lead to similar management strategies. The book is

concluded by [Chapter 24](#), in which Andreas Bohn and Paolo Tonucci give an overview of all the important regulatory constraints, explain their relationship and give an approach to optimise within those constraints.

Based on the feedback on the first edition, this book should be relevant for practitioners working in the field of ALM and related functions in treasury, finance and risk. It should also be relevant for graduate courses in universities and business schools that focus on these topics. Furthermore, we hope that some of the chapters will inspire researchers to advance academic work on the topics presented.

Part I

Introduction

Bank Capital and Liquidity

Marc Farag, Damian Harland, Dan Nixon

Bank capital, and a bank's liquidity position, are concepts that are central to understanding what banks do, the risks they take and how best those risks should be mitigated both by banks themselves and by prudential regulators. As the 2007–9 financial crisis powerfully demonstrated, the instability that can result from banks having insufficient financial resources – capital or liquidity – can acutely undermine the vital economic functions they perform.

This chapter is split into three sections. The first section introduces the traditional business model for banks of taking deposits and making loans. The second section explains the key concepts necessary to understand bank capital and liquidity. This is intended as a primer on these topics: while some references are made to the 2007–9 financial crisis, the aim is to provide a general framework for thinking about bank capital and liquidity. For example, the chapter describes how it can be misleading to think of capital as “held” or “set aside” by banks; capital is not an asset. Rather, it is a form of funding: one that can absorb losses that could otherwise threaten a bank's solvency. Meanwhile, liquidity problems arise due to interactions between funding and the asset side of the balance sheet, when a bank does not hold sufficient cash (or assets that can easily be converted into cash) to repay depositors and other creditors. Appendix A explains some of the accounting principles germane to understanding bank capital.

The final section gives an overview of capital and liquidity regulation. It is the role of bank prudential regulation to ensure the safety and soundness of

banks, for example, by ensuring that they have sufficient capital and liquidity resources to avoid a disruption to the critical services that banks provide to the economy. In April 2013, the Bank of England (“the Bank”), through the Prudential Regulation Authority (PRA), assumed responsibility for the safety and soundness of individual firms, which involves the microprudential regulation of banks’ capital and liquidity positions.¹ At the same time, the Financial Policy Committee (FPC) within the Bank was given legal powers and responsibilities² to identify and take actions to reduce risks to the financial system as a whole (macroprudential regulation) including by recommending changes in bank capital or liquidity requirements, or directing such changes in respect of certain capital requirements. In 2013 the FPC made recommendations on capital that the PRA have taken steps to implement.³

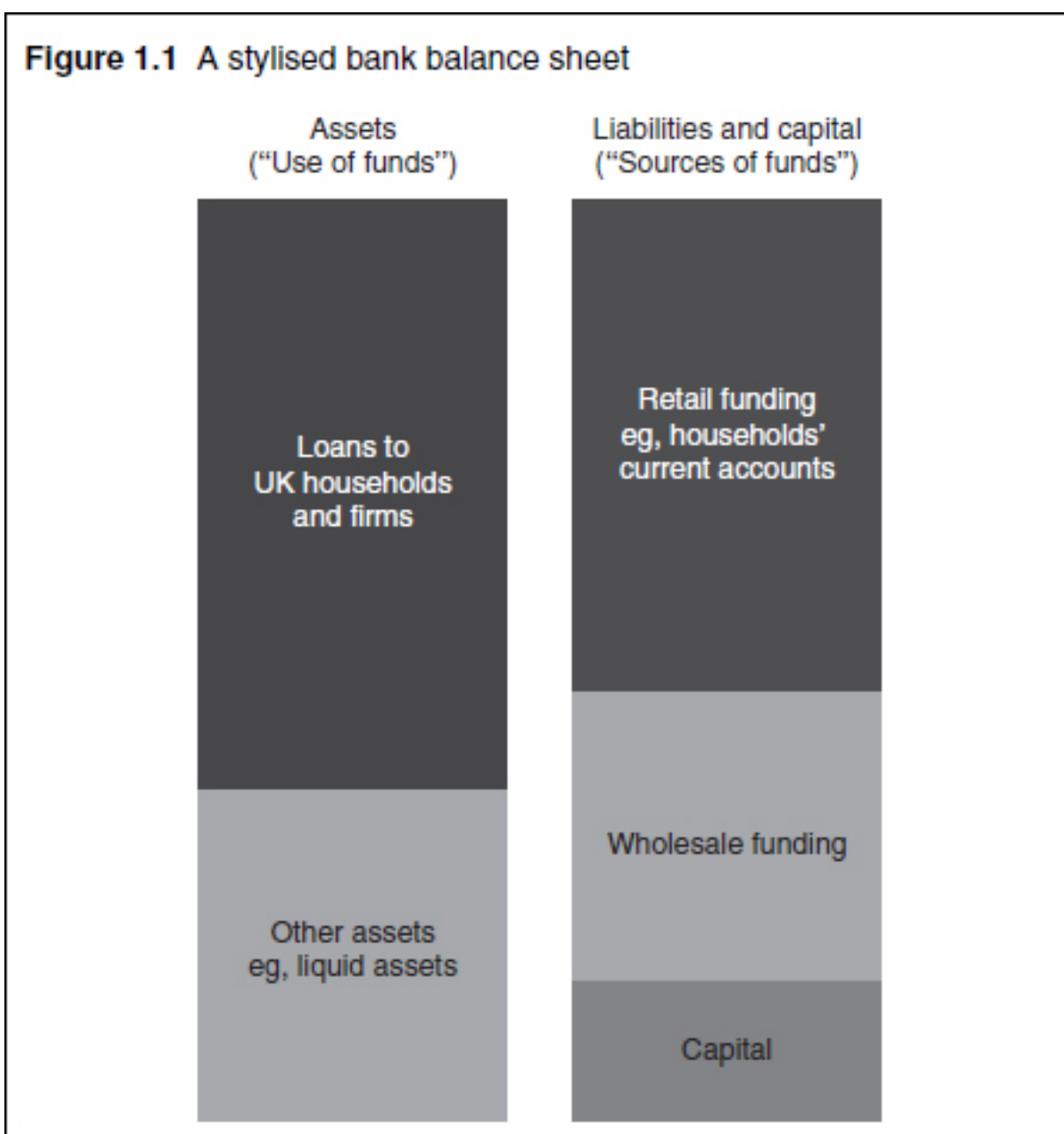
THE TRADITIONAL BANKING BUSINESS MODEL

Understanding why capital and liquidity are important requires an overview of what banks do. This section sets out the traditional banking business model, using a simplified bank balance sheet as an organising framework and highlighting some of the risks inherent in a bank’s business.

Banks play a number of crucial roles in the functioning of the economy. First, they provide payments services to households and companies, allowing them to settle transactions. Second, they provide credit to the real economy, for example, by providing mortgages to households and loans to companies. Third, banks help households and businesses to manage the various risks they face in different states of the world. This includes offering depositors access to their current accounts “on demand”, as well as providing derivatives transactions or other financial insurance services for their broader customer base.⁴

The focus for this chapter is the second function: providing credit to the real economy. Borrowers frequently need sizeable longer-term loans to fund investments, but those with surplus funds may individually have smaller amounts and many want swifter access to some or all of their money. By accepting deposits from many customers, banks are able to funnel savers’ funds to customers that wish to borrow. So, in effect, banks turn many small deposits with a short-term maturity into fewer longer-term loans. This

“maturity transformation” is therefore an inherent part of a bank’s business model.



Banks profit from this activity by charging a higher interest rate on their loans than the rate they pay out on the deposits and other sources of funding used to fund those loans. In addition, they may charge fees for arranging the loan.⁵

Introducing a bank's balance sheet

A useful way to understand what banks do, how they make profits and the risks they take is to consider a stylised balance sheet, as shown in [Figure 1.1](#). A bank's balance sheet provides a snapshot at a given point in time of the bank's financial position. It shows a bank's "sources of funds" on one side (liabilities and capital) and its "use of funds" (that is, its assets) on the other side. As an accounting rule, total liabilities plus capital must equal total assets.⁶

Like non-financial companies, banks need to fund their activities and do so by a mixture of borrowed funds ("liabilities") and their own funds ("capital"). Liabilities (what banks owe to others) include retail deposits from households and firms, such as current or savings accounts. Banks may also rely on wholesale funding: borrowing funds from institutional investors such as pension funds, typically by issuing bonds. In addition, they borrow from other banks in the wholesale markets, increasing their interconnectedness in the process. A bank's capital represents its own funds. It includes common shares (also known as common equity) and retained earnings. Capital is discussed in more detail in the following section.

Banks' assets include all financial, physical and intangible assets that banks currently hold or are due to be paid at some agreed point in the future. They include loans to the real economy, such as mortgages and personal loans to households, and business loans. They also include lending in the wholesale markets, including to other banks. Lending can be secured (where a bank takes collateral that can be sold in the event that the borrower is unable to repay) or unsecured (where no such collateral is taken). As well as loans, banks hold a number of other types of assets, including: liquid assets such as cash, central bank reserves or government bonds;⁷ the bank's buildings and other physical infrastructure; and "intangible" assets, such as the value of a brand. Finally, a bank may also have exposures that are considered to be "off balance sheet", such as commitments to lend or notional amounts of derivative contracts.

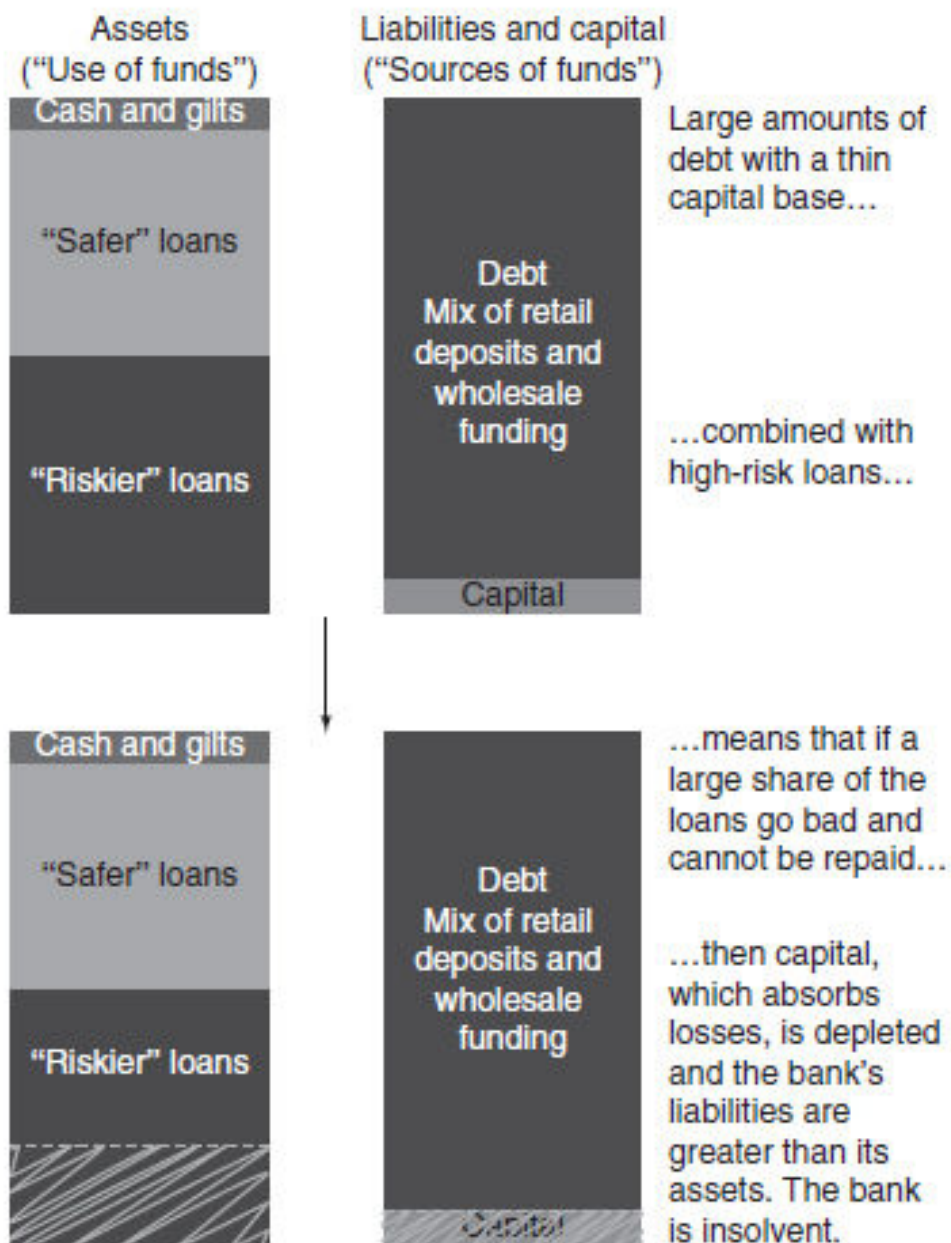
Credit risk, liquidity risk and banking crises

In transforming savers' deposits into loans for those that wish to borrow, the traditional banking business model entails the bank taking on credit risk and liquidity risk.⁸ Credit risk is the risk of a borrower being unable to repay

what they owe to a bank. This causes the bank to make a loss. This is reflected in a reduction in the size of the bank's assets shown on its balance sheet: the loan is wiped out, and an equivalent reduction must also be made to the other side of the balance sheet, by a reduction in the bank's capital. If a bank's capital is entirely depleted by such losses, then the bank becomes "balance-sheet insolvent", that is, its liabilities exceed its assets ([Figure 1.2](#)).

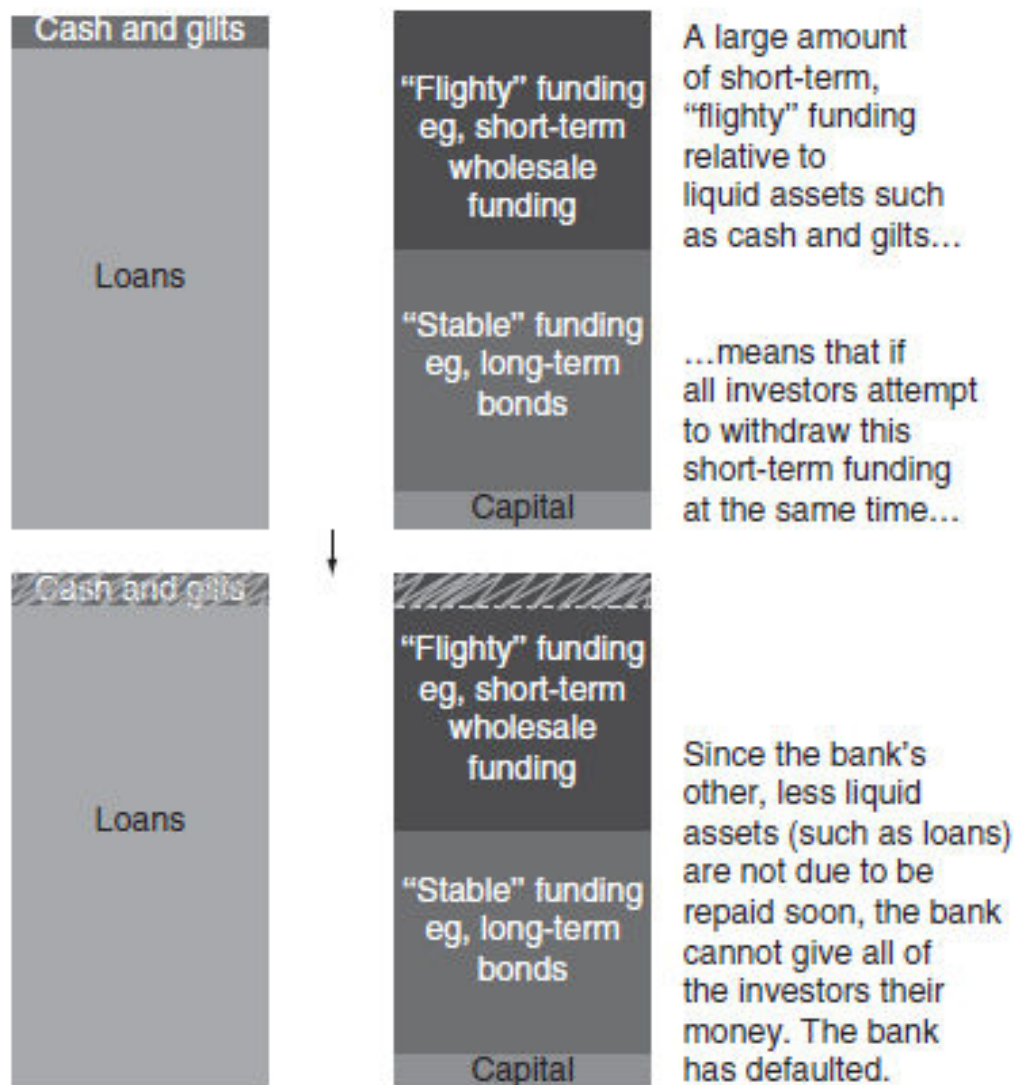
Liquidity risk takes a number of forms. Primarily for a bank, it is the risk that a large number of depositors and investors may withdraw their savings (that is, the bank's funding) at once, leaving the bank short of funds. Such situations can force banks to sell off assets – most likely at an unfavourably low price – when they would not otherwise choose to. If a bank defaults, being unable to repay to depositors and other creditors what they are owed as these debts fall due, it is "cashflow insolvent". This is illustrated in [Figure 1.3](#). A bank "run", where many depositors seek to withdraw funds from the bank, is an extreme example of liquidity risk.

Figure 1.2 Example of solvency problems



The failure of a bank can be a source of financial instability because of the disruption to critical economic services. Moreover, the failure of one bank can have spillover effects if it causes depositors and investors to assume that other banks will fail as well. This could be because other banks are considered to hold similar portfolios of loans, which might also fail to be repaid, or because they might have lent to the bank that has failed.

Figure 1.3 Example of liquidity problems



These risks and others must be managed appropriately throughout the business cycle. The following section considers in more detail how bank capital can mitigate the risk of an insolvency crisis materialising and how a bank’s mix of funding and buffer of liquid assets can help it to prevent or withstand liquidity stresses.

CAPITAL AND LIQUIDITY

The difference between capital and liquidity: an overview

As outlined in the previous section, a bank's capital base and its holdings of liquid assets are both important in helping a bank to withstand certain types of shocks. But, just as their natures as financial resources differ, so does the nature of the shocks they mitigate against. Capital appears alongside liabilities as a source of funding; but, while capital can absorb losses, this does not mean that those funds are locked away for a rainy day. Liquid assets (such as cash, central bank reserves or government bonds) appear on the other side of the balance sheet as a use of funding, and a bank holds a buffer of liquid assets to mitigate against the risk of liquidity crises caused when other sources of funding dry up.

Table 1.1 Key properties of different types of bank funding and assets

Sources of funding (liabilities and capital)	
Seniority:	If a bank becomes insolvent, “senior” liabilities are repaid before “junior” ones. Common equity is the most junior and is the first to absorb any losses.
Maturity:	This refers to the date at which funding can be contractually withdrawn. Some funds can be withdrawn at any time by the borrower (such as current accounts). Others have a fixed term (a two-year bond, say) or are permanent (common shares).
Cost:	The cost is the expected rate of interest that a bank pays on its liabilities or capital. Typically, the more credit risk or liquidity risk a bank takes, the higher the yield it needs to offer in order to attract investors.
Use of funding (assets)	
Liquidity:	This is a measure of the ease with which an asset can be converted into cash. Central bank reserves and “safe” securities like government bonds are considered liquid, while loans to households and firms, or a bank’s buildings, are relatively illiquid assets.
Credit risk:	This is the risk that a borrower will fail to pay what they owe on the due dates. Government bonds (with high credit ratings) are usually considered “low-risk” assets. Loans carry credit risk, the amount varying for different types of borrower and loan product.
Yield:	This is the return (interest and fees) banks earn on their assets. For loans, it is reflected in the interest rate they charge, plus any fees. Typically, lending offers banks a higher yield (but also more risk) than they can get by holding a safer asset.

Importantly, both capital and liquidity provisioning and risk mitigation require the consideration of the details of both the “source of funds” side and the “use of funds” side of the balance sheet. It is useful to consider how the characteristics of various types of typical bank assets and liabilities differ. Some of these characteristics are summarised in [Table 1.1](#).

For instance, if a bank holds more risky assets (such as unsecured loans to households and firms) it is likely to need to hold more capital, to mitigate against the risk of losses in the event that such loans default. And if a bank relies on a high proportion of unstable or “flighty” sources of funding for its activities, such as short-term wholesale funding, then, to avoid the risk of a liquidity crisis, it will need to hold more liquid assets.

The following subsections explain the concepts of capital and liquidity in more detail. While they are considered separately here, in practice, there is often likely to be considerable interplay between risks to a bank’s capital and liquidity positions. Doubts surrounding a bank’s capital adequacy, for example, can cause creditors to withdraw their deposits. Meanwhile, actions that a bank takes to remain liquid, such as “fire sales” or paying more than it would normally expect for additional funds, can, in turn, reduce profits or cause losses that undermine its capital position. Some of the ways in which changes in a bank’s capital position could affect its liquidity position, and vice versa, are discussed at the end of the chapter.

Capital

As noted above, banks can make use of a number of different funding sources when financing their business activities.

Capital can be considered as a bank’s “own funds”, rather than borrowed money such as deposits. A bank’s own funds are items such as its ordinary share capital and retained earnings, in other words, not money lent to the bank that has to be repaid. Taken together, these own funds are equivalent to the difference between the values of total assets and total liabilities.

While it is common usage to refer to banks “holding” capital, this can be misleading: unlike items such as loans or government bonds that banks may actually hold on the asset side of their balance sheet, capital is simply an alternative source of funding, albeit one with particular characteristics.

The key characteristic of capital is that it represents a bank’s ability to absorb losses while it remains a “going concern”. Many of a bank’s activities are funded from customer deposits and other forms of borrowing

by the bank that it must repay in full. If a bank funds itself purely from such borrowing, that is, with no capital, then if it incurred a loss in any period, it would not be able to repay those from whom it had borrowed. It would be balance-sheet insolvent: its liabilities would be greater than its assets. But if a bank with capital makes a loss, it simply suffers a reduction in its capital base. It can remain balance-sheet solvent.

There are two other important characteristics of capital. First, unlike a bank's liabilities, it is perpetual: as long as it continues in business, the bank is not obligated to repay the original investment to capital investors. They would only be paid any residue in the event that the bank is wound up, and all creditors had been repaid. And second, typically, distributions to capital investors (dividends to shareholders, for instance) are not obligatory and usually vary over time, depending on the bank's profitability. The flip side of these characteristics is that shareholders can generally expect to receive a higher return in the long run relative to debt investors.

Expected and unexpected losses

Banks' lending activities always involve some risk of incurring losses. Losses vary from one period to another; and they vary depending on the type of borrower and type of loan product. For example, an unsecured business loan to a company in an industry with highly uncertain future earnings is riskier than a secured loan to a company whose future revenue streams are more predictable.

While it is not possible to forecast accurately the losses a bank will incur in any given period, banks can estimate the average level of credit losses that they expect to materialise over a longer time horizon. These are known as expected losses.

Banks can take account of their expected losses when they manage their loan books. Expected losses are effectively part of the cost of doing business; as such, they should be taken into account in the interest rate that the bank sets for a particular loan. Suppose, for example, a bank lends £1 to 100 individuals and it expects that 5% of its loans will default, and it will receive no money back. For simplicity, it is assumed that the bank has no operating costs and is not paying any interest itself on the £100 of funds that it is lending out. In this scenario, if the bank charges no interest on the loans, then it would expect to receive £95 back from the borrowers. In order to (expect to) receive the full £100 back it would need to charge interest on

each individual's loan. The required interest rate works out to be just fractionally more than the proportion of borrowers expected to default. In this example, then, the bank would need to charge just above 5% on each of the £1 loans in order to (expect to) break even, taking account of expected losses.⁹ Of course, banks are not able to predict future events perfectly. Actual, realised losses will typically turn out higher or lower than losses that had been expected. Historical losses may prove poor predictors of future losses for a number of reasons. The magnitude and frequency of adverse shocks to the economy and financial system, and the riskiness of certain types of borrowers and loans, may change over time. For loans where borrowers have pledged collateral, banks may recover less than they had expected to in the event of default. In the case of mortgages, for example, this would occur if the value of the property falls between the time the loan was made and when the borrower defaults. Or banks may underestimate the likelihood that many borrowers default at the same time. When the economy is unexpectedly hit by a large, adverse shock, such as that experienced during the 2007–9 financial crisis, all of these factors may be at play.

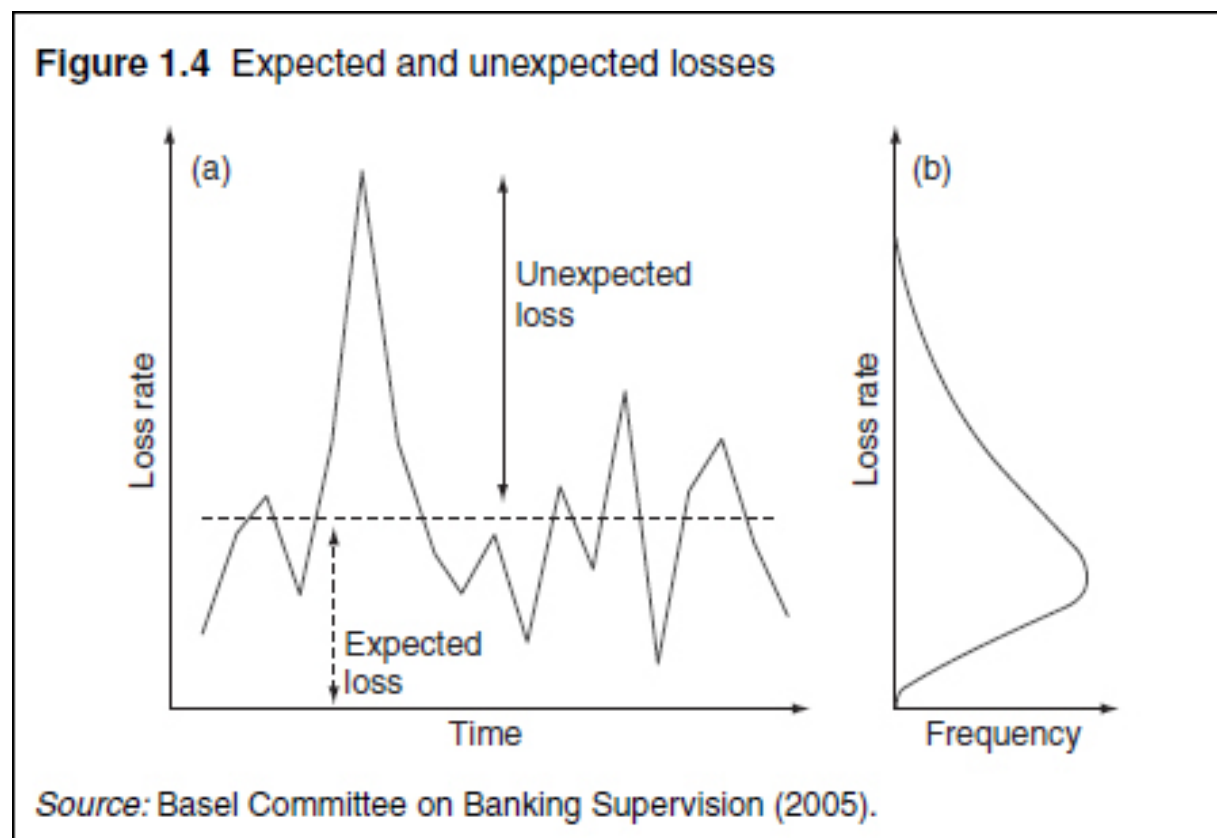
Banks therefore need to take account of the risk that they incur unexpected losses over and above expected losses. It is these unexpected credit losses (the amount by which the realised loss exceeds the expected loss) that banks require a buffer of capital to absorb.

While expected losses can, arguably, be estimated when sufficient past data is available, unexpected losses, in contrast, are by their nature inherently hard to predict. They would include losses on banks' loan books associated with large, adverse shocks to the economy or financial system. [Figure 1.4](#) gives a stylised example of how actual, realised losses can be split into expected and unexpected components. Part (b) shows that for a given period, while the expected loss rate is the expected outcome, in reality losses may be higher or lower than that.

Accounting for losses on the balance sheet

Usually, there is a period between when a borrower has defaulted and when the bank “writes off” the bad debt. When losses on loans are incurred, banks set aside impairment provisions. Provisions appear on the balance sheet as a reduction in assets (in this case, loans) and a corresponding reduction in capital. Impairment provisions are based on losses identified as having been

incurred by the end of the relevant period, but not yet written off. Appendix A discusses developments in the accounting treatment of provisions in more detail. It also explains other accounting principles relevant to understanding bank capital, such as how retained earnings feed into the capital base and the different ways of valuing financial assets.



The leverage ratio

A useful indicator of the size of a bank's balance sheet – and hence potential future losses that a bank is exposed to – relative to its “own funds” (capital) is the leverage ratio. In the context of regulatory requirements, it is usually expressed inversely, as the ratio of capital to total assets.¹⁰ It reflects an aspect of the riskiness of a bank since capital absorbs any losses on the bank's assets: so, high leverage (that is, a low ratio of capital to total assets) is riskier, all else being equal, as a bank has less capital to absorb losses per unit of asset. This could increase the risk of the bank not being able to repay its liabilities. Different definitions of leverage can also include a bank's off-balance-sheet exposures. These include items such as derivatives, security lending and commitments. By capturing these items, the leverage ratio

provides a relatively comprehensive overview of a bank's capital relative to its total exposures. Other metrics for gauging the capital adequacy of a bank, such as the risk-based capital ratio, are discussed in the section on capital regulation.

Liquidity

The concept of liquidity is also intrinsically linked to both sides of a bank's balance sheet. It relates to the mix of assets a bank holds and the various sources of funding for the bank, in particular, the liabilities which must in due course be repaid. It is useful to distinguish between two types of liquidity risk faced by banks (see, for example, Brunnermeier and Pedersen 2008).

- **Funding liquidity risk:** this is the risk that a bank does not have sufficient cash or collateral to make payments to its counterparties and customers as they fall due (or can only do so by liquidating assets at excessive cost). In this case the bank has defaulted. This is sometimes referred to as the bank having become “cashflow insolvent”.
- **Market liquidity risk:** this is the risk that an asset cannot be sold in the market quickly, or, if its sale is executed very rapidly, that this can only be achieved at a heavily discounted price. It is primarily a function of the market for an asset, and not the circumstances of an individual bank. Market liquidity risk can soon result in the bank facing a funding liquidity crisis. Alternatively, with a fire sale, it may result in the bank suffering losses which deplete its capital.

Banks can mitigate these liquidity risks in two ways. First, they can seek to attract stable sources of funding that are less likely to “run” in the event of stressed market conditions. Second, banks can hold a buffer of highly liquid assets or cash that can be drawn down when their liabilities fall due. This buffer is particularly important if a bank is unable to roll over (renew) its existing sources of funding or if other assets are not easy to liquidate. This buffer mitigates both types of liquidity risk.

Liquidity crises: “runs” on banks

A bank “run” is an acute crystallisation of funding liquidity risk and occurs when a significant number of depositors seek to withdraw funding at the

same time. The reason this can happen relates to the “maturity transformation” aspect inherent to traditional banking: short-term liabilities, including deposits, are used to fund long-term loans.

One trigger for a run on a bank is whether creditors have confidence that the bank is “balance-sheet insolvent”, that is, whether it has sufficient capital to absorb losses and to repay its deposits. In this case a depositor who withdraws their funds early will receive all of their money back immediately, while one who waits may only receive compensation up to the £85,000 limit from the Financial Services Compensation Scheme (FSCS) within a target of seven days.¹¹

Liquidity risk can also arise for other reasons. For instance, “contingent risk” arises from scenarios such as an increase in the number of customers drawing down pre-agreed credit lines. In this scenario the bank’s liquid assets are used to meet the contingent commitments to such customers, so that the assets are transformed into loans.

Mitigant (i): stable funding profiles

A bank can adopt a stable funding profile to mitigate against funding liquidity risk and minimise the chances of a bank run happening. Runs are caused by depositors reacting to a fear of losing their money and enforcing their contractual right to withdraw their funding. Stable funding is therefore typically

- diversified across a range of sources,
- sourced from investors or depositors who are less likely to withdraw funds in the event that a bank makes losses,¹² and
- sourced via instruments that contractually lock in investors’ savings for a long period of time.

Banks typically assess the stability of their depositors in three stages: they start with the borrower’s contractual rights, then they assess their behaviour in normal times, and finally they predict behaviour in a stressed market scenario.

In the case of retail deposits (such as households’ current accounts), while account holders may have the contractual right to withdraw on demand, these deposits in normal times may be very stable, not least because retail

depositors have the protection of a deposit guarantee up to £85,000¹³ and are thus less incentivised to monitor the credit quality of the bank. Retail depositors generally withdraw deposits as and when needed, to pay for the goods and services they want to buy. In a stressed environment, such depositors may seek to withdraw their funds to a greater extent due to wider uncertainties. For wholesale unsecured investors, short-term deposits typically have a fixed maturity date. In normal times they would be likely to roll over funding as it matures, but in a stressed market these informed investors are very sensitive to the creditworthiness of the deposit-taking bank and may withdraw substantial volumes of funding.

One measure of a bank's funding profile is its loan-to-deposit ratio. A bank with a high ratio of loans (which tend to be long term and relatively illiquid) to retail deposits could imply a vulnerable funding profile. Although widely used, this is an imperfect assessment of a bank's structural funding profile since certain forms of stable funding, such as long-term debt funding, are excluded.

The 2007–9 financial crisis exposed a number of cases of liquidity and funding problems that resulted from a false assessment of funding stability, especially short-term wholesale funding. And while a maturity mismatch is inherent in the “borrow short term, lend long term” banking business model which plays a vital role in providing credit to the economy, the resulting funding liquidity risk can lead to the failure of a bank. Liquidity regulation, as described later in this chapter, seeks to incentivise the use of stable funding structures and discourage maturity transformation using unstable funding sources.

Mitigant (ii): buffer of liquid assets

The second line of defence against funding liquidity shocks is for banks to hold a buffer of liquid assets. A bank's liquidity resources are cash or assets that the bank can convert into cash in a timely manner and at little cost. They help a bank manage its liquidity risk in two ways. First, they provide a source of liquidity to ensure the bank can meet payments that come due in a stress. Second, their very existence can provide reassurance that a bank will be able to continue to meet its obligations. This reduces incentives for its depositors to “run”.

A bank can convert its buffer into cash either by selling the assets or by pledging them to secure borrowing. In normal times this may be simple to execute, but banks face market liquidity risk so that, in order to be a reliable source of funds across a range of possible market conditions, the buffer should comprise assets that have the best chance of remaining liquid in stressed times. The Basel Committee on Banking Supervision (BCBS) outlines certain characteristics of assets and markets that maximise this chance (Basel Committee on Banking Supervision 2013).

The most liquid assets in the financial system are on-demand deposits at the central bank, also called reserves. They are essentially credit-risk-free and can be used to make payments to counterparties directly. However, they are also low yielding and as such have a significant opportunity cost (that is, representing the “lost” opportunity for income from other, more profitable uses of funds).

Other securities that trade in active and sizeable markets and exhibit low price volatility can also be liquid during a stress, for instance, government bonds and corporate bonds issued by non-financial companies. While these securities may remain liquid, selling such assets during stressed market conditions could entail significant discounts and losses.¹⁴

A key role of the central bank is to provide liquidity insurance to the banking system to help banks cover unexpected or contingent liquidity shocks. Since the crisis, the Bank of England has significantly expanded its Sterling Monetary Framework facilities to ensure that it offers effective liquidity insurance to the banks. At the time of writing the Bank is currently considering further suggestions to improve the efficacy of its liquidity insurance facilities: see the report by Winters (2012).¹⁵

CAPITAL AND LIQUIDITY REGULATION

The previous section explained capital and liquidity and why they are needed to help mitigate the risks that banks take. Building on that, this section provides an overview of the key concepts related to capital and liquidity regulation.

The PRA requires banks to have adequate financial resources as a condition of authorisation. Regulation is designed to help correct market failures and the costs to society that these impose.¹⁶ Specifically, the critical services that banks provide mean that public authorities will provide support

in a crisis, for example by insuring deposits, acting as a lender of last resort, or bailing out banks directly. Expectations of public support in stressed conditions lead to the problem of “moral hazard” whereby banks take on excessive risk, funding their activities with lower levels of capital or liquidity than they would otherwise. Moreover, these expectations mean that depositors and investors do not discipline banks sufficiently, which pushes down on banks’ cost of funding and exacerbates the incentives for banks to take on more risk.

This is a problem because it gives rise to a “negative externality”: excessive risk-taking by banks leads to costs to other parties (the taxpayers that provide for public support). Microprudential regulation seeks to address this negative externality by ensuring that banks manage their activities with sufficient levels of capital and liquidity to reflect the risks that they take.¹⁷ The intention is not to stop banks taking risk – this is an essential part of the economic function that they play – but rather, to ensure that these risks are appropriately accounted for. Consistent with this, the PRA does not operate a “zero-failure” regime: inevitably there will be cases where banks, like other types of firm, fail. In these cases, it is the regulator’s responsibility to seek to ensure that a bank that fails does so in a way that avoids significant disruption to the supply of critical financial services (Bailey *et al* 2012).

In addition to microprudential regulation, which is focused on the specific risks to individual banks, there is also a need to consider the risks stemming from the system as a whole. For example, a buildup in leverage across the system, or an increase in the magnitude of maturity transformation, may increase negative externalities and the riskiness of banks.¹⁸ Examples of such externalities are contagion risks arising through the interconnectedness and common exposures of banks. Building on the microprudential regulatory framework, macroprudential regulation seeks to address such risks (Tucker *et al* 2013; Murphy and Senior 2013).

The following sections provide a high-level overview of the frameworks for capital and liquidity regulation and illustrate how they relate to the risks banks take. Relatively more detail is given on capital regulation since more agreements have been reached regarding the international framework than is the case for liquidity regulation. Typically, regulation takes the form of a requirement specified as a ratio comparing the bank’s financial resources against certain aspects of the bank’s activities, so as to ensure the bank holds what it might conceivably need to stay liquid and solvent. For example, the

ratio could be how much capital banks have relative to their total assets (the leverage ratio outlined above) or the amount of liquid assets that they hold relative to expected outflows as funding expires (a liquidity ratio).

Capital regulation

This section sets out, at a high level, the regulatory framework for capital that is applied to banks in the UK. The framework is embodied in EU law based on internationally agreed “Basel” standards. The EU law had been updated close to the time of writing, reflecting the Basel III standards.

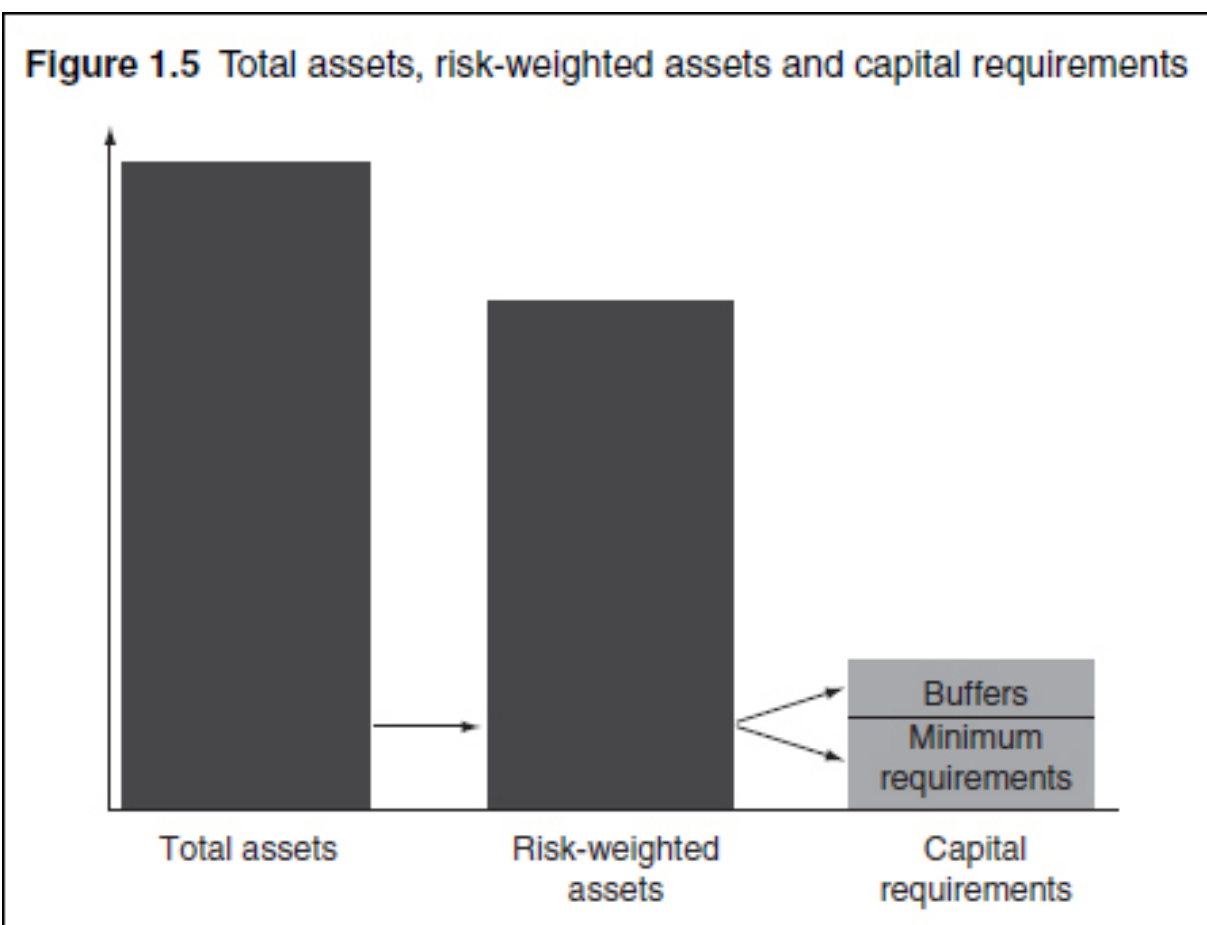
As mentioned above, certain key ratios are useful in thinking about how much capital a bank needs. The previous section defined the leverage ratio as a bank’s capital divided by its total assets. But of course some assets are riskier than others, and each asset class can be assigned a risk weight according to how risky it is judged to be. These weights are then applied to the bank’s assets, resulting in risk-weighted assets (RWAs). This allows banks, investors and regulators to consider the risk-weighted capital ratio, which is a bank’s capital as a share of its RWAs. Another way of thinking about this approach is to consider a different capital requirement for each asset, depending on its risk category.

Banks can alter their ratios by either adjusting the numerator (their capital resources) or the denominator (the measure of risk). For example, they can improve their capital ratio either by increasing the amount of capital they are funded with, or reducing the riskiness or amount of their assets (Tucker 2013). It is common to refer to shortfalls in required ratios in terms of the absolute amount of capital. But altering either the numerator or denominator will change the ratio and reduce this shortfall.

How much of banks’ funding must be sourced from capital?

According to internationally agreed standards (Basel III), banks must fund RWAs with at least a certain amount of capital, known as the “minimum requirements” of capital (Figure 1.5). In addition to the minimum requirements, banks will be required to have a number of capital buffers.¹⁹ These are meant to ensure that banks can absorb losses in times of stress without necessarily being deemed to be in breach of their minimum capital requirements.

Regulatory capital standards comprise three parts or “pillars”. Pillar 1 sets out the capital requirements for specific risks that are quantifiable. Pillar 2 consists of the supervisory review process. It is intended to ensure that firms have adequate capital to support all relevant risks in their business. Pillar 3 complements the other two pillars and includes a set of disclosure requirements to promote market discipline.



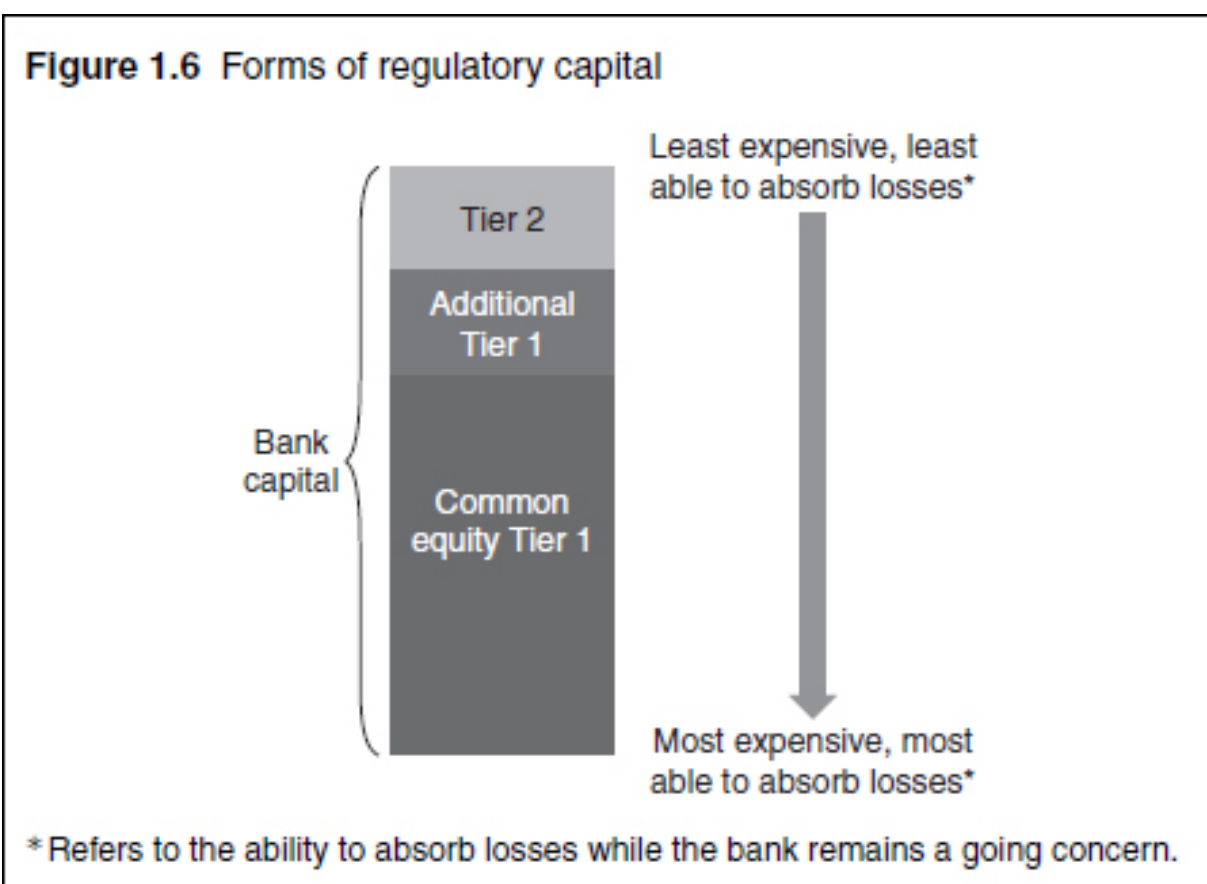
What counts as “capital”?

Banks obtain funding by way of a variety of financial instruments. [Figure 1.6](#) sets out the components of eligible capital resources that correspond to Pillar 1 and Pillar 2 requirements. The main component of a bank’s capital resources is equity, referred to as common equity Tier 1 (CET1). The key aspects of CET1 are the following:

- it absorbs losses before any other tier of capital;
- its capital instruments are perpetual; and

- dividend payments are fully discretionary. Its main constituents are ordinary shares and retained earnings.²⁰

Appendix A explains how retained earnings feed into capital from an accounting perspective. For the purposes of capital requirements, to calculate the amount of CET1, adjustments are made to the accounting balance sheet. For example, items which would give rise to double counting of capital within the financial system, or which cannot absorb losses during stressed periods, are deducted.²¹



Banks can also count, to a limited extent, further instruments in their regulatory capital calculations. So-called Additional Tier 1 (AT1) capital includes perpetual subordinated debt instruments. Basel III standards require that AT1 instruments must have a mechanism to absorb losses in a going concern, for example convertibility into ordinary shares or write-down of principal when capital ratios fall below a pre-specified trigger level.

A bank's regulatory capital resource also comprises "gone concern" capital. Gone concern capital supports the resolution of banks and the position of other creditors such as the bank's deposit customers in bankruptcy proceedings. This includes Tier 2 capital, which is dated subordinated debt with a minimum maturity of five years. In addition, under Basel III, all additional Tier 1 and Tier 2 capital instruments must have a trigger so that they convert into ordinary shares or are written down when the authorities determine that a bank is no longer viable.²²

Liquidity regulation

Microprudential regulation seeks to mitigate a bank's funding liquidity risk – the risk that, under stressed market conditions, the bank would be unable to meet its obligations as they fall due. It aims to achieve this by incentivising – or requiring – banks to have sufficiently stable sources of funding and an adequate buffer of liquid assets. A useful analogy is the risk of a commercial building burning down: regulations require both that the building is built to minimise the risk of fire breaking out (stable funding) and that it has a sprinkler system to extinguish a fire should one occur (liquid asset buffer).²³ In other words: both to reduce the risk of the adverse event occurring and to ensure that, if it does, the harm done is limited.

International liquidity standards have not yet been finalised and implemented. The Basel Committee has agreed the first of two liquidity standards, the liquidity coverage ratio (LCR).²⁴ It is designed to ensure that banks hold a buffer of liquid assets to survive a short-term liquidity stress. A second standard, the net stable funding ratio, is designed to promote stable funding structures and is currently under review by the Basel Committee. The rest of this section characterises the approach of the regulator, although fundamentally this should be closely linked to a firm's own approach in managing its liquidity risk.

Prudential regulators need to consider how adequate a bank's liquidity position would be during a hypothetical stressed scenario. Such a scenario needs to consider the various identifiable sources of liquidity risk in the banking business model, for example: maturing deposits from retail and wholesale customers; triggers for a withdrawal of funds relating to the bank's credit rating; the amount of new lending to customers; and the impact of increased market volatility leading to margin calls and non-contractual

obligations that mitigate reputational risk. The hypothetical stressed scenario is typically of short duration (one to three months) and is a period of time during which the regulator expects each bank to be able to survive with funding from the private markets, without needing central bank support.

Typically, for the stressed scenario, regulators first of all determine the liquidity outflows during the stress period. These depend on the mix of types and maturities of funding that make up the bank's liabilities. Depositors and counterparties are assumed to have varying degrees of sensitivity to the creditworthiness of the bank and behave accordingly. The assumption is that the most credit-sensitive depositors, such as other banks, withdraw funding at a quicker rate than less credit-sensitive ones, such as insured retail depositors. Other liquidity outflows may occur if adverse market movements in respect of derivative positions mean that a bank is obliged to post liquid assets as collateral.

The regulator then defines acceptable liquidity resources, which lie on the asset side of the bank's balance sheet. The regulatory definition of liquid assets stipulates the quality of the liquid assets that banks must hold. The definition in force in the UK regime comprises central bank reserves and high-quality supranational and government bonds. As one bank may lend to another, or hold securities it has issued (unsecured and secured bank debt), the liquid assets of one bank may be liabilities elsewhere in the banking system. These are known as "inside liquidity". In a financial market stress, selling the debt of another bank is likely to prove difficult. Therefore, many regulatory regimes exclude "inside liquidity" from the definition of liquid assets.

The relationship between a bank's capital and liquidity positions

There are a number of ways in which banks can alter their liquidity and capital positions and there is no mechanical link between them. Even so, under certain assumptions, changes in one might affect the other. The purpose of this section is to illustrate some of the ways in which this could happen: in reality, the ultimate impact of a change to one of these ratios will depend on a range of factors.

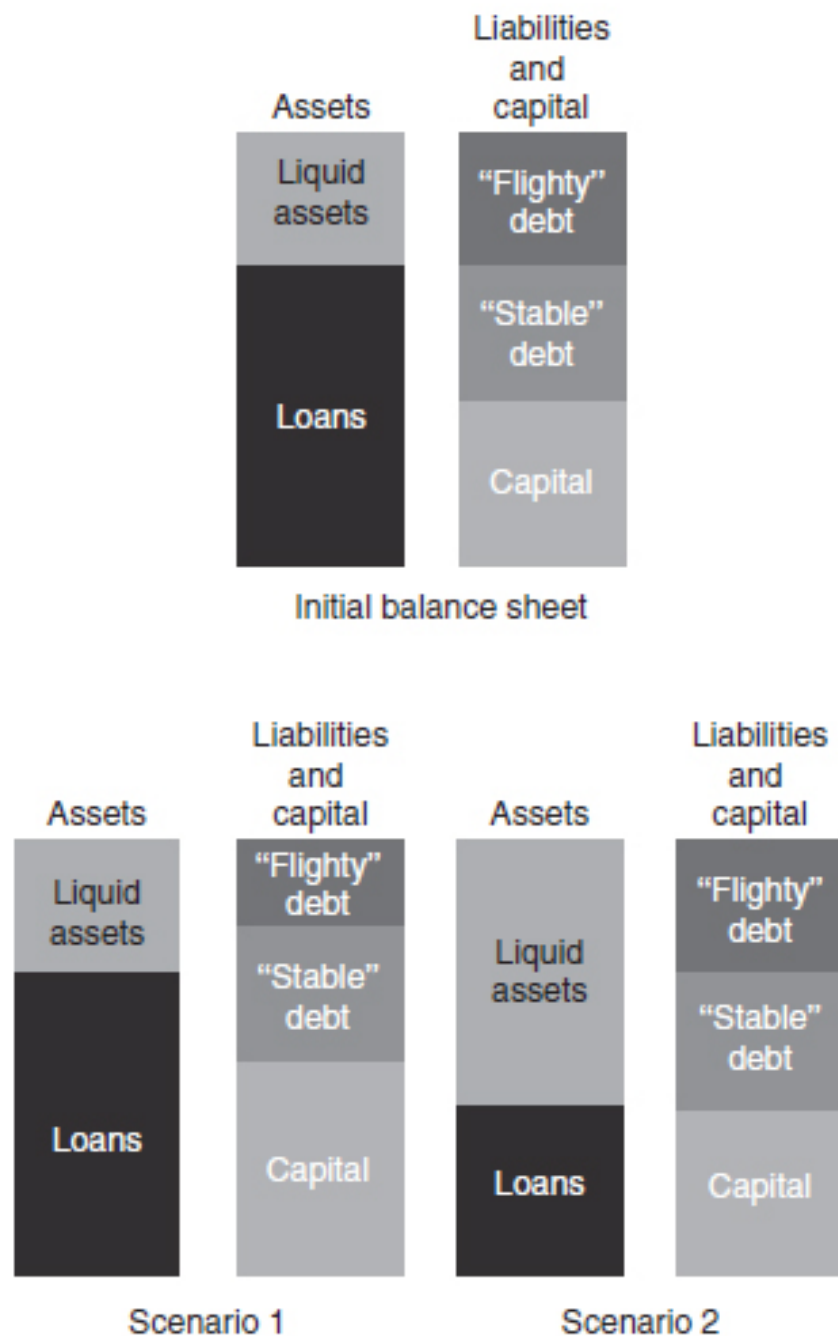
Two scenarios are considered in [Figure 1.7](#). Relative to the baseline case, in scenario 1 the bank increases its risk-based capital ratio (capital as a

share of RWAs). In scenario 2, the bank increases its liquidity coverage ratio (liquid assets held to cover a period of stressed net cash outflows). For both the scenarios considered, changes in the relevant ratios come about via the mix of different types of assets and liabilities, leaving the total size of the bank's balance sheet unchanged:

- **Scenario 1:** the bank increases its risk-based capital ratio by retiring short-term, “flighty” funding from wholesale investors and issuing new equity of the same amount. Its assets are unchanged.

Impact on liquidity: in this scenario, the bank's liquidity position is also improved, since it holds the same amount of liquid assets for a smaller amount of “flighty” wholesale debt. Moreover, as Governor Carney has pointed out (Carney 2013), higher levels of capital gives confidence to depositors and investors who provide funding to banks. With more long-term, stable funding ensured, banks can safely hold fewer liquid assets.

Figure 1.7 Stylised scenarios that represent changes in capital and liquidity ratios



- **Scenario 2:** the bank increases its liquidity coverage ratio by keeping its liabilities unchanged and replacing illiquid loans (once these have been repaid) with liquid assets such as gilts.

Impact on capital: the amount of capital is unchanged but, since the additional liquid assets it now holds are assumed to have a lower risk weight than the loans they are replacing, the capital ratio increases. These examples are intended to be purely illustrative. As mentioned above, the actual impact of a change to one of these ratios will, in practice, depend on a number of factors. If a bank seeks to improve its capital or liquidity position then the total size of the balance sheet may not remain constant, as assumed here. In scenario 1, for instance, if increased capital issuance is associated with a higher aggregate funding cost, then the bank may choose to hold a different amount of loans, either in absolute terms or relative to safer assets. Similarly, scenario 2 assumes that an increase in the liquidity coverage ratio gives rise to an improvement in the capital ratio but one possibility is that, by holding a greater share of low-yield liquid assets, the bank's future profits may be lower (all else equal) and so the potential for future increases in capital via retained earnings would be lower. In addition, the examples do not take account of other important factors such as changes in the perceived riskiness of a bank (and hence its funding costs and profitability) in response to changes in its resilience as proxied by the capital and liquidity coverage ratios.

CONCLUSION

A key function of banks is to channel savers' deposits to people that wish to borrow. But lending is an inherently risky business. Understanding the concepts of a bank's capital and liquidity position helps to shed light on the risks the bank takes and how these can be mitigated.

Capital can be thought of as a bank's own funds, in contrast to borrowed money such as customer deposits. Since capital can absorb losses, it can mitigate against credit risk. In order to prevent balance-sheet insolvency, the more risky assets a bank is exposed to, the more capital it is likely to need. Meanwhile, in stressed market conditions, it is possible that banks find that they do not hold sufficient cash (or assets that can easily be converted into cash) to repay depositors and other creditors. This is known as liquidity risk. A stable funding profile and a buffer of highly liquid assets can help to mitigate this risk.

Banks may prefer to operate with lower levels of financial resources than is socially optimal. Prudential regulation seeks to address this problem by ensuring that credit and liquidity risks are properly accounted for, with the

costs borne by the bank (and its customers) in the good times, rather than the public authorities in bad times.

APPENDIX A: “ACCOUNTING PRINCIPLES 101” FOR UNDERSTANDING BANK CAPITAL

The accounts of a bank are the building block of capital regulation as they present an audited view of its financial condition. This appendix describes some accounting concepts relevant to understanding bank capital, including how provisions and retained earnings feed into the balance-sheet and the capital position.

Balance sheets and income statements

A balance sheet shows a snapshot of the financial condition of a company at a given point in time. A simple example for a bank is shown in [Figure 1.1](#). Assets are recorded in various categories (such as cash and central bank reserves, loans and advances to customers and derivative financial instruments) as are liabilities (for instance, retail deposit accounts and debt securities in issue) and capital (such as ordinary share capital and retained earnings). A balance sheet must balance; resources (assets) must equal the funding provided for the resources (liabilities plus capital). A company's income statement, meanwhile, shows its revenues and expenses (and certain gains and losses) during a given period of time.

Losses, provisions, retained earnings and capital

Accounting rules require that losses on assets such as loans are recognised in the form of impairment provisions as soon as they are incurred, but no earlier.²⁵ Provisions appear in two places in the accounts: on the income statement they appear as an expense, reducing net income; on the balance sheet they appear as a reduction in assets (in this case loans to customers) and a corresponding reduction in capital (specifically, shareholders' equity). The focus on losses arising from past loss events has led to concerns that banks' reported profitability and balance sheets may not reflect adequately the economics of lending. Specifically, a bank recognises the interest income that it receives from a loan as it is earned; but while some of this income will reflect expected future losses that have been “priced in” to the loan (see the

main text for an example), these expected losses are not deducted elsewhere on the income statement; only incurred losses are deducted. This risks overstating the bank's profitability in the period before the losses are incurred.

A recent proposal from the International Accounting Standards Board (IASB) aims to respond to credit deterioration in a more timely fashion by requiring banks to build up provisions earlier in the cycle and in advance of the losses being incurred.²⁶ The proposal recommends a staged approach to establishing loan provisions: from the inception of a loan, provisions would be raised to cover expected losses arising from defaults expected in the next 12 months. This 12-month loss estimate would be updated as the probability of default changes and, where there has been a significant credit deterioration since origination, the provision on the loan would be increased to cover the full lifetime expected loss.²⁷ This approach should result in a more prudent assessment of banks' profitability and capital. As with any forward-looking model, the new approach would rely on some combination of internal models and management's judgements about expected losses.

Along with shareholder equity, retained earnings form a part of a bank's capital base. They also show up on both the income statement and the balance sheet. A simple example helps to illustrate this. Suppose a bank makes a profit of £100 million in a given period, which would be recorded on the bank's income statement. As with other firms, the bank can then choose whether to distribute this money to shareholders (typically in the form of dividend payments) or retain it. If all of the £100 million is retained, then this shows up as an increase in capital resources and, at least in the first instance, as an increase in cash (or central bank reserves) on the asset side of a bank's balance sheet.²⁸

Valuation of financial assets

Financial assets are assets such as cash and deposits, loans and receivables, debt and equity securities and derivatives. The classification of a financial asset held by a bank determines how it is valued on the balance sheet and how it affects the income statement. The loans and receivables discussed above will generally be measured on an "amortised cost" basis with income accrued over time, having deducted any provisions for credit impairment. This is the typical "banking book" treatment. The "trading book" treatment

involves measuring assets on a current market price (that is, “fair value”) basis.

These classifications mean that the market value of a bank’s assets may be lower (or, in some instances, higher) than the amount at which the asset is recorded in the accounts. This can be because there is no requirement to mark the assets to market, although, where the market value is lower, it will also mean the bank has concluded that the fact that fair value is below amortised cost is not evidence that the asset is impaired. In such cases, the accounting equity would overstate the bank’s true capital position and ability to absorb losses.

The authors would like to thank Guy Benn, Stephen Bland and John Cunningham for their help in producing this chapter. This chapter is based on the Bank of England *Quarterly Bulletin* article “Bank Capital and Liquidity” (see Farag *et al* 2013) and as such will not reflect fully the regulatory developments since then.

- 1 The PRA also supervises insurance companies. For more information see Debbage and Dickinson (2013).
- 2 The FPC had existed in interim form since February 2011. See, for example, Murphy and Senior (2013).
- 3 The speech by Governor Carney on August 28, 2013, gives more details and also explores the links between capital and liquidity (Carney 2013).
- 4 For more details on the economic role of banks, see, for example, Freixas and Rochet (2008).
- 5 Of course, other banking activities will also generate income streams and profits. See DeYoung and Rice (2004) and Radecki (1999) for a discussion of some of these other sources of revenues.
- 6 See also Mishkin (2007) for an example.
- 7 Central bank reserves are effectively current accounts for banks. Whereas an individual places their deposits in a commercial bank, a commercial bank keeps its deposits (called reserves) with the central bank. See, for example, Bank of England (2013a).
- 8 While the focus of this chapter is on credit risk and liquidity risk, other risks faced by banks include market risk and operational risk.
- 9 There would of course also be a charge to generate the expected profit on the transaction. For more details on how banks price loans, see Button *et al* (2010).
- 10 For example, in June 2013 the PRA Board asked two firms to submit plans to reach a 3% common equity Tier 1 leverage ratio. See Bank of England (2013c).
- 11 For more information, see <http://www.fscs.org.uk>.
- 12 Deposit protection for retail customers and secured wholesale borrowing are examples of depositors who may face limited losses if a bank fails.
- 13 Per depositor per authorised deposit-taker.
- 14 See Holmström and Tirole (1998) for an exposition on the theory of private and public supply of liquidity.
- 15 The Bank’s response to the Court Reviews can be viewed at http://www.bankofengland.co.uk/publications/Documents/news/2013/nr051_courtreviews.pdf.
- 16 Bailey *et al* (2012) describe the PRA’s role and its supervisory approach.

- 17 For further information on the rationale of prudential regulation, see, for example, Dewatripont and Tirole (1993, 1994) and Diamond and Rajan (2000). Tools for prudential regulation may directly affect the resilience of the financial system to withstand shocks. They may also indirectly affect resilience, through effects on the price and availability of credit; these effects are likely to vary over time and according to the state of the economy. See, for example, Tucker *et al* (2013).
- 18 As discussed in Brunnermeier and Pedersen (2008) and Adrian and Shin (2010), for example.
- 19 While in a general sense capital is said to act as a buffer to absorb unexpected losses, a “capital buffer” may refer to a specific regulatory requirement for a bank to fund its activities with a buffer of capital over and above the minimum regulatory requirements.
- 20 Capital is made up of ordinary shares and reserves. The latter mainly constitutes retained earnings but also includes the share premium account and sometimes other non-distributable reserves. Note that this use of “reserves” as a component of bank capital is distinct from banks’ holdings of central bank reserves (which feature on the asset side of a bank’s balance sheet).
- 21 These include significant investments in the ordinary shares of other financial entities and goodwill.
- 22 For more information on the definition of regulatory capital, see Basel Committee on Banking Supervision (2011).
- 23 See Goodhart and Perotti (2012).
- 24 See Basel Committee on Banking Supervision (2013) for more information on the LCR. The PRA confirmed in August 2013 that it will implement the Financial Policy Committee’s recommendation that banks and building societies should adhere to a minimum requirement of an LCR of 80% until January 1, 2015. This requirement will then rise, reaching 100% on January 1, 2018. See <http://www.bankofengland.co.uk/publications/Pages/news/2013/099.aspx>.
- 25 Note that accountants also use the term “provisions” to describe liabilities for known future expenditures where the exact amount and timing is uncertain, such as mis-selling compensation.
- 26 In March 2013 the IASB – the body responsible for setting accounting standards in the UK – published its third set of proposals to reform the recognition, measurement and reporting of credit impairment losses (“provisions”) on loans and other financial assets.
- 27 This approach could also reduce procyclicality in the system that stems from the current, backward-looking approach, which tends to inflate banks’ balance sheets in upswings and deflate them in downswings. For more details, see Bank of England (2013c, Box 4, pp. 56–57).
- 28 In general, retained earnings will only count as capital for regulatory purposes once they have been audited.

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ALM in the Context of Enterprise Risk Management

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One of the lessons learned by banks during the global financial crisis in 2008–9 was that banks need to have in place a comprehensive risk appetite framework, which is based on the principle that banks should be able to restore their capital and liquidity positions following a stress situation, as it may take years before full access to capital and funding markets is re-established. Regulators picked up on this by implementing, for example, the Capital Requirements Regulation (CRR) and the Capital Requirements Directive IV (CRD IV), which led to new and much stricter capital and liquidity requirements than before. Furthermore, national competent authorities and the European Central Bank have raised the bar significantly with respect to the required quality of the banks' risk appetite frameworks. Therefore, most banks have put a considerable amount of effort in the improvement of their risk and asset and liability management (ALM) processes, frameworks and governance.

In this chapter we provide an insight into the main solvency risks banks are confronted with, and the necessary components of the processes to deal with these. The first section is dedicated to the governance: how can banks make sure that their risk appetite and ALM remains up-to-date, taking into account all the relevant risks given the prevailing macroeconomic and

geopolitical situation, and who should be involved in this process? In the second section, we zoom in on the balance sheet of a typical large bank, and describe the main sensitivities such a bank needs to manage in order to protect itself against potential adverse developments. Then, in the third section, we describe the high-level design of an enterprise-wide risk appetite framework that banks can consider in order to manage these risks, and of the metrics needed for such a framework.

Figure 2.1 The risk management cycle



THE ANNUAL RISK MANAGEMENT CYCLE

Perhaps just as important as the technical design of ALM and risk appetite frameworks, and the conceptual and mathematical approaches used therein, is how the risk process is organised, and who is involved. Risk appetite frameworks need to be maintained and regularly updated as the environment in which banks operate changes and new risks may emerge.

Banks can use a step-by-step risk management approach to identify, monitor, mitigate and manage their financial and non-financial risks. The approach consists of a cycle of five recurrent activities.

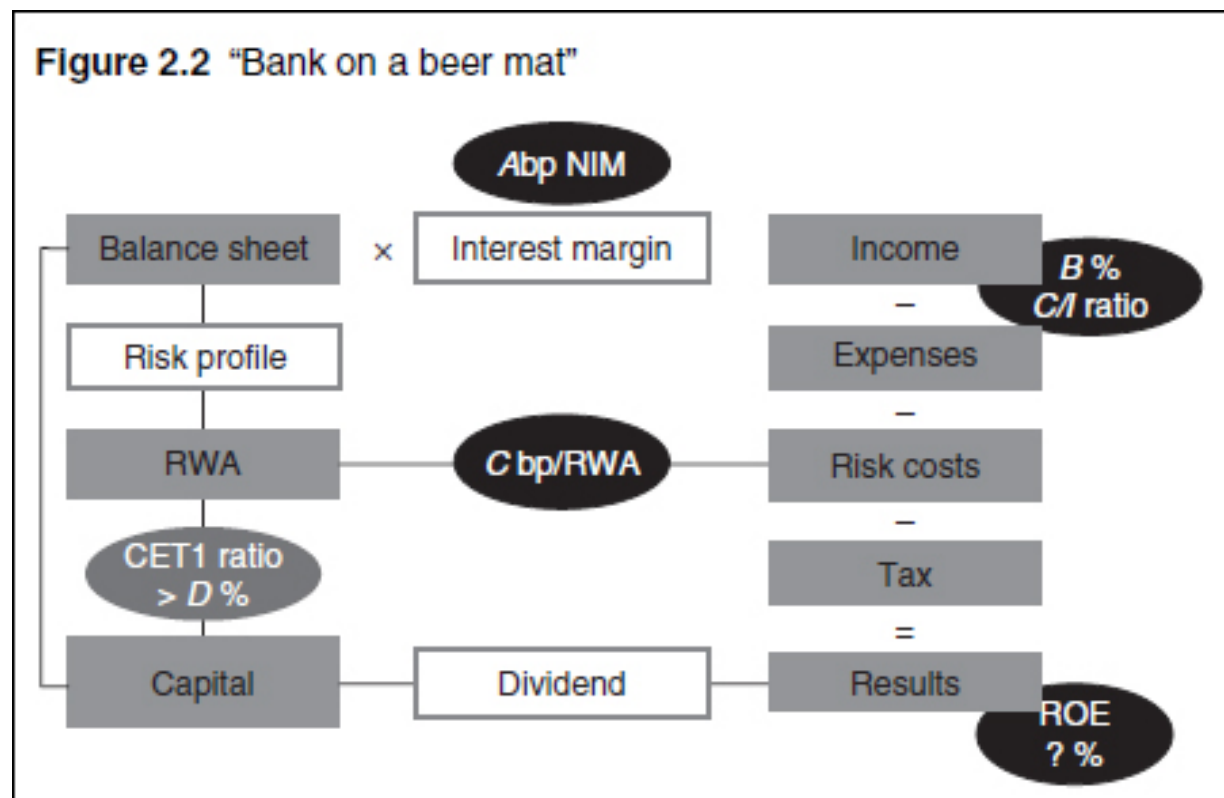
1. Determine the potential risks, by, for example, interviewing a number of senior employees, and/or by reviewing externally published risk assessments.
2. Select the risks that could actually have a significant impact on the capital or liquidity position of the bank.
3. Make sure that the relevant risks are controlled and incorporated into the risk appetite framework. Furthermore, the scenario selection for the stress testing programme should also be based on the risk identification and assessment.
4. Monitor the actual risk profile with respect to the risk appetite.
5. Report the main risk developments to senior management, to enable them to take timely additional measures, if necessary.

The process described in [Figure 2.1](#) recurs in two different ways.

- (i) The identification, assessment and review of the risks and the appetite for these risks are carried out periodically, and potential mitigating measures are updated.
- (ii) The periodical monitoring exercise may indicate that new risks are arising, known risks are changing and assessed risk levels are changing or control measures are not effective enough. Further analyses of these findings may result in renewed and more frequent risk identification and/or assessment and/or a change in mitigating measures.

In each of the steps in the cycle, the involvement of senior management, including board members is of paramount importance. For the risk process and the risk appetite framework to be effective, the board should contribute to, and agree with, the selection, measurement and management of the relevant risks, and the accompanying selection of the stress scenarios to be evaluated. They should also approve the changes to the risk appetite framework, and regularly receive and discuss risk reports. In this way, the board can take informed decisions to further mitigate certain risks, if this mitigation is deemed necessary.

The principle that the design of the overarching risk appetite framework, and the developments of the actual risk profile with respect to the appetite for these risks should be regularly discussed by senior management can be applied in different ways. Various committee structures to achieve this are in place for the various banks, but generally the main risks of the bank are discussed in a dedicated risk committee, a combined finance and risk committee, or in an asset and liability committee (ALCO). In any case, board members are represented in these committees.



THE BUSINESS MODEL AND MAIN RELATED RISKS OF A TYPICAL LARGE BANK

To obtain insight into the main risks and sensitivities with which most banks are confronted, a stylised example of the business model of a typical large bank, such that as shown in [Figure 2.2](#), can be of use.

Usually, the main activity of a bank is to provide loans to customers. To fund these loans, banks either collect deposits from retail customers, or attract funding from the professional funding markets. This leads to a balance sheet as shown in the upper left corner of [Figure 2.2](#).

Normally, banks will ask a higher interest rate to be paid on the loans they grant to banks than the rates they pay on their own funding. This leads to a positive interest margin, which is often the most important source of income for banks. The fact that banks provide loans to customers also means that they accept the risk that clients may not always pay back these loans. To deal with this, banks must quantify their risk profile via risk-weighted assets (RWA), calculated as the exposure multiplied by a risk weight that depends on, eg, the creditworthiness of the client and the amount of collateral received to cover for the loan. Regulators require banks to hold a certain percentage of RWA as capital, to ensure that they have a significant buffer to withstand losses resulting from their activities. These requirements are defined in terms of the minimum Common Equity Tier 1 (CET1) ratio banks need to maintain. This is calculated as the available CET1 capital divided by the amount of RWA.

Returning to the income side: the annual profit and loss (P&L) of a bank will be less than the sum of net interest income, trading and sales income and fee income, because certain expenses and costs need to be deducted from these. Expenses for banks are primarily salary costs, but also include costs related to the necessary infrastructure, as well as those related to non-financial risk events (eg, compensation of clients due to mis-selling, fines from regulators, etc). Furthermore, credit risk costs (ie, costs incurred when clients default on their loans) negatively affect the profit and loss as well. Finally, banks have to pay tax over the gross result

$$\text{gross result} = \text{income} - \text{expenses} - \text{risk costs}$$

which means that the final net profit and loss that can be used to pay a dividend, or to strengthen the capital position is equal to

$$\text{net result} = (1 - \text{tax}\%)(\text{income} - \text{expenses} - \text{risk costs})$$

Finally, a highly relevant indicator of bank performance is the return on equity (RoE), which is calculated as the net results divided by the capital. To remain attractive for investors, banks need to achieve an RoE of approximately 10%, which can be achieved by, eg, making sure that net interest income is high enough and sufficiently stable, the cost-to-income (C/I) ratio and credit risk costs are not too high and required capital does not increase too much.

If everything goes according to plan, a 10% RoE is certainly achievable for a bank with a sound client base and efficient processes. However, there are a number of uncertainties that need to be carefully managed, a process in which the risk management function plays a key role.

Net interest income can, for example, be lower than expected when interest rates remain very low for a prolonged period of time. In that case, maturing assets that originated a long time ago, when interest rates were higher, need to be replaced by new assets with a significantly lower yield. This can be offset by making sure that similar rate reductions occur for the liabilities, but the possibility to fully compensate may be limited if the funding consists mostly of retail deposits. If banks were to make these rates negative, for example, it is likely that a significant percentage of such clients would take their money and put it in their safe or in their piggy bank.

Furthermore, many banks have prepayable assets, such as residential mortgages, on their balance sheets. The related prepayment options can become quite costly for banks in a low interest rate environment, because higher than expected prepayments of loans with higher client rates than the prevailing ones will erode the net interest margin if the related funding is not prepayable.

Sudden increases in interest rates can also be a threat to the net interest income, for example, if banks have to translate these market rate increases into significant deposit rate increases, eg, because new players with no legacy investments enter the market, or because customers may otherwise decide to move their money to other products, such as term deposits.

Interest rate risk can be mitigated by making sure that the interest rate typical maturities of assets and liabilities are to a large extent matched and that prepayment option risk is hedged and properly priced in, and by

reducing the reliance on interest income by, eg, increasing the share of other types of income, such as fee income.

Credit risk costs can be higher than expected due to deteriorating economic circumstances, which will affect the credit risk profile of the bank's clients. Higher unemployment rates will normally lead to a higher default percentage for residential mortgages, for example, while default rates for business lending portfolios typically go up if the economic growth slows down. Furthermore, under the International Financial Reporting Standard 9 rules for loan loss provisioning to be implemented in January 2018, there will be an additional effect from clients for which the credit profile has deteriorated significantly. For these clients, the loan loss provisions must be based on lifetime expected loss instead of one-year expected loss, which means that the provisions must be increased significantly. Obviously, banks will be faced with more clients with a deteriorating credit profile in times of economic downturn.

Credit default risk can be partially mitigated by putting in place sound credit acceptance criteria and policies, while loss risk in the case of a default can be reduced by making sure that sufficient collateral is obtained. Furthermore, credit risk can also be reduced by ensuring a sufficient level of diversification in the credit portfolio, in terms of countries, sectors and asset classes.

Expense risk for banks is primarily related to non-financial risk events. Mis-selling practices and non-compliance with regulations can lead to hefty fines and compensation of clients.

These types of risk can be mitigated by making sure that a bank has developed a strong compliance function and culture.

Not only the numerator of the CET1 ratio (available capital, influenced by, eg, net income) but also the denominator (RWA) is subject to uncertainty. Credit-risk-weighted assets will on average increase in times of economic stress, because deteriorations in the credit profile of the clients will be translated into higher risk weights, and consequently greater RWA. Market-risk-weighted assets will increase as well, as these are influenced by the observed volatility of the market. Furthermore, RWA are subject to regulatory risk, as well as sensitivity to the economic and market circumstances. Since the mid-2010s, the confidence of regulators in the use of internal models for the calculation of RWA has decreased, which has

resulted in proposals for a more standardised approach, with a potentially significant impact on the risk weights for certain asset classes.

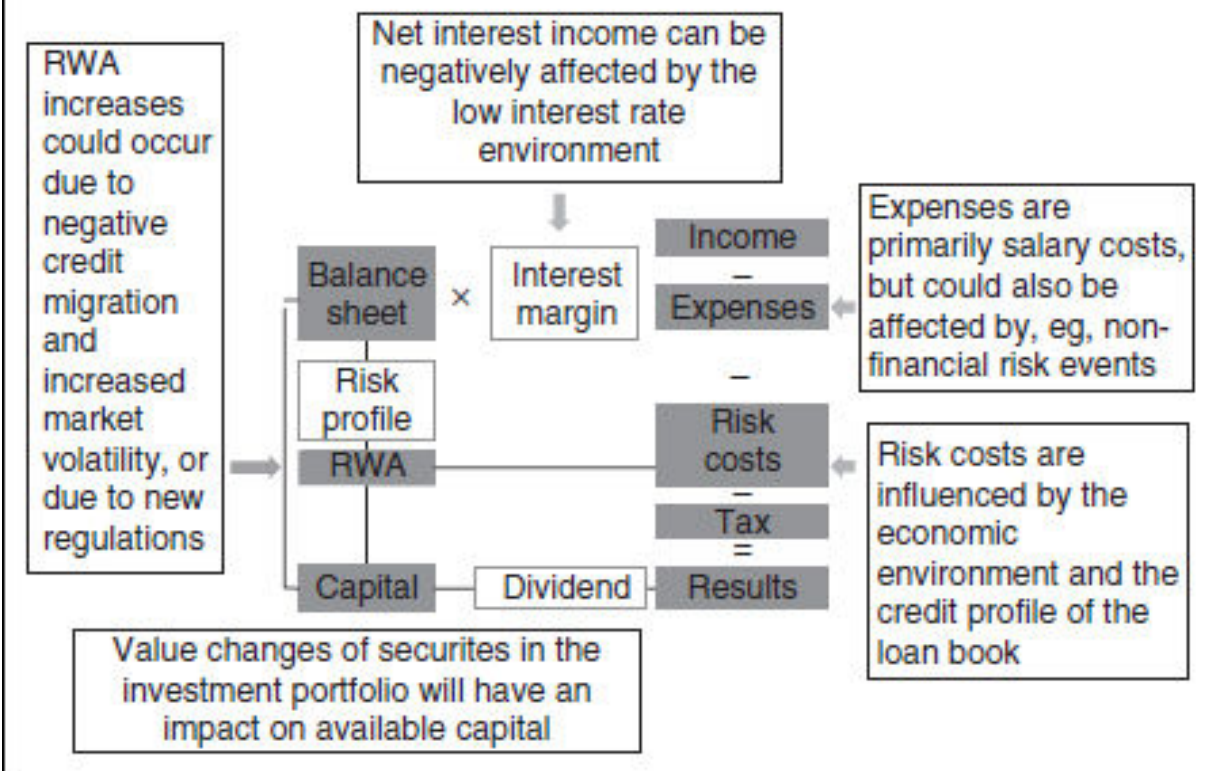
Regulatory risk can be mitigated by reducing the maturities of transactions in asset classes that are likely candidates for significant RWA increases, and by making sure that “originate to distribute” capabilities have been developed that enable banks to offload affected assets from their balance sheets.

Finally, in addition to the risks that could affect the capital ratios, banks are subject to liquidity risk. The liquidity position of a bank can be negatively affected when, eg, the professional funding market dries up for them, or when a significant percentage of customers with a savings account decide to withdraw their money. However, this chapter focuses solely on solvency risk.

A HIGH-LEVEL SOLVENCY RISK APPETITE FRAMEWORK

If the risks mentioned above ([Figure 2.3](#)) are not carefully managed, banks may not achieve their target ROE, may not be able to make the intended dividend payments, and their CET1 ratio may be negatively affected to the extent that regulatory requirements are no longer met. Therefore, it makes sense to implement a risk appetite framework from the starting point that the CET1 ratio should not drop below certain levels in either normal situations or in a (standardised) stress scenario. The evidence, from previous crises, that banks will often not be able to obtain new equity from the capital markets in crisis situations, and that they will have to rely on retained earnings for the restoration of their capital position should be taken into account here.

Figure 2.3 Main risks to the business model of a bank



A possible risk appetite statement for solvency risk could therefore be:

in a standardised 1-in- x years scenario, the CET1 ratio should not drop below $a\%$ throughout the scenario horizon, and should be back at $(a + b)\%$ at the end of the scenario horizon, using retained earnings.

Formulating such a risk appetite statement is of course the first step, but to implement it, and to be able to measure the potential development of the CET1 ratio in a crisis situation, a number of related risk metrics are needed. To determine exactly which risk metrics are needed, it is useful to take a look at the stylised balance sheet of a bank. As will be shown below, the various balance-sheet items influence the CET1 ratio in different ways.

As can be observed from [Figure 2.4](#), a number of different metrics need to be incorporated in order to fully capture the potential impact of a stress scenario on the CET1 ratio.

Figure 2.4 Example of the balance sheet of a bank

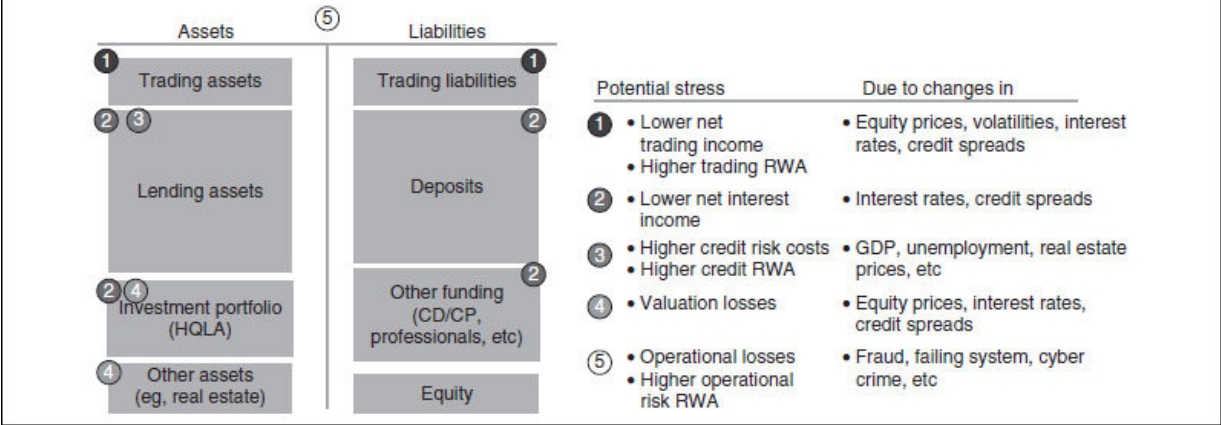
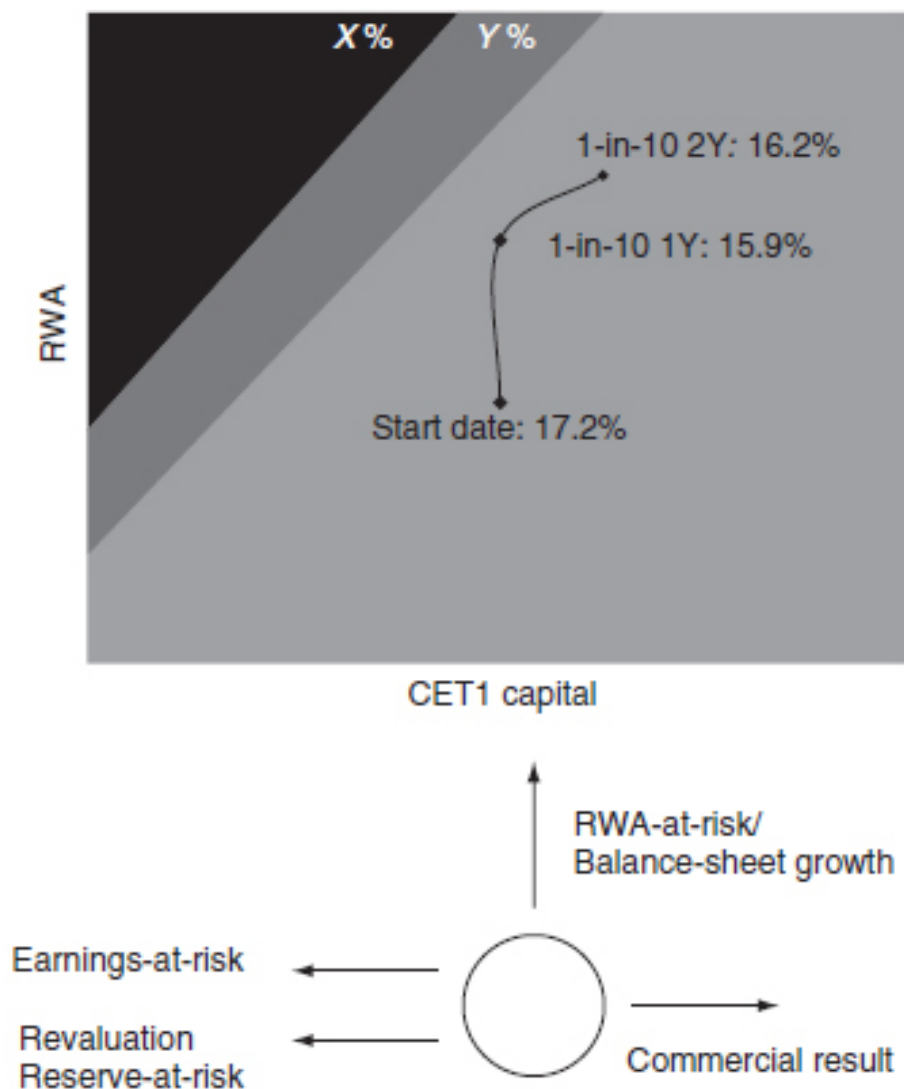


Figure 2.5 Graphical representation of a stress test outcome



The earnings-at-risk value captures the various P&L impacts of a stress scenario, due to, eg, changing interest rates, equity prices, credit losses and operational risk losses. Prevailing accounting rules should be followed for the calculation of these shocks, which means that the calculations for trading books should be based on economic value, whereas impacts for the banking books should be based on book value. This means, for example, that, for the assumed interest rate changes in the stress scenario, the impact on the net interest income rather than the impacts on value should be calculated.

Another metric necessary for the calculation of the impact on available capital is revaluation reserve-at-risk. For the investment portfolio (more

precisely, the part that is classified as “available for sale”), value changes will not affect the P&L, but will affect other comprehensive income via the so-called revaluation reserves. This metric should capture the impact on value of the stress scenario for the assumed interest rate, equity, real estate and credit spread shocks.

As already mentioned, in a stress scenario not only the numerator (available CET1 capital) but also the denominator (RWA) will be affected. Therefore, a risk metric called risk-weighted-assets-at-risk should be included. This should reflect the credit-risk-weighted assets’ increase due to the negative credit migration of the lending assets and the bond portfolio. Furthermore, it should measure the potential value-at-risk increases for the trading book due to increased volatility in the financial markets, which will influence the market risk RWA.

In order not to overestimate the impact of a stress scenario on the CET1 ratio, it is very important that the commercial result is also taken into account. This result is defined as the expected P&L prior to loan loss provisioning, and thus serves as a buffer for all the negative P&L impacts covered in the earnings-at-risk calculations.

In [Figure 2.5](#), the potential outcome of the measurement related to the CET1 ratio risk appetite statement is shown. The graph shows the two dimensions that are relevant for the CET1 ratio: available CET1 capital on the horizontal axis, and RWA on the vertical axis. The curve shows a typical development of the RWA and CET1 capital in a medium severe stress scenario, with RWA going up, and CET1 capital increasing at a slower pace than expected due to negative P&L and valuation impacts. The dark grey, light grey and white areas, respectively, reflect the levels where

- the CET1 ratio should end up at the end of the scenario horizon (white),
- the CET1 ratio may temporarily exist throughout the scenario horizon (light grey),
- the CET1 ratio should not exist at all (dark grey).

Finally, for the risk appetite statement as described in this section to be effective, it should be complemented with a number of more granular supporting risk appetite statements for individual risk types, countries, businesses, etc. Cascading down the overarching risk appetite statements to

all levels of the organisation is an essential part of the risk appetite framework and processes.

Part II

Interest Rate Risk

The New Basel Standards on IRRBB and Their Implications for ALM

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Standard Chartered Group

In this chapter we introduce the technical concepts and definitions applied by interest rate risk in the banking book (IRRBB) managers, describe the evolution of IRRBB regulation and regulatory thinking since the inception of this discipline and, finally, present the most recent regulatory framework set out by the Basel Committee on Banking Supervision (2016a) and some key considerations for its implementation in banking institutions.

First, we show how regulatory philosophies and techniques resulted in the current IRRBB practice and the Basel framework (Basel Committee on Banking Supervision 2016a). Then we show how and why IRRBB has progressively departed from market risk to form a new discipline primarily intended to address structural risk issues. Next we analyse the challenges, considerations and implications of implementing the 12 principles in the Basel framework. These principles are grouped according to their objectives and analysed in terms of different implementation alternatives and their effects on the organisation. Finally, we summarise the key implications of the Basel requirements and the emerging trends at the time of writing. We give some key concepts that are central to understanding the role of IRRBB in a banking organisation.

PILLAR 1, PILLAR 2 AND INTEREST RATE RISK IN THE BANKING BOOK

Risk does not exist “out there”, independent of our minds, waiting to be measured. Instead, it should be seen as a concept that humans have invented in order to understand and cope with the uncertainties of life.

(Slovic 1987)

People learn from experience (Slovic *et al* 2002) in order to evaluate the potential impacts of risks and make decisions such as avoiding those risks, insuring for potential negative outcomes or reducing excessive exposure (Luce and Weber 1986). This leads to two complementary risk management strategies:

1. learning from (third-party) past experience and establishing conservative (risk capital) insurance based on this analysis;
2. pursuing an in-depth understanding of an entity’s specific risk drivers and forward-looking scenarios to define risk mitigating strategies.

When it comes to interest rate risk in the banking book, defined as “the current or prospective risk to the bank’s capital and earning arising from adverse movements in interest rates that affect the bank’s banking book positions” (Basel Committee on Banking Supervision 2016a), regulators have presented these two risk management philosophies for industry consultation (Basel Committee on Banking Supervision 2015) in the form of a standardised Pillar 1 (simplified and conservative) method versus a Pillar 2 (principles-based) approach.

The standardised pillar-1 methodology was designed to protect banks from an interest hike scenario. The regulatory concern was that, as a result of digitalisation, clients may find it easier to switch their current account balances into other products impacting the profitability of banks that lock a long investment tenor for non-maturity deposits. However, at the time of the consultation, one key driver of reduced banking profitability was the low interest rate scenario.

(Virreira 2017)

At the time of the consultation non-interest-rate-sensitive deposits were being invested at progressively lower rates, narrowing the margins of the banking industry and raising concerns on capital origination under negative interest rates (Claessens *et al* 2016).

The principles-based Pillar 2 methodology enhanced the original principles set out in 2004 (Basel Committee on Banking Supervision 2004) and focused on implementing additional governance requirements to derive proprietary models and assumptions, as well as setting out compulsory standardised disclosures and regulatory oversight.

Given the complexity and heterogeneity of banking business models and IRRBB (IIF–IBFed–GFMA–ISDA 2015), any standardised methodology would have resulted in punitive requirements for the banking industry while still failing to address all the potential risks.

After a thorough analysis during the consultation period, regulators and practitioners largely agreed that IRRBB best practice should take the form of a principles-based approach. The Basel Committee on Banking Supervision (BCBS) issued the revised IRRBB principles (BCBS 368) in April 2016 (Basel Committee on Banking Supervision 2016a).

THE EVOLUTION OF INTEREST RATE RISK IN THE BANKING BOOK

By the time BCBS 368 (Basel Committee on Banking Supervision 2016a) was published, IRRBB had already become embedded in the risk management frameworks of most banking institutions. Arguably, IRRBB started as a discipline in 2004 with the “Principles for the Management and Supervision of Interest Rate Risk” (Basel Committee on Banking Supervision 2004). Following their translation of European legislation, the original principles were elaborated in the “Technical Aspects of the Management of Interest Rate Risk Arising from Non-Trading Activities under the Supervisory Review Process” (Committee of European Banking Supervisors 2006).

A similar process took place in other countries, including the publication of: the US Interagency Policy Statement on Interest Rate Risk (FDIC–FRB–OCC 1996), its subsequent financial institution advisory document (FRB–FDIC–NCUA–OCC–OTS–FFIEC 2010); the Australian “Prudential Practice Guide on Interest Rate Risk in the Banking Book” (Australian Prudential Regulation Authority 2008a) and “Prudential Standard: Capital Adequacy for Interest Rate Risk in the Banking Book” (Australian Prudential Regulation Authority 2008b); and the Austrian “Guidelines on Managing Interest Rate Risk in the Banking Book” (Austrian National Bank 2008).

The following decade brought new insight into the IRRBB management practice, constituting a progressive, new risk management discipline that was independent of the original all-encompassing market risk management practice. The European Banking Authority (EBA) evaluated best practice across the industry and prescribed scenarios, stress testing requirements, methodologies for the measurement of IRRBB, governance and capitalisation (European Banking Authority 2015a). At the same time, the BCBS addressed concerns about capital arbitrage across the banking and trading book, defining the boundary between these two books in the first section of the “Minimum Capital Requirements for Market Risk” (Basel Committee on Banking Supervision 2016b).

The responses (IIF–IBFed–GFMA–ISDA 2015; British Bankers’ Association 2015; EBF 2015) to the BCBS IRRBB consultation (Basel Committee on Banking Supervision 2015) marked a turning point in IRRBB thinking. They highlighted the technical challenges in terms of metrics, modelling assumptions and capitalisation that cannot be properly addressed by trading book management methodologies.

IRRBB measurement is highly dependent on assumptions about client behaviours (Leistikow 2014):

[i]n particular, the following balance-sheet items require behavioural assumptions in order to manage their interest rate risk correctly.

- Current accounts and saving deposits: when client rates are not equal to or do not move in line with market rates, changes in market rates will affect the future margin on the deposit portfolio.
- Fixed-rate loans and mortgages: the client’s option to prepay fixed-rate mortgages leads to a reduction in the duration of a fixed income stream.
- Credit card receivables and overdrafts: the ability to raise client interest rates on such products in line with market rates may be limited when market interest rates rise to very high levels. For example, retail overdrafts may be priced differently to corporate overdrafts.
- Fixed-rate loan commitments: the client’s ability to draw on commitments to loans or mortgages with fixed-rate loans may reduce the net interest margin when rates rise. This also applies to loan

commitments where the client's acceptance is uncertain (also known as pipeline or launch risk).

- Capital: different interest rate tenors can be assumed for capital dependent on the business model and the balance-sheet structure of the bank.

To achieve consistency, Basel Committee on Banking Supervision (2015) proposed to use conservative standardised assumptions. The industry contested that

the banks' approaches to IRRBB are adapted to cater for the specificities of, *inter alia*, their different product offerings, market and regulatory environments, business models and customers' behaviour, resulting in justifiably heterogeneous assumptions...[W]hile imposing a standardized methodology leads to comparable numbers in the sense that they are computed in the same way, it does not lead to comparable outcomes.

(IIF–IBFed–GFMA–ISDA 2015)

This discussion resulted in the introduction of Principle 5 (Basel Committee on Banking Supervision 2016a), which sets out the governance of modelling assumptions. This component has profound implications for the management of IRRBB. The key focus moved further from the traditional emphasis on control of exposures towards the management of structural risks under uncertainty by understanding the impact of assumption risk.

Another particularity of IRRBB is that the accrual accounting requires banking book risk management to incorporate earnings sensitivity metrics (in addition to the traditional economic value metrics applied to manage market risk in the trading book). Moreover, IRRBB capitalisation is far from straightforward (see p. 59ff).

Enterprise-wide stress testing (EWST) emerged as an alternative to address the aforementioned complexities by conducting earnings simulation across inter-risk type relationships. Comprehensive Capital Analysis and Review (CCAR) was first introduced in the US Dodd–Frank regulation in 2010 (Board of Governors of the Federal Reserve System 2010) using EWST to analyse capital origination adequacy. At the time of writing, technological constraints were limiting the number of scenarios that organisations could effectively analyse, but technological developments and IT investments (McKinsey 2016) suggested that these methodologies may become more widespread and possibly redefine the boundaries of IRRBB.

THE 12 BASEL PRINCIPLES ON INTEREST RATE RISK IN THE BANKING BOOK

BCBS 368 (Basel Committee on Banking Supervision 2016a) articulated the updated framework in twelve principles (see [Figure 3.1](#)). The technical challenges in prescribing standardised methodologies resulted in the inclusion of Principles 5, 8 and 12. The other nine principles were broadly a new representation of the previous ones (Basel Committee on Banking Supervision 2004).

Figure 3.1 BCBS IRRBB Pillar 2 principles

Principles for banks	1	Identification of IRRBB
	2	Board responsibility
	3	Risk appetite in terms of economic value and earnings
	4	Measurement in terms of economic value and earnings
	5	Validation of modelling assumptions
	6	Accuracy of measurement systems and models
	7	Reporting of structural hedges
	8	Public disclosure of IRRBB
	9	Capitalisation of IRRBB exposures
Principles for regulators	10	Supervisory reporting
	11	Supervisory assessment
	12	Identification of outlier banks

Principles 1–3: risk identification

Principle 1 requires banks' identifying, measuring, monitoring and controlling IRRBB. It effectively mandates including IRRBB as a risk type or sub-type within the organisational risk management framework and setting out policies and procedures that address the different stages in the risk management cycle.

Since IRRBB originates from mismatches in the banking book profile, an extrapolation of this principle requires banks' identifying the drivers of

mismatches in the policies that control business and product approval activities.

- **Gap risk:** this arises from mismatches between the term and repricing profiles of banking book assets, liabilities and off-balance-sheet instruments (the BCBS classifies this driver as parallel or non-parallel risk, depending on the potential losses to different yield curve movements).
- **Basis risk:** this arises from the relative changes in interest rates of financial instruments that may have similar tenor or repricing profiles, but are priced using different indexes.
- **Option risk:** this arises from option derivative positions or the ability of customers to alter the cashflow profile of assets, liabilities or off-balance-sheet instruments (Principle 5 elaborates on the characterisation and treatment of customer options).

Although credit spread risk in the banking book (CSRBB) may be considered beyond the scope of IRRBB (European Banking Federation 2015) as it appears more amenable to traditional market risk controls on price risk (ie, the impact on capital ratios arising from mark-to-market valuation of available-for-sale securities), Principle 1 of this document requires banks to monitor and assess it.

Principle 2 places the responsibility for IRRBB oversight on the board. It enables the governing body to delegate technical management to senior executive committees and requires the third line of defence to conduct independent reviews of the effectiveness of the framework. This implies that the board should have at least one member (or independent advisor) and an audit function who have a good understanding of IRRBB.

Principle 3 requires IRRBB appetite to be articulated in terms of both economic value and earnings metrics. It effectively links the board's responsibilities to the concerns of regulators and investors. Economic value metrics can give an indication of capital adequacy to run interest rate risk mismatches in the current balance-sheet position. Earnings metrics are useful in order to understand potential threats to capital origination and thus to inform investors about the impact of interest rate changes on future

profitability (these two types of metric are further explained in the next section).

Principles 4–6: measuring methodology

Theoretically, it is possible to measure IRRBB with a corporate valuation technique. The value of the firm as an IRRBB metric, however, is challenging to analyse, to report and to articulate into a framework. IRRBB practitioners, therefore, use two simplified metrics – economic value of equity (EVE) and net interest income (NII) – that cover two different perspectives of this comprehensive valuation approach.

- EVE is defined as the present value of assets minus the present value of liabilities. It assumes no ongoing business activity. It does not include cashflows arising from equity, goodwill or fixed assets: it is a simplified gone-concern equity valuation.
- The change in EVE due to an interest rate movement is known as EVE sensitivity.
- NII is the projected revenue driven by the interest rate margin. It incorporates new business origination.
- The change in NII projections due to an interest rate movement is known as NII sensitivity.

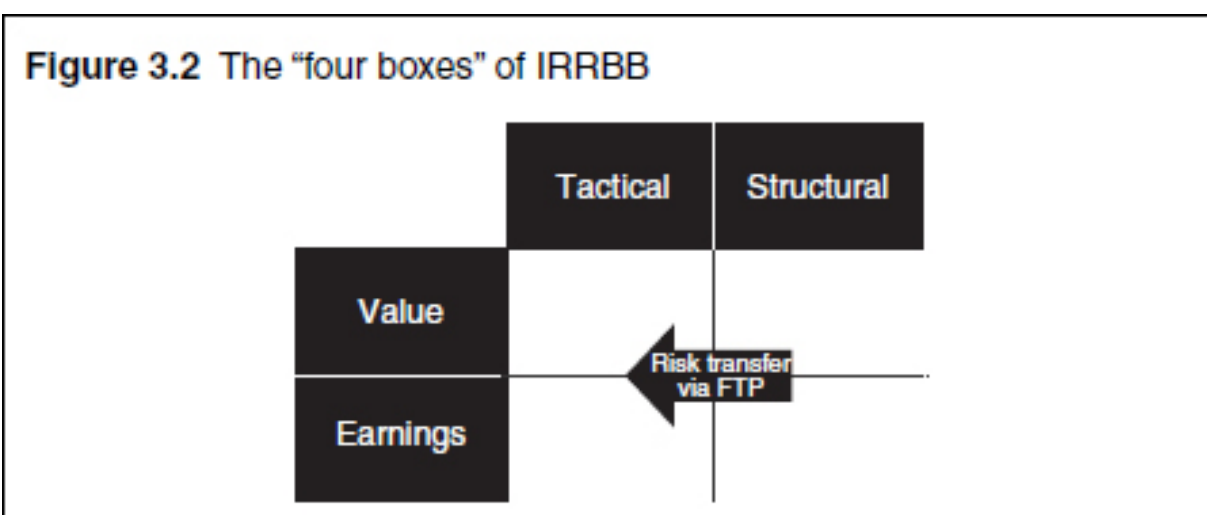
Much discussion has taken place on whether EVE calculation should include cashflows originating from the behaviouralisation of equity (ie, assuming a maturity for the equity), on whether full cashflows or margin-corrected cashflows should be used and on whether risk-free or product-specific discount rates should be applied. These discussions appear irrelevant considering that EVE does not pursue a corporate valuation, but is only a measurement of the interest rate risk profile. EVE sensitivity is useful for understanding the following issues.

- EVE sensitivity is good at capturing interest rate mismatches in the current position, as it incorporates the full balance sheet. It captures the effects of optionality when cashflows are modelled appropriately.

- EVE sensitivity asymmetry is used to analyse optionality. Attribution analysis indicates the products that introduce convexity into the balance sheet.
- EVE sensitivity trends are useful in understanding whether the changes in positions are driven by exposures that can be easily hedged or by structural problems such as increases in convexity.

NII sensitivities, or, more broadly, earning sensitivity metrics including non-NII-accounted effects on revenue, are applied across different horizons, ranging from 12 months (for frequent analysis of IRRBB positions) to 60 or 72 months (for analysis of corporate plans). Earnings metrics capture structural risk drivers such as re-margining issues, and incorporate new production. They therefore are particularly useful for understanding the impact of interest rate movements on the bottom line.

The idea of applying standardised NII metrics has been widely criticised by industry practitioners (IIF–IBFed–GFMA–ISDA 2015). NII sensitivity is useful when it is used in combination with margin strategies incorporating the business into the analysis. When the pricing relationships are understood, NII gives insights into capital origination and future business performance and is possibly the most relevant IRRBB indicator for investors (Préfontaine *et al* 2006).



Principle 4 requires banks to use both regulatory metrics simultaneously across six immediate (currency specific) interest rate shocks and additional

internal scenarios that would incorporate balance-sheet changes, phased interest rate movements, etc.

Practitioners normally place more emphasis on NII (Oliver Wyman 2016) as it enables business strategy and risk management to be aligned. NII requires more assumptions than EVE (NII uses reinvestment and balance-sheet growth assumptions in addition to interest rate scenarios and behavioural balance-sheet profiles), yet it is the analysis of those assumptions what makes NII a powerful tool.

Assumption risk, together with non-repricing balances and optionality or tenor characteristics that are not possible to hedge in the market, originates structural risks (defined here in a wider way than non-repricing balances). Managing structural risks is at the heart of IRRBB, and it is precisely what makes this discipline different from traditional market risk. IRRBB risks can be visualised in terms of four boxes (see [Figure 3.2](#)): the left-hand column reflects risks that can be isolated using funds transfer pricing and can be managed in a specialised function (normally a treasury or balance-sheet management unit) that can actively manage the mismatches of the transferred positions within economic-value and earnings limits. The right-hand column reflects the structural exposures.

Managing the structural exposures requires a cross-functional view: it affects strategic decisions and is normally conducted by a committee, such as the asset and liability committee (ALCO). Typical structural IRRBB decisions include:

- defining the investment tenor of equity, which effectively moves risks from one box to another (increasing investment tenor would normally reduce NII sensitivity and increases EVE sensitivity);
- hedging non-maturing deposits, which effectively locks net interest margins for a defined horizon (and takes into consideration the optionality and uncertainty inherent in these balances);
- securitising or selling portfolios (for example, fixed-rate mortgages generate economic value exposures that may prove difficult to manage due to their prepayment characteristics, and sometimes the easiest way to handle this risk is by securitising these portfolios);
- changing the product mix in the corporate plan (sometimes it is possible to find natural hedges across different commercial products, create new

products or issue debt instruments that mitigate interest rate risks).

Consistent with Principle 5 (validation of models and assumptions), one of the most important tasks for the ALCO is to assess the sensitivity of IRRBB metrics to modelling assumptions. Making structural hedging and strategic decisions requires an understanding of the ranges of EVE and NII depending on the assumptions of the model: understanding the sensitivity of the exposures appears to be at least as important as monitoring them.

Principle 5 also prescribes model analysis and documentation requirements; it places particular emphasis on the modelling of fixed-rate loans subject to prepayment risk, fixed-rate loan commitments, term deposits subject to early redemption risk and non-maturing deposits. Principle 6 complements the controls on model risk requiring independent validation, using the practices set out in EBA guidelines (European Banking Authority 2015a).

The other key components of Principle 6 are rather similar to the requirements on any other risk type: data quality assurance and system controls are natural in information security frameworks, and challenging for organisations, but are not a new topic. Consistent with the regulatory trends at the time of writing (Deloitte 2016), the key challenges in IRRBB concern model risk management, linking risk strategy to business strategy (understanding assumptions and their implications) and linking IRRBB earnings metrics to capital planning and stress testing processes.

At the start of the 2016 financial year, interest rates had been at historically low levels for almost a decade (Claessens *et al* 2016), complicating the calibration of models and their assumptions (Bryman and Bell 2015): historical data did not reflect the scenarios under analysis, and older data did not reflect the social and technological setting, posing a major challenge to BCBS 368 implementation.

Arguably, when complex econometric relationships are difficult to substantiate, the emphasis on dynamic modelling diminishes, and it is worthwhile exploring qualitative techniques (Bryman and Bell 2015). Such a framework, based on uncertain information, would require additional stress of assumptions. This approach faces two important challenges. First, producing qualitative analysis requires involving a wider audience in the IRRBB process (marketing teams and sales forces can make a contribution, but this initiative requires additional training and communication). Second,

stress testing assumptions across several parameters involves running many scenarios and straining IT resources.

At the time of writing, everything indicated that implementing BCBS Principle 5 would change IRRBB practice, pushing the technological limitations to process an increasing number of scenarios and pushing the organisational constraints to communicate them. This would open new business opportunities, aligning business and risk segments (Kotler and Keller 2015), making client behaviour research a common objective across the organisation and maybe coming up with new products that satisfy client needs while creating natural balance-sheet hedges. IRRBB has become mostly about managing structural risks with an organisation-wide perspective.

Principles 7 and 8: reporting requirements

Principle 7 requires the ALCO (or the senior committee in charge) to oversee structural hedging strategies. This concept, as opposed to the broader definition of structural risk discussed in the previous section, normally refers to the investment tenor of non-maturing, non-interest-sensitive liabilities including equity and the non-interest-rate-sensitive portion of non-maturing deposits (NMDs). Banks either transfer via fund transfer pricing at the target tenor or create an ALCO book containing the structural liabilities and their hedges (in both cases applying a ladder profile to enable the hedge to be rolled over on an ongoing basis). BCBS 368 (Basel Committee on Banking Supervision 2016a) prescribes only the principle, leaving banks the flexibility to implement either of these, or other, arrangements.

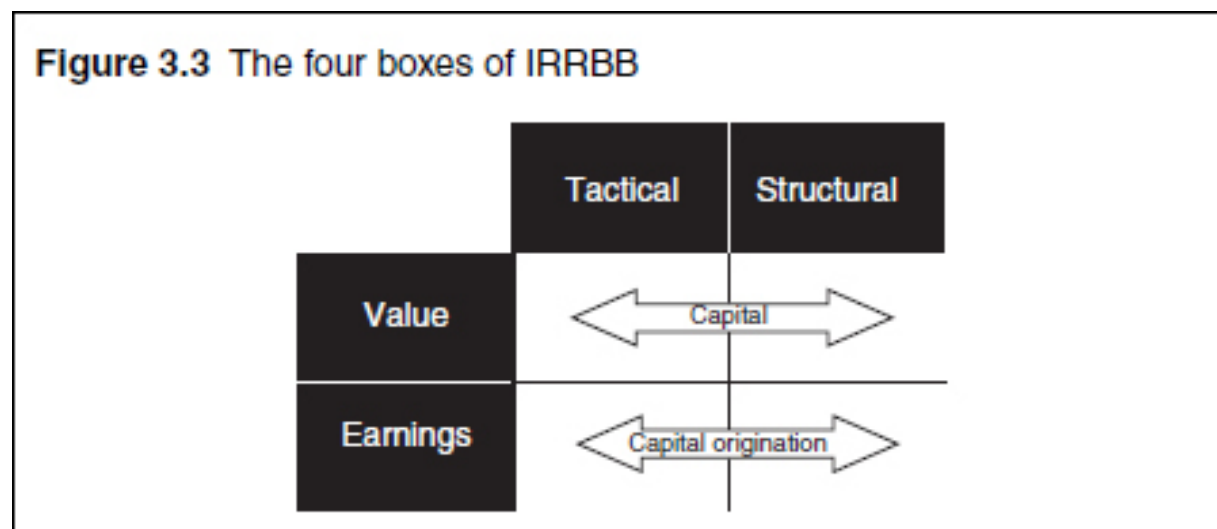
In the BCBS IRRBB consultation (Basel Committee on Banking Supervision 2015), the regulators made explicit their concern with a potential reduction of NMD balances (migration into interest-bearing instruments) under an interest rate up-shift which would result in an economic loss should a bank have invested the NMD portfolio at an aggressively long tenor. Therefore, Principle 7 should be understood in conjunction with Principles 5 and 6: the ALCO (or other senior committee) needs to understand the sensitivity of the behavioural maturity assumed for NMDs before approving the investment tenor or hedging strategy.

Concerning the challenges in calibrating the economic life cycle of NMDs (see p. 49ff), at the time of writing, a common practice was to set a

conservatively short tenor for both the NMD and equities, invest these liabilities consistently and use the assumption to calculate EVE sensitivities, regarding the investment tenors of both equity and deposits in the same way. Arguably, these two assumptions should be different: NMD behaviour is exogenous.

Sophisticated IRRBB frameworks aim to make optionality in the NMD portfolio explicit, assuming their attrition rate and economic life cycle to be interest rate dependent. The key advantages of this philosophy are that it shows the residual EVE risks in the commercial bank, enabling risk managers to monitor the convexity of the balance sheet, and that it helps in understanding whether new products and business initiatives will mitigate or amplify risks.

Both simplified and sophisticated IRRBB frameworks are acceptable under BCBS 368 (Basel Committee on Banking Supervision 2016a). Making residual risks explicit, although it would be impossible to determine them accurately at all times, is a better way to achieve organisational alignment and provoke valuable committee conversations: scenarios, strategies and decisions would effectively move exposures from one “box” to another (see [Figure 3.2](#)) leading to further questions, analysis and discussion.



The introduction of Principle 8 should be seen as an attempt to benchmark EVE and NII exposures, assumptions and governance across the industry: it prescribes a standardised Pillar 3 format that includes these three elements. It requires the disclosure of the average and maximum behavioural maturity of NMDs, the latter being a parameter that affects EVE sensitivities and the

standardised outlier test (SOT) (see p. 60ff). Nevertheless, the granularity of the disclosures is limited, due to the fact that banks would have been required to disclose proprietary or strategic information in other ways (British Bankers' Association 2015). Unfortunately, the standardised disclosures cannot reveal much about the NII sensitivity drivers, the primary focus of investor-analysts (Préfontaine *et al* 2006).

Principle 9: IRRBB capital

Principle 9, Paragraph 74 sets out the considerations for determining IRRBB capital:

[c]apital should be considered in relation to the risks to economic value... for risks to future earnings... banks should consider capital buffers.

This text reflects the traditional Pillar 2a,b framework (Bank of England 2015a) and European Banking Authority (2015a) approach: using economic value metrics to determine capital requirements and earnings metrics to determine capital origination challenges. The IRRBB matrix can be updated to reflect this (see [Figure 3.3](#)).

Apart from the aforementioned IRRBB document for consultation (Basel Committee on Banking Supervision 2015), there are very few sources that attempt prescriptive IRRBB capitalisation methodologies (even Prudential Regulatory Authority (PRA) and EBA regulatory guidance leave room for interpretation). Although the BCBS standard approach was rejected by the industry, the insights provided should be seen as a valuable outcome of this process:

Any capital requirement for IRRBB should consider potential loss of capital not variability risk. (IIF–IBFed–GFMA–ISDA 2015; British Bankers' Association 2015;

European Banking Federation 2015)

In the banking book, accrual streams on customer products are locked in and the economic value of the portfolio is not recognised in [profit and loss] immediately, but rather over the life of the transaction. As a result, the banking book has a stream of accrual flows (embedded value) that will be realised in future periods. When an interest rate shock is applied to the portfolio the resulting change in economic value represents the variability of current earnings, but not the absolute economic value of the portfolio, as it does not capture the embedded value of these locked in accrual flows.

(British Bankers' Association 2015)

These responses do not prescribe a methodology to determine IRRBB capital, but are helpful in guiding practitioners in their IRRBB implementation. Here, these insights highlight the difference between banking book and trading book capitalisation.

Another important consideration when designing IRRBB capital methodologies is the avoidance of double counting exposures in capital requirements and capital buffers.

It was not clear that these issues and their relationships with enterprise-wide stress testing and drivers from other risk types had been addressed at the time of writing: emerging stress test approaches considered top-down and bottom-up processes. European Central Bank stress testing methodologies, for example, stressed different risk types independently (European Banking Authority 2015b) and then recombined them to analyse capital adequacy. Conversely, CCAR's quantitative analysis (Board of Governors of the Federal Reserve System 2017) and the Bank of England's (2015b) stress testing methodologies incorporated the impact of various factors and their interrelationships, combined within scenario forecasts.

Principles 10–12: regulatory requirements and implications

The impossibility of assessing IRRBB appropriately with standardised methods requires regulators to put in place their own processes to analyse risk profiles from different banks. Principle 10 requires regulators to collect granular data, to analyse banks and form their own conclusions. At the time of writing, this principle was expected to create operational and economic challenges for some regulators and some banks. Interestingly, global banks faced a potential requirement to submit very granular information in many different formats to various regulators. The potential operational impacts on different banks depended on the level of flexibility and automation that they incorporated into their processes.

Principle 11 requires supervisors to assess the effectiveness of banks' IRRBB frameworks. This principle was incorporated into regulatory practices well before the issuance of BCBS 368. The EBA produced a comprehensive document (European Banking Authority 2014) detailing all the factors to be analysed, including, for example, business model, organisational arrangements, competition, internal governance, culture, systems and processes, and their consistency. The PRA (Bank of England 2017) elaborated on the EBA document, indicating the rationale for replacing

the internal capital guidance and a regulatory (ie, PRA) capital buffer with additional capital requirements based on the outcomes of the supervisory review. This regulation came into force in 2015 and has been periodically updated.

Following the publication of BCBS 368 (Basel Committee on Banking Supervision 2016a), the European Commission drafted the updated banking prudential requirements directive and regulation known as the Capital Requirements Directive IV (CRD IV) and Capital Requirements Regulation (CRR), respectively (European Commission 2016), placing four requirements on the EBA.

1. To define a standardised methodology (applicable to banks where the proprietary IRRBB framework is deemed unsatisfactory by the regulator, or the bank decides to adopt it) as required by CRD Article 84.4.
2. To update the IRRBB guidelines (ie, an update of European Banking Authority (2015a)), as required by CRD Article 84, Sections 1, 2 and 5.
3. To define standards for IRRBB disclosures as per CRR Article 448.2 (regulatory implementation of Principle 8).
4. To define the standardised regulatory interest scenarios using common assumptions to calculate EVE and supervisory powers to address outlier banks, as per proposed CRD Article 98(5).

The final requirement above is related to Principle 12, which requires regulators to identify “outlier” banks (ie, banks running IRRBB exposures beyond the usual industry ranges). Principle 12 originates from European regulatory practice. The EBA had in place a regulatory SOT (European Banking Authority 2015a), consisting in the ratio of EVE sensitivity (equivalent to 200 basis points or 2% shocks) to equity (net of fixed assets and goodwill). Banks exceeding 20% SOT (ie, the EVE sensitivity exceeds 20% of equity) require additional regulatory analysis.

Principle 12 prescribes a new version of SOT: the worst-case EVE sensitivity across the six regulatory scenarios divided by Tier 1 capital should not exceed 15%.

The use of EVE sensitivity as a regulatory benchmark has been criticised, as cashflow mismatches do not necessarily reflect potential capitalisation

requirements. For instance, EVE sensitivity would

make higher interest rate margin business more capital intensive than low interest rate margin business causing a bank to potentially hold a higher level of capital for no reason apart from having successfully negotiated business at a higher rate than its competitors.

(British Bankers' Association 2015)

Regulatory EVE sensitivity could be prescribed with cashflows net of margins discounted with risk-free rates, but this would still fail to be an indicator of capitalisation requirements due to the difference between variability and loss in accrual accounts discussed on p. 60ff. This simplified EVE metric would deprive the risk management function of some of the structural components that are made explicit with more sophisticated frameworks (defining structural according to the four-box approach presented above).

Ultimately, BCBS 368 opted for a flexible EVE definition, allowing banks to either include or exclude margins from the projected cashflows and to apply either risk-free or product-specific yield curves. Under this definition, SOT is not standardised, however. As the industry and regulators progressively agreed that EVE is only an indicator of cashflow mismatches, the standardisation of the definition became less important: the aim of SOT is only to identify banks that run aggressive mismatches.

Comparing the EVE SOT with the three drivers of IRRBB presented on p. 47ff (gap, basis and optionality risk) shows that this metric ignores some potential risks. Regulators therefore considered metrics that were not included in BCBS 368 in order to set complementary controls relevant to their geographies. For example, the Hong Kong Monetary Authority (HKMA) prescribed a 200 basis point standardised basis risk test in its proposed updated supervisory policy manual (Hong Kong Monetary Authority 2017, Section 4.4.4):

the HKMA assesses the impact of changes in the relationships between key market rates on [Authorised Institutions'] earnings using two hypothetical stress scenarios set out in the Interest Rate Risk IRR Return.... The HKMA will be particularly attentive to those [Authorised Institutions] whose basis risk leads to a significant decline in earnings having regard to the nature and complexity of their activities.

Since capital origination is not captured by EVE, at the time of writing, the inclusion of an NII outlier test was still an option under consideration by

many practitioners. The key challenge in implementing such a metric is the increased number of assumptions underpinning NII sensitivity calculations.

THE IMPLEMENTATION OF INTEREST RATE RISK IN THE BANKING BOOK BASEL PRINCIPLES IN EUROPE

Following the publication of BCBS 368 (Basel Committee on Banking Supervision 2016a) and the subsequent draft of the updated European regulation (European Commission 2016), the EBA released a consultation paper setting out implementation standards (European Banking Authority 2017a). The draft paper presented a clear Pillar 2 approach consistent with Basel's. The consultation provided additional clarity on four topics: calculation of the SOT, review of assumptions, the treatment of CSRBB and the capitalisation of IRRBB.

Calculation of supervisory standard outlier test

Apart from the usual EVE methodology (made explicit in European Banking Authority (2017a, Section 4.5)), EBA proposed clear methodological choices on five topics. EBA assumed that there is a natural non-parallel floor for interest rates: short-term rates can reach –150bp, while 30-year rates are expected to remain positive (medium-term rate floors can be estimated by interpolating). EBA did not give an indication about the rationale behind this assumption, but it is consistent with the minimum 100bp shock presented in Basel Committee on Banking Supervision (2016a). Excluding the plausibility of non-negative long term rates, this methodology may result in IT implementation challenges, depending upon the functionality. Pension obligation and investment asset cashflows are required to be incorporated in the EVE calculation should the interest rate risk not be fully captured in the pension risk framework. The EBA proposed not including positive EVE sensitivities in the currency aggregation methodology (a highly conservative methodology which assumes that losses in one currency will not be partially offset by gains in a different currency). Expected cashflows arising from non-performing loans, net of provisions, are expected to be included in the EVE calculation. In addition to the materiality of this assumption, which varies across the banking industry and geographies, the methodology would require alignment with IFRS 9 implementation. Finally, the EBA rekindled an old discussion about the introduction or exclusion of commercial margins and the

choice of discount rate. Their approach was pragmatic: apply risk-free discount rates, and leave the decision on whether to include or exclude margins up to the banks. Arguably, full cashflows are inconsistent with risk-free curves, but the aim of the metric is to capture mismatches rather than to conduct a portfolio valuation (see p. 54ff).

Review of IRRBB assumptions

As expected, the European Banking Authority (2017a) included several paragraphs requiring controls on modelling assumptions, such as Paragraphs 41, 45, 46, 67, 70, 74, 75, and particularly Paragraphs 102–7, in Section 4, that set out the implementation of Principle 5: “Assumptions on non-maturity-deposits, prepayment risk, fixed-rate commitments and early withdrawal of time deposits”. The same requirements were further elaborated in the EBA stress testing draft paper (European Banking Authority 2017c), which links these elements of control to the stress testing framework and ultimately to the supervisory review draft paper (European Banking Authority 2017b).

Treatment of credit spread risk in the banking book

Oddly enough, Section 2 of the consultative document did not provide a definition of “spread”, which is relevant to understand the scope of credit spread risk in the banking book (CSRBB). Nevertheless, Paragraphs 14, 18, 26 and 67 conveyed that CSRBB is constrained to fair value for “other comprehensive income” instruments as described in Annex I of BCBS 368 (Basel Committee on Banking Supervision 2016a).

Capitalisation of IRRBB

Principle 9, Paragraph 74 of Basel Committee on Banking Supervision (2016a) was reflected in Paragraph 24 of European Banking Authority (2017a) as follows:

In their ICAAP [Internal Capital Adequacy Assessment Process] analysis of the amount of internal capital required for IRRBB, institutions should consider:

- (a) Internal capital held for risks to economic value that could arise from adverse movements in interest rates; and

- (b) Internal capital needs arising from the impact of rate changes on future earnings capacity, and the resultant implications for internal capital buffer levels.

The first part of this paragraph, requires the impact of the fair value of “available for sale” portfolios (including CSRBB) on capital ratios, among other economic loss drivers, to be addressed.

The second part should be understood in conjunction with Section 4.4.4 of the same document (particularly Paragraphs 94, 96, 97 and 100). The EBA effectively requires banks to incorporate IRRBB into their stress testing programme, to identify IRRBB vulnerabilities and to incorporate these drivers into enterprise-wide-stress-testing scenarios and the subsequent results into capital buffers.

To further understand the proposed capitalisation and stress testing approach, it is important to read the IRRBB consultation (European Banking Authority 2017a) in the context of the consultations on supervisory review (European Banking Authority 2017b) and stress testing (European Banking Authority 2017c). The latter defines the IRRBB stress testing requirements in Paragraphs 164–170, and the incorporation of these drivers into scenario design in [Sections 4.6.3–5](#). On the other hand, the Supervisory Review and Evaluation Process (SREP) guidance (European Banking Authority 2017b) requires these stress test results to be used in ICAAP as per Paragraphs 121 and 122.

These three documents, seen together, indicate a convergence of the European and US regulatory views about Pillar II practices (see Board of Governors of the Federal Reserve System 2013).

A FINAL NOTE ON INTEREST RATE RISK IN THE BANKING BOOK

IRRBB is not a new concept; it is a discipline that has been in place for decades and it is still evolving; technology, regulation, economic scenarios and changes in client behaviours are continuously reshaping IRRBB practice and redefining its boundaries.

Unlike interest rate risk in the trading book (market risk), IRRBB metrics are highly influenced by client behaviour assumptions. A good IRRBB framework, therefore, should pay attention to both the metrics and the underlying assumptions behind the calculations. One key change in BCBS

368 is the requirement for banks to set up robust stress testing of assumptions and model validation controls.

IRRBB frameworks can be set up with simplified models and generic conservative assumptions, but this should be done with some caveats: there is no such standardised framework that can capture all the potential IRRBB risks. For example, the BCBS standard approach (Basel Committee on Banking Supervision 2016a) produces conservative results for interest rate hike scenarios, but may underestimate the risks of negative interest rate shocks. There is no substitute for a culture of understanding, analysing and monitoring business and risk drivers at all times.

IRRBB risks may be difficult to measure accurately: historical data may be insufficient or not reflect future scenarios; client behaviours could be difficult to predict, etc. This, however, should not discourage banking institutions from attempting to set up a robust framework: IRRBB is not only about producing a risk measurement, but also about understanding business and risk drivers and their relationships.

At the time of writing, technology and IT platforms were in many cases constraining the ability to cross-analyse IRRBB, budgeting, planning and EWST results (it was believed that these processes would converge, resulting in an in-depth analysis of the balance-sheet dynamics). As this cross-analysis would produce valuable insights for the strategic process, banks may benefit from setting up a robust IRRBB framework that is consistent with other key organisational processes.

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Measuring and Managing Interest Rate and Basis Risk

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Interest rate risk is the exposure of a bank's financial situation to variations of interest rates. It is principally driven by the maturity mismatch embedded in the typical balance-sheet structure of banks.

This chapter will illustrate the main tools for measuring interest rate risk and provide hedging examples.

We first introduce the earnings and economic values approaches to measure interest rate risk, and then illustrate the regulatory treatment according to the Basel Committee framework before introducing basis risk.

Risk measurement techniques will be treated in depth, with a comparison of the relative strengths and weaknesses.

A framework for yield curve construction that takes into account the evolutions following the financial crisis (OIS discounting and importance of basis risks) is given. The chapter concludes with operational examples of hedging with swaps.¹

THE EARNINGS PERSPECTIVE AND THE ECONOMIC VALUE PERSPECTIVE

We can identify two possible perspectives for measuring and managing interest rate risk: the earnings approach and the economic value approach.

The first approach concentrates on the effects of interest rate movements on the bank's net interest income (NII), over short time horizons that could span from one to two years.²

The earnings perspective could fail to indicate the long-term impacts of interest rate movements, as mismatches might be hidden beyond the horizon of the analysis (think of a bank where all interest rate exposures are locked-in for the first year, but a significant fixed-rate funding position of, say, a five-year fixed-rate loan must be rolled over after year 1).

In order to have a comprehensive view of the long-term effects of changes in interest rates, a number of banks adopt the economic value approach, which is based on the present value of all cashflows.

Many banks, especially those of smaller size, give priority to earnings over economic value, as the latter does not essentially affect the financial statements. Even though a synthesis of the two approaches could be problematic, it is advisable that an “optimal” management of interest rate risk includes a combined analysis from both an earnings perspective and an economic value perspective.

Points to consider:

- the sensitivity of earnings directly targets the income statement;
- sensitivity of earnings is simple to implement (for example, gap analysis);
- earnings projections and volatility are one aspect requested under Basel Pillar 1 reporting;
- earnings analysis does not consider effects beyond the projection horizon and may lead to concentration of mismatches in the medium to long term;
- the economic value perspective addresses the effects of interest rate changes in the long term as well as in the short term;
- the sensitivity of economic value could serve as a lead indicator for the impact on prospective earnings;

- the economic value perspective does not focus on the time distribution of cashflows, which are condensed into a single, present-valued figure; it is difficult to allocate economic value effects to prospective income statements.

OVERVIEW OF THE REGULATORY TREATMENT OF INTEREST RATE RISK

The regulatory treatment of interest rate risk depends on whether positions are classified in the banking book or in the trading book according to the applicable regulation. The “minimum capital requirements” published by the Basel Committee in January 2016 identify the instruments and type of activities that qualify as “trading book”.³

For positions belonging to the trading book, a minimum amount of regulatory capital should be held to cover interest rate risk.

The banking book is composed of all positions which do not fall under the trading book definition. In this case, the related interest rate risk would not be subject to a specific capital requirement under Pillar 1, but would be treated under Pillar 2.

The discipline for interest rate risk in the banking book is laid down by the “standards on interest rate risk in the banking book” (Basel Committee on Banking Supervision 2016). As for general principles, risk systems must address the interest rate risk related to all assets, liabilities and off-balance-sheet positions, in particular being capable of measuring risks using both an earnings and economic value approach under a wide and appropriate range of interest rate shocks and stress scenarios. Furthermore, the standards mention explicitly that banks should address interest rate risk in the banking book in their the risk appetite statements, from both the economic value perspective and the earnings perspective. Interest rate risk in the banking book is included in the Internal Capital Adequacy Assessment Process (ICAAP), in the context of which banks are responsible for self-assessing the level of capital they hold and for ensuring its sufficiency to cover such risk. Within Pillar 2, capital adequacy for interest rate risk in the banking book is mostly to be assessed with from the economic value perspective (ie, capital should be sufficient to cover variations in economic value) whereas decreases in future earnings may be addressed by capital buffers. Each national banking supervisor should make public the criteria according to

which they would assess a bank as having an undue exposure to interest rate risk in the banking book (ie, as an “outlier bank”). Additional capital and ad hoc mitigation can be requested from the “outlier” banks by the national regulator. In particular, a warning sign to national supervisors would be the theoretical decline of the economic value by more than 15% of the Tier 1 capital, under the application of six regulator-prescribed interest rate shocks (Basel Committee on Banking Supervision 2016).

BASIS RISK

There are several situations where banks can be exposed to basis risk. Despite the existence of several definitions, in general, basis risk derives from the imperfect correlation between the rates to which different instruments are indexed, even if their coupon structure is similar or identical. Should rates not move in sync, then a mismatch would arise.

Basis risk can first of all appear when instruments refer to different types of rates. One example could be a loan priced on the “prime rate”, but funded by a liability indexed to the Euro Interbank Offered Rate (Euribor) or London Interbank Offered Rate (Libor). The prime rate would be adjusted only by discrete amounts (eg, 50 basis points (bp)) and its differential with money market rates could drift substantially.

Similar problems may occur, for instance, when adjustable rate loans are indexed to the average cost of funding of banks, to the extent that the latter is not reflected promptly in the asset coupon. This is typical when the funding cost is indexed to industry averages that cannot be recalculated daily.

Basis risk can also stem from the liability side of the balance sheet, due to rates on retail customer deposits, typically lower than market rates. Fund transfer pricing (FTP) systems typically represent retail rates as portfolios of market rates that replicate the actual exposures based on historical correlations. While this allows for integrated management of basis-induced mismatches in the context of an interest rate book, the true repricing characteristics should not be hidden by the measurement system.

Finally, basis risk can emerge when banks are exposed to spreads between floating rates indexed to different repricing schedules, or to the same repricing schedule in different currencies. Such spreads are quoted for the related hedging derivatives, eg, a floating–floating swap paying the three-month (3M) and receiving the six-month (6M) Euribor rate, or a cross-currency swap exchanging euro payments with US dollar payments with six

months' floating interest exchanges. The value of such instruments is influenced by the market quotation of the “basis spread” between the reference rates of the two swap legs.

The global financial crisis dramatically increased the volatility of quoted basis spreads which were previously essentially stable. Since mid-2007, basis spreads have become a fundamental variable and a top priority for the management of banking book risks.

Table 4.1 Balance sheet

Assets	(€ bn)	Liabilities	(€ bn)
a. Interbank deposits and reverse repos	2.6	a. Retail deposits	18.3
a1. O/N maturity	1.0	a1. Demand deposits	3.5
a2. One week maturity	0.5	a2. Saving deposits	14.8
a3. Two months' maturity	1.1		
b. Bills and commercial paper	5.7	b. Interbank deposits and repos	10.6
b1. One month maturity	2.6	b1. O/N maturity	3.7
b2. Five months' maturity	3.1	b2. One month maturity	6.9
c. Bonds	5.7	c. Bonds	3.0
c1. Fixed rate: portion maturing below one year	2.6	c1. Fixed rate: portion maturing below one year	0.0
c2. Fixed rate: portion maturing above one year	1.1	c2. Fixed rate: portion maturing above one year	1.0
c3. Floating rate	2.0	c3. Floating rate	2.0
d. Loans	18.6	d. Equity	3.2
d1. Fixed rate: portion maturing or amortising below one year	3.3		
d2. Fixed rate: portion maturing or amortising above one year	5.1		
d3. Floating rate	10.2		
e. Participations	1.7		
f. Building and equipment	0.8		
Total	35.1	Total	35.1

Notes: O/N denotes overnight. Hedging swaps: (1) one 2Y micro hedging interest rate swap receiving fixed and paying floating Euribor 6M (liability hedge); nominal amount: 1.0 bn. (2) One 5Y macro hedging interest rate swap receiving fixed and paying floating Euribor 3M; nominal amount: 3.5 bn.

GAP ANALYSIS

Gap analysis measures the effect of a shift in yield curve on the NII over a short-term horizon: one or two years. Despite a move towards simulation techniques, gap analysis is still widely used, especially in small to medium-sized banks.

A gap is the difference, in one given time bucket, between “interest rate sensitive” assets, liabilities and off-balance-sheet items. An item is said to be “interest rate sensitive” if it matures, if it amortises or if its coupon can change during the time bucket under consideration.

When rate sensitive assets are bigger than rate sensitive liabilities, the gap is positive in the time bucket under consideration (it is said that the bank is “asset sensitive”). If market rates increase, the NII will be positively affected, as more assets than liabilities are repriced at higher rates. A negative gap (“liability sensitive” position) would have the opposite effect.

For example, let us assume we want to prepare a gap analysis, with a one-year horizon, for a bank with the balance sheet in [Table 4.1](#).⁴

We use the following time bucketing: overnight (O/N), O/N–1M, 1M–3M, 3M–6M, 6M–12M, above 1Y.

The first step is allocating assets and liabilities to each bucket in which they are “rate sensitive”. Each item is rate sensitive if its rate may change in one bucket; this is the case if it matures, if it produces a principal payment (eg, amortisation of a loan) or if its floating-rate re-fixes.

Assets are allocated as follows.⁵

- **Interbank deposits and reverse repos:** these are slotted into the bucket corresponding to their residual maturity, when the bank is supposed to reinvest the proceeds at new rates.
- **Bills and commercial paper:** same approach as above.
- **Bonds:** fixed-rate bonds are allocated to their residual maturity. Assuming that €1.4 billion mature in four months and the remaining €1.2 billion in eight months, they will be slotted into the fourth and fifth bucket, respectively. Floating-rate bonds are allocated to the date on which their next floating coupon re-fixes, irrespective of their residual maturity. If the bank has invested in a €2.0 billion quarterly floater and the first coupon re-fixes in 20 days, it will be slotted into the bucket “O/N–one month”.⁶
- **Loans:** these are treated like the bonds. Amortisations are slotted in as if they were “partial maturities”. Assuming that €1.5 billion fixed-rate loans mature in 25 days and €1.8 billion amortise in 205 days, they will be allocated, respectively, to the overnight–one month and to the six months–twelve months buckets. Assuming that floating-rate loans will re-fix their next semi-annual coupon in four months, they will be allocated (€10.2 billion) to the three months–six months bucket, irrespective of amortisations.
- **Participations:** these are not rate sensitive.
- **Building and equipment:** these are not rate sensitive.

Liabilities and equity can be allocated as follows.

- **Retail deposits:** these instruments have undetermined maturity, depending on customers' behaviour, and their rates are not necessarily linked to market rates. The allocation to buckets is model dependent. For illustrative purposes, we assume that 50% of deposits (€9.15 billion) are attributed a one-month maturity, 25% (€4.575 billion) a nine-month maturity and the remainder is considered not to be rate sensitive within one year. In practice, replicating portfolios for retail deposits are often represented with rolling maturities (or “tractors”). In our example, equal slices of the overall 25% with a nine-month modelled maturity are represented with a residual maturity of one month, two months, and so on, up to nine months and allocated to the respective bucket.⁷
- **Interbank deposits and repos:** these are slotted according to their residual maturity.
- **Bonds:** two bonds for a total of €3 billion have been issued with a residual maturity of two years (€1 billion fixed rate and the remainder floating rate 3M). The floating-rate bond will re-fix its next coupon in two months from the analysis date and is slotted into the 1M–3M bucket. The fixed-rate bond has been swapped with an interest rate swap (IRS) paying a floating rate of Euribor 6M, with its first re-fixing due in 10 days from the analysis date. Hence, €1 billion (corresponding to the floating leg of the swap) is slotted into the O/N–1M bucket, and €1 billion, (corresponding to the fixed-rate bond) is slotted into the bucket above one year.
- **Equity:** this is not rate sensitive.

The macro hedging swap has a receiving leg maturing above one year and a floating paying leg. Assuming that the latter re-fixes two months after the analysis date, a €3.5 billion liability is slotted in the 1M–3M bucket.

The total assets and liabilities in [Table 4.2](#) corresponds to the actual assets and liabilities in the balance sheet plus the nominal of the hedging swaps.

The impact on the NII of the bank of a hypothetical parallel shift in the yield curve is a function of the gaps, of the absolute size of the shift and of the remaining time to the end of the horizon of the analysis (one year in the example) according to the formula

$$\Delta \text{NII} = \text{GAP}_i T \Delta r$$

where T is the time to horizon in years and r is the absolute shift in interest rates.

The interest rate shock is typically a parallel shift, but it could be estimated based on several methods, such as historical scenarios, macroeconomic forecasts or expert advice.

The calculation can be implemented based on either the gap or the cumulated gap, as in [Table 4.3](#), where the gap in each bucket is assumed to be concentrated on the middle of the bucket (eg, 60 days for the 1M–3M bucket).

The NII is expected to decrease by €93 million as a result of an increase in interest rates by 1%, all else being equal.

The advantages of gap analysis are as follows.

- **Implementation:** gap analysis is cheap and in many cases gap models are spreadsheet based.
- **Interpretation:** gap models are easy to communicate and explain to senior managers.
- **Uses for limits and hedging:** gap analysis provides a clear representation of the interest rate risk positions of one bank and can be used to set limits and drive hedging actions.

Table 4.2 Interest rate gaps

Interest rate gap	Assets (€ bn)	Liabilities (€ bn)	Gap (€ bn)	Cumulated gap (€ bn)
O/N	1.00	3.70	−2.7	−2.7
O/N–1M	6.60	17.60	−11.0	−13.7
1M–3M	1.10	6.50	−5.4	−19.1
3M–6M	14.70	1.50	13.2	−5.9
6M–12M	3.00	1.50	1.5	−4.4
Above 1Y	10.70	5.60	5.1	0.7
Non-rate-sensitive	2.50	3.20	−0.7	0.0
Total	39.6	39.6	0.0	

Table 4.3 Impact of a 1% rate shift on the net interest income

Bucket	Gap	Time to horizon (30/360)	NII Impact of a 1% shift	Cumulated gap	Time to next bucket (30/360)	NII Impact of a 1% shift
O/N	−2.7	360	−0.027	−2.7	15	−0.001
O/N–1M	−11.0	345	−0.105	−13.7	30	−0.011
1M–3M	−5.4	315	−0.047	−19.1	90	−0.048
3M–6M	13.2	225	0.083	−5.9	135	−0.022
6M–12M	1.5	90	0.004	−4.4	90	−0.011
Total			−0.093			−0.093

The disadvantages of gap analysis are as follows.

- **Over aggregation:** mismatches within each time bucket could be hidden. For example, one gap could be positive (hence suggesting an asset sensitive position). However, should substantial liabilities occur at the beginning of the time bucket and assets at the end, the actual exposure to interest rate movements could be the opposite. As liabilities

are re-fixed in advance of assets, even if the latter exceed the former, income might fall when rates increase.

- **Treatment of embedded options:** gap models assume that all positions react linearly to interest rate shifts. This leads to imperfect treatment of options such as caps or floors embedded in adjustable-rate mortgages or prepayable loans.
- **Treatment of items influenced by behavioural aspects:** gap models can hardly cope with the volume and rate fluctuations of products that are influenced by the behaviour of the counterparty (eg, redemptions of retail deposit).
- **Basis risk:** in its basic form, gap analysis neglects uncorrelated rate movements. This can be addressed by weighting positions indexed to non-market rates with the regression coefficient between their reference rate and the market rates. An alternative approach is to represent such positions with replicating portfolios constructed to minimise the volatility of their rate spread versus the target positions. For example, a loan indexed to the prime rate could be “optimally” replicated by a liability portfolio composed of 25% overnight deposits and 75% rolling deposits with an original maturity of three months. The loan would be represented in the gap analysis based on its replicating portfolio.
- **Scope of analysis:** the evolution of the balance sheet (new business, changing customer base, etc) and fee-based income are not taken into account.⁸

SIMULATIONS AND EARNINGS-AT-RISK

Simulations are used mostly by sophisticated banks. Given their typical focus on earnings risk, simulations can indeed be seen as an improvement over basic gap analysis.

These models simulate the balance sheet under various interest rate, credit spread and business scenarios over multiple time steps in the future, typically ranging from one to three years, in order to measure the expected value and the volatility of the NII.

Simulations require modelling of the specific characteristics of each banking product. Interest and principal cashflows have to be represented, and assumptions about their evolution have to be incorporated in the model. Due

to the number of items on the balance sheet, cashflows of several contracts of the same type are often “compressed” into representative positions.

Alternative interest rate scenarios can be used, involving changes in the level and shape of the yield curve as well as variations of spreads between different interest rates. These can be either applied at single instants of time or allocated through several time steps (“ramp scenarios”). The baseline scenario could consist in constant rates, forward evolution or calibrated around the bank’s expectations.

In “static simulations”, only the cashflows and interest income arising from the bank’s current on- and off-balance-sheet positions are assessed. On the other hand, “dynamic simulations” also include assumptions about the future business evolution in terms of lending activity and related funding mix. Dynamic simulations also aim to capture the embedded options, including (but not limited to) mortgage prepayments and run-off of retail and corporate deposits.

A trait common to static and dynamic models is the need to simulate “integrative positions” that keep assets and liabilities balanced over the various time steps of the simulation, during which the original positions could mature.

Earnings-at-risk (EaR) is the extension of the well-known value-at-risk (VaR) principle to earnings, in that EaR quantifies the maximum potential earnings reduction that can be experienced under a predefined level of confidence (eg, 99%) over a given time horizon, which can range between one and two years (against the typical one-to-ten days of VaR). The representation of positions is the same as in static or dynamic simulations models: the main difference is that earnings are projected not under a limited number of deterministic scenarios, but rather over multi-step stochastic interest rate scenarios, typically Monte Carlo based.

The results provided by dynamic simulations may become a key element in the calibration of the planning process of the bank and within the Pillar 2–ICAAP process.

Simulations can also be used to evaluate the relative attractiveness of alternative hedging strategies. Earnings-at-risk fits quite well in this context, because the alternative hedge strategies could be evaluated under a full range of interest rate curves, in order to minimise portfolio risk in a dynamic framework.

Simulation models show some weaknesses, however:

- significant costs are implied by the setup, maintenance and backtesting of the model, in terms of software development and human resource needs;
- at each model run there is a need to inject updated assumptions about business volumes, maturities prepayment assumptions, etc;
- interpretation of results is problematic due to high dependency on assumptions and to the sheer amount of integrative positions simulated by the system in order to keep assets and liabilities balanced;
- similarly to gap models, simulations focus on a relatively short-term horizon (one to two years).

ECONOMIC VALUE OF EQUITY SENSITIVITY AND DURATION

Economic value of equity (EVE) is calculated by taking the present value of all cashflows of on- and off-balance-sheet items. It aims to estimate the value of the bank as a going concern.

The discounting formula (Equation 4.1) is used, where CF_{ti} are the cashflows occurring at the times t_i , and y_{ti} are the corresponding discounting rates

$$V = \sum_{i=1}^n CF_{ti} (1 + y_{ti})^{-t_i} \quad (4.1)$$

By construction, EVE can be interpreted as the present value of the stream of future net income generated by interest-bearing assets and liabilities, plus the present value of the own funds.⁹ The direct link of EVE with the prospective stream of interest earnings allows use of the former, and its sensitivity measures, as leading indicators of variations of the latter in the short term as well as in the long term.

Assuming the discounting curves in [Table 4.4](#) for assets and liabilities, EVE would be €18.44 million, as in [Table 4.5](#).

A common approach is to calculate EVE sensitivity by parallel shifts in the base rates, in single increments of $\pm 1\%$. [Table 4.6](#) illustrates the effects of such shocks on our previous example. Spreads above the base rate are

assumed to be constant, in order to address the pure interest rate risk sensitivity of the balance sheet.

In our example, an increase in rates by 1% would reduce the EVE by €2.76 million, to a value of €15.68 million, mostly due to the long-dated asset 2, whose present value would decrease by €2.75 million.

Let us assume that the assets of one bank are composed of one €30 million loan maturing three years from now, carrying a fixed rate of 6% (asset 1) and by one €70 million loan maturing five years from now and carrying a fixed rate of 7% (asset 2). The bank has one liability of €80 million at a fixed rate of 4.5%, maturing one year from now.

Table 4.4 Discounting curves

Year	Base rate (%)	Spread asset (%)	Spread liabilities (%)	Rate assets (%)	Rate liabilities (%)
1	3.0	1.0	0.2	4.0	3.2
2	3.8	1.5	0.3	5.3	4.1
3	4.5	1.5	0.4	6.0	4.9
4	5.0	2.0	0.5	7.0	5.5
5	5.4	2.0	0.5	7.4	5.9

Table 4.5 Discounted cashflows

	Year					Total
	1	2	3	4	5	
Asset 1	1.73	1.62	26.70	0.00	0.00	30.05
Asset 2	4.71	4.42	4.11	3.74	52.42	69.40
Liability	−81.01	0.00	0.00	0.00	0.00	−81.01
Total	−74.57	6.04	30.81	3.74	52.42	18.44

Table 4.6 Sensitivities to parallel shifts of discounting curves

	Scenario (%)						
	+3	+2	+1	Baseline	−1	−2	−3
EVE	10.61	13.07	15.68	18.44	21.37	24.48	27.77
Impact	−7.83	−5.37	−2.76	0.00	2.93	6.03	9.32
Impact asset 1	−2.28	−1.55	−0.79	0.00	0.82	1.57	2.55
Impact asset 2	−7.84	−5.36	−2.75	0.00	2.90	5.97	9.20
Impact liability	2.29	1.54	0.78	0.00	−0.79	−1.60	−2.43

The “modified duration” (MD), given by Equation 4.2, is the most well known synthetic measure of sensitivity for a fixed-income product and can be used to approximate calculate the EVE reactivity to interest rate shifts

$$MD = \frac{1}{1+y} \frac{\sum_{i=1}^n t_i CF_{t_i} (1+y)^{-t_i}}{\sum_{i=1}^n CF_{t_i} (1+y)^{-t_i}} \quad (4.2)$$

MD is the absolute value of the first derivative of the value with respect to the discounting rate and enters the first term of the Taylor expansion of the value function

$$MD = \left| \frac{1}{V} \frac{dV}{dy} \right| \quad (4.3)$$

The discounting rate in Equation 4.3 is the yield to maturity, ie, the rate that equates the value of one fixed-income instrument (V) calculated as per Equation 4.1 to the one based on Equation 4.4

$$V = \sum_{i=1}^n CF_{t_i} (1+y)^{-t_i} \quad (4.4)$$

For fixed-income instruments with deterministic cashflows, the value–yield relation can be linearly approximated by

$$\frac{\Delta V}{V} \cong -MD \Delta y \quad (4.5)$$

Based on the example above, the discounted present value of asset 1 is €30.05403 million and its yield to maturity, y , is 5.9327%. The modified duration would be

$$\begin{aligned} MD &= \frac{1}{1 + 0.059327} \frac{1}{30.05403} \\ &\quad \times \frac{1 \times 1.8}{(1 + 0.059327)} + \frac{2 \times 1.8}{(1 + 0.059327)^2} + \frac{3 \times 31.8}{(1 + 0.059327)^3} \\ &= 2.6749 \end{aligned}$$

Should the general level of interest rates (y) rise by 0.50%, the value of asset 1 will decrease by approximately 1.337%

$$\frac{\Delta V}{V} \cong -2.6749 \times 0.005 = -0.01337$$

The MD of one portfolio is the weighted average of the MDs of the single components of the portfolio. The MDs of the assets, liabilities and equity are given by

$$\left. \begin{aligned} MDA &= \frac{\sum_{i=1}^{n^A} V_i^A MD_i^A}{\sum_{i=1}^{n^A} V_i^A} \\ MDL &= \frac{\sum_{i=1}^{n^L} V_i^L MD_i^L}{\sum_{i=1}^{n^L} V_i^L} \\ MDE &= \frac{MDA \sum_{i=1}^{n^A} V_i^A - MDL \sum_{i=1}^{n^L} V_i^L}{EVE} \end{aligned} \right\} \quad (4.6)$$

The relation between the sensitivity of EVE and MD of equity is equivalent to the one illustrated for single financial items in Formula 4.5.

The MDs of the assets and liabilities of the example are as follows.

Asset 1: modified duration = 2.6749
 Asset 2: modified duration = 4.0894
 Liability: modified duration = 0.9690

The MDs of assets and liabilities would be, respectively, 3.66 and 0.97. The MD of equity amounts to 15.49. The decrease in the EVE given a 1% increase in rates would be approximately 15.49%.

The strengths of duration of equity are as follows:

- it provides a synthetic measure of amount and timing of cashflows which can be used as a hedging target to manage the EVE over a long-term horizon;
- it allows identifying which positions are contributing most to EVE sensitivity;

The weaknesses of duration of equity are as follows:

- modified duration is the first derivative of the price–yield relation and therefore only provides an approximation of the value variations due to changes in the discounting curve;
- embedded options (eg, prepayment options and interest rate caps and floors in loans, withdrawal options in deposits) are neglected by duration analysis;
- duration analysis is based on discounting with the yield to maturity and therefore it assumes a flat discounting curve;
- duration analysis is based on actual positions and therefore does not capture assumptions about balance-sheet growth and evolution of balance-sheet figures.

CONVEXITY

A value approximation based only on duration is effective only for relatively small changes in y .

For non-infinitesimal changes in rates, the local approximation provided by duration is not accurate. The “convexity” (C), ie, the second derivative of V with respect to y , divided by the value, has to be used

$$C = \frac{1}{V} \frac{d^2V}{dy^2} \quad (4.7)$$

Convexity is a valuable feature of assets, as it provides protection from interest rate increases, but is an undesirable feature of liabilities for the opposite reason.

The analytical formula for a financial instrument with deterministic cashflows is given by

$$C = \frac{1}{(1+y)^2} \frac{\sum_{i=1}^n t_i(t_i+1) CF_{t_i} (1+y)^{-t_i}}{\sum_{i=1}^n CF_{t_i} (1+y)^{-t_i}} \quad (4.8)$$

and the price sensitivity relation given in Formula 4.5 would extend as follows, in order to take into account the introduction of the second term of the Taylor expansion

$$\frac{\Delta V}{V} \cong -MD \Delta y + \frac{1}{2} C \Delta y^2 \quad (4.9)$$

Accurate measurement and hedging of convexity is a crucial aspect of managing investment portfolios. However, it is more debatable whether the second-order approximation is crucial for the balance sheet of one bank. In the latter case, the most relevant sources of imprecision are modelling assumptions and behavioural characteristics of several banking products.

OPTION ADJUSTED VALUE AND OPTION ADJUSTED DURATION

Compared with option-free assets and liabilities, the price–yield relationship may change drastically with embedded options, affecting the significance of traditional sensitivity measures.

On the bank’s asset side, for instance, mortgages could be disbursed that incorporate the possibility of the client prepaying the loan paying a ‘penalty’, usually expressed as a percentage of the residual outstanding debt.¹⁰ Hence, a call option on the underlying loan is granted to the customer. The duration of the banking book may be compressed in periods of falling interest rates (where duration lengthening would instead be beneficial), due to the incentive to accelerate prepayments in order to refinance the existing mortgages at lower cost. The opposite dynamic would take place should interest rates be on the rise, in which case, all else being equal, prepayments will be deferred.

On the bank’s liability side, there are embedded options granted to the customers in the form of the right to withdraw early from deposits in sight and savings deposits to invest into alternative products that offer a higher yield. This entitles the customer to a put option on the deposit, whose exercise depends on its ‘moneyness’: in the case of rising interest rates the market value of deposits (whose rate is not necessarily linked to market conditions) declines and the put option would tend to be exercised.

More complete pricing approaches (eg, based on interest rate trees and other numerical methods) could be used to determine the option adjusted value (OAV) of those banking products that incorporate embedded options. The related sensitivity measures would be the option adjusted duration (OAD) and option adjusted convexity (OAC).

Most homeowners who exercise prepayment options prepay by refinancing the existing mortgages with a new loan to pay off the outstanding amount of the existing debt.

Assuming that the customers are influenced solely by the level of interest rates in their financial decisions, the prepayment option can be modelled as an American call option on an otherwise identical, non-prepayable mortgage, using a backward induction process based on some form of interest rate tree.

Once the OAV has been calculated based on a baseline scenario which uses the prevailing interest rate curve at the valuation date, the OAD and OAC can be calculated by shocking the interest rate curve upwards and downwards by a small amount, recalculating the OAV and then estimating the percentage sensitivities with the following formulas, where OAV_- and OAV_+ are the values calculated under the scenarios of, respectively, increasing and decreasing interest rates

$$OAD = \frac{OAV_- - OAV_+}{2 OAV \Delta r}, \quad OAC = \frac{OAV_- + OAV_+ - 2 OAV}{OAV \Delta r^2} \quad (4.10)$$

The expressions used for OAD and OAC are based on shocking the entire term structure of interest rates by Δr (or a portion thereof if the aim is to find partial sensitivity measures) and not on the yield to maturity y as in the classical duration and convexity framework.

KEY RATE DURATION

In order to model the impact of changes in its shape, each node of the rate term structure should be shocked. The reference curve can be constructed as indicated in the next two sections, in order to provide a general approach to measuring sensitivities and hedging the related risks.

Focusing on the movements of several nodes allows the OAD approach to be generalised by computing “key-rate durations”, by applying Formula 4.10

to the variations of OAV induced by alternating shocks of non-overlapping buckets of the term structure (r_i), keeping all other rates constant.

Then, the impact on OAV of a non-parallel movement of the term structure would be given by summing the effects of all the different, user-defined shocks Δr_i in Equation 4.11, where OAD_i would be calculated based on Equation 4.10

$$\frac{\Delta OAV}{OAV} = - \sum_{i=1}^n OAD_i \Delta r_i \quad (4.11)$$

There is no general standard for the number of rate buckets. However, between six and ten buckets are normally enough to capture the effects of non-parallel movements, as rates on adjacent nodes are positively correlated. The final choice should be driven by the specific interest rate exposure of the bank.

THE TERM STRUCTURE OF INTEREST RATES: SINGLE-CURVE APPROACH

Let us assume we want to manage an interest rate sensitive portfolio for which we want to assess the value as determined by the interest rate level. We therefore need a tool able to calculate the (changing) present value of our portfolio as a function of the (changing) level of interest rates. This tool is the discount factor curve: the collection of today's values of €1 cashflows to be paid with certainty at predetermined future points in time. To build up a useful discount factor curve, we first need to answer the following question: which market rates do we consider? To answer this question we have to take into account the instruments we shall use to manage the interest rate exposure of our portfolio, that is, the instruments we shall trade to control (hedge) the portfolio's sensitivity to the changes in rates. The choice will depend on the actual nature of the instruments making up the portfolio and on the convenience of using the hedging instruments (correlation with the items to be hedged, liquidity, completeness of the market, transactions costs, etc). In what follows we assume the use of IRSs as hedging instruments.

An IRS, in its simplest form (plain vanilla IRS), is a financial transaction in which two parties agree to exchange, for a predetermined maturity, periodic fixed payments against a stream of cashflows linked to a floating interest rate index. Typically, in euro IRSs one party agrees to pay the other

an annual fixed rate in exchange for semi-annual cashflows indexed to the Euribor 6M. Both (fixed and floating) rates are applied to the same notional amount. At inception, an IRS has no value for either counterparty, ie, no counterparty has to pay the other an upfront cash amount to enter into the transaction. This obviously implies that the two legs (fixed and floating) have exactly the same value

$$R_N \sum_{i=1}^N \alpha_i d(0, i) = \sum_{j=1}^{2N} r_j^6 \beta_j d(0, j) \quad (4.12)$$

where R_N is the fixed rate for maturity N , α_i is the year fraction to be applied to the fixed rate for payment date i , r_j^6 is the Euribor 6M fixed at $j - 1$ and payable at j , β_j is the year fraction to be applied to the Euribor 6M for payment date j , $d(x, y)$ is the discount factor for payment date y as calculated at x . The notional amount is conventionally 1. $2N$ implies that the frequency of the floating leg is twice that of the fixed one and, therefore, for any $j = 2i$, $d(0, i) = d(0, j)$.

The two counterparties, and more generally the market, agree on a fixed rate that satisfies Equation 4.12. In fact the interest rate swap market is made up of professional players that continuously show committed prices (fixed rates) at which they are ready to exchange fixed payments against payments indexed to the Euribor 6M. As a result, fixed-rate quotes are continuously available for a complete set of maturities (usually up to 30 years) and published by financial intermediaries (banks and brokers) via the major information vendors (Reuters, Bloomberg, etc).

This market will be used to calculate the discount factors necessary to assess the present value of our portfolio and to determine its sensitivity to the changes in the price of the underlying market instruments.

To start, consider a one-year swap, and assume that is traded at rate R_1 . From Equation 4.12 we have

$$R_1 \alpha_1 d(0, 1) + d(0, 1) = \sum_{j=1}^2 r_j^6 \beta_j d(0, j) + d(0, j = 2) \quad (4.13)$$

We can add the present value of the notional at maturity to both sides of the equation, without changing its meaning (remember that the indices on the left-hand side represent years, whereas on the right they refer to semesters;

hence, $d(0, i = 1) = d(0, j = 2)$). This alternative formulation represents an IRS as a combination of two opposite positions in, respectively, a fixed-rate bond and a floating-rate bond. Moreover, it allows us to have intuitive insight into the value of the right-hand side of the equation.

The flows of the floating leg of the swaps as described in Equation 4.13 are shown in Figure 4.1. The graph depicts the 6M Euribor payment fixed at 0 and paid at 1, and the 6M Euribor payment fixed at 1 and paid at 2, together with the notional amount.

We can add and subtract the same amount at time 1 without modifying the substance of the figure (Figure 4.2).

It is no accident that we have added and subtracted exactly the notional amount. The positive cashflow at 1 is clearly the total return of the notional amount invested at 0 in a Euribor 6M deposit. Therefore, its value at 0 is equal to 1 (the conventional value we attached to the notional). For the same reason, the value at time 1 of the cashflow paid at time 2 is also equal to 1. As a consequence, Figure 4.1 evolves as described on the left-hand side of Figure 4.3 and collapses to a single cashflow equal to the notional amount at time 0 on the right-hand side of Figure 4.3.

Intuitively, we have demonstrated that the floating leg of our one-year swap is worth 1 at time 0 (later we shall show a more rigorous demonstration).

It would be easy to show that what we have just described is true for any number of payments indexed to Euribor, and hence for any swap maturity. Therefore, Equation 4.13 can be rewritten as

$$R_1 \alpha_1 d(0, 1) + d(0, 1) = 1 \quad (4.14)$$

from which we can easily obtain the first element of our discount factor curve

$$d(0, 1) = \frac{1}{R_1 \alpha_1 + 1} \quad (4.15)$$

Figure 4.1 Floating leg of the swap

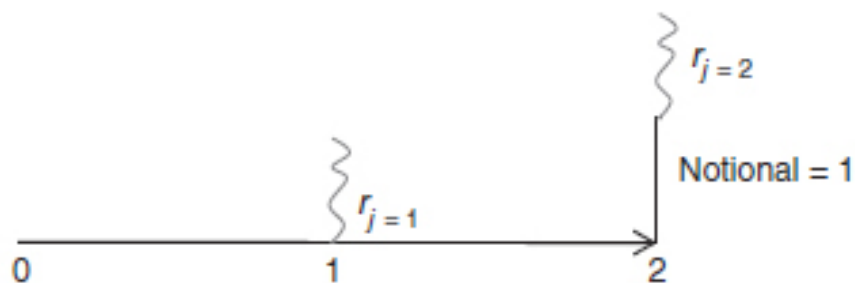


Figure 4.2 Floating leg of the swap (first transformation)

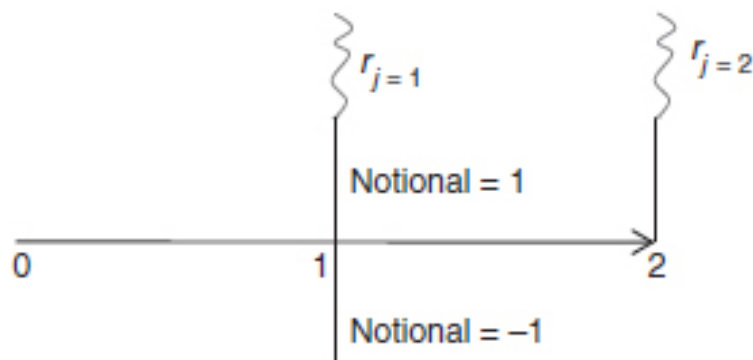
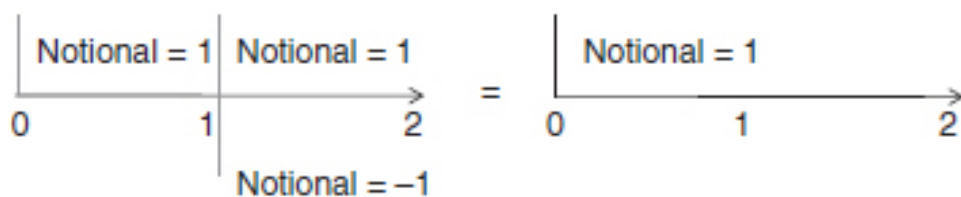


Figure 4.3 Floating leg of the swap (second transformation)



Let us now consider the next available maturity in the swap market: 2Y. Assuming that the market rate is R_2 , the equation representing the equilibrium value of the two swap legs is

$$R_2 \alpha_1 d(0, 1) + R_2 \alpha_2 d(0, 2) + d(0, 2) = \sum_{j=1}^4 r_j \beta_j d(0, j) + d(0, j = 4)$$

or, equivalently

$$R_2\alpha_1d(0,1) + R_2\alpha_2d(0,2) + d(0,2) = 1 \quad (4.16)$$

Recalling that $d(0, 1)$ as calculated in Equation 4.15 is known from the one-year swap, we can solve Equation 4.16 for $d(0, 2)$

$$d(0,2) = \frac{1 - R_2\alpha_1d(0,1)}{R_2\alpha_2 + 1} \quad (4.17)$$

Following recursively the same rationale for all the available maturities, we obtain the curve of discount factors $[d(0, i)]$, whose generic

element is

$$d(0,N) = \frac{1 - R_N \sum_{i=1}^{N-1} \alpha_i d(0,i)}{R_N \alpha_N + 1} \quad (4.18)$$

We obviously assume that the discount factor at zero is $d(0, 0) = 1$ (meaning that €1 today is worth exactly €1) and we consider that suitable interpolation techniques can be used to calculate the discount factors for any time between the i considered in Equation 4.18.

Discount factors can also be described in terms of rates. Since $d(0, T)$ is the value today of €1 payable at T , this euro can be considered as the payout of a zero-coupon bond (a financial transaction that pays the notional amount at maturity without bearing coupons in the meanwhile). The price of the zero-coupon is $d(0, T)$ and its yield to maturity, the zero rate, is the rate representation of the discount factor¹¹

$$d(0,T) = \frac{1}{(1 + zr_T)^T}, \quad zr_T = \left(\frac{1}{d(0,T)} \right)^{1/T} - 1 \quad (4.19)$$

where zr_T is the T -maturity zero-coupon version of the Euribor swap market. Assuming the existence of zr_T (or, equivalently, of $d(0, T)$) is the same as assuming that, at time 0, it is possible to invest $d(0, T)$ in a suitable combination of instruments indexed to Euribor that will return €1 at time T . In the same way, discounting €1 from T to today using $d(0, T)$ implies that our cost of funding is Euribor.

Discount factors are a synthetic and useful representation of the market instruments that we have chosen as a reference in the sense of their use as hedging tools. From them, we can obtain another alternative and equivalent

representation of the same market. Imagine we work for XYZ Bank and a client asks us to quote the rate we can offer on €1 that the client will deposit in one month's time for a six-month period. We need to set the rate today (ie, one month before the actual deposit) and we want to do so without incurring any profit or loss for the bank. We know that in one month we shall receive €1, which will be reimbursed with interest after six months. So we can borrow today an amount of money equal to the value today of €1 payable in one month (we now know that this amount is exactly $d(0, 1 \text{ month})$) and we can reinvest it up until the moment we need to reimburse the money to our client, that is for seven months. The present value of the total return (TR) of this investment is bound to be the investment itself; therefore

$$d(0, 1 \text{ month}) = \text{TR } d(0, 7 \text{ month})$$

Hence

$$\text{TR} = \frac{d(0, 1 \text{ month})}{d(0, 7 \text{ month})}$$

TR is therefore what we can promise our client without making the bank incur any profit or loss. It can be expressed in terms of the rate offered to the client

$$\left. \begin{aligned} 1 + F_{7 \text{ month}}^6 \beta_{7 \text{ month}} &= \frac{d(0, 1 \text{ month})}{d(0, 7 \text{ month})} \\ F_{7 \text{ month}}^6 &= \left(\frac{d(0, 1 \text{ month})}{d(0, 7 \text{ month})} - 1 \right) \frac{1}{\beta_{7 \text{ month}}} \end{aligned} \right\} \quad (4.20)$$

Consistently with the definitions in Equation 4.12, $F_{7 \text{ month}}^6$ is the six-month rate payable in seven months and fixed six months before, ie, in month 1. $\beta_{7 \text{ month}}$ is the year fraction to be applied to the rate payable in seven months.

The rate F calculated as in Equation 4.20 (that is, the rate for a future six-month contract implied by the current market as represented by our discount factor curve) is conventionally called the forward rate. The forward rate of tenor τ payable at j is

$$F_j^\tau = \left(\frac{d(0, j - \tau)}{d(0, j)} - 1 \right) \frac{1}{\beta_j} \quad (4.21)$$

The set of all the forward rates (as generically described in Equation 4.21) calculated from the discount factor curve in Equation 4.18 (and therefore implied by our reference market) is said to be the τ -tenor forward curve $[F_j^\tau]$.

The concept of forward will help us to give a more rigorous demonstration of the value of a swap's floating leg. Let us take the right-hand side of Equation 4.13

$$\sum_{j=1}^2 r_j^6 \beta_j d(0, j) + d(0, 2)$$

This can conveniently be extended as

$$r_1^6 \beta_1 d(0, 1) + r_2^6 \beta_2 d(0, 2) + d(0, 2)$$

We now know that the rates r payable at times 1 and 2 can be replicated with a suitable combination of borrowing and investments. If this is case, the market expectation of these rates is bound to coincide with their forwards

$$F_1^6 \beta_1 d(0, 1) + F_2^6 \beta_2 d(0, 2) + d(0, 2) \quad (4.22)$$

Using Equation 4.20, Formula 4.22 can be developed as follows

$$\begin{aligned} & \frac{1}{\beta_1} \left(\frac{d(0, 0)}{d(0, 1)} - 1 \right) \beta_1 d(0, 1) + \frac{1}{\beta_2} \left(\frac{d(0, 1)}{d(0, 2)} - 1 \right) \beta_2 d(0, 2) + d(0, 2) \\ &= d(0, 0) - d(0, 1) + d(0, 1) - d(0, 2) + d(0, 2) \\ &= d(0, 0) \\ &= 1 \end{aligned} \quad (4.23)$$

which is exactly the same result we obtained using the intuitive argument that led to Formula 4.14.

THE TERM STRUCTURE OF INTEREST RATES: MULTI-CURVE APPROACH

It is important to understand that the relation in Equation 4.23 only holds because so far we have considered a world where only a single curve is relevant. We have assumed the possibility of investing and borrowing at Euribor, and consequently Euribor forward rates are the only function of the

discount factor curve representing the Euribor market. Hence, we value the flows of a Euribor swap (fixed versus floating) using discount factors representing, and derived from, the Euribor market. But is this reasonable? Usually swap dealers receive (post) collateral (cash or highly liquid securities) to guarantee their positive (negative) net present values on outstanding swap transactions. The collateral transferred to the swap counterparty is (at least) equal to the net present value (NPV) of the deal and, in the case of cash, the collateral is remunerated (or financed) at the overnight rate (for euro, the euro overnight index average (Eonia)).¹² Therefore, the basic assumption behind Equation 4.19, that the cost of funding or reinvestment rate is Euribor, is incorrect: in fact the funding (or the reinvestment) of a swap NPV is done at Eonia. Hence, the discounting of the swap cashflows has to be done over the Eonia curve. In fact, until 2008, using the Euribor swap curve for discounting was a reasonable approximation: the basis (difference) between Euribor indexed swaps and overnight indexed swaps was stable and negligible. After the collapse of Lehman Brothers (in September 2008) the confidence of money market players came under question, the cost of the interbank term deposits surged and the basis between Euribor and Eonia widened dramatically: Eonia discounting became a necessity.

An overnight indexed swap (OIS) is a transaction whereby two parties agree to exchange, on a certain notional amount and for a determined maturity, fixed payments against flows indexed to and compounded at Eonia. For an argument similar to that used in Equation 4.14, the OIS floating leg is worth its notional amount (assuming the exchange of notionals at maturity) and therefore an Eonia discount factor curve, with the following generic element, can be calculated

$$d^{\text{OIS}}(0, N) = \frac{1 - R_N^{\text{OIS}} \sum_{i=1}^{N-1} \alpha_i d^{\text{OIS}}(0, i)}{R_N^{\text{OIS}} \alpha_N + 1} \quad (4.24)$$

We can now revisit the IRS equilibrium formula we saw in Equation 4.12. Using the discount factors in Equation 4.24, Formula 4.12 can be rewritten as

$$R_N \sum_{i=1}^N \alpha_i d^{\text{OIS}}(0, i) = \sum_{j=1}^{2N} r_j^6 \beta_j d^{\text{OIS}}(0, j) \quad (4.25)$$

Since $d^{\text{OIS}}(0, i) \neq d(0, i)$, in order to satisfy Equation 4.25, we cannot replace r_j^6 with F_j^6 as implied by Equation 4.22. In other words, when discounting over Eonia, we cannot assume that the market expectation of r_j^6 can be represented by the corresponding forwards implied by the Euribor discount factors: given the swap curve $[R_i]$ and the discount factor curve $[d^{\text{OIS}}(0, i)]$, we need to find a different forward curve $[{}^{\text{OIS}}F_i^6]$, which is, however, coherent with the fundamental assumption that, at inception, a plain vanilla swap (fixed versus 6M Euribor) must be worth nothing for both parties.

Let us start with the case of a one-year swap and assume that the first six-month floating payment is known (ie, that the first six-month Euribor is already known, as indicated by the overbars)

$$R_1 \alpha_1 d^{\text{OIS}}(0, 1) = F_{6*}^6 \beta_{6*} d^{\text{OIS}}(0, 6^*) + {}^{\text{OIS}}F_{12*}^6 \beta_{12*} d^{\text{OIS}}(0, 12^*)$$

where the starred indices refer to months, and the others to years.

Solving for the only unknown forward, we get

$${}^{\text{OIS}}F_{12*}^6 = \frac{R_1 \alpha_1 d^{\text{OIS}}(0, 1) - F_{6*}^6 \beta_{6*} d^{\text{OIS}}(0, 6^*)}{\beta_{12*} d^{\text{OIS}}(0, 12^*)}$$

From the two-year swap we then have

$$\begin{aligned} R_2 \sum_{i=1}^2 \alpha_i d^{\text{OIS}}(0, i) \\ = F_{6*}^6 \beta_{6*} d^{\text{OIS}}(0, 6^*) + {}^{\text{OIS}}F_{12*}^6 \beta_{12*} d^{\text{OIS}}(0, 12^*) \\ + {}^{\text{OIS}}F_{18*}^6 \beta_{18*} d^{\text{OIS}}(0, 18^*) + {}^{\text{OIS}}F_{24*}^6 \beta_{24*} d^{\text{OIS}}(0, 24^*) \quad (4.26) \end{aligned}$$

In Equation 4.26 in fact there are two unknowns. However, we can describe the forward payable in 18 months as a function of the forwards preceding and following it (in 12 months and 24 months, respectively). We shall choose as function a linear interpolating operator¹³

$${}^{\text{OIS}}F_{18*}^6 = {}^{\text{OIS}}F_{12*}^6 0.5 + {}^{\text{OIS}}F_{24*}^6 0.5$$

After rearranging we obtain

$${}^{\text{OIS}}F_{24}^6 = \frac{1}{0.5\beta_{18}d^{\text{OIS}}(0,18^*) + \beta_{24}d^{\text{OIS}}(0,24^*)} \\ \times \left[R_2 \sum_{i=1}^2 \alpha_i d^{\text{OIS}}(0,i) - {}^{\text{OIS}}F_{6^*}^6 \beta_{6^*} d^{\text{OIS}}(0,6^*) \right. \\ \left. - {}^{\text{OIS}}F_{12^*}^6 \beta_{12^*} d^{\text{OIS}}(0,12^*) - {}^{\text{OIS}}F_{12^*}^6 0.5\beta_{18^*} d^{\text{OIS}}(0,18^*) \right]$$

and, generally

$${}^{\text{OIS}}F_j^6 = \frac{1}{0.5\beta_{j-1}d^{\text{OIS}}(0,j-1) + \beta_j d^{\text{OIS}}(0,j)} \\ \times \left[R_{j/2} \sum_{i=1}^{j/2} \alpha_i d^{\text{OIS}}(0,i) - \sum_{s=1}^{j-2} {}^{\text{OIS}}F_s^6 \beta_s d^{\text{OIS}}(0,s) \right. \\ \left. - {}^{\text{OIS}}F_{j-2}^6 0.5\beta_{j-1} d^{\text{OIS}}(0,j-1) \right] \quad (4.27)$$

where j is now the number of semesters.

With Formulas 4.24 and 4.27 we now have everything we need to discount over Eonia fixed payments and cashflows indexed to 6M Euribor. But what about floating-rate transactions indexed to other Euribor tenors, for instance, 3M? Before 2008, market perception was that Euribor genuinely represented the price of interbank lending and its term structure was typical of a quasi-risk-free market. In fact, market players were in principle indifferent, for instance, between a 6M deposit and a 3M deposit rolled over a quarter. In other words, everybody would have agreed to swap one 6M Euribor payment with two 3M Euribor payments: the basis between the two tenors was flat. After Lehman's default, nothing was considered to be risk-free any longer, let alone the interbank money market: tenors started to be considered different from each other, with their own credit and liquidity risk, and the longer the tenor, the higher the risk was considered to be. The basis between 6M and 3M based swaps became substantially positive, and an active swap market started trading the spread between different Euribor tenors.

A 6M versus 3M Euribor basis swap (BS) is a transaction whereby two parties agree to exchange, until a certain maturity, a stream of payments linked to the 6M Euribor for a series of payments indexed to the 3M Euribor. Since the (credit and liquidity) premium of the longer tenor is higher than that embedded in the shorter one, the party receiving the 3M rate requires compensation for paying in exchange the 6M rate: this compensation is the "6M versus 3M basis". Market players quote this basis for a complete set of

annual maturities, usually up to 30 years. We can therefore use these quotes to obtain a curve of 3M Euribor forwards $[^{OIS}F_j^3]$ consistent with the other (OIS-based) curves we have so far calculated. The reader should, by now, be familiar with recursive calculation; therefore, we assume the 3M forward curve up to year $N - 1$ is already known. We denote by S_N the basis for maturity N ; the following equation describes the equilibrium condition of an N -year 6M versus 3M BS

$$\begin{aligned} \sum_{i=1}^{2N} {}^{OIS}F_i^6 \beta_i d^{OIS}(0, i) = S_N \sum_{j=1}^{4N} \gamma_j d^{OIS}(0, j) &+ \sum_{j=1}^x {}^{OIS}F_j^3 \gamma_j d^{OIS}(0, j) \\ &+ (0.75 {}^{OIS}F_x^3 + 0.25 {}^{OIS}F_{4N}^3) \gamma_{x+1} d(0, x + 1) \\ &+ (0.50 {}^{OIS}F_x^3 + 0.50 {}^{OIS}F_{4N}^3) \gamma_{x+2} d(0, x + 2) \\ &+ (0.25 {}^{OIS}F_x^3 + 0.75 {}^{OIS}F_{4N}^3) \gamma_{x+3} d(0, x + 3) \\ &+ {}^{OIS}F_{4N}^3 \gamma_{4N} d^{OIS}(0, 4N) \end{aligned}$$

where $2N$ is the number of semi-annual payments in the N years considered and, similarly, $4N$ is the number of quarterly payments; γ_j is the year fraction applied to the quarterly payment j ; $x = 4(N-1)$ is the number of quarters with respect to which the recursive process has already calculated the 3M forwards; $x+1, x+2, x+3$ and $x+4 = 4N$ are the quarters in year N whose forwards will be calculated by the current iteration; forwards from $x + 1$ to $x + 3$ are calculated from linear interpolation¹⁴ between F_x^3 and F_{4N}^3 so the only unknown is F_{4N}^3 .

By rearranging we get

$$\begin{aligned} \sum_{i=1}^{2N} {}^{OIS}F_i^6 \beta_i d^{OIS}(0, i) = S_N \sum_{j=1}^{4N} \gamma_j d^{OIS}(0, j) &+ \sum_{j=1}^x {}^{OIS}F_j^3 \gamma_j d^{OIS}(0, j) \\ &+ {}^{OIS}F_x^3 \sum_{j=x+1}^{x+3} [1 - 0.25(j - x)] \gamma_j d^{OIS}(0, j) \\ &+ {}^{OIS}F_{4N}^3 \sum_{j=x+1}^{x+4} 0.25(j - x) \gamma_j d^{OIS}(0, j) \end{aligned}$$

and, finally

$$\begin{aligned}
{}^{\text{OIS}}F_{4N}^3 = & \frac{1}{\sum_{j=x+1}^{x+4} 0.25(j-x)y_j d^{\text{OIS}}(0,j)} \\
& \times \left[\sum_{i=1}^{2N} {}^{\text{OIS}}F_i^6 \beta_i d^{\text{OIS}}(0,i) \right. \\
& - S_N \sum_{j=1}^{4N} y_j d^{\text{OIS}}(0,j) - \sum_{j=1}^x {}^{\text{OIS}}F_j^3 y_j d^{\text{OIS}}(0,j) \\
& \left. - {}^{\text{OIS}}F_x^3 \sum_{j=x+1}^{x+3} [1 - 0.25(j-x)] y_j d^{\text{OIS}}(0,j) \right] \quad (4.28)
\end{aligned}$$

which is the generic element of the 3M forward curve we were looking for.¹⁵

USING ADDITIONAL HEDGING INSTRUMENTS

In the preceding section the Euribor market was represented only by the IRS curve and, consequently, the term structures described were developed assuming that swaps are the only instruments that managers can use to take positions on the market. Although the results are valid and do not suffer any loss of generality, it is worth mentioning that other instruments based on Euribor rates exist: deposits, forward rate agreements (FRAs) and futures on deposits. Moreover, using the said instruments over short maturities could be more efficient than using swaps, and therefore their price can – and should – be taken into account when building up discount factor curves.

Deposits are by nature zero-coupon securities. Therefore, calculating the corresponding discount factor is straightforward.

Let r_t be the interest rate borne by a euro deposit maturing at t . The total return of €1 invested in such a deposit would then be $1 + r_t \alpha_t$ (where α_t is the year fraction corresponding to the tenor of the deposit). The corresponding discount factor is therefore $1/(1 + r_t \alpha_t)$.

FRAs and futures are instruments that allow the holder to lock the rate of the underlying deposit at predefined future value dates: in essence, such instruments make it possible to trade the forward rates we discussed in the previous sections. FRAs and futures can be used to secure the total return on a forward deposit, and therefore forward discount factors can be obtained from their prices.

According to their relative liquidity over the maturity spectrum, these instruments can be used to complement swaps in the definition of the term

structures. Typically, deposits are used up to the delivery date of the first available futures, and swaps are taken into account starting from the maturity of the deposit underlying the last future instrument that is regarded as sufficiently liquid.

HEDGING INTEREST RATE AND BASIS RISKS

In the previous section different representations of term structures were calculated from the market prices of the instruments chosen as hedging tools. The term structure will be now used to assess the sensitivity of the value of a hypothetical portfolio to the changes of the interest rates and basis swap curves.

In what follows we shall ignore the changing correlation between the swap market and the financial instruments making up the portfolio. In other words, we shall assume that if a financial instrument is priced at a given spread with respect to Euribor, the spread will never change.¹⁶ Indeed, in most cases, for simplicity but without losing any substance, we shall work with transactions priced at flat Euribor.

Let V be the value of our interest rate portfolio

$$V_0 = \sum_t CF_t d(0, t)$$

This means that the value today ($t = 0$) is the sum of the present values of the future cashflows discounted over the IRS curve. Assuming that this curve consists of maturities from 1 to N years,¹⁷ we define

$$[\Delta V_i] = \left[\frac{\partial V}{\partial R_i} \times 1\text{bp} \right]$$

for any IRS maturity i between 1 and N years (4.29)

for any IRS maturity i between 1 and N years (4.29)

where ΔV_i is the sensitivity of the portfolio's value to a 1bp (0.01%) increase in the fixed rate R_i of the swap maturing in i years from now (all the other swap rates being unchanged).

In fact, the array of sensitivities can be calculated with the following numerical procedure

$$[\Delta V_i] = \left[\sum_t CF_t d^{R_i+1bp}(0, t) - \sum_t CF_t d(0, t) \right]$$

for any IRS maturity i between 1 and N years (4.30)

where CF_t represents the cashflow(s) payable at t and $d^{R_i+1bp}(\cdot)$ is the discount factor curve calculated at the current level of R_i increased by 1bp (the rates of all other swap maturities being unchanged).

Let us assume for the moment that our portfolio consists of a position in a €1 notional of 5Y interest rate swap whereby we pay the fixed rate. We also assume that its fixed rate is at-the-market, ie, it is exactly the same rate used for the calculation of the discount factor curve. It can be demonstrated that, by applying Equation 4.30 to such a portfolio, $[\Delta V_i]$ is zero for any maturity i except the 5Y. This is obviously intuitive: the value of an at-the-market position on an IRS of a certain maturity is not affected by the changes in the rates of the other maturities.¹⁸ Moreover, the only non-zero sensitivity calculated with the procedure in Equation 4.30 is approximately equal to the modified duration of the position multiplied by 1bp. Generalising the argument to the N swap maturities, we obtain the following array of IRS sensitivities

$$[\Delta IRS V_i] \approx [MD_i \times 0.0001] \quad \text{for any IRS maturity } i$$

between 1 and N years (4.31)

where MD_i is indeed the modified duration of the €1 notional paying position in the i -year IRS.

Combining the information in Equations 4.30 and 4.31, it is possible to calculate the hedging positions in the different swap maturities able to offset the portfolio sensitivities and therefore hedge its

Table 4.7 XYZ Bank balance sheet

Assets	€ (bn)	Liabilities	€ (bn)
5Y fix amort. 0.749% pa	10	3Y FRN bullet 3M	13
10Y FRN amort. 3M	10	7Y fix bullet 1.377%	7
6Y FRN bullet 3M	50	Capital	50

Table 4.8 Sensitivities and hedging recommendations

	ΔV	HP
1Y	−202,920.19	−2,009,724,000.00
2Y	−397,752.70	−2,002,502,000.00
3Y	−593,613.87	−1,998,083,000.00
4Y	−785,200.09	−1,994,339,000.00
5Y	−975,473.66	−1,995,307,000.00
6Y	0.00	—
7Y	4,708,715.46	7,000,000,000.00

value V_0

$$[\text{HP}_i] = \left[\frac{\Delta V_i}{\Delta \text{IRS } V_i} \right] \quad (4.32)$$

where HP_i is the notional amount of the hedging position in the i -maturity IRS.

Example 4.1. Consider the XYZ Bank. As can be seen in its balance sheet (Table 4.7), its equity amounts to €50 billion and the other liabilities consist in €13 billion of a 3Y floating-rate note (FRN) paying 3M Euribor and €7 billion of 7Y fixed-rate bonds. On the assets side we have €10 billion of a 5Y fixed-rate amortising loan (annual equal capital tranches), an FRN that pays 3M Euribor every quarter for six years and €10 billion of a 10-year amortising loan paying 3M Euribor on a structure of equal capital tranches.

Applying Equations 4.30 and 4.32 to the described positions (using the term structures displayed at the end of the chapter), we obtain the sensitivities and hedging recommendations in Table 4.8.

Table 4.8 says, for instance, that the value of XYZ Bank loses about €785,000 for each 1bp increase in the rate of the 4Y IRS.¹⁹ To offset this exposure, it is necessary to pay the fixed rate of the 4Y IRS for a notional amount of about €1.994 billion. On the other hand, receiving the fixed rate on

a €7 billion 7Y IRS would neutralise the indicated sensitivity over that maturity.²⁰

Table 4.9 Balance sheet of XYZ Bank after hedging

Assets	€	Liabilities	€
5Y fix amort. 0.749% pa	10,000,000,000	3Y FRN bullet 3M	13,000,000,000
10Y FRN amort. 3M	10,000,000,000	7Y fix bullet 1.377%	7,000,000,000
6Y FRN bullet 3M	50,000,000,000	Capital	50,000,000,000
Swaps (rec. legs)	€	Swaps (pay. legs)	€
1Y float. 6M	2,009,724,000	1Y 0.406% pa	2,009,724,000
2Y float. 6M	2,002,502,000	2Y 0.478% pa	2,002,502,000
3Y float. 6M	1,998,083,000	3Y 0.606% pa	1,998,083,000
4Y float. 6M	1,994,339,000	4Y 0.787% pa	1,994,339,000
5Y float. 6M	1,995,307,000	5Y 0.986% pa	1,995,307,000
7Y 1.377% pa	7,000,000,000	7Y float. 6M	7,000,000,000

Table 4.10 New sensitivities and hedging recommendations

	ΔV^S	HP ^S
1Y	−100,968	−999,315,756
2Y	−201,303	−999,609,214
3Y	3,616,208	12,000,156,189
4Y	−399,271	−1,000,020,218
5Y	−495,805	−1,000,143,866
6Y	−30,091,962	−51,000,223,455
7Y	−682,434	−1,000,254,786
8Y	−772,381	−1,000,230,131
9Y	−858,969	−1,000,154,374
10Y	−942,851	−1,000,031,323

The same rationale used so far can also be applied to define the portfolio's basis spread sensitivity

$$[\Delta V_i^S] = \left[\frac{\partial V}{\partial S_i} \times 1\text{bp} \right]$$

for any basis swap (BS) maturity i between 1 and N years

Table 4.11 Market data

	IRS fixed versus 6M [R_i]	OIS [R^{OIS}]	Basis 3M versus 6M [S]
1Y	0.406	0.079	0.179
2Y	0.478	0.147	0.169
3Y	0.606	0.280	0.162
4Y	0.787	0.474	0.156
5Y	0.986	0.659	0.161
6Y	1.191	0.871	0.146
7Y	1.377	1.054	0.140
8Y	1.542	1.231	0.135
9Y	1.687	1.380	0.130
10Y	1.815	1.512	0.126

Here ΔV_i^S is the sensitivity of the portfolio's value to a 1bp (0.01%) increase in the spread S_i of the BS maturing in i years from now.

In fact, the array of sensitivities can be calculated with the following numerical procedure

$$[\Delta V_i^S] = \left[\sum_t f(^{\text{OIS}}F_t^{3\uparrow})d^{\text{OIS}}(0, t) - \sum_t f(^{\text{OIS}}F_t^3)d^{\text{OIS}}(0, t) \right]$$

for any BS maturity i between 1 and N years (4.33)

where $\sum_t f(^{\text{OIS}}F_t^3)d^{\text{OIS}}(0, t)$ is the present value of all the cashflow functions of the current 3M forward rates curve, and hence of the current 3M versus 6M BS market (recall Formula 4.28); $[^{\text{OIS}}F_t^{3\uparrow}]$ is the forward rate curve calculated at the current level of S_i increased by 1bp (spreads of all other BS maturities being unchanged).

With the same argument as used for market IRSs, we can define the sensitivity of a €1 position in any single 3M versus 6M BS maturity quoted by the market: $[\Delta \text{BS } V_i]$. The ratios of the portfolio's sensitivity to the market BS sensitivity for the respective maturities yield the sought hedging recommendations

$$[\text{HP}_i^S] = \left[\frac{\Delta V_i^S}{\Delta \text{BS } V_i} \right] \quad (4.34)$$

Example 4.2. Imagine that the manager in charge of asset and liability management at XYZ Bank followed the hedging recommendations listed in [Table 4.8](#) exactly. The resulting balance sheet is shown in [Table 4.9](#).

Table 4.12 Term structures

df from IRS [$d(0, t)$]		df from OIS [$d^{\text{OIS}}(0, t)$]		6M forwards [$^{\text{OIS}}F_i^6$]		3M forwards [$^{\text{OIS}}F_i^3$]	
1Y	0.995961	1Y	0.9992	0.5Y	0.396	0.25Y	0.199
2Y	0.990515	2Y	0.997035	1Y	0.393	0.5Y	0.210
3Y	0.982009	3Y	0.99151	1.5Y	0.496	0.75Y	0.221
4Y	0.969012	4Y	0.980888	2Y	0.599	1Y	0.232
5Y	0.951791	5Y	0.966986	2.5Y	0.766	1.25Y	0.295
6Y	0.930712	6Y	0.948014	3Y	0.930	1.5Y	0.357
7Y	0.907364	7Y	0.92716	3.5Y	1.190	1.75Y	0.419
8Y	0.882579	8Y	0.903661	4Y	1.450	2Y	0.482
9Y	0.857118	9Y	0.879674	4.5Y	1.665	2.25Y	0.571
10Y	0.831232	10Y	0.854967	5Y	1.879	2.5Y	0.657
				5.5Y	2.102	2.75Y	0.743
				6Y	2.325	3Y	0.828
				6.5Y	2.445	3.25Y	0.969
				7Y	2.564	3.5Y	1.110
				7.5Y	2.671	3.75Y	1.251
				8Y	2.776	4Y	1.392
				8.5Y	2.861	4.25Y	1.470
				9Y	2.945	4.5Y	1.548
				9.5Y	3.009	4.75Y	1.625
				10Y	3.073	5Y	1.704
						5.25Y	1.880
						5.5Y	2.054
						5.75Y	2.226
						6Y	2.402
						6.25Y	2.399
						6.5Y	2.396
						6.75Y	2.394
						7Y	2.391
						7.25Y	2.484
						7.5Y	2.573
						7.75Y	2.660
						8Y	2.753
						8.25Y	2.774
						8.5Y	2.796
						8.75Y	2.818
						9Y	2.839
						9.25Y	2.881
						9.5Y	2.924
						9.75Y	2.966
						10Y	3.008

Re-running the IR sensitivity analysis of the previous example would now produce $[\Delta V_i] = 0$ for any maturity: the bank is virtually immune from the interest rate risk.²¹ However, the floating-rate positions (genuine and residual after IRS) are linked to both 6M and 3M, exposing the bank's economic value to the fluctuations of the spread between the two indexes.

Applying Equations 4.33 and 4.34 to the positions listed in the new balance sheet, we obtain the sensitivities and related hedging recommendations in Table 4.10.

Table 4.10 says, for instance, that if the 4Y 6M versus 3M basis ($[S_4]$) moves up by 1bp, the bank's economic value loses about €400,000. A €1 billion 4Y basis swap, whereby the bank would pay 3M Euribor against receiving 6M Euribor, would offset such exposure. Other maturities' hedging recommendations can be interpreted accordingly. Tables 4.11 and 4.12 are market information and related term structures used throughout this example.

CONCLUSION

We have summarised the typical approaches used by banks for measuring and managing interest rate risk. The reader should thus be able to compare the different measurement techniques and their relative advantages and disadvantages, and to contextualise them in the regulatory framework. Another learning outcome refers to the construction of an interest rate term structure and the use of swap instruments to hedge interest rate and basis risk exposures from an operational asset and liability management point of view.

The opinions expressed in this paper are those of the authors and do not necessarily reflect the EIB position and practices.

- 1 While credit spreads could also influence the net interest income and economic value of one bank, this chapter does not treat hedging of credit risk.
- 2 Even business lines that produce fee income could be indirectly affected by fluctuations of interest rates. For instance, the volume of assets under custody in the fund administration business could fluctuate depending on the relative level of interest rates, which would have an effect on the level on the overall administration fees. Another example would be the volume of loans serviced in securitisation programmes, which may fluctuate as a consequence of interest rates variations (lower rates imply prepayment acceleration).
- 3 Prior to the final introduction of the new rules in 2016 (the so-called “fundamental review of the trading book”) the Basel Committee provided a more generic definition of a trading book, according to which positions should have been classified in the trading book only in the presence of a “trading intent”.

- 4 In this example, gap analysis is illustrated for the whole balance sheet. Actual gap analyses could be applied only to those financial items in the banking book.
- 5 In line with the most basic gap analysis, only the principal payments are considered, not the interest coupons.
- 6 Floaters, in line with their interest rate sensitivity, are allocated only once, to the bucket corresponding to their first re-fixing date.
- 7 Monthly slices of €0.5083 billion are used in the gap analysis. For example, $€0.5083 \times 3 = 1.5$ billion is allocated to the 3M–6M bucket in order to represent the corresponding maturing portion of the deposits modelled with a nine-month maturity in the replicating portfolio. Similarly, the analysis up to one year would be influenced by those rolling tranches of replicating positions which, despite having an original maturity of more than one year, would “fall due” within the one-year horizon based on their residual maturity: these are not represented in the analysis for simplicity.
- 8 Moreover, interest cashflows are not included in the most basic gap analysis; this could affect results in environments with high levels of interest rates.
- 9 Assuming the own funds are attributed a conventional maturity corresponding to the longest asset or liability in the balance sheet.
- 10 Due to market practice, the penalty could be negligible (with respect to the value of the embedded prepayment option) and the client could benefit from an almost free option to prepay. This is particularly the case in the US. On the other hand, should the penalty exactly make up for the loss of earnings of the bank, there would be no immediate financial impact for the bank.
- 11 Any rate is a formal representation of a value, and as such depends on the use of specific conventions (day count rules, compounding frequencies, etc). In Formula 4.19, an annual compounding has been chosen arbitrarily. Other conventions are obviously possible. For instance, discount factors are often expressed in terms of instantaneous rates. In our case this would give $zrT = -\ln(d(0, T))/T$.
- 12 The same can also be said when posting securities as collateral: in fact, a liquid security can generally be transformed into cash (financed) at the cost of Eonia.
- 13 The choice is not exclusive: other operators could be considered.
- 14 As before, the choice is not exclusive: other operators could be considered.
- 15 It is worth noting that the same rationale and procedure can be applied for calculating the forward rates adjusted for the cross-currency basis swaps.
- 16 In such a way the value of the portfolio is only affected by changes in interest rates and basis spreads, ie, the risk factors that we can hedge with our financial instruments: interest rate and basis swaps.
- 17 Similarly, we assume no cashflow in our portfolio is payable beyond N years
- 18 It is worth noting that this is not valid for IRS position off-the-market. In that case the value of the position is affected by the changes in the market rates of IRSs of shorter maturities.
- 19 Ignoring the convexity.
- 20 The analysis given in Table 4.1 implies that the floating-rate transactions have no interest rate sensitivity. In fact, for simplicity, the assumption is that no coupon has been fixed in any floating-rate loan or note; it is as if the previously fixed coupons had just been paid and the next ones had not yet been re-fixed.
- 21 Ignoring the second-order effect determined by the difference in convexity between the hedged and hedging items of the bank’s balance sheet.

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The Modelling of Non-Maturity Deposits

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Federal Home Loan Mortgage Corporation

Within the banking industry, accurate liabilities or deposit-based expected life modelling is widely considered a prerequisite to sound asset–liability management. Its importance in mitigating interest rate risk is undisputed. However, the techniques associated with this are still evolving. This chapter focuses on establishing a concrete analytic methodology for estimating future expected deposit balance trajectories and their associated expected remaining life. We outline a comprehensive approach to guide the reader on how to define the event variable, establish a robust segmentation scheme, introduce key parameters in a time-series multivariate regression and validate and monitor the model on an ongoing basis to ensure its appropriateness.

THE IMPORTANCE OF MODELLING DEPOSIT BALANCES

Many banks worldwide are funded with non-maturity deposits, and the way in which their average lives are modelled has significant implications when estimating their value and their effectiveness in the management of interest rate risk. For example, modelling non-maturity deposits with too short an average life will subject the bank to rising interest rate risk exposure. Forecasting the expected remaining life of a non-maturity deposit presents

several challenges for modelling the expected remaining life of an asset such as a mortgage. The key difference is that deposit balance behaviour is not necessarily monotonically decreasing (as is the case with an amortising loan or mortgage for example). A savings deposit balance may increase or decrease at any point and its associated fluctuations are a function of multiple factors, including the macroeconomic environment as well as the product/pricing structure relative to both internal and external competition.

All in all, it is the movement within an economic cycle or changes to a product's pricing, fee structure or withdrawal terms and conditions that drive an increase or, conversely, exodus in balances from a bank's book.

PHILOSOPHICAL THEMES THAT DRIVE CONSUMER DEPOSIT BEHAVIOUR

In terms of investing their funds in a savings deposit, there are four key philosophical themes that concern a potential investor:

1. rate of return;
2. liquidity of funds (terms and conditions/policies around withdrawal of funds);
3. safety and reputation of the institution;
4. service level of the institution.

The product's rate of return, relative to similarly structured, competing products on the market, is perhaps the most important driver of a consumer's decision to invest. Theoretically, and controlling for all other factors, the laws of economics suggest that by increasing the rate paid to the consumer, inflows of deposits will in turn increase. The relationship is certainly positively correlated and may exhibit a linear or even exponential relationship over a particular interest rate interval. Of course, the function cannot monotonically increase into perpetuity, as there is a finite supply of liquid cash in the system that deposit takers are competing for. It is the number of customer alternatives or degree of competitive pricing that drives the entry or exit rate of deposit balances. Essentially, bank deposit pricing committees strive to manage their deposit base and profitability landscape through their express understanding of the rate-balance elasticity equation.

Liquidity and terms and conditions around the withdrawal of funds are other key considerations, as banking customers do not wish to pay excessive fees as a result of many withdrawals or receive a lower rate of interest if they do not meet a particular balance threshold.

The safety and reputation of the institution is also of paramount importance: an investor wants to know that their funds are protected and that the institution will not become insolvent.

Finally, the service level of the institution also plays a key role: an investor wants to be informed of product options, have speedy or instant access to their account and be able to do so through multiple channels (branch, Internet, mobile telephony, etc).

MECHANICS OF MODELLING

Defining the event and the expected average life calculation

Defining the “end of life” event

Defining when the deposit account is no longer “alive” is a key consideration in expected life modelling. Dormancy or extremely low balance levels coupled with inactivity over a significant amount of time suggest the account has effectively ended its life. Of course, this threshold is somewhat subjective, but the goal is to draw the line at a balance level and point in time (after non-usage) such that the probabilistic likelihood of deposits flowing into the account is minimal to non-existent. We can then conclusively and confidently say that the account’s life is over. Let us look at a few examples.

In [Figure 5.1](#), the account exhibits a normative pay down period followed by an accelerated reduction in balances. Finally, there is a time period of six months where the account is essentially dormant or at very low balance levels with no inflow/outflow activity. It is at this stage, at 36 months, that we may decide to conclude that the account’s life has ended.

In [Figure 5.2](#), balances decay quite rapidly, appear dormant for a period of roughly four months, and then exhibit a gradual buildup or rebound. It would have thus been potentially premature to assume the life of the account to be over at the 27th month, given the subsequent inflows in balances.

But where do we draw the line? When can we state that the account’s life is over with a high degree of confidence? A decision must be taken that addresses both the threshold of absolute balance levels that are deemed immaterial and the length of time balances are below that level.

Let us assume, for example, that we start by setting these thresholds at £5 and six months, respectively. This means that an account exhibits a balance level of under £5 for six consecutive months. The key question is what is the probabilistic likelihood that “significant” balance inflows into the account will be experienced subsequent to this? Again, the level of “significance” (and the time horizon) must be defined and accepted. Let us assume it is £100 (attained and sustained for three consecutive months) over the next twelve months. We can then measure the rate at which this was achieved for one group (those with a balance level of under £5 for six consecutive months) versus the other (those that did not meet this criterion). If we are satisfied that the event rate is sufficiently low, we may set this as the defining set of thresholds.

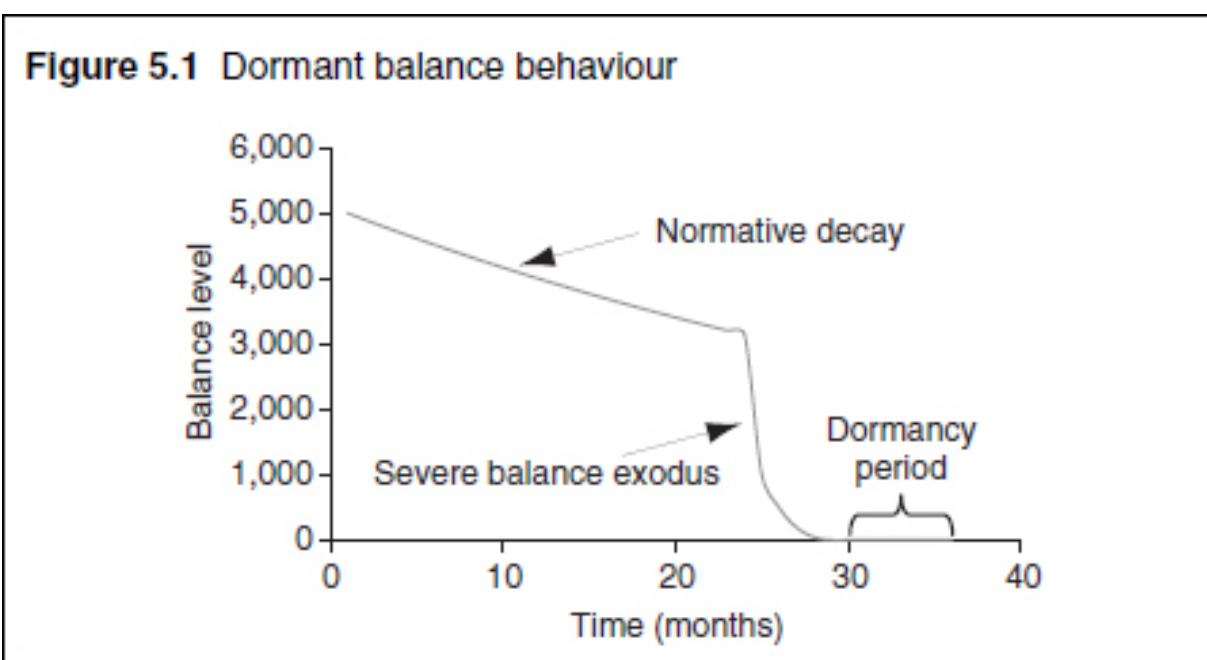
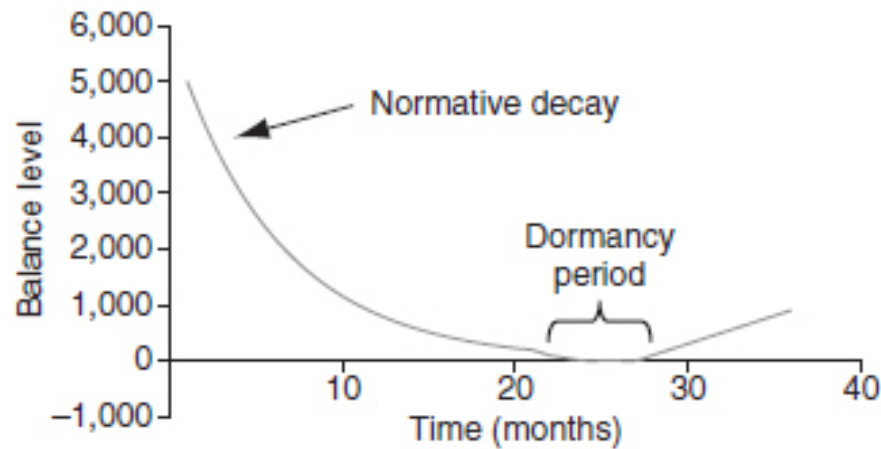


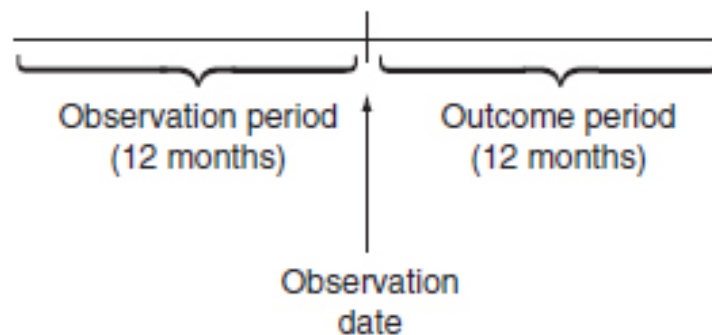
Figure 5.2 Rebounding balance behaviour



Using logistic regression to determine the “end of life” thresholds

A more holistic and appropriate approach would be to build a “logistic” regression model that predicts the likelihood of the event happening. The regression model would yield a set of parameter variables (and their associated estimates) that would feature a significant and strong relationship to the event or dependent variable. These variables could then be used to create a decision logic that will define thresholds that, if breached, will constitute the end of the account’s life.

Figure 5.3 Observation and outcome windows



Let us introduce a variable set that can be used to determine future likelihood of balance inflows:

- balance less than £50 for three consecutive months (B_50_3);

- balance less than £50 for six consecutive months (B_50_6);
- balance less than £25 for three consecutive months (B_25_3);
- balance less than £25 for six consecutive months (B_25_6);
- balance less than £5 for three consecutive months (B_5_3);
- balance less than £5 for six consecutive months (B_5_6);
- percentage change in balance over last three months at observation date, ie, $B(t)/B(t - 3)$ (RCB3);
- percentage change in balance over last six months at observation date, ie, $B(t)/B(t - 6)$ (RCB6);
- percentage change in balance over last twelve months at observation date, ie, $B(t)/B(t - 12)$ (RCB12).

The event we are trying to predict is attained and sustained (for three consecutive months) balance inflows of at least £100 at any time over the next 12 months ($P/100/12$).

We establish an observation and outcome period as shown in [Figure 5.3](#).

The nine independent variables chosen above will be used during the observation period to predict the event variable (during the outcome period) which, in binary form, indicates whether the account has achieved three consecutive months at a balance level of £100. Logistic regression is essentially a technique of maximum likelihood estimation that conforms to the following functional form

$$P(v) = \frac{1}{1 + (\exp(-1 \times v))}$$

where v is a linear combination vector of independent covariates such that $v = a_0 + a_1x_1 + a_2x_2 + \dots + a_9x_9$, indicating nine (x_1 to x_9) independent variables as in the above example and

$$0 \leq P(v) \leq 1$$

(ie, the probability of the event is bounded between 0 and 1).

Six of the nine variables are in the binary form (0, 1), that is, either the event happened in the observation window or it did not. The remaining three

measure percentage change in the level of balances over three, six and twelve months, respectively. By placing these variables within a logistic regression, we can ascertain which variables or combinations of variables (including their interactions) will yield the lowest likelihood of the event occurring (that is, the account attaining and sustaining a balance level of £100 over the next 12 months). Finally, based on the final parameter estimates yielded by the regression equation, we can create a decision criterion logic that will define and set the threshold of when an account's life can be officially pronounced as over.

Figure 5.4 illustrates the above example. We have three deposit accounts that exhibit varying behaviour before and after the observation date.

Account (balance) 1. Balance 1 exhibits a slight decay during the observation period but then rebounds in the outcome period to exhibit three consecutive months at £100 (months 21–23).

Account (balance) 2. Balance 2 exhibits a very sharp drop off in balances during the observation period (prior to month 6) and never subsequently experiences any inflows to meet the outcome period event (three consecutive months at £100).

Account (balance) 3. Balance 3 exhibits somewhat random, non-monotonic behaviour as the balance levels drop and rise significantly throughout the observation and outcome windows. Balance 3, incidentally, meets the event criteria throughout the whole of the observation and outcome windows.

Figure 5.4 Balance movement across observation and outcome windows

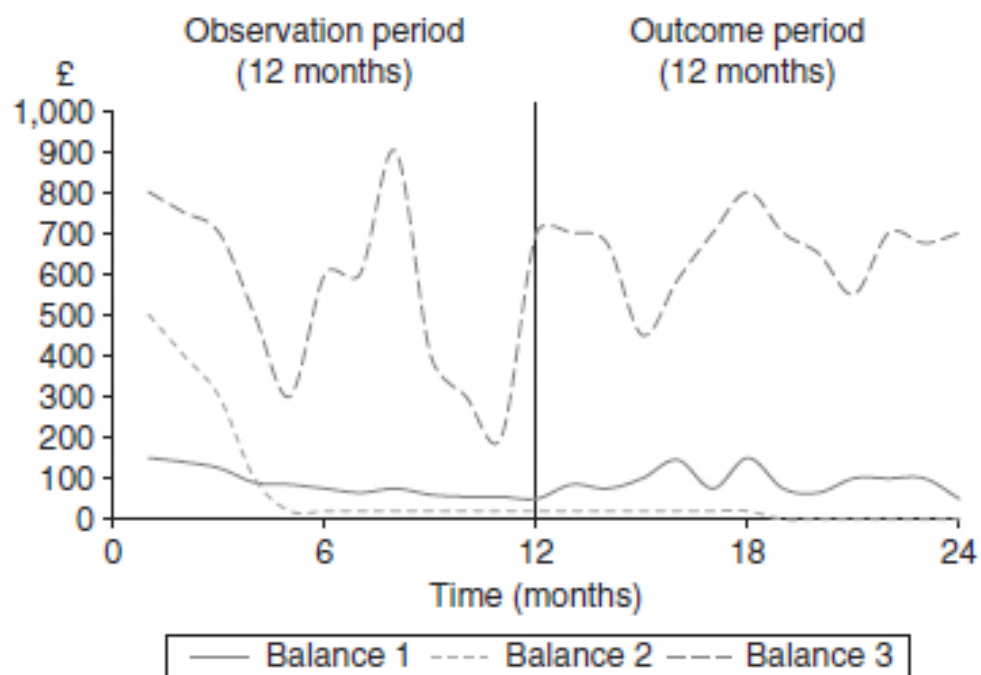
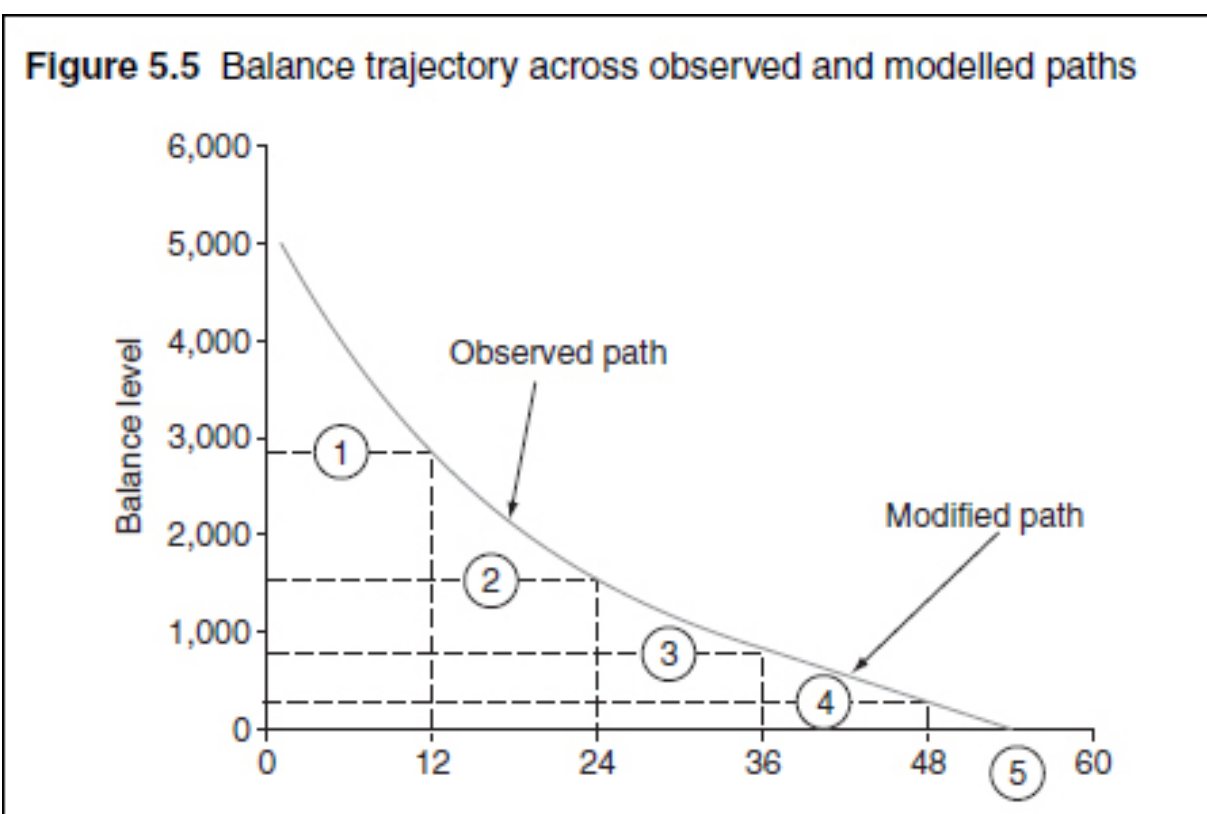


Table 5.1 Event attainment table

Variable	Balance 1	Balance 2	Balance 3
B_50_3	0	1	0
B_50_6	0	1	0
B_25_3	0	1	0
B_25_6	0	1	0
B_5_3	0	0	0
B_5_6	0	0	0
RCB3	0.83	1.00	1.75
RCB6	0.67	1.00	1.17
RCB12	0.33	0.04	0.88
Event attained?	Yes	No	Yes

The key question is which key behaviours were triggered during the observation period and the relationship they have with the event in the outcome window. Table 5.1 shows a summary of the variable set and its relationship to the dependent (event) variable (three consecutive months at £100).

As can be seen from Table 5.1, only Balance 2 did not exhibit three consecutive months at £100. Furthermore, Balance 2 was the only account that triggered four out of the six binary variables and it exhibited the fastest decay over 12 months (whereby the ratio of the balance level at the observation date to the balance level 12 months previously was only 4%).



Based on this very limited example, we could assume that it would be beneficial to include within the decision logic both

- (i) sustained low levels of balances within the observation period, and
- (ii) fast decay rates of balances over a longer (12-month) time period.

Again, it is a subjective exercise as to where to set the thresholds, but it is both intuitive and likely that the presence of both (i) and (ii) above will yield

low likelihood levels of a future inflow of balances. We would thus be able to safely assume that if the above conditions were met, the account's life would effectively be over.

The expected average life calculation

The calculation of expected life is essentially an elementary exercise, but its calculation depends directly on two key items:

1. a continuous function $B(t)$ over an observed and forecasted time period;
2. the number of discrete time intervals that are introduced into the equation.

We will look at two examples, one using annual time intervals and the other using monthly time intervals.

Let us first introduce (Figure 5.5) a deposit balance time path (partly observed and partly modelled on a “go forward” basis; note this is a function of a cohort of accounts that were booked under the same product/pricing construct at the same point in time).

Using annual time intervals

When considering the calculation of expected average life using annual time intervals, let us introduce a series of numbered sections, each corresponding to a respective interval. In Figure 5.5, we have five time intervals corresponding to years 1, 2, 3, 4 and 5, respectively. The calculation of average life is a “weighted” one that factors the proportion of balances that have “survived” at differing cumulative points in time.

Observing Figure 5.5, we have the following balance proportions (as a proportion of the original balance at month 1) across time:

- 54.04% at month 12, which implies that 45.96% of balances exited prior to month 12;
- 29.20% at month 24, which implies that 24.84% of balances exited between months 13 and 24;
- 15.69% at month 36, which implies that 14.13% of balances exited between months 25 and 36;

- 4.65% at month 48, which implies that 11.04% of balances exited between months 37 and 48;
- 0% at month 53, which implies that 4.65% of balances exited between months 49 and 53.

Let us introduce the “midpoint technique”, which essentially assumes that, within each annual time interval, balances, on average, survived only until the midpoint of the interval. As in the above example, we can assume that 45.96% of balances survived only until the sixth month (ie, their “average life” was six months). Extending this logic, we generate the following weighted “average life” equation

$$(0.4596 \times 6) + (0.2484 \times 18) + (0.1351 \times 30) + (0.1104 \times 42) + (0.0465 \times 51) = 18.29 \text{ months}$$

Table 5.2 Balance trajectory table

Months	Year	Balance	Marginal surviving proportion (%)	Months lived	Average life marginal contribution
0	1	5,000			
1	1	4,750	5.00	0.5	0.025
2	1	4,512	4.75	1.5	0.071
3	1	4,287	4.51	2.5	0.113
4	1	4,073	4.29	3.5	0.150
5	1	3,869	4.07	4.5	0.183
6	1	3,675	3.87	5.5	0.213
7	1	3,492	3.68	6.5	0.239
8	1	3,317	3.49	7.5	0.262
9	1	3,151	3.32	8.5	0.282
10	1	2,994	3.15	9.5	0.299
11	1	2,844	2.99	10.5	0.314
12	2	2,702	2.84	11.5	0.327
13	2	2,566	2.70	12.5	0.338
14	2	2,438	2.57	13.5	0.347
15	2	2,316	2.44	14.5	0.354
16	2	2,201	2.32	15.5	0.359
17	2	2,091	2.20	16.5	0.363
18	2	1,986	2.09	17.5	0.366
19	2	1,887	1.99	18.5	0.367
20	2	1,792	1.89	19.5	0.368
21	2	1,703	1.79	20.5	0.367
22	2	1,618	1.70	21.5	0.366
23	2	1,537	1.62	22.5	0.364
24	3	1,460	1.54	23.5	0.361
25	3	1,387	1.46	24.5	0.358
26	3	1,318	1.39	25.5	0.354
27	3	1,252	1.32	26.5	0.349
28	3	1,189	1.25	27.5	0.344
29	3	1,130	1.19	28.5	0.339
30	3	1,073	1.13	29.5	0.333
31	3	1,020	1.07	30.5	0.327
32	3	969	1.02	31.5	0.321
33	3	920	0.97	32.5	0.315
34	3	874	0.92	33.5	0.308
35	3	830	0.87	34.5	0.302
36	4	784	0.92	35.5	0.327
37	4	738	0.92	36.5	0.336
38	4	692	0.92	37.5	0.345
39	4	646	0.92	38.5	0.354
40	4	600	0.92	39.5	0.363
41	4	554	0.92	40.5	0.373
42	4	508	0.92	41.5	0.382
43	4	462	0.92	42.5	0.391
44	4	416	0.92	43.5	0.400
45	4	370	0.92	44.5	0.409
46	4	324	0.92	45.5	0.419
47	4	278	0.92	46.5	0.428
48	5	232	0.92	47.5	0.437
49	5	186	0.92	48.5	0.446
50	5	140	0.92	49.5	0.455
51	5	94	0.92	50.5	0.465
52	5	48	0.92	51.5	0.474
53	5	0	0.97	52.5	0.508
			100.00	Average life	17.76

Notes: all balance and average contribution figures have been rounded for consistency.

For the above example, we conclude that the balance trajectory yields an expected average life of approximately 18 months.

From a generalised standpoint, the average life calculation obeys the following equation

$$\text{average life} = w_1 m_1 + w_2 m_2 + \cdots + w_x m_x$$

where there are x time intervals, $m_1, m_2, \dots, m_x$ constitute the midpoints of their respective time intervals and $\sum_{i=1}^x w_i = 1$.

Using monthly time intervals

Clearly, the use of annual time intervals involves the far-reaching assumption that balance movement within a discrete time interval is linear and uniformly distributed, and thus half of the balance movement is realised by the time the midpoint of the interval is reached. As a result, the annual time interval approach is a very approximate way of estimating average life. A more granular option would be to use the monthly time interval approach (which increases the number of intervals twelvefold in a given year). Although it is more computationally intensive, this approach yields a more accurate estimate of average life. Let us observe how the estimate from the previous example changes with the monthly time interval approach. [Table 5.2](#) shows the data for the balance trajectory in [Figure 5.5](#). We observe that the average life has now decreased from 18.29 years to 17.76 years.

The difference in this example is only 0.53 years, or approximately 2.89%; however, this is a direct by-product of the shape of the balance trajectory in [Figure 5.5](#) and its near linear shape. For balance movements that behave in a much more non-linear fashion, we can expect more material differences between the monthly and annual time interval approaches.

Segmentation considerations

A key critical component in any analytical solution where measurement of rates is involved is having the right segmentation scheme. An appropriate segmentation will allow for a more granular look into what drives consumer behaviour. Additionally, it affords the analyst the opportunity to find pockets of behaviour that drive the overall expected life of the book that otherwise may have been “lost in the mix” if an otherwise “unisegment” approach to

measurement had been implemented. An effective segmentation scheme is designed such that n multiple microsegments or subgroups are formed that exhibit sufficiently different behaviour from the overall average. Indeed, we could surmise that a strong segmentation scheme is designed to maximise the variance of the event that is being predicted across the n segments while simultaneously capturing enough of the behaviour (at the micro-level) that is driving the overall variability in balance movement (and thus reducing the residual error ε_i between actual and predicted values).

Therefore, if we denote the change in the absolute balance of a liabilities book between the first and twelfth month as the ratio $\Delta\text{bal} = \text{balance}_{12}/\text{balance}_1$, we aim to maximise $\text{var}(\Delta\text{bal})$ and minimise ε_i , where

$$\text{var}(\Delta\text{bal}) = \sum_{i=1}^n \frac{(\Delta\text{bal}_i) - (\overline{\Delta\text{bal}})}{n - 1}$$

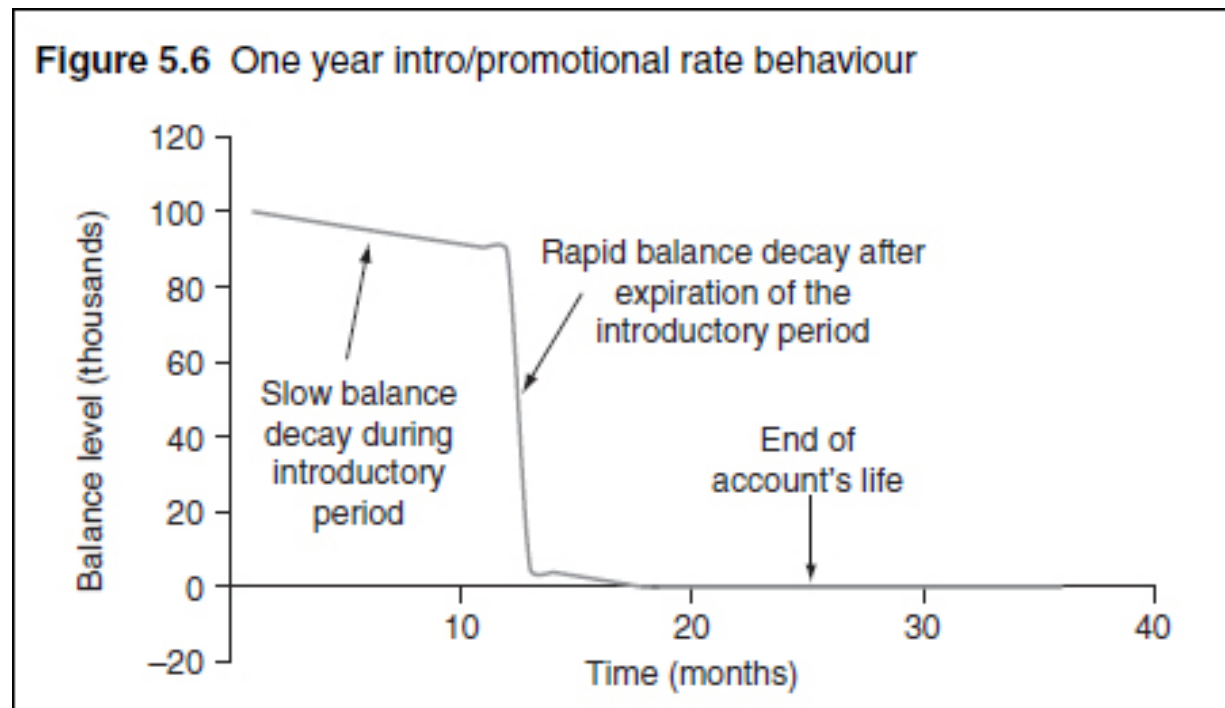
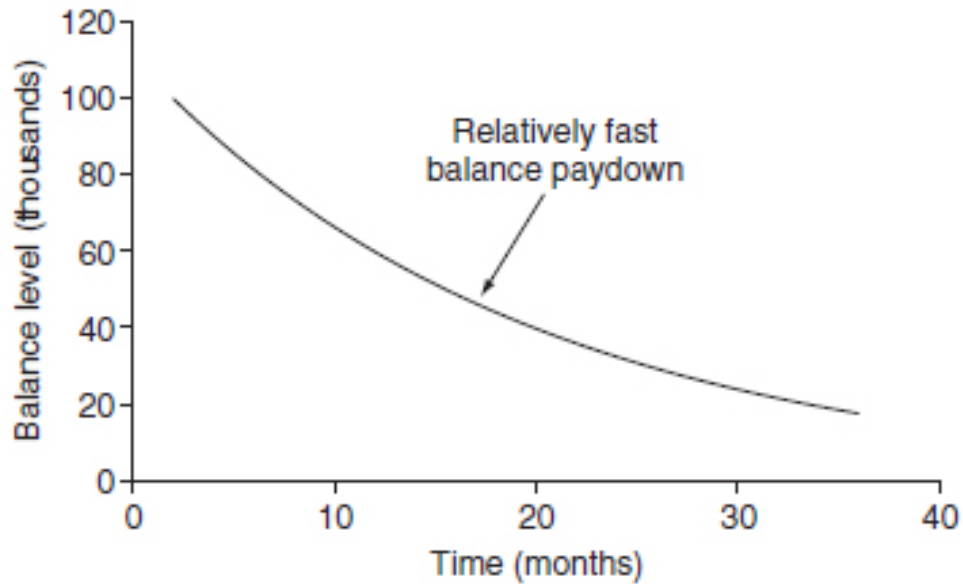


Figure 5.7 Quick balance paydown



and

$$\varepsilon_i = \text{abs}[E(\Delta\text{bal}_i) - (\widehat{\Delta\text{bal}_i})]$$

where the subscript i denotes the i th segment, $E(\Delta\text{bal}_i)$ is the expected value of Δbal for the i th segment and $\widehat{\Delta\text{bal}_i}$ is the predicted value of Δbal for the i th segment. Generally speaking, it is not necessary that this variance is maximised, but it should be sufficiently larger than zero. For example, if the variance of the prediction event of n microsegments was close to zero, we could conclude that the segments were not sufficiently differentiated as evidenced by balance behaviour. Figures 5.6–5.9 illustrate some examples of segments that exhibit different balance trajectories.

Figure 5.8 Increasing balance behaviour

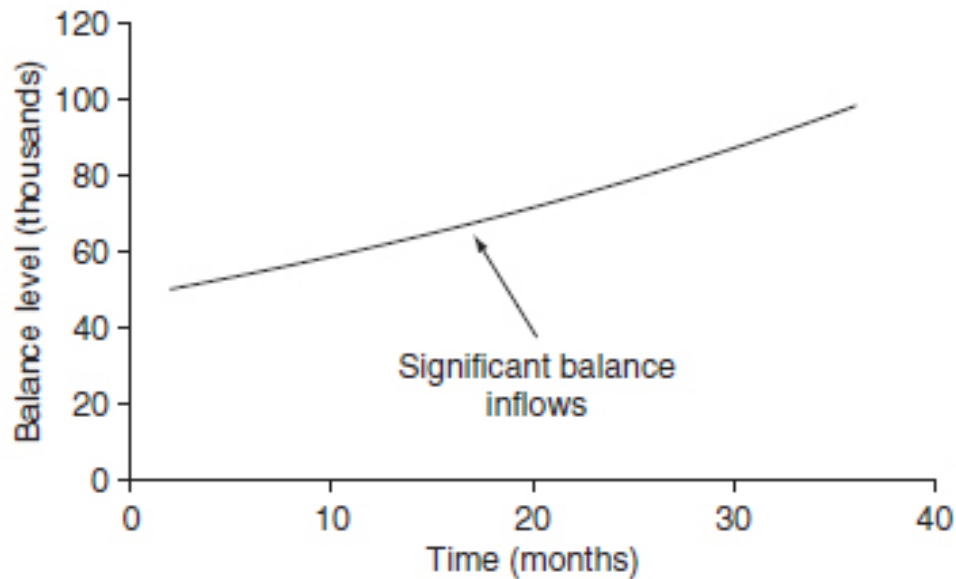


Figure 5.9 Random balance movement

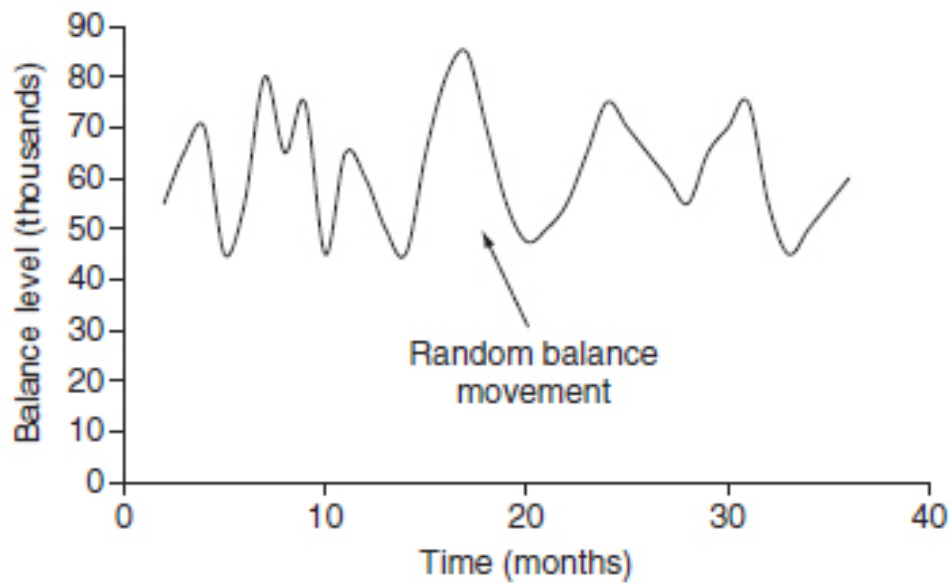


Figure 5.6 shows a typical balance movement associated with customers who are seeking high promotional rates from institutions; once these rates expire, they effectively transfer their balances elsewhere. These high rate seekers do not offer stability to a deposit taker's balance sheet and should be carefully identified or segmented as customers who entered on the balance

sheet under a promotional offer; in the example above we see a normative balance decay followed by a sharp drop post expiration of the intro/promotional offer period.

Figure 5.7 associates balance behaviour with high amount deposit holders. These customers are not necessarily on promotional offers, but do exhibit a relatively high degree of price elasticity and therefore are more prone to move their deposit to other institutions in accordance with the most competitive rate offerings.

Figure 5.8 shows a typical balance growth associated with a customer who is in a savings mode. The near linear growth rate of the savings balance trajectory shows a stable, near constant amount being deposited on a periodic basis.

Figure 5.9 shows random balance movement behaviour across time. This could potentially be associated with a commercial operation whereby the account is being used to deposit large amounts but then is using these amounts to fund future inventory, for example. It is typical balance behaviour of an “operating” account tied to a small business.

Various examples were presented above of significantly differing deposit balance trajectories. Key differentiators among them were

- the level of originating incoming balances and
- whether or not a promotional or introductory rate offer was present.

What other key parameters could drive differentiated deposit balance behaviour? Clearly there are many, but the below list captures several key ones that could be used as key segmentation variables:

- physical age of the customer;
- customers on promotional or introductory rate offers;
- depth and age of relationship with the institution;
- amount of incoming or originating balance;
- origination channel.

Physical age of the customer. The customer’s physical age plays a significant role in their deposit balance behaviour. Typically, physically

older customers are more conservative in their investment philosophy and seek out attractive or high rate, stable savings or money market accounts compared with the equities market. As a result, they are likely to hold large savings and be more price elastic or sensitive.

Customers on promotional or introductory rate offers. As depicted in [Figure 5.6](#), customers on introductory or promotional offers tend to move their balances from one institution to another once the introductory or promotional period expires. We do not mean to suggest that all customers do this; there will remain a residual balance level after the introductory period, but it is very small relative to what the offer initially attracted in terms of overall cohort balance levels. Introductory periods typically range from three to twenty-four months, but most of these offers will behave in a similar fashion, ie, a very slow decay during the introductory period followed by a severe drop-off in balances immediately after it.

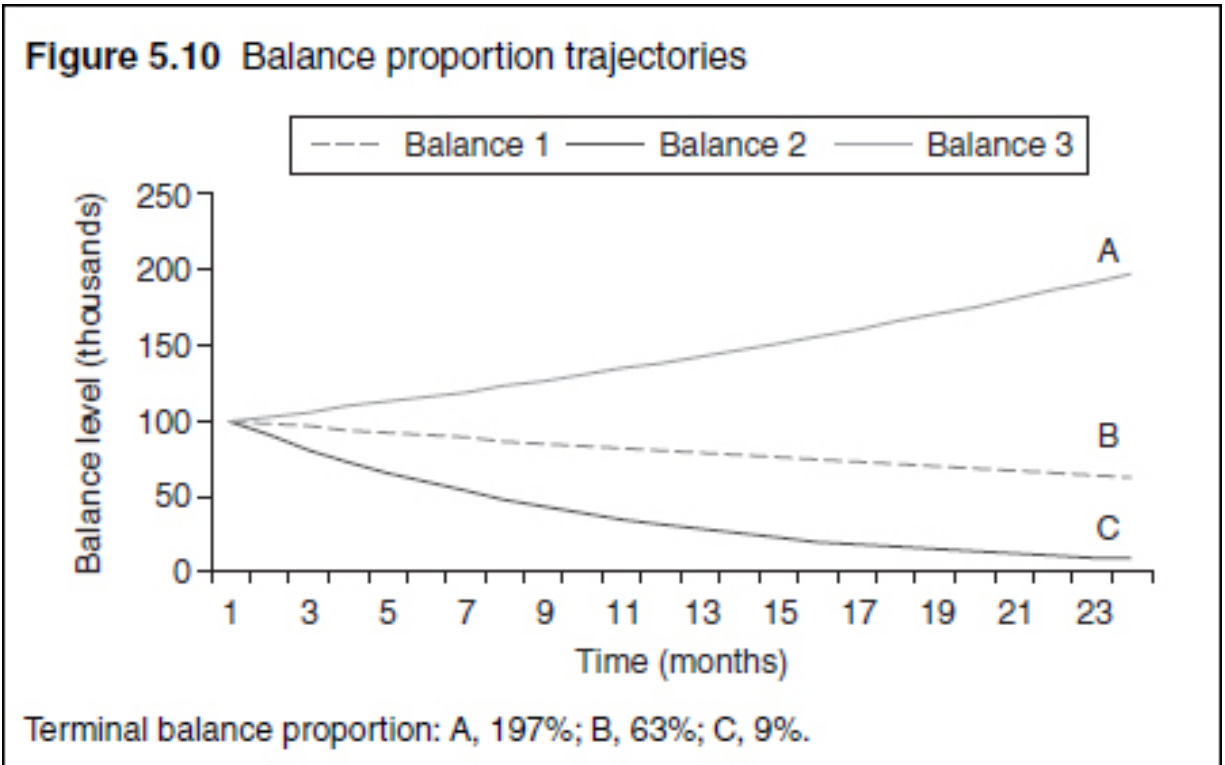
Depth and age of relationship with the institution. Depth and age of relationship with the institution is a key consideration in liabilities modelling, as an institution that has strong ties and service levels with its customers generates a high degree of loyalty. As a result, it is less reliant on competitive pricing as a tool to retain balances for the long term. Customers who have a long-standing tenure and multiple products with an institution will, in all likelihood, exhibit a lower propensity to reduce or move their balances elsewhere.

Amount of incoming or originating balance. The amount of incoming or originating balance can be a key leading indicator insofar as how the deposit balance trajectory will materialise. This is particularly evident in terms of its direction (ie, increasing, decreasing or staying relatively stable). Typically, smaller originating balances exhibit an increasing or balance build-up behaviour, while larger balances tend to decrease over time.

Origination channel. The origination channel (branch, via the Internet, etc) also plays a key role in determining how long customers are likely to keep their balances with the institution. Typically, customers who open savings accounts via the Internet or e-commerce websites are generally more price elastic and may exhibit higher likelihoods of attrition given their “rate-shopping” tendencies.

Designing the segmentation scheme

As discussed earlier, designing an optimal segmentation scheme requires the identification of pockets of differentiated consumer behaviour as exhibited via their balance movement through time. Analytically, there are many ways to compare and contrast balance movement. The question is which metric should be used to ultimately decide that two paths are sufficiently different. Should it be one that looks at the balance movements across time, or simply one that compares final balance levels to initial balance levels?



It is preferable to design the segmentation scheme based on the balance at a point in time relative to the origination date. Let us introduce the following: $B(t)$ denotes balance level at time t . $B(0)$ denotes the balance level at the origination date. The balance level at time t as a proportion of originating balance is given by

$$\text{balance_proportion}(t) = \frac{B(t)}{B(0)}$$

We see that $\text{balance_proportion}(t)$ satisfies the following: $0 \leq \text{balance_proportion}(t) < \infty$. Effectively, it is a ratio with a minimum of zero,

and unbounded above (this, of course, assumes no overdraft facility on the account, which would yield an overdrawn or negative balance).

Figure 5.10 shows an example of various time paths and balance proportion calculations at month 24.

It is evident that these paths exhibit markedly different balance proportion rates at the 24th month. Theoretically, even though balance trajectories 1 and 2 exhibit differing rates of change (balance decay), they can be grouped into one segment due to the directional behaviour of the paths. Customers belonging to Balance 3 would make up their own segment due to the increasing nature of the trajectory.

Clearly, in designing the segmentation scheme, the order of priority must be

- (a) identifying balance movement that is directionally different and
- (b) fine-tuning the segmentation based on the degree of difference at terminal balance proportion levels.

Validating segmentation cut-off points using the two-sample Z-test for proportional differences among unit rates

We have established that the terminal balance proportion metric can be used to establish segmentation cut-off points or levels. The key consideration is whether observed proportional differences are also statistically significant. This cannot be readily assessed using balance proportion, but can easily be quantified using account or unit level proportions. For example, 100 accounts are booked at a point in time t_0 with an aggregate balance level of 100,000. After 24 months, 55 accounts are left with a balance level of 60,000. The balance proportion at time $t = 24$ is $60,000/100,000$ or 60% but the unit proportion (defined as $U(t)/U(0)$) is $55/100$ or 55%.

The two-sample Z-test for proportional differences can be used to assess a unit rate proportional difference at a prescribed level of significance.

1. Introduce a null and alternate hypothesis for two proportions as follows:

$$(H_0) \text{ unit_proportion}_1 = \text{unit_proportion}_2$$

$$(H_1) \text{ unit_proportion}_1 \neq \text{unit_proportion}_2$$

2. Establish a level of significance of $\alpha = 0.01$.
3. Calculate the Z statistic at n degrees of freedom

$$Z = \frac{(\hat{\rho}_1 - \hat{\rho}_2) - 0}{\sqrt{[\hat{\rho}(1 - \hat{\rho})(1/n_1 + 1/n_2)]}}$$

where

$$\hat{\rho}_1 = \left[\frac{U(t)}{U(0)} \right] \Big|_{\text{segment 1}}$$

(unit proportion at time t for segment 1)

$$\hat{\rho}_2 = \left[\frac{U(t)}{U(0)} \right] \Big|_{\text{segment 2}}$$

(unit proportion at time t for segment 2)

$$\hat{\rho} = \left(\frac{n_1}{n_1 + n_2} \right) (\hat{\rho}_1) + \left(\frac{n_2}{n_1 + n_2} \right) (\hat{\rho}_2)$$

n_1 is the number of units at time $t = 0$ (time of origination) for segment 1
and n_2 is the number of units at $t = 0$ for segment 2.

4. Calculate Z .

Decision criteria (Figure 5.11).

- (i) If Z is in the rejection region, reject H_0 .
- (ii) If Z is not in the rejection region, fail to reject H_0 .

As above, the null hypothesis tends to be that there is no difference between the two population proportions; or, more formally, that the difference is zero (essentially that there is no difference between the terminal balance proportion of the two samples).

Example 5.1.

Unit trajectory 1:

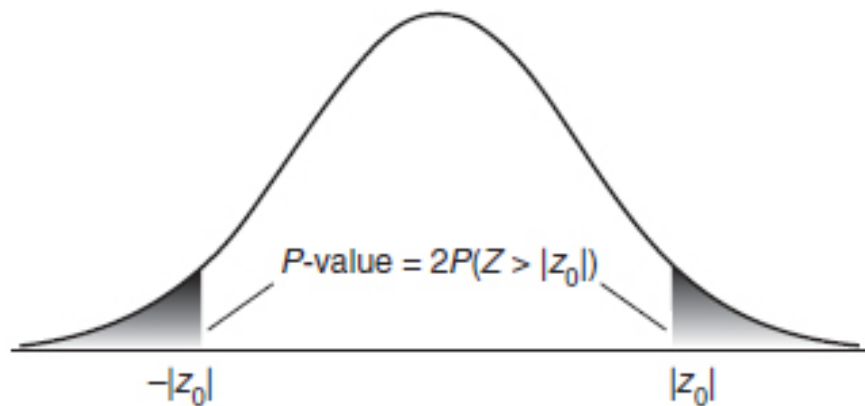
- starts with 100 units and ends after 24 months with 60 units (ie, a survival rate of 60% or an attrition rate of 40%);
- we have $\hat{\rho}_1 = 0.6$;

Unit trajectory 2:

- starts with 100 units and ends after 24 months with 40 units (ie, a survival rate of 40%);
- we have

$$\begin{aligned}
 \hat{\rho}_2 &= 0.4 \\
 \hat{\rho} &= \left(\frac{n_1}{n_1 + n_2} \right) (\hat{\rho}_1) + \left(\frac{n_2}{n_1 + n_2} \right) (\hat{\rho}_2) \\
 &= \left(\frac{100}{200} \right) (0.6) + \left(\frac{100}{200} \right) (0.4) \\
 &= (0.5)(0.6) + (0.5)(0.4) \\
 &= 0.5
 \end{aligned}$$

Figure 5.11 Z-distribution curve



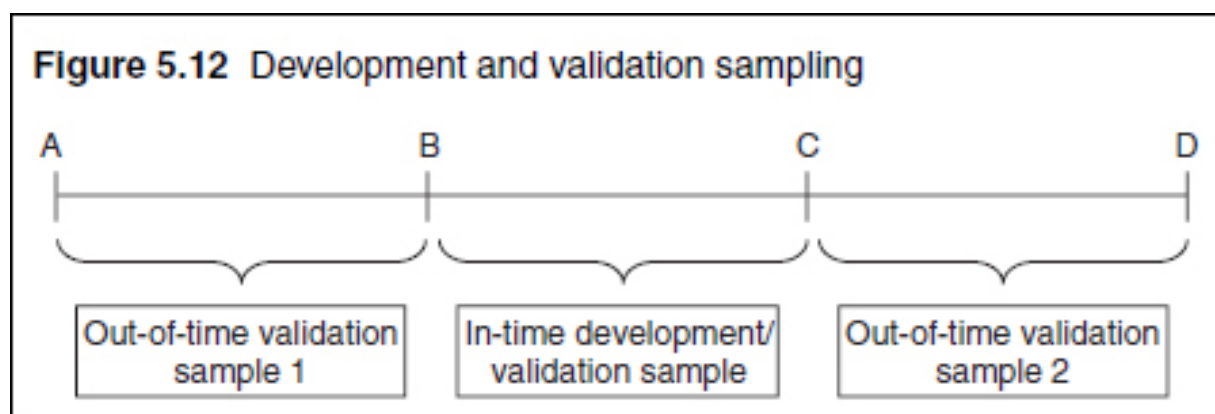
$$\begin{aligned}
 Z &= \frac{\hat{\rho}_1 - \hat{\rho}_2}{\sqrt{\hat{\rho}(1 - \hat{\rho})(1/n_1 + 1/n_2)}} \\
 \Rightarrow Z &= \frac{0.6 - 0.4}{\sqrt{(0.5)(0.5)(1/100 + 1/100)}} \\
 \Rightarrow Z &= \frac{0.2}{\sqrt{0.25(0.02)}} = \frac{0.2}{0.0707} = 2.8284
 \end{aligned}$$

Looking up the value 2.8284 in the statistical Z table, we get a p -value of 0.00466; since this is less than 0.01, the result is significant and we reject H_0 and accept that the unit proportions are different. Therefore, they could potentially be considered as separate segments.

SAMPLING CONSIDERATIONS

Appropriate sampling is a key ingredient in terms of producing a sound and stable predictive model. It is of paramount importance that the sample on which the model is developed is suitable for producing estimates on a go-forward basis. Ideally, we would want to build the model (ie, estimate its parameter coefficients) on a “development” sample and then validate its strength or predictive power on a “hold-out” or “validation” sample. Additionally, we would want to further assess its strength on an “out-of-time” sample or period in time outside of that on which the model was built. The diagram in [Figure 5.12](#) provides an illustration.

Within the development/validation sample, a split of 60%/40% or 70%/30% development/validation will suffice. It is critically important to ensure that the event variable (in this case $B(t)$ or some related transformation) is not statistically significantly different between the development and validation samples (this will ensure samples within the “in-time” period are randomly selected in an appropriate manner).



MODELLING EXPECTED LIFE: THE TIME-DEPENDENT APPROACH

We have discussed the objective of modelling expected lives of liabilities through first identifying and quantifying differentiated pockets or subsegments

of deposit behaviour. Once an intuitive and appropriate segmentation scheme has been defined and established, the actual physical modelling process may begin. The overall goal is to forecast the future trajectory of the liabilities so that the expected remaining life may be estimated. One simplistic way to estimate expected life is to model it singularly as a function of time.

Let us begin by considering one cohort or collection of accounts that all originated or opened at a point in time t_0 . To date, they will have exhibited a balance trajectory $B(t)$ that is observable and that can be estimated through the application of functional forms that are a function of time-dependent factors.

Let us take, for example, [Figure 5.13](#). We see that the negative exponential functional form $B(t) = a_0 \exp(-\beta^*t)$ can be used to model this movement and issue forecasts going forward. Of course, this would not be the most appropriate functional form to adopt, as it becomes asymptotic with the x -axis. One potential solution would be to break the negative exponential function and introduce a linear function (in a piece-wise approach) such that the balance is guaranteed to meet the x -axis ([Figure 5.14](#)).

Let us now look at an example of a liabilities balance that is decaying at a particular rate of change. We wish to estimate the expected remaining life by modelling the decay rate as a function of time and then extrapolating this forward ([Figure 5.15](#)).

Figure 5.13 Negative exponential balance trajectory

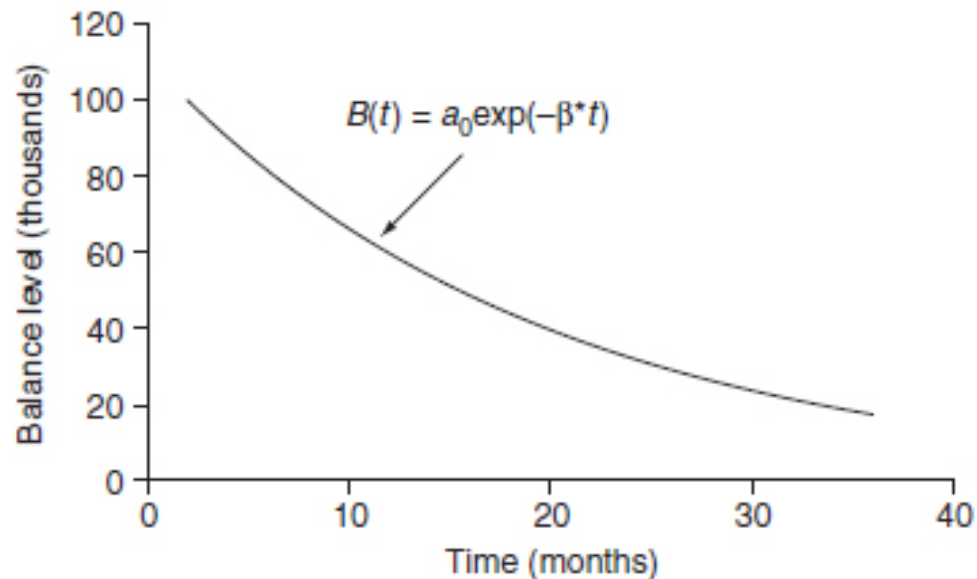
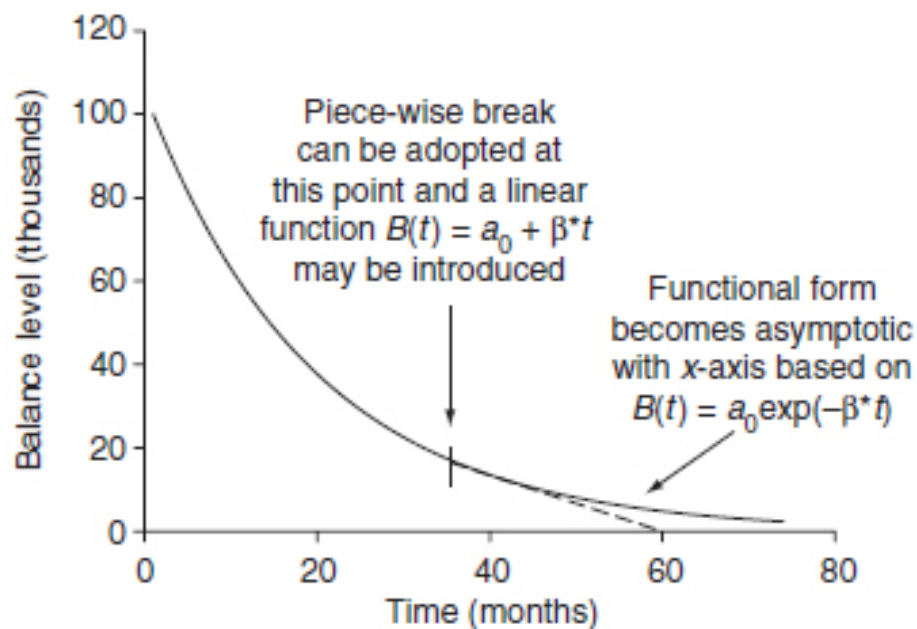


Figure 5.14 Implementing a piece-wise linear break



There are a variety of ways we can extrapolate this balance trajectory forward. We will observe the fit and forward trajectory when a linear or polynomial function is applied.

Linear fit

Using the SAS system, we apply a first-order linear equation and observe the results (Table 5.3).

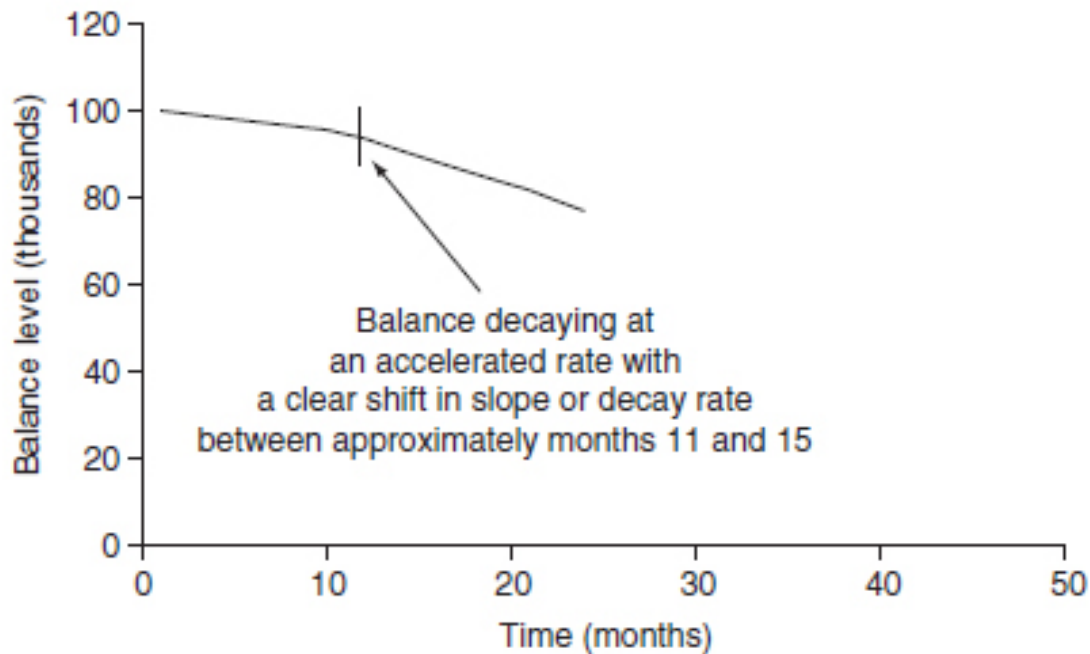
We observe that the parameter estimate of time is significant (p value less than 0.0001) and that the model achieved an R -squared value of 95.83%. Furthermore, the slope is suggesting that the balance is decreasing at a rate of 1009.40166 units per month.

Table 5.3 Analysis of variance: linear fit

(a) Analysis of variance					
Source	DF	Sum of squares	Mean square	F -value	Pr > F
Model	1	1,171,725,468	1,171,725,468	505.97	<0.0001
Error	22	50,947,920	2,315,815		
Corrected total	23	1,222,673,388			
Root mean squared error	1,521.78005	R^2	0.9583	Dependent mean	91,007
Adjusted R^2	0.9564	CV	1.67217		
(b) Parameter estimates					
Variable	DF	Parameter estimate	Standard error	t -value	Pr > $ t $
Intercept	1	103,624	641.20276	161.61	<0.0001
Time	1	−1,009.40166	44.87485	−22.49	<0.0001

Notes: DF, degrees of freedom. CV, coefficient of variation.

Figure 5.15 Accelerated balance decay



Let us now apply a second-order polynomial function (time squared (time^2) is defined as $\text{time} \times \text{time}$) to forward extrapolate the balance trajectory. Again using the SAS system, we observe the results in [Table 5.4](#).

The fit is now improved, with the *R*-squared value at 0.9977, or 99.77%, and all parameter estimates being highly significant (all *p*values less than 0.0006). If we plot these two equations and their associated forward extrapolation, we can observe the shape of the balance run-off ([Figure 5.16](#)).

It is evident that, when applying the predictive estimates to the go-forward extrapolation period, we observe significantly different trajectories. This of course will also have a profound effect on the remaining expected average life. If we apply the expected remaining average life calculation method shown in [Table 5.2](#), we arrive at an expected remaining average life of 39.33 for the linear fit and 15.77 for the second-order polynomial fit. It is here where sound business judgement becomes equally important to statistical methods. Revisiting [Figure 5.15](#), we see that there is a natural break in the rate of change of the balance trajectory at approximately the 12th month. If we were to restrict the modelling data series to the 12th month and later while applying a linear fit, we would arrive at the following equation (as represented in [Table 5.5](#)).

We observe that the parameter estimate of time is significant (p -value less than 0.0001) and that the model achieved an R -squared value of 99.86%. Furthermore, the slope suggests that the balance is decreasing at a rate of 1365.32810 units per month.

Table 5.4 Analysis of variance: polynomial fit

(a) Analysis of variance					
Source	DF	Sum of squares	Mean square	F-value	Pr > F
Model	2	1,219,866,155	609,933,078	4,562.71	<0.0001
Error	21	2,807,233	133,678		
Corrected total	23	1,222,673,388			
Root mean squared error	365.61968	R^2	0.9977	Dependent mean	91,007
Adjusted R^2	0.9975	CV	0.40175		
(b) Parameter estimates					
Variable	DF	Parameter estimate	Standard error	t-value	Pr > t
Intercept	1	100,035	243.94207	410.08	<0.0001
Time ²	1	−33.13255	1.74594	−18.98	<0.0001
Time	1	−181.08796	44.96025	−4.03	0.0006

Notes: DF, degrees of freedom. CV, coefficient of variation.

Table 5.5 Analysis of variance: linear fit post month 12

(a) Analysis of variance					
Source	DF	Sum of squares	Mean square	F-value	Pr > F
Model	1	266,569,279	266,569,279	7,057.53	<0.0001
Error	10	377,709	37,771		
Corrected total	11	266,946,988			
Root mean squared error	194.34733	R^2	0.9986	Dependent mean	84,837
Adjusted R^2	0.9984	CV	0.22908		
(b) Parameter estimates					
Variable	DF	Parameter estimate	Standard error	t-value	Pr > t
Intercept	1	110,095	305.85416	359.96	<0.0001
Time	1	−1,365.32810	16.25214	−84.01	<0.0001

Notes: DF, degrees of freedom. CV, coefficient of variation.

Figure 5.16 Comparison of both approaches

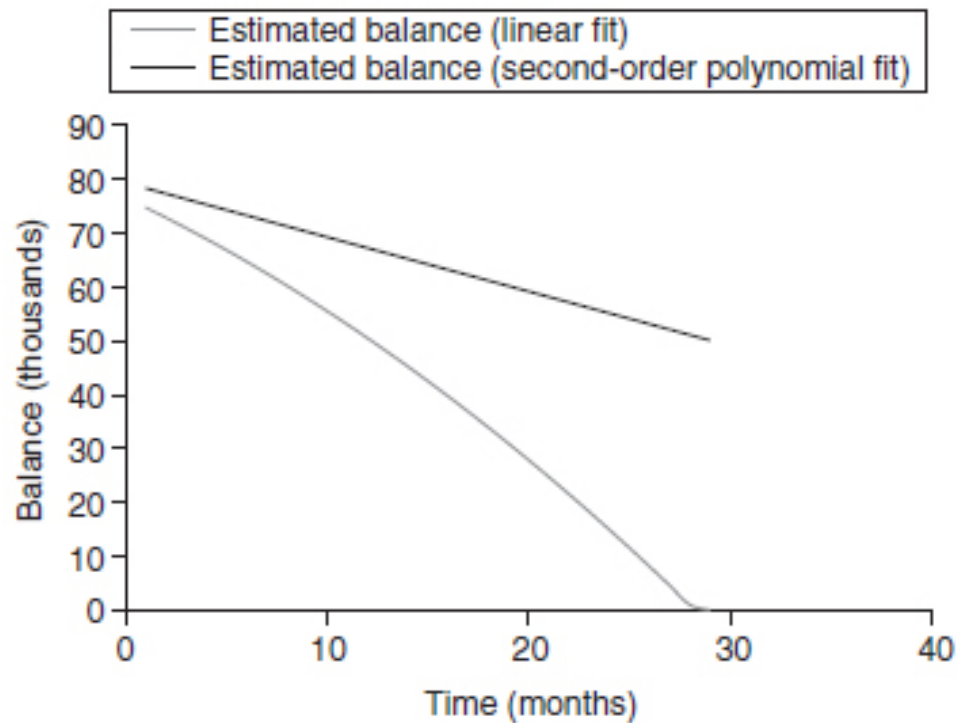
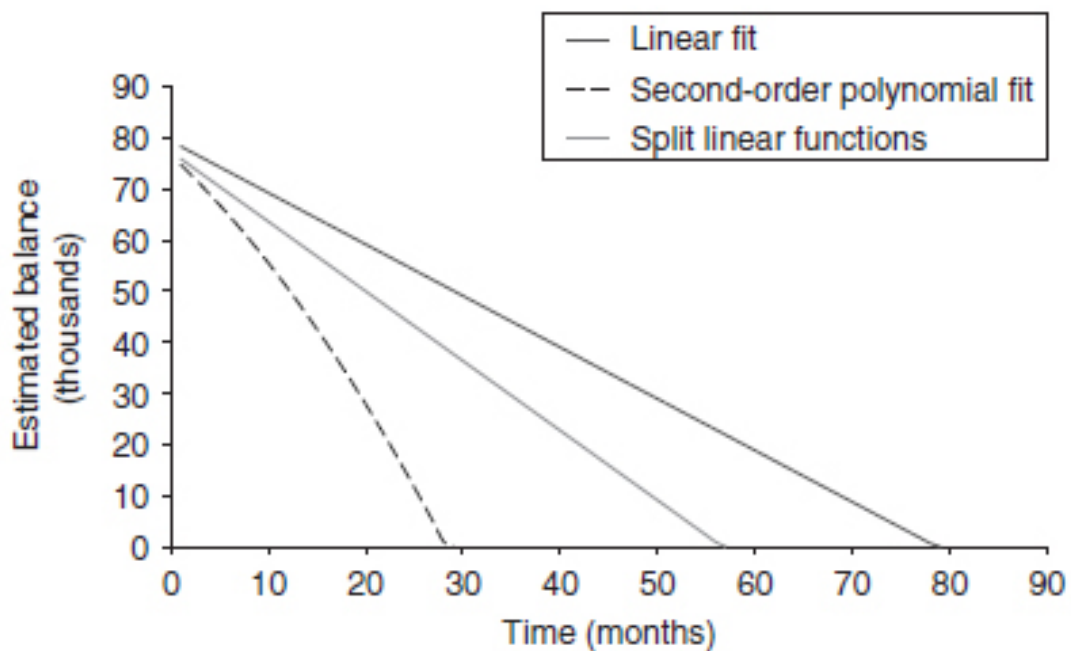


Figure 5.17 Comparing multiple balance run-off trajectories



This fit would yield an expected remaining average life of 27.814 months. [Figure 5.17](#) illustrates how the trajectories of all three techniques compare.

When comparing all three techniques, we see that the second-order polynomial fit crosses the x -axis at the 29th month for an expected remaining average life of 15.77 months; the split linear function crosses the x -axis at the 57th month for an expected remaining average life of 27.814 months; finally, the linear function crosses the x -axis at the 79th month for an expected remaining average life of 39.33 months.

Clearly, expressing B as a function of time, $B(t)$, will deliver an expected remaining life but it will not explain why the rate of change is what it is; as a result, great care must be taken when using this approach (including overlaying portfolio subject matter expertise to explain movements in the trajectory). On an overall basis, we shall see that a much preferred method is the “multivariate” approach.

MODELLING EXPECTED LIFE: THE MULTIVARIATE APPROACH

The previous section introduced a methodology for modelling expected life singularly as a function of time, ie, an approach dependent solely on time factors and their associated variations. While it is true that the passage of time, singularly, can express a proportion of the variation in balance movements (and serve as a rough approximation of what an expected behavioural life would equate to), it is, of course, not the only driver. As discussed previously, there are multiple factors that drive and influence consumer liabilities’ behaviour: the nature and life stage of the consumer, the rate of return, the safety and reputation of the institution, the product structure (promotional rate, fee structures, etc), the liquidity or accessibility to funds and the relationship the consumer has with the institution. Thus, the “multivariate approach” is significantly more sophisticated, as it attempts to describe (through multivariate regression testing) what drives the underlying balance movement or behaviour. It does not rely solely on time-dependent variables, but also includes the testing of a multitude of factors.

Defining the dependent variable

Thus far, we have discussed modelling savings balances over time. Implicit in this was the assumption that the dependent variable being modelled and

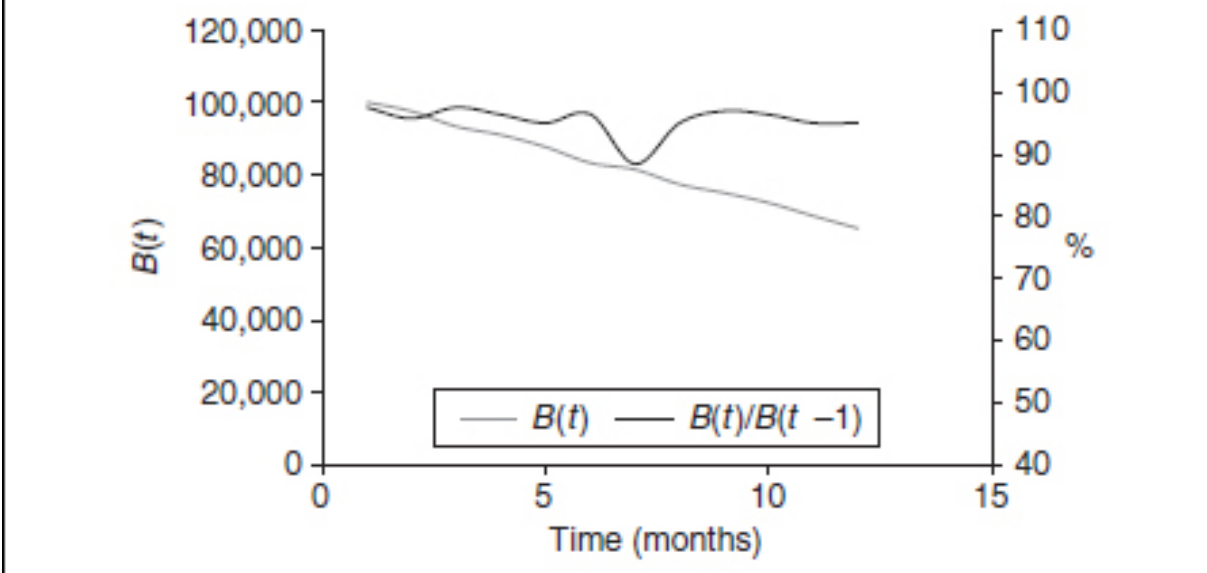
predicted was either

- $B(t)$, or the absolute level in balances at a point in time, t , or
- $\text{balance_proportion}(t)$, or the proportion of balances remaining at a point in time, t , relative to the balances at the origination date.

Table 5.6 Mean and variance comparison

Time	Balance	$B(t) - B(t - 1)$	$B(t) / B(t - 1)$ (%)
1	100,000		
2	97,500	2,500.0	97.5
3	93,335	4,165.0	95.7
4	91,000	2,335.0	97.5
5	87,675	3,325.0	96.3
6	83,291	4,383.8	95.0
Mean	92,134	3,341.8	96.4
Variance	38,294,635	871,468.4	0.0
7	81,450	1,841.3	88.4
8	77,378	4,072.5	95.0
9	75,000	2,377.5	96.9
10	72,250	2,750.0	96.3
11	68,638	3,612.5	95.0
12	65,206	3,431.9	95.0
Mean	73,320	3,014.3	94.4
Variance	34,859,279	700,682.9	0.1

Figure 5.18 Stationarity comparison



Both variables would be measured on a cohort basis (where a cohort is defined as a set of accounts booked under the same product/pricing offer at the same point in time within a microsegment).

Both variables relate to time-series metrics that are continuous variables on an ordinal scale. A key issue that arises when adopting this approach within time-series regression models is that of spurious regression (where ordinary least square regression methods may yield incorrect results) brought on by the potential non-stationarity of the data. The basic principle is that if ordinary least squares regression methods are employed, the time series must exhibit a constant mean and variance over time, that is, to be stationary in nature.

One way to remedy the case of non-stationarity within variables is to model the change in balance from month to month as a ratio. This is where, instead of modelling $B(t)$, we model $B(t)/B(t-1)$ or $\log[B(t)/B(t-1)]$. Let us observe how the trend compares by using the technique for a particular time series (Table 5.6).

We observe that the mean and variance differ considerably when comparing balance against $B(t)/B(t-1)$. Clearly, when looking at the balance movement ratio from month to month, the mean and variance are near constant.

We plot this relationship and confirm it graphically in Figure 5.18.

It is evident that the transformation $B(t)/B(t - 1)$ behaves in a stationary manner; when applying the transformation and creating $\text{balance_proportion}(t)$, the preferred approach thus involves assigning $\text{balance_proportion}(t)/\text{balance_proportion}(t - 1)$ as the dependent variable. Why the change in balance proportion rather than the change in absolute balances? This is due to the fact that when modelling change in proportion, the parameter estimates or coefficients of the model are estimating relative change and are thus not influenced by the amount of incoming origination balances (although this could potentially be a key factor in establishing the initial segmentation scheme). Modelling the change in balance proportions also provides an advantage when comparing results from one model to the next: the parameter estimates are essentially indexed.

Similarly, all time-series independent variables would also have to be transformed in a similar manner; that is, through the use of the “ratio” technique. For example, if we were to introduce GDP per capita as an independent variable we would introduce it as $\text{GDP}(t)/\text{GDP}(t - 1)$.

Key factors driving balance behaviour within the multivariate approach

As discussed previously, there are a multitude of factors that drive balance behaviour within a savings book: both those that are a byproduct of the customer base and those that are more directly related to the competitiveness of the product and its associated pricing within the broader macroeconomic environment (as well as the relative stability and safety of the institution and its associated service levels).

In essence, the segmentation scheme handles the nature and life stage of the consumer (eg, young professional, older wealthy client) and the relationship they have with the institution (number of products held, tenure, etc). Rate of return, product structure, safety of the institution, accessibility to funds and the macroeconomic environment can all be expressed as parameters within the segment-specific model, but how can they be used and what exact form do they take?

Rate of return

Let us first consider the rate of return. The laws of economics dictate that, all else being equal, a higher rate paid out by the institution will attract and

retain deposits more readily than a lower one. But what does a potential investor compare a savings rate to? The answer is quite simple: to other investment products, such as competing savings products, bonds, equities or gold.

If we consider an institution's rate of return r on a savings product, we can formulate an index that compares it to other competitive products. Let us introduce the variable rate_index as the ratio of the savings rate that institution x issues to its customers to the average savings rate of similarly structured competing products on the market. We thus have

rate_index

$$= \frac{\text{savings rate of institution } x}{\text{average savings rate of similarly structured products at competing institutions}}$$

where $0 \leq \text{rate_index} \leq \infty$. A rate_index of less than 1 means that the institution is paying less than the average of the market, and a rate_index of greater than or equal to 1 indicates that the institution is paying the average market rate or more. Intuitively, higher rate_index levels should attract higher levels of incoming balances and promote higher levels of balance retention (thus a longer expected life of the liability).

Example 5.2. Institution x offers a 3% rate on their generic savings product versus the market average of 2%. The rate_index thus takes a value of 1.5. Graphically, we expect a positive relationship between the rate of change of balance with respect to time $(\delta b(t)/\delta t)$ and rate_index . Expressing this mathematically we have Let us now assume we have three competitive institutions offering differing interest rates on their savings products. [Table 5.7](#) allows a comparison.

Table 5.7 Rate index table

Institution	Savings rate (%)	Average savings rate of market (%)	Rate index	Balance trajectory
X	3	2	1.50	1
Y	1.80	2	0.90	2
Z	1.50	2	0.75	3

Figure 5.19 Balance trajectory by rate_index

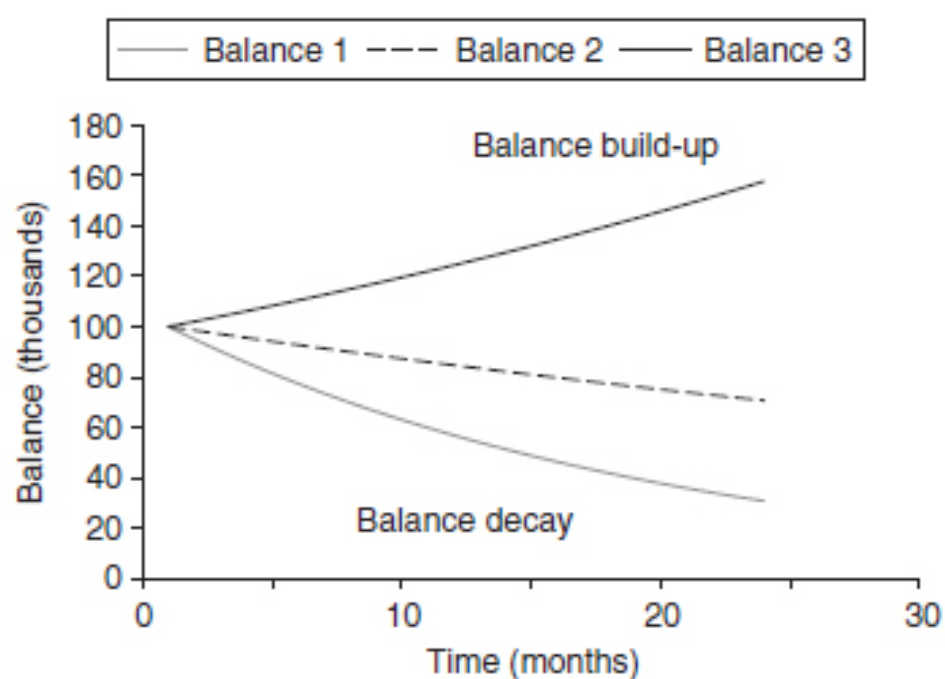
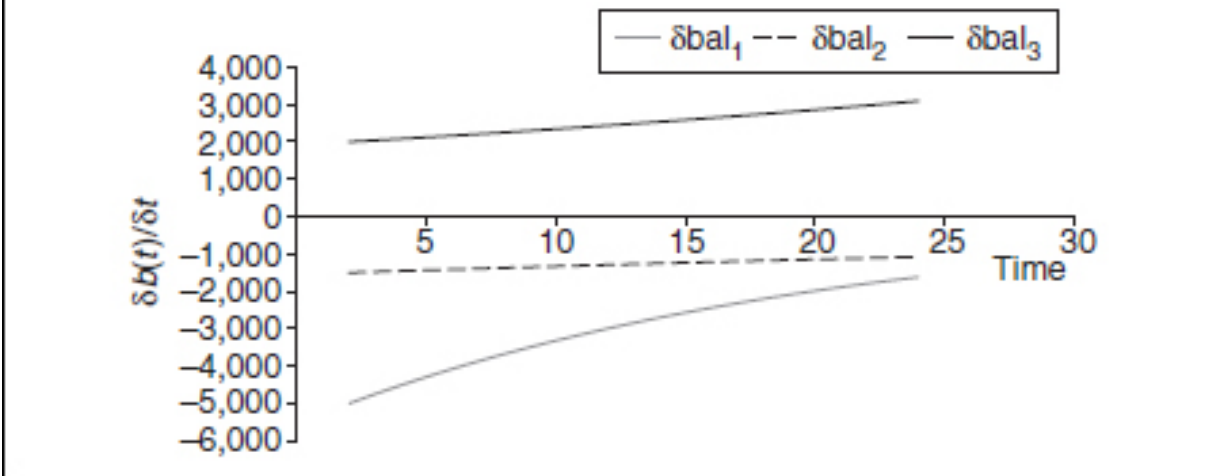


Figure 5.20 Rate of change of balances by rate_index



$$0 \leq \frac{\text{cov}(\text{rate_index}, (\delta b(t)/\delta t))}{(\sqrt{\text{var}(\text{rate_index}))(\sqrt{\text{var}(\delta b(t)/\delta t))}} \leq 1$$

Let us now examine three potential balance trajectories (one for each institution), which are shown in [Figure 5.19](#).

We see from [Figure 5.19](#) that balance trajectory 1 exhibits an increasing trend, balance trajectory 2 shows a slightly decreasing trend and trajectory 3 shows a significantly higher rate of decrease. These patterns of behaviour are intuitive, as we observe the highest rate_index associated with a balance build-up and the lowest rate_index associated with the most significant balance run-off behaviour.

If we observe the rate of change of balance through time for the three institutions' ($\delta b(t)/\delta t$) in [Figure 5.20](#), we can clearly see the positive correlation or association between rate_index and $\delta b(t)/\delta t$.

Introducing the introductory or promo period as a main effect or interaction variable within the regression equation

We have discussed how a price index variable can be created that relates the competitiveness of an institution's savings rate to that of its competitors.

Another key feature of some savings products is the introductory ("intro") or promotion ("promo") period. During this period (which varies from competitor to competitor but typically lasts from three months to one year), an increased rate of return is offered to the depositor, with the objective of

attracting a higher level of deposits or balances from the onset. Subsequent to the intro or promo period, the rate is then reduced. This is often called the “reversion rate”. [Figure 5.21](#) illustrates the above example.

We observe that the introductory period, where the rate paid out to the consumer was 2%, lasts for 12 months. The rate then drops to 0.5% in month 13, leading to a significant exodus of balances.

In the second example ([Figure 5.22](#)), we observe that the introductory period, whereby the rate paid out to the consumer was 1%, lasts for six months. The rate then drops to 0.25% in month 13, again leading to a significant exodus of balances (but at a considerably faster decay rate than the 12-month-intro example).

From a depositor’s standpoint, the ideal scenario would be high promotional or introductory interest rates paid out over a long period of time and then reducing to a similarly high (but obviously not equally high) reversion rate. All of these parameters are thus positively correlated to the likelihood of attracting balances and, additionally, the associated magnitude of the initial investment.

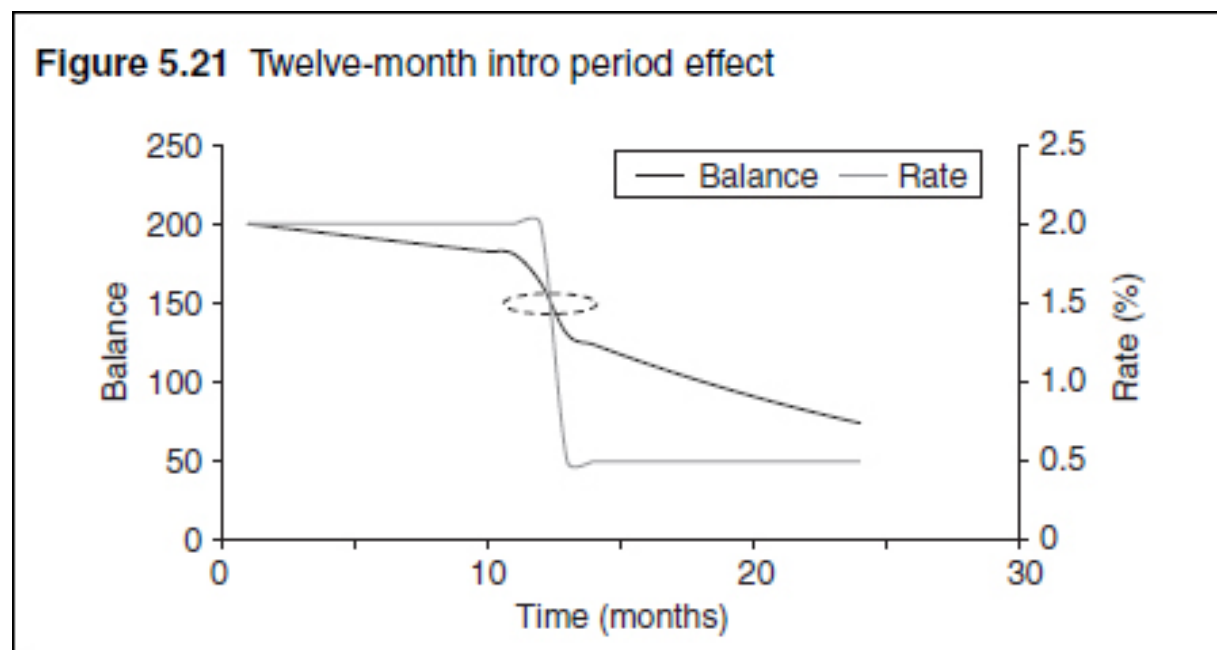
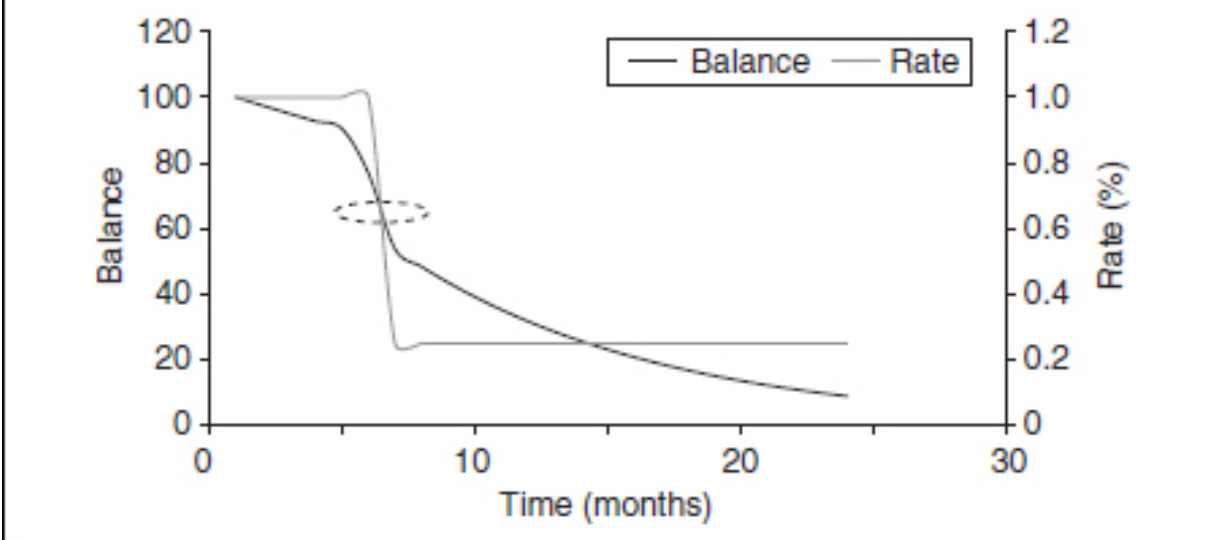


Figure 5.22 Six-month intro period effect



Let us consider how these variables may be used within the modelling exercise. We define the following:

- `intro_rate` = the rate issued during the introductory period;
- `duration` = the length of the introductory period;
- `reversion_rate` = the rate issued post the introductory period.

A key consideration would be to include interaction effects of these variables. Since they are all positively correlated to the attraction and retention of savings balances, certain two-way and three-way multiplicative interactions may be considered and trialled within the regression phase, for example, `intro_rate` $\times$ `reversion_rate` as a two-way interaction or `intro_rate` $\times$ `duration` $\times$ `reversion_rate` as a three-way interaction. Another key theme is how balances are affected once the intro offer expires and the customer moves on to the reversion rate; in this case the ratio `reversion_rate: intro_rate`, or `reversion_rate` – `intro_rate` as an absolute basis points difference, may be used as well.

Comparing a savings product rate of return to other investment vehicles

But it is not only other savings rate products that an investor has to choose from; it is also bonds or treasury bills, commodities such as gold or even the equities markets.

Price index ratios (as depicted in [Figure 5.13](#)) should be created that compare the savings rate of return to

- (i) stock market year-over-year return levels (eg, FTSE 12-month growth rate),
- (ii) 12-month growth rate of the price of gold,
- (iii) one-year, two-year or longer fixed-term bonds or Treasury bills and their associated rates of return.

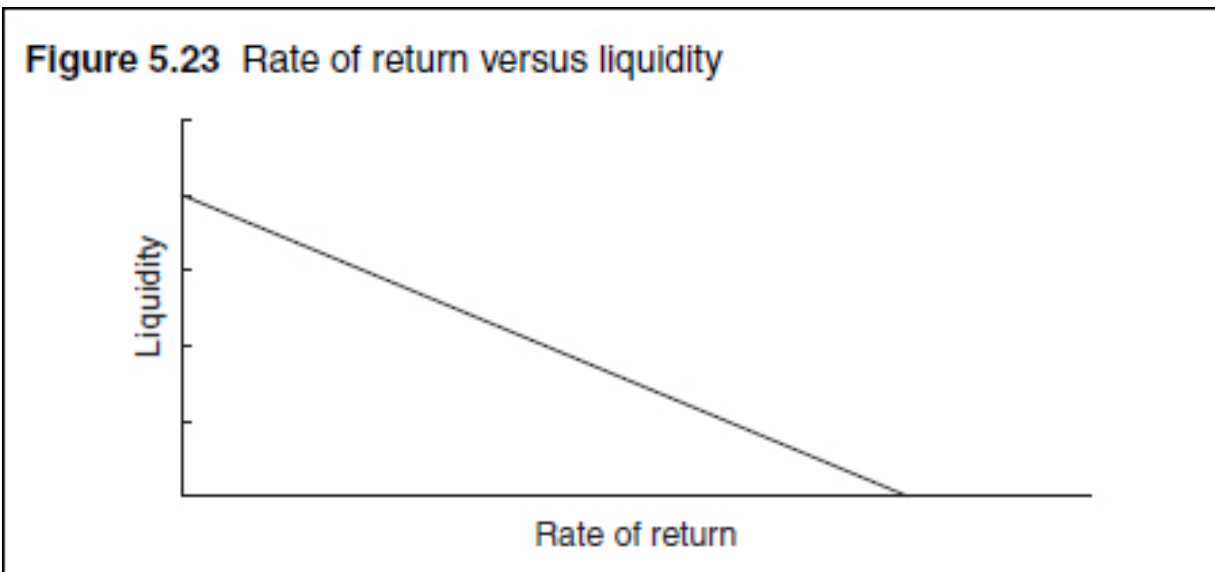
In the case of bonds, it is implicitly understood that interest rates paid out to the depositor are higher in exchange for the depositor “locking in” their funds for an extended period of time. The key question is the following: at what point does the depositor trade-off rate of return for liquidity? Is an investor willing to lock up their funds for a year or more in exchange for a higher rate of return? Conversely, is the depositor willing to take on a lower rate of return in exchange for liquidity (ie, full-time access to their funds)?

We can observe this key relationship in [Figure 5.23](#). Based on this relationship, it is critical that variables are created and tested which measure the degree of difference between savings deposit rates and various bond rates.

Use of macroeconomic data in expected life modelling

Thus far, we have discussed the use of pricing-index-related and duration variables to predict the expected remaining life of a deposit. The balance trajectory, however, is also influenced by the macroeconomic environment. Parameters such as the base rate, unemployment rate and GDP per capita can all influence the inflow or outflow of deposit balances. An economic recession, for example, dominated by high unemployment rates and a low GDP per capita, may have an adverse effect on deposit balances, as savings rate levels typically drop during these times. Interest rate indexes, such as the base rate, can also influence deposit balances; typically, low deposit rates go hand in hand with a low base rate environment and balances tend to flow towards the equities markets in these times. Conversely, a high base rate

tends to lead to high(er) deposit rates, which, in turn, capture investible assets away from the more volatile equities markets.



We may thus introduce an equation that takes the following form

$$\begin{aligned}
 & \log \left[\frac{B(t)}{B(t-1)} \right] \\
 &= f[\delta \text{base rate}, \delta \text{unemployment rate}, \delta \text{GDP per capita}] \\
 &\Rightarrow \log \left[\frac{B(t)}{B(t-1)} \right] \\
 &= a_0 + a_1 \delta \text{base rate} + a_2 \delta \text{unemployment rate} + \dots + a_9 x_9
 \end{aligned}$$

Price related variables (and associated interactions) can then be worked into a potential equation as follows

$$\begin{aligned}
 \log \left[\frac{B(t)}{B(t-1)} \right] &= a_0 + a_1 \delta \text{base rate} + a_2 \delta \text{unemployment rate} \\
 &+ a_3 \text{rate_index} + a_4 \text{duration} \\
 &+ a_5 \text{time}
 \end{aligned}$$

This simple equation combines two macroeconomic factors (the change in the base rate and the unemployment rate), one factor that looks at the competitiveness of the promotional or introductory rate, its associated duration and time.

A product structure's impact on balance behaviour

Another significant driver of balance behaviour is the savings deposit product structure. Of particular interest is the concept of liquidity and its associated cost (on a transaction-by-transaction basis). When considering these themes, several questions arise; for example:

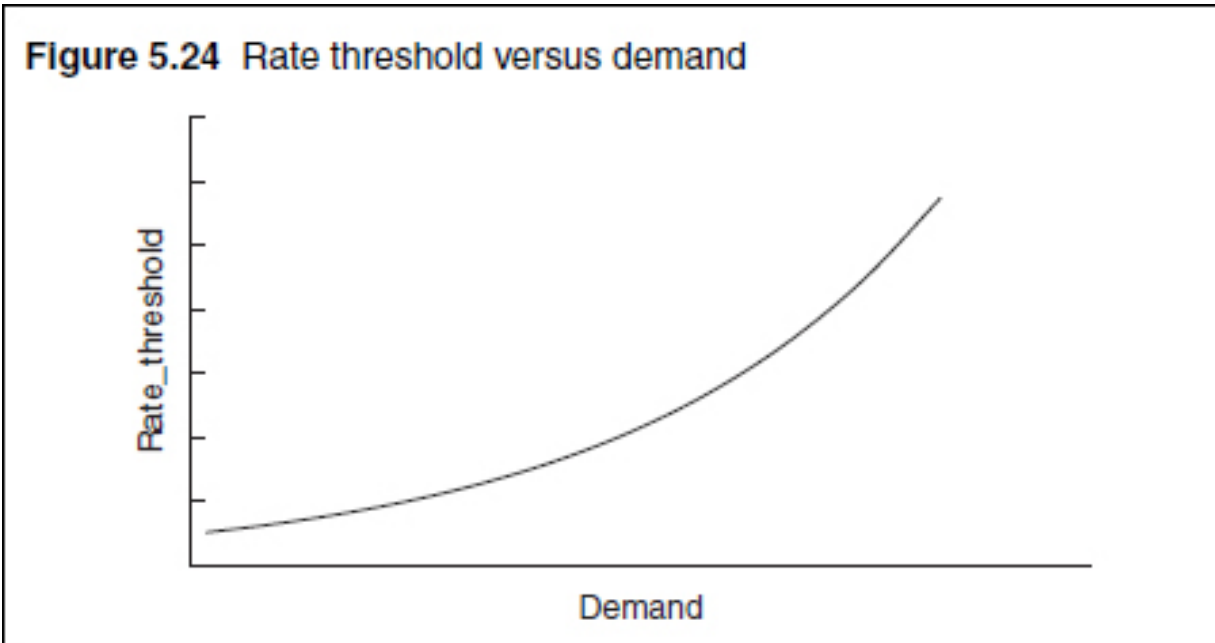
- what is the daily withdrawal limit?
- how many transactions are allowed per month?
- is there a fee if this number is exceeded?
- is there a minimum balance that must be held in the account in order to receive a prescribed level of interest?

From these themes, independent, explanatory variables may be created (similar to the price- or rate-related index variables) that gauge the competitiveness of the product offering on the market. As an example, if an institution issues a rate y but only allows for it to be paid out above a certain balance level z , this serves as a constraint. Ideally, a customer would demand a high interest rate and a low balance threshold associated with it. An interaction variable which looks at the two may be created as follows

$$\text{rate_threshold} = \frac{\text{interest_rate} \times 10,000}{\text{balance_threshold}}$$

High values of this ratio would be appealing to an investor and drive a strong positive correlation with attracting and retaining balances. For example, an interest rate of 3% with a minimum balance threshold of £500 required to achieve it would have a rate_threshold value of $(0.03 \times 10,000)/500 = 300/500 = 0.6$; conversely, an interest rate of 5% with a minimum threshold of £1,000 would have a ratio of 0.5. Generally speaking, we would expect a high ratio (0.6 versus 0.5) to be more appealing. However, is it reasonable that a combination of 3%/£500 would drive higher balance growth and retention than a combination of 5%/£1,000? It is unlikely, and in this case we would have to assign a higher “weight” to the interest rate versus the balance threshold. In order to achieve this we would need to introduce a squared or cubed term to induce this effect

$$\text{rate_threshold} = \frac{(\text{interest_rate} \times 10,000)^2}{\text{balance_threshold}}$$



The combination of 3%/£500 would then yield a value of

$$\frac{(0.03 \times 10,000)^2}{500} = \frac{90,000}{500} = 180$$

whereas the combination of 5%/£1,000 would yield a value

$$\frac{(0.05 \times 10,000)^2}{1,000} = \frac{250,000}{1,000} = 250$$

Using the revised definition of rate_threshold we would expect a positive relationship with balance growth or balance retention ([Figure 5.24](#)).

Safety and security of the institution

As described previously, the safety and security of the institution is of paramount importance to the investor. While several countries have government-supported programmes that insure deposits up to a particular amount, investors that are over these thresholds need assurance that the potential for the financial institution to become insolvent is negligible. The relative safety of an institution and how that safety level attracts and retains deposits is difficult to quantify; nevertheless, variables can be constructed

that look at the change in the institution's capital ratio or the year-over-year movement in its share price, for example. These types of parameters may prove useful in the regression modelling exercise.

Service levels of the institution

Similarly, the customer service level of the institution is a key driver of deposit balance growth and subsequent retention. Financial institutions that allow their customers to communicate with them through different channels and offer them instant access to information about their accounts are providing a positive customer experience and this should correlate positively with deposit balance build-up. Some key variables that can be created under the service theme include the number of branches and automated teller machines (ATMs) the institution has relative to its competitors and the number of options it provides to service multi-channel transactional banking (ie, offering transactions through a mobile telephony channel for example).

The concept of lagged terms

More often than not, the consumer does not react immediately to a change in the macroeconomic environment or a change in a product's pricing, withdrawal policy or terms and conditions. There is often a "lagged effect", where the response to a product or macroeconomic change takes a degree of time to materialise and manifest itself as a "reactionary behaviour". For example, let us assume that a deposit's interest rate is linked to the UK base rate. If the rate increases, we may see an eventual influx of balances; conversely, if the rate decreases, we may see an exodus of balances. These deposit inflows or outflows typically do not happen instantaneously; as described, above there is a "lagged effect". The incorporation of lagged effects is a key consideration when testing the effectiveness of the independent variable set within the regression model. Some examples that may be used are as follows:

- change in unemployment from twelve months ago to six months ago;
- stock market growth rate from six months ago to three months ago;
- interest rate change on a deposit product from three months ago.

When testing for a lagged effect, the model developer will often see a stronger correlation between balance movement at time t and pricing, policy changes or the macroeconomic environment at time $t - \delta t$.

Forecasting

It is important to note that the multivariate regression approach is predicated on explaining variation in historical balance movement as a function of the aforementioned drivers (the macroeconomic environment, price competitiveness, etc). The objective of the model is, of course, to give a forecast. This will necessitate the creation of a set of forward-looking estimates that will serve as inputs for the model. For example, the historical macroeconomic environment and a bank's prevailing pricing position during that time period can explain why the bank's deposit balance trajectories behaved as they did. But what is the macroeconomic environment going to look like going forward? What pricing position or philosophy will the bank undertake? These themes will inevitably have to be assessed; therefore, a series of inputs to the multivariate regression model, such as future GDP and unemployment rates, will have to be estimated and forecast; additionally, the bank's pricing position will have to be estimated.

ASSESSMENT OF MODEL FIT

Once the behavioural model has been specified and regressions run and fitted, the inevitable series of questions arise:

- how close is the fit?
- how will the model perform in the future?
- has the validation been conducted correctly?
- is it stable and sound?
- could it be subject to extrapolation risk?

There are a multitude of tests that must be run in order to assess the fit and appropriateness of the behavioural model. The most important considerations are as follows:

- (i) fit statistics (primarily R -squared value);

- (ii) parameter significance (p -value threshold);
- (iii) test for presence of multicollinearity;
- (iv) residual analysis;
- (v) accuracy levels of the development versus the validation sample;
- (vi) model stability under stressed scenario testing.

Fit statistic

The R -squared value provides the level of fit of the model and is bounded between 0 and 1: higher values indicate a better fit. The following rule of thumb should be considered when interpreting R -squared levels.

0–30%: values in this range correspond to a weak fit.

30–70%: values in this range correspond to a medium fit.

70%+: values in this range correspond to a strong fit.

The R -squared value of the model should also be calculated on the in-time and out-of-time validation segments (see the section on “sampling considerations” on page 128) as well to ensure the model is robust and not necessarily performing well singularly on the data on which it was developed.

Parameter significance

Where to set the statistical significance threshold levels of p -values is often a matter of controversy. Analysts typically have to set a p -value threshold that a variable must meet to be considered for entry into the model. p -values less than 0.0001 are often seen as the preferred threshold level (although these were historically a byproduct of extremely strict testing and significance requirements within the pharmaceutical industry and medicine). For the purposes of behavioural deposit modelling, p -values that are below a limit of 0.05 should be allowed entry into the model.

Test for presence of multicollinearity

Multicollinearity is a key issue within statistical linear regression modelling and it is defined as when two or more of variables exhibit a high degree of

correlation. While the phenomenon of multicollinearity does not explicitly reduce the point estimate capability of the model, it can have ancillary adverse effects. For example, the presence of variables that are collinear can inflate the standard error of their parameter estimates, which could, in turn, give them incorrect coefficient signs or make the model unstable when subjected to a different sample. Models that suffer from a high degree of multicollinearity are also, at times, considered to be “overfit”, in that they contain a high number of superfluous variables (that are in this case also highly correlated to one another).

Table 5.8 Residual analysis table

Time	Actual balance	Estimated balance – linear fit	Actual $B(t)/B(t - 1)$	Estimated $B(t)/B(t - 1)$	Residual
1	100,000	102,615			
2	99,500	101,605	0.995	0.990	0.005
3	99,232	100,596	0.997	0.990	0.007
4	98,777	99,586	0.995	0.990	0.005
5	98,100	98,577	0.993	0.990	0.003
6	97,809	97,568	0.997	0.990	0.007
7	97,500	96,558	0.997	0.990	0.007
8	96,999	95,549	0.995	0.990	0.005
9	96,514	94,539	0.995	0.989	0.006
10	95,980	93,530	0.994	0.989	0.005
11	95,020	92,521	0.990	0.989	0.001
12	94,500	91,511	0.995	0.989	0.005
13	93,465	90,502	0.989	0.989	0.000
14	92,800	89,492	0.993	0.989	0.004
15	91,575	88,483	0.987	0.989	–0.002
16	90,700	87,474	0.990	0.989	0.002
17	90,650	86,464	0.999	0.988	0.011
18	89,999	85,455	0.993	0.988	0.004
19	88,549	84,445	0.984	0.988	–0.004
20	87,219	83,436	0.985	0.988	–0.003
21	85,710	82,427	0.983	0.988	–0.005
22	84,250	81,417	0.983	0.988	–0.005
23	82,100	80,408	0.974	0.988	–0.013
24	80,458	79,398	0.980	0.987	–0.007

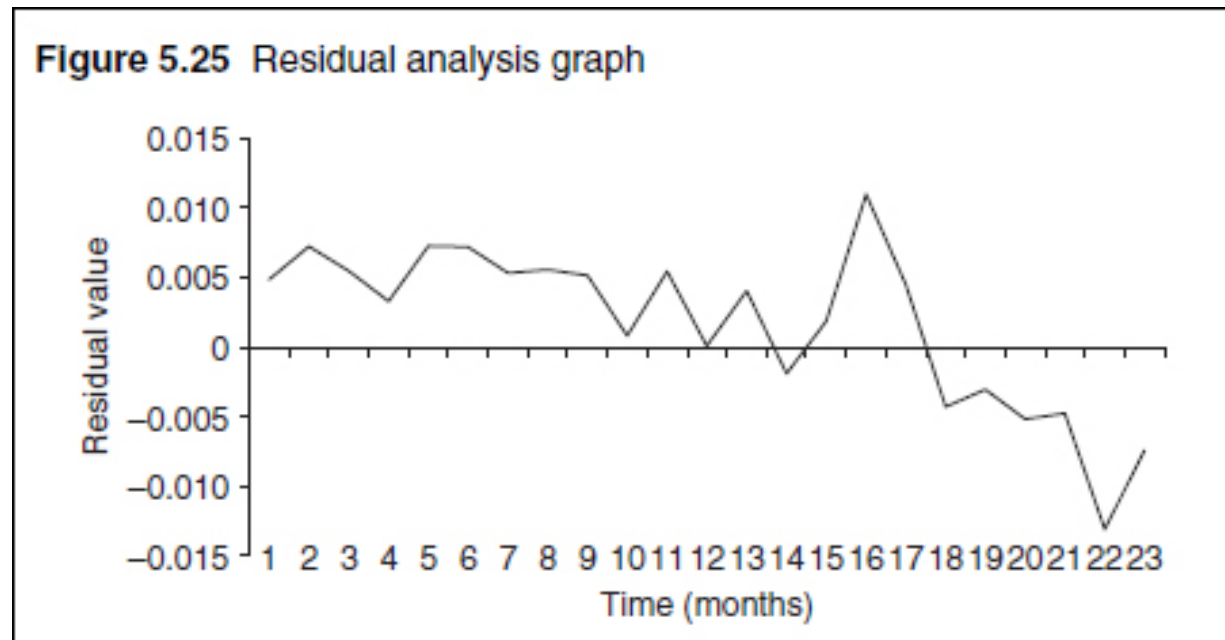
Notes: at $t = 15$ residual values begin to cross over and take on negative values.

Ridge regression, principal component analysis and correlation variable reduction exercises can all be employed to reduce multicollinearity.

Typically, variance inflation factors of variables that exceed levels of 5–10% should be candidates for closer inspection and potential exclusion from the model.

Residual analysis

Once the model has been built, residual analysis may be conducted by observing actual versus expected deviations or residuals. Assume we assign $E\{B_t\}$ as the expected value of the balance at time t and $A\{B_t\}$ as the actual value at time t . We may then compute the residual at time t as $\varepsilon_t = [A\{B_t\} - E\{B_t\}]$. Theoretically, ε_t , where $1 \leq t \leq n$ (n being the latest month in the development data time series), may exhibit both positive and negative values (see the example in [Table 5.8](#)).



In [Table 5.8](#) and [Figure 5.25](#), we observe that the residuals get smaller and smaller in the course of time and are negative by the 19th month. This is considered a “systematic error” and these residuals would be considered “heteroscedastic” versus a preferred “homoscedastic” distribution where no discernible pattern is observed (ie, they are distributed randomly). What this essentially signifies is that a key term is missing from the model and, as a result, the residuals are correlated with time. Residual analysis should always be undertaken to determine the distribution of the errors; thus, the

errors should always be randomly distributed in order to ensure a sound and stable predictive model.

Accuracy levels of the development versus validation samples

When comparing accuracy levels (actuals versus the predicted estimates) of the model, it is imperative that they do not vary significantly between the development and validation samples. It is, of course, expected that the accuracy of the model on the validation samples is somewhat weaker, but not significantly weaker. This would indicate that the model may have been too finely tuned to the development sample (ie, essentially “overfitted”) and thus not robust enough for predictive usage on other population samples. When a substantial accuracy differential is detected across the development and validation samples, it is important that the model is reviewed for any potential superfluous variables and retested against the validation samples until an adequate level of accuracy is established.

Model stability under stressed scenario testing

Another key consideration that requires significant focus and attention is how the model will estimate or predict balance movements when presented with a future stressed macroeconomic scenario. Let us assume that the model was built over a historical time series where the base rate ranged from 5% to 0.5%; what if it is now asked to provide an estimate where the base rate is stressed to a 10% level? It is here where the model is subjected to “extrapolation risk” or, more simply, it is asked to provide a prediction given a series of input data whose range does not constitute a subset of the data range upon which the model was developed. When this happens, the model will issue estimates that either are far out of the range of historically observed results or appear to be counter-intuitive.

Models that are especially subjected to “extrapolation risk” are usually those that contain

- (i) variables that entered the model but featured strictly monotonically increasing or decreasing values, eg, base rate decreasing from 5% to 0.5% but not increasing ever again or unemployment increasing from 5% to 10% but not decreasing ever again,

- (ii) variables that feature second- or third-order terms (ie, squared or cubic terms).

For example

$$\log \left[\frac{B(t)}{B(t-1)} \right] = a_0 + a_1 (\delta \text{base rate})^2 + a_2 (\delta \text{base rate})^3 + \dots$$

Models that are under development and will knowingly be subjected to “extrapolation risk” must take this into consideration. As a result, the model developer should (if possible) avoid second- or third-order terms to express relationships and additionally potentially temper the parameter coefficients to induce a differing rate of change (for ranges outside those the model was built on).

MODEL MONITORING/CALIBRATION FOR MAINTAINING ACCURACY

Once the model has been built and put into production, balance trajectory forecasts must be measured and calibrated on a timely basis to ensure the model is producing sound estimates of the expected remaining life. Residual analysis must be conducted on a continuous basis to ensure the model is accurate, within acceptable error tolerance thresholds, and not generating systematic errors (ie, consistently over- or underestimating balance movement). Depending on the distribution of these errors and their associated absolute magnitude, a decision can be taken on whether to

- (i) recalibrate the model, due to systematic and consistent but minimal over or underestimation, or
- (ii) rebuild the model, due to significant errors being exhibited.

SUMMARY AND CONCLUSION

In summary, building a sound statistical model to predict the expected remaining life of a deposit involves a multitude of key steps. Significant consideration must be given to

- how the event variable is defined,

- sampling techniques determining the development and validation data sets,
- how the segmentation scheme is established,
- the set of explanatory predictors that are chosen (price related, macroeconomic related, time functions, etc),
- appropriately transforming the dependent variable,
- testing that the model is fit for purpose by assessing fit statistics, collinearity diagnostics, residual error distributions and accuracy levels over both the development and validation samples, and
- ensuring the model is (post-implementation) subject to frequent monitoring to assess its predictive ability and soundness.

The appropriate execution of the above steps will yield a useful measurement/simulation tool for a bank's treasury function while ensuring that predictions made regarding a deposit's expected remaining life can be made in confidence.

Any views expressed are solely those of the author and do not necessarily represent the opinions of Freddie Mac or its Board of Directors.

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Modelling Non-Maturing Deposits with Stochastic Interest Rates and Credit Spreads

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This chapter introduces an approach to hedging non-maturing deposits under stochastic deposit volumes, interest rates and credit spreads. The method described allows outflows to be captured from the unexpected weakening of the creditworthiness of a financial institution. Furthermore, it allows modelling of the negative convexity from margin compression risks, which are the risks of market rates falling close to or below a floor for interest rates paid to clients. The approach allows the management of the present value – also referred to as economic value – of client deposit portfolios.

Following this introduction, a brief overview of the scope of the deposits product is provided. Subsequently, a static approach to constructing replicating portfolios is given. We then describe the simultaneous modelling of deposit volumes, interest rates and credit spreads and illustrate simulation results. The specific topics of hedge ratios for the margin compression risk and applications to decay models follow. A summary concludes the chapter.

The approach described in this chapter is similar to the approach to hedging the present value of demand deposits presented by Jarrow and van Deventer (1998) and the method of hedging the present value of the interest margin from demand deposits by Elkenbracht and Nauta (2006). However, our approach focuses on the deposit run-off profile resulting from stochastic

modelling of deposit volumes, credit spreads and interest rates. The approach also differs from that of Kalkbrenner and Wiling (2004), due to the addition of credit spreads.¹

Our approach allows the reader to implement a hedging strategy for sight deposits that allows the margin compression risk due to very low interest rates to be captured. Margin compression risk is the risk of the net interest margin being reduced due a floor in client rates while the hedge rate drops with market rate levels.

The approach presented provides an asset–liability management (ALM) manager the opportunity to adjust hedge ratios in an environment of high interest rates, not just when rates are low and the net interest margin is directly threatened. Furthermore, it allows us to reflect the risk of withdrawal of clients' money due to deterioration in their credit standing and to define a risk tolerance with respect to liquidity risk and convert this into a hedging strategy.

THE MAIN TYPES OF NON-MATURING DEPOSITS

Non-maturing deposits are deposits without a clearly specified contractual end date. The interest rate paid on such accounts may vary; for non-interest-bearing accounts the interest rate is zero. Such conditions mainly apply for current accounts or clearing balances.

Managed-rate deposits are deposits for which the rate paid to the client is set at the discretion of the deposit-taking institution. The rate set by the institution may depend on factors such as the development of the interbank money market, external reference rates and the competitive environment. As a consequence, the rate paid to clients may not be fully correlated with benchmark interest rates.

Deposits with client rates directly linked to an external index usually pay a spread below or above a certain index for a specified period of time. The index can be a market benchmark rate (eg, Federal Reserve funds, Euro OverNight Index Average, Sterling OverNight Index Average), a central bank rate (eg, European Central Bank refinance rate, Bank of England base rate) or any other reference rate.

The stability of non-maturing deposits may depend on the purpose of such deposits as well as on sensitivity to changes in the credit rating of the bank. The following types of non-maturing deposits are most common.

Current account balances

Current account balances are balances that individuals, corporates or other institutions hold on accounts established to facilitate regular credit and debit payments. Current accounts allow immediate disposal of balances and usually facilitate a wide range of access points. The interest paid on credit balances is usually lower than that for alternative investment products. Clients tend to hold some balances on current accounts for the following reasons.

- Credit balances provide a buffer against unexpected and extraordinary payments that yield overdraft charges or may exceed limits.²
- The opportunity costs of holding balances on current accounts versus alternative accounts are low, particularly in times of low interest rates, such that they do not exceed monitoring and potential transaction costs for alternative investments.
- Deposits for retail and corporate clients are generally subject to deposit insurance schemes up to a certain limit (€100,000 in the eurozone), which may lead to a greater acceptance of lower client rates for credit balances up to this threshold.³

Depositor preference, which gives depositors (particularly retail depositors) preferred status over other senior creditors in the event of bankruptcy, may provide a source of additional stability for deposits.⁴

Savings deposits

Savings deposits can broadly be characterised as follows:

- they are maintained by a retail financial institution and offer an interest rate that compensates for the liquidity provided to the financial institution;
- they cannot be used for direct payments in the sense of a medium of exchange;
- savings deposits are subject to a form of documentation;
- the ability to withdraw such deposits is usually limited to a number of withdrawals within a certain time period or requires a specified notice

period.

As with current accounts, savings deposits in most countries are covered, up to a certain threshold, by deposit insurance schemes.⁵ Also as with current account deposits, depositor preference may be a source of additional stability.

Clearing balances

Clearing balances can originate either from cash or from securities clearing. Balances from clearing activities for corporates and non-bank financial institutions typically remain on a bank's balance sheet for a longer period of time in order to provide a liquidity buffer throughout the payment cycle. This liquidity buffer can be left in the current account with the additional convenience of deposit insurance schemes, which are valid for corporate customers in most cases.

Stability can be inferred from the time and effort it would cost a corporate to change bank accounts and settlement instructions. Cash clearing balances from banks end up on the loro accounts of a bank offering nostro services to other banks. Even if the correspondent bank actively manages down end-of-day nostro balances, it is unlikely that all balances can be cleared before the cut-off times of the respective clearing systems. Balances from securities clearing are usually maintained in order to fund cash payments in a securities settlement process or are a result of such a process.

As part of their liquidity stress test, banks must assess the sensitivity of client deposits not only with respect to systematic stress scenarios but also with respect to firm-specific stress scenarios. Such stress scenarios may include a worsening of the creditworthiness of the bank unless it can provide some indication of quantitative modelling of the credit sensitivity of non-maturing deposits.⁶

HEDGING NET INTEREST INCOME WITH REPLICATING PORTFOLIOS

This section revisits the basic hedging principles for deposits with respect to interest rates and credit spreads in a deterministic world. The net interest income $NII(t)$ from deposits at a certain point in time $t + 1$ can be expressed from the financial institutions' perspective by

$$NII(t) = D(t)[r(t) - i(t)]$$

with $D(t)$ representing the notional of deposits at the beginning of the previous period, $r(t)$ representing the rate earned on the deposits and $i(t)$ representing the rate paid to clients. Variations in deposit balances are not taken into account, as they do not represent an economic value in their own right (apart from the funding value). Rather, it is the potential net interest margin that can generate the economic value for the financial institution. The present value of the interest received and paid can be determined as

$$PV_{NII}(0) = \sum_{t=0}^{T-1} \frac{D(t)[r(t) - i(t)]}{B(t)} \quad (6.1)$$

where $B(t)$ is the value of a money market account established at time $t = 0$ with

$$B(t) = \prod_{\tau=0}^t [1 + r(\tau)]$$

which is equivalent to the discount factor for t in a world of static interest rates. The rate paid to clients $i(t)$ can in some way be dependent on the actual market rate $r(t)$. T represents the time horizon over which the deposits are expected to remain on the balance sheet. The income stream can be compared with the payout of an exotic interest rate swap with an amortising (or expanding) principle.⁷

The 2007–9 global financial crisis highlighted the fact that, despite the general stability shown by client deposits, the risks of deposit outflows need to be taken into account.⁸ In order to account for the implied basis risks of potential client balance withdrawals, Equation 6.1 is adjusted so that the hedgeable volume of deposits is reflected by $D(t)$, while the hedged amount is reflected by $A(t)$

$$PV_{NII}(0) = \sum_{t=0}^{T-1} \frac{A(t)r(t) - D(t)i(t)}{B(t)}$$

$A(t)$ is dependent on the hedge decision of the deposit-taking institution; $D(t)$ depends on the client behaviour, which should be regarded as uncertain. It is

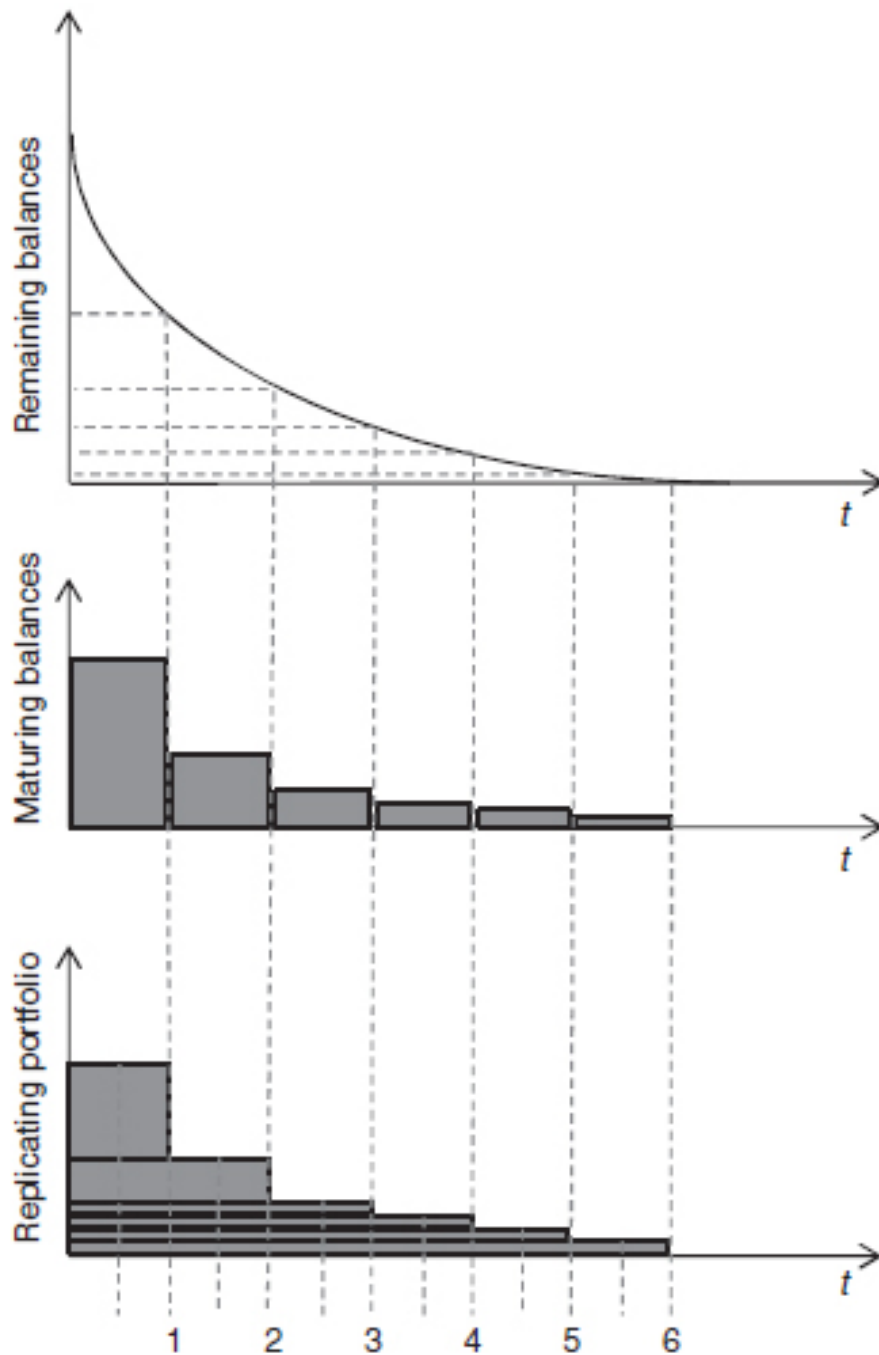
assumed that the financial institution defines its risk appetite for deposits falling below the respective hedge amount.

As uncertainty with respect to the future development of deposit balances exists, a stochastic process

$$D(t) = D(0) + \eta t + \sigma_D \varepsilon(t)$$

for the development of deposit volumes is assumed. At the same time we limit the probability of the deposit balance, $D(t)$, falling below the assets invested, $A(t)$, by a value ϕ such that

Figure 6.1 Replicating portfolio with static interest rates



$$p[(D(t) < A(t)] \leq \varphi$$

The factor ϕ represents the risk aversion of the bank with respect to liquidity shortfalls. In order to ensure that the above equation holds, the replicating

portfolio $A(t)$ must comply with the following rule

$$A(t) \leq D(0) + \lambda t - N^{-1}(\phi, \sigma_D) \sqrt{t}$$

Here $N^{-1}(\phi, \sigma_D)$ is the inverse normal distribution with a ϕ confidence level and a volatility of σ_D . Confidence levels between $\phi = 0.01$ and $\phi = 0.1$ appear in line with a conservative risk appetite. A level of $\phi = 0.5$ implies an unchanged balance development, resulting in a perpetual replicating portfolio. The steps towards the actual implementation of a replicating portfolio are depicted in Figure 6.1. The upper graph depicts the remaining balances of the replicating portfolio $A(t)$, which are determined by the above formula.

Table 6.1 Numerical example for Figure 6.1 (values in percent)

	Months							
	0	12	24	36	48	60	72	84
$A(t)$	100	60	44	31	20	11	2	0
$A(t) - A(t + 1)$		40	17	13	11	9	9	2
12M tranche		23						
24M tranche		4	4					
36M tranche		2	2	2				
48M tranche		1	1	1	1			
60M tranche		1	1	1	1	1		
72M tranche		7	7	7	7	7	7	
84M tranche		2	2	2	2	2	2	2

The middle graph in Figure 6.1 depicts the difference in the remaining balances $A(t) - A(t + 1)$. This difference shows that the replicating portfolio must be constructed in such a way that respective investments mature between t and $t + 1$.

The lower graph depicts the construction of the replicating portfolio. The replicating portfolio has to reflect the fact that at any point in time the outflow profile of $D(t)$ can be reassessed but – if the parameters do not

change – will maintain its original shape. Consequently, the replicating portfolio will need to be rebalanced constantly. This is best achieved if the “vertical” view of the replicating portfolio is converted to a “horizontal” view, as depicted in the lower graph. Here the horizontal bars represent “tranches” of the replicating portfolio, which can be rolled over continually.

Table 6.1 shows a numerical example for this approach. It is assumed that σ_D is estimated to be 7%, the risk appetite ϕ is 5% and $\beta = 0$. The remaining balances of the replicating portfolio $A(t)$ are given in the first row. The respective maturing balances $A(t) - A(t+1)$ are depicted in the second row. The other rows of the table depict the construction of the actual tranches representing a replicating portfolio that can be rolled over on a continuous basis. All tranches in the lower part of the table sum to the overall notional $A(0)$. While for this example the rollover frequency is 12 months, any higher rollover frequency (half yearly, quarterly or monthly) can be chosen.

The derivation of the replicating portfolio based on a 100% notional hedge implicitly assumes zero elasticity of the client rate $i(t)$ with respect to changes in interest rates, ie, the client rate is either zero or has a constant value throughout the interest cycle. When the client rate elasticity with respect to changes in the market rate is higher, the notional amount of the replicating portfolio needs to be adjusted, as will be seen in the following section.

SIMULTANEOUSLY MODELLING DEPOSIT BALANCES, INTEREST RATES AND CREDIT SPREADS

The aim of this section is to carry over the above framework from a world of static interest rates and credit spreads into a world of stochastic deposits, interest rates and credit spreads. Consequently, an economy with three stochastic factors is modelled. These factors are the short-term interest rate, the credit spread of the deposit-taking institution and the deposit volume itself. The interdependencies between these factors are determined by the correlation matrix

$$\text{CORR} = \begin{bmatrix} 1 & \rho_{r,c} & \rho_{r,D} \\ \rho_{r,c} & 1 & \rho_{c,D} \\ \rho_{r,D} & \rho_{c,D} & 1 \end{bmatrix}$$

In this matrix, $\rho_{r,c}$ represents the correlation between changes in interest rates and credit spreads, $\rho_{r,D}$ is the correlation between changes in interest rates and deposit volumes and $\rho_{c,D}$ represents the correlation between changes in credit spreads and deposit volumes. As the correlation parameters cannot be derived from market prices, they are best estimated from historical time series.

In order to represent this correlation matrix in the formulation of the stochastic processes for interest rates, credit spreads and deposits, a Cholesky decomposition of the correlation matrix is applied. This yields the matrix such that $GG^T = \text{CORR}$. The elements of the matrix G are used for the specification of the stochastic processes for interest rate, credit spreads and deposit volume in the following.

$$G = \begin{bmatrix} 1 & 0 & 0 \\ \rho_{c,D} & \sqrt{1 - \rho_{r,c}^2} & 0 \\ \rho_{r,D} & y & \sqrt{1 - \rho_{r,c}^2 - y^2} \end{bmatrix}, \quad y = \frac{\rho_{c,D} - \rho_{r,c}\rho_{c,D}}{\sqrt{1 - \rho_{r,c}^2}}$$

Table 6.2 Example withdrawal matrix

$c(t)$	$\lambda(t)$ (%)
Up to 100	0
100–200	10
200–300	20
300–500	50
500 and higher	80

The short rate is defined by a generalised Vasicek model with the process

$$dr(t) = \theta_r(t)[\mu_r(t) - r(t)]dt + \sigma_r dW_r(t)$$

The distribution for $r(t)$ can be assumed to be normal, lognormal or a blended distribution. The process for credit spreads is set as

$$dc(t) = \theta_c(t)[\mu_c(t) - c(t)]dt + \sigma_r \rho_{r,c} dW_c(t) + \sqrt{1 - \rho_{r,c}^2} \sigma_c dW_r(t)$$

Here θ_c is a mean reversion parameter and μ_c represents the mean value for the credit spread. The volatility of the credit spreads is denoted by σ_c and

can be inferred from the implied volatilities of options on credit default swaps.

The model also allows for withdrawals of customer balances due to severe downgrades of the deposit-taking institution. The extraordinary withdrawals are defined by a factor λ , which is a function of the credit spread $c(t)$. The withdrawal behaviour of clients with respect to changes in credit spreads is not trivial. The estimation can be based on expert judgement or on external evidence of other banks in distress. The relative creditworthiness of the institution compared with its competitors needs to be taken into account. An example withdrawal matrix is given in [Table 6.2](#). $\lambda(c(t))$ denotes the proportion of balances that is withdrawn in the case of a spike in credit spreads.

Deposit volumes are modelled via a stochastic process specified by

$$dD(t) = \theta_D(t)\mu_D(t) + \rho_{D,r}\sigma_D(t) dW_r(t) + \gamma\sigma_D dW_c(t) + \sqrt{1 - \rho_{D,r}^2 - \gamma^2}\sigma_D dW_D(t) - \lambda(c(t))$$

with

$$\gamma = \frac{\rho_{c,D} - \rho_{r,c}\rho_{c,D}}{\sqrt{1 - \rho_{r,c}^2}}$$

This formula is subject to the condition that deposit volumes cannot assume negative values. As there is no tradeable index on the development of deposit volumes, both the drift and the volatility need to be estimated from historical data.

The approach depicted here only simulates the evolution of the aggregated deposit volumes. It may be more appropriate to perform simulation for different cohorts of deposits, such as those clustered by product (eg, as listed above), deposit size or client group. Furthermore, the analysis should be carried out separately for different currencies and legal entities.

The interest rate paid on deposits $i(t)$ is defined⁹ as a function of the short rate

$$i(t) = \max[\alpha + \beta r(t), \delta]$$

This function implies a deterministic dependency between the absolute level of client rates and market rates. Parameters of this function can be

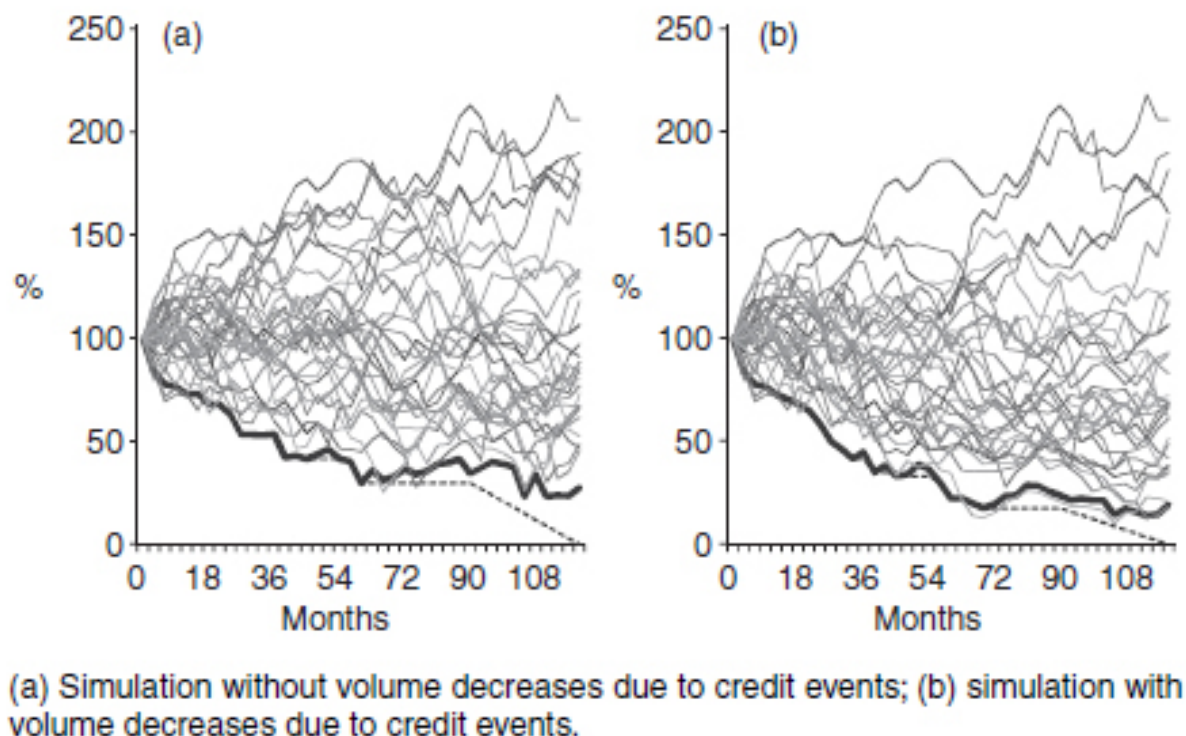
determined by regression of time series. Furthermore, it is assumed that the customer rate cannot drop below a predefined floor, δ . Other specifications for $i(t)$ where the client rate is dependent on the three-month rate or a blend of longer rates could also be defined.

SIMULATION OF DEPOSIT VOLUMES

Figure 6.2 shows an example of Monte Carlo simulation results for deposit volumes.¹⁰ For every path of the deposit volume, corresponding paths for the credit spread and the interest rates are determined. Part (a) depicts simulation results for a normal distribution without the impact of sudden outflows due to an increase in credit spreads. Part (b) shows the results of the same simulation, which reflect the simultaneous development of credit spreads. The impact of sudden spikes in the credit spread and subsequent outflows can be observed in a more downward-skewed distribution of future deposit volumes.

The thick black line reflects the ϕ th percentile of deposit volumes at each point in time (we denote the respective value for each t by $\phi(t, \psi)$).¹¹ The confidence level in this example is chosen as $\phi = 0.001$. A higher confidence level with a lower ϕ will result in a steeper downward-sloping thick black line and a shorter hedge profile. Part (b) of Figure 6.2 shows that the $\phi(t)$ line has generally lower values on the graph, reflecting outflows from higher credit spreads than in part (a), which does not reflect such outflows.

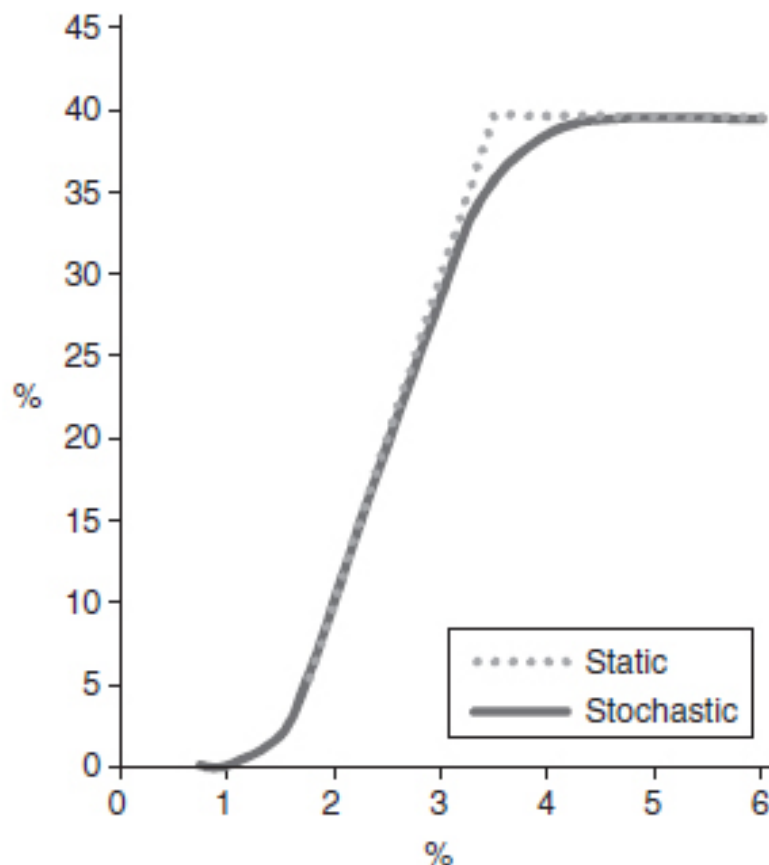
Figure 6.2 Monte Carlo simulations



The $\varphi(t, \psi)$ line may be amended for practical reasons. The dashed lines depict two amendments. The first amendment is to disregard increases in volumes in order to allow for a monotonic decreasing volume function; the second is to define a point in time from whence volumes decrease linearly, such that they will reach zero at a predefined point in time.

The $\varphi(t, \psi)$ line does not reflect the replicating portfolio as outlined above (see the section on deposit types on p. 156). This line is the basis for the cashflow calculation. Hedge ratios are derived on the basis of changes in the present value due to shifts in the underlying interest (and forward) rates. This procedure is chosen to capture the “margin compression risk” inherent in deposit portfolios. The margin compression risk arises when decreasing interest rates affect not only the discounted value of cashflows from deposits but also the magnitude of the cashflows themselves. As an example, a deposit product paying a specific reference rate $r(t)$ minus a spread α will decrease in value if the reference rate $r(t)$ decreases below α , although the respective discount factor increases.

Figure 6.3 Present value of net interest income with static and stochastic interest rates

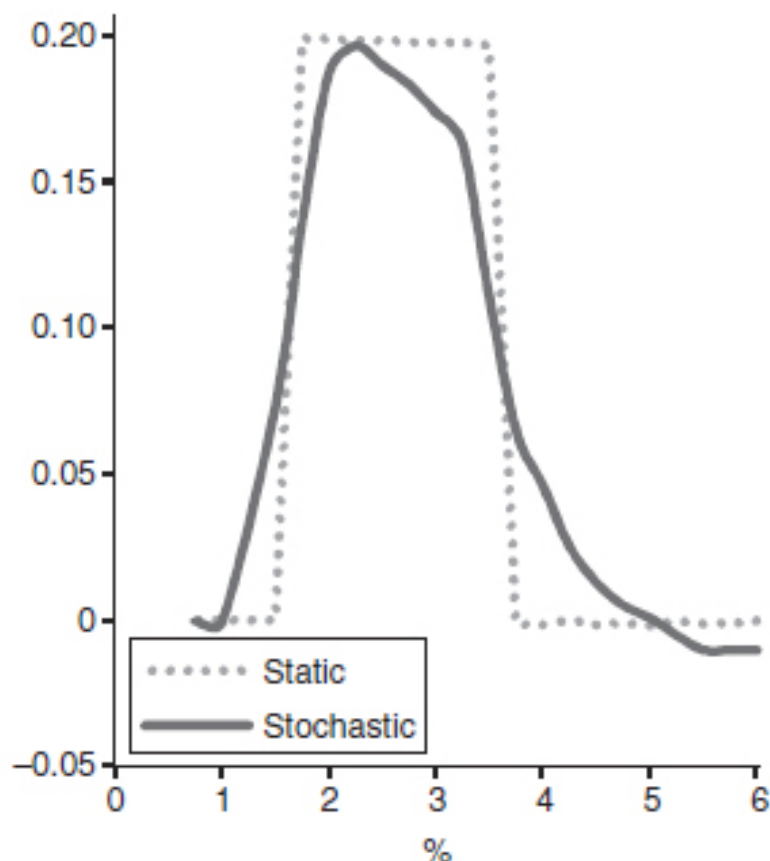


One disadvantage of using stochastic factor models to determine the profile on the basis of a confidence level is the reduced ability to backtest such models. In order to perform backtesting of a run-off model as described above, the historical behaviour of such deposit balances relative to the estimated volatility needs to be assessed. Similarly to value-at-risk models, an outlier test may be applied. This means that, over a given period of time, a permitted number of outliers is determined theoretically in order to allow the model to be accepted at a certain confidence level. This means that a limit on the probability of a type 2 error (ie, the model is incorrect but is accepted) needs to be determined.¹²

HEDGE RATIOS WITH RESPECT TO CHANGES IN INTEREST RATES

Hedge ratios can be calculated with respect to the sensitivity in interest rates based on the derived deposit outflow profile. The sensitivity with respect to interest rates can differ significantly, depending on whether the analysis has been carried out in a stochastic or a deterministic interest rate environment. The hedge ratio in the stochastic environment is calculated by performing two Monte Carlo simulations for deposits, interest rates and credit spreads and calculating the $\varphi(t)$ lines simultaneously: one for the original interest rate curve plus a one basis point (1bp) shift, and another for the original interest rate curve minus a 1bp shift. The present value of a basis point (PV01) is calculated by dividing the difference in the discounted value of cashflows by the shift in the interest rates in basis points. In this way the change in the net interest margin due to margin compression can be captured in a very low interest rate environment. Hence, the implied optionality of deposit portfolios with respect to changes in interest rates, which is caused by the floor in client rates, can be quantified; this is not possible by first calculating the $\varphi(t)$ line and then shifting interest rates.

Figure 6.4 Changes in present value of net interest income per basis point with static and stochastic interest rates



An example of this effect is depicted in [Figure 6.3](#). In an environment with high interest rates, both static interest rates and stochastic interest rates derive the same NII present value. As rates decline towards a level close to the net interest margin, the potential margin compression starts to affect the present value. This is captured by the decline in the present value in the stochastic rate calculation occurring earlier than the static interest rate calculation. As rates decline further and margin compression materialises, both methods again yield the same present value.

When it comes to hedging the risk of margin compression, the hedge ratios for the stochastic simulation approach also deliver superior results, as can be seen in [Figure 6.4](#), which depicts the sensitivities of the profiles in [Figure 6.3](#) with respect to interest rate changes of 1bp. The delta short positions (expressed by positive numbers in [Figure 6.4](#)) increase at higher interest

levels, such that the deposit-taking institution is in a position to readjust its hedges earlier.

In general it can be concluded that a simulation with stochastic interest rates allows the capture of the implied short convexity position due to the floor for client interest rates, while the deterministic view does not. This also means that the rollover strategy for a replicating strategy as described earlier (see the section on hedging starting on p. 158) cannot be applied in all cases. As suggested by [Figures 6.3 and 6.4](#) it will only deliver sufficient hedges if the market level of interest rates is sufficiently above the threshold for margin compression risk to materialise. As the market level of interest rates decreases, the hedge ratio will be too low until the margin compression risk actually materialises. Consequently, the hedging strategy – whether it is a rollover strategy of bonds or of swaps – needs to be adjusted dynamically (at least monthly) and will entail higher notional hedges if the market level of interest rates is above the average margin of the underlying deposit portfolio. Hence, the replicating portfolio derived in a world of static interest rates will only deliver correct results in an environment where market rates are significantly above the average margin on the deposit portfolio. As soon as the rates are observed to approach the interest rate floor, the hedge ratios need to be adjusted upwards.

APPLICATION TO DECAY MODELS

The method for deriving hedge ratios described in the previous section can also be applied to decay models based on logarithmic regressions, as described in [Chapter 5](#) in this volume. In this case the change in deposit balances from one period to another is measured as a function of age and cohort as well as a function of economic variables.¹³ Such economic variables can be a short- or long-term interest rate. The credit spread of the respective institution, and further macroeconomic and monetary variables, such as money supply, inflation, economic growth or stock market returns can also be added to the model. It is important that such variables are captured by the Monte Carlo simulation as described in [Chapter 5](#), or are taken into account in additional scenario analysis or stress tests.

SUMMARY

In this chapter an approach to simultaneously model deposit volumes, interest rates and credit spreads was introduced in order to model and hedge cashflows from non-maturing deposits. The higher the volatility of deposit volumes, the shorter the tenor of the cashflow profile. It is further shortened by deposit outflows due to significant worsening of the credit standing of the financial institution. The approach requires some input and calibration of market parameters and is particularly useful for deposit portfolios with limited information on the history of singular accounts where a portfolio-based assessment is necessary. It may be useful to break down the deposit portfolio by volume in order to isolate concentration accounts or break down the deposit portfolio by product type.

The inclusion of stochastic interest rates allows us to capture the margin compression risks for deposits due to market rates approaching or falling below a predefined floor for client rates. Thus, the hedge ratio for deposits needs to be adjusted at higher level of market interest rates so that a higher economic value is obtained when interest rates really approach the floor for client rates.

The model can be combined with other approaches or extended. For example, the run-off profile could be determined via survival periods of individual accounts ([Chapter 3](#)) rather than via a confidence level. Also, the outflows due unexpected to changes in the creditworthiness of a financial institution could be examined further.

- 1 See also Castagna and Fede (2013) for a general description of stochastic factor models for risk management of client deposit balances.
- 2 See Diamond and Dybvig (1993), who argue that depositors are looking for a highly liquid investment while leaving the macroeconomically important maturity transformation function to banks.
- 3 In some cases, current account balances have to be non-interest-bearing by regulation, in order to be eligible for a deposit protection scheme. For example, Section 343 of the Dodd–Frank Wall Street Reform and Consumer Protection Act (Dodd–Frank Act) provides temporary unlimited deposit insurance coverage for non-interest-bearing transaction accounts (NIBTAs) at all Federal Deposit Insurance Corporation insured depository institutions (IDIs) from December 31, 2010, to December 31, 2012 (the Dodd–Frank Deposit Insurance Provision). See Federal Deposit Insurance Corporation (2012).
- 4 See Hardy (2013) and Clifford Chance (2011) for an overview of depositor preference.
- 5 See BCBS–IADI (2009) for an overview of deposit insurances.
- 6 With the introduction of the liquidity coverage ratio and net stable funding ratio, regulators have given some guidance on stability assumptions for deposits (Basel Committee on Banking Supervision 2013). Bank-specific liquidity stress tests that are reviewed by regulators usually include more detailed assumptions on deposit outflows due to downgrades.
- 7 See Jarrow and van Deventer (1998) for a derivation of this representation.

- 8 The Basel III liquidity ratios acknowledge the outflow risks associated with deposits. The liquidity coverage ratio determines the liquidity buffer to be held against deposits, while the net stable funding ratio limits the amount of deposits that can be assumed to stay on the balance sheet for longer than one year.
- 9 A similar function for the client rate is specified by Elkenbracht and Nauta (2006). Kalkbrener and Willing (2004) follow a similar three-factor approach with stochastic deposit balances, market rates and client rates while leaving credit spreads constant.
- 10 The Monte Carlo simulation depicted in the graphs is based on 50 runs. It is suggested that a significantly higher number of runs (at least 5,000) be applied in practice.
- 11 This method is similar to the approach suggested by Kalkbrener and Willing (2004), who call the $\varphi(t)$ line the “term structure of liquidity”.
- 12 For a discussion on backtesting of value-at-risk models see Jorion (2006).
- 13 See also the analysis by Wahl (2014) on application of logistic regressions to deposit accounts.

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Managing Interest Rate Risk for Non-Maturity Deposits

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For many banks, non-maturing deposits represent a significant part of funding. However, there remains no commonly accepted approach to managing such deposits' interest rate risk. We introduce two dynamic hedge strategies to stabilise the margin between investment return and client coupon. As extensions of Jarrow and van Deventer's (1998) model, these strategies can be used for both interest rate risk management and funds transfer pricing.

An important goal in modelling non-maturing deposits¹ is to find an investment strategy² that stabilises the margin independently of interest rate movements. Sales departments prefer a stable margin to help them project and manage their income accurately.

Historically, the most commonly used replicating portfolio model targeted stabilising margins. However, this model contained a static investment rule that did not take into account current markets, which limited its performance.

A step forward was taken by Jarrow and van Deventer (1998), who developed a dynamic investment rule aimed at stabilising the value of non-maturing deposits. However, a stable value does not imply a stable margin.

In this chapter, we provide an extension of Jarrow and van Deventer's model that is aimed at stabilising the margin instead of the value. First, we briefly describe the goals of an interest rate risk model for non-maturing deposits and the well-known replicating portfolio model. Next we describe the model developed by Jarrow and van Deventer and our extension, where we introduce our margin concept. Based on this concept, we derive two new models that differ in how volume growth or decline is treated. Both models result in a dynamic investment strategy. Finally, we compare our model with the replicating portfolio model for an example situation. Although this chapter focuses on non-maturing deposits, the methodology is also applicable to fixed term deposits.

THE PROBLEM

At first glance, the non-maturing deposit product looks rather simple. On more careful examination, this product's characteristics appear difficult to capture when managing interest rate risk on the balance sheet. This is due to two features not found in most of the "usual" (for example, fixed-rate loan or term deposit) products:

- the customer has the option to adjust the notional at any time;
- the bank has the option to adjust the interest rate at any time.³

Without a defined maturity date, a defined notional and a defined interest rate, incorporating this product correctly into measures used for balance-sheet management poses a challenge.

Some banks incorporate this funding into their balance sheets as if deposits were made on an overnight basis. This approach is very conservative from a liquidity risk perspective. From an interest rate risk perspective, this approach is neither risk-free nor conservative, as, generally, the bank will pay more or less than the overnight rate. When the non-maturing deposits are invested in overnight funds, the risk measures could indicate no risk. In reality, however, a gain or loss that varies over time could occur.

GOAL OF THIS CHAPTER

To manage our balance sheet, we would like a methodology to replace the volume of non-maturing deposits by a portfolio of products with defined

characteristics such that:

- their behaviour in relation to interest rates is reflected as accurately as possible,
- their investments are directed to minimise interest rate risk,
- risk indicators (for example, net interest income at risk or duration) give appropriate values,
- a mechanism transfers interest rate risk from the business to the asset and liability management (ALM) department,
- the non-maturing deposits business is rewarded appropriately,
- the method offers flexibility with respect to pricing policy changes, which can be incorporated by adjusting the portfolio.

As we attempt to minimise the interest rate risk, we are not trying to find the optimal investment of non-maturing deposits in terms of risk versus return, as in Markowitz theory.

THE REPLICATING PORTFOLIO MODEL

A replicating portfolio is an investment portfolio of bonds of various start dates, tenors and coupons. Each month, maturing bonds and new non-maturing deposit volumes are invested according to a fixed investment rule. Such a fixed rule could be: invest 30% in three-month deposits, 30% in one-year bonds and 40% in five-year bonds. The business margin is the coupon proceeds of the imaginary investment portfolio minus the customer coupon.

The maturities and the percentages applied in the investment rule are to be determined by optimisation such that the rule performs “best” over a historical period. For example, “best” is defined as the lowest standard deviation of the margin, or it relates to a one-sided risk measure such as the maximum loss that will be incurred on the investment portfolio in a volume decrease/interest increase scenario, or it represents a combination of these factors.

This method represents an improvement over the inclusion of non-maturing deposits on an overnight basis. There is, as with the overnight method, a fixed investment rule, which is now tailored to minimise risk, at least over

the estimation period. Furthermore, this method offers a rule for both investment and the margin.

We have a few concerns, however.

- The risk was minimised over the estimation period, but will this investment rule be the “best” for the future? For example, when a business wants to change its customer coupon pricing strategy, this will be hard to accommodate.
- Investments are fixed and therefore independent of the current level and shape of the yield curve. For instance, consider when the client rate is set, taking into account current market rates. When the yield curve is upward sloping, we may expect to pay larger client coupons in the future than when the curve is inverted. However, the replicating portfolio model prescribes the same investments in the long run for these two situations.
- The investment portfolio at a certain time depends on the past, which may lead to results that are difficult to interpret. Note that the proceeds of an investment portfolio are determined by the combined development of interest rates and volume over the past five years (where a five-year bond is the longest maturity in the portfolio). However, a business might prefer to let the client rate follow market rates closely.
- This method would not account for a new type of non-maturing deposit account for which there is no history. Instead, we would prefer a method that results in a stable margin independent of the specific interest/volume path that we followed historically and takes into account the business volume and customer coupon outlook.

We have addressed both our general goals and the issues specific to the replicating portfolio model by developing an alternative approach based on Jarrow and van Deventer’s model.

THE JARROW AND VAN DEVENTER MODEL

Jarrow and van Deventer (1998) approach the value of non-maturing deposits as if this product were a standard market instrument: trying to predict all cashflows and discount. To predict the cashflows, they require a

volume and client rate model. We shall use the following simple models: an exponential volume model

$$V(t) = V(t_0) \exp(\alpha t) \quad (7.1)$$

and a linear client rate model, for example

$$c(t) = ar_n(t) + b \quad (7.2)$$

with $V(t)$ the volume at time t , $V(t_0)$ the current volume, α a fitted exponential growth rate, $c(t)$ the client rate at time t , $r_n(t)$ the n month market rate at time t , a the fitted client rate elasticity and b the fitted constant part. These simple models are used in our historical backtest discussed below.

This methodology allows for more complicated models. For example, a client rate could depend on multiple market rates with different tenors, moving averages thereof, rates that were fixed in the past and liquidity spreads. Another model accounts for the volume depending on the client rate, market rates, and client specific variables like, for example, account size. In these more complex models, estimating parameters and calculations may become more difficult, but the principles remain the same. Even with simple client rate and volume models, as in Equations 7.1 and 7.2, good results can be achieved.

With future interest rates taken as forward interest rates from the current yield curve, we define the value of non-maturing deposits by discounting the expected cashflows

value($c(t)$)

$$= V(t_0) + \sum_{i=0}^{N-1} d(t_{i+1}) \left[\Delta V(t_{i+1}) - \frac{V(t_i)c(t_i)}{12} \right] - d(t_{N+1})V(t_N) \quad (7.3)$$

where $d(t_i)$ is the discount factor for time t_i and $V(t_{i+1})$ is the change in volume between times t_i and t_{i+1} . We have to choose an end date t_{N+1} , when the funds are assumed to be returned to the customer.

The factor $\frac{1}{12}$ multiplying the client rate indicates that we have chosen a time step of one month. The client rate in Equation 7.3 uses Equation 7.2 based on the forward rate. Equation 7.3 is an approximation of the complete

expression that includes a convexity correction. For client rates that are based on a market rate with a tenor shorter than one year, the convexity correction is small. We neglect the convexity correction in our following models, which has the advantage that the value in Equation 7.3 can be evaluated completely, given the current yield curve and no volatilities are required (for further discussion of Equation 7.3, see the appendix). Equations 7.2 and 7.3 are based on a single-curve framework that is used in Jarrow and van Deventer (1998) and in the rest of this chapter. Following the financial crisis, multiple curves are increasingly being used for derivatives' valuation and ALM. We briefly point out how the value in Equation 7.3 can be extended in a multiple curves framework. First, the client rate should reference the appropriate tenor; this could be the three-month London Interbank Offered Rate for $n = 3$. Also, the curve used for the discount factors needs to be determined. Since the client can adjust the notional at any time, the liquidity tenor is overnight. Therefore, the proper discount curve for non-maturing deposits is the overnight index swap (OIS) curve. Hence, in general, two curves are required for determining the value: an index curve for the n -month market rate and the OIS-curve for discounting.

Based on the above, Jarrow and van Deventer designate the investment portfolio as the portfolio that hedges the value, defined from the sensitivities to the interest rates on the yield curve.

The ALM unit pays, on a monthly basis, the modelled customer coupon $c(t)$ to the non-maturing deposits business, who pays out the actual customer coupon. In addition, initially the ALM unit rewards the business by paying value $c(t)$.

OUR APPROACH TO DEFINING MARGIN

We would like the business to be rewarded on an accrual basis for non-maturing deposits. To reflect this business model, we have adapted Jarrow and van Deventer's model as follows.

We define the margin as the increase in customer coupon that sets the net present value of the non-maturing deposits equal to zero

$$\text{value}(c(t) + \text{margin}) = 0 \quad (7.4)$$

The business will receive $c(t) + \text{margin}$ from the ALM department and pays the customer coupon. This setup transfers the interest rate risk to the asset

and liability department. The business earns the margin as long as the volume follows Equation 7.1 and the customer coupon follows Equation 7.2; thus, the model risk for Equations 7.1 and 7.2 resides with the business. We have chosen an investment methodology that tries to stabilise, or hedge, the margin as much as possible.

Therefore, an analogy to the “fair value” concept in derivatives pricing is that this margin could be considered as the “fair margin”, since a dynamic hedge strategy can be developed to guarantee this margin.

ONE PROPOSED APPROACH: HEDGING THE MARGIN

To hedge the fair margin as defined in Equation 7.4, we can hedge the value while adding the margin to the customer coupon. This approach creates an investment strategy such that the gain/loss in market value on the investment portfolio cancels the gain/loss in value of the non-maturing deposits (including the margin)

$$\text{value}(c + \text{margin}) + \text{P\&L}(\text{investment portfolio}) = 0 \quad (7.5)$$

This approach consists of the following steps, that, eg, can be executed monthly:

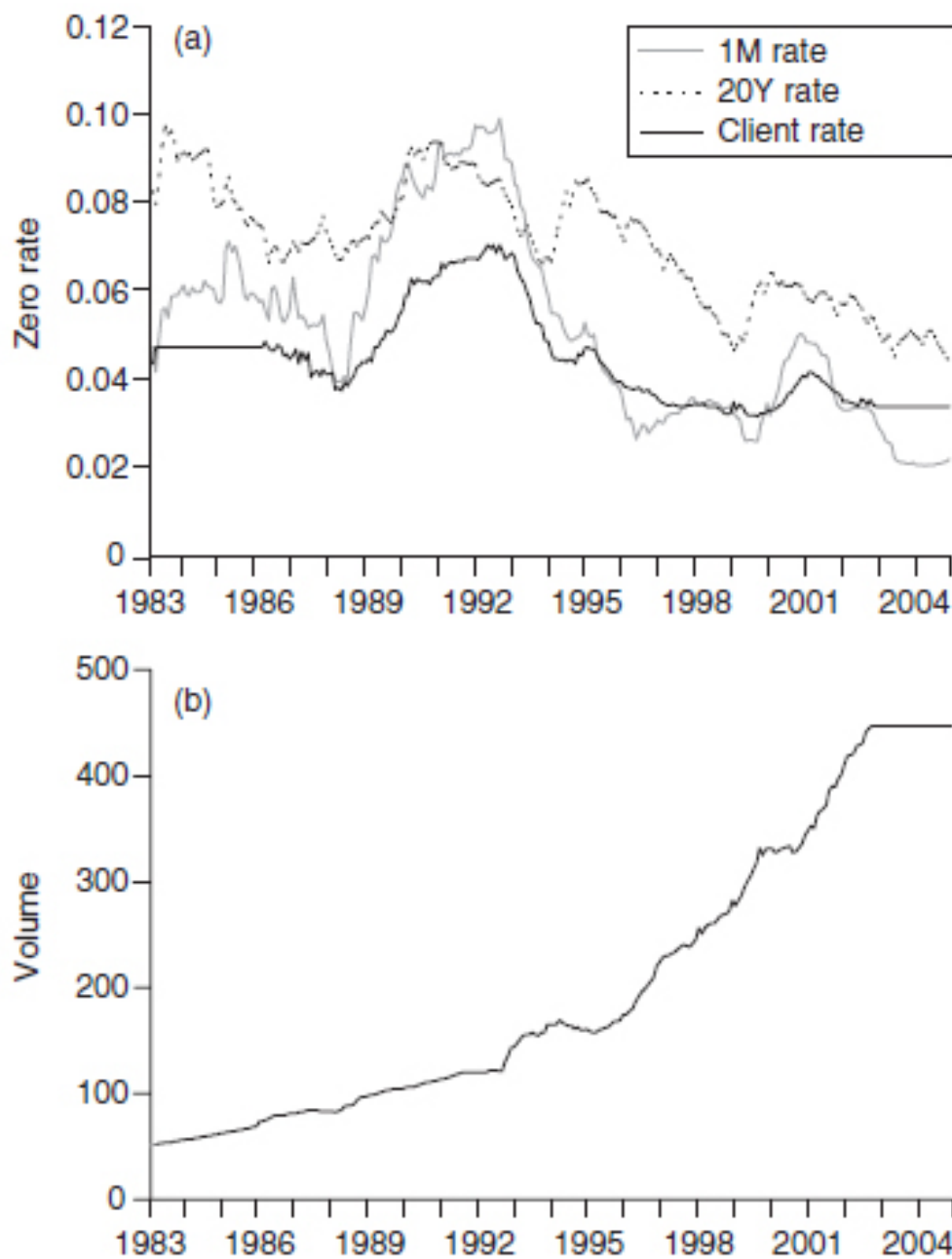
1. calculate the sensitivities of the value of the non-maturing deposits, including the margin, to the interest rate curve;
2. determine the amount of zero-coupon bonds required in each bucket to hedge the value;
3. invest or borrow the surplus or shortage in the shortest maturity in the investment portfolio (for example, one month);
4. after one month, calculate the new market value of the investment portfolio, subtract the customer coupon and margin that has to be paid out and calculate the profit or loss made since inception/last month;
5. given the accumulated profit or loss, calculate the new margin from Equation 7.5, which should be very close to the previous margin.

The main reason we cannot get a completely stable fair margin is that we have to set an end date. We only hedge cashflows until that end date, and each month that date has to be rolled forward.⁴ The volume model and client

rate model are estimated using simple functions, as in Equations 7.1 and 7.2, over the same period. These results are therefore optimal for these simple models. The models are used only to determine the margin and the investment portfolio. The client coupon that is paid out is given by the actual historical data. In this test, we have taken 20 years as the longest maturity in the investment portfolio. The results of the backtest can be found in [Figure 7.2](#).

The margin stays remarkably stable, especially when compared with the difference in the client rate and market rates shown in [Figure 7.1\(a\)](#). The margin remained stable when we estimated the volume and client rate model only over the first half of the historical period and performed the same test on the second half of the period. The stability of the margin is even more remarkable given the simplicity of the model for the volume (Equation 7.1) and client rate (Equation 7.2).

Figure 7.1 One-month and twenty-year market rates and client rates and volume, January 1983 to December 2004

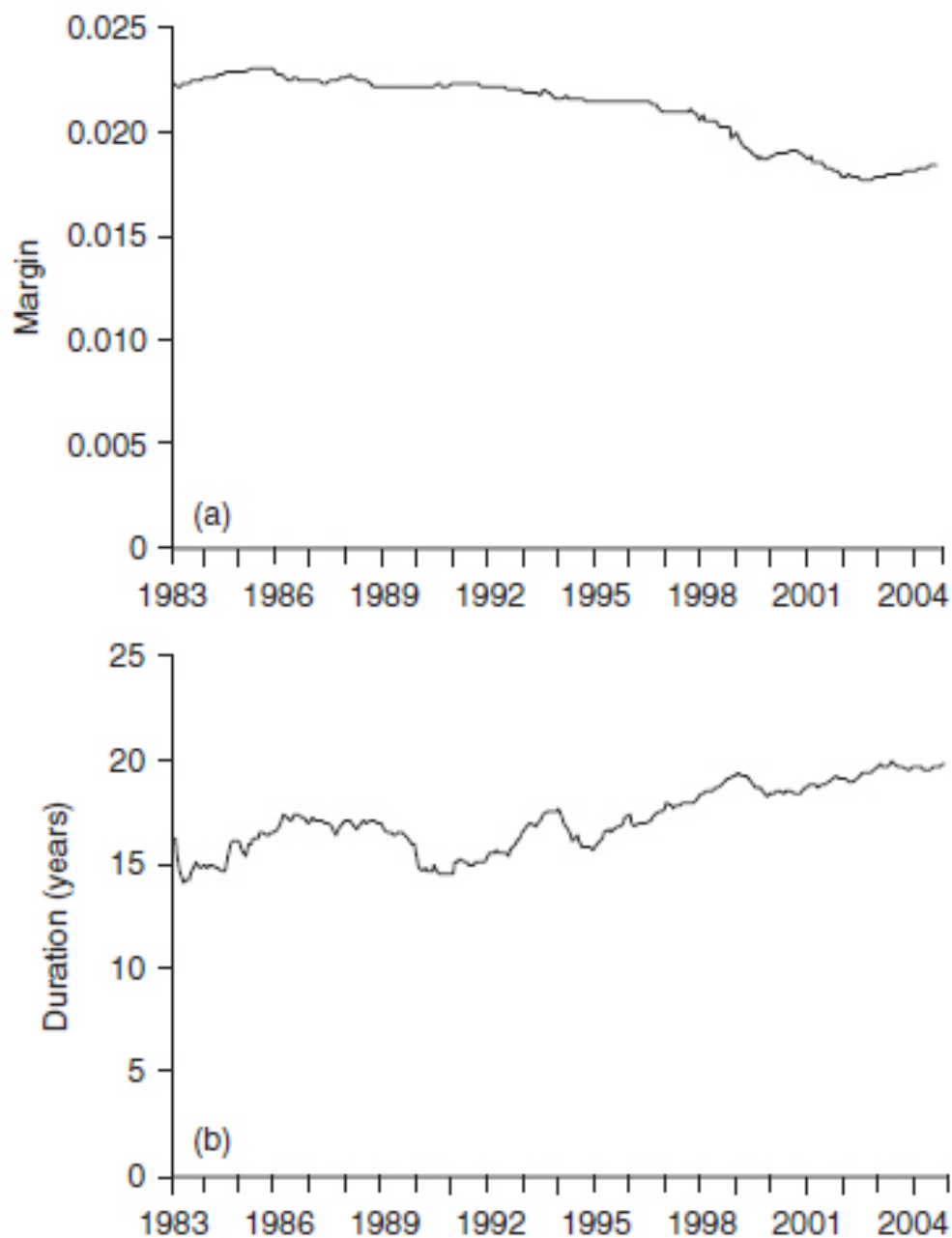


(a) 1M and 20Y market rates; (b) volume.

The margin dips at the end of the period because the cashflows are hedged out to “only” 20 years. In the historical period, the difference between market rates and the client rate narrows near the end of the period. If we had started

hedging at the end of the period, December 2004, the margin would have been approximately 50 basis points (bp). Since only cashflows out to 20 years are hedged, the investment portfolio cannot prevent the margin from declining slightly at the end.

Figure 7.2 Results for the margin and duration in a historical backtest, January 1983 to December 2004



(a) Margin; (b) duration.

The long duration resulted from two characteristics of this example:

1. the customer coupon has a small elasticity coefficient (the smaller the elasticity coefficient, the larger the duration; the fixed component b of $c(t)$ only influences the margin and not the duration);
2. the volume was estimated to grow exponentially by 10% a year.

In this situation, the method puts so much volume in the longest (20-year) bucket that we borrow significantly in the shortest (one-month) bucket.

In this approach we have included future volumes from existing and new clients in order to determine the interest rate risk of the existing non-maturing deposits portfolio. Including future volumes, in combination with a small elasticity coefficient in the client rate model, significantly increases the duration, which is highly sensitive to the chosen end date.

AN ALTERNATIVE APPROACH: HEDGING CURRENT VOLUME ONLY

As an alternative, we could hedge just the current volume. Additional volumes beyond replacing deposits that are withdrawn are treated as a new product that is sold, and the resulting position is hedged from that point on. This approach resembles the way fixed-rate loans or mortgages are often treated: not all the expected future volume is hedged – only the loans and mortgages that are currently in the bank's portfolio.⁵ The following advantages result from using this method:

- non-maturing deposits are treated similarly to other items on the balance sheet;
- we do not need additional borrowing;⁶
- results are less sensitive to the (arbitrary) end date;
- we can distinguish between margin on net volume increase/decrease and margin on total volume.

The investment portfolio is defined as the portfolio that provides the hedge for the following value:

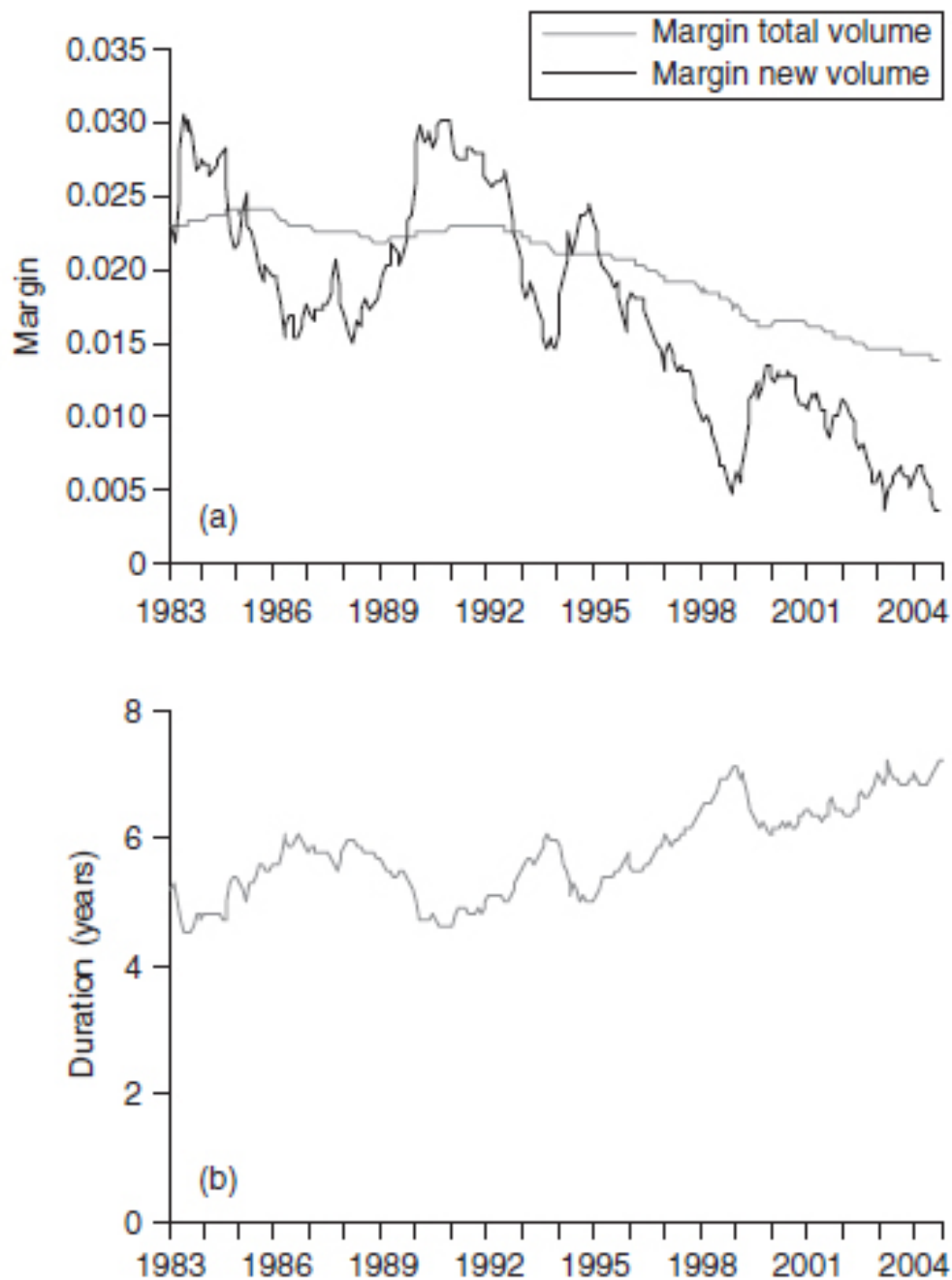
$\text{value}(c(t) + \text{margin})$

$$= V(t_0) - \sum_{i=0}^{N-1} \frac{d(t_{i+1})V(t_0)[c(t_i) + \text{margin}]}{12} - d(t_{N+1})V(t_0) \quad (7.6)$$

which is similar to Equation 7.3, but with the margin included and the volume kept constant. Instead of assuming a constant volume, the methodology also allows for a more complex model capturing the amortisation schedule of the portfolios present volume (client withdrawal behaviour) in terms of interest rate levels.

For now we assume a constant volume. The margin is again determined by Equation 7.5. The whole procedure for determining the margin and investment strategy is the same as that indicated in the steps following Equation 7.5. Additionally, we can calculate the margin on the new volume, which is determined by Equation 7.4.

Figure 7.3 Results for the hedge of only the “original” or “current” volume

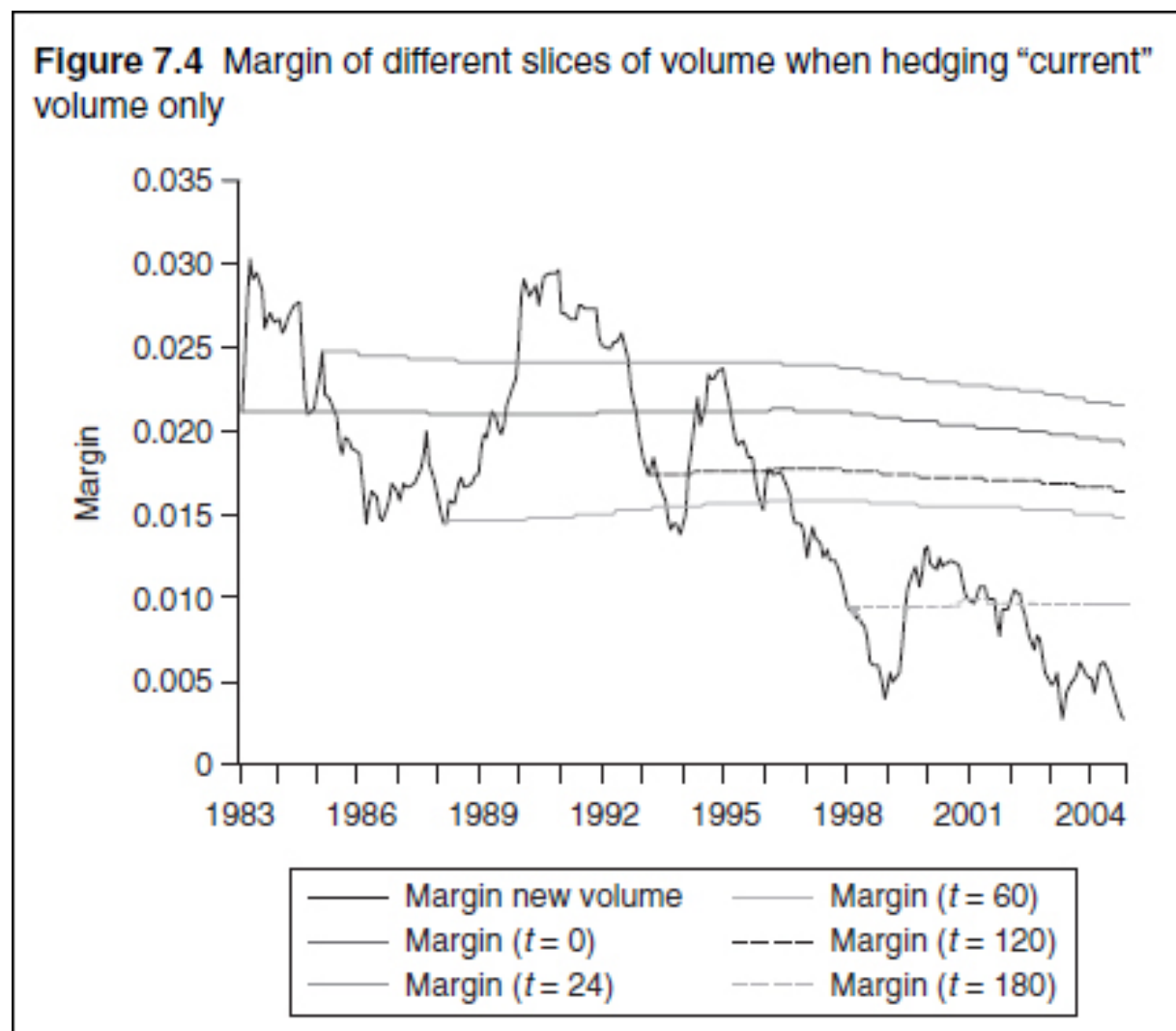


(a) Margin; (b) duration.

The margin on the total volume can be viewed as a weighted sum of margins for different slices of volume. That is, the margin of the initial

volume plus the (weighted) margin of the net volume increase/decrease after one month, plus the (weighted) margin of the net volume increase/decrease after two months, etc.

We performed the same test as the previous case, in which the total projected volume was hedged. Results are shown in [Figure 7.3](#).



The margin on the total volume is less stable than in the previous case, in which the margin of the total projected volume is hedged. But the total margin is much more stable than the margin on the new volume. The duration is much shorter than in the previous case, ranging from four-and-a-half to seven years. These results for the duration depend largely on the client rate model that we use, especially the elasticity.

We also checked how this method keeps the margin stable for a slice of volume. The following results show the margin on additional volume (“margin new volume” in [Figure 7.4](#)), the margin on the initial volume (“margin($t = 0$)”), the margin on the net volume increase/decrease after 24 months (“margin($t = 24$)”), etc. The margin for a slice of volume is very stable, especially when compared with the difference in client rate and market rates over the same period.

[Figure 7.4](#) shows the importance of timing when starting using the model. The level of the margin predominantly depends on the market rates when starting hedging. The timing is discretionary and is not determined by the model. In particular, when we start to hedge in a low interest rate environment this would mean stabilising the margin at a low level, possibly at a slightly higher level than not hedging, but abandoning the possibility of having a higher margin when interest rates increase. We can consider starting to hedge according to the model in a phased approach in an attempt to ultimately receive an average margin over a business cycle.

COMPARISON OF PROPOSED MODELS WITH REPLICATING PORTFOLIO MODEL

To compare the two value-based models and the replicating portfolio⁷ in different circumstances, we have chosen four different scenarios. We used two up/down rates scenarios that incorporate a parallel shift in rates of $\pm 200\text{bp}$ during the first year. In addition, we used two up/down volume scenarios of 10% growth or decline for two years ([Figure 7.5](#)).

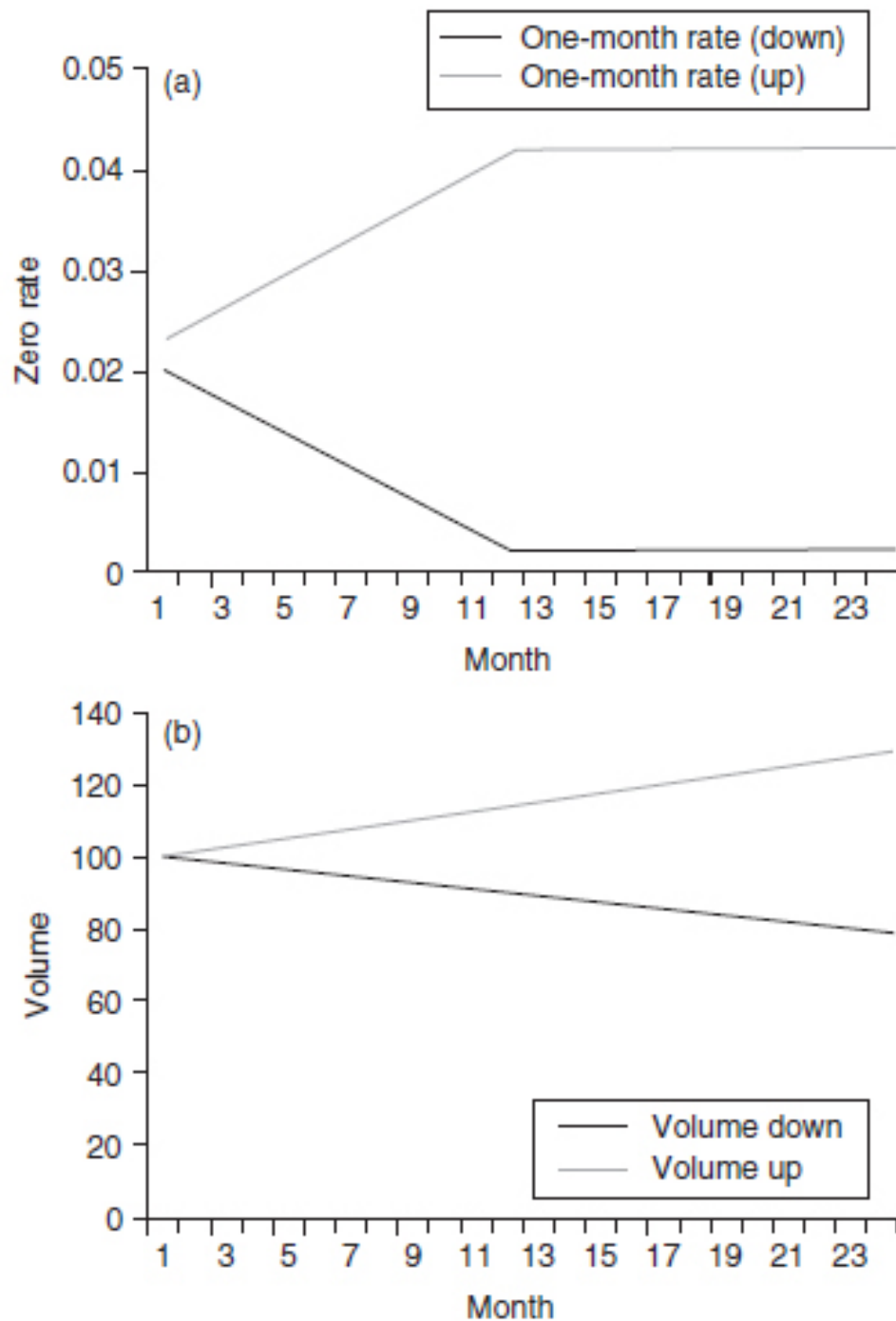
In [Figure 7.6](#), we compare two risk measures: the stability of the margin⁸ over the two-year period (on the vertical axis) and the profit and loss on the investment portfolio after two years (on the horizontal axis).

In addition to stable margins, it is desirable to reduce the profit and loss risk on the investment portfolio, which is closely linked to the duration. Profit and loss risk corresponds to the risk of a concrete loss for the investment portfolio that occurs when interest rates rise. In our proposed methods, such a loss will be compensated by an increase in the value of the non-maturing deposits, but only to the extent that the volume follows the volume model (for our first method) or the volume stays constant (for our “hedging current volume only” method). When liquidity risk materialises and the volume grows less (method I) or decreases (method II), a net loss will

occur. This risk can be reduced by limiting the profit and loss risk on the investment portfolio.

In fact, these two risk measures conflict, and we need to find an optimal balance between them. An easy way to influence the profit and loss on the investment portfolio (and the duration) in our two value-based investment strategies is to vary the end date (t_{N+1} in Equations 7.3 and 7.6). The results that correspond with an end date of five years and 20 years are shown in [Figure 7.6](#). The 10-year and 15-year points are in between. In the rate increase graphs particularly, the trade-off between these risk measures is evident.

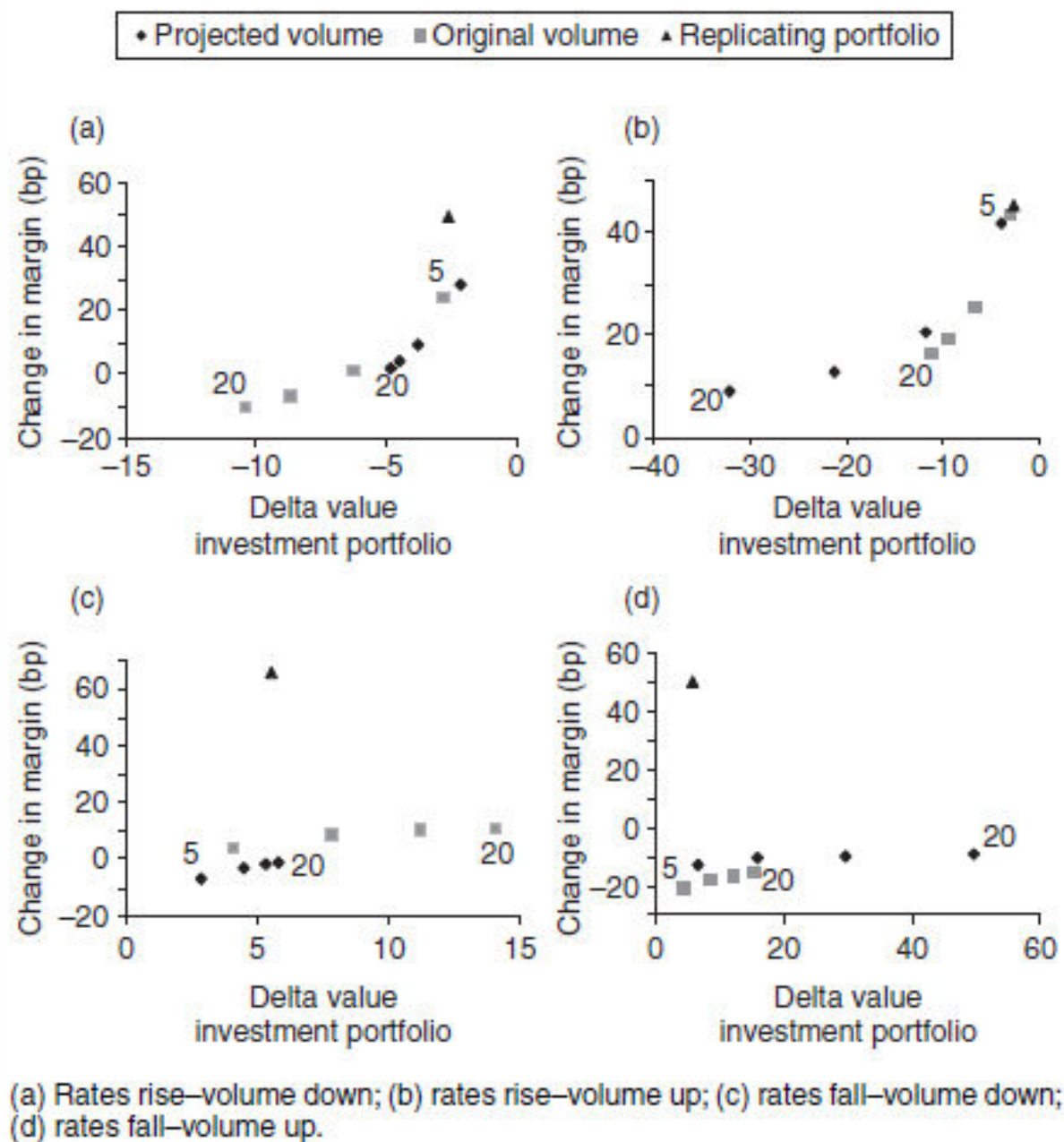
Figure 7.5 The two rate scenarios and two volume scenarios used in the comparison of the three models



(a) Rate scenarios; (b) volume scenarios.

Without going into too much detail in these results, from [Figure 7.6](#) it is clear that in some scenarios, especially the “rates down” scenarios, the value-based hedges significantly outperform the replicating portfolio. We believe this is a general feature, although the displayed results are for the volume and client rate model at hand. When we restrict the loss on the investment portfolio to the maximum loss of the replicating portfolio, the end date should be chosen at (approximately) five years (see the “rates rise” scenarios in [Figure 7.6](#)). Given this five-year end date, margin stability for the value-hedge approaches resembles the replicating portfolio in the rates rise–volume up scenario. Margin stability is significantly better in the other scenarios, especially the “rates fall” scenarios.

Figure 7.6 Comparison of models in four different scenarios



CONCLUSION

We have introduced two investment strategies intended to stabilise the margin and replicate the interest behaviour of non-maturing deposits based on the value-hedge approach of Jarrow and van Deventer (1998). The first method requires a model for future volume evolution. With this method, the

margin on the total volume can be stabilised quite well. In our alternative model, only the current volume is hedged. This model leads to a less stable margin on the total volume, but each slice of volume has its own fairly constant margin.

Both models meet the following characteristics formulated in our problem statement: the risk indicators of the non-maturing deposits can be obtained from the investment portfolio (eg, for duration) or the projected client rate plus margin (eg, for net interest income sensitivity). The non-maturing deposits business can be rewarded appropriately, and the interest rate risk is transferred to the ALM unit. Additionally, explicitly depending on a client rate model (and volume model when all projected volume is hedged) allows for a quick adjustment of margin and investment strategy when the non-maturing deposits business changes its pricing policy.

A comparison of the different models shows that both value-based investment strategies in an example market situation outperform the replicating portfolio in maintaining margin stability. An end date can be chosen to limit potential losses in the investment portfolio, thereby balancing stability of margin and stability of value in the investment portfolio.

APPENDIX

Here, we make the connection between results in Jarrow and van Deventer (1998) and Equation 7.3. We also estimate the size of the convexity correction, which was neglected in Equation 7.3. We start with Equation 8.5 of Jarrow and van Deventer

$$\text{value} = V(t_0) + \mathbb{E} \left[\sum_{i=0}^{N-1} D(t_{i+1}) [\Delta V(t_{i+1}) - \frac{1}{12} V(t_i) i(t_i)] \right] - \mathbb{E}[D(t_{N+1}) V(t_N)] \quad (7.7)$$

Here, the expectation is taken with the risk-neutral measure (with the money market account as numéraire). Further, $D(t_{i+1})$ denotes the stochastic discount factor that satisfies $\mathbb{E}[D(t_{i+1})] = d(t_{i+1})$, where $d(t_{i+1})$ denotes the discount factor that can be obtained from today's zero rate curve. $V(t)$ is the volume of non-maturing deposits at time t , and $\Delta V(t_{i+1}) = V(t_{i+1}) - V(t_i)$.

The rate $i(t)$ is the client rate plus servicing costs. In our approach, we replace $i(t)$ by the client rate $c(t)$, since the stable margin transferred to the

sales business should include compensation for the servicing costs.

Using the models in Equations 7.1 and 7.2 for the volume and client rate, the value defined in Equation 7.7 can be calculated. Since the volume does not depend on the market rates, the only complication arises from the expectation value $\mathbb{E}[D(t_{i+1})r_n(t_i)]$. In the result (Equation 7.3), we have used the approximation

$$\mathbb{E}[D(t_{i+1})r_n(t_i)] = d(t_{i+1})r_f(t_i, t_{i+n}) \quad (7.8)$$

with the forward rate satisfying

$$\left[1 + \frac{1}{12}r_f(t_i, t_{i+n})\right]^n = \frac{d(t_i)}{d(t_{i+n})}$$

(we use monthly compounded rates). This results in

$$\text{value} = V(t_0) + \sum_{i=0}^{N-1} d(t_{i+1})[\Delta V(t_{i+1}) - \frac{1}{12}V(t_i)\zeta(t_i)] - d(t_{N+1})V(t_N) \quad (7.9)$$

with $\zeta(t_i) = ar_f(t_i, t_{i+n}) + b$.

For a client rate proportional to the one-month rate, the resulting value (Equation 7.9) is exact. In this case, $n = 1$, the relation $D(t_i) = D(t_{i+1})[1 + \frac{1}{12}r_1(t_i)]$ implies that Equation 7.8 holds exactly.

For $n > 1$, there is a convexity correction that we neglected in Equation 7.8. In the following, we estimate the size of this correction. This is more conveniently done in continuous time for the expectation value $\mathbb{E}[D(t)r_n(t)]$. Henceforth, we consider simply compounded rates (n -month compounded rates). Assuming a lognormal dynamics of the forward rate under the forward measure for the maturity $T = t + nT$, where T denotes one month, we obtain (see, for example, Brigo and Mercurio 2001, Section 10.1)

$$\begin{aligned} \mathbb{E}[D(t)r_n(t)] &= d(t)r_f(t, t + n\Delta T) \\ &\times \left[1 + \frac{n\Delta T r_f(t, t + n\Delta T)}{1 + n\Delta T r_f(t, t + n\Delta T)}(e^{\sigma^2 t} - 1)\right] \end{aligned} \quad (7.10)$$

Here, σ denotes the volatility of the forward rate on time interval $[t, t+nT]$. The second term between square brackets in Equation 7.10 is the relative correction. Since the time t can reach 20 years and the (caplet)

volatility can be of order 20%, the factor $(e^{\sigma^2 t} - 1)$ cannot be considered to be small, but is rather of order 1. The suppression of the correction comes from the factor $nTr_f(t, t + nT)$. In our case we have chosen $n = 3$, so that

$$n\Delta Tr_f(t, t + n\Delta T) = \frac{1}{4}r_f(t, t + n\Delta T) \approx 1-2\% \quad (7.11)$$

The relative size of the convexity correction is 1–2%, which we have neglected.

If the convexity correction were important and unjustly neglected, this incorrect evaluation should appear in the historical backtest.

This chapter is an updated version of Elkenbracht-Huizing and Nauta (2006). The views expressed in this chapter are the authors' and not necessarily those of ABN AMRO or DNB.

- 1 This chapter concerns deposit account types that lack a contractual maturity date. Examples are demand deposits, transaction deposits, negotiable order of withdrawal accounts, savings and money market deposit accounts.
- 2 In this chapter the resulting investment portfolio is an imaginary portfolio, which is used to replace the non-maturing deposits on the liability side of the balance sheet for interest rate risk management. On the asset side the “real” investment portfolio, consisting of, eg, the loans the bank has originated, is used.
- 3 Although we focus on account types that pay interest, our approach (setting the client rate to zero) will also result in a stable margin for non-interest-bearing accounts.
- 4 When implementing this method, hedging the last buckets and rolling forwards should be carefully considered. It can be considered to spread the end date over a period. Choices can affect results considerably.
- 5 Even if the bank hedges a part of the loans and mortgages to which it has made a commitment (the “pipeline”), this differs from hedging all projected volume for, say, the next 20 years.
- 6 Some borrowing might occur in the end buckets of the investment portfolio due to solving the “rolling forwards” problem. These can be netted with nearby buckets.
- 7 The specific replicating portfolio model that we use in these tests has a longest tenor of 10 years. The fixed investment rule was obtained by minimising the standard deviation of the margin over a historical period.
- 8 Since the margin for the replicating portfolio can go up and down during the two-year period, the maximum difference in margin during two years is taken. For the two value-based approaches, the maximum difference is the difference in margin at the beginning and end of the period. In this case, the end margin minus beginning margin is plotted; for the replicating portfolio, the absolute value of the maximum difference is plotted.

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Replication of Non-Maturing Products in a Low Interest Rate Environment

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The risk management of non-maturing products is an important challenge for most banks, particularly for those with significant retail business. This task is complicated by the inherent options of these products: clients may add or withdraw volumes any time, and the product rate can be adjusted by the bank as a matter of policy. Both properties make future cashflows uncertain.

Usually, non-maturing products are replicated by a portfolio of fixed-maturity instruments. Popular approaches are based on static investment or funding rules where the portfolio weights are determined with the aim of minimising the volatility of the margin between the average portfolio rate (or opportunity rate) and the product rate. In the past it was common to determine these weights based on the analysis of historical data. Due to the unprecedentedly low level of interest rates in the aftermath of the financial and the European debt crisis, it is often advocated that the portfolio composition should be adapted to a scenario of future interest rates and product data.

We provide examples showing that static rules are inefficient in both cases. As an alternative, we propose a novel method that we call “dynamic replication”. Here, new decisions are periodically made on the allocation of maturing tranches, corrected by volume changes, that are determined by a multistage stochastic optimisation model. We describe this approach

conceptually before illustrating its performance in a case study. Technical details are documented in the appendixes.

CHARACTERISTICS OF NON-MATURING PRODUCTS

The balances of most banks typically contain a significant portion of so-called non-maturing assets and liabilities (NoMALs). These are positions without contractually specified maturity, and individual clients can always add or withdraw investments or credits without penalty and without (or possibly only with a short) notification period. Common examples include savings and sight deposits on the liability side, as well as overdraft and credit card loans on the asset side. The clients' option to determine the amortisation scheme only makes sense if the issuer can adjust the product rate at any time. For example, if after an increase in the level of market rates a bank does not also raise the product rate paid on savings deposits, its customers are likely to switch to alternative investment opportunities, but the bank's option to adjust the product rates makes customers more reluctant (Bardenhewer 2007).

Typically, the rates on non-maturing products exhibit a very particular dynamic. Although they mostly follow the direction of changes in money and capital market rates, an adjustment (often in discrete steps) is made only after larger changes in the latter, and with some delay. This can be explained by the administrative and other costs involved, but to some extent banks might also "optimise" their margins by delaying the pass-through of higher market rates to their deposit rates, or by postponing the decrease in variable mortgage rates until after market rates have dropped. The extent to which a bank might exercise its market power may also depend on the particular product, the availability of alternative products for clients and the (local) competition (Paraschiv 2013).

The clients' prepayment/withdrawal option and the banks' option to adjust the product rates are linked in a non-trivial way. Both depend on the current and expected future term structure. Clients react to changes in the product rates and in the relative attractiveness of alternative investment or financing opportunities. For example, during a period of high interest rates depositors tend to substitute their savings by long-term fixed-rate investments to "lock in" the high level of yields. This leads to a loss of cheap funding opportunities for banks, because deposit rates are usually significantly below market rates. When interest rates are low, the deposit rate becomes more

attractive than for other investments, which attracts additional savings volume. This applies particularly in the environment of unprecedentedly low (or even negative) market rates at the time of writing. On the other hand, deposit rates usually do not fall below zero.

As a consequence of the client behaviour described above, the volumes of non-maturing products often fluctuate significantly over time, but the magnitude of the variation may depend on many factors, such as the product type or a bank's client structure. For example, savings or non-maturing mortgages can easily be substituted by corresponding fixed-rate products. A sight account may also show high volume fluctuations, but these are less related to changes in interest rates, since clients use this product for transactions. On the other hand, certain pension accounts may formally be non-maturing but are actually considered as long-term investments by clients (due to incentives in the form of tax benefits) and show therefore a stable upward volume trend.

Clients as a whole exercise their prepayment or withdrawal option either for asset products or for liability products, but not for both simultaneously, and generally they exercise it to the bank's disadvantage. A non-maturing product on the liability side is not a good hedge for a corresponding product on the asset side because their volumes fluctuate out-of-phase. For example, savings deposits are withdrawn when interest rates go up, and the bank might suffer losses on fixed-rate assets if these must be squared to compensate the drop in volume.

In summary, the embedded options of non-maturing accounts pose a significant challenge for risk management, as the future cashflows from interest paid or received, as well as from amortisations, are uncertain. In contrast to fixed- or adjustable-rate products with defined maturity, the associated risk from changes in market rates cannot be quantified directly. The application of conventional hedging techniques such as duration matching therefore requires additional assumptions about the magnitude and timing of future cashflows. A common approach to overcome this problem is the definition of a fixed maturity profile for the non-maturing position. In this way, the uncertain cashflows are transformed into (apparently) certain ones, so that the usual risk management techniques for fixed-rate positions can be applied. This is achieved by construction of a replicating portfolio that consists of traded standard instruments and mimics the cashflows of an underlying non-maturing position, ie, the paid or received product rate plus

margin and volume increases or decreases. In theory the non-maturing product is then immunised against the risk of interest rate changes.

REPLICATING PORTFOLIOS

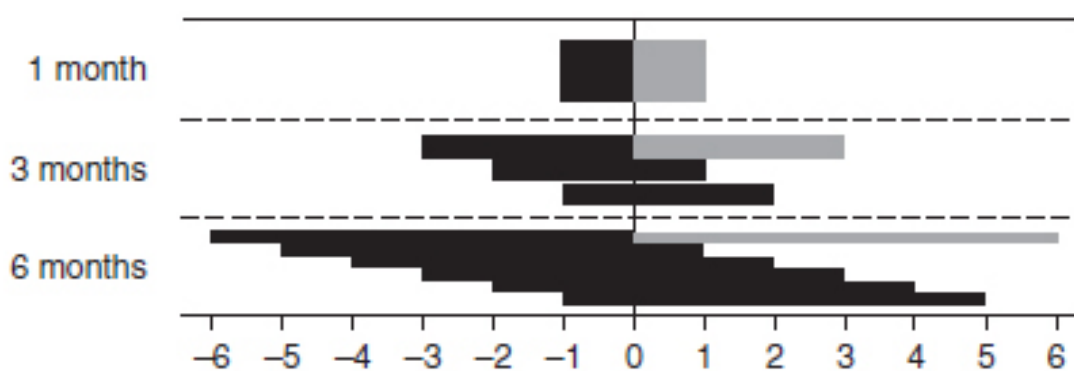
The replicating portfolio also defines the transfer price at which the margin is split between the business unit retail that acquires the position and the treasury, as the bank's central unit for the management of interest rate risk (fund transfer pricing). For fixed-maturity positions, the margin contribution of the retail unit is simply the difference between the product rate and the interest rate on the money or capital market for the corresponding maturity. For non-maturing products, the latter is replaced by the average rate of the positions in the replicating portfolio, the so-called "opportunity rate". Obviously, an "accurate" determination of the portfolio composition in terms of an exact replication of (product rate) payments from/to clients and cashflows due to changes in the notional is essential. Otherwise inefficient hedging decisions may result, and the profitability of non-maturing products will be measured incorrectly. In addition, the regulators demand that the assumptions underlying the determination of a replicating portfolio are based on a comprehensive analysis and are well documented.

In practice, the construction of a replicating portfolio requires the specification of an investment (or funding) rule. We assume for the moment that the volume is constant; the consideration of volume changes would require additional corrections. A common approach is to split the total volume into different time buckets (eg, 20% in one month, 10% in three and six months, 20% in one year) that consist of several tranches; the number of tranches corresponds to the maturity of the bucket. For example, a three-month bucket consists of three tranches, and a six-month bucket of six tranches, which are all equally weighted within their bucket. Each month one tranche matures and is then renewed at its original maturity. This mechanism is illustrated in [Figure 8.1](#) (see Bardenhewer 2007), where only time buckets up to six months are shown for simplicity. In practice, time buckets up to one year are replicated with money market instruments, and buckets above one year with par-coupon bonds.

The average rate of each time bucket is simply the moving average of the corresponding market rate. For the five-year bucket this would be the equally weighted average of the five-year rates in the current month and the previous fifty-nine months. The opportunity rate is then a weighted mix of these

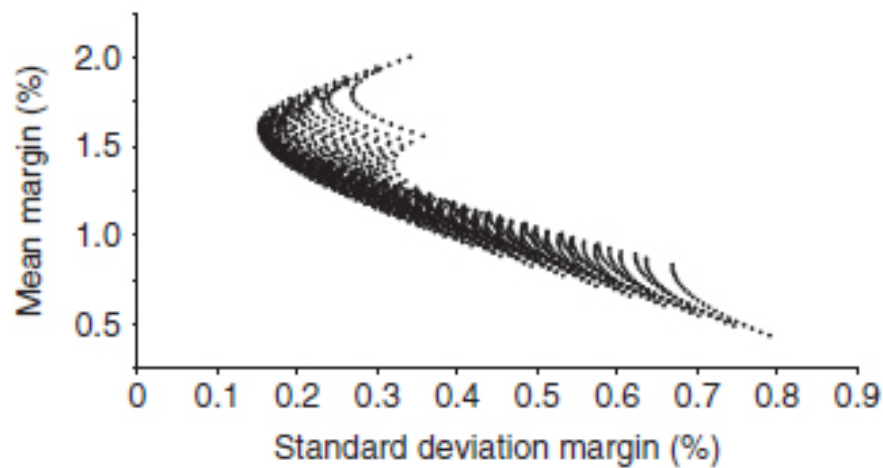
moving averages. For liability products, the difference between the average portfolio rate and the product rate is the margin. (For asset products the margin is the difference between the product rate and the average portfolio rate.) The concrete maturities of the considered time buckets and their percentages in the portfolio are determined so that it performs “optimally” over a historical sample period. Usually, “optimal” means that the standard deviation of the margin is minimised, although other risk measures are also possible. For example, each point in [Figure 8.2](#) represents the average of the margin (shown on the y -axis) and its standard deviation (x -axis) for a certain composition of a replicating portfolio. The combination at the tip of the efficient frontier represents the portfolio that provided the best replication over the sample period in terms of lowest volatility.

Figure 8.1 Replicating portfolio with different time buckets



Source: Bardenhewer (2007).

Figure 8.2 Means and standard deviations of margins of different replicating portfolios



The outlined approach enjoys great popularity in the banking industry, particularly in the German-speaking countries. However, there are some concerns from a theoretical point of view (Elkenbracht and Nauta 2006; Frauendorfer and Schürle 2007).

- The optimisation of the portfolio composition (weights of the time buckets) was performed for a single historical scenario. But will this portfolio also perform well in the future, eg, if the shape of the yield curve changes from normal to inverse after calibration of the portfolio weights?
- It can often be observed that the portfolio weights change significantly when the length of the sample period is varied. What then is the “true” replicating portfolio, since there is no “canonical” sample period? Instead, the sample period is often simply determined by the available historical data of a non-maturing product.
- Since the portfolio rate is constructed from the moving averages of many (historical) market rates, it exhibits a certain tardiness that reflects the characteristic slow adjustment behaviour of common non-maturing products. However, if a business unit wants to design a product where the client rate follows market rates more closely, similarly to the deposit accounts offered by banks, the weights of short maturities may dominate, which leads to smaller margins.

- Sometimes use of (larger) changes in the portfolio rate as “triggers” is also recommended for an adjustment of the product rate, to keep the margin stable. However, the bank’s product rate policy is then actually dictated by the replication method.
- It is possible that the margin may even become negative for the portfolio with the smallest volatility (after deduction of other costs). Since this is equivalent to a sure loss, such a portfolio cannot be viewed as “risk-minimal”, which questions the application of the standard deviation as risk measure. This aspect became more relevant under the central banks’ policy of low interest rates after the outbreak of the financial crisis in 2008, as we saw market rates close to or even below zero in some countries. On the other hand, banks cannot implement negative deposit rates, so margins are put under pressure.

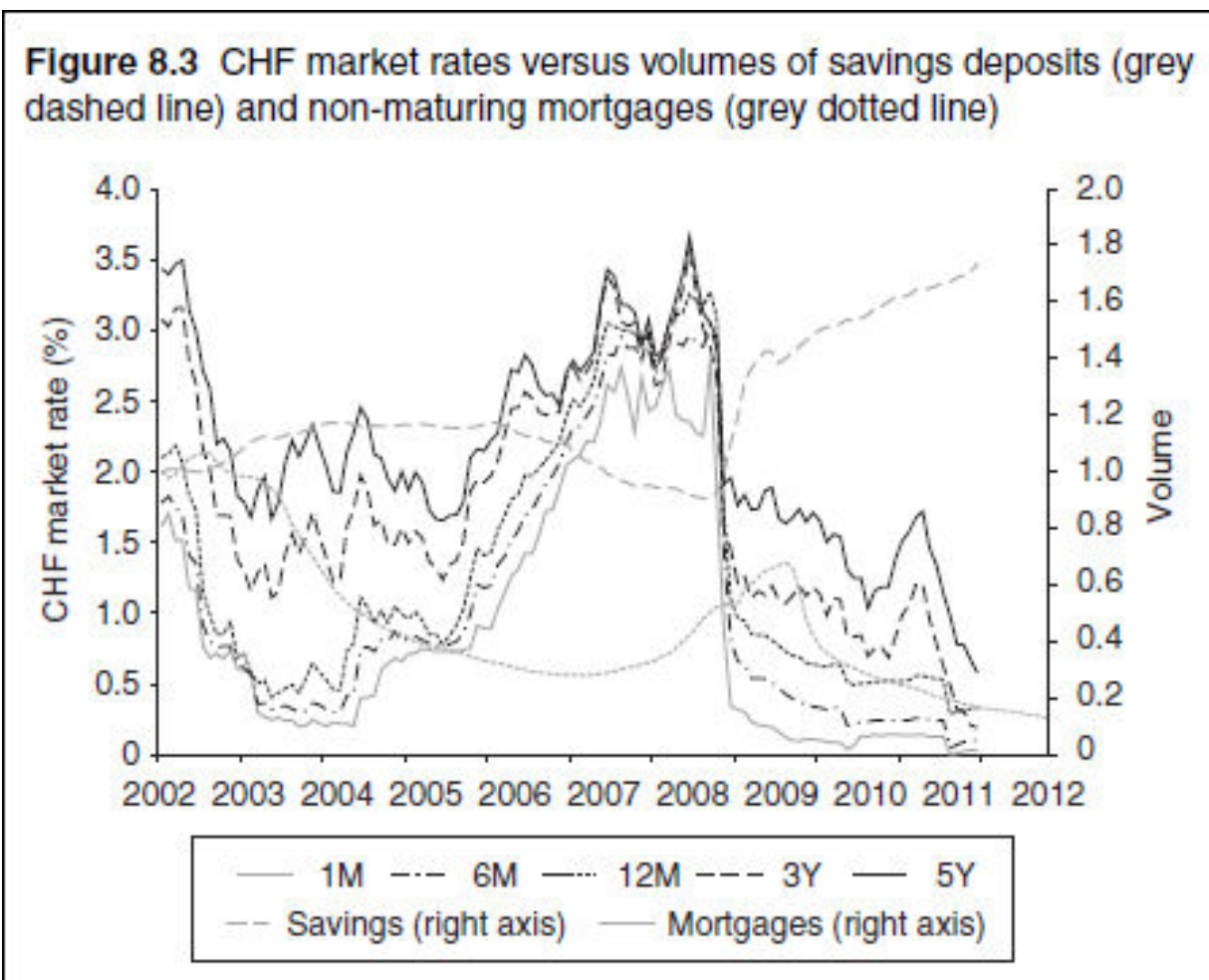
Aside from these concerns, the main problem with the above approach is that in reality the volume is not stable. Since the weights of the time buckets (and also the weights of the tranches within each bucket) are kept constant over time, all portfolio positions must be increased or decreased proportionally when the volume of the non-maturing product rises or falls. This implies that instruments must be bought or sold, but their coupon rates are in general no longer consistent with the current market rates of their remaining time to maturity. Therefore, the bank might realise a profit or suffer a loss from these transactions. The latter is more likely, as clients exercise their options when it is unfavourable for the bank, as outlined earlier. In the following section, we discuss two common approaches for handling volume changes and investigate their impact on margins more closely.

COMMON APPROACHES AND THEIR SHORTCOMINGS

Without corrections to the opportunity rate, the treasury would take over the costs or profits caused by volume fluctuations. This is inconsistent with the idea of fund transfer pricing, as the retail unit is responsible for the acquisition of the positions. Therefore, the opportunity rate must be adjusted by the economic effects of changes in the product volume to reveal the effective margin of the client business. This aspect became particularly important after most banks experienced a significant growth in their savings

volume in the low interest rate environment at the time of writing, but this increase can only be reinvested at much lower rates than the current moving average. Basically, there are two popular approaches to correct this.

1. Existing positions are increased or decreased at historical rates, which means that instruments in the portfolio are (virtually) bought or sold at their current value. Because in general the market price of these positions has changed since their acquisition, the differences between their original (ie, nominal) and their present values are immediately charged to the margin. Apart from these compensation payments, the opportunity rate is purely the weighted moving average of the rates of the maturities that correspond to the time buckets considered in the portfolio construction.



2. The opportunity rate is adjusted by a so-called rebalancing portfolio, which consists of investment or financing transactions with the same

time to maturity as the existing positions in the portfolio, but at current market rates. For example, assume that 100% of the replicating portfolio for a deposit is invested in the five-year time bucket. In the case of an increase in the product volume, one-sixtieth of the change is additionally invested at the current one-month rate, one-sixtieth at the current two-month rate and so on. Finally, the maturing five-year tranche is reinvested at its original volume plus one-sixtieth of the change in the non-maturing position. Analogously, in the case of a decrease, tranches in this example are squared proportionally by (virtual) funding transactions: one-sixtieth of the change is borrowed at the current one-month rate, another sixtieth at the two-month rate, etc. The reinvestment of the maturing five-year tranche is now reduced by one-sixtieth of the loss in the product volume. As a consequence the opportunity rate is composed of the current and previous rates of many different maturities.

Table 8.1 Replicating portfolios for Swiss savings deposits (2002–11)

Savings deposits	Volume changes considered	Weights time buckets (%)				Margin statistics (%)			
		3M	12M	3Y	5Y	SD	Mean	Min	Max
1 Uncorrected	—	5	25	10	60	0.152	1.59	1.18	1.97
2 PV changes	<i>Ex post</i>	5	25	10	60	0.464	1.39	–1.18	2.52
3 Rebalancing portfolio	<i>Ex post</i>	5	25	10	60	0.191	1.44	0.95	1.68
4 PV changes	<i>Ex ante</i>	0	45	35	20	0.363	1.19	–0.55	2.18
5 Rebalancing portfolio	<i>Ex ante</i>	0	25	20	55	0.183	1.47	0.95	1.74

Notes: PV denotes present value. SD denotes standard deviation.

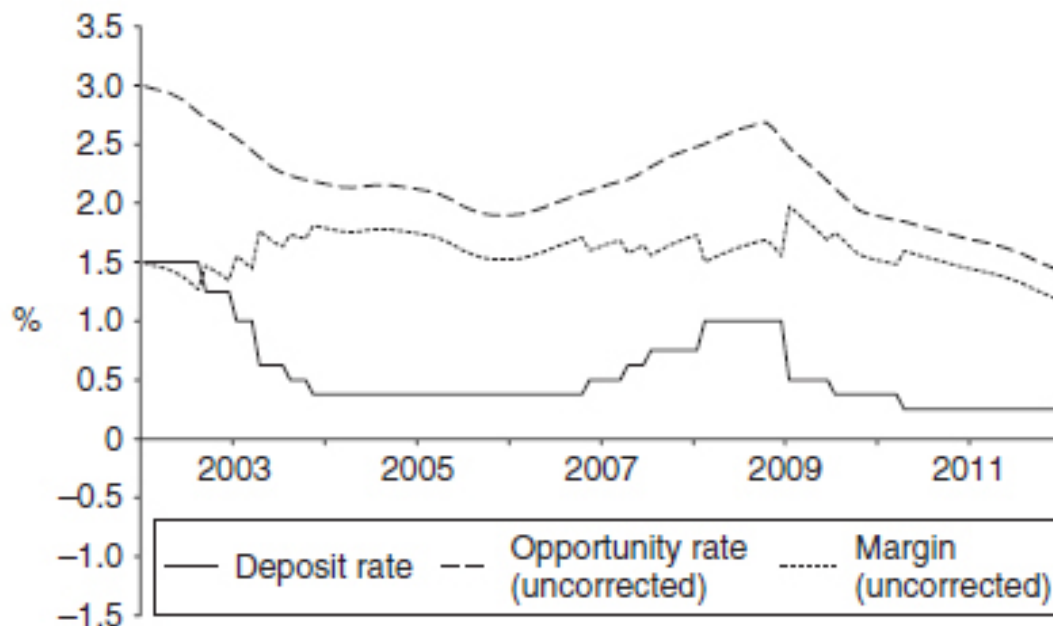
Table 8.2 Replicating portfolios for non-maturing mortgages (2002–11)

Non-maturing mortgages	Volume changes considered	Weights time buckets (%)				Margin statistics (%)			
		3M	12M	3Y	5Y	SD	Mean	Min	Max
1 Uncorrected	—	0	25	0	75	0.164	0.89	0.43	1.23
2 PV changes	<i>Ex post</i>	0	25	0	75	0.988	1.71	0.79	7.95
3 Rebalancing portfolio	<i>Ex post</i>	0	25	0	75	0.769	-0.21	-1.39	1.19
4 PV changes	<i>Ex ante</i>	0	60	40	0	0.619	1.76	0.54	4.80
5 Rebalancing portfolio	<i>Ex ante</i>	40	15	15	30	0.350	0.90	0.34	1.87

Notes: PV denotes present value. SD denotes standard deviation.

The first approach is very popular, for example, among German savings banks, because of its apparent simplicity for the calculation of the opportunity rate; the second is more common in Switzerland. The following examples illustrate that the margin can become highly volatile when it is corrected *ex post* by the present values of the increased or decreased tranches. We consider a real savings deposit and a non-maturing mortgage position of a Swiss bank over a period of 10 years between January 2002 and December 2011. (We chose this period since it covers an increase in market rates and a decline after the financial crisis.) [Figure 8.3](#) shows the evolution of market rates for the Swiss franc and the volumes of the two positions relative to the initial value at the beginning of the sample period. In both cases we can clearly see a correlation between volumes and market rates (positive for mortgages, negative for savings) as described earlier. In particular, the volatility of the non-maturing mortgage volume is large, as it can easily be substituted by alternative (fixed-maturity) mortgages, which is beneficial when the interest level is low. Note also the significant increase in the deposit volume at the end of 2008 because the bank considered could attract large volumes during the market turbulence after the Lehman insolvency.

Figure 8.4 Deposit rate versus opportunity rate of savings without correction for volume changes



Impact of volume changes

The replicating portfolios of both products that minimise the standard deviation of the margin during the sample period are documented in [Tables 8.1](#) and [8.2](#). In each case the first line, “uncorrected”, refers to the characteristics of a portfolio that is derived by minimising the volatility of the margin, defined as the difference between product rate and the weighted moving average of the rates of the four time buckets considered without corrections for volume changes. For both products the weight of the longest maturity bucket is large, which reflects the tardiness of the product rate. [Figure 8.4](#) shows the evolution of the uncorrected opportunity rate versus the deposit rate for savings. The resulting margin is indeed relatively stable, except for a reduction at the end of the considered time interval, when market rates drop further in the context of the financial crisis, but the deposit rate is forced to stay above zero, even if market rates become zero or negative. [Figure 8.5](#) shows the same for the mortgage product.

Figure 8.5 Product rate versus opportunity rate of non-maturing mortgages without correction

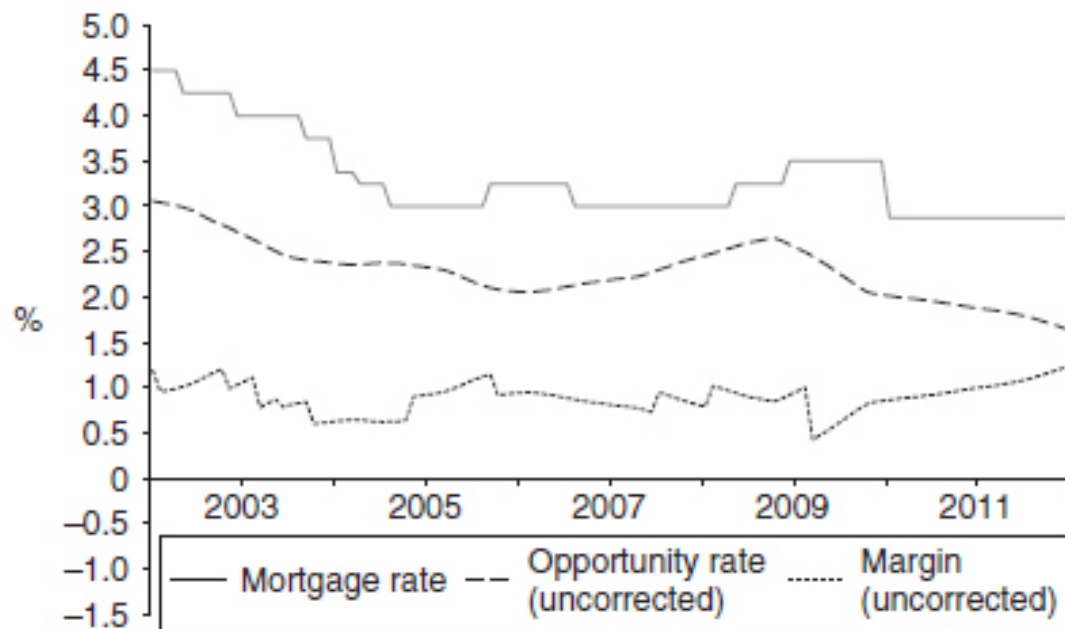
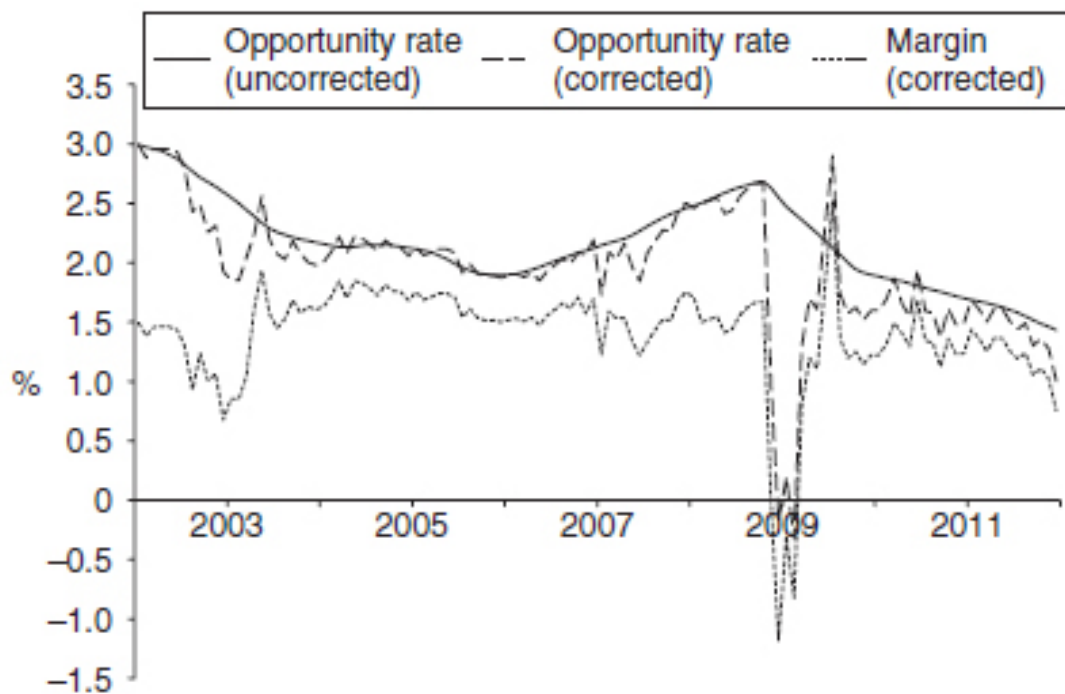


Figure 8.6 Deposit rate versus opportunity rate savings after correction for present value effects



Now the opportunity rates of the portfolios obtained in this way are corrected by the present value effects caused by volume changes using the first method. As can be seen in [Figure 8.6](#), the resulting rate is not only more volatile but also systematically below the rate before correction, particularly after the significant volume increase at the end of 2008. The compensation payments for the higher market value of the positions that must be increased to keep the portfolio composition stable wipe out the margin. The margin volatility becomes even more severe for the non-maturing mortgages (see [Figure 8.7](#), where the extreme values have been truncated). This case corresponds to the second row in the tables, when the effects of volume changes are considered *ex post*.

Figure 8.7 Product rate versus opportunity rate of non-maturing mortgages after correction for present value effects

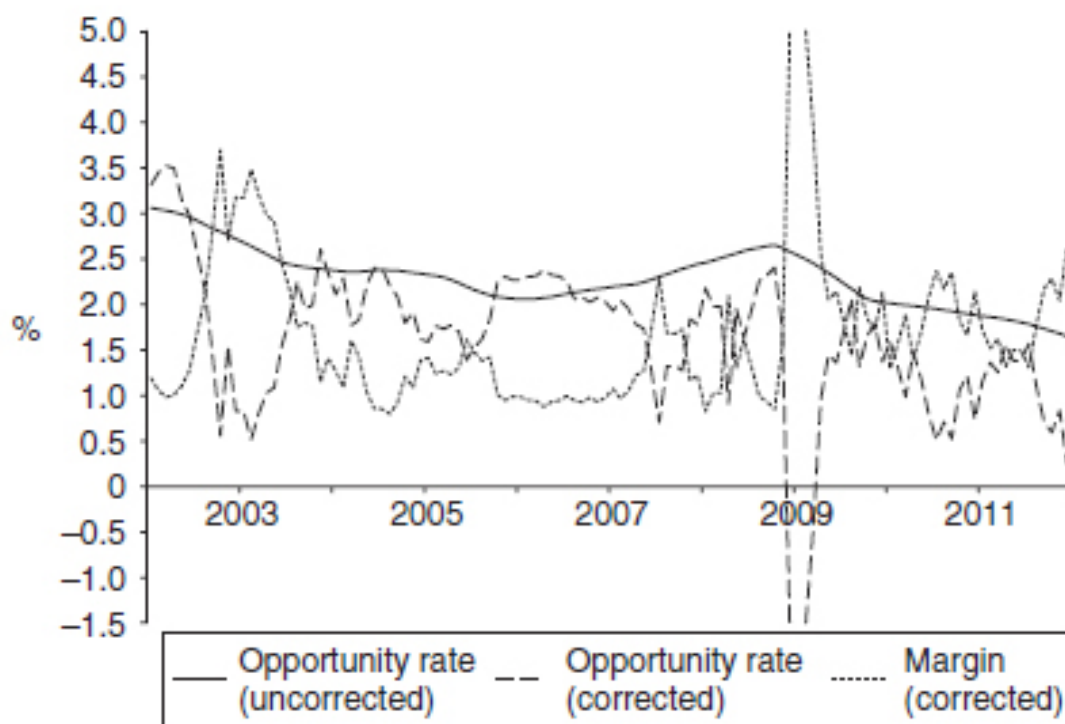


Figure 8.8 Deposit rate versus opportunity rate of savings with rebalancing portfolio

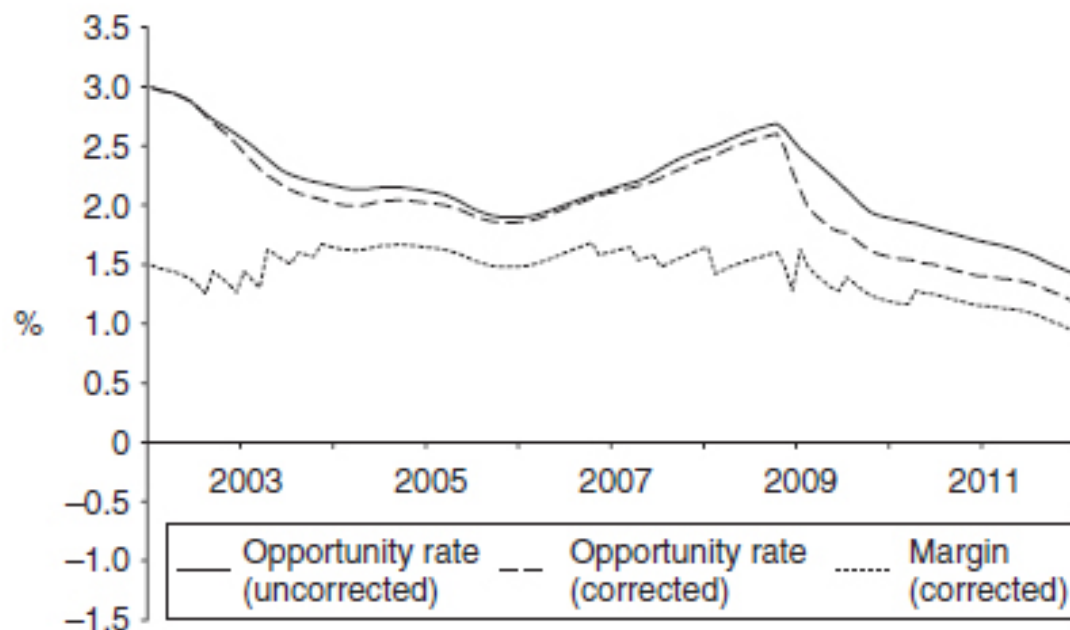
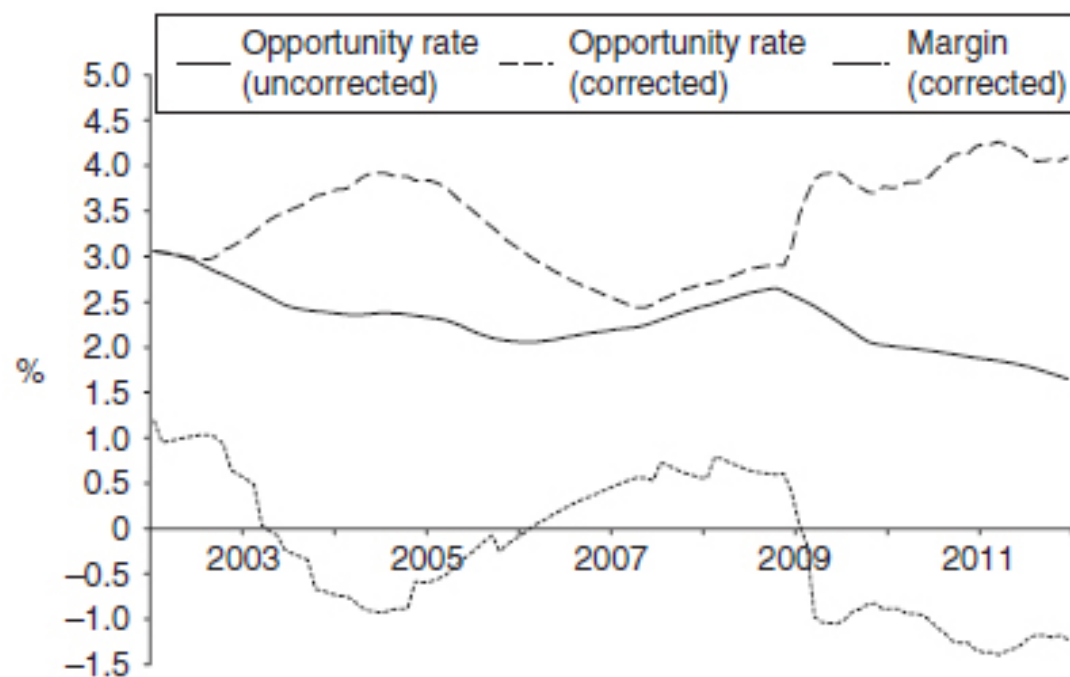


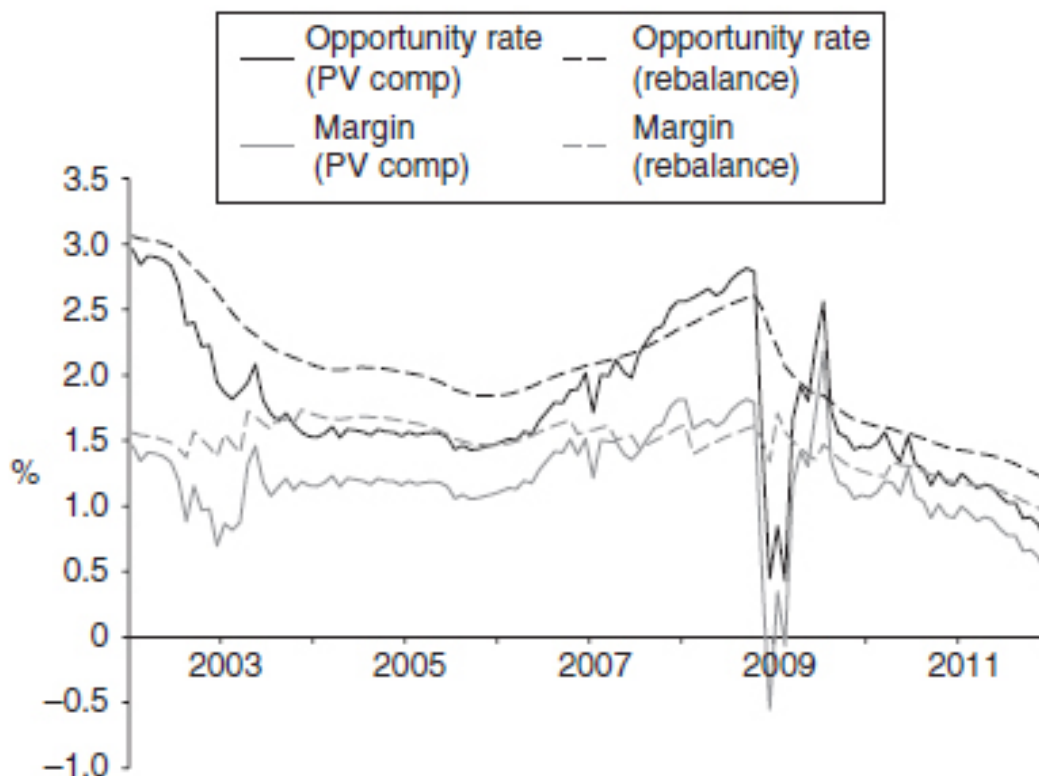
Figure 8.9 Product rate versus opportunity rate of non-maturing mortgages with rebalancing portfolio



Figures 8.8 and 8.9 show the same analysis for the second method with the rebalancing portfolio. The opportunity rate for the savings is much smoother, since here the cashflows from the correcting transactions are distributed over the remaining time to maturity of the positions, while in the former approach they occur only once. However, the opportunity rate is still systematically below the uncorrected weighted moving average. This is a consequence of the clients' withdrawal option that is always exercised against the bank's interest. For the non-maturing mortgages, the opportunity (funding) rate including the rebalancing portfolio is above the product rate after the observed significant reductions in volume, which leads to negative margins because the position was previously financed at higher interest rates. The corresponding tranches remain in the portfolio but are squared by investments at lower rates. Since the non-maturing mortgage volume drops to a fraction of its earlier values, the volume of the rebalancing portfolio may become greater than the position actually replicated. The corresponding key values of these portfolios can be found in the third row of the tables.

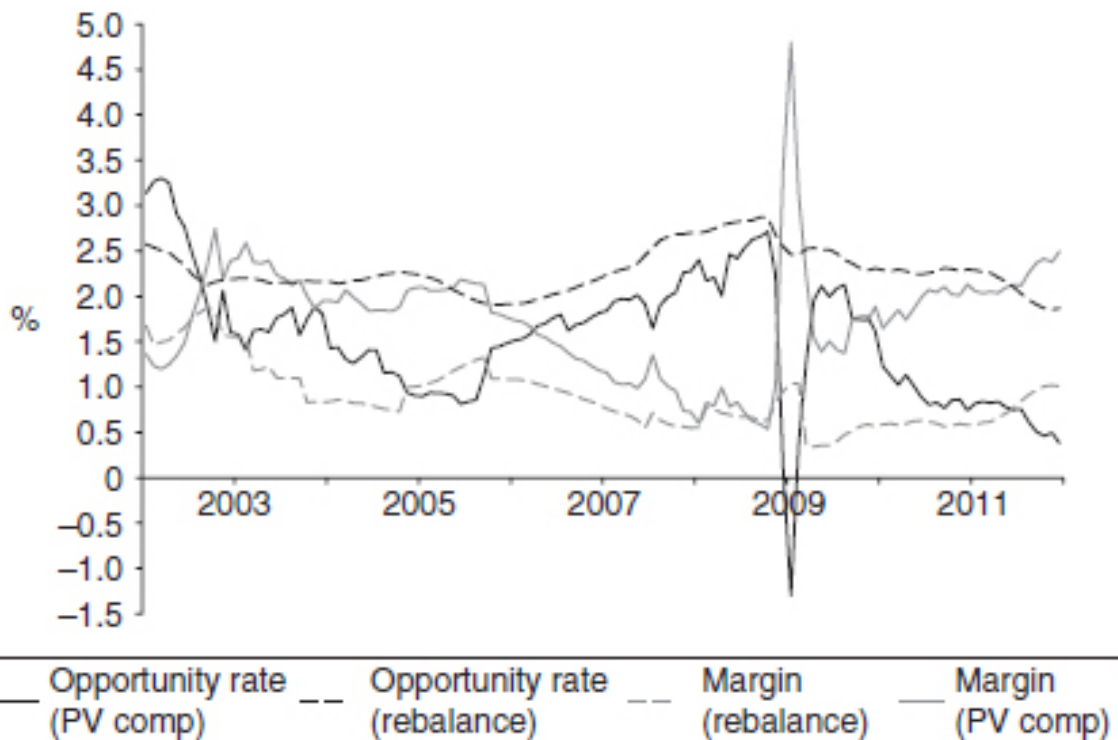
The negative margin that occurred for mortgages with the rebalancing portfolio method results from too high a percentage of longer maturities in the portfolio. The increased margin volatility observed for the first approach (present value (PV) changes are charged to the opportunity rate) has the same cause, since the values of instruments with long maturities are more sensitive to changes in interest rates. This implies that the effects of volume fluctuations should already be taken into account for the determination of the portfolio weights, ie, *ex ante*, which is often ignored in practice. Instead many banks (and software products for the management of non-maturing products) determine the portfolio weights for a constant volume and calculate corrections only *ex post*.

Figure 8.10 Opportunity rate and margin of savings when corrections caused by volume changes are taken into account *ex ante*



The resulting portfolios when volume changes are already taken into account for the determination of the weights and the key values for these portfolios are shown in rows 4 and 5 of the tables. Figures 8.10 and 8.11 show the margins of savings deposits and non-maturing mortgages that result when the standard deviation is minimised with respect to the corrected opportunity rate. The rebalancing approach (solid lines) leads to a higher margin for the savings deposit because the portfolio has a longer duration. Here the cashflows of the correction transactions are distributed over a longer time, while in the other method with compensation of PV changes they occur at once. This reduces the percentage of longer maturities since their prices are more sensitive to interest rate changes. The margins of the mortgages are no longer systematically negative and, compared with the previous situation, became relatively stable for the rebalancing portfolio approach.

Figure 8.11 Opportunity rate and margin of non-maturing mortgages when corrections caused by volume changes are taken into account *ex ante*



In summary, the examples show the significant impact of volume fluctuations on the realised margins and underline the importance of taking them into account for the determination of the portfolio composition. If this is done, the rebalancing portfolio method, which calculates corrections based on current interest rates, leads to higher and less volatile margins than the other approach with compensation of PV changes. In the latter method the compensation payments reduce the percentage of long-term instruments too much due to their high sensitivity to interest rate changes. However, the application of different approaches can lead to very different resultant portfolios and margin characteristics; this has direct implications for the risk management and assessment of the products' profitability.

In practice, replicating portfolios are frequently tested for their effectiveness. For instance, a bank might annually re-estimate the weights of the time buckets based on new data for market rates, product rates and volumes. If this leads to a new composition, then the portfolio must

rebalanced, ie, existing tranches are squared and invested (or financed) in new positions. But it is not obvious which bank unit – retail or treasury – must take over the possible profits or losses caused by the corresponding transactions. Eventually, these profits or losses result only from the adjustment of a “calculation rule” and are not related to the performance of the units involved.

Overcoming the difficulties with stochastic models

One disadvantage of the replicating portfolio approach discussed so far is that it leads to static investment rules, as weights remain constant over time and tranches are distributed uniformly within time buckets. This ignores the current market situation, which limits the performance, and it does not sufficiently take into account the effects of the inherent options that clients may exercise to the bank’s disadvantage. We have seen that the correction for volume changes can have a clear impact on the profitability of non-maturing products. The question is why a non-maturing product must be replicated at all by a portfolio with constant duration, although its volume changes significantly over time and with a certain dependency on the interest level.

Since the late 1990s, stochastic modelling approaches have been proposed to overcome the above-mentioned shortcomings of static replicating portfolios. A major contribution was made by Jarrow and van Deventer (1998), who define the value of a non-maturing liability L as the sum of the expected discounted future cashflows

$$PV_L = V_0 + \mathbb{E}^Q \left[\sum_{t=0}^{T-1} d_{t+1} (\Delta V_{t+1} - \frac{1}{12} V_t c_t) \right] - \mathbb{E}^Q [d_{T+1} V_T] \quad (8.1)$$

where Q denotes taking expectation under the risk-neutral probability measure, as is common in valuation models. V_t defines the volume, c_t denotes the client rate (which is here assumed to be paid monthly) and d_t is the discounting factor for time t . According to this equation the liability value consists of the initial balance plus any volume changes over time and minus the costs for holding the position, which are basically payments to clients and other non-interest (eg, administrative) costs that are not explicitly stated here. In practice, the expectation in Equation 8.1 can be calculated by generating scenarios of the short-term rate (to determine the discount factor) as well as the future product rates and volumes. Therefore, the planning horizon is

truncated at time T and the terminal nominal volume is (virtually) repaid, which motivates the last term. The approach is much more complex than static replication methods because the stochastic dynamics of product rates and volumes have to be modelled.

Elkenbracht and Nauta (2006) adapt this framework for an alternative construction of replicating portfolios. First, the liability value is made dependent on some margin m

$$PV_L(m) = V_0 + \mathbb{E}^Q \left[\sum_{t=0}^{T-1} d_{t+1} (\Delta V_{t+1} - \frac{1}{12} V_t (c_t + m)) \right] - \mathbb{E}^Q [d_{T+1} V_T]$$

Then a “fair” margin m^* is defined as the spread that must be added to the client rate to set the present value of the liability to zero, ie, $PV_L(m^*) = 0$. The approach allows the calculation of the sensitivity of the non-maturing position including the margin with respect to changes in interest rates. Then a portfolio can be identified which hedges the margin in such a way that its profits and losses compensate changes in the value of the non-maturing liability, ie

$$PV_L(m^*) + P\&L(\text{portfolio}) = 0$$

The transactions required so that the above equation holds must frequently be recalculated; this leads to a dynamic investment strategy. Elkenbracht and Nauta (2006) report that the margins obtained with their approach are remarkably stable compared with static replicating portfolios. However, for a positive volume trend large amounts are assigned to the longest time bucket and the model borrows significantly up to the shortest maturity, ie, the portfolio itself performs some term transformation.

This method provides replication strategies that react to changes in the current yield curve. If the models for the evolution of product rate and volume also reflect their dependencies on interest rates, then the value of the inherent options is also considered appropriately. This makes the approach useful for measuring interest rate risk. But investment strategies are in a certain sense myopic, since future decisions are not taken into account. For instance, the positions of today’s “perfect” hedge or replication are possibly squared again tomorrow, which would lead to unnecessary transactions. Frauendorfer and Schürle (2007) propose to overcome potential inefficiencies that arise from a myopic view by also taking into account

future decisions and their impact on today's strategy. To this end, the problem is formulated as a multistage stochastic optimisation model.

In the following we present an updated version of this model, which differs from the description in Frauendorfer and Schürle (2007) with respect to the following points. First, in their model the decision-maker can specify a desired margin; then the shortfall with respect to this target is minimised. Here we consider the margin as a model result rather than as an input parameter; profit goals may be taken into account in the new model by optimising a weighted mix of risk and return. Second, optimisation models are restricted to a limited number of stages. In order to obtain a planning horizon of several years, decisions were made only at yearly time steps in the previous model, and only instruments with maturities of one year or a multiple of one year could be considered. Our approach extends the number of stages so that investment decisions can be made monthly. Finally, we apply advanced models for product rates and volumes, as well as a scenario-generation procedure that allows a better match with the observed kurtosis of interest rate changes.

DYNAMIC REPLICATING PORTFOLIO APPROACH

In a nutshell, multistage stochastic programming is a framework for modelling optimisation problems that involve uncertainty. This is obviously the case for many financial applications. A naive approach to deal with uncertainty would be to use a forecast of the problem-specific data (here market rates, product rate and volume) for some finite time horizon and to determine over that period investment decisions that best satisfy a given optimisation criterion. But the resulting policy would only be appropriate for this particular scenario of the future and may be inefficient (or infeasible) if a different evolution is realised. It may also turn out that decisions are highly sensitive to different forecasts.

Stochastic programming methods instead use a large set of scenarios for possible evolutions of the risk factors. This requires that the probability distributions of the problem-specific data are known or can be estimated. For instance, if we have a model for the joint (conditional) distribution of our risk factors, then scenarios can be generated by "sampling" possible future outcomes at discrete time steps. All these scenarios are integrated in a large-scale optimisation problem. Its solution provides a strategy that is feasible for all (or almost all) scenarios and satisfies an optimisation criterion such

as maximisation of expected profits, minimisation of risk or a trade-off between risk and expected return. The resulting strategy may be seen as the “best compromise” for a large set of possible scenarios, and is therefore more robust against changes in the input parameters.

An important feature of multistage problems is that decisions are made not only in the first stage, but also at later time points. In this way, the ill effects of earlier decisions can be compensated for once outcomes of the uncertain future data have been realised, eg, the portfolio structure can be rebalanced in later stages if it is too risky in certain scenarios, or new volumes or tranches maturing at later time points can be allocated. However, although the decisions in subsequent stages are basically necessary to quantify the impact of later corrections on the first-stage decision, only the first-stage decision will be implemented in reality. When a new action must be taken after, eg, one month, scenarios are generated, again based on the latest observations of the relevant data on the market, and a new optimisation problem is solved. Thus, the term “dynamic replication” can be understood from two perspectives: first, the method is based on multistage (ie, dynamic) stochastic programming. Second, instead of keeping the portfolio composition constant, as in the traditional static replication approach, we periodically determine a new reallocation of maturing tranches plus or minus a volume change.

SPECIFICATION OF THE STOCHASTIC OPTIMISATION MODEL

In the following description of the optimisation model for the determination of dynamic replicating portfolios we restrict ourselves to liability products, such as deposit accounts, to keep the explanation simple. The formulation of a corresponding model for non-maturing asset products is equivalent and can immediately be derived when the terms “investing” and “borrowing” are interchanged. We assume that all uncertain coefficients are functions of a few stochastic factors. Their joint evolution can be modelled by a multi-dimensional stochastic process ω in discrete time. For simplicity we drop the dependency of coefficients on this process in the notation. The same applies to decision variables that also depend on realisations of the risk factors. At this time we do not make any specific assumptions regarding the

(joint) process for the risk factors. Examples of appropriate models will be presented in a subsequent section.

Notation

At a fixed frequency, eg, monthly, the model is used to decide about the reinvestment of maturing tranches plus or minus a change in the total volume. $D = \{1, \dots, D\}$ denotes the set of dates when fixed-income instruments in the replicating portfolio mature, where D represents the longest available maturity. The maturities of standard instruments that can be used for investment transactions are given by the set $D^S \subseteq D$. Alternatively, positions held in the portfolio may be squared prior to maturity, which is modelled as borrowing funds with maturities in D^S .

Decisions on the allocation of instruments are made at stages $t = 0, \dots, T$. Although the replication of a non-maturing position is actually an application with an infinite planning horizon, the number of time steps must be truncated for practical reasons (limited computational recourses). In order to account for the effects of transactions made at times $t = 0, \dots, T$ beyond the end of the planning horizon, an additional stage, $T + 1$, is added, where the present values of the remaining portfolio positions are calculated. This may be seen as a virtual sale of the portfolio where the resulting profits or losses are charged to the margin.

The problem-specific (stochastic) coefficients in the optimisation model for $t = 0, \dots, T$ are: $r_t^{d,+}$, the bid rate per period for investing up to maturity $d \in D^S$; $r_t^{d,-}$, the ask rate per period for borrowing in maturity $d \in D^S$; c_t , the client (product) rate paid per period; and v_t , the volume of the non-maturing account.

For time $t = 0$ the values of these coefficients are known, but for $t = 1, \dots, T$ they depend on the history of realisations of the (joint) risk factor process ω . It is assumed that interest rates are paid periodically, ie, if the model's period is one month, then the values of the coefficients $r_t^{d,+}$ and $r_t^{d,-}$ are obtained by dividing the annualised market rate for maturity d by 12 (after correction for a bid–ask spread). The coefficient for the client rate c_t is obtained analogously. Additionally, the calculation of the present values of outstanding cashflows in the terminal stage $T + 1$ is based on the stochastic coefficients $PV^{t,d,+}$, the present value of cashflows, resulting from the

investment of \$1 at maturity $d \in D^S$ at time t , that occur after the end of the planning period (the coefficient is calculated based on the term structure in $T + 1$); $PV^{t,d,-}$ is defined analogously for borrowing.

At each time point $t = 0, \dots, T$, decisions are made on the transactions at each maturity for the allocation of maturing tranches (which are in general not renewed in the same maturity) plus or minus the change in volume. This requires the following decision and state variables: $x_t^{d,+}$, the amount invested at maturity $d \in D^S$; $x_t^{d,-}$, the amount financed on maturity $d \in D^S$; x_t^d , the total nominal amount with time to maturity $d \in D$ (non-negative); x_t^S , absolute surplus defined as income from the replicating portfolio (coupon payments) minus costs for holding the account (client rate payments and other non-interest costs).

Optionally, the transaction amounts $x_t^{d,+}$ and $x_t^{d,-}$ may be bounded by limit values $\ell^{d,+}$ and $\ell^{d,-}$. The total invested amount x_t^d for all maturity dates d must be non-negative, since the portfolio should replicate only the underlying non-maturing position and must not perform a term transformation itself, eg, by taking short positions in the money market and investing the corresponding amount in the capital market. Positions in the existing portfolio, which result from decisions in the past, must also be taken into account since they represent a certain risk profile; a variable x_{-1}^d with negative time index refers to a nominal value in the initial portfolio with maturity $d \in D$. The aggregated interest cashflow from positions in the initial portfolio that accrue at time d is denoted by cf_{-1}^d ; it is deterministic since only fixed-income securities are taken into account for the portfolio construction. For consistency with the decision variables introduced above, the superscript refers to the remaining time to maturity, eg, x_{-1}^1 (cf_{-1}^1) represents a nominal (interest) cashflow that accrues at time $t = 0$. Finally, p_{v-1} denotes the present value of cashflows from the initial portfolio that accrue after the end of the planning horizon. It is stochastic since discount factors used in the calculation are based on the term structure at time $T + 1$.

Specification of constraints

At each stage t , budget constraints must hold that update the nominal volume with maturity date $d \in D^S$ by the corresponding transaction amounts

$$x_t^d = x_{t-1}^{d+1} + x_t^{d,+} - x_t^{d,-}, \quad t = 0, \dots, T \quad \text{for all } d \in \mathcal{D}^S \quad (8.2)$$

For non-traded maturities the nominal amounts simply equal the corresponding value at the previous stage

$$x_t^d = x_{t-1}^{d+1}, \quad t = 0, \dots, T \quad \text{for all } d \in \mathcal{D} \setminus \mathcal{D}^S \quad (8.3)$$

The sum of all positions in the portfolio must match the current volume of the non-maturing account at all times

$$\sum_{d \in \mathcal{D}} x_t^d = v_t, \quad t = 0, \dots, T \quad (8.4)$$

The previously listed constraints are necessarily required to ensure the feasibility of investment and borrowing decisions. Additionally, some optional constraints may be specified if the decision-maker also wants to observe other criteria, eg, liquidity thresholds that can be enforced by restrictions on the portfolio structure. For instance, we may introduce limits for the percentage of certain time buckets. Let w_i^l and w_i^u be lower and upper bounds for the percentage of the i th bucket defined by the subset of maturity dates $D^{w_i} \subseteq D$, $i = 1, \dots, k$, where k is the number of time buckets for which such a restriction applies

$$w_i^l v_t \leq \sum_{d \in D_i^{w_i}} x_t^d \leq w_i^u v_t, \quad t = 0, \dots, T, \quad i = 1, \dots, k \quad (8.5)$$

Corresponding amounts for absolute limits instead of percentages may be defined analogously. Also, limits for the transaction sizes are possible, eg, the amount of sales can be restricted to a drop in volume

$$\sum_{d \in \mathcal{D}^S} x_t^{d,-} \leq \max\{0, v_{t-1} - v_t\}, \quad t = 0, \dots, T \quad (8.6)$$

Without such a constraint the model can decide to rebalance the portfolio actively by selling existing tranches and investing them at different maturities, ie, to square positions to a greater extent than required to compensate withdrawals only. With Equation 8.6 the volume of squared positions is limited to a comparable magnitude as transactions in the static approach in order to compensate a volume loss.

Definition of surplus

The ultimate goal is the optimisation of some trade-off between risk and return. Before this is specified formally in terms of an objective function, we have to define the surplus that results from the replicating portfolio. It consists of the periodic income from the portfolio (coupon payments) minus the costs for holding the account

$$\begin{aligned}
 x_t^S = & \sum_{\tau=0}^t \sum_{d \in \mathcal{D}^S, d > \tau} r_{t-\tau}^{d,+} x_{t-\tau}^{d,+} \\
 & - \sum_{\tau=0}^t \sum_{d \in \mathcal{D}^S, d > \tau} r_{t-\tau}^{d,-} x_{t-\tau}^{d,-} + \text{cf}_{-1}^{t+2} - (c_t + \alpha_0) v_t, \\
 & t = 0, \dots, T \quad (8.7)
 \end{aligned}$$

The first term on the right-hand side of Equation 8.7 corresponds to the interest received for investments in traded standard instruments. The second term is the costs paid for squaring positions. Here all the transactions that were made up to the current stage (summation from 0 to t) and are not yet matured (ie, their maturity d is greater than the difference τ from the time when they were made) are taken into account. Note that the constraint summarises cashflows that result from transactions at the current stage t , although in reality they accrue one period later. This must also be taken into account when the cashflows from positions in the initial portfolio cf_{-1}^{t+2} are added. The superscript notation emphasises that these cashflows actually accrue at time $t + 1$ (see the definition in the section on notation).

These cashflows are relevant for the optimal strategy since the model takes into account the risk of not covering the overall costs for holding the non-maturing position given by the last term. They consist of product rate payments to depositors in absolute terms (“client rate multiplied by volume”); furthermore, non-interest expenses α_0 for managing the account may be added. We focus on absolute profits instead of the margin (which would be obtained if the right-hand side were divided by v_t). In this way scenarios with higher volumes have also a greater impact on the determination of the optimal portfolio.

In order to quantify the effect of a truncation of the actually indefinite planning horizon, the portfolio is virtually sold at time $T + 1$ to repay the

deposit volume. To this end, all outstanding cashflows are discounted based on the term structure observed at $T + 1$

$$x_{T+1}^S = \sum_{\tau=1}^T \sum_{d \in \mathcal{D}^S, d > \tau} PV^{T-\tau, d, +} x_{t-\tau}^{d, +} - \sum_{\tau=1}^T \sum_{d \in \mathcal{D}^S, d > \tau} PV^{T-\tau, d, -} x_{t-\tau}^{d, -} + pv_{-1} - v_t \quad (8.8)$$

Recall that the model description is restricted to the case of liability products. For asset products the signs of all terms on the right-hand side of Equations 8.8 and 8.9 must be inverted. In addition to the variables defined for the surplus per stage, we introduce the corresponding accumulated revenues

$$z_t = \sum_{\tau=0}^t x_{\tau}^S, \quad t = 0, \dots, T + 1 \quad (8.9)$$

In the following we abbreviate the sequence of revenues over all stages to $z := (z_0, \dots, z_{T+1})$.

Model objective

As already outlined earlier, there are various options for defining an optimality criterion for portfolio construction in general. A common choice in stochastic programming is to maximise the expected value (mean) of the overall revenues, ie, to maximise $E[z_{T+1}]$. This is beneficial from a technical point of view, since “expectation” leads to a linear objective. Taking this, together with the above constraints (which are all linear in the decision variables), results in a linear optimisation problem, which is important because we deal here with large-scale problems due to the large number of scenarios. For large linear problems, efficient algorithms are available; otherwise, the numerical solution may become more difficult to obtain.

However, maximisation of expected revenues is not appropriate for the determination of a replicating portfolio, since risk in the sense of margin (or surplus) variations is not reflected. An alternative is to include a risk measure in the objective and to optimise a trade-off between risk and return

$$\min \gamma \varrho(z_0, \dots, z_{T+1}) - (1 - \gamma) \mathbb{E}[z_{T+1}]$$

where the weighting factor $\gamma \in [0, 1]$ depends on the decision-maker's preference. The function ρ denotes a multiperiod risk measure, ie, it is applied to all time steps to allow for a dynamic perspective of risk. This is different from a consideration of the revenues z_{T+1} only at the terminal stage, because the variations in the earlier values $z_0, \dots, z_T$ are also relevant for risk management.

Obviously, the specifically chosen function ρ should exhibit certain properties to justify its use as a “risk measure”. A simple requirement is that a large value of $\rho(z)$ indicates a higher risk than a small (or negative) value. Various authors have postulated many other properties that a risk measure should or should not have. For instance, widely accepted among practitioners are the so-called “coherent” risk measures for single-period problems introduced by Artzner *et al* (1999). For the multiperiod case the situation is more complex; a discussion would go far beyond the scope of this chapter. Mainly for technical reasons, we have selected a function ρ from the class of the so-called polyhedral risk measures, which are a subclass of coherent risk measures (Eichhorn and Römisch 2005).

Recall that linear optimisation problems are tractable by numerical solution methods. But, if the objective function incorporates a risk measure, it is no longer linear because risk measures are nonlinear by nature (Eichhorn and Römisch 2008). Common risk measures such as value-at-risk even lead to non-convex problems, which is the worst possible situation from an optimisation perspective, since we may not find the “best” solution (technically speaking, only a local optimum is attained instead of the global one). The nice feature of polyhedral risk measures is that they maintain the linearity structures of the optimisation problem although they are nonlinear functions. This is achieved by the introduction of additional variables and linear constraints that transform the nonlinear risk measure ρ into a linear objective.

Among several alternatives discussed in the literature, we have chosen here a risk measure that is based on the conditional value-at-risk for some probability level $\alpha \in (0, 1)$

$$\varrho(z) := \text{CVaR}_\alpha(\min\{z_0, \dots, z_{T+1}\}) \quad (8.10)$$

$\text{CVaR}_\alpha(x)$ is a widely accepted coherent risk measure. It can be interpreted as the mean of the distribution of x below the α -quantile, ie, it specifies the expected revenue in the worst $\alpha \times 100\%$ cases. Here we focus on the lowest revenues over time, since the possibility of not covering the costs for holding the non-maturing account should be minimised. The choice of this risk measure is also motivated by extensive tests of different alternatives with real data, where Equation 8.10 showed the best performance.

Complete optimisation problem

With this objective the complete stochastic optimisation reads as follows

$$\min y \text{CVaR}_\alpha(\min\{z_0, \dots, z_{T+1}\}) - (1 - y)\mathbb{E}[z_{T+1}]$$

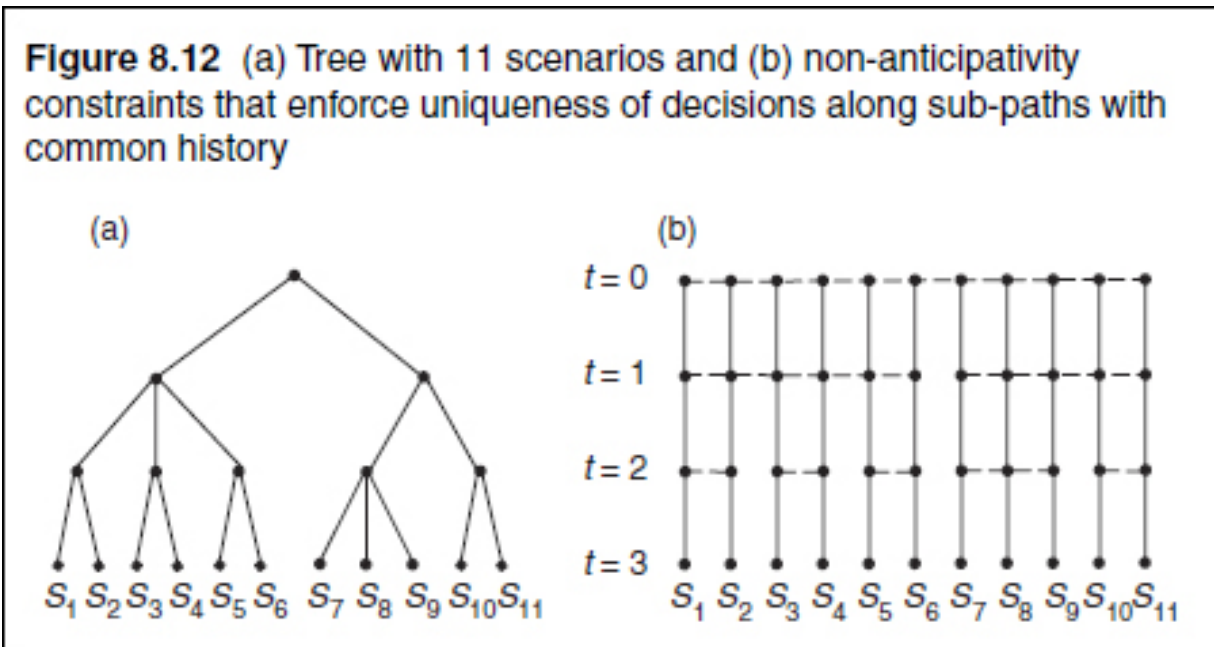
subject to the constraints in Equations 8.2–8.9

$$\begin{aligned} 0 &\leq x_t^{d,+} \leq \ell^{d,+}; \quad 0 \leq x_t^{d,-} \leq \ell^{d,-}, \quad t = 0, \dots, T, \quad \forall d \in \mathcal{D}^S \\ x_t^{d,+} &\geq 0, \quad x_t^{d,-} \geq 0 \text{ non-anticipative}, \quad t = 0, \dots, T, \quad \forall d \in \mathcal{D}^S \\ x_t^d &\geq 0 \text{ non-anticipative}, \quad t = 0, \dots, T, \quad \forall d \in \mathcal{D} \\ x_t^S &\in \mathbb{R}, \quad z_t \in \mathbb{R} \text{ non-anticipative}, \quad t = 0, \dots, T + 1 \end{aligned}$$

The constraints specified earlier must hold for all realisations ω_t of the stochastic process at time $t = 1, \dots, T$. Moreover, investment policies at any stage have to be found independently of the future outcomes of the uncertain problem-specific data (here interest rates, product rate and volume). In other words, decisions must not anticipate any information that becomes available only in the future. These requirements are expressed by the so-called “non-anticipativity constraints” in the last three lines of the optimisation problem. Formally, their meaning is that decisions at time t depend only on the history of random data $\omega_1, \dots, \omega_t$, not on future realisations $\omega_{t+1}, \dots, \omega_{T+1}$, since the latter are not yet known.

In practice non-anticipativity of decisions can be implemented as follows: the scenarios that represent possible realisations of the risk factor process are organised in the form of a non-recombining tree, as in the example in [Figure 8.12\(a\)](#). Suppose that all decision variables and constraints are duplicated for each scenario, which obviously leads to a large optimisation problem if many periods are taken into account. This is also the reason why we stressed above that the problem should remain linear; otherwise, the

numerical solution can become greatly challenging. The structure of the resulting optimisation problem is illustrated in Figure 8.12(b): here each node is equivalent to a set of decision variables and constraints for the corresponding scenario and time point.



Although we obtain in this way different variables for the decisions in scenarios $s_1, \dots, s_{11}$, it is logical that the decisions at time $t = 0$ must be identical because they all refer to the root node of the original scenario tree. In other words, the decision made today must be unique, because it is not known which evolution will be realised. Therefore, the decision variables at time $t = 0$ are linked by equality constraints, which are illustrated by the horizontal lines in Figure 8.12(b). Analogously, scenarios $s_1, \dots, s_6$ are identical up to time $t = 1$, and the corresponding decision variables must be linked again by additional constraints. The same applies to the decision variables along all paths with a common history in the scenario tree up to a certain node. In this way, a decision at a particular node is based only on the (discrete) probability distribution of the future risk factors (implied here by the sub-paths that branch from this node), not on the foresight of their future realisations.

Typically, the scenario tree branches several times at each node, which leads to an exponential growth in the problem size when the number of stages increases. In lattice-based models, which are applied in option pricing, the

problem size can be controlled by construction of a recombining tree (at least for a small number of underlying risk factors). This is not possible here because the tree of a stochastic optimisation problem reflects not only realisations of the risk factor process but also the history of decisions. Therefore, limitations in the available computational resources (time and memory) restrict the number of time stages at which the tree can branch. This problem is addressed in [Appendix A](#) in the context of the particular scenario-generation method used here.

RISK FACTOR MODELS

Now we describe some comparatively simple approaches for modelling the evolution of the relevant uncertain data: market rates, product rate and volume. Nevertheless, they perform sufficiently well, as results from practical applications indicate, which we demonstrate later. It must be emphasised that our optimisation framework is not restricted to the specific models below. Instead, a bank should choose models that fit best to its specific situation, ie, the relevant market (currency) or product characteristics, which might differ from the assumptions made here. In contrast to valuation models, where risk-neutral probabilities are usually used, we have to base our models on the statistical probability measure because they are applied to decision-making; the use of the risk-neutral probability measure would lead to biased policies (Geyer *et al* 2010).

Market rates

The market rate model describes the evolution of the rates of the fixed-income instruments that are taken into account for the construction of the replicating portfolio. It should reflect the essential characteristics of interest rates such as the following.

- **Mean reversion:** interest rates cannot rise indefinitely like stock prices, but fluctuate within a limited range with a tendency to revert to a long-term mean. This must be taken into consideration by the choice of appropriate stochastic processes.
- **Multiple factors:** the yield curve can show a variety of different shapes in reality; also, changes in the yield curve may have a complex pattern. For instance, the rates of short and long maturities can move in different

directions and the yield curve may become more or less steep. On the other hand, the rates of maturities close to one another are more highly correlated.

By means of a principal component analysis it is revealed that most of the variation in the term structure can be explained by two or three factors. These may be associated with changes in level, steepness and curvature of the term structure. Inspired by the approach of Reimers and Zerbs (1999), we model the evolution of the latent factors by mean-reverting processes. Interest rates are obtained using the sensitivities with respect to these factors that are derived from a principal component analysis. We restrict ourselves to the consideration of two factors; this takes into account more than 90% of the variability of the term structure. The dynamics of the two factors η_1 and η_2 are described by the following stochastic processes in continuous time

$$d\eta_j = -\alpha_j \eta_j dt + \sigma_j d\omega_j, \quad j = 1, 2 \quad (8.11)$$

The parameter α_j specifies the speed of adjustment from the current value of factor j to the long-term mean zero, σ_j measures the volatility of the factor and $d\omega_j$ represents a random fluctuation, ie, the variation of a Wiener process (or Brownian motion). This specification implies that η_1 and η_2 are normally distributed. Let the term structure be defined by a set of n rates r_i , $i = 1, \dots, n$, and denote by r_i^∞ the long-term mean of the corresponding rate. Then the interest rates are derived from the factors by the relation

$$r_i = r_i^\infty + \sum_{j=1}^2 \beta_{ij} \eta_j, \quad i = 1, \dots, n \quad (8.12)$$

where β_{ij} is the sensitivity of rate i with respect to changes in factor j . The sensitivities are determined by principal component analysis (they correspond to the eigenvectors of the covariance matrix of observed interest changes; for details see Reimers and Zerbs (1999)).

According to Equation 8.12, interest rates depend linearly on factors; thus, they are also normally distributed. In this respect our approach differs from Reimers and Zerbs (1999), who use another type of relation, which leads to lognormally distributed rates in order to avoid negative values. Our

specification allows the consideration of negative interest rates, as observed, eg, in Switzerland as a response to the European debt crisis after 2011.

Product rates

A simplistic approach for a product rate model may be based on the assumption of an equilibrium relation between the client rate c_t at time t and a market rate r_t of the form

$$c_t = a + br_t$$

The coefficient b defines the extent to which changes in the market rate are passed to clients, while the constant a reflects any bank administrative costs. This specification implies that there is a unique equilibrium product rate for each level of the market rate; the product rate responds instantaneously to changes in the market rate and the relation between them is linear. However, empirically observed product rates typically follow changes in market rates with some lag. Hawkins and Arnold (2000) propose a characterisation of the delayed response that is inspired by models in physics which describe the anelastic response of a material to an applied stress; they propose that a wide range of dynamics can be described by the differential relationship

$$\frac{dc}{dt} + \eta(c - a) = b_U \frac{dr}{dt} + \eta b_R r$$

Here the adjustment of the client rate to the market rate is determined by the two components on the right-hand side: an immediate reaction of the product rate to a change in the market rate (controlled by b_U) and an asymptotic long-term adjustment to the current level of the market rate (controlled by b_R). The parameter η denotes the rate at which the product rate moves towards the equilibrium level, and a is the long-term spread between both rates. The estimation of the model parameters is based on observations in discrete time, and therefore a discretisation of the above equation is required. A simple approach is the Euler discretisation

$$c_t = \eta a + c_{t-1} + b_U(r_t - r_{t-1}) + \eta(b_R r_{t-1} - c_{t-1})$$

which is a single-step method, as the new value is lagged only on the previous one. To better reflect the tardiness of the client rate and its delayed

reaction, it can be useful to involve more temporal lags. In computational methods for the solution of differential equations it is also common to apply a multistep approach where some information from previous steps is preserved in order to increase the accuracy. An example is the so-called Adams–Bashford three-step method

$$c_t = \eta a + c_{t-1} + b_U(r_t - r_{t-1}) + \frac{23}{12}\eta(b_R r_{t-1} - c_{t-1}) - \frac{16}{12}\eta(b_R r_{t-2} - c_{t-2}) + \frac{5}{12}\eta(b_R r_{t-3} - c_{t-3}) \quad (8.13)$$

which involves three lags. The reason for the choice of the coefficients in Equation 8.13 is a good approximation of the original differential equation, although in a general approach they may also be fitted to the data and more lags can be included. More insights into the motivation for this approach and other more general specifications, which are beyond the scope of this chapter, are given in detail by Hawkins and Arnold (2000). In order to keep the presentation of our dynamic replication approach simple, we apply the model in Equation 8.13 with the three-month rate as a representative market rate for the case study examples presented below.

However, different and more complex models have appeared in the literature to reflect the observed characteristics of typical non-maturing retail products, such as asymmetric adjustment (when market rates go up, banks are usually more reluctant to adjust deposit rates, while a decrease in the interest level is passed on much more quickly to clients), adjustments in discrete steps or other types of nonlinear dependencies on market rates, etc. A detailed discussion of the characteristics of product rates of typical deposit products can be found, for example, in Paraschiv (2013), where different modelling approaches and their corresponding estimation techniques are also introduced. These are tested using data from real products from various banks. The empirical results show clear differences in the pricing policies between individual institutions; these can be attributed to characteristics such as bank size or dependency on retail business. Therefore, the applied model must always be adapted to the specific situation and tested using the data from the particular bank and product.

Volume model

The model for the product volume follows Kalkbrener and Willing (2004). They assume that volumes $v_t = f(t) + \xi_t$ are defined as the sum of a linear

trend

$$f(t) = a + bt$$

and an Ornstein–Uhlenbeck process

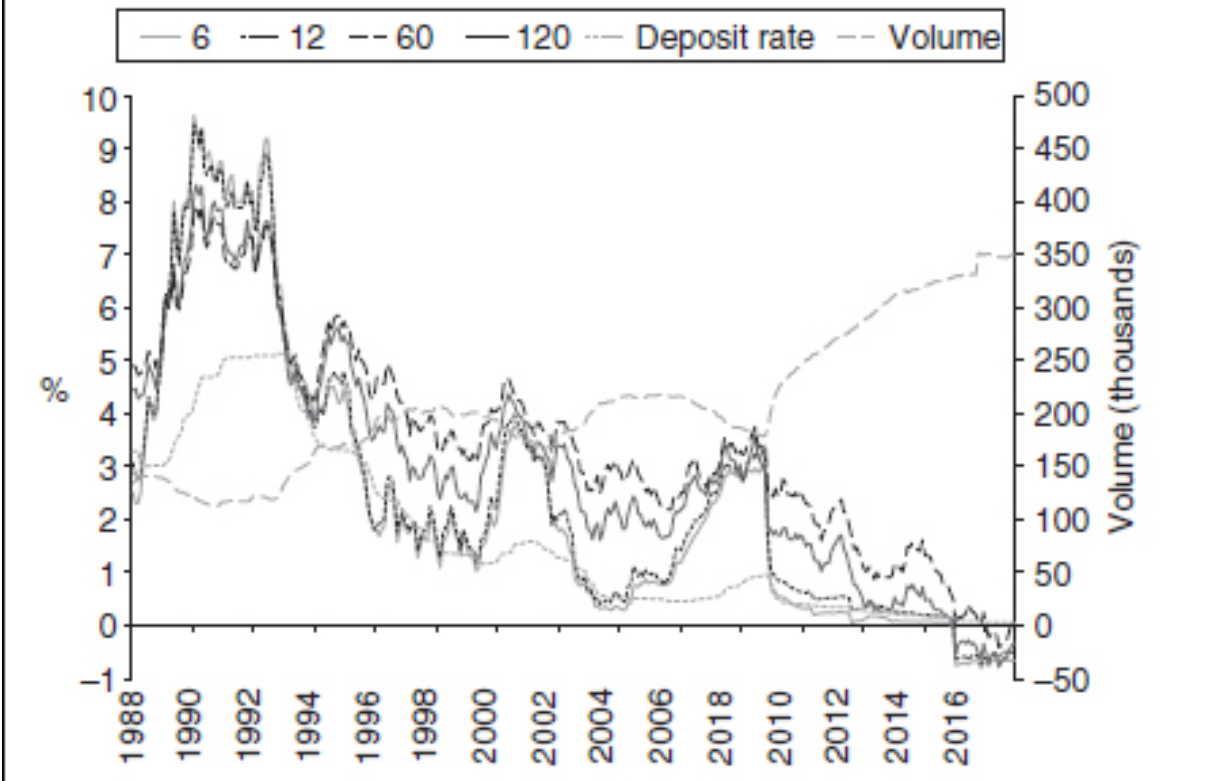
$$d\xi = -\alpha_V dt + \sigma_v d\omega_v \quad (8.14)$$

that reflects fluctuations around the long-term trend. The dependency on interest rates is taken into account by correlations ρ_1 and ρ_2 between ω_V and the Wiener processes ω_1 and ω_2 of the term structure model.

CASE STUDY

The dynamic replication approach based on the multistage stochastic optimisation model is tested with data for savings deposits from the statistical database of the Swiss National Bank (SNB).¹ We use the total volume of savings deposits of all banks in Switzerland and Liechtenstein from the monetary aggregate statistics of the SNB, together with the average of published deposit rates. Market rates for the Swiss franc are obtained from Thomson Reuters. This data is shown in [Figure 8.13](#) for selected maturities. The volume shows a positive long-term trend with fluctuations that are clearly negatively correlated with interest rates. The deposit rates exhibits the typical pattern of lagged adjustments. Note that market rates for maturities up to five years fell below zero at the beginning of 2015, when the SNB abandoned the euro peg, while deposit rates remained positive, which further increased the pressure on banks' margins on deposits. The advantage of using the aggregated SNB data is that this is available for a long sample period, allowing model performance to be tested in different interest rate regimes (high, low and negative levels).

Figure 8.13 CHF market rates, deposit rate and volume for the case study



Ideally, the test of the optimisation model and the static replication approach as a benchmark should be performed “out-of-sample”, ie, a decision regarding the update of a portfolio is determined using only the information available at the time of making the decision. In other words, for the estimation of the models for market rates, client rate and volume, only historical data from before the time of a portfolio adjustment was used, which requires that we also have historical data before the beginning of the actual test period.

We start with the case study in January 1998 and estimate the risk factor models with historical data from the previous 10 years; market rates for earlier periods were not available. An initial portfolio was given that consisted of 40% 12-month, 30% five-year and 30% ten-year tranches according to the construction rule for static replication (this corresponds to the optimal “in-sample” portfolio determined for the whole period from 1998 to 2016). A first decision was determined by the multistage stochastic optimisation model described earlier, and the portfolio was updated by

implementation of the resulting transactions. In the next step the portfolio was optimised for the market rates, deposit rate and volume for February 1998, and so on. After 12 months the risk factor models were re-estimated by replacing the oldest observations in the historical sample with the new available information (a rolling ten-year time window). In this way, a new (updated) investment policy was determined for each month up to December 2016.

The set of maturities used by the optimisation model for transactions was $D^S = \{6, 12, 24, 36, 48, 60, 84, 120\}$ (values given in months); transaction costs were not taken into account. Squaring of positions was only allowed in order to compensate a loss in volume by imposing the constraint in Equation 8.6. In order to observe strictly risk-averse policies, the weight λ of the risk term in the objective function was set to 100%. A scenario tree was generated over a five-year horizon with branches in the root node and after 6, 24 and 60 months (see the appendixes for details).

A static replicating portfolio served as the benchmark with time buckets of six and twelve months and five and ten years. We chose the rebalancing portfolio approach for the correction of volume changes that turned out to be more efficient in our previous tests than the variant with compensation of PV changes. We tested two alternatives to determine the portfolio composition: using a historical sample; and fitting the weights to a scenario of future data. For the historical estimation, we used the market and product data from the previous ten years and determined the portfolio with the smallest margin variation during the sample period. However, during the first ten years of the study (1998–2007) we could not use historical data for the estimation since the market rates from the previous ten years were required for initialisation of the moving averages (earlier data was not available). Therefore, decisions up to the end of 2007 had to be made with perfect knowledge of future data (in-sample test), which might have biased the results to some extent in favour of the static approach. Out-of-sample testing was performed from 2008 onwards, and the resulting weights remained constant for the next 12 months. The estimation was then updated (again for a rolling ten-year time window), and the portfolio rebalanced.

It can be argued that a historical analysis is inappropriate for the current market environment. Therefore, it has become popular for practitioners to fit the weights of the replicating portfolio to a scenario (forecast) of future market rates, client rates and volumes. However, the results then may depend

to a large extent on the choice of such a forecast. Therefore, we chose the following design of an objective approach for a future-based estimation: 1,000 scenarios were generated over a ten-year horizon with the risk factor models introduced earlier (again out-of-sample testing with yearly parameter updates). For each scenario path the means and variances of the margin were determined for all feasible portfolio compositions. Then, the combination was selected for which the sum of variances over all scenarios becomes minimal. It turned out that, particularly towards the end of the test period, ie, in an environment with significantly negative market rates, the average margin of the lowest-volatility portfolio also becomes negative for a large number of scenarios. This means that a loss must be expected from the implementation of this “minimum risk” policy. We therefore excluded all the portfolio combinations where a negative mean occurred in more than 5% of the scenarios, which led to better risk–return characteristics in the overall *ex post* evaluation.

Figure 8.14 shows the resulting evolution of the margin for the dynamic replication, based on the multistage stochastic programming model, and compares it with the two static approaches. Both are clearly dominated by the dynamic approach. However, the future-based estimation of the weights shows a slight improvement over the use of historical data. The margins of the static portfolios follow the drop in market rates in 2002 and, more distinctly, the drop in 2008, when tranches with initially high rates in the moving averages subsequently expired. The outperformance by the dynamic approach results from the greater flexibility of reinvestment policies, since tranches do not have to be distributed uniformly within time buckets.

Figure 8.14 Margin evolution for the dynamic replication based on the stochastic optimisation model versus the static replication

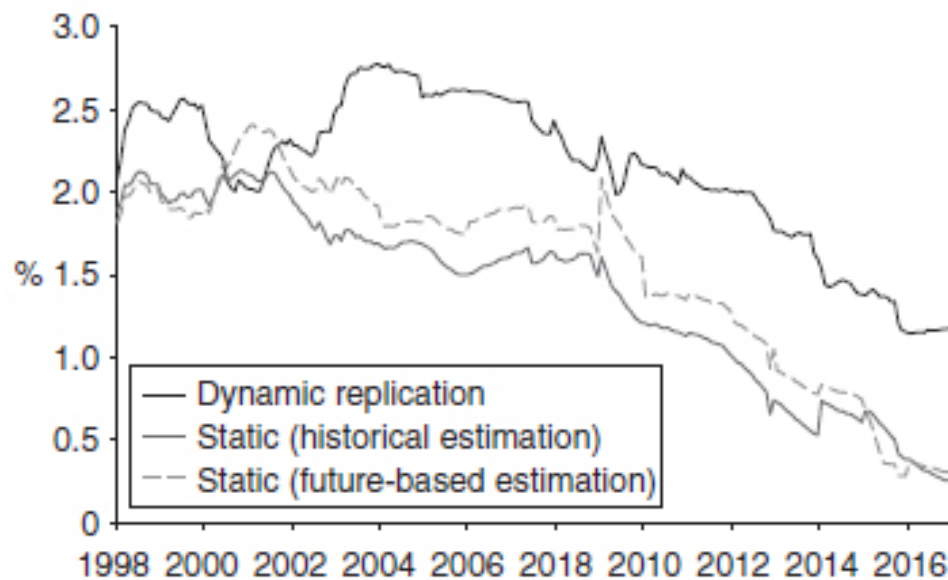


Figure 8.15 Composition of the dynamic replicating portfolio over time

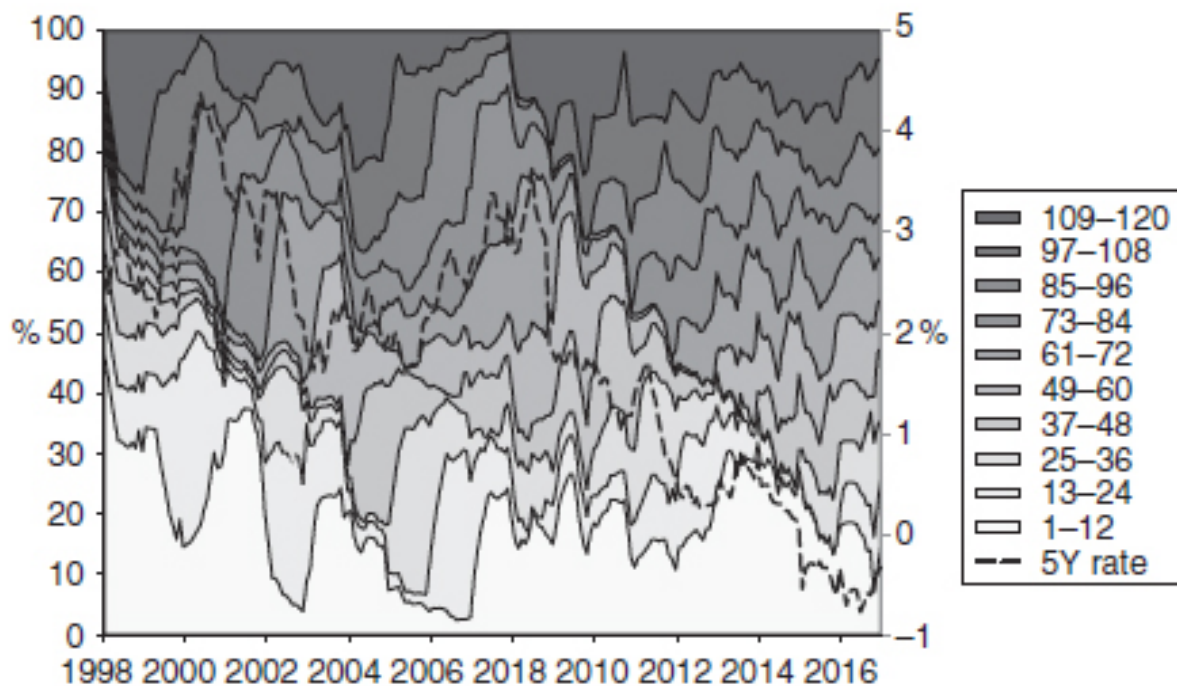


Figure 8.15 illustrates the evolution of the portfolio composition, namely the fraction of tranches with certain time to maturity. The 5Y rate is also shown as the representative level of interest rates. The model extends the share of longer maturities when market rates are above average and start falling, in order to lock-in their high value, while the fraction of short and medium maturities is reduced. This flexibility helps to stabilise the margin for a longer time. On the other hand, there is a higher concentration on medium maturities when increasing rates are expected. Table 8.3 summarises the average margins and their standard deviations for the different approaches over the whole 228-month test period. In addition, the improvement of the dynamic replication compared with the two static benchmarks is reported. It shows that an extension of the average margin could be achieved and its volatility simultaneously reduced, which implies that the resulting dynamic strategy not only is more efficient but gives a better replication.

Table 8.3 Margin characteristics for the optimisation model and the static benchmark

	Average margin (%)	Diff (bp)	Standard deviation (%)	Diff (bp)
Dynamic replication (MSP)	2.14	—	0.453	—
Static (historical estimation)	1.39	75.7	0.547	−9.5
Static (future-based estimation)	1.55	59.2	0.577	−12.4

Notes: MSP denotes multistage stochastic programming.

CONCLUSIONS

The problem of calculating a replicating portfolio for non-maturing positions is highly relevant for banks, particularly for those with significant retail business. Common approaches based on static investment (or funding) rules have several shortcomings and must be applied with care. Often the economic effects of volume fluctuations are ignored, which leads to highly volatile margins. This particularly applies to the approach where PV effects are charged to the opportunity rate, which shows a significantly larger margin volatility than the rebalancing portfolio method. As these margin fluctuations

are caused by volume changes, the resulting correction payments may be taken into account in estimating the portfolio weights that minimise the standard deviation of the margin, but this generally leads to a smaller percentage of the more interest sensitive longer maturities, and hence to a lower opportunity rate. The reason for the popularity of this approach is therefore incomprehensible given the problems that occur in the presence of volume fluctuations. A specific problem in the low interest rate environment at the time of writing is that the levels of the moving averages cannot be realised in order for the increased volumes to be invested. The approach appears to be inappropriate for the application for which it is designed: the replication of non-maturing products. The shortcomings are less pronounced for the rebalancing portfolio method. However, our study implies that it should be combined with a future-based estimation of the weights.

Stochastic approaches take the inherent options that are generally exercised to the banks' disadvantage into account more appropriately. These are based on models for the evolution of uncertain data (market rates, product rates and volumes) and consider the dependencies between them. Even simple models, which can be easily calibrated and communicated, are useful for this purpose. Investment (or funding) policies are then determined for many future scenarios. The resulting portfolios are frequently adjusted to the latest market observations and information on client behaviour. In this way dynamic investment rules are derived. While the focus of stochastic valuation models for non-maturing products proposed in the literature is clearly on interest rate risk management, the dynamic replicating portfolio approach also allows an extension to liquidity risk management. On the one hand, it generates a portfolio that can be implemented directly with standard fixed-maturity instruments and provides a stable margin. On the other hand, we may also define liquidity constraints for the optimisation. The approach also uses the real-world probability measure, which is appropriate for liquidity risk.

APPENDIX A: SCENARIO GENERATION

In summary, the relevant risk factors are driven by three stochastic processes defined in Equations 8.11 and 8.14. For the scenario generation at time steps $t = 1, \dots, T$ we consider the corresponding Euler discretisation for the step size δ , which here is one month (ie, $\delta = \frac{1}{12}$)

$$\tilde{\eta}_{j,t+1} = \tilde{\eta}_{j,t}(1 - \alpha_j \delta) + \sigma_j \Delta \tilde{\omega}_{j,t}, \quad j = 1, 2 \quad (8.15)$$

$$\tilde{\xi}_{t+1} = \tilde{\xi}_t(1 - \alpha_v \delta) + \sigma_v \Delta \tilde{\omega}_{v,t} \quad (8.16)$$

$$\Delta \tilde{\omega}_{v,t} = \rho_1 \Delta \tilde{\omega}_{1,t} + \rho_2 \Delta \tilde{\omega}_{2,t} + \sqrt{1 - \rho_1^2 - \rho_2^2} \Delta \tilde{\omega}_{3,t} \quad (8.17)$$

$$\Delta \tilde{\omega}_{j,t} \sim N(0, \delta), \quad j = 1, 2, 3 \quad (8.18)$$

Note that the tilde over the above stochastic variables is used to distinguish the discrete-time approximation from the original continuous-time processes introduced earlier. At first glance an obvious way of generating a scenario tree might be to simulate the Brownian motions $\Delta \tilde{\omega}_{j,t}$, $j = 1, 2, 3$, at times $t = 0, \dots, T$ and to update the values of the risk factors according to Equations 8.15–8.18. The initial values for $\tilde{\eta}_{1,0}$, $\tilde{\eta}_{2,0}$ and $\tilde{\xi}_0$ (which can be derived from the observed interest rates and volume after calibration of the corresponding models) are assigned to the unique root node of the scenario tree. Then n_1 samples for each component of the three-dimensional Brownian motion are chosen, which results in n_1 different outcomes for $\tilde{\eta}_{1,1}$, $\tilde{\eta}_{2,1}$ and $\tilde{\xi}_1$. The latter are used to construct the successor nodes of the root at time $t = 1$. Now the procedure is repeated for each of the n_1 nodes: a sample of size n_2 for the realisations of the three Brownian motions is generated, the risk factors are updated and a corresponding number of successor nodes is obtained for which the same procedure is applied until the terminal stage is reached. As a consequence the total number of nodes at stage $t > 0$ is $N_t := n_1 \times \dots \times n_t$.

Unfortunately, Monte Carlo simulation requires large sample sizes to keep the associated sampling error low (in the sense of small confidence intervals and robustness of the solution with respect to a “contamination” of the sample, eg, by using a different initial value for the random number generator). For instance, if for each node 10 samples are chosen to generate its successors (which is still not a sufficient sample size for crude Monte Carlo simulation), the resulting scenario tree for a planning horizon of six months has one million scenarios. On the other hand, the manageable size of a multistage problem is only around a few hundred or thousand scenarios. Therefore, a wide range of “smarter” methods than (naive) Monte Carlo simulation has been developed for stochastic programming; often they are adapted to very specific characteristics of the underlying application. A comprehensive overview goes far beyond the scope of this chapter; the interested reader is referred to Shapiro *et al* (2009) as a starting point.

Approximation with Platonic solids

In Frauendorfer and Schürle (2007) an earlier version of the model for dynamic replication used a relatively simple but effective approach for the discretisation of the continuous joint distribution of the risk factors, conditional on the outcomes in a given node. The multivariate normal distribution implied by the underlying stochastic models was approximated by a multinomial distribution in combination with some transformations to match its expectation and (co)variances.

In the new model presented in this chapter we follow an idea that was suggested originally by McCarthy and Webber (2001) for pricing applications. Here the discretisation points are derived from the vertices and midpoints of Platonic solids, which are convex polyhedrons such as the tetrahedron, cube, octahedron and icosahedron. Compared with the discrete outcomes used by Frauendorfer and Schürle (2007) in the multinomial approximation, the vertices of these solids have greater distances from the midpoint (which represents the expectation in the approximated distribution). This allows more extreme events to be taken into account. Furthermore, it introduces the flexibility to match the kurtosis of the resulting discrete distribution to the empirically observed one. In this way the “fat tails” evident in financial data can be better reproduced.

For instance, in terms of rectangular Cartesian coordinates, an icosahedron with edges of length 2 has the set of vertices

$$\mathcal{V} = \{(0, \pm\psi, \pm 1), (\pm 1, 0, \pm\psi), (\pm\psi, \pm 1, 0)\}$$

where $\psi = 2/(\sqrt{5} - 1) \approx 1.618$ is the golden ratio. Consider now a vector of three uncorrelated standard normally distributed random variables $(\omega_1, \omega_2, \omega_3)$. Suppose we want to use the points in the set $\mathcal{A} = \mathcal{V} \cup \{(0, 0, 0)\}$ as discretisation points for the approximation of its distribution. Together with some scaling factor s , this leads to the outcomes

$$(\hat{\omega}_1, \hat{\omega}_2, \hat{\omega}_3) = s \cdot a, \quad a \in \mathcal{A} \quad (8.19)$$

Let p_0 be the probability of $(0, 0, 0)$ and $p = \frac{1}{12}(1 - p_0)$ be that of the other points. Obviously, the expectation of the discrete distribution obtained in this way is zero for all components. According to McCarthy and Webber (2001), the value of the scaling factor s required to match the variance is given by

$$s = \sqrt{\frac{3}{(1 - p_0)(1 + \psi^2)}}$$

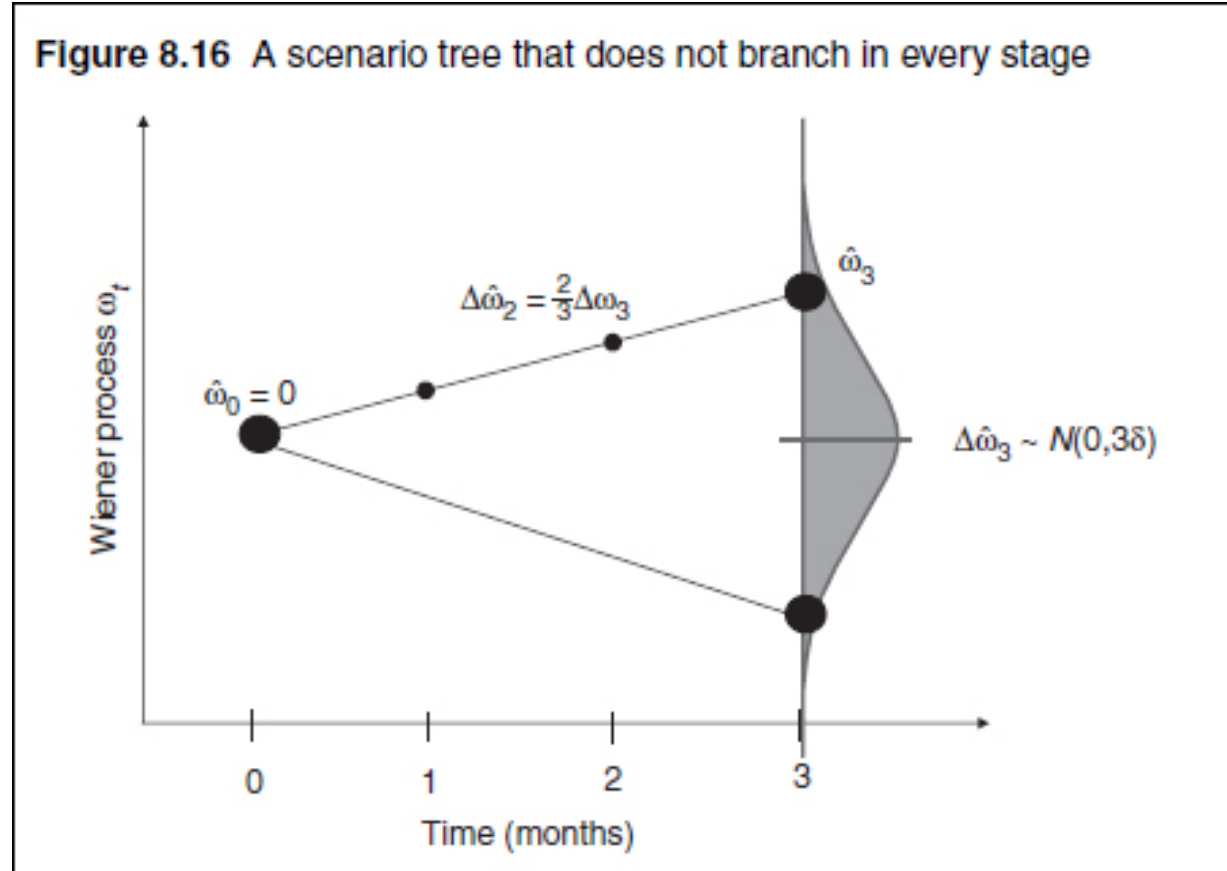
If extreme outcomes should also be considered in the approximation, we can control their distance from the midpoint by controlling s , which itself depends on the choice of p_0 . For example, with $p_0 = 0.5$ we have $s = 1.29$. While expectation and variance are matched, for $p_0 = 0.5$ the excess kurtosis in each factor becomes 0.6. This comes close to the empirical kurtosis of the error terms that are derived from the estimation of the time-discrete risk factor processes (Equations 8.15 and 8.16). The multiplication by the inner circle of the icosahedron, $R = \sqrt{3}(3 + \sqrt{5})/6 \approx 1.51$, shows that all points obtained by Equation 8.19 are at least 1.95 standard deviations away from the origin. In other words, there is a 95% probability that the possible outcomes of the original (three-dimensional standard normal) distribution are contained in the polyhedron.

To summarise the scenario tree generation procedure: in the previous description we have to replace the term “samples” (obtained from Monte Carlo simulation) by “outcomes” (obtained from the approximation with the method illustrated here). The probability of a node is the product of the probabilities of all its predecessors up to the root node multiplied by the probability of the new outcome (ie, p if obtained from a vertex; p_0 if obtained from the midpoint). An icosahedron has 12 vertices. Together with the midpoint, this leads to 13 branches in each node. To reduce the growth of the tree, in later stages ($t > 0$) we apply bodies with fewer vertices, eg, the tetrahedron (four vertices). The parameters p_0 and s must be adjusted correspondingly to match the moments of the joint risk factor distribution.

Reduction of tree growth

Even if we apply a discretisation method with a relatively moderate branching factor, the scenario tree will still grow much too fast if we perform the approximation in each time step for a period of one month. For example, with only two branches per node we can hardly achieve a planning horizon of one year, since $2^{12} = 4,096$ scenarios would result. In some applications this problem might be tackled by choosing larger time steps. This is not possible here since, to determine the replicating portfolio based on standard contracts each month, a decision on the reinvestment of maturing tranches, corrected by the volume change, has to be made. As a consequence, the tree

cannot branch at each stage to reduce its growth. Therefore, we define a subset of time points $T \subseteq \{0, \dots, T\}$ where the tree branches. Realisations of risk factors at these time points are obtained using the discretisation procedure; outcomes in stages $t \notin T$ are interpolated.



For instance, assume that the tree should branch at the root node and then after three months. By definition, the variance of a Wiener process $\omega(t)$ is proportional to time, which is measured here in years, ie, for a time interval of one year the variance is 1.0. Over an interval of three months the Wiener processes in Equation 8.18 are normally distributed random variables with $\omega_j(0.25) \sim N(0, 0.25), j = 1, 2, 3$. Their joint distribution is now discretised using the points from the set defined in Equation 8.19 to obtain an approximation of a three-dimensional standard normal distribution; then the outcomes are multiplied by the standard deviation $\sqrt{0.25} = 0.5$. In this way we obtain a set of nodes at stage $t = 3$ of the scenario tree. Let $\hat{\omega}_{j,3}$ be a realisation of $\omega_j(0.25), j = 1, 2, 3$, in a certain node at this stage. Then a path with intermediate nodes is constructed from the root to this node and the

values for component j of the Wiener process are interpolated, ie, $\omega_{j,1} = \frac{1}{3} \omega_{j,3}$ and $\omega_{j,2} = \frac{2}{3} \omega_{j,3}$ (Figure 8.16). From these interpolated points the monthly changes $(\Delta \omega_{1,t}, \Delta \omega_{2,t}, \Delta \omega_{3,t})$ are derived for $t = 1, 2, 3$, which are used to obtain scenarios for the risk factors of the market rate and volume model by Equations 8.15 and 8.16. The procedure is then repeated analogously for the nodes at time 3, starting with a discretisation of the multi-dimensional Wiener process at the subsequent branching stage.

Decision rules

The use of a sparse tree has the disadvantage that the predictability becomes too high, ie, in nodes with only one successor the outcomes of the future risk factors are known with certainty up to the stage with the next branch. This contradicts the requirement that decisions should be taken without perfect foresight. In particular, the model might decide to borrow money at low (short-term) rates and invest it at higher (long-term) rates. Without any additional constraints (eg, the restriction of total nominal amounts $x_t^d, t = 0, \dots, T, d \in D$) to non-negative values, the optimisation problem could even become unbounded. But in the presence of such constraints the resulting decisions would also be biased. As a remedy for this shortcoming we require that decisions on the portfolio composition in stages $t \notin T$ (where the tree does not branch) are made in a previous stage $t' < t$ with $t' \in T$, where the tree branches. Then these decisions are based on the distribution of future risk factors instead of foresight of their known values.

Denote by $d_1 < d_2 < \dots < d_m$ the maturities in D^S . Then the time buckets are defined in the following way: the first bucket consists of positions with remaining time to maturity up to d_1 , the second bucket of positions with time to maturity between $d_1 + 1$ and d_2 , etc. Further, define by $[t] := \max\{\tau \in T \mid t \geq \tau\}$ an operator that assigns to a time point t the index τ of the immediate predecessor stage where the tree branches. The volume of the individual time buckets in stages $t \notin T$ is determined by the constraints

$$\begin{aligned} \sum_{\delta=d_{i-1}+1}^{d_i} x_t^\delta &= y_{[t]}^{t,0,i} + \sum_{\tau=0}^{t-[t]-1} y_{[t]}^{t,1,i,\tau} \tilde{\eta}_{1,t-\tau} \\ &+ \sum_{\tau=0}^{t-[t]-1} y_{[t]}^{t,2,i,\tau} \tilde{\eta}_{2,t-\tau} + \sum_{\tau=0}^{t-[t]-1} y_{[t]}^{t,3,i,\tau} \tilde{\xi}_{t-\tau}, \quad i = 2, \dots, m \end{aligned}$$

The right-hand side defines a “decision rule” that models the volume of a time bucket as a linear function of risk factors; the weights of the latter are additional decision variables that “technically” belong to a stage $t \in T$. In other words, a bucket i at time $t \in T$ depends linearly on the history of observations of all three risk factors after the last branch of the tree up to now, ie, over the time interval $[t] + 1$ and t . The decisions $y_{[t] + 1, i}^{t, j, i}$ on the contribution of the outcome of the j th factor at time $\tau \in \{[t] + 1, \dots, t\}$ have already been made at a previous time point, denoted by the index $[t]$, where realisations of uncertain data have not yet been revealed. In this way the portfolio composition is determined without knowing the future and depends on the distribution of the two factors that drive the evolution of the whole term structure, which can be associated with level and spread, plus the volume factor.

As outlined above, the continuous distribution of the risk factors is approximated by a discrete one using the points given by Equation 8.19. The latter are vertices of a convex set that contains 95% of the possible realisations of the original distribution (or any other probability, depending on the choice of p_0), ie, 95% of the possible outcomes are convex combinations of the vertices with positive weights. The constraint above defines a linear transformation from risk factors to portfolio compositions; therefore, the linear constraints in Equations 8.2–8.5 are also observed in 95% of the cases.

APPENDIX B: FINITE STATE SPACE REPRESENTATION

Earlier the optimisation model was presented on a more conceptual level. It is now specified in a finite state-space formulation taking into account the modifications discussed in [Appendix A](#). The scenario generation provides a tree with node set $N = \{0, 1, \dots, N\}$. The levels of the tree correspond to decision stages. N_t denotes the set of nodes at level $t = 0, \dots, T + 1$; the last level N_{T+1} contains the leaves of the tree and can be identified with the scenario paths. The tree structure represents the process showing how information on the realisation of risk factors is revealed over time. Relations between nodes are specified by the following notation: a node n at time t has the unique predecessor $n_{(\tau)}$ at the earlier stage $t - \tau$. By convention, the root

node is denoted by 0 and represents the present state. Each node $n \in N$ has a probability $\pi_n \geq 0$ with

$$\sum_{n \in \mathcal{N}_t} \pi_n = 1$$

for all points in time t .

Since decisions are related to nodes, in the notation for decision and state variables we now replace the time index by a node index, eg, we denote by $xn^{d,+}$, $xn^{d,-}$ and x_n^d the investment and borrowing transactions as well as the total amount with time to maturity d in a node $n \in N$. The same applies to realisations of uncertain coefficients. Finally, we introduce some new variables: u_0 , $u_{n,1}$, $u_{n,2}$ and $u_{n,3}$. These are required to model CVaR as a linear optimisation problem (for details see Eichhorn and Römisch 2008). Then the complete linear optimisation problem reads

$$\min y \left(u_0 + \frac{1}{\alpha} \sum_{n \in \mathcal{N}_{T+1}} \pi_n u_{n,2} \right) - (1 - y) \sum_{n \in \mathcal{N}_{T+1}} \pi_n z_n$$

subject to

$$\begin{aligned}
x_n^d &= x_{n(1)}^{d+1} + x_n^{d,+} - x_n^{d,-}, & n \in \mathcal{N} \setminus \mathcal{N}_{T+1}, & \forall d \in \mathcal{D}^S \\
x_n^d &= x_{n(1)}^{d+1}, & n \in \mathcal{N} \setminus \mathcal{N}_{T+1}, & \forall d \notin \mathcal{D}^S \\
\sum_{d \in \mathcal{D}} x_n^d &= v_n, & n \in \mathcal{N} \setminus \mathcal{N}_{T+1} \\
w_i^1 v_n &\leq \sum_{d \in \mathcal{D}} x_n^d \leq w_i^u v_n, & n \in \mathcal{N} \setminus \mathcal{N}_{T+1}, i = 1, \dots, k \\
\sum_{\delta=d_{i-1}+1}^{d_i} x_n^\delta &= y_{n(t-\lfloor t \rfloor)}^{t,0,i} + \sum_{\tau=0}^{t-\lfloor t \rfloor-1} y_{n(t-\lfloor t \rfloor)}^{t,1,i,\tau} \tilde{\eta}_{1,n(\tau)} + \sum_{\tau=0}^{t-\lfloor t \rfloor-1} y_{n(t-\lfloor t \rfloor)}^{t,2,i,\tau} \tilde{\eta}_{2,n(\tau)} \\
&\quad + \sum_{\tau=0}^{t-\lfloor t \rfloor-1} y_{n(t-\lfloor t \rfloor)}^{t,3,i,\tau} \tilde{\xi}_{n(\tau)}, & n \in \mathcal{N}_t, t \notin \mathcal{T} \\
x_n^{d,-} &\leq \max\{0, v_{n(1)} - v_n\}, & n \in \mathcal{N}_t, t \in \mathcal{T} \\
x_n^S &= \sum_{\tau=0}^t \sum_{d \in \mathcal{D}^S, d > \tau} r_{n(\tau)}^{d,+} x_{n(\tau)}^{d,+} - \sum_{\tau=0}^t \sum_{d \in \mathcal{D}^S, d > \tau} r_{n(\tau)}^{d,-} x_{n(\tau)}^{d,-} \\
&\quad + c f_{-1}^{t+2} - (c_n + \alpha_0) v_n, & n \in \mathcal{N}_t, t = 0, \dots, T \\
x_n^S &= \sum_{\tau=1}^T \sum_{d \in \mathcal{D}^S, d > \tau} \text{PV}_n^{T-\tau,d,+} x_{n(\tau)}^{d,+} \\
&\quad - \sum_{\tau=1}^T \sum_{d \in \mathcal{D}^S, d > \tau} \text{PV}_n^{T-\tau,d,-} x_{n(\tau)}^{d,-} + \text{pv}_{-1,n} - v_n, & n \in \mathcal{N}_{T+1} \\
z_{-1} &= 0 \\
z_n &= z_{n(1)} + x_n^S, & n \in \mathcal{N} \\
u_{0,2} - u_{0,3} &= 0 \\
u_{n,2} - u_{n,3} - u_{n(1),2} &= 0, & n \in \mathcal{N} \setminus \{0\} \\
u_{n,1} - u_{n,2} - u_0 &= z_n, & n \in \mathcal{N}
\end{aligned}$$

and

$$\begin{aligned}
0 &\leq x_n^{d,+} \leq \ell^{d,+}, \quad 0 \leq x_n^{d,-} \leq \ell^{d,-}, \quad x_n^d \in \mathbb{R}, & n \in \mathcal{N} \setminus \mathcal{N}_{T+1} \\
x_n^S &\in \mathbb{R}, \quad z_n \in \mathbb{R}, & n \in \mathcal{N} \\
u_{n,j} &\geq 0, & n \in \mathcal{N}; j = 1, \dots, 3 \\
u_0 &\in \mathbb{R} \\
y_n^{*,0,i} &\in \mathbb{R}, \quad y_n^{*,1,i,*} \in \mathbb{R}, \quad y_n^{*,2,i,*} \in \mathbb{R}, \quad y_n^{*,3,i,*} \in \mathbb{R}, \\
&& n \in \mathcal{N}_t, t \in \mathcal{T}, i = 2, \dots, m
\end{aligned}$$

¹ See <http://data.snb.ch>.

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Managing Mortgage Prepayment Risk on the Balance Sheet

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Mortgage loans form one of the largest asset classes on bank balance sheets. In most markets, they not only bear interest rate risk, but also introduce convexity and prepayment (model) risk to the balance sheet. In this chapter we focus on convexity and prepayment risk in the US mortgage market. Many of the results and ideas, however, can be applied to any mortgage market.

We first describe the different product forms that carry the prepayment risk that can be found on a balance sheet. Then we illustrate some empirical relations between prepayments and the market variables that are typically part of prepayment models. We briefly discuss how to compute value and sensitivities for mortgage products. We conclude by discussing how to incorporate these products into typical balance-sheet metrics.

BALANCE-SHEET PRODUCTS WITH PREPAYMENT RISK

In this section we discuss the different balance-sheet products that are affected by prepayment risk.

Mortgage loans

The most popular mortgage in the US market is the fixed-rate annuity mortgage. Common terms are 15 and 30 years and there are no prepayment penalties. This is the example that we shall follow. We assume a monthly compounding note rate, i . Given a maturity of N months and a loan amount, L , the monthly payment will be equal to

$$A = \frac{Li}{1 - (1 + i)^{-N}} \quad (9.1)$$

and the balance B_k at month k ($k = 0, \dots, N$) will be equal to

$$B_k = L \frac{1 - (1 + i)^{-(N-k)}}{1 - (1 + i)^{-N}}$$

Next we define a stochastic prepayment process that produces monthly prepayment rates equal to λ_k in month k . The prepayment rate is a function of loan characteristics and economic variables such as interest rates and home prices, as we shall discuss below.

Given the prepayment rates we can define a “survival” factor F_k that represents the probability that the loan still exists at month k

$$F_k = \prod_{j=1}^k (1 - \lambda_j)$$

Now given the observed path of prepayment rates the cashflow of a mortgage loan in month k can be written as

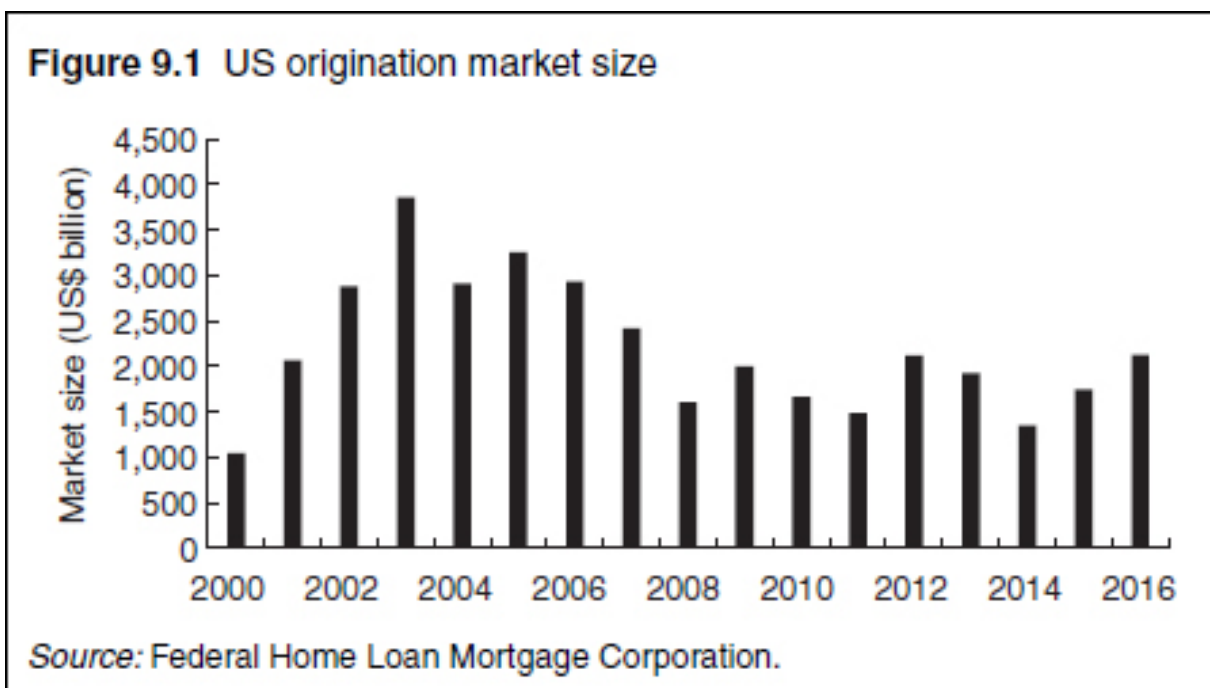
$$cf_k = F_{k-1}A + \lambda_k F_{k-1}B_k$$

Mortgage-backed securities

A mortgage-backed security (MBS) represents a pool of loans. A pool of fixed-rate loans has a coupon rate that is a multiple of 0.5%. The difference between the note rate on the loans in the pool and the coupon of the MBS consists of a servicing fee retained by the servicer of the loan and a guarantee fee retained by the agency providing the credit risk guarantee. Let us define the monthly servicing fee as s and the monthly guarantee fee as g . Then (assuming the pool has loans with identical coupon) the cashflow on an MBS pool can be written as

$$cf_k = F_{k-1}A + \lambda_k F_{k-1}B_k - (s + g)F_{k-1}B_{k-1}$$

In addition to securities for which the payments on a pool are passed through to the investor directly, there are collateralised mortgage obligation securities that pass the cashflows of a pool (or collection of pools) to different tranches based on more complicated cashflow rules. The latter are beyond the scope of this book.



Mortgage-servicing rights

Mortgage-servicing rights (MSRs) are created via the securitisation process. Once a pool of loans is sold to investors as part of an MBS, the servicer will collect cashflows and receive a servicing fee for that work. In exchange for the servicing fee, the servicer will incur various types of costs and receive other forms of income (eg, float income on tax and insurance balances). A good overview of servicing cashflows can be found in Aldrich *et al* (2001). For simplicity, in the formula below we shall focus on the most important cashflow: the servicing-fee income; many of the other cashflows can be approximated by adjusting the servicing fee. With this in mind, the cashflow in month k on an MSR can be written as

$$cf_k = sF_{k-1}B_{k-1}$$

Mortgage origination profits

Mortgage origination generates fee income and is not always captured in a bank's balance-sheet model. However, as we shall see, it is highly correlated with prepayment activity in the market. Origination income can be viewed as volume multiplied by net margin. Figure 9.1 shows the size of the US origination market over time. An originator's production volume will be its market share times the size of the market. Obviously the size of the market in a given year is highly correlated to the amount of prepayments of existing mortgages.

It is more difficult to look at the public data for an originator's margin, but we can look at the spread between the primary mortgage rate and the rate in the secondary market as a proxy that is highly correlated to the size of the margin.

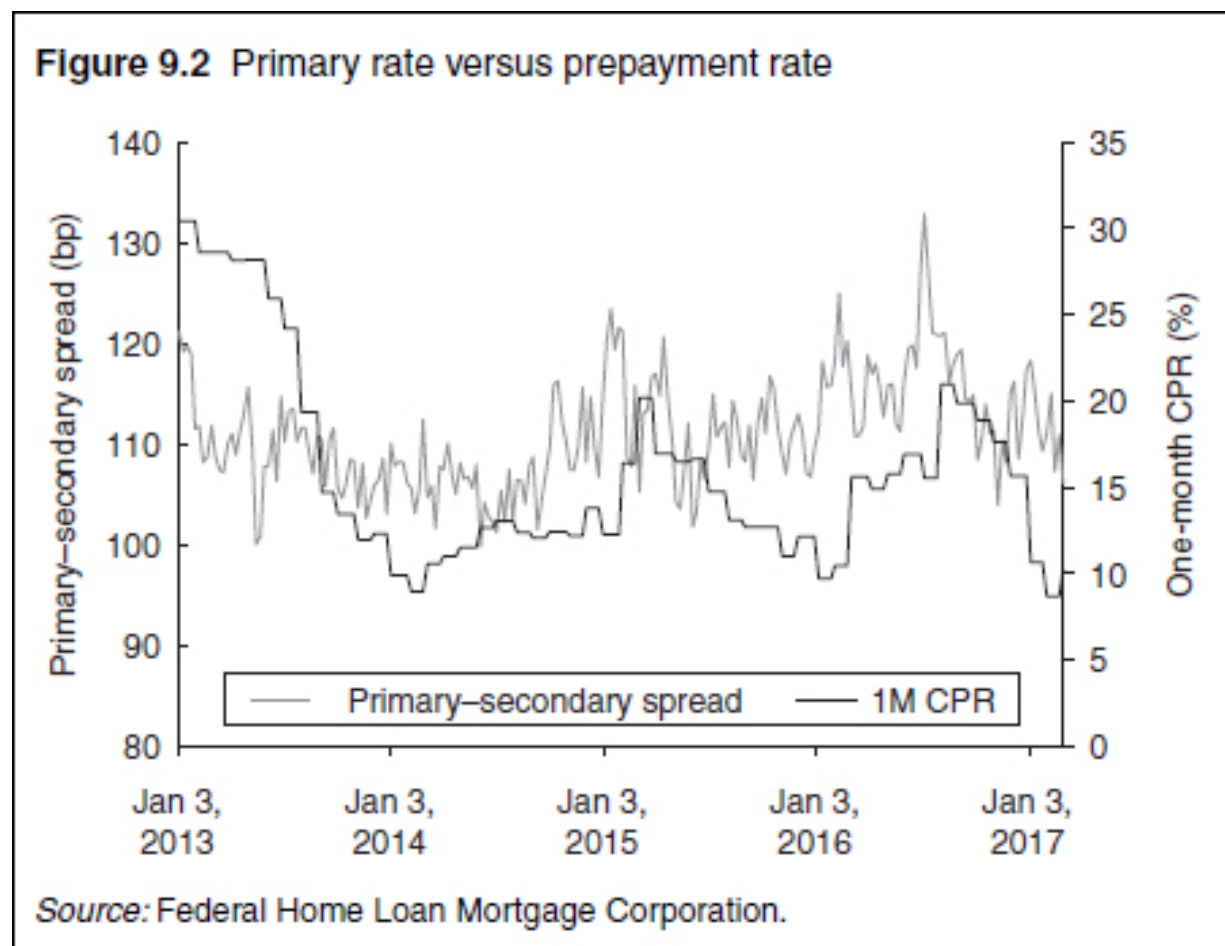


Figure 9.2 compares the level of the primary-secondary spread with the one-month CPR (annualised prepayment rate) on 30-year (30Y) fixed-rate

collateral and shows a significant correlation: in a market where prepayments increase, capacity will be constrained and margins will go up.

With this we conclude that origination earnings are highly correlated to the level of prepayments in the market, and origination can therefore provide a natural hedge to MSRs or other assets that are negatively affected by prepayments.

PREPAYMENT MODELS AND EMPIRICAL RELATIONS

Now that we have seen how various mortgage-related cashflows are affected by the prepayment rate, let us take a look at how we can model prepayments.

Figure 9.3 shows prepayments on 30Y Federal Home Loan Mortgage Corporation (FH or Freddie Mac) loans between 2000 and 2017, and, as we can see, monthly prepayments varied from about 0.5% to 9%.

Prepayments can occur for a variety of reasons. In prepayment models we typically differentiate between the following types:

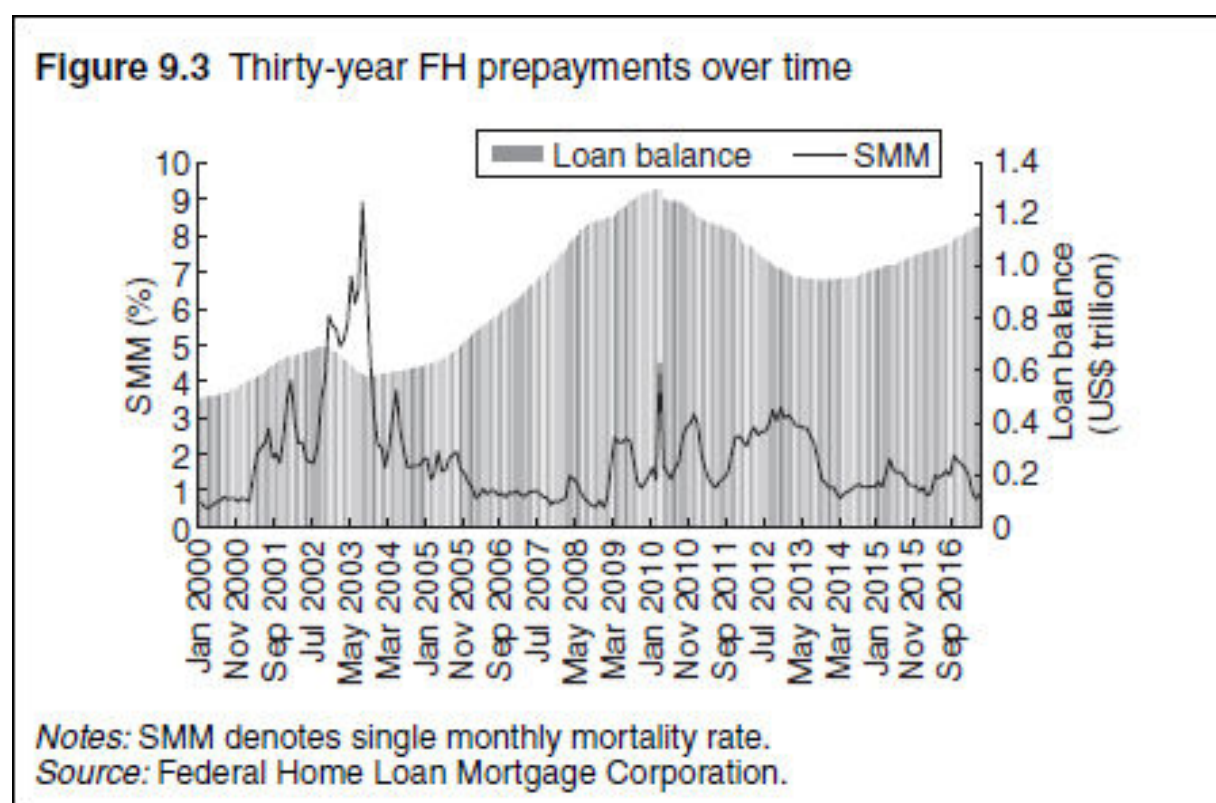
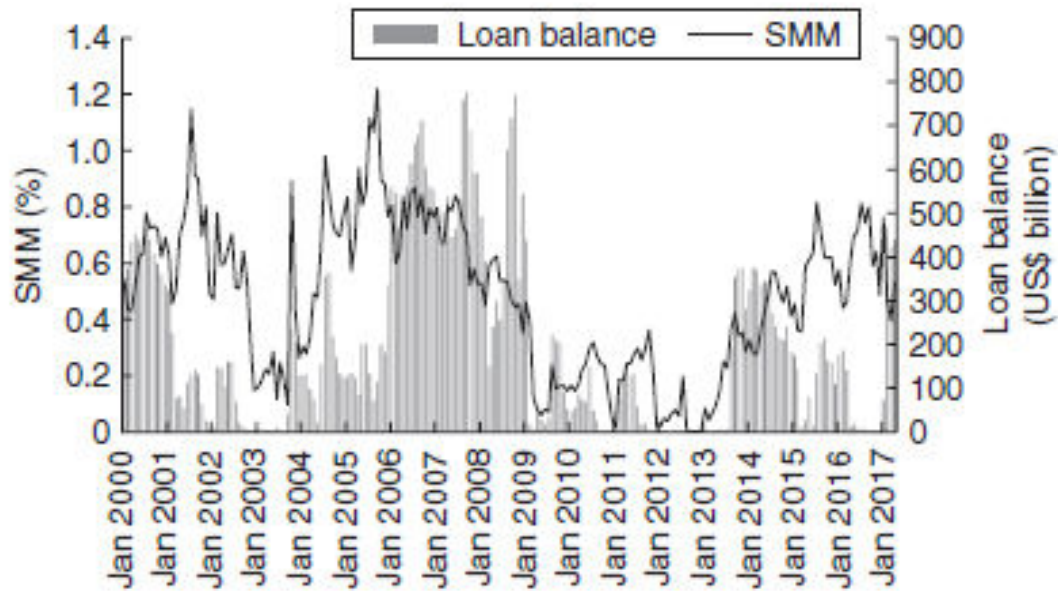


Figure 9.4 Thirty-year FH turnover prepayments over time



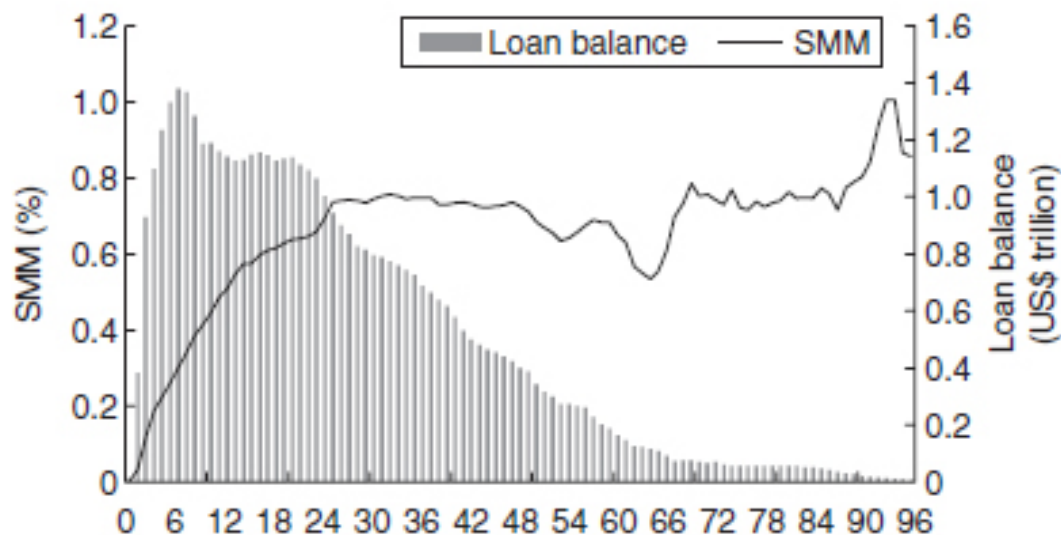
Notes: SMM denotes single monthly mortality rate.

Source: Federal Home Loan Mortgage Corporation.

1. turnover prepayments;
2. rate-driven refinancing prepayments;
3. cash-out refinancing prepayments;
4. default prepayments.

Below we discuss each of them in more detail.

Figure 9.5 Thirty-year FH: turnover ageing curve



Notes: SMM denotes single monthly mortality rate.

Source: Federal Home Loan Mortgage Corporation.

Turnover prepayments

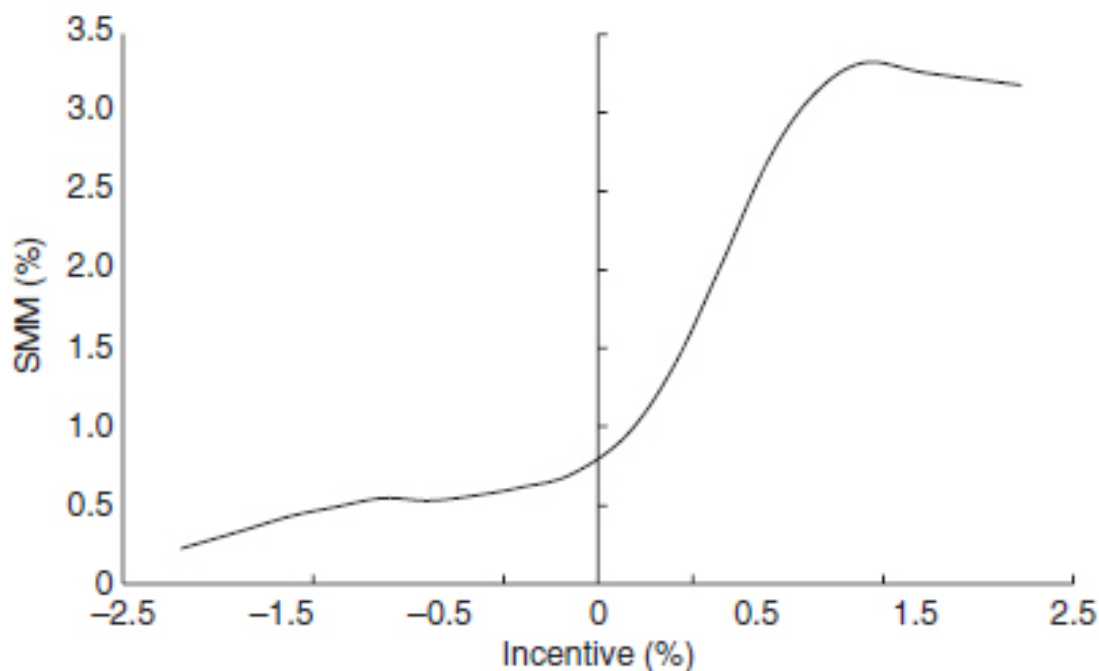
Turnover occurs when a borrower prepays a mortgage due to relinquishing the property. This can be difficult to observe in a typical data set, and one way to estimate prepayments from turnover is to only consider loans that get prepaid with a negative incentive to refinance. Figure 9.4 shows estimated turnover speeds by looking at loans with negative prepayment incentive. Note that in some periods there were no loans with negative incentive, since rates were at their all-time low.

Below we discuss a few drivers that affect turnover speeds on a portfolio of loans.

- **Loan age:** an example of a turnover housing curve is given in Figure 9.5. Once borrowers buy a house, they are likely to stay in it for a while, so turnover on brand new mortgages is going to be low. This creates the ageing ramp, in which it takes about 18–24 months for turnover speeds to reach their normal level (about 0.7% per month for 30Y FH loans over the period 2000–16; see Figure 9.5).

- **Seasonality:** turnover prepayments are seasonal, as expected with housing turnover. More people relocate in the summer, when weather is good and schools are on a break, than in the winter.

Figure 9.6 Thirty-year FH prepayments: refinance incentive versus single monthly mortality rate



Notes: SMM denotes single monthly mortality rate.

- **“Lock-in” effect:** if mortgage rates are high, relocating will increase a homeowner’s borrowing costs, causing them to be less likely to relocate.
- **Loan size:** smaller loans typically turn over faster, as, for example, first-time home buyers upgrade to a larger home.
- **Home prices/current loan-to-value:** in a strong housing market, it is more likely for a borrower to turn over and, eg, realise the appreciation of their house. In a weak market, on the other hand, the loan-to-value (LTV) of their loan may become too high, making it more difficult to pay the down-payment on a new home.

Rate-driven refinancing prepayments

Rate-driven refinancing prepayments are triggered by borrowers refinancing their loan to benefit from a lower market rate. As expected, the larger the potential savings from refinancing, the more likely will it be for a borrower to refinance his mortgage. [Figure 9.6](#) shows the relation between prepayment speed and incentive (calculated as the note rate on the current mortgage (unknown char) the market rate). This curve has a typical S-shape.

As we can see, prepayments increase fastest for an incentive of around 50 basis points (bp), and start to level off when the advantage is about 150bp. [Figure 9.6](#) also shows the turnover lock-in effect for negative incentives.

In addition to incentive, there are many factors that drive refinancing prepayments. We list some examples below.

- **FICO score:** borrowers with lower credit scores will find it harder to find a new loan and are less likely to refinance.
- **Loan-to-value:** similarly, borrowers with a high LTV will be less likely to refinance.
- **Loan size:** larger loans will need a smaller incentive to recapture the cost of refinancing and will be prepaid faster.
- **Burn-out:** certain borrowers are not very sensitive to incentive to refinance. Once a borrower has not exercised refinance opportunities in the past, modelers will assume that they are less likely to do so in the future.
- **Media effect:** when rates are at a low point, refinancing opportunities may get more publicity, triggering extra prepayments.
- **State effects:** there are differences in the costs of refinancing between US states, and this triggers differences. For example, New York state is known as a “slow” state.

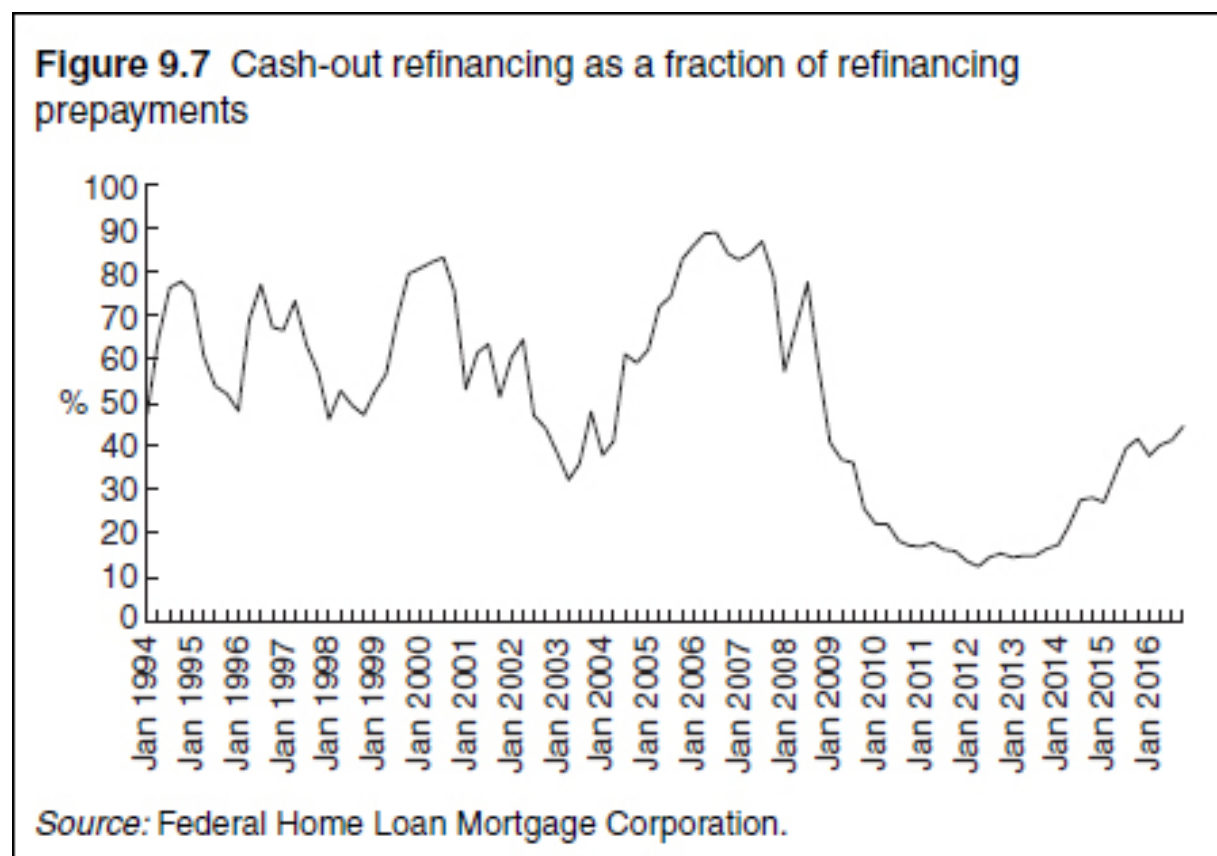
Cash-out refinancing prepayments

Cash-out refinancing is driven by borrowers who want to monetise the appreciation of their home price by refinancing into a mortgage with a higher balance. These prepayments are harder to observe in a typical data set as both the original loan and the new loan need to be seen. Freddie Mac reports

estimated ratios of refinancing prepayments that took cash out (defined as the new loan being more than 5% larger than the original loan).

As we can observe in [Figure 9.7](#), during the housing market bubble about 90% of refinancing prepayments took out cash. This dropped to a level of 15% after the 2007–9 financial crisis, but increased steadily from 2013, to a level of over 40% by the end of 2016.

Cash-out prepayments are driven by similar factors to the other prepayment types, and they cannot be observed separately. Often they are modelled jointly with turnover and rate-driven refinance prepayments.



Default prepayments

A final category of prepayments occurs when the borrower defaults. Depending on the LTV at the time of default, the lender (or guarantor) may incur a loss. Default prepayments are a function of loan age, FICO score and LTV, among other factors.

Obviously, since the “Great Recession” in the late 2000s and early 2010s much work has been done on modelling mortgage credit risk. Since this book

focuses on interest rate risk we shall not cover this in detail.

Modelling of prepayments

Now that we have seen some of the drivers of prepayments, the next step will be to build a model. One question is whether to model prepayments at a loan level or at a pool level. Historically, loan-level data was only available to banks and servicers for their own portfolio. At the time of writing, however, all agencies have started to report loan-level information on their mortgage-backed securities. Therefore, the development of loan-level prepayment models has become increasingly popular. At the same time, for an investor in MBSs, it is often still infeasible to analyse all pools at the loan level, and it is unclear how much benefit a loan-level prepayment model will give in these cases.

A variety of statistical approaches can be used to estimate prepayment models. For loan-level models a logistic or survival model is a common approach (see, for example, Deng *et al* 2000). However, various other approaches, both parametric and non-parametric, are also used (for an example see Hayre *et al* 2000). Data mining techniques are also becoming more popular. However, care must be taken that variable relations that are found make intuitive and economic sense.

Once we have a prepayment model we can use it in our risk management, and below we discuss a couple of approaches to do that.

VALUING MORTGAGE PRODUCTS

Once we have a prepayment model at our disposal, we need to link it to an interest rate model to value and compute sensitivities on mortgage products. For this we can use standard interest rate derivatives pricing models such as the London Interbank Offered Rate (Libor) market model (LMM). The modelled prepayments can then function as the payoff function of the interest rate instrument that we are pricing. Generally, the prepayment model uses primary mortgage rates as inputs and a formula to convert the Libor coming out of the LMM into mortgage rates needs to be defined. For this, often statistical relations between primary or secondary mortgage rates and swap rates are applied.

Another feature of prepayment models is that they are usually path dependent, and we need to resort to Monte Carlo simulations to value the

mortgage product. Finally, an option-adjusted spread needs to be added to the discount rate to make sure that the model price matches observed market prices. The combination of these features makes mortgage products one of the most challenging products to model and risk manage.

The practical implications of the ability to prepay are that, like standard callable bonds, mortgage loans and securities have negative convexity. When rates drop, expected prepayment rates increase, and therefore the duration of a mortgage loan shortens. In a hedged position for a long mortgage, this means that we must buy back part of the hedge. This will occur at a higher price, since rates have dropped, introducing a loss.

Mortgage-servicing rights, on the other hand, have highly leveraged negative durations. At the same time, convexity is negative when rates are close to or above the coupon and positive when rates are low. How the shape of the prepayment model's refinancing curve affects the duration and convexity of MSRs is discussed in Boswinkel and Westerbeck (2008).

Table 9.1 Overview of sign of duration and convexity

	Duration	Convexity
Whole loan/MBS	$\pm$ SMM level	$-$ SMM slope
MSR	$-$ SMM slope	$\pm$ SMM slope change
Origination	$+$ SMM slope	$+$

Table 9.1 summarises the sign of duration and convexity for the various mortgage instruments (first line) and its relation to the prepayment model refinancing S-curve (second line).

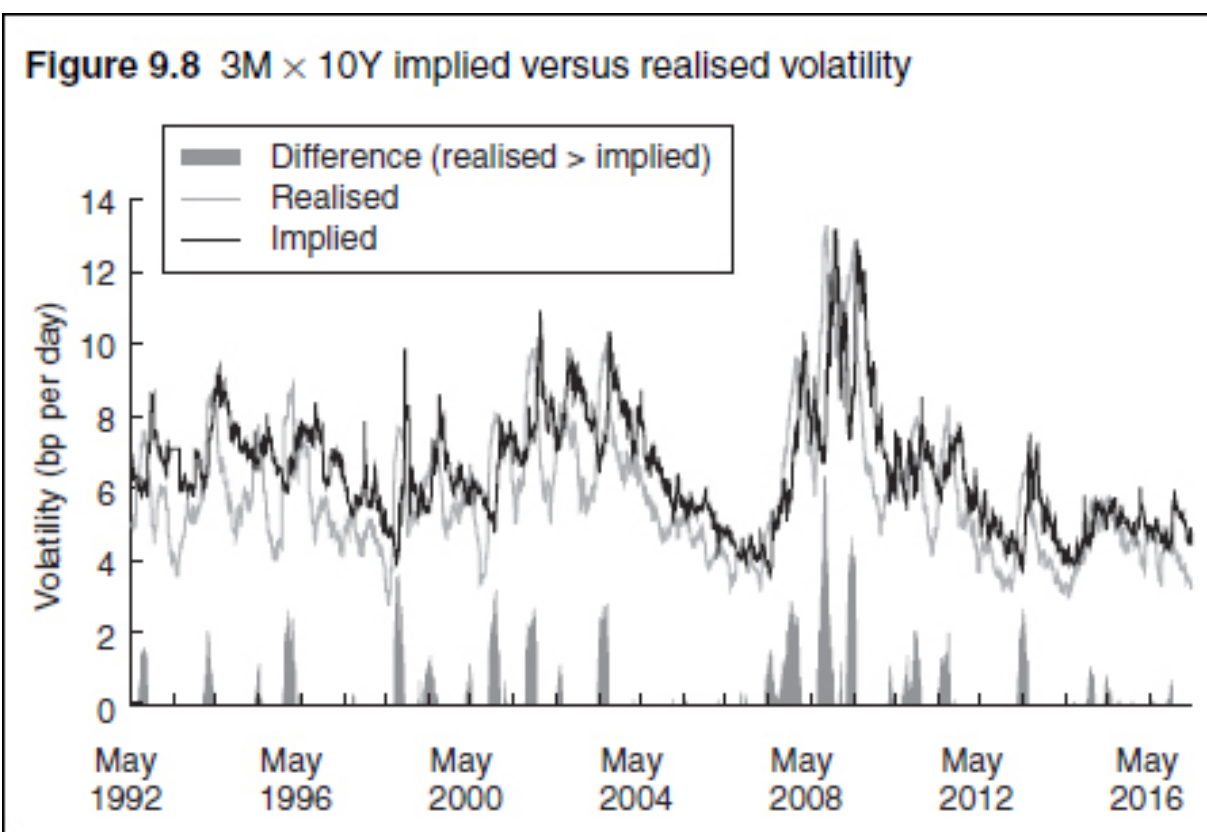
Of those three asset types, origination is the most challenging to model. It is affected not just by prepayment uncertainty, but also by uncertainty in other factors driving the size of the mortgage market (eg, home prices, share of home ownership), the originator's market share and margin predictions.

However, it can provide important offsets to the mortgage-servicing asset, not only in duration and convexity, but also in prepayment model error. While higher-than-expected prepayments will hurt the mortgage-servicing market value, they will also lead to higher-than-expected origination fee income.

While we may attempt to include some origination value in market value at-risk metrics, the asset is not marked-to-market and is more commonly included in earnings-at-risk metrics. This will be the topic of the next section.

BALANCE-SHEET MANAGEMENT OF PRODUCTS WITH PREPAYMENT RISK

While market value risk metrics are more suitable to handle mortgage products or products with embedded options in general, it is important to get meaningful results out of traditional earnings-risk-based metrics too. Here we discuss how to include those products in gap analysis and earnings-at-risk measurement.



As discussed by Gentili and Santini ([Chapter 4](#) of the present volume), gap analysis does not work well for products with embedded options. One approach that allows us to represent a mortgage position in a gap report is to convert the key-rate duration profile (see, for example, Ho 1992) of the position in the replicating partial differential hedge portfolio. This will result

in a portfolio of zero-coupon bonds with the same key-rate duration profile as the mortgage position and can easily be reflected in a gap report.

In earnings-at-risk measurement we look at earnings generated by a portfolio over a period of a few years. One advantage of this approach is that earnings from non-marked-to-market assets, eg, mortgage origination, can easily be included. However, one of the largest risks in portfolios with significant negative convexity is the hedge rebalancing costs that occur over time. This makes it important to model the dynamic adjustment of hedges as it occurs while the scenario progresses. This will include rebalancing of options that hedge convexity, or the simulation of dynamic delta hedging. Profit and loss from delta hedging, in particular, depends on the relation between realised and implied volatility.

There is generally believed to be a risk premium on implied volatility. Looking back at the $3M \times 10Y$ implied swaption volatility over the 25 years up to 2017 (Figure 9.8), the average realised volatility is about 5.8bp per day, versus 6.45bp for the implied volatility. This suggests that, over time, delta hedging may be less costly than buying option hedges. However, as with any risk factor that carries a risk premium, in adverse markets dynamic delta hedging can trigger large losses. Therefore, many mortgage hedgers choose to cover all or part of their convexity risk with options.

Table 9.2 Daily 10Y volatility for 2017 US Federal Reserve Bank CCAR scenarios

	Basis points
Baseline	1.0
Adverse	0.6
Severe adverse	4.9

Returning to earnings-at-risk management, the problem is that often in both typical bank earnings scenarios and regulatory stress test scenarios, the realised volatility is very low (see, for example, Table 9.2 for the 2017 US Comprehensive Capital Analysis and Review (CCAR) scenarios, where even in the severe adverse scenario the realised volatility was not higher than 4.9bp per day), resulting in an understatement of delta hedging costs. In order to address this, the best approach for instruments with embedded

options is stochastic earnings-at-risk. In this approach we simulate a large number of scenarios with realistic levels of realised volatility. To determine earnings-at-risk, the balance-sheet risk manager can then look at the distribution of earnings over these scenarios, rather than, for example, the difference between two low-volatility scenarios.

CONCLUSION

In this chapter we described the main assets that carry prepayment risk on a bank's balance sheet: mortgage loans and securities, mortgage servicing rights and mortgage origination. We discussed some of the drivers as well as the modelling of prepayment risk. We also talked about how to treat these assets in the balance-sheet risk management process. While a whole book would not be sufficient to discuss all the aspects of managing mortgage risk, we hope to give the reader some basic ideas to start developing their own mortgage models and/or apply some of this in their balance-sheet management framework.

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Considerations for ALM in Low and Negative Interest Rate Environments

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Asset and liability management (ALM) includes the management of interest rate risk arising from a bank's business activities. In cases of very low or even negative interest rates the behaviour of bank risks presents new challenges. Usually, ALM functions are also tasked with stabilising net interest income (NII) banking book business revenues; in an environment of negative yields this objective becomes difficult to achieve. Customers may behave differently, which challenges product modelling assumptions, and certain products reveal inherent optionalities that need to be looked at carefully. This chapter illustrates objectives of ALM functions in banks and its difficulties in low-yield environments with a focus on the impact on interest rate risk management. Liquidity management aspects are left aside. We look at the impacts on modelling of products with behavioural aspects and countermeasures to deal with margin erosion. Finally, we illustrate the different views of regulators and developments in legislation, and discuss their effects on banks' risk management. The chapter concludes with an analysis of market changes caused by technological and competition changes.

INTEREST RATE RISK IN THE BANKING BOOK MEASUREMENT CONCEPTS

The typical role of a commercial bank is the intermediation between lenders and borrowers. The supply and demand of its lenders and borrowers differ in terms of maturities. Hence, banks could be faced with a structural interest rate risk position, which occurs whenever the interest rate maturity profile of the assets on the balance sheet is not refinanced with a matching liability maturity profile. In addition to the interest rate risk that could arise out of customer business and interest rate modelling, financial institutions must decide how to treat their own equity in terms of the inherent interest rate risk; the maturity of own equity is not contractually defined and banks have to decide in which maturity buckets the equity needs to be invested to adequately reflect the profile from an interest rate risk management perspective.

Banks usually have two different approaches to measure the interest rate risk in the banking books: the change in economic value of equity (ΔEVE) and the change in the net interest income (ΔNII). (In the literature ΔNII is also referred to as earnings-at-risk or gap analysis.) The first concept is the measurement of changes in the economic value of (the bank's) equity. The second is the measurement of the change in NII over a certain time period (in practical terms between 12 and 36 months). The results of the two key measurements are used to actively manage or protect against the interest rate risk in the banking book (IRRBB).

NII is defined as interest income minus interest expenses. The difficulty in IRRBB management is that a bank cannot easily protect against the ΔEVE impact and simultaneously stabilise the NII. Banks have to balance both risk measures and decide which one is the overarching target for their IRRBB management. It is obvious that the regulators expect comprehensive interest rate risk management from banks and that banks comply with the regulatory expectations and legally binding requirements.

During the low interest rate environment in the eurozone, banks with a large “euro” portion of the balance sheet were exposed to some additional challenges in IRRBB management. The same holds for all other monetary areas with a low or negative interest rate environment.

To demonstrate the key IRRBB measurement concepts and potential challenges in a low or negative interest rate environment we now introduce a simplified hypothetical bank.

We shall show the baseline numbers for EVE and NII for a balance sheet that is immune to changes in the bank's EVE and NII and discuss the results across different interest rate shock scenarios and any additional impacts due to low or negative rates, respectively.

Therefore, we assume a static balance sheet of 100. The hypothetical bank is funded with deposits and equity, which contractually were overnight (O/N) products in terms of their interest rate duration. The bank can invest the liabilities in two different types of asset. The first investment opportunity is a central bank overnight account paying O/N interest rates. The second investment opportunity is a 10-year (10Y), long-term asset. All positions of the balance sheet are accrual positions; accounting effects are not taken into account and it is assumed that the interest rate environment is "normal" with a positive slope of the curve.

For the deposit base of the hypothetical bank, we have to differentiate between different types of deposit. Deposit customers (depending on the product) have different price sensitivities to changes in the market rate. One concept of deposit price sensitivity is called "deposit beta" (deposit- β). This is a measure that expresses the deposit rate changes (ie, product rate paid out to the customer) relative to the market interest rate changes

$$\text{deposit } \beta = \frac{\text{change in product rate}}{\text{change in market rate}}$$

To explain the mechanism of deposit- β we give examples of the most extreme cases: deposits with a beta of 1 and deposits with beta of 0. The former are highly sensitive to changes in the market interest rate. Hence, a one basis point (1bp) increase in the market rate causes a 1bp increase in the customer rate, otherwise the customers will shift their deposit into other products or withdraw it. For the latter, eg, current accounts, the customers are insensitive to changes in the market rate.

Balance sheet A is constructed to reduce the impact on the bank's NII results due to changes in interest rates. We assume the bank is funded with 80 O/N deposits with a deposit- β of 1, 10 O/N deposits with a β of 0 and 10 equity. The bank invests all O/N deposits with a β of 1 in O/N assets. The deposits with a β of 0 and the equity are invested in a 10-year asset paying a coupon of 5% in order to reflect the deposit clients' and shareholders' behaviour in terms of interest rate sensitivity. The baseline NII is the interest income minus the interest expenses and in this example should be 0.9 at t_0

(interest income of 1.9 and interest expense of 0.9). The baseline EVE at t_0 should be 10.

Balance sheet B is managed in such a way as to reduce the EVE volatility. The bank invests all liabilities (deposits with a deposit- β of 0 or 1) and the bank's equity in O/N assets paying O/N interest rates. The hypothetical bank's baseline NII is lower, due to the fact that the deposits with a deposit- β of 0 and equity are not modelled/termed out at longer tenors. The interest income is 1.0 and total interest expenses are 0.9. Hence, the baseline NII is 0.1 and the EVE is 10 at t_0 .

As a next step in the development of our simplified model, we apply two different instantaneous interest rate shock scenarios (a parallel up shock of 100bp and a parallel down shock of -100bp) and calculate the impact on the bank in terms of ΔNII and ΔEVE results.

According to the composition of balance sheet A, all O/N assets and deposits with a β of 1 re-price immediately. We assume for deposits with a β of 0 that the interest rate stays unchanged at 1% and the 10-year fixed rate asset continues to pay a coupon of 5%. In our interest rate stress scenarios, the ΔNII impact is approximately 0 (our total NII of 0.9 remains unchanged). In contrast, the ΔEVE results are affected, because the future cashflows of the 5% asset need to be discounted with a new interest rate curve, which leads to a significant impact on the present value of the asset and consequently on the EVE of the bank. For simplicity, we calculate the ΔEVE impact in a two-step approach. The first step is to calculate the future value (FV) of the 10Y asset, which was traded at 100 (the present value (PV) of the asset is 20)

$$\text{FV} = \text{PV} \left(\frac{1}{(1+r)^n} \right)^{-1} = 20 \left(\frac{1}{(1+0.05)^{10}} \right)^{-1} = 32.58$$

Thereafter, we continue by discounting the future value with the stressed interest rates and build the difference to the present value of 20

$$\begin{aligned} \text{PV} &= \text{FV} \times \frac{1}{(1+r)^n} \\ \text{PV} + 100\text{bp} &= 32.58 \times \frac{1}{(1+0.06)^{10}} = 18.19 \\ \text{PV} - 100\text{bp} &= 32.58 \times \frac{1}{(1+0.04)^{10}} = 22.01 \end{aligned}$$

$$\Delta\text{EVE} + 100\text{bp} = \text{PV} + 100\text{bp} - \text{PV} = 18.19 - 20 = -1.81$$

$$\Delta\text{EVE} - 100\text{bp} = \text{PV} - 100\text{bp} - \text{PV} = 22.01 - 20 = 2.01$$

The ΔEVE impact in the case of a positive parallel upshift of 100bp is about -1.81 , and in the case of a parallel downshift of 100bp is about $+2.01$.

The impact of the interest rate shifts on balance sheet B is totally different. The ΔEVE values are approximately 0, because all liabilities are invested in O/N assets, and therefore the impact on the present value due to shifts is relatively negligible. The NII values change because all O/N assets and O/N deposits with a deposit- β of 1 re-price immediately, while the interest expenses for deposits with a deposit- β of 0 do not change at all. The bank's NII is therefore sensitive to changes in the interest rates. The new NII in the positive scenario is 0.3. The higher interest income (2% on 100) is not fully eroded by higher interest expenses (2% on 80) because the deposits with a deposit- β of 0 and interest expenses for equity remain at unchanged interest rate levels. The same calculation could be applied to the negative curve shift and ends in an NII of -0.1 .

As an interim conclusion, we can state that there are different compositions to reduce NII or EVE volatility that are effective in a “normal” interest rate environment (ie, interest rates are positive across all tenors under stress assumptions).

In the next step we apply the same interest rate shocks to comparable balance-sheet compositions but with the crucial difference that the O/N yield is already at 0%, and the long-term interest rates are 2% for the 10-year tenor and therefore more comparable to a situation in the European monetary environment, where at the time of writing rates were around 0% ([Figure 10.1](#)).

Balance sheet A was explicitly constructed to reduce NII volatility. The calculation shows no ΔNII impact in the case of a positive interest shift, but a significant ΔNII impact of -0.8 (0.4 at t_0 and -0.4 at t_1) in the case of a -100bp interest rate shock. As we saw in the first example (see also [Figure 10.2](#)), the balance sheet composition to reduce NII volatility works in a positive interest rate environment but not in the case of zero or negative interest rates. The driver for the result is that banks with a large portion of euro retail deposits cannot charge customers with negative interest rates. On the other hand, the deposit base is typically invested in money market products or central bank accounts that can trade at negative interest rates. In

the simplified example, ΔEVE impact appears relatively unchanged in comparison with the first example; however, to be precise, the ΔEVE impacts in a low or negative interest rate environment are slightly higher, with less convexity Balance sheet B was constructed to reduce EVE volatility. The ΔEVE impact continues to be approximately 0, and therefore the composition of the balance sheet was effective, but the ΔNII impact of -1 is much higher than the first example due to an asymmetric payout profile in a low or negative interest rate environment as outlined above.

Figure 10.1 Interest rate shocks in a low interest rate environment

	Balance sheet A – NII volatility reduction		Balance sheet B – EVE volatility reduction	
	Assets	Liabilities	Assets	Liabilities
	Value 80 O/N 0%	Value 80 deposit (beta 1) 0%	Value 100 O/N 0%	Value 80 deposit (beta 1) 0%
		Value 10 deposit (beta 0) 0%		Value 10 deposit (beta 0) 0%
	Value 20 10Y 2%	Value 10 equity		Value 10 equity
Interest income/expense	0.4	0	0	0
Baseline NII	0.4		0	
Baseline EVE	10		10	
NII + 100bp	0.4		0.2	
Delta EVE + 100bp	-1.86		~0	
NII – 100bp	-0.4		-1	
Delta EVE – 100bp	2.07		~0	

$$FV = PV \left(\frac{1}{(1+r)^n} \right)^{-1} = 20 \left(\frac{1}{(1+0.02)^{10}} \right)^{-1} = 24.38$$

$$PV + 100bp = 24.38 \times \frac{1}{(1+0.03)^{10}} = 18.14$$

$$PV - 100bp = 24.38 \times \frac{1}{(1+0.01)^{10}} = 22.07$$

$$EVE + 100bp = PV + 100bp - PV = 18.14 - 20 = -1.86$$

$$EVE - 100bp = PV - 100bp - PV = 22.07 - 20 = 2.07$$

Figure 10.2 Example balance sheets reducing NII or EVE impacts

	Balance sheet A – NII volatility reduction		Balance sheet B – EVE volatility reduction	
	Assets	Liabilities	Assets	Liabilities
	Value 80 O/N 1%	Value 80 deposit (beta 1) 1%	Value 100 O/N 1%	Value 80 deposit (beta 1) 1%
		Value 10 deposit (beta 0) 1%		Value 10 deposit (beta 0) 1%
	Value 20 10Y 5%	Value 10 equity		Value 10 equity
Interest income/expense	1.8	0.9	1.0	0.9
Baseline NII	0.9		0.1	
Baseline EVE	10		10	
NII + 100bp	0.9		0.3	
Delta EVE + 100bp	10		10	
NII – 100bp	0.9		–0.1	
Delta EVE – 100bp	2.01		~0	

From a practical point of view, one question might be which of the two interest rate measurement and steering approaches is better for IRRBB management in a low or negative interest rate environment. In practice, there is no clear recommendation, because the answer depends on the business models and risk appetite of the bank. To cope with low interest rates, banks need to balance the effects of ΔNII and ΔEVE . In the following sections we discuss the embedded options in customer business, the impacts on the model and possible countermeasures to manage IRRBB for low interest rates.

PANEL 10.1 RIGHT OF THE BORROWER TO GIVE NOTICE OF TERMINATION

- (1) The borrower may terminate a loan contract with a pegged lending rate, in whole or in part
 1. if the pegging of the lending rate ends prior to the time determined for repayment and no new agreement is reached on the lending rate, observing a notice period of one month to end at the earliest at the end of the day on which the pegging of the lending rate ends; if an adjustment of the lending rate is agreed at certain intervals of up to one year, the borrower may only give notice to end at the end of the day on which the pegging of the lending rate ends;
 2. in any case at the end of ten years after complete receipt, observing a notice period of six months; if, after the loan is received, a new agreement is reached on the repayment period or the lending rate, the date of this agreement replaces the date of receipt.
- (2) The borrower may terminate a loan contract with a variable rate of interest at any time, giving three months' notice of termination.

Source: Bürgerliches Gesetzbuch, Section 489. URL: https://www.gesetze-im-internet.de/englisch_bgb/englisch_bgb.html#p1758.

INTEREST RATE RISK IN THE BANKING BOOK MANAGEMENT CONSIDERATIONS

Embedded options

Banking book products can contain different types of embedded options on the customer side that have an impact on a bank's interest rate risk management.

For assets, the most prominent example is most likely to be prepayment rights, particularly if they do not allow the bank to get a prepayment penalty

(even though this moves missed future earnings to earlier periods). These can be either contractually agreed (eg, optional annual prepayments) or legally fixed (eg, as in the German civil code (see Bürgerliches Gesetzbuch (BGB), Section 489 (1–2); see Panel 10.1)). These play an important role, especially for fixed-rate mortgages, which usually have a long duration and consequently a greater interest rate risk (ie, a larger PV01). Hedging of these prepayment options is of course an option, but requires a dedicated model and expectation on prepayments. Consumer loans often have a permanent prepayment right. Nonetheless, their duration is usually shorter and thus they carry less risk.

The question then is what impact low interest rate levels will have on these options. Three aspects to consider are

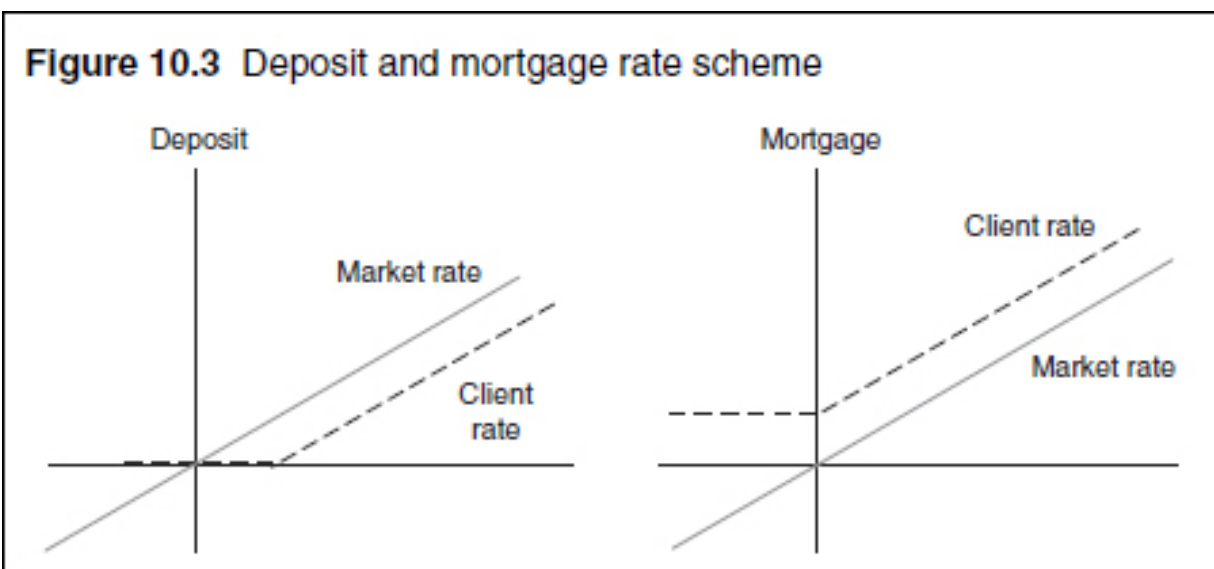
1. the prepayment probability,
2. the NII impact,
3. the EVE impact.

In general, it might be more attractive for customers with a fixed-rate mortgage to prepay their mortgage and enter into a new contract (as long as they do not have to pay a penalty), as they might be able to either reduce their interest burden and therefore repay faster or reduce their instalments. From an NII perspective, a prepayment means the opportunity loss of a constant interest income over time. From an Δ EVE perspective the impact might also be negative, even though this is not immediately visible. Assuming that a mortgage is hedged with its contractual tenor, prepayments leave the bank with an overhedged position and a negative Δ EVE.

In addition to prepayments, the era of negative interest rates brought to light a new but crucial embedded option partially embedded in floating rate mortgage contracts (these mortgage contracts are particularly common in eastern and southern Europe). The rate to be paid by the customer for a floating rate mortgage is expressed as reference rate+spread. Assume a bank has a hedged interest rate risk position, offset by retail deposits priced at the same reference rate (eg, a three-month London Interbank Offered Rate (Libor)); while this seems to be a perfect hedge, this only holds for Libor rates of at least zero, as retail deposit rates cannot be priced below zero (see below and also [Figure 10.3](#)). Consequently, this means that each basis point

below zero will eat up revenues and could lead to a constant negative carry on the trade. While all banks will most likely have adopted their (new) credit contracts, ie, flooring at least the floating rate at zero, some old contracts might face the issues mentioned above.

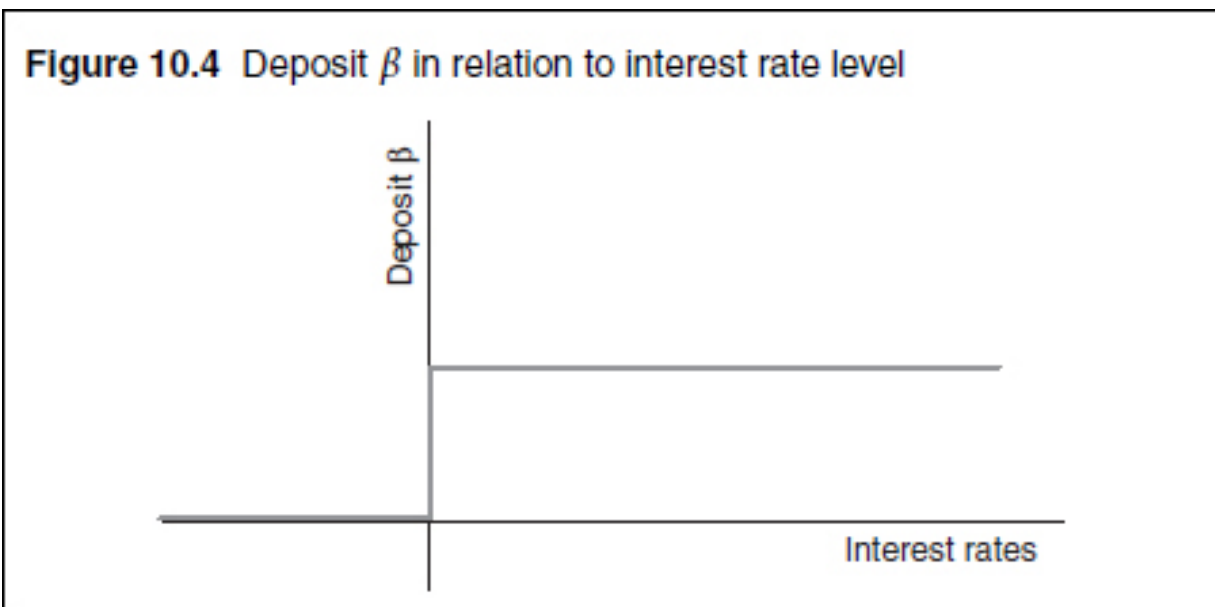
Switching to the liability side of the balance sheet, a negative short-term interest rate environment “activates” a natural floor, at least on retail customer deposit products. While there are still various opinions on the question of whether and under what circumstances (the burden of) negative rates can be passed on to (ultra)high-net-worth individuals or corporates, this has (so far) been a taboo for classical retail depositors with small volumes (ie, under €100,000). While we shall look at the impact on banks’ internal models in the next section, the impact on NII could be very imminent. Even if the customer rate has been set to “0” (ie, the natural floor), each additional euro of deposit creates negative interest income, to distinguish it from the external expense (the interest paid to the customer). Here, we again ignore the funding/liquidity ratio aspects and consider only NII. Banks may be able to partly overcome this issue, as they usually apply some form of maturity transformation (MT; see also the section on countermeasures starting on p. 262). Nevertheless, it challenges their funds transfer pricing, which forms the basis of active balance-sheet management by setting incentives. Here, other factors such as stress outflows or aspects of the regulatory liquidity ratios (eg, net stable funding ratio or liquidity coverage ratio) have to be considered as well.



Model impacts

Many aspects of ALM are based on models. Examples are the calibration of a bank's deposit- β (ie, the pass-through factor for market rate changes on deposits) or the calibration of loan prepayment expectations.

Let us take a deeper look at the issues with β calibration. The most common model for such an estimation tends to be some form of historical regression analysis, which compares customer rates with market rates. A long-term low-interest-rate (or a long-term “no-move”) scenario creates the problem that the correlation between market and customer rate changes is most likely close to zero. This is the case (for negative rates), for example, because even if market rates fluctuate below zero, retail client rates will most likely not, leading to zero correlation. Thus, the risk of an incorrect β is much greater in a steady/low-rate environment than when it is calibrated before the low-rate period (or might have been estimated by expert-judgements). While too low a β would lead to underestimation of costs when rates rise, too high a β estimate might mean a more volatile (and lower) NII due to insufficient hedging in the low-yield environment based on the model.

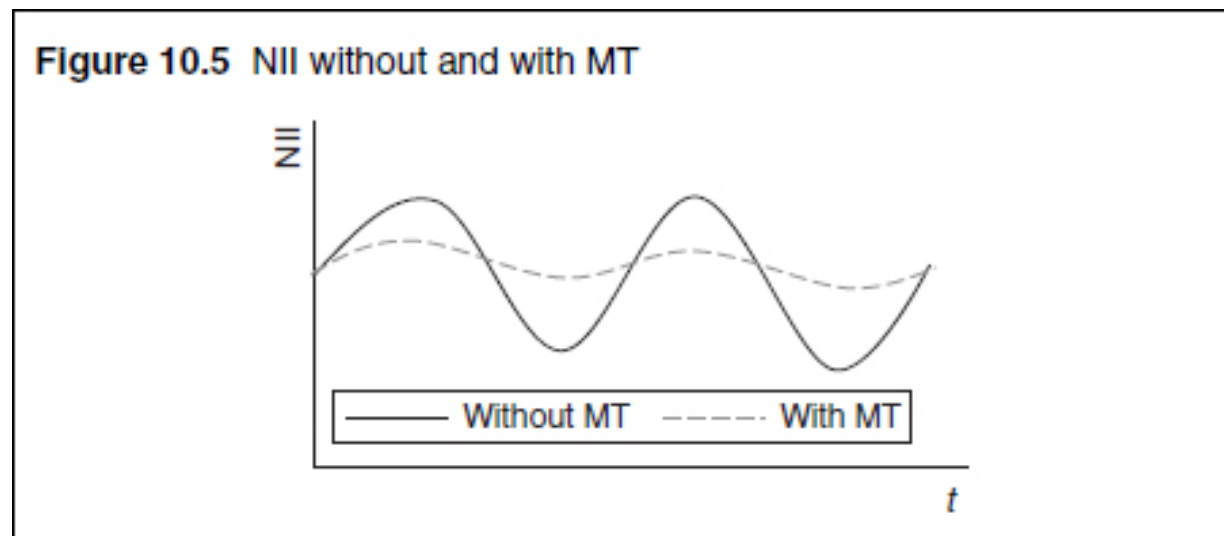


Furthermore, negative rates in particular trigger an interesting discussion on the application of a β . Let us assume that

$$r_t^c = r_{t-1}^c + (\Delta r^m \beta)$$

with r^c being the customer rate at a point in time t and Δr^m being the delta between market rates in the current period and the previous period.

In a normal (positive) interest rate environment, a 10bp increase in rates would lead to 1bp increase in customer rates (for $\beta = 0.1$). This approach can produce unreasonable results, as it does not take into account the absolute market level. When market rates increase from -40bp to -30bp and the client rate was previously zero, we would not expect the customer rate to rise until market levels are at least back to positive values. Therefore, a β model, taking the yield level into account, is required for correct NII estimation. Figure 10.4 shows an illustrative β in relation to the absolute interest rate level.



Countermeasures

Different types of countermeasures can be taken (and have been observed in the market) to overcome the reduced NII in a long-lasting low-rate environment. Some examples are given below.

- The introduction of (higher) fees, even though (accounting-wise) they are not NII, and pass-through of negative rates to big corporates and very large volumes have all been visible in the market.
- The impact of margin erosions could be mitigated via extension of MT and asset re-pricing. Depending on banks' objectives and risk strategies, the ALM functions could be forced to extend the duration of assets when deposits are invested. Depending on their ability to assess

deposit behaviour properly, this leads to higher returns and therefore mitigates revenue pressure. However, a dampening effect will also affect revenues in the case of rising rates, ie, leaving income lower for longer. [Figure 10.5](#) illustrates NII with and without MT. Such an approach could come with higher model risk and negative mark-to-market values of positions if these are fair valued.

- Following an objective to generate stable NII for a bank, ALM departments could also keep overall margin stable by increasing the margins on assets, ie, forcing business areas to charge higher rates for consumer loans and mortgages. While this effect seems to be counterintuitive, it could be observed when Denmark introduced negative rates for the first time. In markets with a high level of competition, transparency and client mobility, this approach could be expected to be diminished after a short time.
- Asymmetric derivatives might be a suitable instrument to (partially) protect NII from decreasing rates. Buying floors or receiver swaps could be a hedge to a low-rate environment, but is not visible in the market. It should be borne in mind that the use of derivatives could create an upfront cost; and that they usually need to be measured at fair value if no hedge accounting measurements are available.

REGULATORY VIEWS

The field of IRRBB received increasing regulatory attention in the years after the global financial crisis. In particular, the review of IRRBB by the Basel Committee on Banking Supervision (BCBS) aimed to align global standards for this area of risk management. In addition to this regulatory framework, many national legislative and supervisory bodies are influencing banking book risk management. In particular, institutions acting across boundaries are subject to different economic and legislative environments.

The specific situation with very low interest rates in some economies, and even negative rates for short- and mid-term tenors in some countries, has created further challenges for banks' risk management: while regulators are mainly concerned about negative effects on market values in the case of rising rates, earnings pressure due to low rates can affect the whole business model.

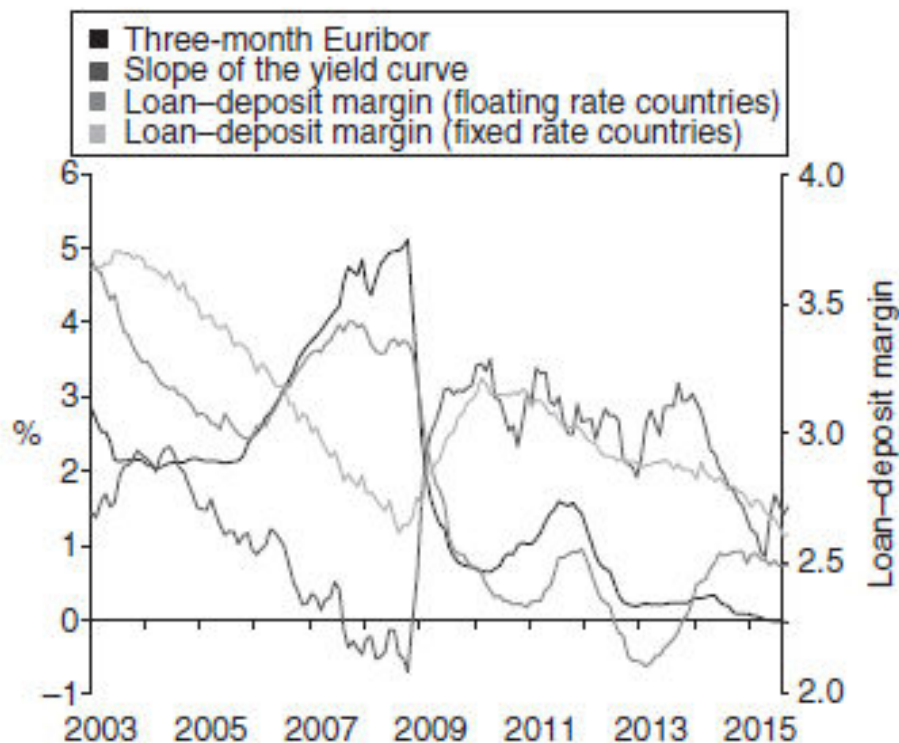
Legal requirements in a sample of European countries

Under normal circumstances the ALM function manages banking book business to generate stable margins. Business functions usually require a stable NII margin to earn running costs (see [Figure 10.6](#) for interest rate and margin development in Europe between 2003 and 2015).

During “normal” interest rate environments, where rates are above 0%, the margin for deposits will be positive when interest paid to clients is at or close to 0%.

When interest rates become close to 0% or negative, the institution’s ability to generate sufficient margin becomes challenging. For reputational reasons and/or even due to legal requirements, banks hesitate to, or are unable to, charge negative rates on client accounts (Panel 10.2 shows a sample of German law (BGB 488(1)), which is interpreted as prohibiting charging negative rates, at least for retail clients).

Figure 10.6 Net interest margin compression due to low interest rates



Note: Loan-deposit margins are defined as the volume-weighted lending rates to households and non-financial corporations minus the volume-weighted deposit rates on deposits from households and non-financial corporations. Weights are based on outstanding amounts. Euribor is the Euro Interbank Offered Rate.

Source: ECB Financial Stability Review 2015, Chart A, p. 65.

PANEL 10.2 SAMPLE FOR LAW SETTING IN GERMANY

Typical contractual duties in a loan contract.

1. The loan contract obliges the lender to make available to the borrower a sum of money in the agreed amount. The borrower is obliged to pay interest owed and, at the due date, to repay the loan made available.

Source: Bürgerliches Gesetzbuch, Section 488. URL: https://www.gesetze-im-internet.de/englisch_bgb/englisch_bgb.html#p0741.

Most of the deposits are collected from retail clients, which in many countries are protected from negative rates. Legal cases have also restricted the introduction of fees linked to deposit volumes, as this is seen as close to a negative rates charge.

While in the past deposit business was revenue generating before any additional ALM measures (MT, transfer pricing to support asset business), the ability to generate positive margins for deposit business has since diminished. Institutions are now looking to stabilise net interest margins across stable business balance sheet.

Regulators have focused on setting up a comprehensive consistent framework. However, as this has been developed during phases of high interest rate volatility (both before and after the global financial crisis), one of the regulators' major concerns is rising rates and a bank's ability to withstand a run on deposits. Rising client rates impacting their ability to earn positive margins and negative MTM values on risk management positions are seen as a concern by regulators. As a result, regulators are looking at simplified interest rate shock scenarios. For example, the standard outlier test as required by the German Federal Financial Supervisory Authority (BaFin) requires 200bp up- and downside shocks. The downward shock, however, is floored at 0%. Therefore, the regulatory quantitative assessment is misleading and does not transparently show the risk of falling rates. The following actions taken by regulators aim to address this issue:

- the European Central Bank (ECB) stress test on IRRBB included various interest rate scenarios that had negative rates;
- updated European Banking Authority guidelines are expected to include negative rates for the standard scenario assessment;
- regulators are consequently asking banks to consider the main concerns arising from rising rates on EVE and low rates on NII.

Economic impact and requirements for ALM

Negative interest rates are jeopardising parts of banks' long-lasting stable business model: borrowing short term at low interest rates, lending longer term at higher rates. ALM functions of banks across the globe have developed more comprehensive models to properly manage interest rate risks within banking books and to generate stable income streams for the

institutions. However, this simple concept has usually been the basis. Independent of a bank's ALM management approach this was the basis of a stable margin generation as net interest margin (NIM) across the balance sheet. Alternatively, banks treat liabilities separately, placing them into treasury's books at a short-term rate and then lending out to borrowers at a higher rate. In particular, the revenue contribution of deposits only becomes flawed in an environment of very low or negative rates.

This impact is usually not seen immediately for two major reasons:

1. banks are applying MT, ie, investing stable amounts of deposits in the longer term, which usually allows them to benefit from curve steepness, ie, as long as the average investment rate is still above the client rate, the NIM contribution is still positive;
2. for NII results, as long as the average return of the existing assets (funded via the stable deposits) is higher than current deposit costs, the impact will not be seen in financials, as banks are usually aiming to manage this business area on an accrual basis.

While the objective of ALM is to stabilise risk and revenue management of banking books, it could also lead to a dilution of risk awareness. Banks need to be aware of different revenue contributions across the balance sheet and monitor changes closely.

In particular, the quite long-lasting low or negative rates in big economies such as the EU, Switzerland and Japan at the time of writing raises questions about the suitability of existing non-maturing deposit models. A rise in interest rates has not been seen for some time, although the environment did change significantly with globalisation, digitalisation, client mobility, transparency and online banking options. This environment requires the ALM experts to carefully assess behavioural aspects as well as the dynamics of interest rates and behavioural risk management. It requires dynamic approaches and regular model validation and calibration to keep risks controlled.

Another significant banking book risk management area affected by low interest rates is options within the bank's asset base, ie, a client's right to prepay their loans at their discretion; in an environment of low rates the probability of clients repaying earlier is expected to be higher than in one of

higher rates. This will have negative effects on both the EVE and NII measures:

- NII is expected to be lower, as those higher yielding assets still need to be re-priced/reinvested at lower rates (earlier than expected);
- EVE is expected to be affected negatively, as hedges that banks might have in place (where banks pay fixed coupons via derivatives or liabilities) have a negative PV without the opposite asset in the case of an early prepayment.

Bank's ALM departments usually look at these types of interest-rate-related prepayment risks. However, the model risk might increase in scenarios of long-lasting low or negative rates. The hedging strategies that banks apply need to be managed dynamically to cover the dynamics of markets as well the characteristics of client behaviour. Interest rate options are seen to best cover the economic profile of this risk, while the model risk needs to be measured closely. In addition, as option strategies do cost banks upfront fees, these need to be incorporated into the initial pricing of these loan and mortgage products.

While the characteristics of prepayment options differ across products, legislation in many jurisdictions gives clients such prepayment options. Legal options can differ significantly across jurisdictions, and can also include different behavioural aspects, particularly when they are given to retail clients as part of a product. Typical examples are

- annual prepayment rights up to a certain amount within mortgages,
- legal prepayment rights within mortgages (eg, after 10.5 years in Germany, as given by BGB Section 489), and
- prepayment rights within consumer finance loans.

The interest rate sensitivity of these options is usually very widespread, as many factors need to be included. However, particularly in very low-rate environments, greater awareness by clients should be expected.

Negative interest rates in stress testing

Stress testing is a tool to assess and analyse the financial impacts of risk on banks under different scenarios. Risk taking in the context of interest rate risk is a crucial function of a bank. Therefore, IRRBB stress tests are organised by the regulators and internal risk functions on a regular basis.

As outlined above, IRRBB is measured via two different concepts. The main drivers of the ΔEVE and ΔNII results are the interest rate scenarios. Usually, banks must calculate the EVE and NII values for a given scenario or set of scenarios. It is obvious that the scenarios which are negative for the NII of the bank (a decrease in interest rates) are beneficial for its EVE (discount future cashflows with a lower zero curve lead to higher PV).

The best known regulatory interest rate scenarios are parallel up 200bp and parallel down 200bp. In general, the application of positive interest rate shock scenarios is not critical for EVE and NII calculation. Additional guidance from regulators is required for a conceptually sound application of negative shocks in a low to negative interest rate environment. Guidance regarding the application of floors in interest rate scenarios is most important. The application of a floor at 0% interest rates or at any other level could lead to significantly different results. The different options for flooring need to be considered when comparing the stress test results across the industry. In the past, most regulators applied 0% floors in stress test scenarios, because negative interest rates are beneficial for the ΔEVE measure and lead to a positive contribution, which was obviously unintended. Nevertheless, negative interest rates could significantly affect a bank's NII results and thus stress a bank's profitability. Regulators have subsequently removed the methodological weaknesses, run unfloored scenarios in stress tests (as in the 2017 ECB IRRBB sensitivity analysis) and discuss floors at negative interest rates.

A further weakness of some IRRBB stress tests is the static-balance-sheet assumption, especially in a low-interest-rate environment. The overarching target of an external stress test is comparability across the industry, and regulators are aiming to benchmark banks against their peer groups to identify weaknesses. Nevertheless, if a negative interest rate scenario, eg, –200bp, is realised, banks would shift between different funding sources and apply different countermeasures, such as fees.

Technological features and industry changes

Negative interest rates changed the overall banking industry landscape. Technological features that had to be considered include the following:

- interest rate models had to be adjusted to cover negative rates;
- the importance of margin measurement and management significantly increased due to shrinking NII margins;
- data management and analysis became critical for suitable model calibration and risk management;
- new market participants entered funding markets with new technologies and different utility functions.

Globalisation and digitalisation have led to the development of new firms entering the financial markets. New financial technologies (fintechs) have been founded, and existing institutions have been developing and selling financial services (usually in very special market niches), while they leverage expertise and knowledge through new distribution channels or new market and product areas. In areas where fintechs have needed to generate funding, eg, to fund their activities through debt raising, this was comparably easy at the time of writing, as liquidity was ample and funding costs were historically low. In summary, under the existing market conditions, fintechs remain in a development phase, with low yields and with available liquidity; funding competition by traditional financial institutions has not been seen. Funding competition could even be seen as an alternative employment of cash, where cash would otherwise generate negative margins.

However, under changing conditions, further growth of alternative institutions and/or a change in interest rates and liquidity, the competition for funding be viewed differently. Fintechs' increasing need for funding could lead to higher interest rate sensitivity of client liquidity; while traditional banks are expected to use a rise in rates to increase their margin, alternative institutions might need to look for funding and could increase the market-wide costs for funding. In addition, the expected stability of deposit funding could be lower than expected, as clients look for more profitable alternatives. Technology helps clients to assess their opportunities and to transparently assess risk–return profiles that affect their behaviour. Assessing this behaviour accurately is one of the tasks of the bank's ALM departments.

SUMMARY AND CONCLUSIONS

We have summarised some of the challenges for ALM functions in a low or negative interest rate scenario. The reader should thus understand that ALM needs to reflect and dynamically adapt to changes in markets and their environment. While traditional approaches and techniques are at least partly obsolete, the alternative options we have illustrated can and should be assessed and adopted.

The opinions expressed in this paper are those of the authors and do not necessarily reflect the Deutsche Bank position and practices.

Credit Spreads

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ALM risk management has traditionally focused on management of the interest rate risk, but standards for interest rate risk in the banking book (IRRBB) published in 2016 by the Basel Committee on Banking Supervision (BCBS) includes credit spread risk as another piece of the puzzle, as changes in the credit spreads could amplify the risk already arising from the IRRBB.

Incorporating the credit risk factor in the calculation of market value of equity (MVE) is often very complex and challenging. First, it is necessary to consider whether it is more appropriate to model this risk through a future cashflow estimation (assuming a probability of default and recovery) or through a credit spread. As we shall show in this chapter, this decision could affect some metrics, such as the duration of the portfolio, in different ways. Second, the challenge is to build the spread curves (or, if a probability of default (PD) and a recovery is needed, how to calibrate them), and to determine the relationship between these parameters and those used for other purposes within the bank: the credit risk area might estimate different PDs for different purposes, while the counterparty risk area might calculate credit spread curves for credit valuation adjustment (CVA), and market risk area could calculate funding spread curves for funding fair value adjustment (FFVA).

Within the credit spread risk in the banking book (CSRBB), the BCBS standards consider only the market credit risk and the liquidity risk (but not other risks) as the idiosyncratic credit risk or the duration risk. Therefore, a third challenge arising could be to create a credit spread free of other components, as usually it is impossible to distinguish these elements in the market prices of some products.

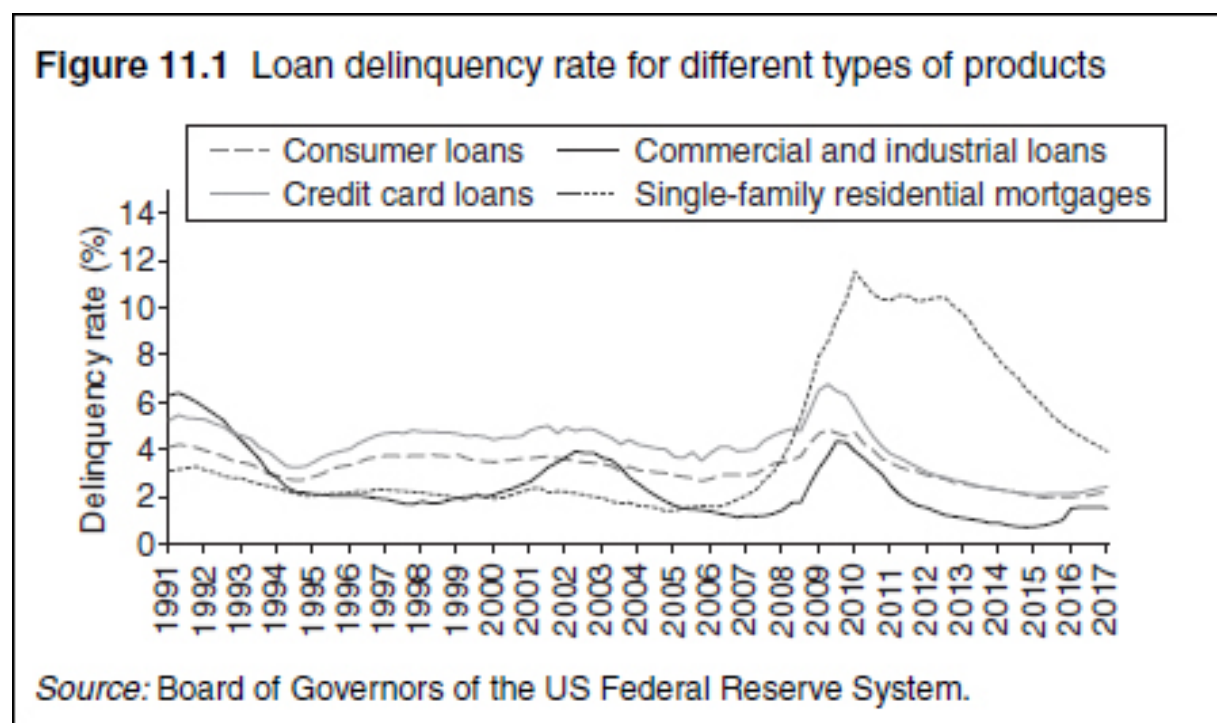
Regarding the balance sheet, credit spreads have often been considered in the valuation of the investment portfolio, usually using a specific discount curve adapted to the characteristic of the issuer. In fact, a broader definition of credit spread could include the effect due to changes in credit spreads (downgrades as well as default). These effects could be an important source of risk at times of economic crisis.

As several studies have shown,¹ default rates can be volatile, especially for lower rating bonds or during periods of crisis. Several authors have analysed the correlation between default rates and interest rate movements. Examples can be found in the work of Jarrow and van Deventer (1998) or Grundke (2005).² On the other hand, it may be a common practice for some entities to value other types of assets in the balance sheet by assuming a risk-free discounting curve, especially when the default probability is small. But assets' cashflows reflect the actual interest rate that the bank charges to its customers, and thus incorporate commercial, credit and other kinds of spread that have to be added to the bank's own marginal cost of funds to reflect credit and other risks. This means that if the cashflows are simply discounted with a risk-free-like interest rate curve, the economic value of the assets in the balance sheet may be overestimated. In addition, asset products can be exposed to default risk, which needs to be modelled via cashflow estimation, either considering the default rate when the cashflows are estimated or via credit spread adjustment in the discount rate. [Figure 11.1](#) shows the evolution of delinquency rates for different types of loan in the US market.³ The figure shows an increase in delinquency rates during the 2007–9 global financial crisis, especially for single-family residential mortgages.

Depending on how the default risk is modelled, the effect on the expected duration of the portfolio could change, although the different approaches provide the same market value, as shown later in this chapter.

A common market practice when valuing bonds is to consider an average credit spread structure for bonds with similar characteristics – rating, sector and geography – to build the discount factor. On the other hand, for structured

derivatives, the valuation usually incorporates valuation adjustments (XVAs: credit and debt valuation adjustment, FFVA, etc) instead of using specific discount curves. These adjustments are calculated at portfolio level to take into account netting effects at counterparty and portfolio levels. Loan valuation through an XVA lacks sense, especially for retail portfolios. In this case it is possible to consider that the credit diversification is almost perfect for a homogenous segment (if the contribution of each loan to the total is small, and the number of individual loans is large enough). This means that the expected losses given the default probability and conditional loss of the individuals will be an accurate estimation of the *ex post* realised credit losses, even though some unexpected losses can arise due to systemic events.



Consideration of the future credit behaviour of held assets is a key aspect of the forthcoming International Financial Reporting Standard 9 (IFRS 9) accounting requirements. Here provisions are forward-looking and include considerations of future economics, in contrast to the existing accounting framework (International Accounting Standard 39), where provisions are backward-looking and focused on already impaired assets.

Under IFRS 9, adjustments will occur where there is a weakening in the economic outlook. This will be a function of key variables used in modelling, which may include the prevailing level of interest rates.

Irrespective of the precise parameterisation of such models, the shift in accounting standards raises the question of consistency in behavioural assumptions for general IRRBB purposes, as well as how best to manage and hedge any interest rate risk inherent in any such provisions.

To address this, the valuation method for the provision should be compared against the framework used for measuring the MVE. Whereas the profit and loss (P&L) from the cashflows on such assets are generally accounted for on an accrual basis, the provisions will be calculated forward in time, which is more akin to a fair value approach. This ties in to the general philosophy of MVE, which also looks to incorporate the prevailing market conditions. However, if MVE results have already been adjusted to account for the credit quality of the borrower, this overlap with IFRS 9 renders the provision effectively already incorporated. This is complicated somewhat if a backward-looking PD is used for IRRBB purposes with a forward-looking view for IFRS 9, as there will be a timing mismatch between the two with regard to when provisions are raised compared with when defaults occur.

Nevertheless, if these metrics are both live, the consequence of hedging the interest rate risk in the IFRS 9 provisions as well as a credit adjusted valuation would result in increased valuation volatility rather than a reduction, due to this double counting. Hence, any hedging should reflect either the IFRS 9 provisions without consideration of credit-spread-adjusted MVE figures or the credit-spread-adjusted MVE figures without consideration of the IFRS 9 provisions.

Another point to consider when taking credit spreads into account is the need for consistency with the modelling assumptions of the prepayment, whether or not a prepayment option exists, and the consideration of the recovery rate. For example, if the recovery behaviour is considered jointly with the prepayment in the cashflow estimation, then the credit-default-adjusted model should assume a zero recovery.

VALUATION

Since the value of the expected cashflows of an asset position in the balance sheet can be calculated in terms of the equivalent bond, the value of an expected cashflow that incorporates default risk can be valued as the equivalent defaultable bond. In this context, credit spread could be defined

as the difference (in terms of yield) between a risk-free bond and a debt security with the same characteristics but with default risk.

In the literature on credit risk there are several papers on default bond valuation, with two main approaches: structural models and reduced-form models. The structural models approach (see, for example, Merton 1974; Nielsen *et al* 1993; Longstaff and Schwartz 1995) relies on the value of the bond issuer's assets and the subordination structure of the issuer's liabilities. Due its nature, this is difficult to extend to the valuation of the asset balance-sheet items.

On the other hand, the reduced-form models can naturally be extended to the valuation of this type of product. This approach started with the work of Jarrow and Turnbull (1995), Lando (1998) and Duffie and Singleton (1999),⁴ and ignores the mechanism that makes a company default. The model only considers the moment an event takes place using a random variable, subject to stochastic modelling.

Under the reduced-form approach it is possible to estimate the value of a defaultable bond in terms of the value of a non-risky bond. The formulation could be different depending on the recovery assumption (ie, recovery of face value, recovery of market value or recovery of treasury value). For simplicity, to express the value of the risky bond, the recovery is considered here in the form of market value (Duffie and Singleton 1999) as a percentage of the pre-default market value of debt.

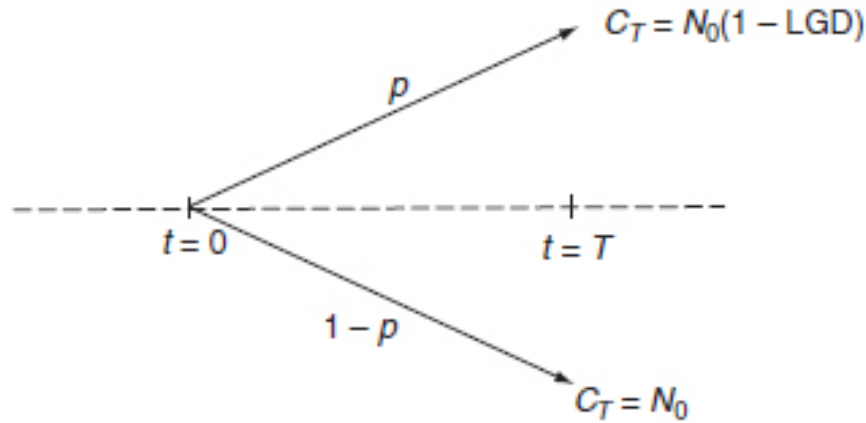
If the value of a generic fixed-income asset (without default risk) is considered, the value could be defined by the amortisation schedule, the outstanding principal and the fixed interest rate that determines all coupon payments. In the case of a floating-income asset, the coupons should be determined by the forward interest rates. In both cases, the value is usually expressed as the present value of the future cashflows, and the value of a zero-coupon bond could equivalently be expressed as the discounted cashflow at the risk-free rate.

If a defaultable bond is considered, the issuer may default with probability p , and the investors who purchased the bond receive an amount of the recovery (1 minus the loss given default (LGD)). If, on the other hand, the issuer does not default, the investors receive the full amount N with probability $(1 - p)$ (Figure 11.2).

Under the risk-neutral measure Q , the valuation of a non-defaultable claim X may be written in terms of risk-neutral valuation as

$$V(t, T) = \mathbb{E}_t^Q \left[X \exp \left\{ - \int_0^T r_t dt \right\} \right] \quad (11.1)$$

Figure 11.2 Default diagram



Notes: $V_t = pe^{-r_t}(1 - \text{LGD}_t)N_0 + (1 - p)e^{-r_t}N_0$.

On the other hand, the valuation of a risky claim may be expressed as the weighted average of the present value of the expected payment at t (ie, X) if the claim survives, and as the recovery amount in the case of default, where the weights are equal to the probability of default or survival

$$V(t, T) = \mathbb{E}_t^Q \left[S(T)X \exp \left\{ - \int_0^T r_s ds \right\} \right] + \mathbb{E}_t^Q \left[F(T)Z_\tau \exp \left\{ - \int_0^T r_s ds \right\} \right] \quad (11.2)$$

Here $S(t)$ is the survival probability, $F(t)$ represents the probability of default on or before time t , Z_τ is the recovery function and τ is the time of default.

Assuming a recovery, as a percentage of the market value prior to default, the above formula may be simplified to

$$\begin{aligned} V(t, T) &= \mathbb{E}_t^Q \left[X \exp \left\{ - \int_0^T (r_s + L_s \lambda_s) ds \right\} \right] \\ &= \mathbb{E}_t^Q \left[X \exp \left\{ - \int_0^T R_s ds \right\} \right] \end{aligned} \quad (11.3)$$

where λ_t is the hazard rate (the conditional probability at time t of a default between t and $t + 1$ given no default by t), and L_t is the LGD ($1 - L_t$ is the percentage recovered in the event of default).⁵

The price of a claim with default risk can be expressed as the present value of the promised payoff (X) treated as if it were default-free, discounted by the default-adjusted short rate R_t , as in standard valuation models for default-free securities, but discounted with the default-adjusted rate $R_t \approx r_t + \lambda_t L_t$.

Assuming that, under the risk-neutral probability of default, our valuation only needs to consider the credit spread and no other spread (such as liquidity or funding) to model the cashflows, considering a default rate when the expected cashflows are calculated or incorporating a credit spread in the discount should be equivalent.

To calculate the spread when an implicit hazard rate cannot be calibrated, it is usual (see, for example, Hull 2000, Chapter 26) to approximate the credit spread via historical data with the calibrated PD and recovery given by

$$Ne^{-R_t t} = (1 - \text{PD})Ne^{-r_t t} + \text{PD}N\text{recovery}e^{-r_t t} \quad (11.4)$$

$$\text{PD} = \frac{1 - e^{-(R_t - r_t)t}}{(1 - \text{recovery})} \cong 1 - \exp \left\{ -\frac{(R_t - r_t)t}{1 - \text{recovery}} \right\} \quad (11.5)$$

$$\text{spread}_t = R_t - r_t = \ln(1 - \text{PD}) \frac{(\text{recovery} - 1)}{t} \quad (11.6)$$

Finally, with respect to the bond portfolios (and as is usually the case for the investment portfolio), a more sophisticated model could be considered to take into account the rating migration risk along with default risk and the correlation between defaults. In fact, losses caused by credit migrations combined with the widening of credit spreads were more common during the 2007–9 crisis than the losses raised for actual defaults. A model based on Monte Carlo or other numerical methods could be developed to analyse the effect of the credit spread based on different systematic factors. This type of model could be especially useful for stress testing, to analyse the sensitivity of the market value of the portfolio under different scenarios.

Although modelling the default correlation in the case of mortgages or consumer loans may be complicated and less accurate, it may be necessary to consider a correlation between interest and credit risk, especially in the

context of stress tests for products with a high default rate. For this purpose, the default rate considered in the cashflow estimation, or the credit spread directly, should depend on different variables to take into account the relationship between the risks and their dynamics. Some examples of how to measure credit and market risk for the whole portfolio can be found in Barnhill and Maxwell (2002), Barnhill *et al* (2002), Jobst and Zenios (2001), Jobst *et al* (2003, 2006), Alessandri and Drehmann (2010) and Drehmann *et al* (2006).

A COMPARISON OF DIFFERENT APPROACHES TO INCORPORATE DEFAULT RISK

The impact of credit spreads on the market value of equity may be modelled with several approaches. One approach is to use a hazard rate model, adjusting the cashflows by the defaults and recovery as shown in the previous section. An alternative would be to simply adjust the discount rate by the corresponding credit spread. These may be made equivalent by converting the probability of default and recovery via Equation 11.6, assuming a one-year PD.

There is typically a high recovery rate on prime mortgages, resulting in a reduced equivalent credit spread. Within the model, a post-default recovered asset is set at par, and any unearned interest payments (assuming funding rates are less than mortgage rates) are forfeited. Consequently, the MVE is broadly similar for the two approaches. However, interestingly, the corresponding sensitivities will differ. Within a cashflow model, a default results in the termination of the mortgage, reducing the outstanding balance and resulting in a shorter duration. When modelled on a credit-spread-adjusted basis, the reduction in spread from the high recovery rate reduces this impact, resulting in a longer profile.

This relationship is displayed in [Table 11.1](#), which shows how the duration will vary as a function of recovery rates and PDs for a fixed-rate mortgage. The duration of the cashflow model remains fairly static with respect to the recovery rate, but varies more in relation to the PD. Recoveries here will increase both the value and sensitivity of the asset, but will reduce the duration, as a greater proportion of its value is paid off earlier.

As the credit-spread model simply incorporates any recovery as a reduced spread, it fails to capture the reduced lifetime of the asset (as it ignores any lost interest). Thus, due to this omission the credit spread model overstates the sensitivity. Therefore, the duration grows as recovery increases, reflecting the fact that this method does not distinguish the reduced lifetime of the asset.

Of course models may be refined further by adjusting for the correlation between the credit variables and interest rates. When reviewing the macroeconomic components of default drivers the key variables are unemployment rates, changes in house prices (through to loan-to-value (LTV)) and changes in GDP. Fixed-rate mortgage borrowers are generally less affected by any changes in interest rates than variable rate payers, who are more exposed. As interest rate exposure is larger for fixed rate products than for variable ones, this inverse relationship means the introduction of a rate-dependent PD has a smaller impact on IRRBB. However, changes in interest rates may arise alongside changes in other variables.

Table 11.1 Modified duration as a function of recovery/PD

(a) Modified duration (years) of cashflow model					
Recovery (%)	Probability of default (1Y)				
	2%	4%	6%	8%	10%
20	6.03	5.54	5.09	4.69	4.32
40	6.00	5.50	5.04	4.63	4.26
60	5.98	5.46	4.99	4.58	4.21
80	5.95	5.42	4.95	4.54	4.18
100	5.93	5.39	4.92	4.51	4.14

(b) Difference in modified duration (years): credit spread model minus cashflow model					
Recovery (%)	Probability of default (1Y)				
	2%	4%	6%	8%	10%
20	0.12	0.22	0.30	0.36	0.40
40	0.24	0.45	0.62	0.75	0.85
60	0.37	0.68	0.94	1.16	1.33
80	0.50	0.92	1.28	1.59	1.85
100	0.62	1.16	1.63	2.05	2.41

Notes: constant prepayment rate = 5%; mortgage rate = 5%; funding rate = 4%; maturity = 20 years; rate shift = 10bp.

For instance, higher interest rates may result in a contraction in house price growth, denting the LTV on the property, which decreases the borrower's disincentive to avoid default, as well as undermining any subsequent recovery by the creditor. In addition, changes in interest rates may well occur as a monetary policy response to contractions in the economy, staving off adverse changes in unemployment rates, or as a measure to counteract currency movements, with unclear consequences.

Thus, when incorporated together with lagging effects (which are generally significant), the value of factoring in such a relationship between interest rates and credit data for estimating MVE sensitivity is questionable. However, interest rate changes are more important for stress testing requirements, and for determining any impact on capital, particularly in the eventuality of IRRBB being deemed within the scope of a Pillar 1 capital charge.

Table 11.2 OAS as a function of CPR/CDR plus recovery

(a) OAS (%): recovery 20%					
CPR (%)	CDR				
	2%	4%	6%	8%	10%
2	1.54	2.91	4.24	5.50	6.71
4	1.48	2.71	3.88	5.01	6.08
6	1.43	2.53	3.59	4.60	5.57
8	1.38	2.38	3.33	4.25	5.13
10	1.35	2.25	3.12	3.96	4.76
CS	1.62	3.27	4.95	6.67	8.43
Gradient	-2	-8	-14	-19	-24
(b) OAS (%): recovery 80%					
CPR (%)	CDR				
	2%	4%	6%	8%	10%
2	0.51	0.86	1.16	1.42	1.64
4	0.56	0.88	1.15	1.38	1.59
6	0.60	0.89	1.13	1.35	1.54
8	0.64	0.90	1.12	1.32	1.50
10	0.67	0.90	1.11	1.30	1.46
CS	0.40	0.82	1.24	1.67	2.11
Gradient	2	0	-1	-2	-2

Notes: mortgage rate = 5%; funding rate = 4%; maturity = 20 years.
CPR, constant prepayment rate. CDR, constant default rate.

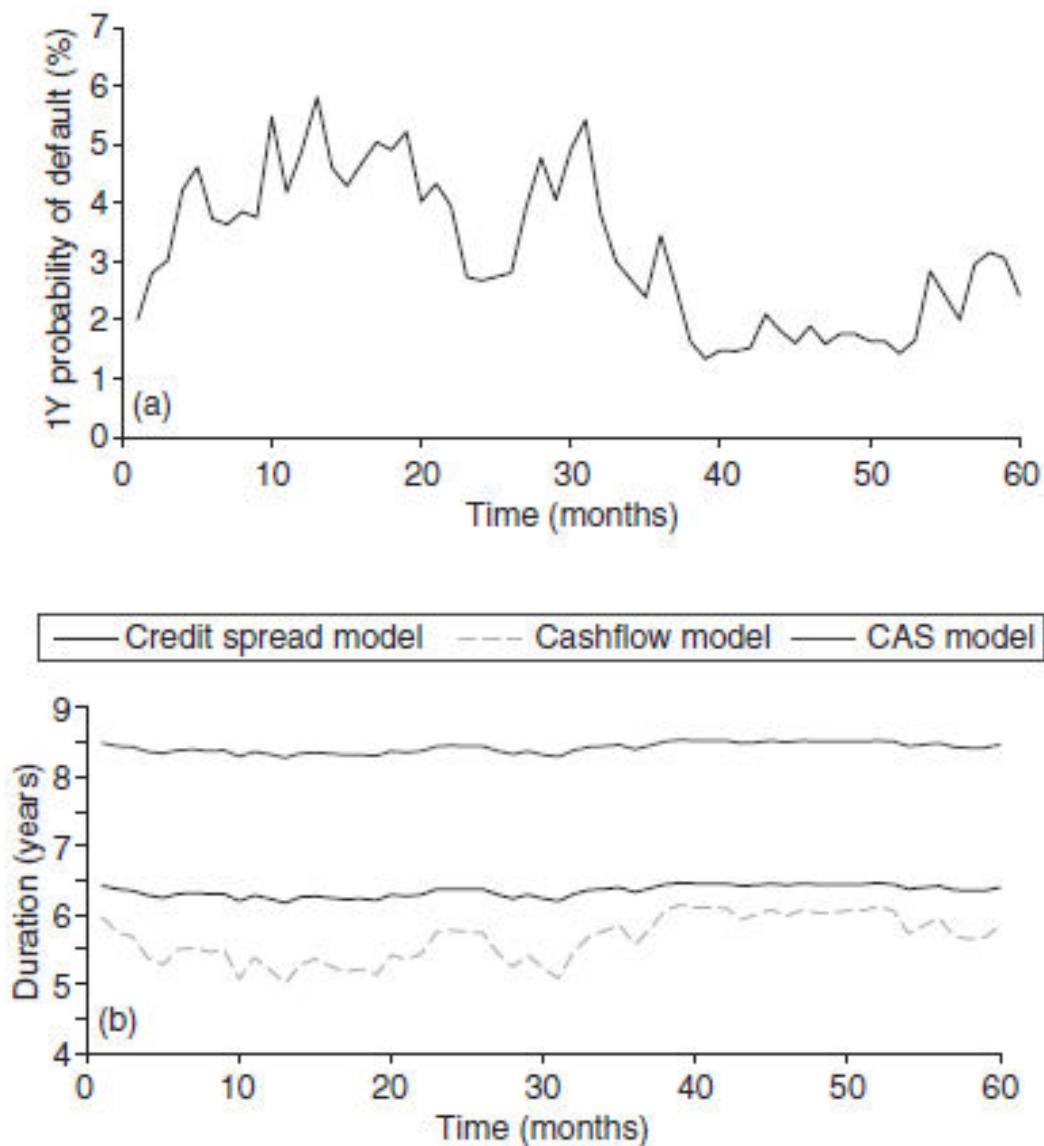
Alternatively, these results may be compared with an option-adjusted-spread (OAS) model framework. Here, the spread will incorporate both credit and prepayment aspects only via an adjustment to the discount rate on the contractual cashflows. Table 11.2 shows how the OAS may vary with prepayment/default rates. A monotonic relationship between default and OAS may be readily identified, reflecting the deterioration in credit quality that requires compensation via an increased OAS. When viewed against prepayments, the OAS will compensate for any forfeited margin, reflecting the differential between funding and mortgage rates. However, when viewed against defaults, this effect is cancelled out, as prepayments additionally mitigate against credit losses, resulting in a non-monotonic relationship.

Table 11.3 Comparison of modelling frameworks

Model	MVE (%)			Modified duration
	Base	−(0.1%)	+(0.1%)	
Non-credit	106.64	0.70	−0.70	6.55
Credit spread adjusted	103.20	0.67	−0.66	6.42
Cashflow PV	102.97	0.62	−0.61	5.96
OAS model	102.97	0.88	−0.87	8.48

Notes: PD = 2%; constant default rate = 2%; constant prepayment rate = 5%; mortgage rate = 5%; funding rate = 4%; recovery = 75%; maturity = 20 years; rate shift = 10 bp.

Figure 11.3 Impact of simulated PD on duration of a 20-year mortgage



Notes: (a) Simulated probability of default. (b) Simulated duration by model. Constant prepayment rate = 5%. Mortgage rate = 5%. Funding rate = 4%. Recovery rate = 75%.

As the OAS model only reflects the forfeited income or P&L consequence of a default, such a model is not sensitive to the reduction in lifetime that prepayments or defaults generate. Thus, the OAS model has a substantially longer duration, as highlighted in [Table 11.3](#), which compares the three modelling frameworks.

To compare the relative performance of the different modelling frameworks we simulated a hypothetical time series of PDs. Assuming the other market data remained constant, the duration of a fixed-rate mortgage was assessed for credit-spread-adjusted, cashflow (hazard rate), stochastic and OAS models, as shown in [Figure 11.3](#).

As identified via the above results, there is reduced variation in both the OAS and credit spread models over time, with greater variation in the cashflow model approach. This additional sensitivity may result in more noise when consolidated into the portfolio-level reported IRRBB metrics.

CONCLUSION

Valuation of the CSRBB poses several challenges to the effective incorporation of the effects of default risk. We showed that the choice of method to capture the credit risk may lead to different duration values. The preferred implementation could depend on the availability of the data and/or the tolerance for variation in the metrics. For example, for a product with clear market prices, the best approach would be an OAS, while for most products without market prices a more practical approach would be to use credit spread curves for sector, rating or geography. However, for portfolios with a high default risk, the cashflow estimation method may be the most suitable, as the tolerance should be smaller than that for products with a low default risk.

The procedure to calibrate the additional spread incorporated over the risk-free rate presents further challenges. In the absence of market prices, one possibility could be to use the historical probability of default and recovery. Another proxy could be the prevailing market spread of new business. Additionally, incorporating credit events into any modelling framework suffers from the problem of partitioning the portfolio according to the corresponding credit quality. This may be feasible via LTV bands tied to pre-existing credit reports (where possible), or via inference from traded residential mortgage-backed securities data. In the absence of such data, an alternative approach may be to use the prevailing lending rates as a proxy; these should be adjusted by the appropriate issuer spread.

Recall that, while credit spreads are important in the valuation of the balance sheet, other factors, such as funding spreads, which may have a material impact, are also pertinent. Thus, reflecting the cost of credit on the

balance sheet onto assets without considering the impact on adjustments to liabilities may introduce additional volatility.

Furthermore, particular care should be taken to avoid any double counting with existing credit capital requirements on any assets introduced into the capital framework, but these metrics can be used to help assess or monitor CSRBB.

The views and opinions in this chapter are those of the authors and may not necessarily reflect those of their employer.

- 1 See, for example, Standard & Poor's "Annual Global Corporate Default Study and Rating Transitions" or Moody's "Annual Default Study: Corporate Default and Recovery Rates". The 2016 studies are available at <https://www.spratings.com/documents/20184/774196/2016+Annual+Global+Corporate+Default+Study+And+Rating+Transitions.pdf> and <https://www.moody.com/researchdocumentcontentpage.aspx>, respectively.
- 2 Jarrow and van Deventer (1998) show that, in terms of hedging a bond portfolio, both credit and interest rate risk must be taken into account. Grundke (2005) finds that significant errors are made when the correlated nature of rating transitions, credit spreads, interest rates and recoveries is ignored.
- 3 Delinquent loans are considered to be those which have a payment that has been overdue for more than, say, three months.
- 4 Several later works extended structural and reduced-form models to incorporate stochastic volatility, jumps, rating migration and stochastic recoveries (see, for example, Schönbucher 2003; Duffie *et al* 2003; Lando 2009). Although the literature has more recently focused on the CVA perspective, some authors have considered this type of model from alternative angles (see, for example, Egami *et al* 2013; Bielecki *et al* 2014; Sanjiv and Kim 2015; Tapiero and Vallois 2017).
- 5 Although the recovery rate could be stochastic, for simplicity it is usually considered to be constant; thus, under this approach it is not possible to estimate the recovery rate and the hazard rate function separately, and the calibration requires one fixed variable, usually the recovery.

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Hedge Accounting

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The hedge accounting principles are special accounting rules that were incorporated into International Accounting Standard 39 (IAS 39) to eliminate valuation asymmetries resulting from the specific valuation principles in the four categories of financial instruments in the original standard. Hedge accounting is applicable for hedges of financial instruments in the IAS 39 categories “loans and receivables”, “held to maturity” and “available for sale” with derivative instruments as hedging instruments. Although hedge accounting was introduced to consider risk management effects in financial accounting, it was characterised by many administrative obligations and limitations that restricted the use of the hedge accounting rules in banks.

In 2009 a project was set up to replace the IAS 39 by International Financial Reporting Standard 9 (IFRS 9). Hedge accounting rules in IFRS 9 were the third part of the replacement of IAS 39 after classification and measurement and the replacement of the impairment rules by expected credit loss. The final version of the hedge accounting rules for IFRS 9 was published on July 24, 2014, and endorsed by the European Union on November 29, 2016 (European Commission 2016, henceforth IFRS 9).

Classification of financial assets in IFRS 9

To understand the changes in hedge accounting it is necessary to give a short overview on the classification rules in IFRS 9. The new standard distinguishes four categories of financial assets (see IFRS 9, Paragraph 4.1).

1. Debt instruments recognised at amortised cost (comparable to former loans and receivables).
2. Debt instruments at fair value through other comprehensive income (OCI): these instruments cumulate gains and losses in OCI. Gains and losses are reclassified to the profit and loss (P&L) account upon derecognition (comparable with the former available for sale).
3. Debt, equity instruments and all derivatives measured at fair value through the P&L account (trading).
4. Equity instruments measured at fair value over OCI without recycling in the P&L account.

A category for bonds such as “held-to-maturity” is not part of the IFRS 9 classification.

The classification of financial assets in IFRS 9 is based on two criteria (see IFRS 9, Paragraph 4.1.1):

1. the bank’s business model for managing the financial assets;
2. the individual contractual cashflow characteristics of the financial assets.

A debt instrument, which in most cases will be the hedged item within the hedge accounting application, can be recognised at amortised cost or at fair value through OCI. The objective of holding debts at amortised cost is to collect the contractual cashflow. The cashflows are solely payments of principal and the interest on the principal amount that is outstanding.

A debt instrument is normally measured at fair value through OCI when the assets are held in a business model with the object of collecting contractual cashflows and selling financial assets. The contractual terms of the financial assets give rise solely to payments of principal and interest on the principal amount outstanding.

For hedge accounting, only financial instruments recognised at amortised cost or at fair value through OCI can be designated hedged items. Derivative

instruments other than those in IAS 39 can also be part of a group of hedged items.

Liabilities

No separate category exists for liabilities (see IFRS 9, Paragraph 4.2), which generally have to be measured at amortised cost. Exceptions are trading liabilities and liabilities at fair value options. These instruments have to be measured at fair value through P&L. If the bank's own credit spread is considered in the fair value measurement, its effect on the change in fair value must be disclosed separately. The measurement of liabilities according to IFRS 9 is the same as that under IAS 39.

Hedge accounting

The changes in the hedge accounting rules were driven by the need to simplify them and to apply them to risk management methods. These were accompanied by

- an extension of hedged items (ie, hedging of a group of hedged items including derivatives) to include real hedges in a documented hedge account, which was not permitted under IAS 39,
- a simplification of the process (*ex post* efficiency tests are no longer required, and the 80–120% range for hedge effectiveness was removed),
- recognition of the hedging costs in OCI as well as the P&L account, which reduces the volatility of the P&L account,
- documented hedge accounting for product groups and net position,
- the integration of hedge accounting into the high-level risk strategy (violation of the risk strategy may compromise hedge accounting),
- more disclosure requirements than for IAS 39 hedge accounting.

But why does the hedge accounting process need to be dealt with in a handbook on asset and liability management? Of course, the principles for the recognition and measurement of financial instruments are also the principles of accounting. But, the reclassification of the categories under

IFRS 9 is very limited, and this means the treasury function must also be aware of the classification principles for financial instruments, and bear in mind the key steps in the hedge accounting process.

Furthermore, IFRS 9 focuses on the rule-based use of hedging instruments and techniques as a requirement for a documented hedging account. This moves more responsibility to the treasury function as an influential part of the hedging process. The requirements to align hedge accounting with the real risk management in a bank (or vice versa) are far more stringent than under IAS 39. For example, the voluntary termination of a documented hedge relationship is no longer permitted without severe changes in the risk management strategy.

In this chapter we explain the most important principles for designated hedge accounting according to IFRS 9 and give some examples of their application in a bank. The changes in the financial markets with respect to risk assessment and valuation principles during the global financial crisis had a considerable effect on the regulations for hedge accounting.

IFRS 9 distinguishes a general hedge accounting model, which is valid without any restrictions. Although there are fundamental changes in the hedge accounting rules under IFRS 9, the general mechanics remain unchanged from IAS 39 (BDO 2014, p. 6):

- the new standard retains accounting models for fair value hedge, cash flow hedge and net investment hedge;
- hedge effectiveness should be measured and any ineffectiveness must be recognised in P&L;
- hedge documentation is still required;
- hedge accounting will remain optional;
- only external deals can be designated hedged instruments.

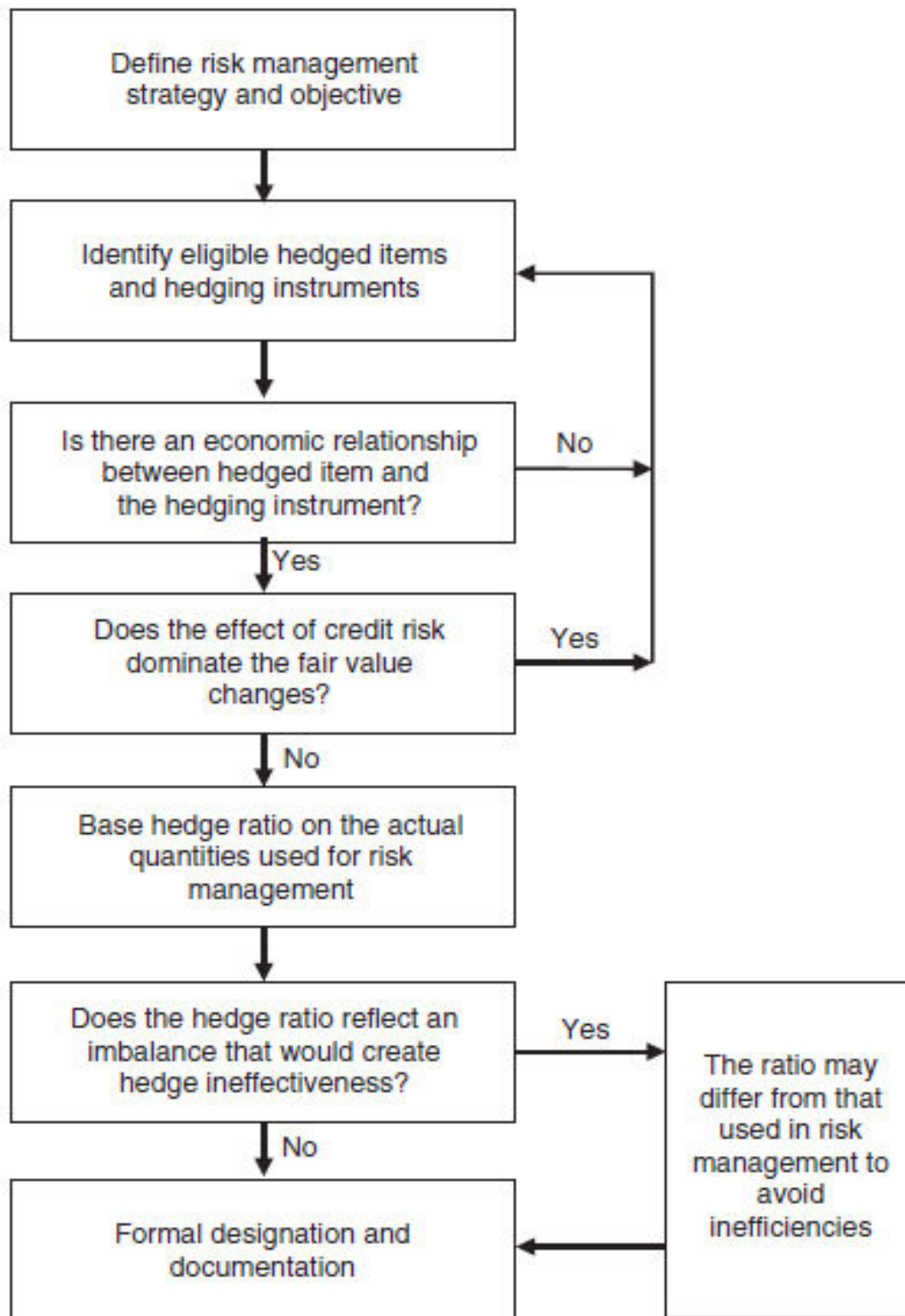
At the time of writing, the macro-hedging model (where amounts of both the hedging instrument and the hedged item can change constantly) was still being deliberated by the International Accounting Standardising Board (IASB). In this chapter, we therefore focus on the final “general hedge accounting model”. [Figure 12.1](#) gives an overview of how hedge accounting can be achieved.

RISK MANAGEMENT

Risk management strategy

All enterprises are exposed to risks that affect their cashflows and the market value of their financial instruments and liabilities. For the purpose of hedge accounting it is necessary to describe the expected or experienced risks and their respective risk management in a risk management strategy at the highest management (board) level (Ernst & Young 2014b, p. 10). Enterprises fix their risk management strategy in the form of risk management guidelines or policy within the body of their rules and regulations. The documentation of types of risk and their management should be permanent but should contain some elements of flexibility to adapt to changes in the environment.

Figure 12.1 Achieving hedge accounting



Source: Ernst & Young (2014a p. 31).

Table 12.1 Interaction between risk management guidelines and the risk management objectives

Risk management strategy	Risk management objective
The amount of fixed-rate loans should not exceed 40% of the total loan amount	Designation of a payer swap to hedge the fair value of a fixed-rate loan with maturity of five years (€100 m)
All currency risks over a 12-month horizon have to be hedged into the base currency (euro)	Designation of a cross-currency swap to turn sterling payments later than 12 months into euro to hedge the currency risk of a shipping loan

Source: Ernst & Young (2014b, p. 10).

The risk management guidelines are also part of the disclosure according to IFRS 7 and enable shareholders and stakeholders to review the risk management strategy and objectives.

Risk management objectives

The risk management objectives are defined for every hedge relationship. Each hedging objective has to be aligned to the high-level risk strategy described in the risk management guidelines. A risk management objective that does not conform to the risk management strategy can prevent the accounting of the documented hedge or, in existing documented hedge groups, a change in the risk management objective can force the bank to terminate a documented hedge relationship that affects the P&L account.

Examples of the interaction between risk management guidelines and the risk management objectives are shown in [Table 12.1](#).

As a result of all the changes from IAS 39 to IFRS 9, the hedge accounting rules have moved closer towards risk management techniques. Risk managers (eg, the treasurer) are responsible for meeting the strict requirements of the risk management strategy and the risk management objectives to allow the bank to take advantage of the new and more liberal rules fixed in IFRS 9.

A broad range of hedged items

One of the most important improvements is the extended range of hedged items, which allows a lot more hedge relationships to be documented under the hedging account in order to reduce the P&L volatility. The main changes are hedging of risk components, of groups of hedging items, of net positions including derivatives and of aggregated risk positions. Furthermore, under certain circumstances, hedging of inflation risk and credit risk is allowed.

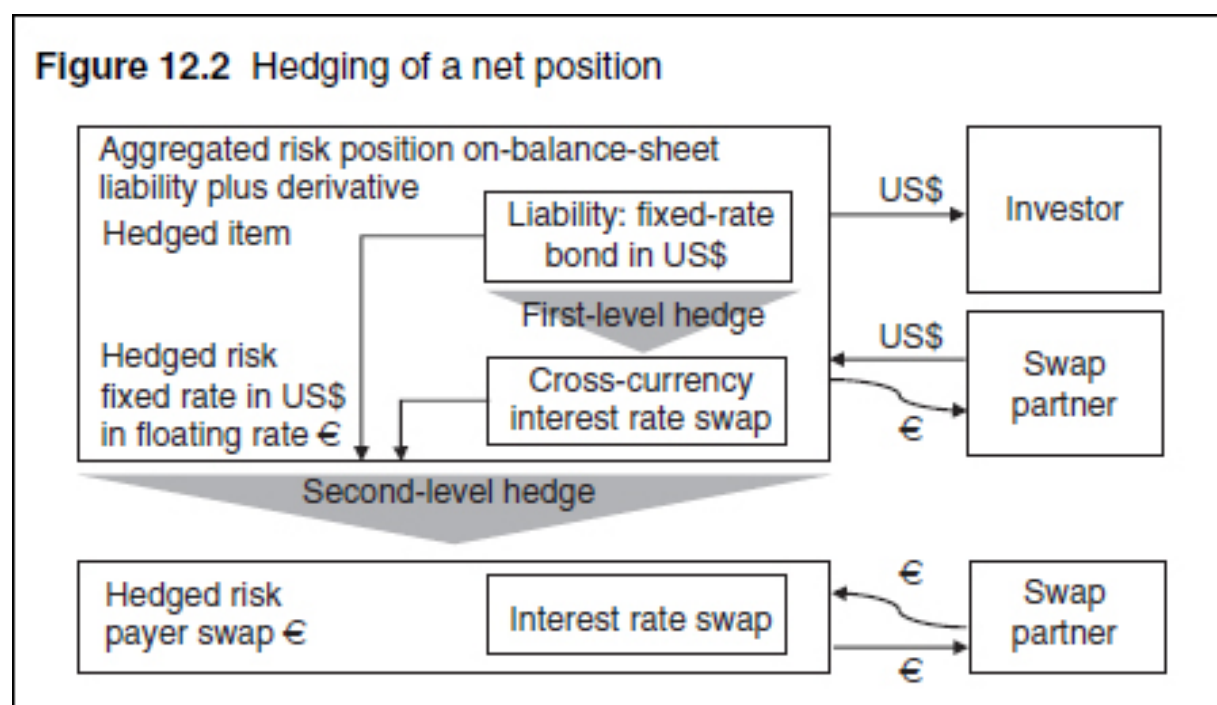
More than one pricing factor

For contracts with more than one pricing factor it is possible to hedge a single specific risk of one pricing factor without hedge inefficiency (IFRS 9, Paragraphs B6.6.1–10). If the hedged item is exposed to more than one risk factor, a hedging with derivatives causes hedge ineffectiveness because the fair value of the hedged item fluctuates more than the value of the derivative due to the former's exposure to risk factors that are not hedged by the derivative. Risk managers hedging different types of risk separately often relinquish hedge accounting due to expected poor hedge effectiveness.

Under IFRS 9 it is possible to assign the specific risk from the hedged item to the hedging instrument to reach a better hedge effectiveness with a corresponding reduction in P&L volatility. Every component that is intended to be hedged needs to be separately identifiable and reliably measurable. In the case of a contractual market factor, the relationship between the change in the risk factor and the fair value change of the hedged item is fixed in the contract. If a non-contractually specified risk component is used in a documented hedge accounting, an analysis of the link between the risk component and the hedged item will be necessary to prove the impact of the risk component on the fair value of the hedged item.

Risk factors can be assigned individually if they are separately identifiable and reliably measurable. The basis risk inherent in the valuation of cross-currency swaps exemplifies the effect of a second pricing factor when hedging an on-balance-sheet position in a foreign currency with a cross-currency basis swap. In this case the basis represents the difference between the liquidity premiums of the two currencies involved and is added to the floating rate of one of the legs. The basis risk is driven by liquidity premiums rather than interest rate differentials. Under IAS 39 basis risk in cross-currency basis swaps could cause hedge ineffectiveness. The

designation of basis risk as a second price factor facilitates the cross-currency basis swap in documented hedge accounting.



Net positions including derivatives: aggregated risk position

Risk managers often manage their interest rate risk and foreign exchange (FX) risk separately. With the introduction of hedging of aggregated risk positions, including derivatives, individual risk management is now widened to include documented hedge accounting in order to further reduce P&L volatility.

The function of hedging an aggregated risk position is illustrated in [Figure 12.2](#) and by the following example.

Example 12.1.

1. A European bank with euro as its local currency issues a fixed-rate bond denominated in US dollars with a fixed-rate coupon of 5% per year and with a maturity of 10 years.
2. The bank wants to hedge the currency risk from the fixed US dollar coupon payments. The bank expects decreasing interest rates and decides to swap the fixed-rate US dollar coupons into floating-rate payments to reduce interest payments when market rates decline.

3. Consequently, the bank enters into a 10-year cross-currency swap with the result that the it receives annual fixed US dollar payments and pays floating interest in euro. This is the first level of risk hedging.
4. After two years, the bank's interest rate expectations change and it now expects rising euro rates, which will increase its interest payment.
5. With a payer interest rate swap, the bank turns the floating-rate interest payment from a bond into a fixed-rate interest payment. This is the second level of risk hedging.
6. The hedged item is a net position from a fixed-rate liability in US dollars and a cross-currency swap to turn fixed US dollar interest payments into floating euro interest payments.

In this case the second-level risk hedging is set up on the net position, consisting of the US dollar bond in combination with the cross-currency interest rate swap. The hedge of aggregated risk positions moves hedge accounting over to real risk management, and banks are able to reflect their risk management activities better in the financial statement.

The designation of a net position and hedging on an aggregated level is preferred when all risky transactions are netted in the first level and the remaining risk position is hedged in the second level. Under IFRS 9 it will no longer be necessary to identify any single transaction as a substitute transaction representing a hedged item in order to apply documented hedge accounting (Ernst & Young 2014b, p. 17).

Group hedging

Hedging accounting for a portfolio of securities with an index contract according to IAS 39 often failed because the securities in the basket of the index did not show the required price move in proportion to changes in the index price. In IFRS 9 the requirements have been lowered. A gross position of securities can be hedged by an index contract in the case when every item (including components of the item) of the portfolio may be an accepted hedged item, and the portfolio items will be managed as an entire group for the purposes of risk management.

It is no longer necessary that the prices of the individual securities move proportionally to the group or index price. The hedge of an equity portfolio by shortening an equity index contract can be designated as documented

hedge relationship and reduce P&L volatility. Any recycling of gains and losses from hedging instruments into P&L is presented as a separate line item. Accordingly, the risk management of the whole group, the gains and losses are not recorded for the related individual line items.

Hedging of layers: hedging of tranches

Another new rule in IFRS 9 (see Paragraphs B6.6.11–12) is the allowance to hedge only parts of a total position (layer). The hedged part of the total volume can be nominated as a percentage of the total position. In this case it is necessary to specify the total position. Another method is to quantify the volume part, which should be hedged as a nominal amount. Derived from the rule above, in the latter case it is not necessary to specify the total position itself.

For banks this rule gives the opportunity to hedge only that part of a total loan position for which the bank has shortened a call option, or for which the bank expects a prepayment independent of the call rights for the borrower. This allows a bank to hedge call rights in the loan volume for only a part of the total loan volume, and to hedge linear interest rate risk with plain vanilla interest rate swaps. Both parts are appropriate for documented hedge accounting and will reduce P&L volatility. The following example explains the hedging of a layer to limit risk from loan prepayments (see Ernst & Young 2014b, p. 29).

Example 12.2.

1. A bank grants a five-year fixed-rate loan of €100 million. Thirty percent of the loan amount has a call right to prepay the loan at the fair value (€30 million). The bank expects that the call rights will be exercised only for one-third of the volume for which call rights exist (€10 million). The funding of the loan is a floating-rate liability.
2. To hedge the interest rate risk from the loan, the bank enters into a payer swap for that portion of the loan for which a prepayment is not expected. This is 90% of the total loan volume (€90 million).
3. For €20 million of this hedge the bank has an open option position, because this part of the loan can be prepaid. But, the bank does not expect that prepayment will occur on this volume tranche. To hedge the

risk from the shortened option (call right) and to hedge future interest income in case the call rights are exercised, the bank enters into a swaption for €20 million to retain the interest income for the periods after the call rights have been exercised and the loan is partly repaid.

4. The top layer of €10 million remains unhedged because the bank expects early redemption payments.
5. All derivatives can be integrated into the documentation for hedge accounting. The volume of the swaption corresponds to the volume of the borrower's call rights. The interest rate payer swap retains the interest rate margin for the volume layer, which is not expected to be prepaid at all.

Hedging inflation risk

Inflation risk can be considered as a separate risk factor if contractually agreed. The hedge of inflation risk is then part of the hedge accounting documentation. But even in the case that inflation risk is not contractually fixed, inflation can be designated as a separate risk factor for markets or countries where a broad range of inflation adjusted securities is available, which enables the impact of inflation on the market interest rates to be measured separately. For example, hedging of inflation risk can be useful for banks in the eurozone, where the interest rate market curves are very similar but inflation rates differ considerably between countries (see Ernst & Young 2014b, p. 23).

Hedging credit risk

Under certain circumstances IFRS 9 enables documented hedge accounting for credit positions and credit derivatives. To designate a documented hedge accounting relationship, it is necessary that credit risk be managed within the risk management strategy of the bank. In this environment, loans or committed lines can be measured at fair value. If the name of the debt issuer and the seniority or range of the asset in the credit derivative is exactly the same as that of the hedged item, the credit derivative can be used as a hedging instrument in the hedge accounting process. However, the measurement of the credit derivative will not be changed because, as a derivative, the credit derivative will be measured at fair value in any case.

As an alternative, the bank could draw the fair value option for the managed item and have the hedge effects from the hedged item measured at fair value and the credit derivative directly in the P&L account. Nevertheless, hedge accounting has some advantages: it can start after the first recognition of a deal and does not have to set up at first recognition; the choice of integration into hedge accounting can be applied only to a part of the position; and the fair value measurement can be stopped if desired (Ernst & Young 2014b, pp. 38–9).

Hedging basis risk

Hedging an on-balance-sheet position in foreign currencies with cross-currency swaps often causes inefficiencies due to the use of basis risk as a pricing factor in cross-currency swaps, which is not found in the foreign currency on-balance-sheet position. The basis risk is the differential of the liquidity premium that is usually added to one of the floating legs. Under IAS 39, the basis risk was often a source for inefficiency when hedging currency risk with cross-currency swaps, with the result that some hedging relationships had to be terminated. Depending on the market value of the on-balance-sheet position, the unwinding of a documented hedge affects the P&L account. Basis risk is regarded as the cost of hedging and, if the risk management of the position is strictly in line with the risk management strategy and risk management objectives, the hedge inefficiencies from changes in basis risk can be recognised in the OCI and will not lead to further P&L volatility.

Some new rules for hedging instruments

The changes for hedging instruments in IFRS 9 compared with IAS 39 are less extensive than those for hedged items. Regarding the range of hedging instruments two amendments should be mentioned.

1. It is no longer the case that only derivative instruments are permitted to be used as hedging instruments. All non-derivative financial instruments measured at fair value through P&L are applicable as hedging instruments, independent of whether the fair value measurement is mandatory or opt-in by fair value option. Non-derivative instruments are now allowed to hedge another risk, not just FX risk. A combination

of a non-derivative financial instruments and derivative instruments is also allowed, as is a partial designation of both instruments.

The only two exceptions are the following.

- Liabilities measured at fair value, for which the price effect of the bank's own credit spread is recognised in OCI: for these debt instruments the full fair value change is not captured in the P&L account, and therefore they cannot be used as hedged items.
 - The net position of written options: these options can be used as hedging instruments only against a long option position.
2. Embedded derivatives no longer have to be separated from the basis instrument. For this reason embedded derivatives cannot be used as hedging instruments. On the other hand, financial basis instruments including the embedded derivative can be designated as hedge instruments when measured at fair value through P&L.

For other hedging instruments such as options, forwards and cross-currency swaps, the booking logic has changed. Under IFRS 9 options are allowed as hedge instruments, but in IAS 39 only the intrinsic value of an option could be designated as hedge instrument. The corresponding time value had to be recognised directly in the P&L account. From this treatment, long running option contracts with a large portion of the price allocated to the time value results in a poor hedging performance. According to IFRS 9 the time value of an option as a hedging instrument can be amortised over the maturity of the option to the P&L account. Subsequent changes in the time value will be recognised by OCI and will not increase the P&L volatility (see IFRS 9, Paragraphs B6.5.29–38). Only the change in time value to the extent that it relates to the hedged item has to be booked entirely in OCI, under the condition that the critical terms of the option (nominal amount, expiry date, underlying instrument) match the hedged items. If the critical terms do not match, the difference between the time value of the option and the aligned time value has to be taken into the P&L account.

In the same way, forward points (time value from forward contracts) can be booked in the OCI as a cost of hedging, instead of creating hedge inefficiencies in the P&L account.

Basis risk from cross-currency basis swaps also are accepted as the cost of hedging with the same effect, being recognised in the OCI instead of the P&L.

ONLY PROSPECTIVE HEDGE EFFECTIVENESS IS MEASURED

More important than the new rules for hedging instruments are the amendments and the easing of the hedge accounting process, particularly for efficiency tests and booking rules. With the removal of retrospective effectiveness testing and of the 80–125% of fair value changes for the derivative in relation to the fair value change for the hedged item, one the toughest hurdles was eliminated. Retrospective effectiveness has been replaced by the requirement for an ongoing assessment of whether the hedge continues to meet the hedge effectiveness criteria, including an assessment whether the hedge ratio remains appropriate.

Financial institutions will have to ensure that the hedge ratio is aligned with that required by their economic hedging strategy (or risk management strategy). A deliberate imbalance is not permitted. This requirement is to ensure that entities do not introduce a mismatch of weightings between the hedged item and the hedging instrument in order to achieve an accounting outcome that is inconsistent with the aim of hedge accounting. The hedge relationship must not be perfect. The weightings of the hedging instruments and actual hedged item used should not be selected to introduce or to avoid accounting ineffectiveness.

A retrospective test is still necessary, but only in order to identify hedge inefficiencies that are booked directly to the P&L account or to get an indication of whether a rebalancing of the hedge is appropriate.

Hedge effectiveness is defined as the extent to which changes in the fair value or cashflows of the hedging instrument offset changes in the fair value or cashflows of the hedged item. The three requirements for entering into hedge accounting are (see IFRS 9, Paragraphs B6.4.1–11):

- economic relationship;
- credit risk;
- hedge ratio.

Even without a formal test on prospective effectiveness, an economic relationship between the hedged item and the hedging instrument must exist. There must be an expectation that the value of the hedging instrument and the value of the hedged item move in opposite directions as a result of the common underlying or hedged risk. There are no guidelines on how strong the economic relationship has to be. To prove an economic relationship is sufficiently strong, the bank can use qualitative or quantitative criteria. For financial instruments with linear risks a critical term match for notional amount, maturity, coupon size and payment frequency can be enough to indicate the economic relationship. For structured financial instruments or options, a quantitative simulation of expected market developments can demonstrate the hedge quality (see IFRS 9, Paragraph B6.4.13). A change in the credit risk of the hedging instrument or the hedged item must not be so large that it dominates the value changes resulting from that economic relationship. This requirement must not only be checked when hedges are designated for hedge accounting. If credit risk has a greater effect during the lifetime of the hedge, this can also be a reason for the termination of the hedge. An increase in credit risk impact on a financial instrument could be perceived when a hedged bond suffers a severe downgrade and the credit spread increases considerably. As a result, the bond's fair value is more dependent on the changes in the credit spread than on the market interest rates.

The hedge ratio is defined as the relationship between the quantity of the hedging instrument and the quantity of the hedged item in terms of their relative weighting. IFRS 9 requires that the hedge ratio used for hedge accounting purposes should be the same as that used for risk management purposes. Selecting hedge ratios in such a manner that inefficiencies must occur is not permitted. This is not the intension of hedge accounting.

The bank should check the requirements of hedge efficiency when designating the hedge and at every reporting date, or in any cases for which it is necessary to re-check the hedge relationship.

TWO METHODS FOR HEDGE ACCOUNTING: MOSTLY UNCHANGED

Fair value hedge

If the bank decides to hedge the fair value of a hedged item, a fair value hedge (see IFRS 9, Paragraph 6.5.8) should be applied. For the hedging of interest rate valuation risk for bonds or fixed-rate loans, the banks will usually use interest rate swaps (payer swaps). A hedging of the fair value of a fixed-rate liability (originally measured at amortised cost) can be executed by a receiver swap. In both cases the changes in the fair value of the fixed-rate cashflow of the hedged item will be offset by the changes in market value from the fixed-rate leg of the respective interest rate swap.

Two examples will illustrate the general function of a fair value hedge.

Example 12.3. The bank granted fixed-rate loans and intends to hedge the interest rate risk using payer swaps. Without applying hedge accounting, the fair value changes for the derivatives will be booked in the P&L account, while the fixed-rate loans are recognised at amortised cost. If market rates fall, the bank will disclose a loss in the P&L account from the payer swap. The economic fair value increase for the loan will not be recognised. With a documented hedge accounting, the fair value gain of the fixed-rate loans that has been calculated in respect to the move in the benchmark curve (swap curve) will be booked in the P&L account and will compensate the market value loss of the payer swap. A prerequisite is the proved prospective effectiveness of the hedge.

Example 12.4. The bank holds bonds as a liquidity reserve and is not willing to carry the pure interest rate risk from these bonds. The interest rate risk should be reduced by using payer swaps. Without hedge accounting, the bonds have to be measured at fair value, and the fair value changes have to be recognised in OCI (after adjusting the periodic amortisation, which is booked in the P&L account). The market value change in the payer swaps will be booked directly in the P&L account. When interest rates fall, a valuation mismatch will appear between the fair value losses from the swaps in the P&L account and the fair value increase of the bonds in OCI. Applying hedge accounting will partly compensate this mismatch, because the fair value increase of the bonds, which is caused by the decline in the benchmark interest rate curve (swap curve), has to be booked in the P&L account and will compensate the market value decrease in the swaps. The difference in the fair value increase for the bonds due to the change of the benchmark

curve and the market value increase observed on the market has to be booked in comprehensive income.

Cashflow hedge

If the hedged item is a floating cashflow because the bank wanted to hedge the variable funding cost for a fixed-rate asset or the placement of fixed-rate funds in a three-month money market deposit on the interbank market, then the cashflow hedge (see IFRS 9, Paragraph 6.5.11) can be applied. Cashflow hedges are also very often used to hedge FX risks with cross-currency swaps.

A bank that wants to hedge the funding rate for a certain asset over time can use a cashflow hedge. A derivative with the same cashflow structure as the original one can reduce the volatility of the future funding cost. The bank can enter into a payer swap as a hedging instrument for hedging the funding cost for a fixed-rate bond. The bank receives the floating leg of the swap, which compensates the funding cost of the fixed-rate asset. Economically both the fair value hedge and the cashflow hedge have the same result. However, the booking is slightly different.

The following example will highlight the function of a cashflow hedge and the differences in booking compared with the fair value hedge. A bank has a liquidity portfolio with fixed-rate bonds, which is funded by a floating three-month Libor.¹ The bank expects rising interest rates and wants to hedge the funding cost against increasing money market rates with a cashflow hedge. In the first step a perfect hedge against the funding stream will be constructed (hypothetical derivative, no real contract!). The perfect hedge matches exactly the terms of the hedged item's notional amount, interest rate, payment conventions and maturity. The fair value for this perfect hedge will be calculated using the benchmark curve (swap curve). Any fair value change in the perfect hedge will be measured in OCI (closed over capital). The cumulative fair value changes will neutralise over time until maturity. This treatment describes the recognition of the effective part of the hedge in OCI. In the second step the ineffective part of the hedge is calculated as the difference in the fair value change of the real hedge instrument (the payer swaps) and the fair value change of the perfect hedge (the hypothetical deal). The ineffective part is recognised directly in the P&L account.

FX hedges are often designated as cashflow hedges. The handling of the cashflow hedge with respect to the effectiveness measurement is easier than

for the fair value hedge. It is only necessary to define a hypothetical derivative to determine the effective part of the hedge. Furthermore, fair value changes due to FX volatility are generally higher than those for interest rate changes. A high volatility of fair value changes due to the hedged risk reduces the problem of ineffectiveness due to small market movements for fair value hedges. Moreover, accrued interest and non-zero credit spreads will have much a smaller impact on the effectiveness results for cashflow hedges than those for fair value hedges (see details below).

A cashflow hedge can also be applied for a highly probable foreseeable transaction. The bank has to evidence the future transaction. Arguments for the execution of the future transaction can be the frequency of comparable transactions in the past, or the willingness and the ability to execute the foreseeable transaction.

Limitations of the application of cashflow hedges

The application of cashflow hedges has practical limitations if the cashflow of the hedged item is not linked to a market rate such as Libor or the Euro Interbank Offered Rate (Euribor), but is instead linked to an index such as Federal Reserve (Fed) funds or the European Central Bank (ECB) funding rate. From the market's point of view the rate changes for Fed or ECB funds are discretionary on decisions made by the central banks. There are no standard over-the-counter (OTC) derivatives available on the market to hedge the rate volatility of Fed funds and ECB funding rates. If financial instruments linked to Fed funds and ECB funding rates are hedged with standard interest rate swaps, there is great potential for the hedge to fail the effectiveness test, and a large number of the fair value changes of the swap must be recognised in the P&L account.

The same poor effectiveness test results will be expected if a bank hedges financial products for which the interest rates are not linked to market index rates but can be changed for standard interest rate swaps at the discretion of the bank (for example, sight deposits or savings deposits). As in the earlier case there is great potential for an ineffective hedge.

Another limitation for applying hedge accounting is the lack of fixed tenors for financial instruments on the asset side (revolving loans, rollover loans), especially when the borrower has the flexibility to decide how often and in which currency (multiperiod and multicurrency revolving loan facilities) to

make withdrawals and the time interval during which they wish to make the withdrawals.

Even if banks estimate an average maturity for a loan facility and/or an expected average amount of withdrawal of the loan, in many cases a hedge of revolving loan facilities with an interest rate swap will not qualify for hedge accounting due to the hedge accounting restrictions, which refer to loan contract parameters and not to estimations.

Do cashflow hedge results affect the regulatory capital?

The fair value changes for the effective part of cashflow hedges are booked in OCI. The time value points from forward contracts and time value of options are also recognised in OCI, which is part of the capital (in regulatory terms it is part of Core Tier 1 capital). In most banks the regulatory capital value is derived by using national Generally Accepted Accounting Principles (GAAP). In this case, the position OCI is not affected by results from IFRS regulations.

If banks switch to international accounting standards (IFRS) to derive their regulatory capital, the position OCI, beside other effects, includes fair value changes in products not recognised in the P&L account and fair value changes in cashflow hedges. Additionally, fair value effects from changes in own credit spread for liabilities measured at fair value can be part of OCI. The change of classification and the lapse of the “available for sale” category will change the effects in OCI.

To avoid short-term valuation effects from other comprehensive income positions increasing the volatility of the regulatory capital, the Committee of European Banking Supervisors (CEBS) published their “Guidelines on Prudential Filters for Regulatory Capital” (Committee of European Banking Supervisors 2004). Prudential filters eliminate certain valuation effects from OCI for the purpose of calculating regulatory capital. The aim of prudential filters was to maintain consistency in the definition and quality of regulatory capital for those institutions applying IFRS and using national GAAP. In order to create a level playing field across the European Union and other G10 countries, the CEBS proposals on prudential filters were in line with the Basel Committee’s former work on the same subject (Basel Committee on Banking Supervision 2011, p. 28).

In any case, value adjustment due to expected credit loss or impairment rules is not filtered out by prudential filters. All impairment bookings will

affect the regulatory capital.

The prudential filters have been implemented across the different countries by their national regulators. However, there is no harmonised application of the CEBS guidance on prudential filters for regulatory capital across EU jurisdictions.

DOCUMENTATION OF A HEDGE GROUP

Every designated hedge group has to be documented in detail before entering into the hedge, and has to be updated at every valuation date (see IFRS 9, Paragraph B6.4.12). Before the designation of a hedge group, a master document has to describe all the master file data of the hedge.

- The purpose of the hedge (eg, elimination of fair value changes due to benchmark curve movements) and the aim of risk management are in line with the risk management guidelines.
- The type of hedging instrument (fair value hedge, cashflow hedge, portfolio hedge).
- The nature of hedged risk, eg, a fair value change of a fixed-rate loan with respect to the benchmark curve with a maturity of five years. The benchmark curve is the swap curve. Other hedged risks can be interest rate cashflows, basis risks, FX rates, inflation, credit risk or option risks.
- Each single hedging instrument with all relevant attributes (eg, a payer swap with a five-year maturity, fixed leg with annual coupons, a floating leg tenor six-month Libor, notional amount, fixed coupon, maturity, reference number, counterparty, payment schedule, hedge ratio), plus other derivative or non-derivative instruments.
- Every hedged item with all relevant attributes (eg, fixed-rate loan with quarterly interest payments, loan volume, redemption at maturity, maturity, interest binding period, fixed coupon). This can be a single hedged item or a net position of hedged items, maybe including derivatives, a group of hedged items or hedging of a layer with a description of the total position.
- The effectiveness test (eg, method of testing prospective effectiveness) and the impact of credit risk.

As of every valuation date the hedge documentation has to be updated with the description and the result of the retrospective and prospective effectiveness tests and the booking of the effective and the ineffective parts. For the retrospective effectiveness test the 80–120% range of present value changes has been removed.

Compared with IAS 39 there are fewer requirements in IFRS 9, as formal retrospective effectiveness tests have been abandoned. But the easing will be overcompensated by the requirements for additional documentation from aligning hedge accounting with risk management strategy, documentation of net positions, layer designation, recognition of the cost of hedging in OCI when aligned to risk management strategy and objectives, critical term matching or analytic analysis, handling of more than one risk factor, group hedgings, the impact of credit risk, rebalancing, and so on. The prospective effectiveness test must still be performed at each reporting date.

REBALANCING AND DISCONTINUATION

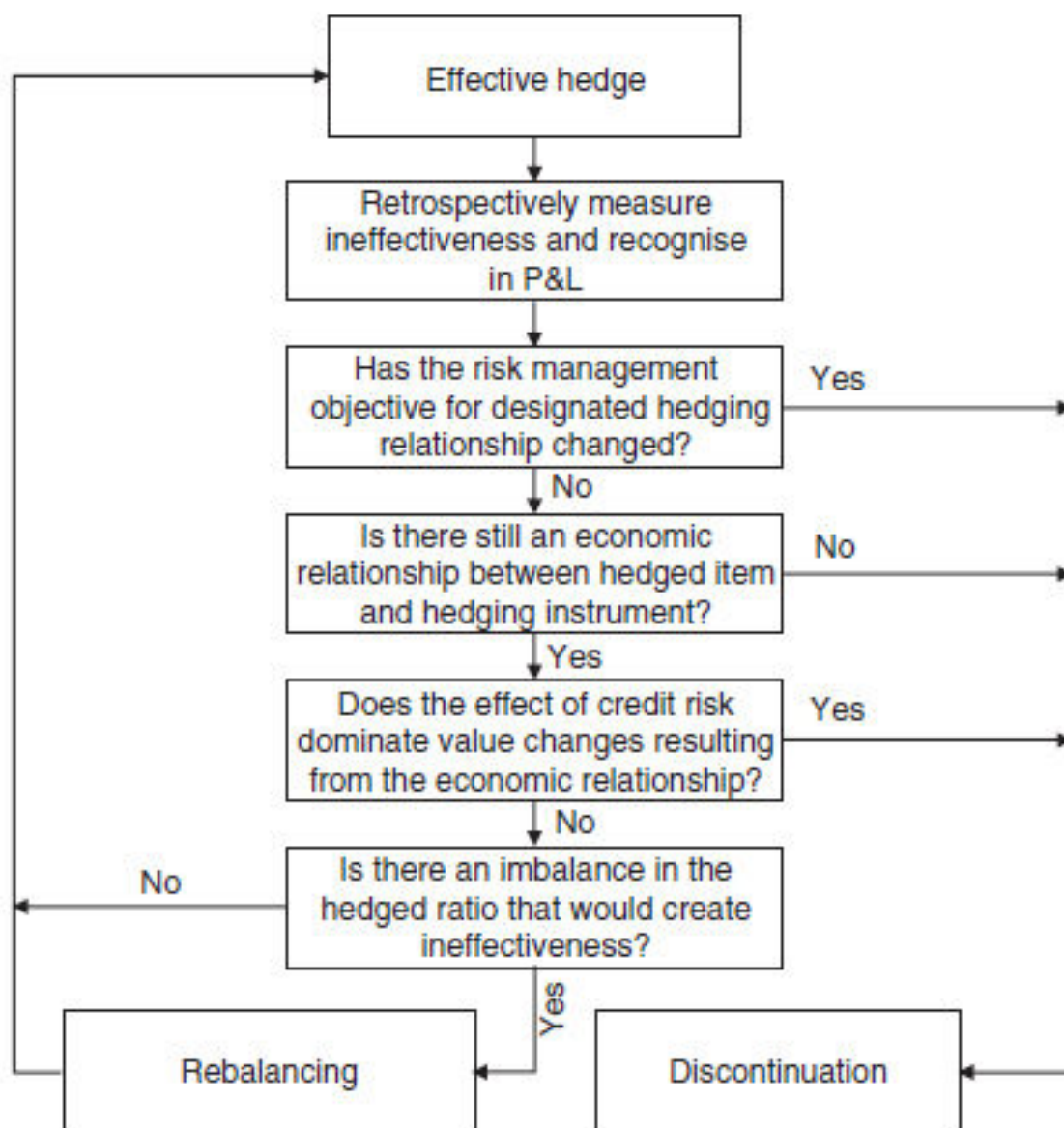
The concept of rebalancing an existing hedge introduced in IFRS 9 links the measurement in accounting with real risk management. [Figure 12.3](#) shows the steps towards rebalancing or discontinuation. If a treasurer holds a hedged position over a longer period, the hedge ratio has to be adjusted from time to time due to (adverse) market movements that could not have been foreseen for a longer time period. The treasurer improves the effectiveness of the hedge. The adjustments to hedge ratios during the run time of a hedge are one of the basics of risk management. But, rebalancing in hedge accounting terms means only those recalibrations of hedges that adjust the hedge ratio in line with risk management objectives (see IFRS 9, Paragraphs B6.5.7–21). After the removal of the 80–125% range for the ratio of fair value changes, the real risk management methods can now be adequately recognised in the accounting scheme.

The removal of the quantitative retrospective effectiveness test does not mean that hedge accounting remains independent of the effectiveness of the hedge. A prospective effectiveness test not only is required at the inception of the hedge but should be performed on an ongoing basis (at each reporting time as a minimum). The rebalancing is not voluntary: if the hedge becomes ineffective, the hedge ratio has to be adjusted.

The rebalancing can be achieved by several actions (see IFRS 9, Paragraphs B.6.5.16–21).

- If the hedge ratio has to be decreased:
 - increase the volume of the hedged item;
 - decrease the volume of the hedging instrument.

Figure 12.3 Rebalancing and discontinuation



Source: Ernst & Young (2014a, p. 39).

- If the hedge ratio has to be increased:

- increase the volume of the hedging instrument;
- decrease the volume of the hedged item.

The adjusted volumes, ie, hedged items or hedging instruments, do not leave the bank. They remain on the balance sheet and generally are measured as they would be without any hedge accounting.

An increase in the volume of the hedged item does not change the measurement of the already hedged part of the total volume. For an increase in volume, the fair value changes for the new designated part should be compared with the hedging instrument's fair value changes starting on the date of rebalancing. A decrease in the volume of hedged items should be treated like a discontinuation for this part.

Changes in hedged instruments to increase or decrease the hedge ratio will not change the measurement.

Under IAS 39, documented hedge accounting could be terminated voluntarily at any time. The fair value effects had to be recognised in the P&L account. The opportunity for voluntary termination was an advantage over the fair value option. The latter can be drawn only at the first recognition of the financial instrument, and is irrevocable for the lifetime of the financial instrument on the bank's balance sheet.

IFRS 9 introduced the concept of rebalancing, and subsequently restricted the possibilities of terminating documented hedges. A financial institution cannot "de-designate" a hedge relationship and thereby discontinue a hedging relationship that (see IFRS 9, Paragraphs B6.5.22–28) still meets the risk management objective and continues to meet all other qualifying criteria (after taking into account any rebalancing, if applicable).

In order to discontinue a hedging relationship, it is necessary to understand the distinction between the notions of "risk management strategy" and "risk management objective" and their impact on any decision to continue a hedge relationship.

Discontinuation of a documented hedge relationship is only possible if

- the entity's hedge objective has changed;
- the hedged item or hedging instrument no longer exists or is sold;
- there is no longer an economic relationship between the hedged item and the hedging instrument;

- the effect of credit risk starts to dominate the value changes that result from the economic relationship.

MACRO HEDGE ACCOUNTING

“Macro hedge accounting” will not be available at the full adoption of IFRS 9 starting in 2018, including hedge accounting. The IASB will circulate discussion papers and expect to integrate this in the next release of IFRS 9. Until this takes place the rules for portfolio fair value hedging from IAS 39 remain valid and will not be integrated into IFRS 9 rules. So the non-inclusion of “macro hedge accounting” had no impact on the endorsement of IFRS 9 in the European Union.

CONCLUSION

IFRS 9 moved the accounting rules for hedge accounting significantly towards those for real risk management techniques. Accounting hurdles such as the 80–125% range were removed. The hedging of net positions including derivatives as hedge items, the designation of non-derivative financial instruments as hedge instruments, the hedging of groups and more than one risk factor will allow a bank to recognise results from real risk management in accounting figures, which will make hedge accounting easier to process and far more understandable by bank stakeholders.

¹ Libor is the London Interbank Offered Rate.

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Part III

Liquidity Risk

Supervisory Views on Liquidity Regulation, Supervision and Management

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Liquidity risk has been around since the first notes and coins were used for trading so many years ago. The risks were simple at that time: if you brought too many coins with you, then you ran the risk of losing them in a robbery, and if you did not bring enough, then you would not be able to have a decent meal that night. The tradeoff was rather clear: you had to estimate how many coins (liquidity) you needed on a certain day to purchase all desired goods. On most days this would be quite predictable, as the same merchant would visit your village every month or so with a steady supply of goods. However, after a bad harvest, accidents, some medieval violence or simply due to pick-up in demand for the goods, the prices may turn out to be higher and you need more cash this month. Being a smart trader, you would bring a second purse, well hidden, with a stock of extra cash just in case you needed more than you expected. While I am pretty sure no one at the time would have called it a liquidity buffer, in my view this is exactly what it is. Even before the time of coins you might have taken your boy with you to the market, so he could run back to the farm to grab an extra chicken in case you found something interesting to exchange it for. The closer the farm, the more comfortable you would be that you would be able to get it in time. This relates to the “maturity mismatch risk”: the longer it takes you to get a good

you can turn into something you need (monetise it), the higher the risk. So even though many people in today's world really started to rethink liquidity risk in the light of the 2007 market disruptions, we have learnt to live with and manage liquidity risk for many generations.

PANEL 13.1 BEST PRACTICE GUIDANCE: REGULATION IS NOT THE ANSWER TO EVERYTHING

Even though regulation has grown in length tremendously, I advise keeping in mind that it cannot, and should not, replace sound management.

- **Having regulation does not guarantee the absence of risks:** even the best written piece of liquidity regulation cannot take away the risk. Nor should it aim to. Regulation helps to mitigate risks by setting direct boundaries (such as the liquidity coverage ratio (LCR)) and by pointing banks in the right direction on how to best manage the risks (such as the Internal Liquidity Adequacy Assessment Process (ILAAP)). With a model of borrowing and lending to different maturities, there will always be the risk that liabilities run out before assets are returned. When banks get into trouble, the public sometimes forgets that supervision is not intended to avoid all kinds of problems; rather it should reduce the risks of problems to what is *ex ante* deemed an acceptable level. *Ex post*, this assessment is not always remembered or accepted retrospectively.
- **Regulation cannot take over sound liquidity management by the bank:** with the multitude of developments in regulation in the field of liquidity risk at the time of writing, banks and supervisors should be mindful that regulation can never be a checklist for sound management. Merely complying with liquidity requirements on paper is no guarantee that banks will be safe. Even a bank with a 200% LCR could run out of liquidity tomorrow when faced with stress. Strikingly enough, there are many reasons why this may happen: outflows being much higher than assumed under the LCR; expected inflows not being received as assumed; liquid assets turning out to be not so liquid when needed; or a mismatch between in- and outflows in the first 30 days, while the LCR aggregates all cashflows.

Historic trends in regulation

Even though liquidity risk has been at the forefront of the minds of risk managers and boards since 2007, the history of regulation goes back quite a bit further. Of course the question that immediately comes to mind is whether that regulation turned out to be futile (we had the global financial crisis) or whether it actually worked (it could have been much worse). There are guaranteed to be different views on this, but I would like to think both are true. The regulation on liquidity was very basic in 2007, and was hugely

extended over the years that followed, to soften the impact of future (market) stress events. I have included two lessons I consider fundamental with respect to the role of regulation in Panel 13.1.

Banks' management should start by ensuring they are properly informed about the liquidity risk their bank runs and how it is managed. Then comes the question of regulatory compliance, which, of course, is also one that should be managed properly both now and in the future.

So, after a relatively modest pace of liquidity regulation – mainly publications by the Basel Committee on Banking Supervision (BCBS) and European Banking Authority (EBA) (actually the Committee of European Banking Supervisors (CEBS) at the time) with best practices and guidance – the 2007–9 crisis years spurred on a change in approach. The content changed (hard metrics on top of qualitative principles), the enforcement changed (regulation instead of guidance) and the level of detail changed (more guidance on what to do). All of this could suggest that banks did not manage their liquidity properly and regulation is the answer to make them do so. Again, I personally believe the answer lies somewhere in the middle. With hindsight, I do think many banks ran liquidity risks much higher than they should have, and in many cases higher than they (or at least the executive board) were really aware of, partly because of lapses in reporting (the risk was known somewhere, but not reported adequately) and partly because risks were perceived differently (they were not measured, or the measurement, often a stress scenario, underestimated the true risk). But it was not just the bankers that received a wake-up call on where and how liquidity risk can surface; the supervisors also had to adjust their approach to supervising (monitoring to start with, then setting minimum requirements for) liquidity risk. In the end, the new regulations were aimed as much at banks (getting them to a higher level of liquidity risk management) as at the supervisors (to harmonise efforts to better supervise liquidity and funding risks). Managing liquidity is costly, so harmonisation in the supervisory approach is essential to ensure a level playing field. Having the same regulation does not mean having the same supervision. The translation from regulation into supervisory methodology, handbooks and day-to-day work of supervisors is susceptible to divergence. Striking the right balance between being overly prescriptive and leaving enough room for expert minds to figure out how to best identify and manage risks is crucial.

PANEL 13.2 BEST PRACTICE GUIDANCE: THE BANK'S BOARD UNDERSTANDS NOT ONLY THE LCR LEVEL, BUT ALSO ITS SENSITIVITY TO THE ASSUMPTIONS

In a universe of continuous probability curves, the likelihood of exactly one specific event happening is zero by default. The key lesson from econometrics is to think in intervals instead of single outcomes. Having a good estimate helps, but looking into the effects that changes in assumptions have on outcomes is crucial, especially for the LCR, for which stress assumptions are key. Therefore, metrics should be used as a steering mechanism for the bank rather than a value that needs to be optimised. This requires a proper methodology in order to make this sensitivity to stress assumptions visible, to clarify the categories to which the bank is especially sensitive due to the high degree of difficulty in predicting counterparty behaviour and providing the managing body with quantitative insights into the effects of the different assumptions.

LIQUIDITY COVERAGE RATIO AND NET STABLE FUNDING RATIO: METRIC OR STEERING WHEEL?

The two key metrics that were introduced by the BCBS as minimum requirements are the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR).¹ The aim of these metrics is to ensure that there is an appropriate amount of market-liquid assets to refinance maturing obligations during a stress event and that the balance sheet is robustly funded, meaning liabilities and assets have roughly similar maturity profiles. (I would like to emphasise “roughly” here.)

Since there have been many documents produced on the calculation of the LCR,² I will not go into details of the mathematics here, but rather will focus on what these metrics aim to do. In the literature, as well as in daily interaction between supervisors and banks, the main point of discussion is calibration of these metrics, especially when it comes to less well-defined items such as retail deposits subject to higher outflow rates and operational deposits or derivatives. The “behaviour” of these liabilities under severe (bank-specific and market) stress is difficult to predict. But still the metrics need a number, so there is some guidance there but also scope for interpretation. And this scope is much debated, which is of course good in itself. However, it distracts from the key goal of these metrics: to ensure the bank has proper liquidity buffers and stable funding. This is a high-level goal and should be fundamental in driving the bank's management (see also Panel

13.2). The exact calculation is a tool for reaching this goal: a metric that gives an indication. By no means is it the absolute truth.

As a final note, I recognise of course that metrics such as the LCR and the NSFR are also increasingly used in banks' publications and estimates made by analysts, so the world is watching and rating the banks on such metrics. While transparency is a great virtue, it also affects the choices banks make with respect to calibration. This means that being more conservative in the calibration "to be on the safe side" can work against you in peer comparisons when only the final outcome is compared. This is an important difference between market practice and supervisory practice (in general supervisors are happy with more conservatism). This trade-off leads to more accurate, rather than conservative, LCR metrics. This is in itself not a problem, but it is not enough to only use LCR and NSFR to steer the liability structure of the bank. Internal stress testing is a key tool to complement these metrics, as will be explained further below.

LIQUIDITY COVERAGE RATIO AND NET STABLE FUNDING RATIO VERSUS INTERNAL LIQUIDITY ADEQUACY ASSESSMENT PROCESS

In addition to the introduction of the LCR and NSFR metrics, supervisors have increased the emphasis on ILAAP. While it will be hard to find the term in BCBS publications, the concept of ILAAP introduced by EBA and De Nederlandsche Bank (DNB) in 2012 was strongly based on the BCBS "Principles of Sound Liquidity Risk Management and Supervision"³. Since the introduction of ILAAP through the EBA Supervisory Review and Evaluation Process (SREP) guidelines was still relatively recent at the time of writing, practices varied considerably across countries and banks. Since the inception of the Single Supervisory Mechanism (SSM) in 2014, the ECB has focused on harmonising the Internal Capital Adequacy Assessment Process (ICAAP) and ILAAP expectations. In November 2016, the ECB launched draft ICAAP and ILAAP guidelines in advance of its consultation on its new expectations in relation to ICAAP and ILAAP, seeking increased dialogue with the banks to foster harmonisation of banks' internal practices. In parallel the ECB was working on harmonisation of the internal assessment approach for liquidity and funding risk, and in particular, for the assessment of ILAAP. The key point here is that the methodology was still in

development at the time of writing, so it can be useful to closely follow such developments and take a critical look at your own ILAAP to see if it is truly fit for purpose.

PANEL 13.3 BEST PRACTICE GUIDANCE: A SOUND TEST OF THE ILAAP IS TO VERIFY THE MANAGEMENT BODY IS FULLY COMFORTABLE WITH ITS INFORMATION FLOW ON LIQUIDITY

Ultimately, the management body should be able to make an informed decision on the liquidity adequacy of the bank and take any measure necessary to improve its liquidity position if they decide it is not at the desired level. This will then be documented in the liquidity adequacy statement, which is one of the key documents with which supervisors start their ILAAP evaluation. To assess whether the liquidity information flow is adequate, I would expect the management body to ask itself questions such as the following.

- Is it clear where liquidity risk comes from for this bank?
- Is it clear how liquidity risk is measured and what assumptions are crucial?
- Is the information complete (all entities, countries, currencies, etc, are included)?
- Is the information granular enough (I can see the main entities, currencies, actual liquidity position versus limits between reporting dates, etc)?
- Can the information be trusted to be accurate (is proper quality assurance in place)?
- Does the information link the actual positions to the risk appetite (limits)?
- Does the information provide a clear conclusion in terms of well-defined metrics?

So why do we supervisors think ILAAP is so important now we also have LCR, and NSFR is around the corner? The answer is relatively simple: we think the banks are in a better position to identify, measure and manage liquidity risks than we are. From a macro perspective, it is simply impossible to design a metric that accurately captures the risks of all banks in all countries. Such metrics are not very adaptive, as it took around 10 years to get them introduced (assuming NSFR will be) as minimum requirements, making it highly unlikely they will be recalibrated in a year or so. But banks' balance sheets change, as do market patterns and consumer behaviour, so adaptability is important. From a micro perspective, the direct supervisors of a bank are there to challenge the internal assumptions made by the bank, not

to make them up themselves. Finally, and possibly most importantly, the ILAAP is the complete process of liquidity risk management: it encompasses not only a wide set of metrics such as LCR and NSFR, but also internally run stress tests, resulting in survival periods, time-to-central-bank metrics,⁴ etc. The choice of metrics, calibration, risk coverage, risk appetite limits, internal reporting, key strategies (eg, contingency plans, targets for the liquidity buffer size and composition, funding strategy), etc, is all part of the ILAAP. All the components of the ILAAP should be integrated into one process that identifies, measures and reports on risks to the appropriate (management) body and ensures all policies and escalation procedures are in place should a crisis hit the bank. See Panel 13.3 for further guidance on the role of the management body in determining the appropriateness of the ILAAP.

In contrast to LCR and NSFR, the internal ILAAP metrics are not generally made public. This enables banks to include very severe scenarios with prudent assumptions and longer stress horizons, in order to assess their resilience to prolonged severe stress and to identify (early) potential vulnerabilities in the liquidity profile that are not captured by the LCR. As the ILAAP is an internal process, it permits a wider view on what assets can be used to obtain liquidity during stress. Obviously, banks should not count solely on the central bank for liquidity support in the case of a stress event. The LCR is designed to avoid such reliance, at least in the short term. However, it has two drawbacks:

1. it does not account for stress events lasting longer than 30 days; and
2. it assumes the buffer can be drawn to zero without any further consequences.

PANEL 13.4 BEST PRACTICE GUIDANCE: THE USE OF A SUFFICIENTLY WIDE AREA OF STRESS SCENARIOS AND METRICS IS CRUCIAL, LIMITING ONESELF TO ONLY ONE TYPE OF SCENARIO OR METRIC WOULD CREATE THE RISK OF MISSING IMPORTANT RISK DRIVERS

In a best practice ILAAP, I would expect at least the following scenarios and metrics to be taken into account.

- **Market disruption:** the bank does not face direct (reputational) stress, but markets are severely disrupted, limiting the availability of funding to the bank.
- **Idiosyncratic stress:** a bank-specific stress event reduced the ability to attract funding and may cause outflows in, eg, deposits, or may trigger different counterparty behaviour (eg, more collateral calls, counterparty buyback requests, etc). Markets are otherwise normal.
- **Combined market and idiosyncratic stress:** a combination of the two scenarios above.

The following metrics can be used under each scenario.

- **Time-to-LCR-breach:** a stress test that measures the period of time the bank would still have an LCR greater than 100% while being under stress. This combines the absolute distance (eg, a management buffer of 10% is better than 5%) with the maturity profile of the balance sheet (high amount of maturing debt in the short term, shortens this ratio). The risk appetite should be set based on how quickly the bank knows it is under stress (data lag, escalation procedures, etc) and on the importance of the LCR for the stability of its liabilities (eg, wholesale investors will be more likely to react to a strong drop in LCR than retail clients).
- **Time-to-central bank:** a stress test that measures the period of time the bank can survive stress based only on its market-liquid assets. In principle this is at least 30 days, but banks may consider using more severe internal stress tests than the LCR stress scenario for this purpose. Also, having a 100% LCR does not actually guarantee a bank will make it through a 30-day LCR scenario, as there may be mismatches within the 30-day period.
- **Combined survival period:** a stress test that measures the period of time the bank can survive a stress event using its HQLA as well as its contingency liquidity buffers. When all risk drivers are appropriately captured, this metric gives the best insight into how much time the management of the bank has to take action and react to the root causes of the stress affecting it. The more difficult it is to accurately predict what will happen (eg, highly uncertain client behaviour or large amounts of both in- and outflows), the higher the target for the survival period should be. One key choice the bank needs to make is what to do with assumptions on inflows from assets, with best practice guidance to investigate both
 - a “run down of the bank” scenario, where you assume contractual inflows are indeed returned to the bank, implying there is no rollover of loans, which means the bank is closed for new business,
 - limited assumptions on inflows, recognising that management would like the bank to still be in business when the stress resides, and thus some rollover of loans and new production is assumed (meaning lower inflows compared with the contractual view).

The global financial crisis has clearly shown that, in certain markets and for a large number of banks, liquidity stress lasting longer than 30 days is not just a theoretical concept. Markets as well as supervisors (and let’s not forget the new kids on the block, resolution officers) may not be very happy with a bank with a 0% LCR. So having “something extra” can be considered

a sound strategy. Partly, this can take the form of a management buffer on top of LCR, so you do not breach the LCR if someone decides to buy a new car; this buffer should be a reflection of the volatility in the LCR due to client/market behaviour (on both sides of the balance sheet!). Furthermore, banks should look into the size of their contingent liquidity buffer as part of the ILAAP. This buffer is generally made up of assets that are eligible at a central bank, but do not meet the requirements to count as LCR high-quality liquid assets (HQLA). While banks should of course try to manage their liquidity in markets as much as possible, the role of the central bank as lender of last resort should not be completely ignored. Banks should not just count on extraordinary measures being available to them overnight. While such measures are always at the discretion of the central bank, the likelihood of such measures increases the longer market disruptions persevere. And of course many central banks have facilities for monetary operations (such as the long-term refinancing operations and marginal lending facility at the ECB⁵). Access to such facilities has a place in liquidity risk management and thus in ILAAP, but it just comes on top of the minimum level of stress-resilient liquid assets that can be used in markets.

In practice, when a bank has both LCR HQLA and central-bank-eligible assets, it has a choice once a crisis hits: either it goes to markets first and reduces HQLA, potentially reducing LCR, or it can use non-HQLA in markets or with central banks via repo. While central bankers and supervisors may each have their preferences, in reality what is actually best to do is very case dependent. Banks therefore should think about this during benevolent times and use metrics to determine their likelihood of breaching LCR or being forced to go to the central bank. It is important to distinguish between scenarios (such as market- or bank-specific stress) and metrics (such as survival periods). The appetite to breach a certain level of any metric can (and in my opinion should) be different under different scenarios. For example, breaching the LCR due to a market disruption would be worse (reputation wise) than breaching it under a combined market and idiosyncratic shock (see Panel 13.4 for more details on what scenarios and metrics I think should be considered as a minimum).

TRENDS IN REGULATION: WHERE NEXT?

Even though regulation in the field of liquidity and funding risk is still developing and has not seen the end when it comes to requirements and supervisory expectations, regulation cannot (or at least should not) overtake sound liquidity management by the banks. Many banks responded to the increasing number of expectations by creating new documents, overviews and processes with the aim of complying with the new regulation. Since these expectations became available piecemeal, many banks were now confronted with a large set of documents in which the internal logic was not always clear. While more guidance on what supervisors expect was still to be issued, the time is ripe for banks to also take a critical look at the documentation they themselves have created. When it comes to expectations on ILAAP, the most pronounced one is that the process is for internal use, meaning the process should logically connect to the activities of the bank and be a key tool for informing the appropriate level of management and steering the bank. A cleanup of the documentation produced in the past may be very worthwhile in order to ensure it is actually practical for its purpose and is not just created for the supervisor (with the exception of the readers' manual that is part of the compulsory ICAAP/ILAAP information package). This will help to improve the effectiveness of liquidity risk management, can remove inefficiencies due to double documentation and should be viewed more positively by supervisors. Creating an internal structure and logic increases its use, transparency, ease of maintenance and added value for risk management. Doing this should also create more scope to apply proportionality, a principle mentioned in almost all regulations but the hardest to apply in practice. Finally, I expect new supervisory guidance and regulations will focus on this cleanup and establish consistency with the various liquidity and recovery regulations already in place (see Panel 13.5 for additional best practice guidance on the overarching logic for the (documentation of the) ILAAP).

PANEL 13.5 BEST PRACTICE GUIDANCE: THE ELEMENTS OF THE ILAAP SHOULD BE IMPLEMENTED IN LOGICAL CONSISTENCY WITH OTHER (RISK) MANAGEMENT ELEMENTS

As a rough guide, banks could think about the ILAAP starting from identification of risks (based on business model, clients, markets, regions and products relevant for the bank) to select and calibrate the appropriate tools for measuring their risks. From this they can set the desired risk appetite (limits) and create internal reporting and escalation procedures consistent with this appetite. Of course, proper quality assurance (validation of models and assumptions, audits, robustness checks, etc) should be included. This should lead to (timely) discussion by the board, resulting in a liquidity adequacy statement and action to be taken when needed. Potential actions to mitigate a liquidity stress event should be written down in a liquidity contingency plan, which should be consistent with recovery measures determined within the bank. The policies governing both liquidity contingency and recovery actions should make clear how these are linked; in particular, they should clarify what actions are likely to have already been taken before recovery actions are triggered. Detailed best practice guidance on the relationship between liquidity contingency and recovery actions as well as at a risk appetite level was yet to be developed at the time of writing.

Looking further ahead, the availability of data on a more granular level with a higher frequency is a clear trend. The more quickly robust data becomes available, the more accurately liquidity risk metrics can be used to identify risks and help banks to react in a timely manner to changes in their balance sheets and in the markets. Banks should invest in the right data sources and processing software to make optimal use of the data to promptly identify developing risks or (market) stress and react to it appropriately. Supervisors will be interested in following this closely, especially when markets are under stress. Investing in improving these reporting capabilities in the short term will help avoid the pressure and costs involved in the organisation delivering more frequent and detailed reports during stress events.

However, more and better data is not the only change we can expect in the near future. The existing approach to liquidity risk is very much only from a prudential view: there are new minimum requirements and banks manage their balance sheets to ensure they meet them (at the time of writing, that is, in a period of relatively calm markets). The whole idea of the LCR is that the HQLA buffer can be used during periods of stress. But for how long can you really use it? When will markets react more severely? When will resolution authorities consider the bank ready for resolution? Will they wait until all (contingent) liquidity is gone or intervene earlier? Perhaps banks could ring-fence part of the balance sheet that can be turned into liquidity (through covered bonds, for example)? There is no clear answer to all of these questions (yet), but banks can bring together things already in their risk appetite. When they measure the liquidity position under stressed conditions,

they can show what management actions could be taken, against what cost (eg, to capital) and with what gain in terms of liquidity. A risk appetite could be set for the level of LCR the bank does not want to fall below during stress, which would be commensurate to the bank's assessment of how much liquidity it could generate to recover and avoid resolution. Linking all the elements of liquidity risk management in a logical and robust way will be the main challenge in the short-term, with two crucial elements: stress testing and a robust risk appetite framework.

The views expressed in this chapter are the personal views of the author and do not represent the formal policy stance of either the DNB or the ECB.

- 1 At the time of writing, the LCR had been introduced as a binding minimum requirement within the European Union, while the regulation introducing the NSFR as a binding requirement was still being drafted.
- 2 In addition to the text of the BCBS Basel III framework and LCR Delegated Act, see, for example, the “frequently asked questions” published by the BCBS and EBA (available at <http://www.bis.org/publ/d406.htm> and <http://www.eba.europa.eu/regulation-and-policy/liquidity-risk>, respectively).
- 3 See <http://www.bis.org/publ/bcbs144.htm>.
- 4 That is, measuring the time a bank can survive using market-liquid assets without using central bank facilities. See below for further details.
- 5 See <https://www.ecb.europa.eu/mopo/implement/html/index.en.html>.

Measuring and Managing Liquidity and Funding Risk

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ABN AMRO

Credit institutions define liquidity management objectives as part of their strategy execution plans. Liquidity management activities are typically delegated to asset and liability management (ALM) and/or treasury functions that identify, measure and manage the liquidity position of the bank in a robust framework based on a defined risk appetite. This implies that ALM and/or treasury need to have an insight into the current and projected liquidity profile of the balance sheet (for both on- and off-balance-sheet positions) and manage the liquidity mismatch that results from the bank's maturity transformation role. Credit institutions will also have an independent risk function in place to identify the inherent liquidity risks and provide assurance that these risks are (and will continue to be) effectively managed and mitigated within the risk appetite of the credit institution.

DEFINITION OF LIQUIDITY RISK

Liquidity risk can be divided into two subcategories.

1. **Funding liquidity risk:** this is the risk arising from the potential inability of the bank to meet both expected and unexpected current and

future cashflows and collateral needs. Funding liquidity risk can be further differentiated into:

- the liquidity risk resulting from the inability of the treasury function to access capital markets to attract funding or generate liquidity in relevant currencies (funding risk);
- a liquidity risk due to a change in liquidity costs (spreads) resulting in a liquidity maturity mismatch between cash inflows and outflows (funding spread risk or repricing risk) that has an adverse impact on the profit and loss (P&L) of the bank.

2. **Market liquidity risk:** this is the risk of being unable to generate liquidity in less efficient or disrupted financial markets. The liquid assets held to generate liquidity in stress situations might not be sold at a reasonable market price, and/or reverse repo transactions might be executed with unanticipated high haircuts or might not provide the expected liquidity due to inadequate market depth or market disruptions. Some types of market liquidity risk are the following.

- Insufficient market liquidity of the liquid assets held in the portfolio with the treasury function: this risk arises when the assets have insufficient market liquidity in a stress situation. It is thus related to market risk.
- Foreign exchange (FX) convertibility risk arises when a bank has payment obligations in currencies in which it does not have available funds or collateral. FX positions are actively managed, and relevant FX instruments are traded to convert the available funds into the currency of the committed payments. Inadequate market depth or market disruption introduces the risk that the treasury cannot trade sufficient FX instruments to make the required currency available, and possibly fails to make the payment when it is due.

The aforementioned liquidity risks can occur at any moment, including intraday. Intraday liquidity risk refers to the risk that a bank fails to meet payment obligations in a timely manner, taking into account relevant cut-off times in different time zones.

Liquidity risk management framework

To obtain insight into these risks and manage them effectively a bank needs to have an overarching liquidity risk management framework to identify, measure and manage liquidity across time zones, and for different currencies, organisational units (such as business lines), countries and legal entities. Senior management is responsible for an adequate organisational setup of the liquidity management function, and must also set the liquidity risk appetite of the bank. The risk appetite should clearly define the level of liquidity risk the bank is willing to accept. It consists of quantitative statements that can be monitored frequently to ensure the risk appetite is adhered to. These statements should include all the liquidity risks identified by the institution.

MEASURING THE LIQUIDITY POSITION

Liquidity risk can be measured in a number of different ways. Most indicators focus on funding liquidity risk, although some metrics also contain an element of market liquidity risk. A distinction can be made between regulatory indicators and a bank's internal metrics. With the introduction of Basel III, the following regulatory liquidity indicators have been established.

- The liquidity coverage ratio (LCR) ensures that a credit institution holds enough liquid assets to cover their net liquidity outflow under stressed conditions during the next 30 days. This should ensure the liquidity buffer is sufficient to face a possible imbalance between liquidity inflows and liquidity outflows. As a result, banks should actively manage not only the portfolio of liquid assets but also the projected net liquidity outflows. This metric covers not only short-term funding liquidity risk, but also market liquidity risk with the application of haircuts on the liquid assets in the liquidity buffer. An LCR calculated for a specific currency would address the FX convertibility risk.
- The net stable funding ratio (NSFR) ensures that a credit institution's long-term obligations are adequately met with stable funding instruments, such as stable customer deposits and long-dated debt instruments. The NSFR requirements are expressed through the weighting of available and required stable funding. The NSFR restricts the use of less stable short-term funding sources. It also sets higher requirements for assets that are used in transactions to generate funding (eg, mortgages that are used in covered bonds). The NSFR forces banks

to adopt a view on the optimal composition and funding structure of their balance sheets. The NSFR addresses the long-term funding liquidity risk of a bank.

The concept of maturity mismatch

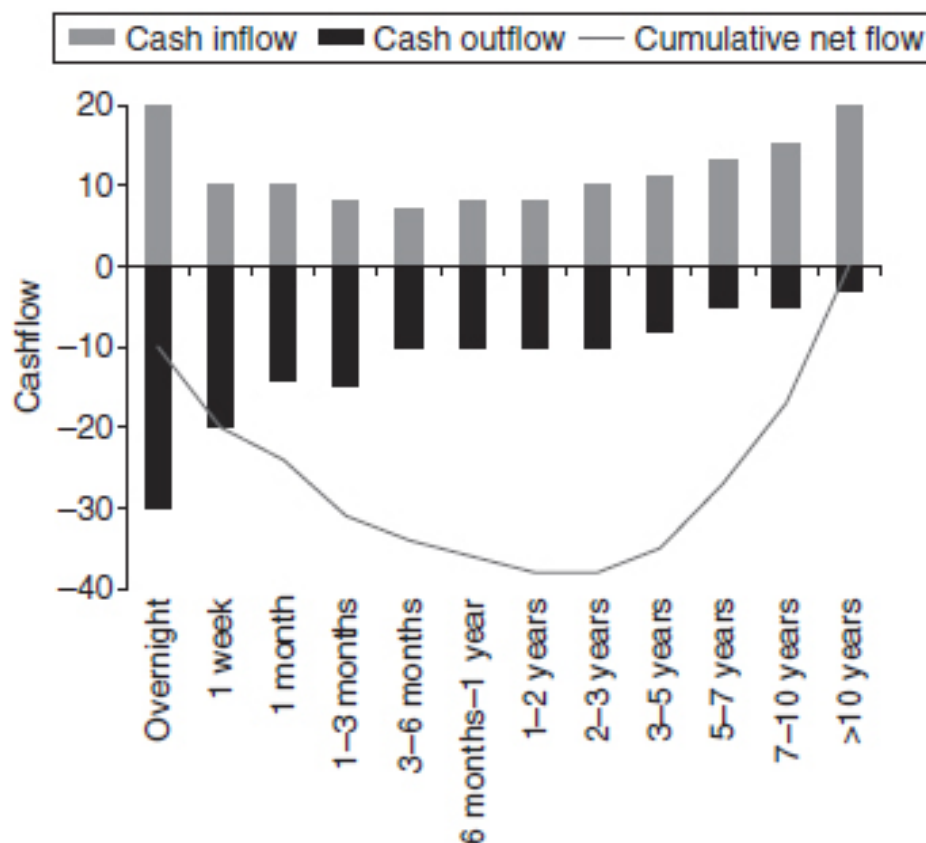
In addition to the regulatory metrics, banks should develop their own internal view on liquidity risk. An important internal metric to measure funding liquidity risk is the liquidity mismatch profile of the bank. Liquidity risk arises since the maturity dates of assets and liabilities are not perfectly matched. The most common situation for a bank is a position in which long-term assets are funded with shorter-term liabilities. This is called “maturity transformation”, and is the principal role of a bank. The bank benefits from this maturity mismatch when the yield curve is upward sloping, meaning that the interest rates on short-term liabilities are lower than those on the long-term assets. This position introduces funding liquidity risk for the bank, since the liabilities have to be renewed before the assets mature. Funding liquidity risk has two aspects: the risk that the bank is not able to repay its liabilities when they come due; and the risk that new funding can only be raised against higher rates. When these rates are higher than those received on the assets, this could lead to a loss for the bank. The latter is also known as repricing risk.

Contractual maturity calendar

To effectively manage this liquidity risk, the bank should have a clear overview of the maturity profile of its assets and liabilities. This profile can be derived by aggregating over all the contracts and distributing them to different time buckets based on their maturity and/or drawing date. This results in a so-called “liquidity gap profile” or “contractual maturity calendar”. This overview shows the contractual cash inflows and outflows per maturity bucket, and therefore shows the time buckets in which these in- and outflows are not matched. An example is given in [Figure 14.1](#). The overview demonstrates that a substantial fraction of the liabilities are mapped to the overnight maturity bucket. These are mainly demand and saving deposits that have no fixed maturity date and that clients can withdraw every day. It also shows that a large fraction of the assets mature only after 10 years. These are mainly residential mortgages with an initial maturity of

30 years. Most of these mortgages, however, will be repaid earlier because clients move to new houses or refinance their mortgage.

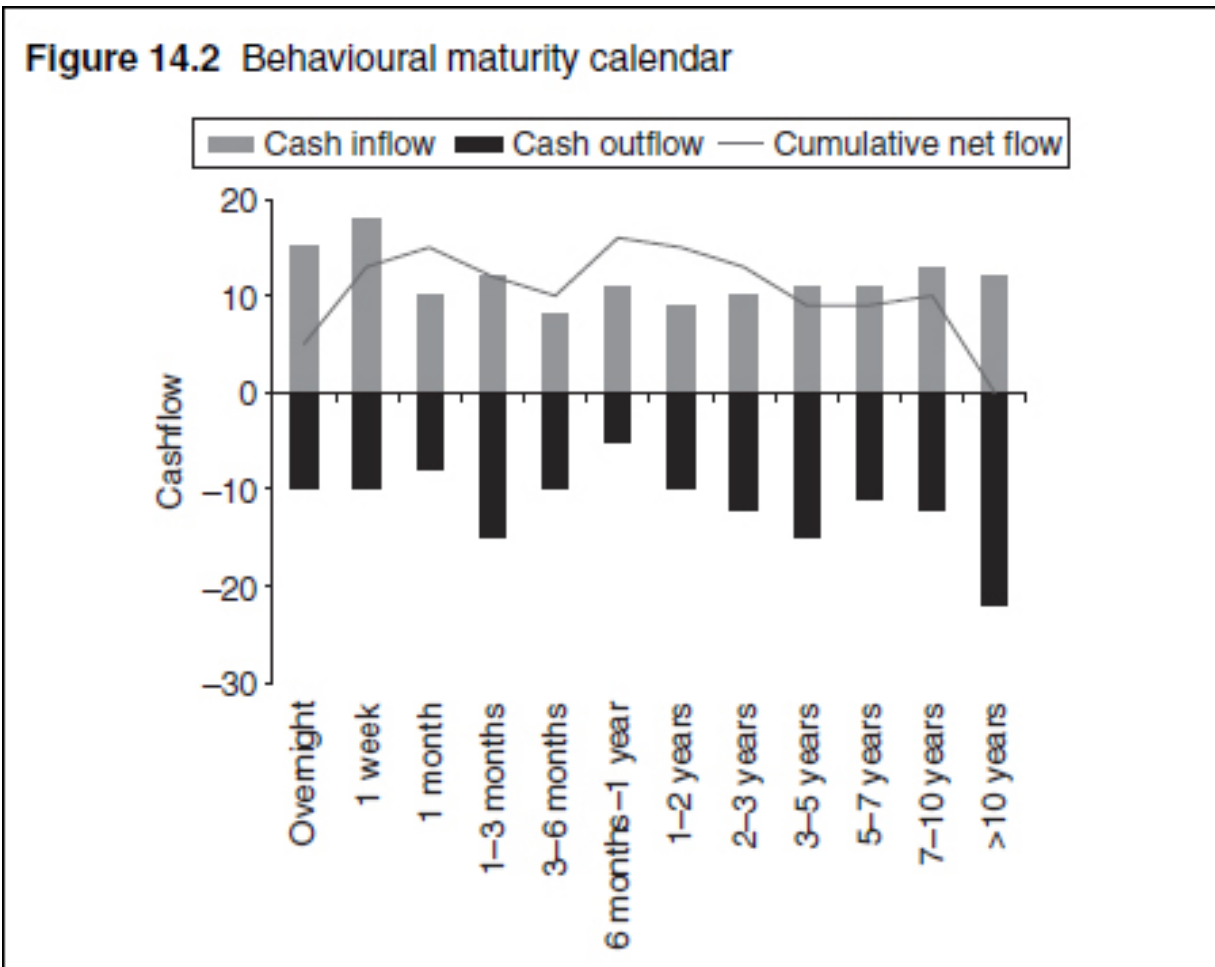
Figure 14.1 Contractual maturity calendar



Behavioural maturity calendar

As the example above shows, a contractual gap profile alone is not sufficient to manage the liquidity profile of the bank. The contractual profile can show a large gap caused by liabilities with short-term maturities and assets with long-term maturities. This might look like a very large liquidity risk. In reality, however, the maturity of the liabilities is much longer because only a small amount of deposits will be retrieved by clients on a specific day. In addition, the maturity of the assets might be shorter than the contractual maturity because of significant loan prepayments. This means that, instead of a contractual maturity calendar, it is more important to look at the behavioural maturity calendar, in which the expected client behaviour is incorporated. An example of a behavioural maturity calendar is presented in

Figure 14.2. This calendar shows that the cash inflows are expected before their contractual maturity date, while the cash outflows are expected after their contractual maturity. As a result, the net cumulative cashflows are positive in the behavioural maturity calendar, whereas they were negative in the contractual maturity calendar.



To create their behavioural maturity calendar, the bank should analyse how real client behaviour deviates from the contractual arrangements. This difference can occur because most of the products that the bank offers to its clients contain (implicit) optionality. One example is a demand deposit, which gives the client the option to withdraw the amount they have deposited at any time in the future. In addition, the client can also increase the amount on deposit. Another example is a residential mortgage, where the client has the option to prepay, change the amortisation profile or change the interest rate term.

Liquidity risk models

To capture actual client behaviour, banks have been attempting to model it. Traditionally, banks have a strong track record in risk modelling, which was inspired by Basel II and focused mainly on credit and market risk modelling. Since the global financial crisis, this focus has been extended to liquidity risk modelling. For the latter, the level of sophistication has increased significantly. Modelling techniques have been extended from simple fixed prepayment rates or expert-assumption-based approaches to statistical models using macroeconomic explanatory variables. In addition, the scope of products that are modelled from a liquidity risk perspective has increased because banks have acknowledged that more products contain (implicit) optionality.

In the next section we describe the main products modelled from a liquidity risk perspective in more detail. There are, however, some general aspects that apply to all products, which we shall discuss first.

Run-off versus static versus dynamic balance sheet

The balance sheet of a bank is dynamic by nature. New assets and liabilities are generated every day, and a proper forecast of the bank's future balance sheet should take this into account. However, modelling the assets and liabilities on the balance sheet usually starts with the current balance sheet. The characteristics of the assets and liabilities that are on the current balance sheet are known, and the future behaviour of these assets and liabilities can be estimated based on these characteristics. This results in a run-off profile of the current balance sheet, ie, for all assets and liabilities an expected maturity date is available. The run-off profile shows how much of the original amount is still expected to be outstanding in each future period. Note that this can be either an asset or a liability.

Another way to examine the bank's balance sheet is the "static-balance-sheet assumption": in this case it is assumed that all maturing assets and liabilities are replaced with new assets and liabilities with similar characteristics. This results in a constant balance sheet, which gives no real insight for managing liquidity risk. However, this assumption is often applied for solvency stress testing purposes.

The third way to examine the balance sheet is the dynamic view, in which the business forecast is incorporated. In this option, maturing assets and

liabilities are rolled over and new business initiatives leading to an increase in the assets and liabilities are taken into account.

To analyse the liquidity risks of a bank, the run-off profile is often used as a starting point, since the properties of the bank's current assets and liabilities are known and their behaviour can be modelled and forecasted. New business can be taken into account on top of this run-off profile.

Modelling at portfolio or contract level

Credit risk models are usually estimated on a contract level because client and contract characteristics are important drivers for credit risk, and the results of the credit models are used to make decisions on a client level. Ideally, liquidity risk models are also developed on a client or contract level, because liquidity behaviour can be influenced by specific client or contract characteristics. However, such models are used differently from credit risk models, since liquidity is managed centrally within a bank and the results of the models are not directly used to make decisions on a client level. Therefore, liquidity models are sometimes estimated on a portfolio level. Typically, the behaviour of clients in different segments is modelled and aggregated and reflected in the behavioural model of a portfolio of clients and/or products.

Non-maturing savings (saving deposits and current accounts)

The most important product category for any bank that is funded by deposits is demand and saving deposits. Demand deposits are current accounts that are used by clients for their daily cash management. Clients can be either retail clients, who receive their salary on their account and pay out for their mortgage and daily grocery shopping, or corporate clients, who use their account for all their incoming and outgoing payments. Balances of individual clients can fluctuate substantially, but from a bank perspective these deposits are a stable source of funding, since such fluctuations are often offset at the aggregated bank level. Saving deposits are those in which clients save their excess cash. The number of transactions on these accounts is limited. When these deposits are part of a deposit guarantee scheme, they are also seen as a stable source of funding for the bank.

To be able to construct the behavioural maturity calendar we need to know when a deposit matures. However, there might not be a single date on which

a client withdraws the total deposit. It is more likely that the balance on the account fluctuates over time and that the total amount is only fully withdrawn when the client moves to another bank. Therefore, modelling the behaviour of these deposits can be defined as answering the following questions:

- what is the expected average balance of the deposits, at any future point in time, given the current balance?
- what is the expected number of clients that are still with the bank at any future point in time?

As discussed above, these questions can be answered on a client level or on a portfolio level. The combination of the expected number of clients and expected average balance gives the expected future cashflows.

There are a number of drivers that can affect the behaviour of the clients and thus are drivers for the liquidity risk. These can be divided into client- or contract-specific characteristics and macroeconomic drivers. Examples of the former are: type of client (retail clients behave differently from corporate clients); size of the deposit (clients with higher balances behave differently from clients with only a small deposit); sales channel (Internet-only deposits are considered to be more volatile); and whether the deposit falls under a deposit guarantee scheme, as these are more stable than other deposits. The interest rate on the deposit can also be an important driver, especially when compared with the rates of the bank's competitors.

Macroeconomic drivers or the state of the economy in general can also have an impact on client behaviour. High unemployment can indicate that clients need to use their reserves, which will lead to a decrease in their savings. On the other hand, a booming economy might lead to increased investments by corporate clients, reducing their excess savings. The impact of these macroeconomic drivers can differ per portfolio, and statistical analysis should evidence these assumed correlations when modelling their behaviour.

Non-maturing assets (current accounts including credit cards)

Non-maturing assets are loans that do not have a fixed maturity date, and the client has the option to decide when to repay the amount borrowed. There are

different categories of such loans. The most common is a current account that allows the client a debit position. These are used by retail clients that have a small limit, in order to be able to absorb fluctuations in their payment patterns, and by corporate clients that use these accounts for their working capital. Note that these products can in fact be the same as the non-maturing savings product mentioned above, ie, when the balance on the account is positive it becomes a non-maturing saving. Instead of modelling non-maturing assets and non-maturing savings separately, banks can model them as one product with either a positive or a negative balance.

A similar type of product is a revolving credit, which allows the client to withdraw up to a certain limit. These products can have different amortisation schedules, where the client pays a fixed percentage of the outstanding amount or a fixed percentage of the limit, or be interest-only accounts where the client pays only the interest.

Another example of a non-maturing asset is a credit card. Here, the client has the option to make payments up to a specified limit. Some clients (transactors) use a credit card only for payments and repay the amount every month, while other clients (revolvers) use it as a source of credit and repay only a small amount each month.

Non-maturing assets introduce two types of liquidity risk. When contracts have no fixed maturity date the bank does not know when these loans will be repaid and what the maturity of the funding raised to fund these loans should be. More important, however, is the risk that clients draw on their credit lines more than anticipated. When clients only use 50% of their facility, they have the option to withdraw the other 50%, which means that the bank needs to have the funds available to accommodate such a request. When modelling non-maturing assets we therefore need to answer similar questions to those for the non-maturing liabilities:

- what is the expected debit balance on the account, for any future point in time, given the current balance?
- which clients are still expected to be with the bank at any future point in time?

The drivers that influence the behaviour on non-maturing assets can again be either client or contract specific or macroeconomic. The age of the contract, for example, can give an indication of the future expected drawings.

One macroeconomic driver that can influence behaviour is the general state of the economy. When the economy is booming companies might need more credit to finance their activities and as a result the usage on their credit lines will increase. The same holds for consumers that are more optimistic about their financial situation and therefore use additional credit to finance their spending. On the other hand, in a recession the demand for credit might reduce, probably with the exception of clients that face financial difficulties and therefore need the credit facility to meet their normal obligations.

Residential mortgages

Residential mortgages have a long maturity, and therefore introduce liquidity risk because the liabilities to fund these mortgages are usually shorter. Most of the cashflows resulting from a mortgage contract are known when the mortgage is originated, and depend on the type of mortgage. However, in a mortgage contract the client has several options to adjust the mortgage, and these will affect the projected cashflow schedule. Clients can make extra repayments, increase their mortgage or they can repay in full when they move to a new house. The last option in particular has a significant impact on the behavioural calendar. If, on average, clients move every 10 years, the average mortgage will not be 30 years but only 10 years, indicating that the required funding for these mortgages only needs to be 10 years.

When modelling the behaviour of residential mortgages we can answer the following question: what is the probability that the client repays its mortgage, partly or in full, in the next period, for any future point in time, given the current outstanding mortgage? Note that this modelling approach assigns probabilities to a limited number of possible events that can occur.

There are several drivers that influence the behaviour of clients with a mortgage; these can be client or mortgage-contract specific or macroeconomic. Examples of client-specific drivers are the age and the credit score of the client. Younger clients have a higher probability of moving to a new house and will therefore have a higher prepayment rate. The opposite can apply for clients with a poor credit history, who cannot easily find a mortgage with another bank. Mortgage-specific drivers affecting client behaviour can be the type of mortgage (annuity, linear, interest only) or the time until the next interest reset date.

Macroeconomic parameters can also have a significant impact on client behaviour. When interest rates are low it might be beneficial for clients to

refinance their mortgage, and lock-in these low rates for a long time. When interest rates are rising or when the general economic circumstances do not motivate people to move (high unemployment, low GDP growth) prepayment rates can be reduced, indicating that the average liquidity maturity date will be extended.

The prepayment rate estimated by a liquidity risk model for residential mortgages is an important element for any bank with a significant mortgage portfolio. Since the initial maturity of the mortgages is very long, small changes in prepayment rates can have a significant effect on the mortgage portfolio after 10 years or thereafter. This has an impact on the amount of long-term funding the bank has to raise, and also affects the repricing risk for the bank, especially when low mortgage rates are locked in for a long period.

Term loans

Term loans, which can be loans to retail, small and medium-sized enterprises or corporate clients, have a similar liquidity risk to residential mortgages. Although the latter have a fixed contractual maturity date, the client often has the option to prepay the loan at an earlier date. Term loans, however, have an additional liquidity risk component, which is called extension or rollover. For corporate clients in particular, a large number of term loans have a short maturity (for example, three or six months) but the client will often roll over the loan. This is not usually an explicit option in the contract, and the bank will have to make a new credit decision. However, clients often see it as an implicit option, and when their creditworthiness is not a problem these loans will normally be extended. From a liquidity risk perspective this means that the actual date that the cash returns to the bank will be later than the original contractual maturity of the first loan.

Modelling term loans can be summarised by answering the following question: what is the probability that, in the next time period (eg, next month), the outstanding amount of the term loan will be (partly) repaid, follow the contractual cashflow schedule or be extended? As with the other models, risk drivers can be either client or contract specific or macroeconomic.

Term deposits

Term deposits are those with a fixed maturity date. From a liquidity risk perspective they are preferred over non-maturing deposits because clients

cannot easily withdraw them. Therefore, they often receive a higher interest rate. Client behaviour regarding these deposits does not often deviate much from the contractual behaviour. Clients need to pay a penalty when withdrawing their deposit before maturity, so they hardly ever do this. On the other hand, if banks assume that clients withdraw their deposits at maturity and clients roll over the deposit into a new deposit or move the funds to a non-maturing deposit, the funding remains within the bank, which benefits its liquidity position.

Modelling term deposits answers the following question: what is the probability that the client will withdraw their deposit in the next period, given a certain time until maturity, or will roll over the deposit into a similar type of product at maturity? As with the other models, risk drivers can be client or contract specific or macroeconomic.

Collateral

The models discussed above concerned “normal” banking products, such as loans and deposits. For a bank that has a substantial collateralised derivatives portfolio, another source of liquidity risk is in unexpected collateral calls. These derivatives can have a contractual cashflow schedule, but such cashflows are often limited compared with the collateral postings required when the market value of the derivatives substantially changes due to adverse market (interest rate) movements. Banks should therefore prepare for these potential collateral calls by regularly analysing the potential impact of market movements on the collateral position of the bank and ensure that sufficient liquidity is available to cover these potential adverse scenarios.

Liquidity models and stress testing

In the liquidity models discussed above, the relationship between the behavioural cashflows and client/contract and macroeconomic drivers is modelled. The use of macroeconomic drivers permits different behavioural calendars to be created for different economic scenarios. In their planning process, banks often consider different economic scenarios; eg, a base case scenario, a positive economic scenario and an adverse economic scenario. By generating liquidity mismatch calendars for these different scenarios, banks can analyse the impact on their liquidity position and take these insights into account in their planning process.

MANAGEMENT OF THE LIQUIDITY POSITION

The behavioural maturity calendar explained in the previous section forms an important input into a bank's liquidity management process. It is used to determine the long-term liquidity requirements and the long-term funding transactions that are included in the funding plan. The latter are based on projected balance-sheet growth and the composition of the required balance sheet. The treasury function maintains access to financial markets and executes long-term funding transactions to manage the liquidity mismatch on the balance sheet. A long-term liquidity mismatch will typically be managed with funding instruments such as long-dated debt instruments. Cross-currency swaps are used to manage the liquidity mismatch per currency, and therefore help to address the FX convertibility risk. For timing long-term funding transactions certificate of commercial paper and certificate of deposit (CP/CD) products, interbank deposits and FX swaps can be used to cover temporary liquidity needs if long-term funding transactions are scheduled for a later date.

To manage short-term liquidity positions the treasury function has the required oversight to manage the positions. This ensures that the liquidity position of the bank remains within its defined risk appetite. Traditional activities for daily management of short-term liquidity are:

- management of the liquidity positions at the start of the day to meet obligations in all currencies;
- management of positions of unencumbered collateral at central banks to cover the minimum intraday buffer requirement allowing settlement of transactions;
- management of nostro positions to minimise the liquidity drawn under available facilities of correspondent banks;
- management of collateral positions and cash buffers under the control of the treasury to mitigate liquidity risk arising from unexpected outflows;
- management of the liquidity position by issuing CP/CD or using short-term interbank deposits.

LIQUIDITY STRESS TESTING

Liquidity risk stress testing is performed to measure the impact of infrequent but plausible stress scenarios on the liquidity position. The key input to stress testing is a complete set of identified liquidity risk drivers, both on and off the balance sheet. Typical risk drivers are deposit run-offs, reduced access to funding markets, unexpected collateral calls and additional drawings on committed or uncommitted credit lines. The objective of liquidity stress testing is to measure the development of these liquidity risk drivers under normal and stressed conditions. The sensitivity of the liquidity position is related to the defined liquidity drivers. The results provide a basis for defining risk appetite statements for liquidity risk. These will include the limits for an acceptable liquidity mismatch, guidance on diversification of funding sources, a funding profile, refinancing risk indicators, the size and composition of the portfolio of liquid assets and net cash outflow limits.

For many banks the most important risk drivers in a liquidity stress scenario are the run-offs of customer deposits in combination with reduced access to wholesale funding markets to compensate for the run-offs. Estimating possible deposit run-offs obviously remains a challenge for banks who have not experienced a bank run before, and it is difficult to translate the situations from other failed banks (eg, Northern Rock, DSB) to any specific bank. In the development of liquidity stress scenarios the link with solvency stress tests will become more important, as there is a clear interaction between the two. A deteriorating solvency position can lead to a liquidity outflow, which potentially can only be mitigated against higher costs, putting further pressure on the bank's solvency position.

LIQUIDITY-GENERATING CAPACITY

Credit institutions establish investment mandates for portfolios of liquid assets that can be used to generate liquidity in a liquidity stress situation. To provide assurance on the liquidity-generating capacity, and thus to manage the market liquidity risk, the liquidity buffer of financial investments is subject to an assessment of the characteristics of the security and liquidity in cash markets and securities finance markets. To align the investment mandate with the risk appetite, relevant market and credit risk limits are established. A diversified mix of financial investments will allow the treasury function to generate liquidity in financial markets via the selling of bonds outright, reverse repo transactions or by negotiating secured facilities with a diverse

group of counterparties. To provide assurance on the liquidity value of financial investments the liquidity assessment is embedded in the investment process and includes the following:

- the characteristics/features of the financial instrument;
- the denomination in a convertible currency with low foreign exchange risk;
- traded volumes, to determine liquidity in the market for the financial instrument;
- bid–offer spreads and quoted volumes;
- listings on a developed and recognised exchange;
- listing in general collateral schemes (widely accepted collateral);
- the relative value to other financial instruments;
- the assessment of potential wrong-way (highly correlated) risk;
- eligibility criteria for the European Central Bank to pledge financial instruments;
- high quality liquid asset assessment of the instrument, in order to report it for liquidity metrics.

This assessment supports decisions to invest in selected financial investments. Additional assurance on the liquidity-generating capacity of the financial instruments is provided in day-to-day management of liquidity, as these financial investments are used by other market participants in (reverse) repo transactions against securities or cash to cover short positions and facilitate client business. In line with regulations, a sample of financial instruments are used for financial transactions to allow risk management to assess the liquidity-generating capacity.

The required size of a portfolio of liquid assets per currency is based on the bank's current balance sheet, forecasts, liquidity stress tests and risk appetite. The liquidity stress test scenario that results in the highest outflows is used to determine both the asset mix in the portfolio and a minimum required amount of cash. To meet obligations in the few first days under stress, an amount of cash must be readily available to match cash outflows

within the settlement time frame of securities finance transactions and/or outright sale of financial investments. Additional cash will be generated by reverse repo and/or outright selling of bonds from the earliest settlement date onwards. The asset mix of a portfolio does not need to change significantly unless new insights from stress tests or changes in balance-sheet composition support decisions to revisit it in the future. To arrive at the currency composition of the buffer, a forecast, liquidity stress test and risk appetite statement are required for material currencies, or at least for the euro- and US dollar-linked currencies.

INTRADAY LIQUIDITY MANAGEMENT

The risk metrics discussed so far do not address intraday liquidity risk. The latter is a key element of the treasury function, where both liquidity and collateral are actively managed to meet payment and settlement obligations throughout the day, over different time zones. Insight into scheduled in- and outflows (including time-critical obligations) is obtained, and gross liquidity inflows and outflows per currency during the day (including the matched scheduled in- and outflows) are presented. This information allows the treasury function to anticipate intraday funding needs. Intraday funding can be raised and collateral can be mobilised to obtain such funds.

In addition, the resilience to withstand a significant increase in intraday liquidity demand by the business lines and/or increased intraday collateral needs should be assessed. This can be captured by a risk appetite measure, where maximum intraday liquidity need is compared with the available intraday funding. We can consider using a simulation approach: a stochastic model can be used to simulate the impact of intraday stress events on the intraday liquidity management process. The fundamental assumption is that incoming payments can be considered as stochastic variables (in amount and timing) and that the underlying probability distribution can be derived from a set of historical intraday evolutions of payments for a currency or an account. For the timing of each inflow and outflow, the time difference between flows is drawn from the distribution of times, which is derived from the historical distribution. This permits the aggregation of historical data into a single representative day, which can be used to perform analysis on both business as usual and stress testing for specific intraday liquidity risk events and scenarios.

The liquidity management processes within financial institutions have been enhanced due to standards for payment notifications from SWIFT,¹ large-value payment systems such as the Single Euro Payments Area (SEPA) and the correspondent banking network, allowing continuous insight into nostro accounts, and thus monitoring and reporting more accurate position information during the day. Updating positions continuously and managing liquidity on an intraday basis mitigates some of the risks that were observed in the midst of the global financial crisis, when participants did not meet their payment and settlement obligations in a timely manner. The strong dependencies between financial institutions that had an impact on the financial system during the crisis are also partly mitigated. These dependencies can only be mitigated further as and when monitoring, reporting and management of liquidity positions on an intraday basis are further enhanced. The above-mentioned initiatives mitigate risks in the existing payment systems. At the time of writing a new initiative had been launched by the European Payment Council to meet client demand with regard to the timely processing of payments and stronger liquidity management (European Payment Council 2017). This requires payment service providers to upgrade their payment handling capabilities by May 2019 in order to reduce the processing time of payments from ten seconds to around five seconds. This initiative will introduce new use cases and challenges to manage associated risks. Both financial institutions and regulators acknowledge the need for continuous development of intraday liquidity risk management.

The monitoring and reporting tools for intraday liquidity risk (BCBS 248) complement the previously existing tools for liquidity and capital metrics (Basel Committee on Banking Supervision 2013). However, the requirements for monitoring and reporting are different from all metrics reported so far. The intraday liquidity management requires position information based on aggregated retrospective liquidity measurements for the day. This type of position information is not readily available for all credit institutions, and can be a challenge to retrieve, as it is not typically stored for accounting or risk management purposes.

CONTINGENCY PLANNING

A bank's contingency funding plan should address its strategy for handling liquidity stress situations. It describes the framework for analysing and responding to liquidity stress situations. Different risk factors should be identified, stress tests, scenario analysis and potential outcomes signalling stress defined, and related risk drivers and metrics monitored to assess the severity of a liquidity crisis and/or market disruptions. These include a set of early warning indicators that help to identify possible liquidity stress at an early stage. To support decisions to activate the contingency funding plan, the treasury function should make use of the output of stress tests and scenario analysis and/or observed severe intraday disruptions. In addition to monitoring to identify liquidity stress, a list of possible liquidity-generating actions should be prepared.

Governance on how to activate the contingency funding plan is reflected in delegated mandates for the key groups of individuals who will coordinate the actions to mitigate the liquidity risks. The contingency funding plan should be defined for the whole organisation. However, selected business line and entity-specific contingency plans should be prepared if these are more exposed to liquidity risk.

CONCLUSION

Liquidity risk management in European banks has improved substantially since the global financial crisis, driven by the Basel III regulations and the introduction of the Internal Liquidity Adequacy Assessment Process (ILAAP). Significant improvements have been made in liquidity risk measurement, liquidity modelling and liquidity stress testing. However, at the time of writing, not all banks had fully implemented the principles of sound liquidity management (Basel Committee on Banking Supervision 2008). We expect new challenges to emerge, be these market driven through new business models, enhanced payment systems, changes in the regulatory landscape or in competition from financial technologies. Therefore, liquidity risk management will have to develop further to address the rapid developments in the financial world.

¹ See <http://www.swift.com>.

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Managing Reserve Assets

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From the 1950s onwards, the asset side of a bank's balance sheet would usually show a portfolio of sovereign bonds and bills. This changed in the first decade of the 21st century: with the flawed thinking that market liquidity can be taken for granted, the practice of holding assets of sovereign debtors fell into disuse in favour of holding higher yielding bank bonds and corporate bonds. This would have been attractive from the return point of view, as government debt carries lower returns than bank debt (Choudhry 2012, p. 622). It turned out, however, that these investments were less liquid than government debt.

Prior to the 2007–9 financial crisis, the assumption of guaranteed liquidity was somewhat correct: financial markets were liquid, and funding was easily available at low cost, but the emergence of the crisis showed how rapidly market conditions can change, leading to a situation where several institutions, regardless of their capital levels, experienced severe liquidity issues, forcing either an intervention by the central bank or a shutdown of the institution (Bonner and Eijffinger 2016).

Analogously to their thinking about market liquidity, banks did not consider proper liquidity management to be a crucial part of their daily operations, but rather had a somewhat pragmatic approach to measuring and managing their liquidity (Baretzky 2012, p. 62). This resulted in a more-or-

less unsystematic view of liquidity risk as part of a bank's asset–liability management.¹ As a logical consequence of this, the financial crisis showed that sustainable liquidity management is crucial for a bank's survival (Bodemer 2011, p. 282). The crisis emphasised the importance of a proper liquidity management for financial institutions as well as regulators (Hull 2012, p. 385).

If market turmoil can bring the global financial system to its knees, then it is important to enhance our understanding of the mechanisms of liquidity and manage the respective risk properly (Fecht *et al* 2011, p. 6). In 2007–9 many banks relied heavily on wholesale deposits and faced serious trouble as investors lost confidence in markets and financial institutions. For this reason many banks found that many instruments for which there had previously been a liquid market could only be sold at fire-sale prices (Hull 2012, p. 385). Even under “normal” market conditions, the liquidity needs of a financial institution are somewhat uncertain. Therefore, banks must assess a worst-case liquidity scenario and make sure that they can endure such a scenario by either borrowing cash externally or converting assets into cash (Hull 2012, p. 385). The latter is the primary purpose of a bank's liquidity reserve or, more specifically, a bank's liquid asset buffer. Corresponding to the events in 2007–9, we shall outline the importance of liquidity management with particular focus on the risk management of the liquidity reserve of a financial institution and on the strategies related to such an approach.

The chapter is organised as follows. The following section deals with the basics of asset and liability management and liquidity management. The third section discusses several strategies for managing a bank's liquidity buffer. The fourth section concludes.

BANKS' LIQUIDITY MANAGEMENT AND LIQUIDITY REGULATION

In the following, we provide a brief overview of banks' asset and liability management, attempt to define the term “liquidity” and show how banks can manage their liquidity in general. We then discuss the regulator's view of banks' liquidity management, particularly the liquidity ratios introduced under the Basel III framework.

A brief overview of the principles of liquidity management

Liquidity management is part of a bank's asset–liability management (ALM). This is a generic term for the high-level management of a bank's assets and liabilities and the risks that arise from them. It is a strategy-level discipline but operates at a business-line level and is also a strategic level and tactical issue. The principal function of an ALM desk, or treasury desk in general, is to manage the bank's interest rate risk and liquidity risk (Choudhry 2011, p. 144). Good ALM addresses mismatched risks in their two primary forms: interest rate risk and liquidity risk. For the latter, ALM gives an overall picture of a bank's short- and long-term liquidity and its profile in all relevant currencies (Bessis 2010, p. 268).

The meaning of the term “liquidity”, however, is not commonly agreed: several definitions can be identified both in the literature and in practice (Heidorn and Schäffler 2011, p. 310). Within the financial system, three broad types of liquidity can be distinguished: central bank liquidity, funding liquidity and market liquidity; these capture the workings of the financial system sufficiently on an aggregate level.² The links between these liquidity types are dynamic, complex and strong. Hence, they can have positive or negative effects on the stability of a financial system and the financial institutions that operate within it. In uneventful financial periods the effects are positive and help to redistribute liquidity within the financial system efficiently and freely, so that, overall, liquidity does not matter (Nikolaou 2009, pp. 42ff). While funding liquidity and market liquidity are crucial elements of bank's liquidity management, relying heavily on the bank's business model, and therefore intrinsically linked to both sides of the bank's balance sheet, economic or central bank liquidity is measured by money supply and is influenced by a country's economic growth and stability, monetary circulation and monetary policy (Schäffler 2011, p. 12; Farag *et al* 2014, p. 36).

Funding liquidity and market liquidity relate to the mix of assets the bank holds and its various funding sources, particularly the bank's liabilities, which must be met when they become due (Farag *et al* 2014, p. 36). Therefore, a bank's funding liquidity relies on both the idiosyncratic liquidity risk arising from its operations and market liquidity risk, and both risks are connected to several other risks as well. In this context, a misallocation of the liquidity reserves leads to illiquidity and therefore the bank's insolvency. This is, however, a fundamental difference from other types of risk, such as market risk, credit risk or operational risk. While the latter can be covered

by the bank's capital, they are limited to funding liquidity risk. Crucially, the risk of illiquidity and insolvency threatens the existence of a financial institution. While the other risks may also generate tremendous losses, which a financial institution is normally able to neutralise over the remainder of a business year or back with its own equity, liquidity risk does not permit such a period of grace: illiquidity and therefore insolvency is sudden and irreversible (Heidorn and Schäffler 2011, pp. 313ff; Schäffler 2011, pp. 11ff; Nikolaou 2009, pp. 10ff). It is the purpose of the liquidity reserve to counter potential illiquidity.

In this chapter, liquidity is regarded as the ability to fund obligations immediately. Consequently, a bank is illiquid if it is unable to settle obligations on time: in such a case, the bank defaults (Bodemer 2011, p. 282). Given this definition, it can be said that a bank's funding liquidity risk is driven by the possibility that, over a specific horizon, the bank does not have the ability to meet its obligations when they become due (Choudhry 2012, p. 590).

Positive maturity transformation (ie, transformation of short-term deposits into long-term loans) is the fundamental role of banks, but it exposes them to liquidity risk, ie, the risk that demands for repayment outstrip the capacity to raise new liabilities or liquefy assets (Basel Committee on Banking Supervision 2008a). To mitigate liquidity risk, banks have two options: they can attract stable funding sources, which are less likely to flow out during crises, and they can hold a portfolio of highly liquid assets as well as a certain amount of cash. The latter can be used when the bank's liabilities fall due. This portfolio of liquid assets is particularly important if a bank is unable to roll over or substitute its current funding sources or if other assets are not easy to liquidate (Frag *et al* 2014, p. 36).

Predicting future cashflow requirements properly is quite a challenge even under favourable market conditions, as it requires the ability to use information from the bank's various operations to assess the impact of external events on the availability of funding liquidity. This challenge increases during stressed conditions, as the assumptions underlying liquidity risk may change, notably through changes in counterparty behaviour and market conditions that affect the liquidity of financial instruments and the availability of funding (Basel Committee on Banking Supervision 2008a).

Liquidity risk is usually associated with a funding gap, ie, excess assets over liabilities. But it can also be the other way around: excess liabilities

over assets. So, there's a liquidity risk in both cases: either funding must be obtained, or surplus assets must be sold off. The ALM desk must be aware of its future funding or excess cash positions and act accordingly (Choudhry 2011, p. 151ff). Lastly, under its liquidity management mandate, it is also the ALM desk's duty to maintain liquidity at times of crisis, and, more specifically, to maintain crisis prevention and crisis survival. Potential liquidity crises should be addressed by a conservative assumption in stress tests to ensure that the bank's liquidity reserves consist of qualitatively and quantitatively sufficient assets.

The regulatory view on liquidity management

The global financial crisis was caused by uncertainty over the solvency of financial institutions, and primarily took place in the wholesale funding markets (Gatev and Strahan 2006; Huang and Ratnovski 2011). The Basel III framework seeks to address this liquidity risk through the liquidity coverage ratio (LCR), a liquidity requirement to promote short-term resilience (Ratnovski 2013, p. 3; Basel Committee on Banking Supervision 2013a, p. 1).

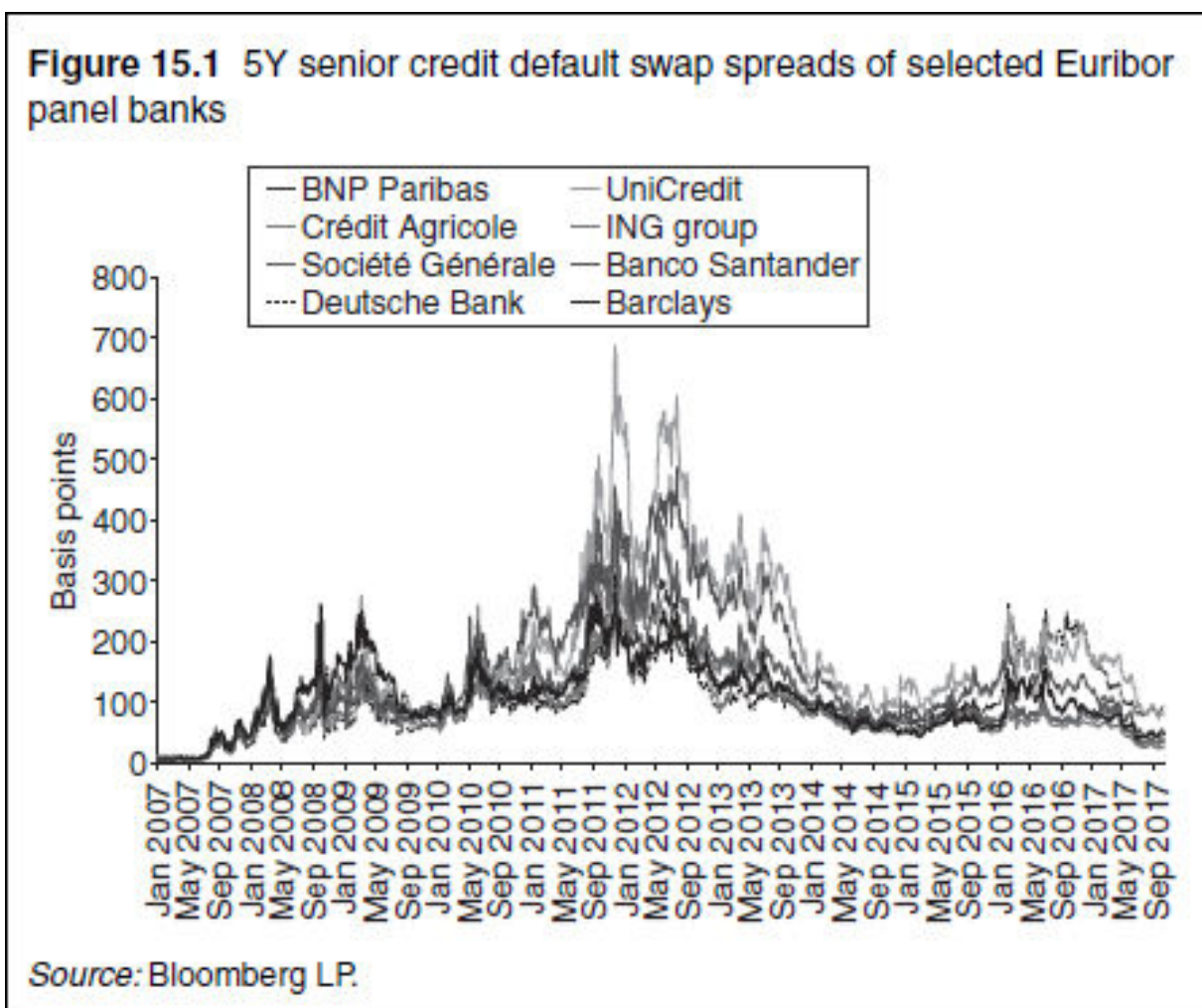
ALM desks run different currency exposures, eg, transferring liquidity from a currency that is available to one which is needed, and liquidity risk is measured, monitored, reported and managed in various currencies (Matz 2011, pp. 294, 506). However, the LCR is a single-currency liquidity model. By focusing on a single currency, which is likely to be the bank's home currency, this ratio primarily refers to liquidity risks arising from certain products and counterparties. The introduction of the LCR under Basel III challenged banks' ALM (Kleffmann *et al* 2011, p. 1).

The financial crisis and new (inter)national liquidity standards

The financial crisis was a catalyst for significant bank regulation reforms, as the pre-crisis regulatory framework turned out to be inadequate for coping with large financial shocks. The Basel III framework envisioned a rise in bank capital requirements and the introduction of new liquidity requirements such as the LCR (De Nicolò *et al* 2012, p. 2).

Prior to the financial crisis, interbank markets were among the most liquid in the financial sector. They played a key role in a bank's liquidity management and, as implied above, in highlighting the relationship between

economic and funding liquidity in the transmission of monetary policy. As the financial crisis worsened in September 2008, the interbank market's liquidity dried up as banks preferred to hoard cash instead of lending it out, even at short maturities.³ Central banks' massive injections of liquidity did little to restart interbank lending. The failure of the interbank market to redistribute liquidity became a key feature of the crisis. But it was not just the interbank market that showed massive turbulence; as [Figure 15.1](#) illustrates, at the same time, the banks' funding costs leapt to unprecedented highs (Heider *et al* 2009, p. 7).⁴



Due to the economic consequences of the financial crisis, national regulators felt impelled to overhaul their respective liquidity frameworks. We shall briefly address the German regulatory view of liquidity in this

analysis.⁵ Moreover, we shall discuss the measures regarding liquidity introduced in Basel III.

In Germany, banks' liquidity management is primarily regulated by the "minimum requirements for risk management" (Mindestanforderungen an das Risikomanagement, or "MaRisk"), which set out the qualitative requirements of Paragraph 25a of the German Banking Act ("Kreditwesengesetz" or "KWG") in greater detail. Under the MaRisk requirements, banks must ensure that they are able meet their financial obligations when they become due.⁶ More specifically, module BTR 3.1 of MaRisk describes general liquidity requirements that must be met by all financial institutions in Germany.⁷ In addition, module BTR 3.2 regulates publicly traded banks; to safeguard their solvency, these institutions must maintain an adequate liquidity reserve consisting of cash and highly liquid assets for at least one week, as well as using other assets for at least one month (BaFin 2013).

The Basel Committee for Banking Supervision (BCBS) drafted a new regulatory framework (Basel III) from 2008 onwards in order to achieve a more stable and less vulnerable banking system in response to the financial crisis and to specify short- and long-term liquidity requirements as key concepts reinforcing banks' resilience to liquidity risks (Bonner and Eijffinger 2016). The LCR was developed to promote the short-term resilience of the liquidity risk profile of banks (Basel Committee on Banking Supervision 2013a, pp. 3, 25).

The globally harmonised liquidity standards of Basel III should replace national liquidity regulations for the foreseeable future (BaFin 2013). The LCR has to be reported to the respective regulator, on a monthly basis at least (Seifert 2012, p. 311). This is a unique supervisory step and there is a wide consensus about the rationale and merits of the new liquidity requirements and the LCR in particular (Borio 2009, p. 8; Basel Committee on Banking Supervision 2008a, p. 4; Bonner and Eijffinger 2016).

Short-term liquidity: liquidity coverage ratio

The LCR is a short-term measurement, which requires financial institutions to hold an amount of high-quality liquid assets (HQLA) at least equal to their net cash outflows over a 30-day stress period (Basel Committee on Banking Supervision 2013). The LCR metric not only promotes short-term resilience to liquidity shocks and ensures that a sufficient amount of HQLA are

maintained by the bank to offset cash outflow in a stressed market environment but can also be used to identify the amount of unencumbered HQLA required to offset net cash outflows arising in a short-term liquidity stress scenario. A regulatory limit for the LCR ensures that banks meet this requirement at all times (Choudhry 2012, p. 664). The official introduction of LCR encouraged banks to strengthen their liquidity reserves (Bohn and Tonucci 2014, p. 61). Lang and Schröder (2015) investigated banks' demand for sovereign debt between 1999 and 2013 and showed empirically that banks' demand for sovereign debt was mainly driven by both BCBS's capital regulation and by liquidity regulation. They further showed that banks reshuffle their portfolios towards the marketable debt of their home sovereign in times of market turmoil.

Table 15.1 LCR liquid assets

Level 1 liquid assets	Level 2 liquid assets
Cash	Sovereign or agency assets qualifying for 20% risk-weighting
Central bank reserves	
Marketable securities representing claims on sovereign, central banks and similar institutions	Corporate and covered bonds
Government or central bank debt issues in the domestic currency	

Source: based on Hauschild and Buschmann (2014, p. 337).

The stress scenarios specified by national regulators contain both institutional or idiosyncratic stresses and systemic shocks. These stress scenarios are based primarily on experience from the financial crisis. A time horizon of 30 days⁸ was chosen by national regulators with the assumption that during this period of time a stressed bank, as well as regulators and the

central bank, will take sufficient measures to overcome the liquidity shortage (Basel Committee on Banking Supervision 2013b, p. 4).⁹ This is important because the outflow value denominator drives the liquidity reserve's size requirement.¹⁰ Under Basel III the LCR is calculated as

$$\frac{\text{stock of HQLA}}{\text{stressed net cash outflows over a 30-day period}} > 100\% \quad (15.1)$$

The rules on how to define an asset as high quality and liquid and to construct a stressed cash outflow are specific, pretty detailed and governed by the following principles. First, the stock of HQLA should have a low credit risk and a low market risk, to permit easy and confident evaluation. The stock of HQLA is divided into two subgroups ([Table 15.1](#)): level 1 assets (cash, central bank reserves), and level 2 assets, of which the stock should be composed of at least 60% of level 1 assets; the remainder can be level 2 assets (Choudhry 2012, p. 664).

Table 15.2 Stressed net outflow provisions of LCR

Funding sources	Run-off-assumption within 30 days
Retail deposits, unsecured funding from small business customers	Stable: 5%; less stable: 10%
Unsecured wholesale funding provided by non-financial corporate, sovereigns, central banks and agencies	25% operational relationship; 75% of "others" where deposits mature or are callable
Unsecured wholesale funding provided by others (including financial institutions)	100% where deposits mature or are callable
Secured funding arrangements	0% backed by sovereign or central bank debt meeting the same criteria as Level 1 assets; 100% for all other liabilities maturing
Embedded credit-rating downgrade triggers	100% of posted collateral in the three-notch credit rating downgrade
Derivatives and associated collateral valuation changes: collateral that is required following changes in derivatives mark-to-market values to be determined by national regulators	20% add-on is required where derivative collateral does not comprise cash or high-quality sovereign debt
Asset backed commercial paper conduits and similar vehicles	100% where debt matures or is callable
Required liquidity cover for bank liquidity and back-stop credit lines given non-financial corporations, sovereigns and financial institutions	100% liquidity facilities; 10% credit facilities; 100% for both provided by financial institutions

Source: based on Hauschild and Buschmann (2014, p. 338).

Second, the main assumption in the denominator is that the reason for the 30-day period is an idiosyncratic and a market-wide liquidity shock. This assumption has to be included in a bank's stress-test scenarios (Choudhry 2012, p. 664). As can be seen from the preceding remarks, the LCR increases banks' liquidity-risk-bearing capacity during short-term liquidity shocks (Brzenk *et al* 2011, p. 6).

Although the composition of the HQLA portfolio depends on the characteristics of the comprised securities, the calculation of the stressed net cash outflow is subject to certain provisions shown in [Table 15.2](#).

As can be seen in [Tables 15.1](#) and [15.2](#), the LCR calculation applies certain weighting factors for the HQLA as well as the stressed cash outflow to keep the evaluation scope of the single positions as small as possible. In addition, several other restrictions have to be taken into account, such as cutting cash inflows by 75% of the total cash outflows, so that there is always an imputed liquidity gap. To close this gap, a liquidity reserve is required (Kleffmann *et al* 2011, p. 2).

This standard aims to ensure that, to meet its liquidity needs within a 30-calendar-day liquidity stress scenario, a bank has an adequate stock of unencumbered HQLA consisting of cash or assets that can be converted into cash at little or no loss of value in private markets (Basel Committee on Banking Supervision 2013b, p. 1) The LCR metric indirectly provides short-term protection to liquidity shocks and identifies the necessary amount of unencumbered, high-quality highly liquid (HQHL) assets required to neutralise short-term liquidity stress-scenario-driven net cash outflows (Choudhry 2012, pp. 663ff).

STRATEGIES FOR THE MANAGEMENT OF THE LIQUIDITY RESERVE

In general, banks experience liquidity stress when actual cashflows differ from the expected ones. This is notably through changes in counterparty behaviour and market conditions affecting the liquidity of financial instruments and the availability of funding (Matz and Neu 2007, p. 103). One of the clearest lessons from the financial crisis was that many types of assets hitherto considered to be liquid were, in fact, not truly liquid. During the last quarter of 2008 many banks could not sell or repo parts of their assets. Therefore, most funding instruments banks used were either limited or

unavailable (Heidorn and Schäffler 2008, p. 24). As the previous section implies, banks need a liquidity reserve to be truly liquid and capable of being used to generate funding liquidity under all market circumstances (Choudhry 2012, p. 622). In such market turmoil a bank's liquidity reserve is the most reliable source of funding (Matz and Neu 2007, p. 103). In the following, we shall explain the overall concept of a liquidity reserve, what it should be composed of and how its size may be properly calculated. Finally, we shall give some strategies for managing it efficiently.

The liquidity reserve

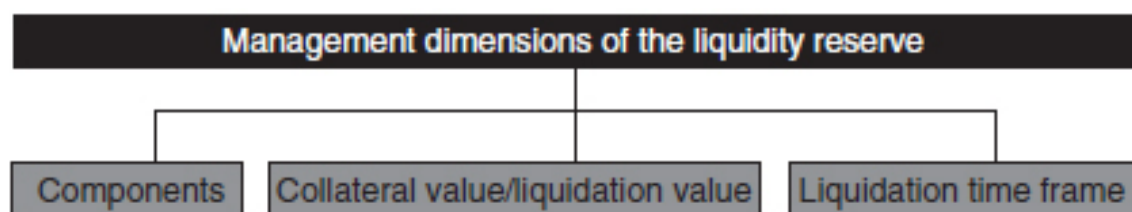
Overview and concept

In a liquidity crisis, banks are strictly required to honour their obligations at any time, and therefore it may be, depending on the intensity of the stress, that liquidity is more important than profitability and therefore greater financial losses have to be taken into account when liquidating assets to ensure sufficient liquidity (Müller and Wolkenhauer 2008, p. 244). According to bank-specific liquidity risks, the liquidity reserve should limit a bank's funding risk. Therefore, a bank should hold a quantitatively and qualitatively sufficient liquidity reserve (also known as the "liquid asset buffer", "liquidity reserve", "liquidity portfolio" or "portfolio of reserve assets"), which should be used to acquire funding on a sudden and reasonably short-term basis. However, we view all these terms as synonymous. Hence, the liquidity reserve has to be thoroughly and progressively managed to ensure that, to the maximum extent possible, assets will be available in times of financial stress (Bohn and Tonucci 2014, p. 62). Funding generated through use of these assets is also called "crisis liquidity". The overall amount of crisis liquidity limits a bank's funding risk and should be sized individually for any financial institution (Heidorn and Schäffler 2008, p. 27).

The allocation of liquidity reserves is driven by both internal and external factors. Internal factors are a bank's risk–return considerations, capital charges on the assets held in the liquidity reserve and how much accounting volatility the bank wants to have on its books. The latter clearly depends on the IFRS categories in which the bank booked these assets.¹¹ However, given the purpose of the reserve assets, they should be booked under "available for sale"; this IFRS category ensures that such assets are constantly valued with their market value, "marked-to-market" and can be sold without additional

losses in a liquidity crisis. Within national accounting regimes, a bank must use an accounting category that ensures reserves assets are free of any hidden liabilities/losses. The bank's risk appetite also plays an important role. External factors are mainly driven by market conditions: here, we must focus on the market's credit cycle, eg, how expensive is cash in comparison to bonds, or how expensive are covered bonds and corporate bonds in comparison to sovereign bonds or agency bonds?

Figure 15.2 Management dimensions of the liquidity reserve



Source: based on Hauschild and Buschmann (2014, p. 344).

When composing the liquidity reserve, we must take these factors into account and make a trade-off between the liquidation time frame and liquidation value of the single components (Figure 15.2). A high liquidation time frame is associated with high(er) value losses. So, liquidation time frame and liquidation value are negatively correlated (Müller and Wolkenhauer 2008, p. 240). The main criterion when selecting of the components of the liquidity reserve is optimal liquidisation. This can be achieved by using publicly traded securities with high market volumes and low bid–ask spreads as well as cash, central bank deposits, other central bank eligible assets or committed credit and liquidity lines (Heidorn and Schäffler 2008, p. 27). By generating additional crisis liquidity via the liquidity reserve, a bank gains time to trigger further contingency measures, such as reorganising its business model or reshuffling its funding structure (or both). Therefore, the size and structure of the liquidity reserve determines how quickly a bank needs to act (Bessis 2010, p. 286; Matz 2011, p. 62).

According to the Committee of European Banking Supervisors (CEBS), the liquidity reserve is defined as the excess liquidity available outright to be used in liquidity stress situations within a given short-term period (Table 15.3). In other words, it is the availability of liquidity that obviates the need

to take any extraordinary measures (Committee of European Banking Supervisors 2009, p. 10).

Table 15.3 Guidelines on liquidity reserves and survival periods

	Guideline	Description/contents
Liquidity reserve	1	A liquidity reserve represents available liquidity, covering the additional need for liquidity that may arise over a defined short period of time under stress conditions.
	2	Institutions should apply three types of stress scenario: idiosyncratic, market specific and a combination of the two. The core of the idiosyncratic stress should assume no rollover of unsecured wholesale funding and some outflows of retail deposits. The market-wide stress should assume a decline in the liquidity value of some assets and deterioration in funding-market conditions.
Time horizons	3	A survival period of at least one month should be applied to determine the overall size of the liquidity buffer under the chosen stress scenarios. Within this period, a shorter time horizon of at least one week should also be considered to reflect the need for a higher degree of confidence over the very short term.
Composition of the reserve	4	The liquidity reserve should be composed of cash and core assets that are both central bank eligible and highly liquid in private markets. For the longer end of the reserve, a broader set of liquid assets might be appropriate, subject to the bank demonstrating the ability to generate liquidity from them under stress within the specified period of time.
	5	Credit institutions need to manage their stocks of liquid assets to ensure, to the maximum extent possible, that they will be available in times of stress. They should avoid holding large concentrations of particular assets, and there should be no legal, regulatory or operational impediments to using these assets.
	6	The location and size of liquidity reserves within a banking group should adequately reflect the structure and activities of the group in order to minimise the effects of possible legal, regulatory or operational impediments to using the assets in the reserve.

Source: based on Hauschild and Buschmann (2014, p. 343).

Because all banks are subject to the same regulation, they are likely to hold the same assets as liquidity reserves and therefore be equally affected when a market-wide stress occurs. This leads to another type of liquidity risk, which we have not mentioned before: so-called “liquidity black holes” (Duttweiler 2009, p. 6). These arise when several market participants want to sell (buy) the same assets at the same time, with the consequence that there are no longer any buyers (sellers) in the market. Thus, liquidity dries up very quickly and these assets lose their original purpose. In this situation banks can only generate liquidity by taking heavy losses on their fire-sold assets; therefore, it is crucial that the liquidity reserve is broadly diversified (Hull 2012, p. 398).

Asset allocation and size of the liquidity reserve

When composing the liquidity reserve, it is useful to think of liquidity and illiquidity in terms of how much sellers might lose if they need to sell immediately as opposed to engaging in a costly and time-consuming search for buyers (Fabozzi *et al* 2010, p. 174). With the focus on constant funding liquidity, the ALM desk will define the size of the liquidity reserve itself with respect to the bank’s business model, the intensity of the assumed market

disruptions and, of course, regulatory requirements. Hence, by the size and composition of the liquidity reserve, a bank defines its own ability to sustain an idiosyncratic liquidity stress, a market-wide disruption or both (Bessis 2010, p. 286). Consequently, most of the assets in the liquidity reserve are HQHL assets (Heidorn and Schäffler 2008, pp. 27ff). Most relevant for the asset allocation of the liquidity reserve are what we call “off-balance-sheet liquidity consumers”, which do not require continuous funding, but might generate huge and unexpected liquidity gaps.¹²

In a liquidity stress scenario, the liquidity reserve must be able to offset both the tremendous losses in funding during a bank run¹³ and unforeseen liquidity gaps that may be inherent in the bank’s balance sheet. With this in mind, the composition of the liquidity reserve depends on the assets’ ability to generate liquidity during crises. This is primarily done by broad and deep markets with assets of high creditworthiness. In this way, the composition itself determines how much liquidity can be generated during a crisis. This volume is also determined by the potential haircut of the single assets; this haircut is reciprocal to the credit quality of the asset: the better the creditworthiness, the smaller the haircut, and therefore the better the fungibility (Heidorn and Schäffler 2008, pp. 27ff).

Liquid asset holdings are ruled by a tight regulatory regime. Therefore, based on market experience as well as common sense, we show several regulatory requirements that are somewhat similar in general, but differ in detail: according to module BTR 3.1 of MaRisk, financial institutions are expected to check whether they can meet their liquidity needs in a stressed market environment. In particular, banks must check the fungibility of their held assets. Moreover, they have to check how reliable their funding sources are. To offset sudden and unexpected liquidity deterioration, banks are required to hold an individually sufficient and sustainable liquidity reserve of highly liquid and unencumbered assets. This administrative directive applies for all banks in Germany (BaFin 2013). For publicly traded banks MaRisk’s module BTR 3.2 specifies that these banks have to overcome a liquidity shortage of at least one week by holding cash and highly liquid assets that are central bank eligible and can be sold in a stressed market environment without any significant losses. The bank potentially has to survive this period without any assistance by the central bank. For a time frame longer than a week, other assets can also be used, as long as they can be liquefied within one month (BaFin 2012).

Securities of the liquidity reserve

On the basis of the previous remarks, it is obvious that the liquidity reserve should be composed of cash and highly liquid assets. A bank must be able to sell or repo the latter under stressed market conditions. These assets have to be of the kind that will not be affected by a large downwards valuation akin to a fire sale. They should be as credit-risk-free as possible and should not have any correlation with the financial sector (Choudhry 2012, p. 627).

Consequently, the question is which securities are truly liquid. In the financial crisis only high-quality sovereign bonds had this characteristic. But here, the liquidity-black-hole problem arises: when all banks hold the same sovereign securities, markets might be trapped in the liquidity black hole when the next liquidity crisis emerges. That is why regulators allow limited a number of securities other than sovereign securities in the liquidity reserve. According to the BCBS, these securities should be traded in large, deep and active repo or cash markets. Moreover, they should have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions with individually defined haircuts over a predefined period of time (Basel Committee on Banking Supervision 2013b, pp. 12ff).

Table 15.4 Liquidity measures and proxies

Measure	Liquidity
Bid–ask spread	The smaller the bid–ask spread, the higher a bond's liquidity
NDC	The smaller the non-default component, the higher a bond's liquidity
Proxy	Liquidity
Age	The “younger” the bond, the higher its liquidity
Tenor	The shorter the tenor, the higher a bond's liquidity
Issue volume	The greater the issue volume, the higher a bond's liquidity
Trading volume	The larger the trading volume, the higher a bond's liquidity
Yield	The smaller the yield, the higher a bond's liquidity
Price volatility	The smaller the price volatility, the higher a bond's liquidity

Source: based on Hauschild and Buschmann (2014, p. 347).

Unfortunately, the degree of liquidity changes with market conditions, and it is a matter of observable historical record that the only assets that remained liquid under all market conditions were sovereign bonds. Given that the liquidity of other types of assets changes according to market conditions, banks should estimate the pertinent level of liquidity at any time: an assessment that will help them determine the level of liquidity of their non-government assets (Choudhry 2012, p. 625).

Liquidity can be measured directly and indirectly. There are several direct measures, such as the bid–ask spread or the so-called non-default-component

(NDC) of the asset–swap (ASW) spread,¹⁴ as well as indirect proxies such as age and tenor, issue and trading volume or yield and price volatility. The liquidity measures are generally accepted by market participants, whereas liquidity proxies are primarily based on empirical evidence (Table 15.4).¹⁵

The composition of the securities or products of the liquidity reserve are pretty simple to explain. As shown in the previous sections, the regulatory requirements are very similar and limit the universe of eligible securities. Given that the Basel III provisions are adapted by national regulators, we believe that banks will be expected to hold a sufficient number of the securities described in the numerator of the liquidity coverage ratio (Figure 15.3). The composition of the liquidity reserve with a given regulatory framework can be considered as a passive management strategy.

When composing the liquidity reserve, a benchmark has to be set from a return point of view. A liquidity reserve's benchmark is the bank's own sovereign bond (eg, Bunds in the case of a German bank, or obligations assimilables du Trésor (OAT) treasury bonds for French banks). The reasons why government bonds of the bank's home country are used as the benchmark are pretty simple: they are, by definition, risk-free for such banks and, according to the Basel III regime, they have the lowest haircut. Therefore, a bank's liquidity reserve is always home biased, and its returns clearly depend on the home country's yields because they presumably consist mostly of sovereign bonds.

Another problem arises from this composition, however: systemic risk. The BCBS liquidity regulation applies for each and every bank, which means that, as implied above, banks have the same kinds of government bonds, most likely those of their respective sovereigns, in their liquidity reserves. Since sovereign debt remained liquid in the financial crisis, the Basel liquidity regulation omits a sovereign stress in the assumption of the LCR. Even though banks are protected against a liquidity shock or liquidity shortfalls, they are still exposed to the liquidity risk stemming from distressed sovereign debt. Buschmann and Schmaltz (2017) show how this unaddressed risk might translate into a system-wide liquidity shock and describe, as seen in the sovereign debt crisis, how deteriorating sovereign debt can lead to an overall liquidity squeeze and non-compliance with the Basel III liquidity regulation.

Figure 15.3 Composition of the liquidity reserve according to Basel III/LCR in detail

List of securities		Haircut	
1	(a) Cash (b) Central bank reserves (including required reserves)	0%	Level 1 assets
2	(a) Marketable securities representing claims on or guaranteed by sovereigns, central banks, supranational organisations (BIS, IMF, ECB, EC), multilateral development banks (b) Sovereign or central bank debt issued in domestic currency in the country in which the liquidity risk is being taken or in the bank's home country for sovereigns with a non-zero Basel II risk weight	0%	
3	Marketable securities representing claims on or guaranteed by sovereigns, central banks, PSEs or multinational development banks with 20% risk weight under the Basel II standard approach	15%	Level 2A assets
4	Corporate debt securities with a long-term credit rating of $\geq$ AA- and not issued by a financial institution	15%	
5	Covered bonds with a long-term credit rating of $\geq$ AA- not issued by the bank itself or any of its affiliated entities	15%	
6	RMBs with a long-term credit rating of $\geq$ AA not issued – and the underlying assets not originated – by the bank itself or any of its affiliated entities	25%	Level 2B assets
7	Corporate debt securities with a long-term credit rating between A+ and BBB- not issued by a financial institution	50%	
8	Common equity shares that are exchange traded and centrally cleared and not issued by a financial institution or any of its affiliated entities are a constituent of the major stock index in the home jurisdiction or where the liquidity risk is taken and denominated in the domestic currency of a bank's home jurisdiction	50%	

Notes: Although this listing of assets meeting the theoretical liquidity characteristics initially seems to be quite logical, it was outpaced by market reality: European Banking Authority (2013, p. 24) showed that, based on EBA's empirical and principles-based analysis, equity shares, among others, are insufficiently liquid for their intended purpose.
Source: based on Hauschild and Buschmann (2014, p. 349).

In addition to home bias, the composition of the liquidity reserve mostly relies on the requirements set by the Basel liquidity regulation. But here another problem arises: the divergence of regulatory assumptions and economic reality. For example, under Basel III, Greek government debt does not require any capital charges, while debt issued by the government of Singapore, a country with far better public finances and a higher rating than Greece, requires capital charges. These capital charges make Singaporean sovereign debt unattractive as part of the liquidity reserve.

In principle, when selecting the securities, the bank has to choose the maturity of the reserve assets. There is no optimal tenor for the liquidity reserve. Therefore, it is advisable to choose maturities that best fit the requirements for reserve assets. Regarding liquidity characteristics, the benchmark of every liquidity reserve is cash. Cash has no tenor and no risk or return. However, in countries with negative central bank rates, such as in the euro area, it might cost the deposits rate. Consequently, the duration of the liquidity reserve's assets depends on its optimal added returns over cash but primarily on the assets' liquidity. As implied in [Table 15.4](#), shorter maturities are more liquid than longer ones; thus, shorter maturities should be preferred over longer ones. In addition, market depth plays an important role: when composing the liquidity reserve, assets with greater market depths should be selected, and these have, by nature, shorter maturities.

The risk management department might apply an asset maturity limit of up to two years, five years or even ten years. These limits should guarantee a proper cost-efficient mixture in the liquidity portfolio. First, this mixture reduces the negative maturity transformation. As we shall explain later, a liquidity reserve should be funded long-term, but, as shown in [Table 15.4](#), shorter tenors are more liquid than longer ones. Here, certain maturity limits prevent a maximum negative maturity transformation. Second, since shorter maturities do not feature any significant spreads (or even negative ones), a mixture between shorter and longer maturities ensures a greater average spread of liquidity reserve and therefore reduces the cost of the maintaining it.

Calculating the size of the liquidity reserve

How long the bank will last in a liquidity crisis is a crucial question when determining the size of the liquidity reserve. As implied, this reserve should be composed of assets that other financial institutions and cash lenders would accept as collateral in a financial crisis, which suggests only sovereign issuers. Thus, one benefit of holding government bonds, on both individual and aggregate levels, is that this forces banks to develop their liquidity risk management ability and run a tighter ship with respect to their liquidity policy. This is because regulators will insist on a greater liquidity reserve as a proportion of total assets for those firms with structural liquidity problems or following a poor liquidity policy; in other words, and as a logical consequence of this, the more liquidity risk a bank runs, the larger the liquidity reserve should be (Choudhry 2012, p. 622).

The exact proportion of a bank's balance sheet that is held as the liquidity reserve is a function of the bank's business operations and their resulting liquidity risk on both sides of the balance sheet, ie, for both lending business and funding. As discussed above, the given length of the survival period and percentage of overall long-term funding determine the size of the liquidity reserve: the greater the amount of long-term funding available and the shorter the set survival period, the smaller the liquidity reserve.¹⁶ Moreover, the liquidity reserve is also a function of the type of bank funding: retail funding is regarded as more stable than wholesale funding; therefore, the more a bank relies on wholesale funding, the larger the liquidity reserve should be (Choudhry 2012, p. 631).

As a consequence of the remarks above, on setting the framework for calculating the liquidity reserve, a bank needs to determine their actual retail and wholesale funding together with their respective maturities, whereby the ALM desk sets the desired survival period in accordance with regulatory requirements. Once these features have been set, a bank needs to ensure that their IT systems are able to produce projections of expected cashflows, broken down into time buckets, financial instruments and business lines, and all sources of liquidity risk. These cashflow scenarios need to be accompanied by a description of the alternative funding sources available to meet liquidity needs. This will include the liquidity reserve, which will be the first port of call for the bank's counterbalancing capacity and therefore exists to enable the bank to continue its normal business operations during an idiosyncratic or market-wide stress. By running a conservative liquidity management approach, a bank has a greater chance of surviving a liquidity stress (Choudhry 2012, pp. 631ff).

Bank-specific stress-test results are the prime driver of the size of the liquidity reserve. The largest cashflow gap determines the required size of the liquidity reserve. The cashflow should be analysed at a granular level, so that, when determining cashflows and counterbalancing capacity, a bank identifies contractual and behavioural flows and applies a conservative assumption of the liabilities' behaviour when estimating its liquidity position. When assessing cashflows, the bank calculates the sum of the expected outflows and subtracts this from the sum of expected inflows (Choudhry 2012, p. 632).

Moreover, the proportion of the bank's balance sheet that is dedicated to the bank's liquidity reserve is a function of a number of factors, including the composition of its funding and structural limitations in its ability to raise liabilities (Choudhry 2012, p. 623). The size of the liquidity reserve should be determined according to the funding gap under stress conditions over a specified survival period. This time horizon, as well as the related liquidity reserve, should not supersede other measures taken to manage the net funding gap and funding sources, and the institution's focus should be on surviving well beyond the stress period. Therefore, the survival period should only be the period during which an institution can continue operating and still meet all its obligations without needing to generate additional funds (Committee of European Banking Supervisors 2009, p. 10).

As funding liquidity risk is a bank-specific characteristic, financial institutions are expected to tailor their liquidity management, stress tests and, of course, their liquidity reserve to their specific business model. This does not preclude approaches aiming to capture liquidity risk factors that are common to all banks. The combination of a tiered market structure and the concentration of activity imply that the potential severity of contagion is higher for banking groups than for small banks at the fringe of the market. Liquidity risk varies across credit institutions, and the underlying risk should be properly reflected in the size of a bank's liquidity reserve. All material sources of liquidity risk should be included under any approach, regardless of their nature as liabilities or assets, on-balance-sheet or off-balance-sheet, currency denomination and others (Committee of European Banking Supervisors 2009, p. 11). So, in general, from an individual perspective, the ideal size of the liquidity reserve can be determined by the marginal benefits of maintaining a portfolio of reserve assets and its marginal costs. Cost and benefits need to be equal (Bohn and Tonucci 2014, p. 62).

Figure 15.4 Simplified balance sheet before liquidity stress

Assets	Bank's balance sheet	Liabilities
Liquidity reserve		Deposits/repurchase agreements
Loans		Long-term funding

Source: based on Hauschild and Buschmann (2014, p. 353).

Figure 15.5 Simplified balance sheet after liquidity stress

Assets	Bank's balance sheet	Liabilities
Loans		Long-term funding

Source: based on Hauschild and Buschmann (2014, p. 353).

Managing strategies

As argued in the previous section, the composition of the liquidity reserve within a certain regulatory framework can be interpreted as a passive management strategy. In this section we present some active management strategies: the funding strategy and related rationales, and the liquidity reserve's related spread strategies.

Funding strategy

Although short-term funding is cheaper than long-term funding, the liquidity reserve has to be funded on a long-term basis, as [Figure 15.4](#) shows. This figure is highly simplified and assumes that a bank holds only loans and securities. The latter serve as the bank's liquidity reserve. These assets are diversified and cost-efficiently funded by interbank deposits, repos for the loans and medium- and long-term funding for the liquidity reserve.

Whatever the nature of the liquidity stress is, it is quite likely that, as we saw in the financial crisis, the unsecured interbank funding will dry up, and only limited funding via repos is available (Müller and Wolkenhauer 2008, p. 240). In such a scenario, securities will be sold in the market and the liquidity reserve will be depleted. As [Figure 15.5](#) shows, this is accompanied by balance-sheet contraction and safeguarding of the loans' funding in the medium to long term.

The example given above shows two essential aspects of how a bank's liquidity reserve works.

First, the actual liquidity reserve is generated on the liability side of a bank's balance sheet by issuing long-term debt, eg, senior unsecured debt. The generated is "parked" through HQHL securities on the bank's asset side.

Therefore, the liquidity reserve serves as some kind of liquidity repository and shows that bank's liquidity is clearly linked to both sides of the balance sheet. As long as there is no liquidity stress, the bank is likely to run a *pro forma* negative maturity transformation by financing intended short-term assets in the long term. Despite using securitised debt to generate long-term liquidity, banks can also use internal funding models that roll over on a long-term basis. Which long-term strategy is chosen depends on the bank's liquidity management strategy. In a stress environment, this long-term funding will be used to fund the bank's normal lending business. This simple example shows that funding of the liquidity reserve by repos is impractical because the liquidity reserve's purpose is to generate crisis liquidity. This can only be achieved by using long-term debt. By using repos, someone gives a collateralised loan: the bank receives cash and gives away a bond. This is not liquidity creation in a narrower sense, as the bond must somehow be funded and therefore no additional liquidity will be generated from the bank's perspective. Moreover, repos will not be renewed in an idiosyncratic or market-wide liquidity stress.

Second, through its negative maturity transformation and due to its high fungibility and good rating with low short-term yields, crisis liquidity is a relatively expensive form of liquidity. So, a trade-off between liquidity and return has to be made (Hull 2012, p. 393). This trade-off is inevitable. By holding a large enough liquidity reserve, banks can buy time until the liquidity stress ends. The negative carry from, or the funding costs of, these assets can be seen as an insurance premium for the resulting contribution to the bank's liquidity (Matz 2011, p. 288). As previously stated, by using its liquidity reserve, a bank buys sufficient time to reshuffle its business model and adjust its funding strategies to the new market environment. Therefore, the setup of a new liquidity reserve has to reflect new market conditions, eg, some securities may be left out, whereas others might be included because they remained liquid in a preceding market stress.

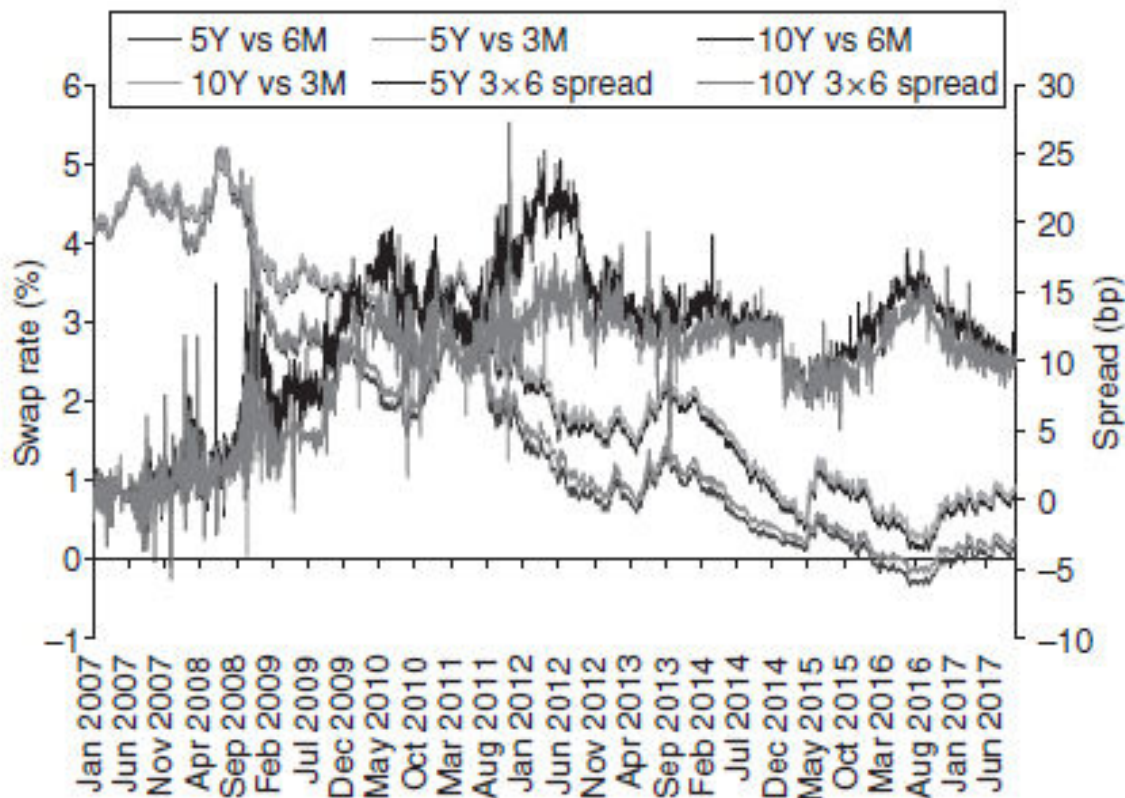
As implied and due to negative maturity transformation, it is quite likely that holding the liquidity buffer generates some basis risks. In addition, loans contain a bundle of risks such as credit risk and interest rate risk (Wall and Shrikhande 2000, p. 1). Even though they are tradeable, bonds can be seen as a loan, and therefore create such risks. The adequate management of these risks will be discussed in the following sections.

Managing basis risk

In ALM, another important source of interest rate risks is basis risks, which arise from rates earned and paid on different instruments with similar repricing characteristics but whose correlation is imperfect, meaning such rates differ by a certain spread. When interest rates change, these differences can give rise to unexpected changes in the cashflows and earnings spread between assets and liabilities and off-balance-sheet instruments of similar maturities or repricing frequencies (Leistikow 2014, p. 5). There are several situations in which banks are exposed to basis risks. Despite the several definitions in the literature, we think that basis risk derives from imperfect correlations between two rates, eg, benchmark rates such as the Euro Interbank Offered Rate (Euribor) and London Interbank Offered Rate (Libor), to which financial instruments are linked. Therefore, basis risk can emerge when banks are exposed to spreads between floating rates indexed to different repricing schedules or to the same repricing schedule in different currencies. Such spreads are quoted for the related hedging derivatives, eg, a floating–floating swap paying the three-month (3M) rate and receiving the six-month (6M) Euribor, or a cross-currency swap exchanging euro payment with US dollar payments with six month floating interest exchanges (Gentili and Santini 2014, p. 88).

As a consequence of the financial crisis, many different anomalies appeared in the interest rate market. One of these was basis spreads. These appeared for exchanging floating payments with different tenors between single-currency interest rate instruments (Morini 2009, p. 2; Amentrano and Bianchetti 2009, p. 3). The crisis increased the volatility of the quoted basis spreads, which were previously essentially stable. Since mid-2007, basis spreads have become a fundamental variable and a top priority for bank's ALM (Gentili and Santini 2014, pp. 88ff). Before the financial crisis these basis spreads were negligible. They appeared when swap rates of the same tenor but different reference rates/money market indexes diverged ([Figure 15.6](#)).

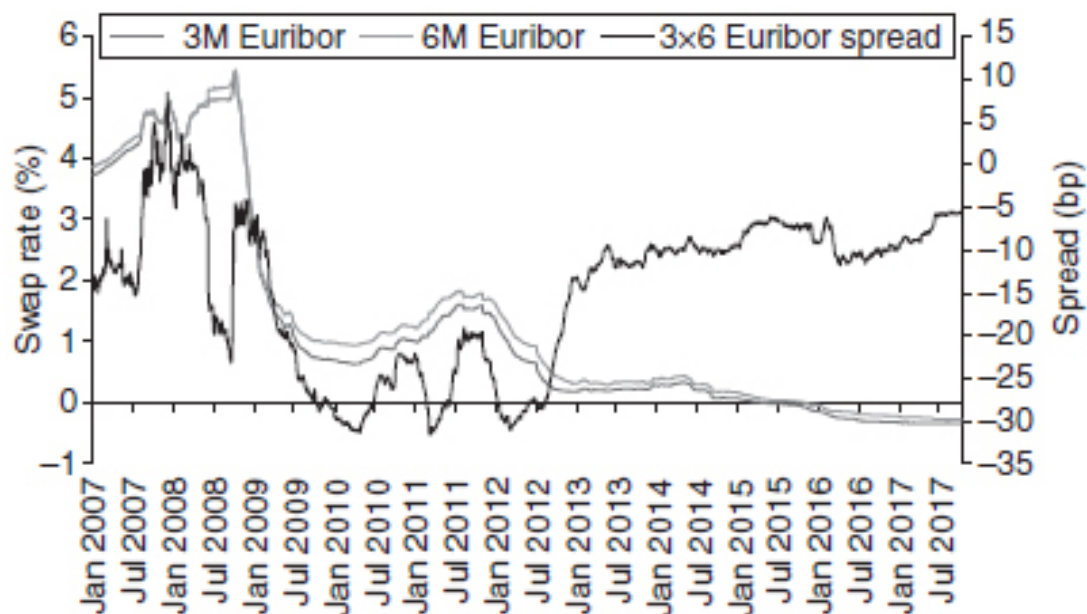
Figure 15.6 Spreads of 5Y and 10Y swap rates references to 3M and 6M Euribor



Source: Bloomberg LP.

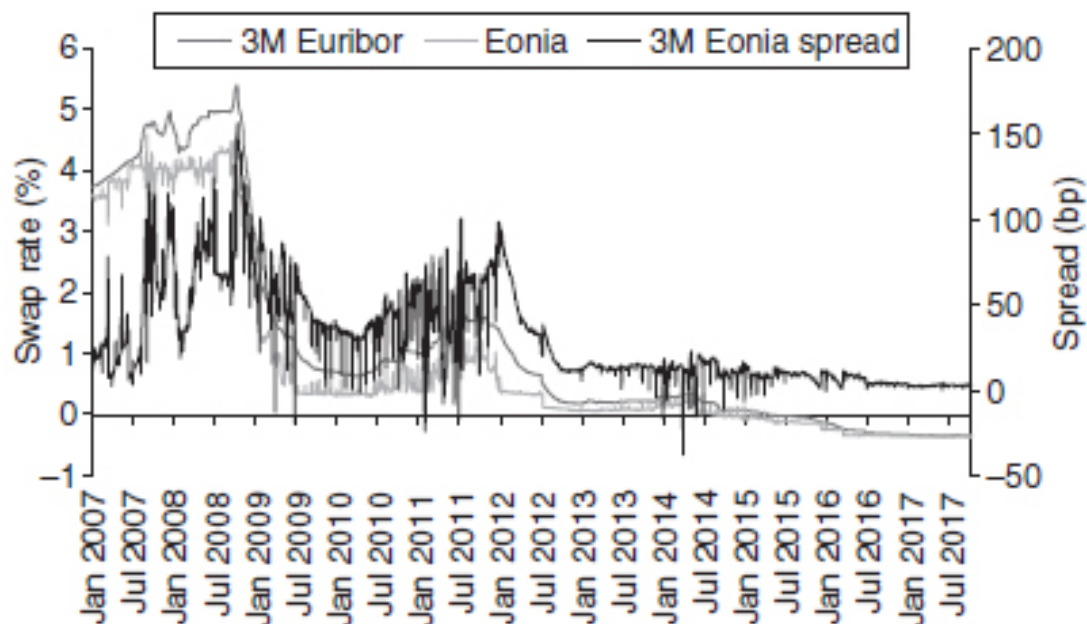
The basis swap emerged from these basis spreads. In contrast to a “conventional” interest rate swap, a basis swap has two floating legs that are linked to two different money market benchmarks. A basis swap should eliminate the bank’s basis risk between the bank’s income and expense cashflows. In Europe, most basis swaps are linked to Libor or Euribor, but with different maturities, eg, one leg might be at the three-month tenor and the other at the six-month tenor. In such a swap the basis and the payment frequency are different: one leg pays interest on a quarterly basis, whereas the other pays on a semi-annual basis (3 × 6 basis swap). By having different payment frequencies one party has a higher level of counterparty risk and hence, a higher credit risk this materialised in the financial crisis (Choudhry 2007, p. 656).

Figure 15.7 3M and 6M Euribor and the corresponding spread



Source: Bloomberg LP.

Figure 15.8 3M Euribor fixing versus 3M Eonia swap rate



Source: Bloomberg LP.

As [Figure 15.7](#) shows, the volatility of the 3×6 Euribor basis spread reflects the anticipated liquidity risk in the money market and the corresponding preference of banks for receiving payments with higher frequency, eg, quarterly instead of semi-annually. In addition, there are other indicators of regime changes in the interest rate markets, such as the divergence between deposit rates and overnight indexed swap (eg, Eonia swap) rates with the same maturity ([Figure 15.8](#)).¹⁷

These interest rate differentials are not completely new in the market: non-zero basis swap spreads were already quoted and understood before the crisis, but they were very small and therefore traditionally neglected (Amentrano and Bianchetti 2009, p. 3).

The interbank money market is an unsecured and short-term market. With the emergence of the financial crisis banks were uncertain about forthcoming losses, which caused them to be reluctant to lend to each other in money markets and to fear counterparty risks. As a result basis spreads of interbank short-term interest rates widened (Hirvelä 2012, p. 1). So here the observed money market basis swap can be seen first as a built-in credit premium (the credit premium built into one particular rate index differs from that built into another (Tuckman and Porfirio 2003, p. 3)) and second as a liquidity premium or, better, a liquidity spread (by being unsecured and short-term, money market deposits clearly affect the LCR's denominator). The longer the maturity of the trade, the larger the spread. Therefore, banks have become more keen to relieve the LCRs that subsequently determine the spread: the extent of the spread complies with its LCR relief. Thus, we believe that these observed spreads will definitely remain.

Since their (observable) emergence, basis spreads have been quoted by the swap desks of market participants, and basis swaps have become a real hedging instrument of basis risks. If assets are floating rate, there is less concern over interest rate risk because of their frequent resets. This also applies for floating-rate liabilities but only insofar as these match the floating-rate assets. Floating-rate liabilities issued to fund fixed-rate assets create forward risk exposure to rising interest rates. Even if both assets and liabilities are floating, they can still generate interest rate risk: this is simply a basis risk that could be inherent in a bank's liquidity reserve, and presumably arises from the bank's internal transfer price curve.

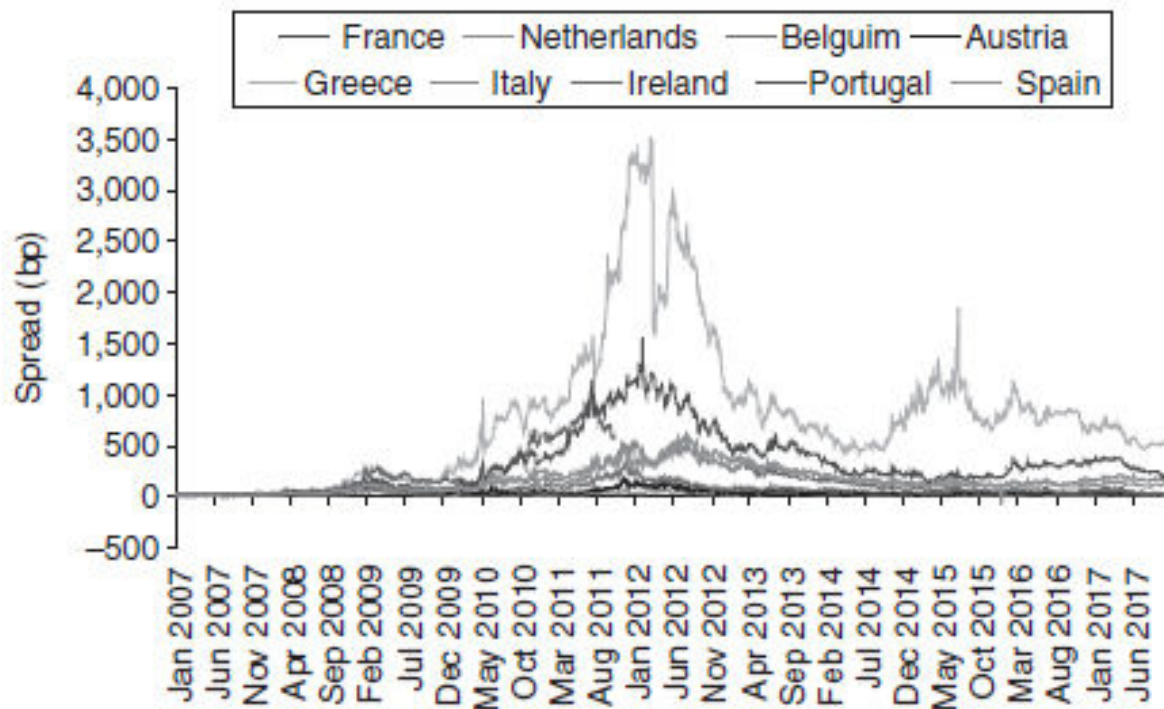
In Europe, securities are normally traded against 6M Euribor or 6M Libor. So, when buying an asset for the liquidity reserve in an ASW package, which

eliminates most of the potential interest rate risk, it is quite likely that the reserve asset will be swapped against one of these benchmarks.¹⁸ Assuming that liquidity reserve is funded according to the bank's internal transfer price curve on a quarterly basis (versus 3M Euribor), an interest rate spread risk will arise according to the explanations given above: if assets pay 6M Euribor and matching term liabilities are referenced to 3M Euribor, there is a basis risk. Here, liquidity risk is eliminated but interest rate spread risk remains (Choudhry 2012, p. 360).

The liquidity reserve will benefit if the basis between 3M Euribor and 6M Euribor increases, because the portfolio's 6M Euribor asset fixing will gain a relative advantage over the 3M Euribor liability fixing. Here, it would be useful if the ALM desk hedged a broader basis or, in other words, the performance of the liquidity could be stabilised by using basis swaps. The risk the bank faces is that the spread between the six-month and three-month rates will change. The bank can use basis swaps to make floating-rate payments on a semi-annual basis (because this is the rate determining how much the bank receives on a bond) and receive floating payments on a quarterly basis (because this is the rate determining the bank's funding cost); see Fabozzi *et al* (2010, p. 619). This hedging strategy can be implemented by using overlay hedges on the overall basis risk structure of the liquidity reserve. Here, the fixings of the portfolio determine these hedges. Properly implemented, these hedges can generate extra return in the liquidity reserve and hence minimise its costs.

In addition to above example of 3 × 6 basis risk, another basis risk arises when reverse repos are used in the process of managing the liquidity reserve: due to the different bases between repos (which are overnight indexed rates, eg, Eonia) and their funding versus the money market index (eg, 3M Euribor), there will be an Eonia–Euribor spread risk as well. In contrast to the example given above, a widening of this spread would adversely affect the liquid asset buffer's performance because funding would increase, compared with a shrinking Eonia-based income from the reverse repos. Therefore, the negative carry will increase further. To limit the negative carry and to stabilise earnings from the liquidity portfolio, a certain spread is necessary. For this purpose the ALM/treasury desk can use money market futures or forward rate agreements versus forward–Eonia swaps as short-term instruments or a combination of longer-term 3 × 6 basis swaps and Eonia–Euribor swaps with the same maturity.

Figure 15.9 Ten-year European sovereign spreads versus German Bunds



Source: Bloomberg LP.

Managing credit risk

Assuming from the previous section that the ALM/treasury desk has hedged all interest rate risks, credit risk still remains. Keeping in mind that the liquidity reserve should consist primarily of sovereign bonds, the management of the liquidity reserve faces a real challenge: historically, sovereign bonds of developed countries have been considered a safe and almost default-risk-free asset. With the introduction of the euro, European investors largely diversified their portfolios by investing in non-domestic but euro-denominated bonds. The stability of the euro area between 2000 and 2007 explains this phenomenon: a European investor could prefer Italian Buoni del Tesoro Poliennali (BTPs) over German Bunds, because they offered a better return for an incremented default risk that was considered to be negligible. Before the sovereign debt crisis, bond management in the euro area was principally explained by the search for better spreads ([Figure 15.9](#)). But the sovereign debt crisis in Europe led to a rediscovery of

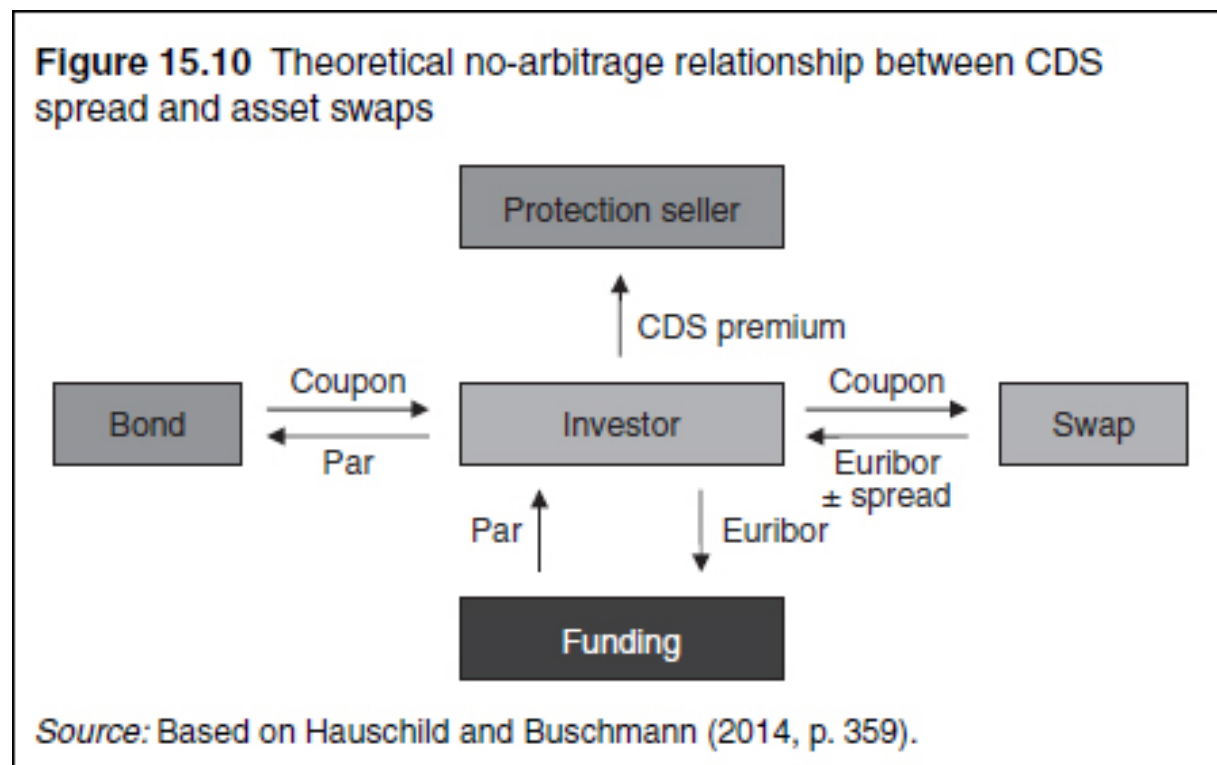
sovereign credit risk and led to a rethink about the management of bond portfolios by placing more emphasis on their credit risk management (Bruder *et al* 2011, p. 2).

The debt crisis played havoc with the conception that the debt of major developed countries is almost free of credit risk. So, the creditworthiness of sovereign issuers is increasingly scrutinised and bond investors face a new challenge (Bruder *et al* 2011, p. 2). Empirical evidence suggests that diversification can indeed reduce credit risk and that the best way to achieve this is through cross-border investments. Smaller benefits are also obtainable through diversification across other dimensions, namely, industry sector, maturity and credit rating (Varotto 2003, p. 36). But in running a liquidity reserve consisting of sovereign bonds, credit risk management is a real issue for a bank's ALM desk.

Asset swaps are a common form of derivatives written on fixed-rate bonds and it is common practice that banks buy bonds in an asset swap package. By doing so, banks separate the credit risk from the interest rate risk that is embedded in a fixed-rate bond. Effectively, the interest risk of the bond is transferred from the investor to its swap counterparty, leaving the credit risk with the bond holder. Thus, asset swaps are mainly used to create positions that are similar to cashflow and risk exposure of floating-rate notes with only little interest risk remaining (Bomfim 2005, pp. 53ff). To offset mathematically the value of all the fixed- and floating-rate payments during the swap's lifetime, the so-called asset-swap spread is calculated. This reflects the difference between the bond's yield and the yield of the maturity-matching benchmark in the same currency: the swap curve assumed to reflect the average rating of the banking sector (Betz 2005, p. 46). Under the no-arbitrage assumption we can assume that investing in a floating-rate note or in a credit-risky bond bought in an ASW package has the same economic risk profile as selling protection via credit default swaps (CDSs).

A CDS is a bilateral financial contract in which one counterparty (protection buyer) pays a premium (expressed in basis points (bp)) on an agreed notional amount in return for a contingent payment by the other counterparty (protection seller) following a so-called credit event of the reference entity (JP Morgan 1999, p. 12). As a result, the no-arbitrage assumption implies that the CDS premium should reflect the Euribor spread on an asset swap from the same credit-risky entity. Without going further into theoretical and mathematical detail and, for simplicity, disregarding

collateral postings or counterparty risk, we assume that the ASW spread and the CDS premium should be the same (De Wit 2006, p. 5) to avoid arbitrage between cash bond markets and the derivatives market.



As implied in Figure 15.10, a credit risk hedge of the liquid reserve is somewhat unfavourable from an earnings point of view. As stated before, crisis liquidity is the most expensive form of liquidity. The liquidity reserve generates earnings to the amount of the ASW spread on a gross basis, and only to the difference between the ASW spread and its related funding cost on a net basis. Often this carry is negative: as they have better credit ratings than banks, sovereign bonds often have negative ASW spreads; therefore, an additional hedge cost for the liquidity reserve's credit risk would be unjustifiably high.

Even though an outright credit risk hedge is unfavourable, the liquid reserve's credit risk can be managed on a relative-value basis. Credit spread risk arises when the spread of a bond performs differently from the respective swap curve (eg, changes in demand for a certain asset classes) or from its peer group (eg, federal states, cantons or provinces versus central government, or covered bonds versus agencies). Moreover, spread

anticipation by market participants and changes in the regulatory environment also cause credit spread risks.

The sovereign debt crisis caused a change in investor behaviour: investors are less willing to buy (government) bonds from Europe's formerly most crisis-prone countries, and are keener on buying bonds with shorter maturities. This causes certain concentration risks. To manage these concentration risks, banks' risk management departments formulate certain credit risk limits for the ALM desk. These are largely dependent on two factors: the debtor's size (and therefore the bond market this debtor constitutes), and the debtor's rating. While the former is relatively simple to understand because large issuers, eg, Germany or France, establish large market segments in which their debt can be traded, the latter is not so easy to understand due to the statistical problems that accompany sovereign debt ratings. Globally, there are only around 100 sovereign issuers for which a reliable rating exists. Given the low default rates of these issuers, it is pretty difficult to make reliable default assumptions and derive default probabilities.

On this basis, banks' risk management departments often define so-called "country limits", which include every issuer from a particular country: both public entities, such as central or regional governments, and private entities, such as corporations. Within these country limits, sublimits are assigned for sovereigns and for the liquidity reserve. Nevertheless, the question remains whether banks receive compensation for the additional risks taken. For example, does a credit spread compensate for the additional risk taken, in comparison to, eg, the home sovereign? This compensation should include the additional credit risk taken as well as additional capital charges. Although sovereign debt issued by a euro area country official does not have any capital charges, the internal credit risk model may apply a capital charge to sovereign debt. Here, the home sovereign will be set as the best available credit quality, and every other sovereign or private issuer will be assessed in comparison with this benchmark. The additional credit spread must compensate for the additional risk taken.

From a market point of view, the absolute interest rate level influences credit spread. In a low interest rate environment, the credit spread performance by risk-free assets is limited but possible: after spread narrowing, there is an opportunity to lock in the existing credit spread level without changing the portfolio structure, by, eg, selling bonds or using CDSs.

Again, this is done by overlaying hedges to a certain debtor in the portfolio. This hedge position will make a profit when credit spreads widen again. The profit amounts to the difference between the locked-in spread and that observed in the market. Like basis spread hedging, the credit spread can be used to minimise the portfolio costs.

Credit spread movements also have an impact on the repo market: credit spread widening or narrowing influences repo rates relative to their overnight interest rates, eg, Eonia. Due to the lack of alternatives, say, an excessive anticipated credit risk in the unsecured money market might cause excess demand in secured funding. As can be seen in the above examples, a bank's ALM desk needs to manage the credit (spread) on a relative-value basis rather than by completely hedging credit risk through the use of CDSs. When managing the liquidity reserve on a relative value basis, the treasury desks need to consider two interrelated dimensions – credit risk and return – within the framework described above. This simply means that when two assets or securities with the same credit are available, the one with the higher return will be bought. On the other hand, when two assets have the same return, the one with the lower credit risk will be chosen. In addition, we believe that market saturation (how many more bonds by one issuer can be absorbed by the market) must also be taken into account.

Post-crisis markets

The management strategies described above worked well in the functioning interbank and debt markets before the global financial crisis and during the sovereign debt crisis. However, these crises provoked European central banks to adopt unconventional steps to cope with the debt markets. The unconventional monetary policy measures undertaken by the European Central Bank (ECB) to curtail the sovereign debt crisis were a major game changer for financial markets in the euro area.

Throughout the crisis years 2007–9, there was relatively little concern about sovereign debt from euro area countries, and sovereign debt markets remained relatively calm. At that time, the ECB primarily tried to address the global financial shock by counterbalancing. However, the focus changed from the stability of the banking system to country-specific fiscal risks, since the global financial crisis placed a heavy fiscal burden even on formerly “low indebted” countries such as Ireland or Spain. The disclosure of flawed budgetary numbers in Greece led to a loss in confidence in the rules of the

European Monetary Union (EMU) and, as shown in [Figure 15.9](#), caused a rise in sovereign spreads towards German Bunds and an effective rise in European countries' funding costs. After the crisis, due to recessionary economic conditions in some parts of the EMU (notably Greece, Italy, Ireland, Portugal and Spain), banks located there became less stable, and their funding conditions worsened. This development caused doubts about these countries' ability to bail out their banking sectors (Lane 2012).

This overall development provoked the ECB to undertake extraordinary measures to restore confidence in the financial market and to ease banks' funding conditions. The ECB's first step was to restore banks' funding conditions by conducting two long-term refinancing operations (LTRO) at the end of 2011 and the beginning of 2012. Both operations had a tenor of three years, an interest rate of 1.00% and full allotment. As Archaya and Steffen (2015) have shown, with the liquidity the ECB injected into the financial system, banks committed themselves to giant carry trade behaviour and used the relatively cheap funding of EMU government debts, predominantly those from their home sovereign to earn a spread between 1.00% and the much higher yielding government debt. Although this behaviour strengthened banks' income basis, it tightened the so-called "sovereign-bank nexus", and increased the inter-dependencies between banks and their home sovereigns (Horvarth *et al* 2015).

In a second step, various asset purchase programmes were implemented. The first was the so-called "securities market programme", which was conducted between autumn 2010 and spring 2012. This was replaced by the outright monetary transaction (OMT) programme in autumn 2012; the OMT programme was itself replaced by various *de facto* ECB asset purchase programmes (APPs) starting in spring 2015. According to Borio and Zabia (2016), the measures undertaken by the ECB have influenced financial conditions, and could have a lasting impact on bond yields, various asset prices and exchange rates. While these programmes were quite favourable for the issuers of the purchased assets, they made investment decisions rather tough for investors.

With its purchase programmes, the ECB has effectively changed market sentiment and created an artificial bond market in which it acts as the largest single buyer: bond prices no longer reflect investors' demand and supply considerations and the EMU interbank market, in which banks traded liquidity surpluses and deficits before the financial crisis, was effectively

destroyed when the ECB aborted the various purchase programmes. Since the types of HQLA-eligible assets presented above are bought under the APPs, banks are faced with a shrinking investment universe that offers much lower yields than were available before the crises. Consequently, liquidity reserves managers must consider the tradeoff between yields on EMU government debt (measured in ASW spreads) and the ECB's deposit rate (which has been below zero since June 2014) to ensure a viable risk–return ratio. Banks shifted the composition of their liquidity reserves from government debt (55.0% in June 2011 versus 41.5% in September 2016) towards cash (31.4% in June 2011 versus 38.5% in September 2016)¹⁹ because cash, even with a negative central bank deposit rate, gives higher yields than government debt.

This development forced investors and managers of liquidity reserves to actively search for spreads, and the big question is in which securities (and at what maturities) are these still available. The answer seems to be challenging, since banks prefer mostly expensive, shorter term maturities for their liquidity reserve, while, in a low interest rate environment, issuers prefer to issue long-term debt. While it is not a problem at all to purchase long-term debt, the purchase of expensive short-term debt seems to be quite a dilemma for risk-averse banks: the shorter the asset's maturity, the more expensive the asset and the greater the loss in maintaining crisis liquidity.

In 2012 liquidity reserves mainly consisted of government debt and cash. At the time of writing, liquidity reserves seemed to be more diversified than in the past. Although in smaller portions, they include: debt with 20% risk weights, BBB government debt, covered bonds, debts of non-financial corporations and even equity shares (Bank for International Settlements 2012, 2016). However, this is not a major change in portfolio composition. Since an asset's liquidity is mainly driven by its issue size (see [Table 15.4](#)), the only truly liquid assets are government debt. Since ECB's APPs re-established pre-crisis conditions, which means that all spreads are equal, banks need to use the smallest spread differences. This can be achieved in four different ways.

1. Banks might swap euro-denominated excess liquidity into other, higher yielding currencies and buy short-term non-euro-denominated government debt. Since this government debt must be as safe as possible, only that of advanced economies is suitable for this type of

investment. For example, a bank can use parts of its euro-denominated excess liquidity to swap into Japanese yen or Swiss francs and purchase short-term Japanese government bonds or Swiss sovereign debt to gain a higher spread than from EMU government debt.

2. Banks might reduce their existing asset swap positions: mean positions are sold as soon as there is a considerable spread tightening. Assume, for example, that an asset was purchased with an ASW spread of -10bp towards Euribor and now trades at -25bp . The banks can sell the asset and realise a gain of 15bp (the difference between the purchase price and the sell price). However, the question of how to reinvest the means still remains.
3. While assets can be sold when there is a spread tightening, they can also be bought when there is a spread widening. As with spread tightening, this strategy gains the greatest yields by making opportunistic use of market development. Before the French Presidential election in May 2017, spreads in French government bonds (OATs) rose considerably. In hindsight, this would have been a profitable investment, since spreads tightened afterwards. This implementation of this strategy, however, depends on how much risk a bank wants to take.
4. Banks might purchase bonds outright. This means that they deliberately set aside interest rate swaps to use for hedging interest rate risk. Using this strategy banks can earn the full coupon on their bond holdings, but are unprotected as soon as interest rate levels rise again. Further, banks have to invest in long-term HQLA to earn a reasonable coupon.

From the remarks above, it can be seen that managing a liquidity reserve in an EMU-located bank can be very challenging. Unlike in the past, the main challenge is no longer to obtain cash but is the limited investment universe: even though there are some assets with attractive spreads, these are not available in larger amounts, and thus in liquid secondary markets. However, larger purchases of corporate debt are also not a true alternative to government debt. Hence, liquidity reserve managers should keep a lid on the costs of the liquid assets rather than make them as profitable as possible.

CONCLUSION

Managing liquidity risk is a crucial part of a bank's asset–liability management, ensuring the bank's short-term solvency and long-term structural liquidity in line with both market prices and regulatory requirements (Bodemer 2011, p. 306). Within its liquidity risk management mandate, an ALM desk is expected to maintain the bank's solvency at all times and in any imaginable market condition. To achieve this goal, national regulators expect banks to hold a portfolio of high quality, highly liquid assets as part of their liquidity reserve. In an idiosyncratic or market-wide liquidity stress, these assets will be sold to generate crisis liquidity to enable a financial institution to honour its obligations when they become due.

In this chapter we briefly described the role of ALM relating to the liquidity issues of a banking organisation. Several types of liquidity were presented, and we described how these are interconnected and how they might affect a financial institution's liquidity risk. As a logical consequence of this, the regulatory provisions and Basel III framework are discussed. The latter part the chapter is dedicated to the liquidity reserve itself: both its purpose and functionality and the components and adequate size of the liquidity reserve were discussed. We presented various funding and risk management strategies and showed their effects on liquidity reserve's performance. We also remarked on the challenges of the post-crisis market.

We examined the role of management of the liquidity reserve into a broader context, with the aims of filling a gap in the literature. When managing the liquidity reserve and its assets, those responsible should take the following into account: the banking organisation itself, with its business model, funding structure and related types of risk; national and international regulatory requirements; market and market participants' behaviour.

As we implied in this chapter, and the global financial crisis impressively showed, “liquidity” is not an absolute characteristic: it is relative. We believe that the biggest risk arises from managing the liquidity reserve itself: an asset hitherto assumed liquid can suddenly turn out to be illiquid the next day (Matz 2011, p. 254), either due to non-acceptance as a part of the liquidity reserve by national regulators or due to secondary market activity drying up. Because such assets would produce a calculated, expected, measured cash outflow in the LCR, it is quite likely that they must be sold at a loss. By coping with these factors, we believe that a bank's liquidity reserve can be properly managed.

In conclusion, management of the liquidity reserve is a continuous process with adjustments to be made when necessary: liquidity that arises from normal banking operations should be monitored in a similar way to the given regulatory framework and overall financial market conditions.

- 1 Before the financial crisis, liquidity and liquidity risk were regarded as concomitant with other types of risk such as market risk, credit risk or operational risk, even by the literature (Schulte and Horsch 2004, p. 52; Leistenschneider 2008, p. 172).
- 2 Often, the term “economic liquidity” refers to central bank liquidity, and “funding liquidity” is often called institutional liquidity. The terms “central bank liquidity” and “economic liquidity” as well as “funding liquidity” and “institutional liquidity” are used equivalently in this chapter (see also Heidorn and Schäffler 2011, p. 310).
- 3 In particular, maturities longer than a few days experienced considerable pressures. The way secured/collateralised money markets operate has changed significantly since the crisis. Haircuts have changed, and lower-rated assets have become more difficult to borrow against. Central banks have introduced a wide range of measures to try to improve the functioning of the money markets (see Allen and Carletti 2008, p. 2). For a complete overview of the development of the financial crisis, see, for example, Bank for International Settlements (2008, pp. 99ff; 2009, pp. 16ff).
- 4 Here, we assume credit default swap spreads as a good proxy for a bank’s funding costs.
- 5 Other national regulatory requirements for banks’ liquidity management can be taken from Bergner *et al* (2014), Bouwman (2013) as well as Hauschild and Buschmann (2014).
- 6 The MaRisk’s provisions also include various conclusions from international regulatory discussion papers (Basel Committee on Banking Supervision 2008b; Committee of European Banking Supervisors 2009, 2010).
- 7 This includes the preparation of a liquidity overview, the performance of appropriate stress tests, the preparation of contingency plans and the incorporation of liquidity-related cost–benefit considerations in the management of the institutions’ business activities (BaFin 2013).
- 8 This 30-day time horizon is often called the “survival period” or “survival horizon” and it is included in existing regulatory regimes: even though it is not mentioned explicitly, MaRisk requires a survival period of one month, with the further requirement that a bank survive a liquidity stress of seven days without assistance by the central bank. The term “survival period” itself was introduced by the Committee of European Banking Supervisors (see BaFin 2013; Committee of European Banking Supervisors 2009, p. 5; Matz 2011, p. 62).
- 9 In contrast to the Basel Committee, other national regulators, such as the British Financial Services Authority (FSA), have stipulated a 90-day test period when calculating the LCR (see Financial Services Authority 2008, pp. 38ff).
- 10 On the reasonably safe assumption that a liquidity stress will last more than 30 days, we believe that the FSA standard is more conservative than the Basel Committee’s approach, but also more expensive and therefore difficult to adopt in practice, even though some others believe that the FSA standard should be adopted as an industry standard (Choudhry 2012, p. 664). In addition, even though the LCR is an international liquidity measure, it has not been implemented in the US in the same way as in Europe. For the interaction of LCR with the US Dodd–Frank Act, see Bouwman (2013, pp. 30ff).
- 11 International Accounting Standard 39 (“Financial instruments: recognition and measurement”; IAS 39) splits financial instruments into five different categories: loans and receivables (L&R), held to

- maturity (HtM), held for trading (HfT); available for sale (AfS); other liabilities (OL); see Subramani (2009, p. 6).
- 12 Some good examples of liquidity consumers are margin calls from an exchange or compensation from protection sold on credit default swaps when the reference entity defaults. See Heidorn and Schäffler (2008, pp. 27ff).
 - 13 For a detailed description of bank runs and related liquidity management constraints see Bergner *et al* (2014). For the mechanics of failures of large banks see Duffie (2011).
 - 14 The non-default component is the difference between the asset–swap spread and a maturity-matching credit default swap spread of the same entity. It quantifies the non-credit-risk-related part of any asset–swap spread and can be simplified when seen as a liquidity measure. See Heidorn and Rogalski (2010, pp. 8ff).
 - 15 Examples of papers that deal with diverse liquidity measurement methods are Chakravarty and Sarkar (1999), Houweling *et al* (2002), Chordia *et al* (2003) and Jankowitsch *et al* (2002).
 - 16 Here, there is a difference in the lengths of the regulators’ specified survival periods: while MaRisk and Basel III require a survival period of a month and 30 days, respectively, the FSA recommends 90 days. See Hauschild and Buschmann (2014, p. 350).
 - 17 The Euro OverNight Index Average (Eonia) represents the effective one-day (overnight) interest rate in the euro area.
 - 18 We believe that it is absolutely inappropriate to willingly take outright interest rate risk in the liquidity reserve. If a bank really needs to use the liquidity reserve to generate additional liquidity, the asset’s (fire) sale has to proceed with a minimum of P&L and accounting effects. The only acceptable remaining risk is basis risk.
 - 19 See Bank for International Settlements (2012, 2016).

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Instruments for Secured Funding

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Inter-American Development Bank; European Investment Bank

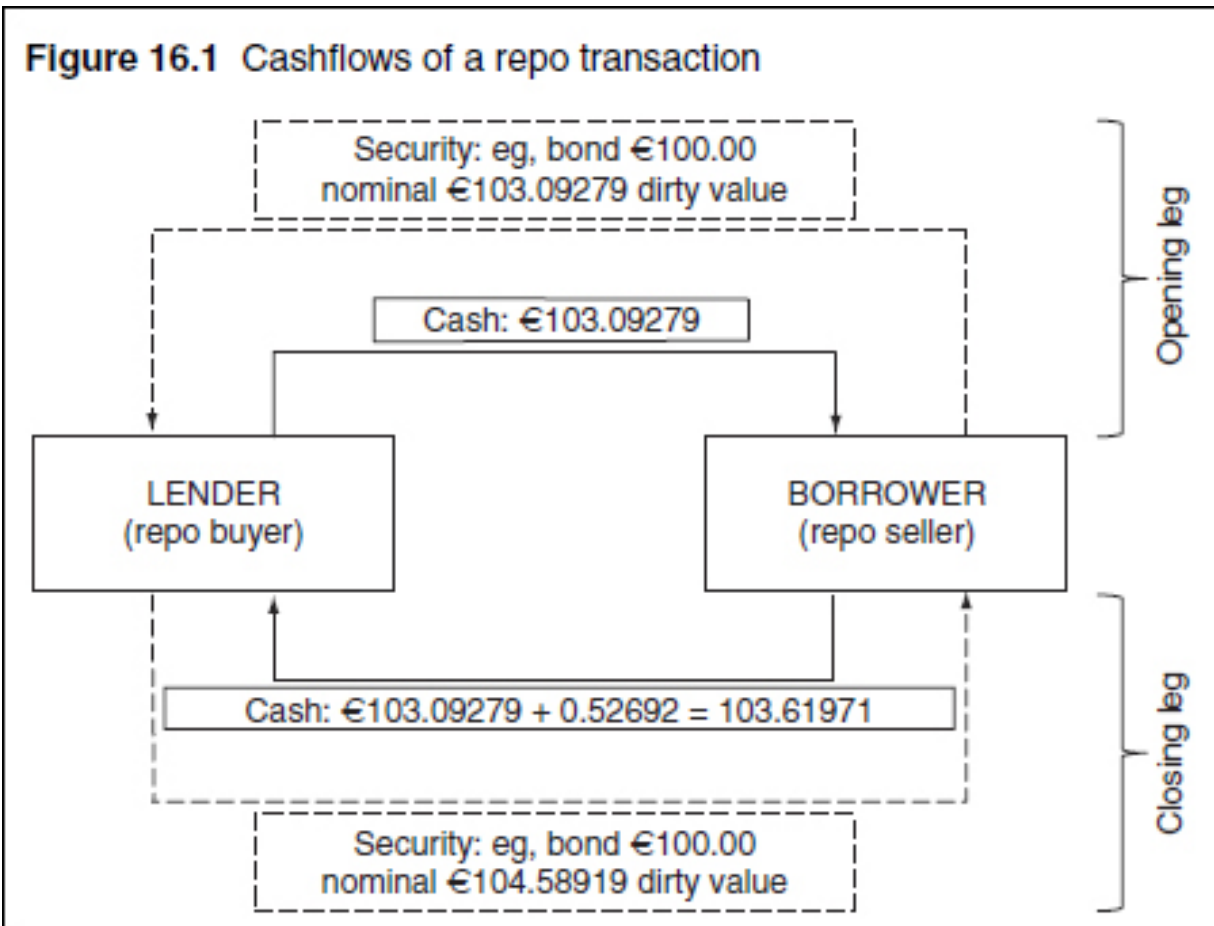
At the time of writing, a paradigm shift had taken place for bank wholesale funding. While retail deposits continued to function very similarly to the way they did prior to the 2007–9 financial crisis, markets, central bank and regulatory action were shifting banks towards secured funding. This applied to the money markets, where the traditional Libor-based unsecured lending was concentrated in overnight transactions and repurchase agreements (repos) had become the norm at longer terms,¹ and also concerned medium- and long-term funding, with asset-backed securities (ABSs) and covered bond issuance complementing senior bonds, especially as the latter became vulnerable to bail-in regulations. There also continued to be significant reliance on central bank facilities, to the point where ABSs were being issued and “retained” by the same originators to be used for central bank refinancing.

In this chapter we explain the new paradigm, which covers short-term instruments (particularly the repo), medium- and long-term instruments, covered bonds and ABSs. In keeping with the aim of the book, these instruments are treated from the point of view of the issuer rather than the investor. The concluding section outlines the issue of asset encumbrance, a natural consequence of the new trend for secured funding at short, medium and long-term maturities.

SECURED SHORT-TERM INSTRUMENTS AND MARKETS

The repo instrument

A repo is a short-term operation secured by financial collateral. At inception, the borrower provides securities as collateral to its counterparty (lender), with the opposite exchange at maturity.



The repo volume on the European market as of December 2016 was estimated to be €5.6 trillion, with almost 56% having less than one-month maturity and approximately 19% having overnight maturity. In the US, volume was estimated to be about US\$3.4 trillion, of which a substantial amount had overnight maturity (International Capital Market Association 2017; Baklanova *et al* 2015).

The securities (mostly bonds) are sold and repurchased with full transfer of property title. The borrower and the lender are called, respectively, the “seller” and the “buyer” of the security in a repo transaction. The first exchange of securities is settled against cash on the settlement date and is

called the “opening leg” of the repo, while the eventual, opposite, exchange is called the “closing leg”.² The cost of the borrowing for the cash-taker is represented by the agreed “repo rate”.

The deal is called “repo” from the point of view of the borrower (ie, funding deal) and “reverse repo” from the point of view of the lender.

As the legal title on the security is transferred from the borrower to the lender, during the life of the deal the latter is entitled to some legal rights such as voting rights. However, dividends and coupon payments produced by the security are transferred back from the lender to the borrower, who retains the economic benefits of the security as well as its credit and market risk. It will be the borrower who ultimately bears a loss should the underlying security fall in value. If its counterparty defaults, the lender can sell the security on the market in order to offset its credit loss.

Figure 16.1 illustrates the cashflows of a repo with a maturity of 92 days and a rate of 2% (Act/360), collateralised by a bond (€100 nominal) with a clean value of €101.3 and an accrued interest of €1.79279.

The cash transferred on the settlement date is the “repo selling price”, ie, the dirty value of the security.³ The repo rate is computed as simple interest, with the prevailing market convention (Act/360 for EUR, USD, CHF, JPY and Act/365 for GBP). The final payment is as follows.

At maturity, the security (€100 nominal) is transferred back from the lender to the borrower. Assuming that the clean price has increased to €101.9 and the accrued interest is €2.68918, the dirty value on the closing date would be €104.58919, but the cash movement from borrower to lender would still be €103.61971 as implied by the agreed repo rate.

Repos sometimes take the form of “sell/buy-back”. While the economic substance is the same as the example above (“classic repo”), a sell/buy-back is composed of a spot purchase of a security and a simultaneous sale at maturity, for a forward price. The interest rate is not set explicitly, the remuneration for the lender being the difference between the spot and forward price. Coupon and dividends on the security are not passed through immediately to the borrower, but are deducted from the forward price at maturity.

Buy/sell backs exist for operational reasons (they can also be dealt in the lack of position-keeping and settlement systems adequate for repo transactions) as well as legal reasons (it is sometimes problematic to enter into classic repos due to uncertain treatment in some jurisdictions). In

Europe, classic repos represent the vast majority of the estimated overall market. Sell/buy-backs are typical in Italy, Spain and many emerging markets.

Despite the legal transfer of title that actually takes place on the repoed securities, from an accounting perspective the securities underlying a repurchase agreement or a sell/buy-back remain in the assets of the borrower and a corresponding liability represents the debt due to the lender.⁴

Use of repos in the money and capital markets

Banks use repos to obtain short-term funding secured by assets they have in their proprietary treasury portfolios.

On the other hand, those financial institutions that are structurally on a cash long position, such as money market funds, insurance companies, sovereign wealth funds, pension funds, endowments and cash rich corporates, can use reverse repo as a short-term investment. They can also benefit from the fact that their securities portfolios can be mobilised on the repo market.⁵

Repos are also employed by market-makers in order to support their intermediation business on the capital markets. Hence, despite their short-term nature, repos play a role in facilitating the activities on the long-term capital markets. On the primary markets, repos are used to finance underwriting of newly issued securities. On the secondary markets, they are employed to either fund outright securities inventories or hedge short positions in the portfolios of market-makers.

For instance, assume that an insurance company wants to sell a bond held in its investment portfolio. A market-maker will purchase the bond, booking it in its inventory, in order to eventually sell it to another client. The purchase will be financed by a repo where the bond is given as collateral, for instance, to a money market fund for an overnight maturity. The repo will be rolled over (either with the same or with other counterparties) until the market-maker can sell the bond to another investor. The proceeds from the bond sale will be used to close the overnight repo with the mutual fund.⁶

Alternatively, should the insurance company want to purchase bonds that are not held by the market-maker, a reverse repo would be used by the latter to cover the short position in its inventory. The reverse repo would need to be rolled over until the market-maker buys the bond on the market and delivers it to its counterpart, to close the reverse repo transaction.

Central banks also use repos, often on a public auction basis, as the primary tool to implement their monetary policy and provide liquidity support to the banking sector.

Repo rates

Borrowers can raise funds with repo at a cost cheaper than unsecured interbank deposits. Particularly for longer maturities, the repo rate depends on the quality of collateral and its availability on the market.

The market is segmented into “general collateral” (GC) repos (those that are entered into due to a pure secured funding or investing need) and “special” repos (dealt to obtain a specific security, for instance, for the purposes of hedging a short inventory position).

In GC repos, the collateral initially delivered can be substituted with other GC throughout the life of the transaction. Under normal conditions, GC is traded at a rate (the “GC rate”) that is not a function of the nature of the underlying collateral.

“Special” securities are those for which the market expresses an exceptional peak of demand. In order to obtain a “special” security, counterparties accept they will receive a rate below (sometimes significantly below) the GC rate prevailing at the time. In other words, the cost of “reversing” the security is represented by a low return on the money placed, all else equal.⁷

In general, a bond goes “special” when a “short base” exists on the cash market, often linked to arbitrage activities. Should several market participants short a bond that is considered overpriced, a concentration of demand for the specific bond could appear when the short positions need to be covered.

Specific special securities are the “on-the-run” issues of government bonds (especially US). Each time a new on-the-run bond is issued, investors tend to switch previously purchased bonds (now “off-the-run”) with the new on-the-run. As a result, market-makers face pressure from the cash market and cover with reverse repos their short positions in the on-the-run issue, which commands a repo rate lower than GC.⁸

The “special rate” is the equilibrium price expressed by the interaction of the repo market demand and supply for the security. Its spread against the GC rate indicates the “specialness” of the underlying security. Desks can run

books aiming at extracting the specialness of a security on which they are long by repoing out the special collateral and investing the received cash in a reverse repo yielding the higher GC.

For instance, assuming that security X can be repoed out for 30 days at a special rate of, say, 1.5% (Act/360) and the rate on GC collateral on the same maturity is 2% (Act/360), the desk could enter into a deal where it funds itself for, say, €5,000,000 at a cost of 1.5%, giving the special bond X as collateral, simultaneously investing the cash in a GC-based reverse repo yielding 2%. In this case, the repo instrument would be used as a profit-enhancing tool rather than a pure financing tool. The total profit would be $€5,000,000 \times (2.0\% - 1.5\%) \times 30/360 = €2,083.33$.

The financial crisis significantly dislocated the money markets, often replacing regularities that could normally be observed before mid-2007 with a more complex reality. The GC rate has started to be influenced by factors other than the general conditions of money markets. The spread between unsecured markets rates and GC rate, which had been relatively stable beforehand, has spiked as a result of the increased risk perception and general unwillingness of counterparties to lend on an unsecured basis.

Moreover, during the financial crisis the market started to differentiate between the credit quality of the bonds issued by the various eurozone sovereigns, leading to different GC repo rates and undermining the notion of GC itself in the eurozone area.

Tri-party repos

In a tri-party repo, two counterparties negotiate the repo among them, but aspects such as settlement, collateral management, margining and custody are outsourced to a “tri-party agent”.⁹

The collateral is specified in “baskets” pre-agreed by the counterparties. Dedicated templates specify the eligible assets, the haircuts by rating, the acceptable maturities as well as the applicable valuation criteria (eg, quoted prices only). Concentration limits are also often specified on the acceptable collateral (eg, maximum 10% same issuer securities can collateralise one repo).

Once a repo has been transacted and associated to one basket, the collateral can be substituted on an intraday basis with other acceptable collateral. Collateral selection can be done by the tri-party agent with

automated systems that optimise the collateral allocation in terms of rating and haircut across all trades by the borrower.¹⁰ The tri-party agent independently values the collateral (intraday) and issues corresponding margin calls.

For small banks, cash rich corporations and “buy side” institutions, tri-party repo is appealing, as these entities may lack resources to manage the collateral process internally. For bigger banks, tri-party repos facilitate funding portfolios composed of many individual securities that could be held in small positions.

It is important to bear in mind that, unlike repos negotiated with a Central Clearing Counterparty (CCP) the tri-party agent does not interpose itself between the two original counterparties.

According to ICMA, as of December 2016, tri-party repo represented 12% of the European repo market. In the US the proportion was estimated to be more than 50% of the total. The market is very concentrated (based on European Central Bank (ECB) data, the top 10 counterparties accounted for 90% of the European tri-party volumes as of year-end 2015).

Collateral

The role of the repo collateral is to contain the impact on the lender of a possible default of the borrower.

However, collateral value is exposed to market risk, as it could fluctuate immediately after the default of the repo counterparty, especially if collateral is denominated in a different currency from that of the underlying transaction. It is customary to limit the maximum maturity of eligible bonds, in order to limit their interest rate sensitivity, and/or to require haircuts that increase as a function of the maturity. Collateral provided in the form of equity could be accepted (and haircuts set) on the basis of its volatility.¹¹ Currency risk could be mitigated by limiting the portion of collateral that can be provided in a currency different from the one of the related transaction and by imposing additional haircuts.

The protection provided by collateral in case of a counterparty default is also influenced by liquidity risk, due to the price impact on the non-defaulting counterparty in the case of forced liquidation of collateral. This can be mitigated by imposing a minimum outstanding size for bonds, by excluding private placements and by applying concentration limits on the

percentage of a given issue that can be posted against one repo. Finally, the nature of the collateral used for repo transactions can itself be relevant: during the financial crisis it suddenly became almost impossible to use ABSs as repo collateral; these were previously perceived as safe and hence acceptable. This aspect should be carefully taken into consideration not only by collateral takers but also by collateral givers, who should not rely excessively on a concentrated stock of securities that could suddenly be perceived as ineligible by the other participants in the repo market.¹²

Credit risks on collateral are typically addressed by modulating the related haircut as a function of the rating. Moreover, concentration limits should target the wrong-way risk due to the credit risk of the repo counterparty and that of the collateral issuer being correlated. For instance, the default of a major bank could have an impact on the value of the sovereign bonds of its country of incorporation.

Repo collateral is principally represented by government bonds. As of December 31, 2016, approximately 86% of the total repo volume on the European market was backed by government bonds (mostly rated above AA-).¹³

The picture is quite different when the focus was restricted to European tri-party repos, where government bonds accounted for only 42% of the collateral. On the other hand, 24.8% of tri-party repo volume was collateralised by covered and corporate bonds and 14% by equity. The use of equity is facilitated by eligibility being specified as a generic basket belonging to a main equity index.

US sovereign and agency paper represent most of the collateral on the US repo market. As of June 2015, these securities accounted for approximately 85% of the tri-party collateral (of which approximately 18% were agency MBSs) (Baklanova *et al* 2015).

Overcollateralisation and margining

Repos are typically overcollateralised, to constitute a buffer for possible declines in value which, upon default of the borrower, could lead to collateral sale proceeds insufficient to cover the exposure. In theory, haircut levels should be set only on the basis of the collateral risk. The credit quality of the repo counterparty should not influence haircuts, being instead priced

into the repo rate. In practice, however, haircuts are also set as a function of the repo counterparty rating.

Overcollateralisation can be expressed in the form of either a “haircut” or “margin ratio”. While economically similar, the two differ in the way margin calls are calculated. The mechanism can be specified either in the overall GMRA signed between two counterparties, or in the trade confirmation document for a specific transaction.¹⁴

A “haircut” is defined as the percentage difference between the dirty market value of an asset and the price at which it is sold in the repo transaction, according to the following formula

$$\text{haircut} = 1 - \frac{\text{repo selling price}}{\text{dirty market value}} \quad (16.1)$$

A haircut of 3% would mean that a bond with a dirty price of €103.09279 can be repoed against an initial cash payment of €100. Hence, the margin at inception of the trade would be €3.09279.

As the overcollateralisation is to be maintained on an ongoing basis, additional margin will be called whenever the exposure is not covered any longer by the collateral provided. The net exposure at any date includes the interest accrued at the repo rate and is given by

$$\begin{aligned} \text{net exposure} \\ = \text{repo selling price} \times \left(1 + \text{repo rate} \times \frac{\text{actual days}}{360} \right) \\ - \text{dirty market value} \times (1 - \text{haircut}) \end{aligned} \quad (16.2)$$

Let us assume a repo transacted on the July 9, with a settlement date of July 11 and maturity date of October 11, with a haircut of 3%, selling price €100 and 2% rate. As of August 13 (33 days from settlement), should the collateral dirty value fall to €101.55, the net exposure to be covered by additional margin would be €1.679833.

Additional margin can be provided as cash or securities, upon agreement of the counterparties. If cash is used, the amount would actually be €1.679833. If securities are used, the appropriate haircut would apply.

The “margin ratio method” is an alternative way of expressing the haircut, based on the following formula

$$\text{initial margin} = \frac{\text{dirty market value}}{\text{repo selling price}} \quad (16.3)$$

Based on the margin ratio method, the additional margin call would be based on

$$\begin{aligned} &\text{net exposure} \\ &= \text{repo selling price} \times \left(1 + \text{repo rate} \times \frac{\text{actual days}}{360} \right) \\ &\quad \times \text{initial margin} - \text{dirty market value.} \end{aligned} \quad (16.4)$$

The “haircut” and the “margin ratio method” are applied on different calculation bases. Assuming that the initial margin is set to 103%, our repo of 100 units would be collateralised by 103 units of security, not €103.09279, as it would be with the “haircut” method.¹⁵

Haircuts are set by market participants in different ways, ranging from a “rule-of-thumb” qualitative approach (depending on repo counterparty quality, client relations and market competition) to VaR and stress-testing. The latter methods aim at measuring the decline in value that a collateral could experience upon default of the repo counterparty, at a given confidence level (eg, 95–99%) and liquidation time horizon (eg, 10 days) with specific add-ons for illiquid securities.¹⁶ Several banks apply the standard regulatory haircuts provided by the Basel Committee for calculating capital requirements on repos.

It is quite difficult to indicate general levels for haircuts, because these figures depend on the risk appetite of each repo counterparty, on the general market sentiment as well as on the market specifically considered (eg, bilateral versus tri-party repos). However, typical values are reported to be between 0.5% and 3% for highly rated government bonds, 1% and 8% for covered bonds, 8% and 20% for investment grade senior unsecured bonds, reaching up to 40% for sub-investment-grade bonds. Haircuts for equity could hover between 15% and 25% for developed markets (Committee on the Global Financial System 2010).

Regulators and the repo industry debate whether haircuts may have a procyclical role in financial crises. The argument is that market participants tend to set low haircuts in “booms” and raise them “busts”, suddenly increasing the amount of collateral that repo borrowers would need to provide. Such a herding behaviour would first lead banks to build excessive

leverage, and subsequently trigger a deleveraging process, thus contributing to a general loss of confidence in the liquidity of the banking system.

According to some industry bodies, it is unclear whether haircuts have been a material driver of deleveraging, especially in Europe. With a view towards reducing procyclical forces in the money markets, in 2015 the Financial Stability Board recommended that haircut floors be applied to collateral other than government bonds on non-centrally cleared repos, if cash is provided by a bank to a non-bank or by a non-bank to a non-bank. At a Europe-wide level, at the time of writing, the European Commission was expected to consider whether the FSB recommendations were suitable for the EU markets, possibly opening up the possibility that haircut floors would be introduced in the Securities Financing Transaction Regulation (SFTTR).

The role of Central Clearing Counterparties

CCPs provide clearing services to the financial sector, including the repo market. The following description applies also to other CCP-cleared instruments such as derivatives. CCPs take the central position between lenders and borrowers, becoming the actual counterparty to each of them. In the case of default of one participant, the CCP continues to perform on its deals with all the other participants, thus protecting them from the direct consequences of the default. Prior to trading, the CCP requires all the participants to post margin and other contributions, which would be used only in the case of insufficient collateral and margin being posted by a defaulting counterparty. CCPs are believed to reduce systemic risk on the market, as they reduce dependencies between institutions and provide an orderly system of loss sharing in case of need, according to a waterfall structure where the “defaulter pays first”.

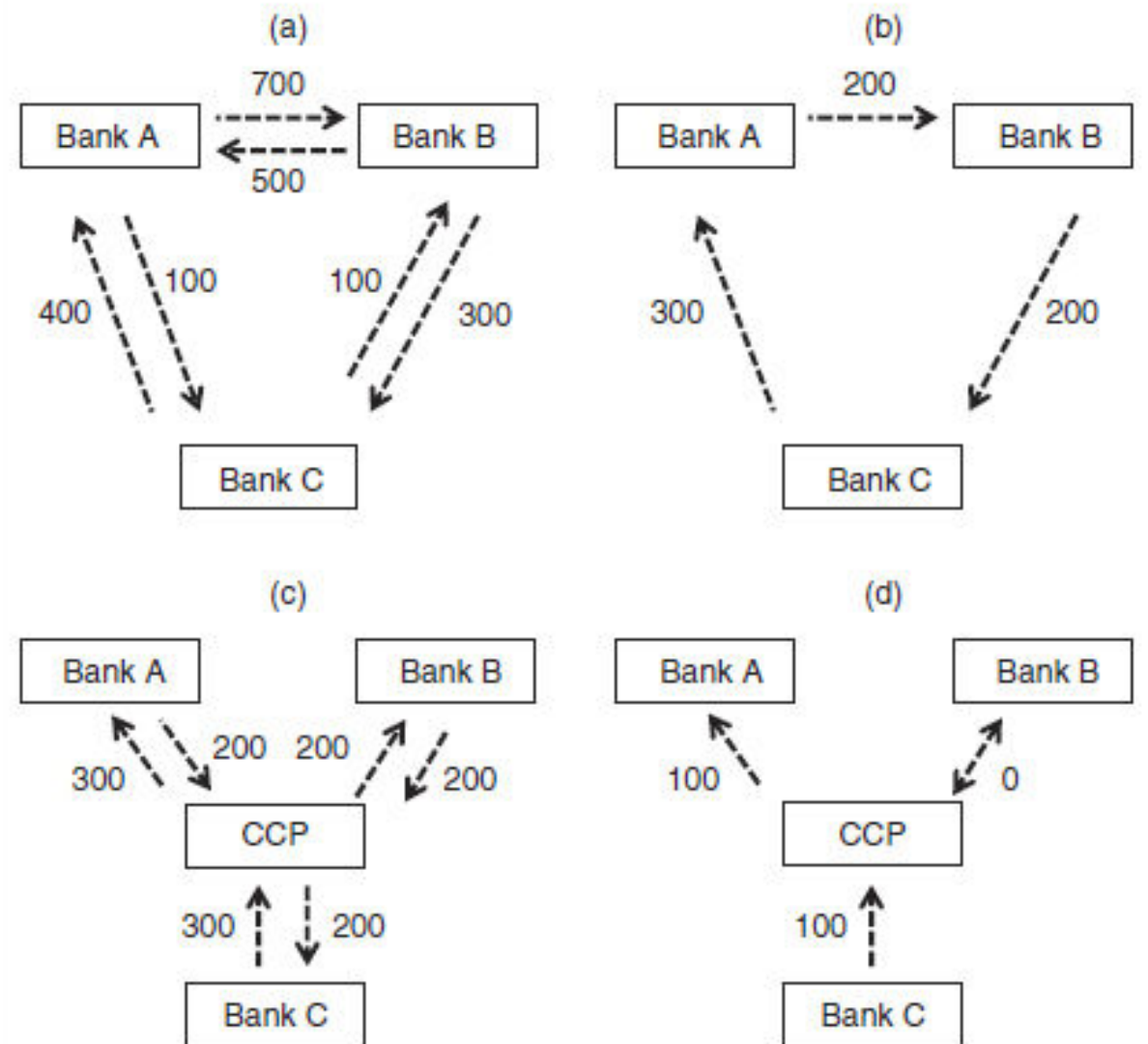
CCPs can be accessed directly only by “clearing members” (typically large banks), while “non-clearing members” (often buy-side institutions such as mutual funds) would have their trades on the CCP intermediated by one participating clearing member.

Once a transaction is executed between two counterparties,¹⁷ the CCP matches trades and substitutes itself in the transaction involving the two original parties, becoming the new counterparty to each of them. From a legal perspective, the original counterparties “novate” their rights to the CCP, which then enters into one contract with the original borrower as well as a

separate contract with the original lender. The original contract is cancelled and each party remains exposed only to the CCP.

In order to mitigate its counterparty risk, the CCP requires from participants variation margin, initial margin and a contribution to the “default fund”.¹⁸ As seen in the context of classical repos, the variation margin is requested by the CCP in order to cover its exposure versus one counterparty due to intraday changes in collateral values. The initial margin, which aims to cover the possible loss on each specific transaction in a worst case scenario,¹⁹ is a function of the volatility of the transaction and the rating of the counterparty. The default fund aims to cover losses from a participant default in case its margin and default fund quota have been exhausted. The contribution of each counterparty to the default fund is a function of all its outstanding trades. Upon a participant default, in the case of default fund insufficiency and missing additional contributions by the remaining participants, the CCP’s own funds would finally be hit.

Figure 16.2 Multilateral portfolio netting



(a) Original exposures, no CCP. (b) Bilateral netting, no CCP. (c) Multilateral netting with CCP. (d) Final position with CCP. The figure shows cash exposures only. Collateral movements are not indicated.

A crucial feature of CCPs is multilateral portfolio netting, which allows minimisation of mutual exposures between participating counterparts. In [Figure 16.2](#), Bank A is the ultimate cash-taker from the market, while Bank C is the ultimate cash-giver.

In the European market, as of December 2016, approximately 27% of repo volumes had been cleared through CCPs. During the financial crisis, European CCPs (mainly Eurex Clearing AG and LCH-Clearnet) facilitated

access to secured funding to those banks whose direct repo lines were being reduced or cut by other market participants (often as a result of a sovereign downgrade) also thanks to the anonymous access to the automated trading systems cleared by CCPs and the increase in the number of banks participating to CCPs.

Eurex Clearing operates the Eurex Euro Repo Market and the Eurex GC Pooling market, with anonymous trading on the Eurex Repo platform.

At the time of writing, participants in the Euro Repo Market can trade repos on 23 different baskets of euro-denominated GC (divided by country of issuer and sector) as well as special repos.

On the GC Pooling market, collateral is represented by

- (i) the ECB Basket, composed of around 3,000 ECB-eligible securities,
- (ii) the ECB Extended Basket, composed of around 14,000 ECB-eligible securities and
- (iii) the Equity Basket, composed of approximately 100 shares selected from the AEX 25, CAC 40, DAX and Euro Stoxx 50 indexes.

LCH.Clearnet Ltd (UK) operates the Repoclear market, where repos are tradeable on electronic trading platforms or bilaterally, collateralised by bonds of several European countries. Repos based on euro-denominated GC are based on the Euro GC Baskets, which contain, split by AAA, AA and A rating, eurozone government debt eligible for the Eurosystem. The Pound GC basket contains UK Government bonds eligible for the monetary operations of the Bank of England. Moreover, under the euro-denominated GC Plus clearing service, two collateral baskets are available, composed of ECB-eligible collateral.

Regulators promote the use of CCPs to reduce the overall counterparty risk on the financial market, benefit from the efficiency of centralised settlement and increase transparency. Prudential capital requirements for deals cleared by CCPs attract favourable capital requirements.²⁰

However, risks and unintended consequences of CCPs should also be taken into account, in the light of their increased use to clear derivative transactions.

1. Concentration risk: despite the benefits of multilateral netting and loss sharing, CCPs could become themselves the most systemically relevant financial institutions.
2. Procyclicality: by enforcing restrictions to collateral eligibility and haircut increases in a potentially simultaneous fashion, CCPs could exacerbate volatility in the money markets and represent a source of procyclicality.
3. Funding cost and market access: the specific margining approaches used by CCPs (initial margin, contribution to default fund) could increase the collateral needed for repo transactions, hence increasing the average funding cost and operational burden.

SETTLEMENT FAILURES AND LIQUIDITY SHORTAGES

As with any other market, that for repos is subject to some issues that could hamper its regular functioning.

The first is related to “settlement failures”, which occur when one party to a repo trade fails to deliver the securities underlying the trade on the agreed date. This can happen either in the opening leg of the repo (failure to deliver by the seller) or in the closing leg, when it is the buyer who may fail to (re)deliver the security. Settlement failures are particularly problematic when they occur on “special” repos, since in this case the buyer enters into the trade with the main intention of getting hold of the specific security (eg, to cover a short on the primary market). Should the repo fail to be settled on time, the buyer would not reap the ultimate benefit that the trade was intended to produce.

According to the European Securities Market Authority the share of failed settlement instructions in the EU financial markets between 2014 and 2016 stood, on average, slightly below 6% for equities, at around 3% for corporate bonds and around 1.5% for government bonds (European Securities and Markets Authority 2017).

Since there will naturally tend to be an impact on the reputation and client relationships of the party failing to settle on time, most settlement failures are not caused by the failing counterparty deliberately choosing not to deliver securities on time, but rather are attributable to the process of securities settlement. That said, we cannot exclude the fact that some repo

counterparties may choose not to deliver a security according to the agreed terms of the trade, when the consequences in terms of relationship and interest flow (especially with very low or negative interest rates) are considered to be bearable.

Common technical reasons for settlement failures are linked to: the imperfect interoperability of security transfers between national and international central securities depositories; inefficiencies within financial institutions (such as insufficient investments to update and maintain back-office systems); outsourcing of settlement tasks to overseas teams in an effort to cut administrative costs. Another reason mentioned by market participants is the trend towards smaller trade sizes than in the past and the consequent multiplication of the number of tickets and settlement instructions for the same overall market volume (Hill 2017a).

The second issue is related to the liquidity squeezes of the repo market (occurring, in particular, on quarter end dates), which significantly reduce the depth of the market and dislocate repo rates.

These events are well known by market participants and tend to be linked by the natural tendency of banks and other cash-rich institutions to scale down the volume of repos when the year end approaches, due to general balance-sheet management. Since this reduces the overall size of the market, the fluidity of the collateral and the volatility of the repo rates are both affected.

It cannot be excluded that the trend to reduce trades at year end is accentuated by the introduction of the Basel III prudential risk regulations, such as those for the liquidity coverage ratio (LCR), net stable funding ratio (NSFR) and leverage ratio. In particular, since the cash leg of a repo is reported on the balance sheet, regulated entities tend to reduce the amount of repos traded close to the year end in order to manage the leverage measure. Beyond prudential regulation, the existence of fees and levies calculated on quarter end dates is reported to exacerbate the recurrent shrinkages in the repo market: for example, the contributions that EU banks have to make to the Single Resolution Board (SRB) are based on balance-sheet figures.²¹

SECURED LONG-TERM INSTRUMENTS AND MARKETS

The funding of long-term assets requires liabilities of similar terms. Historical cases of mismatch have led to crisis and costly bailouts. As policy rates were hiked up dramatically in the early 1980s, the savings and loans associations (S&Ls) saw their cost of funding rise while being unable to reset the interest on long-term mortgage assets. The resulting losses wiped out the capital and ultimately led to the closure of hundreds of S&Ls. Two decades later, large commercial banks placed medium-term ABSs in special investment vehicles (SIV) funded via short-term commercial paper. When the credit quality of the assets deteriorated, they were unable to roll over liabilities in a timely manner and were forced to take back billions of assets on their balance sheets.

The following sections illustrate two popular classes of secured financial instruments for medium and long-term funding – covered bonds and ABSs – which mitigate the aforementioned mismatch.

The covered bond instrument

Covered bonds first appeared in Germany and Denmark over two centuries ago and remain an instrument used prevalently in Europe. From the point of view of the issuer, the covered bond has similar characteristics to senior indenture and is typically issued with fixed coupon and bullet maturity. The key difference with respect to a senior bond is from the point of the view of the investor, because of a specific legal framework giving access to a segregated pool of collateral in case of issuer insolvency. Because it offers a “second way out” to the investor, a covered bond commands higher ratings than senior bonds and thus results into a lower cost of funding.

The European Covered Bond Council (ECBC) sets out on its website what it considers to be the essential features of covered bonds.²²

Covered bonds are characterised by the following common essential features that are achieved under special-law-based frameworks or general-law-based frameworks:

1. the bond is issued by – or bondholders otherwise have full recourse to – a credit institution which is subject to public supervision and regulation;
2. bondholders have a claim against a cover pool of financial assets in priority to the unsecured creditors of the credit institution;

3. the credit institution has the ongoing obligation to maintain sufficient assets in the cover pool to satisfy the claims of covered bondholders at all times;
4. the obligations of the credit institution in respect of the cover pool are supervised by public or other independent bodies.

According to the *2017 European Covered Bond Fact Book* (European Covered Bond Council 2017), there were €2.5 trillion equivalent outstanding at the end of 2016, with two-thirds of the total accounted for by Germany, Denmark, France, Spain and Sweden.

A quality initiative, the Covered Bond Label, launched in 2012, responded, according to its website, “to a market-wide request for improved standards and increased transparency in the European covered bond market”.²³ A key tenet under the Covered Bond Label Convention is the restriction of cover pools to comprise mortgage, public sector or ship assets as these are in principle characterised by better performance in secondary markets. According to the *Fact Book*, mortgages account for over four-fifths of all outstanding covered pools.

Asset-backed securities

ABSs differ from covered bonds in several aspects, the two most important being bankruptcy remoteness from the originator and the ability to tailor the seniority of claims to the investor’s risk appetite. These securities are issued by special purpose vehicles (SPVs), which use the proceeds to purchase the title over an underlying asset pool. The originator of the asset pool fully transfers the assets (“true sale”), and the service of principal and interest on the bonds is based exclusively on the cashflows generated by the pool, irrespective of the solvency of the originator (“bankruptcy remoteness”). Multiple classes of securities are issued by the SPV with different degrees of seniority over the cashflows originating from the assets. Thus, for instance, cashflows are dedicated in priority to the service of the senior notes, and only after that to the mezzanine and junior claims. In principle, this allows a full decoupling of the rating of the notes from the rating of the originator and, for well-structured pools in non-distressed economies, it is not uncommon for the most senior notes to be granted the highest rating in the scale. On the other hand, mezzanine and junior claims often receive non-investment-grade

and even highly speculative ratings. That said, they are characterised by higher yields that make them attractive investments to specialist investors.

ABSs are significantly more complex to issue, manage and assess than covered bonds, as their performance is intimately related to that of the underlying pool. For example, they are generally amortising and contain prepayment features, are issued at variable rates and require underlying swaps and special liquidity lines to mitigate the mismatches between assets and liabilities. The rating frameworks are also more complex, as they need an explicit modelling of the underlying cashflows. In a nutshell, each ABS needs to be looked at and managed as if it were a standalone financial institution; hence the recurrent reference to the “shadow banking system”. Inevitably, liquidity is also more limited and this, together with the underlying complexity and more limited investor base, implies that even highly rated ABSs often do not present a funding cost advantage compared with covered bonds. As we shall see below, the rationale for their issuance often rests on the regulatory capital relief that comes from being able to transfer assets and liabilities off the balance sheet of the originating institution.

According to AFME (2017), there were €1.3 trillion of ABSs outstanding in Europe at the end of 2016, down from almost €2 trillion at the end of 2011. Not unlike covered bonds, mortgages represented two-thirds of the collateral. The US market outstanding amounted to €8.8 trillion, three-quarters of which, however, were represented by agency mortgage-backed securities (agency MBSs).

Agency MBSs are in fact neither covered bonds nor ABSs. They are issued by the government-sponsored agencies in the US, and differ significantly from private-label pass-through MBSs because their repayment does not generally depend on the underlying mortgages. This was confirmed during the crisis, when the two major agencies benefited from government support and continued to service their obligations, while private label MBSs experienced downgrades and defaults. The agencies purchase mortgages from financial intermediaries and fund these purchases by issuing bonds on a regular basis; these continue to be characterised by high liquidity and creditworthiness and represent by far the largest portion of the market for medium-long-term secured instruments.

While covered bonds did well during the crisis and their popularity increased, significant declines in value and outright defaults in ABSs were

considered by many to be at the heart of the 2007–9 turmoil. Closer examination, however, reveals that underperformance was concentrated in ABSs backed by sub-prime residential mortgages and in more complex structures involving resecuritisation. In response to public scepticism on this asset class both private and public-sector initiatives sought to establish distinguishing criteria for high-quality securitisation. Prime Collateralised Securities (PCS) provided investors with quality labels for transactions meeting specific eligibility criteria. The Basel Committee on Banking Supervision and the International Organization of Securities Commissions (IOSCO) jointly proposed criteria for identifying and granting beneficial capital treatment to “simple, transparent and comparable” securitisations.²⁴ The European Commission launched an “EU framework for simple, transparent and standardised securitisation” in 2015.²⁵

As a final note, it is useful to clarify that this chapter looks only at ABSs that are used for funding purposes, sometimes referred to as “true sale”. These differ from “synthetic” securities that are aimed at capital relief purposes and are better categorised as credit derivatives. For instance, synthetic collateralised debt obligations (synthetic CDOs) may reference a basket of credits that remain on the balance sheet of the institution. Investors in CDOs will receive a fee as long as those assets perform, and will have to cover defaults whenever these occur. From the point of view of the originator, synthetic CDOs provide a capital relief, but not funding.

Capital relief, central bank eligibility and Basel III

To gauge the relative attractiveness of covered bonds versus ABSs from the point of view of a bank’s funding, it is useful to go beyond the pure asset and liability management (ALM) properties and to consider ancillary characteristics, in particular the potential for ABSs to provide capital relief, and the wider eligibility for covered bonds as collateral accepted by central banks (Tables 16.2 and 16.3). Covered bonds also benefit from a special status under Basel III liquidity ratios. Another aspect, which goes beyond the scope of this chapter, is the treatment of covered bonds and ABSs under the recovery and resolution regimes for systemic banks. Covered bonds were excluded from proposals of bail-in, that is, the possibility of writing down the value of unsecured creditors as a means of avoiding the liquidation of a struggling bank. The same applies, by construction, to ABSs.

Table 16.1 Simplified balance sheet of a financial intermediary

(a) On-balance-sheet items	
Assets	Liabilities
Covered pool Unencumbered assets	Covered bonds Deposits Senior bonds Equity
(b) Off-balance-sheet items/SPVs	
Securitised assets	ABSs

Capital relief for ABSs

In the case of covered bonds both assets and liabilities remain on the balance sheet of the originating institution and influence cashflows and rates in the same way as any other asset or liability. The key factors from the point of view of the issuer are the desired maturity and achievable pricing, just like the case for other senior facilities. There is no capital relief for the assets, as all underlying risks remain with the issuer.

The ABSs are instead typically set up as a separate balance sheet, characterised by true sale and bankruptcy remoteness from the originator. Thus, they should in theory be outside both the ALM and the capital management of the originator of the securities, as the assets are sold to the SPV and the securities are serviced exclusively out of the cashflows of the assets. The simplified balance sheet in [Table 16.1](#) illustrates the point.

There are, however, several practical ways in which ABSs maintain an umbilical cord to a financial institution. Indeed, it is often neither possible nor advisable to sell all of the securities to investors. There is, in particular, a concern for the potential for moral hazard and adverse selection in the choice of the assets that are securitised. The issuer is typically required to retain the lowest ranked category of bonds, known as the “first loss piece” or “equity tranche”. Because of the way securities are structured, losses are allocated in priority to these lower tranches. In this way, senior liabilities carry a lower risk profile, higher ratings and ultimately lower pricing. Equity tranches attract high capital charges under banking regulation, and this

contributes to the alignment of interest between the originator and the investors.

So much for the theory: this is not how it worked in practice in the run-up to the 2007–9 crisis. Hedge funds became keen buyers of equity tranches because of the contractual high rates of return provided. However, paying high rates on lower ranked securities meant that contractual rates had to be lower for the senior tranches, since both securities had to be serviced from a finite cashflow coming from the assets. Banks were initially so keen to obtain the capital relief coming from being able to sell the equity piece that they were also prepared to retain the senior notes – exactly the opposite of what theory predicts would happen. To finance the retained notes, they often made recourse to short-term asset-backed commercial paper (ABCP), which became a very popular form of investment for money market funds and corporate treasurers.

The unravelling came from two concurrent dynamics. First, being able to sell the first loss piece eliminated alignment of interest to the detriment of loan underwriting practices, especially in the sub-prime space. Second, long-term securities were being financed with short-term ABCP liabilities. As investors started to nurture doubts about the quality of the underlying loans being securitised, they also moved away from ABCP, and the issuers had to consolidate the securities on the balance sheet.

Central bank eligibility and Basel III ratios

The attractiveness of securities to investors has traditionally been a function of their eligibility as collateral for central bank liquidity facilities. Following a series of crisis-induced reforms, the eligibility to satisfy the regulatory liquidity ratios, in particular the liquidity coverage ratio (LCR) in Basel III, has also played a role. The two sets of eligibility are in fact related, as higher quality collateral receiving a more favourable treatment for LCR purposes is also typically subject to smaller haircuts in central bank repo operations.

Covered bonds and agency MBSs have historically been the most liquid asset classes and thus consequently recognised as higher quality collateral under bank regulation. It is important to take this into account for ALM purposes, as it may reduce the cost of funding while also satisfying risk management requirements.

Table 16.2 Types of marketable and non-marketable collateral accepted by the Eurosystem and the other central banks

Collateral	Euro	BoE	Riksbank	SNB	Fed	BoJ
<i>Marketable assets*</i>						
Central governments	✓	✓	✓	✓	✓	✓
Central banks	✓	✓	✓	✓		
Public sector institutions other than central governments	✓	✓	✓	✓	✓	✓
Supranational institutions	✓	✓	✓	✓		✓
Credit institutions (covered bonds)	✓	✓	✓	✓		
Credit institutions (exc. covered bonds)	✓	✓		(✓)		
Corporations (other than credit institutions)	✓	✓	✓	✓		✓
ABSs	✓	✓				
Equities						
Money market funds						
Gold						
<i>Non-marketable assets</i>						
Credit claims (bank loans)	✓					✓
Retail mortgage-backed debt instruments	✓		✓			
<i>Cash as collateral</i>						
Cash incl. fixed-term deposits from eligible counterparties	✓					

Notes: * Debt instruments issued by type. Euro denotes Eurosystem; BoE denotes Bank of England; Riksbank denotes Riksbank (Sweden); SNB denotes Swiss National Bank; Fed denotes US Federal Reserve System; BoJ denotes Bank of Japan.

Source: European Central Bank (2013).

The Eurosystem has in place a relatively broad eligibility for collateral, with a key feature being that both marketable and non-marketable assets are eligible, albeit with different haircuts. The former are defined as those admitted to trading on a regulated market, and also certain non-regulated ones specifically recognised by the ECB. Covered bonds and ABSs are broadly eligible only if marketable, with a partial exception granted for retail mortgage-backed debt instruments. In the case of the Federal Reserve, given the importance of the aforementioned agencies, only certain covered bonds and ABSs are eligible (in particular, Jumbo Pfandbriefe and ABSs to the extent they are rated AAA). [Tables 16.2](#) and [16.3](#) summarise key eligibility requirements for collateral across the main central banks.

Another important regulatory ratio in Basel III goes to the heart of ALM. The net stable funding ratio (NSFR) imposes a balance between longer term assets and liabilities measured in terms of available versus required stable funding over a one-year horizon. Both covered bonds and ABSs help to meet this requirement in full as they mature beyond the horizon. For some institutions, the cost of funding may also be lower than for senior bonds and make this type of issuance attractive. There is, however, an inherent limit to the amounts that can be issued, as explained in the next section.

Covered bonds or ABSs? A primer for issuance

In researching the material for this chapter, it became evident that most analysis of long-term secured issuance is written from the point of view of the investors, not the issuer. In [Table 16.4](#) we summarise the key considerations in choosing different instruments from the point of view of the issuer.

Each category of indentures has characteristic advantages and disadvantages, and may appeal to different distinct categories of issuers, depending on the geographical area of activity. For instance, a specialised mortgage lender in Germany or Denmark may prefer a covered bond, while a similar institution in the US would opt for either selling mortgages to an agency or issuing a pass-through mortgage bond. Financial institutions with a diversified business model are likely to find it advantageous to tap all three sources of medium-to-long-term funding illustrated in this chapter.

We conclude this section with an example of ABSs and covered bonds backed by mortgages, as these dominate the markets in the US and Europe,

respectively. The interested reader can find additional information in International Monetary Fund (2011) and Carbó-Valverde *et al* (2011).

Following the origination of a mortgage pool, a bank could decide to ring-fence it as a segregated portion of its balance sheet and issue a covered bond for an amount that is typically smaller than the nominal value of the book value of the ring-fenced pool. The same legal base granting privileged access to the mortgages in the case of insolvency of the bank will determine parameters such as the maximum loan-to-value ratio and other creditworthiness criteria for the underlying mortgages, the degree of overcollateralisation, eligibility criteria for replenishment and substitution of the pool, disclosure requirements of asset quality, stratification and other risk parameters. There is thus a certain need for active management of the pool. The bank will thus choose this form of issuance when its key ALM objectives are the cost and duration of funding.

Table 16.3 Criteria and requirements for the marketable assets accepted by the Eurosystem and the other central banks

Collateral	Euro	BoE	Riksbank	SNB	Fed	BoJ
<i>Type of issuer/debtor</i>						
Central governments	✓	✓	✓	✓	✓	✓
Government agency	✓	✓	✓	✓		
Regional and local governments	✓		✓	✓		✓
Corporate	✓	✓	✓	✓		✓
Bank	Excluding own issue	For uncovered bank bonds, a central or local government guarantee is required	Securities issued by domestic banks and their subsidiaries abroad are not eligible	For uncovered bank bonds, a central or local government guarantee is required		✓
<i>Type of issuer/debtor</i>						
Supranational	✓	✓	✓	✓		✓
ABSS	Only if there is a true sale of assets and an SPV is bankruptcy remote from originator	Only if there is a true sale of assets and an SPV is bankruptcy remote from originator				Only if there is a true sale of assets and an SPV is bankruptcy remote from originator
<i>Issuer residence</i>						
Domestic	✓	✓	✓	✓	✓	✓
Foreign	Issuer ¹ : EEA or non-EEA G10 countries. Debtor: EEA. Guarantor ¹ : EEA.	AT, AU, BE, CA, CH, DE, DK, ES, FI, FR, IR, IT, JP, LU, NE, NO, NZ, PT, SE, SI, US	For securities denominated in foreign currencies: CH, EU or EEA	EEA, JP, CA and US	Confined to sovereign issuers	Only central government debt of UK, US, DE or FR
<i>Seniority</i>						
Senior	✓	✓	✓	✓	✓	✓
Subordinated			✓			
<i>Credit standards</i>						
Minimum credit threshold for issuer or asset (ratings often used for reference only)	Minimum BBB–. ABSs under standard framework require AAA/Aaa rating at issuance and single A– rating during the life of the security ²	Accepted specified sovereign debt and highest credit quality for accepted ABS types (broadly equivalent to AAA)	Minimum single A for securities denominated in CHF. AA for securities denominated in foreign currencies	Minimum A– ³	N/A	Minimum single A; foreign government bonds require AA and ABSs, AAA (use as indicator only)
<i>Settlement</i>						
Domestic	✓	✓	✓	✓	✓	✓
Foreign		✓		✓		✓
<i>Currency</i>						
Domestic	✓	✓	✓	✓	✓	✓
Foreign	Not accepted under standard framework ⁴	EUR, USD, CAD, AUD, SEK, CHF, JPY, DKK, NZD, NOK	EUR, USD, GBP, DKK, SEK, NOK	DKK, EUR, GBP, JPY, NOK, SEK, USD		USD, EUR, GBP for government debts

Notes: ¹Corporate bonds without ratings must be issued in the euro area. ²For eligibility criteria under the temporary framework see Guideline ECB/2013/4. ³There will be a limit on the share of collateral with creditworthiness corresponding to a rating of less than AA. This limit is being phased in, starting on January 1, 2014. ⁴For eligibility criteria under the temporary framework, see Guideline ECB/2013/4.
Source: European Central Bank (2013).

Table 16.4 Characteristics of senior bonds, covered bonds and ABSs

	Senior bonds	Covered bonds	Senior, high grade, ABS
Collateral pool	N/A	Restricted by law	Broad eligibility
Maturity	Longest term	Medium term	Pass-through
Pricing	Benchmark	Lower than benchmark	Lowest if liquid
Capital	No impact	No impact	Limited impact
Leverage	Increase	Increase	Decrease
Management	Simplest	Simple	Complex
Disclosure	No additional	Additional	Specific

Alternatively, the bank could sell the mortgages to an SPV, financed through issuance of bonds (the case of a pass-through mortgage-backed security). A securitisation law, where applicable, or other specific legal arrangements will ensure the necessary transfer of title for the underlying mortgages and associated real estate. The composition of the pool will be dictated not by law but rather by the ultimate rating that the bank wants to achieve for the securities, together with the required capital release. For instance, riskier mortgages would require a larger first loss piece for the senior bonds to achieve the same rating, and therefore the capital release would be smaller. The SPV administrative requirements could be significant, and require identification of parties for trustee, paying agent, servicer of the loans, etc. At the time of writing, banks have found it advantageous to retain part of their own ABS issuance for repurchase with the central bank.

CONCLUSION: IS ASSET ENCUMBRANCE BECOMING AN ISSUE?

Funding risk took a on new dimension during the 2007–9 crisis and the landscape is still evolving. The shift to repo, originally restricted to less creditworthy counterparties, has now become the norm in short-term funding. At longer maturities, covered bond issuance has been particularly dynamic, encouraged by both investor sensitivity to heightened counterparty risk and favourable regulatory treatment. ABSs initially came to a halt, though selected asset classes are making their way back, particularly mortgages. At

the time of writing, policymakers were supporting the use of SME loans as collateral for ABSs, in particular, and the interested reader should pay attention to the development of these markets, given the importance of SME lending for ALM purposes.

In this chapter we have outlined key considerations for a bank using secured instruments for short-, medium- and long-term funding. There is, however, one aspect that is raising important questions, as the increased use of secured funding eats into the recovery rate of senior bonds and uninsured deposits in the case of bank failure. Our concluding thoughts provide a few pointers on this issue.

The increased role of repos for short-term funding and the International Swaps and Derivatives Association requirement for collateral posting in the case of intermediaries active in derivative markets have led market observers to warn of a potential collateral shortage, though others disagree (Fender and Lewrick 2013; Committee on the Global Financial System 2013). We are not aware of studies that consider the impact of collateral posting for unsecured creditors of individual institutions, but this is an area which warrants close examination and appropriate disclosure to investors.

In a similar vein, while covered bonds and ABSs may lower the cost of funding for an intermediary, a trade-off needs to be considered, as excessive issuance may end up raising the cost of senior bonds and uninsured deposits significantly. Indeed, in the case of an issuer default, unsecured creditors will have any recourse limited to assets that are both on the balance sheet (thus excluding securitised assets that are bankruptcy remote) and unencumbered (thus excluding assets that are legally ring-fenced because they are tied to covered bond issuance). Given the requirements for overcollateralisation and asset quality that benefit covered bonds in particular, unsecured creditors will start to charge a premium, which at some point could balance out the advantage of secured funding. ABS issuance may also contribute to augment the risk for senior bondholders in the sense that, while securitised assets are no longer on the balance sheet, the first-loss risk may be retained.

At the time of writing the European Banking Authority did not detect significant increases in asset encumbrance (European Banking Authority 2017). Bohn and Tonucci (2013, 2014) note that most banks operate at a level of encumbrance of 10–30% and that this level is unlikely to affect unsecured recoveries. Also, financial institutions characterised by higher levels of encumbrance tend to be specialised lenders that do not issue

liabilities other than covered bonds. While rating agencies make some reference to encumbrance, they also note that it matters only in a liquidation scenario and therefore would not affect the probability of default. Because the probability of default is the key variable in rating decisions, encumbrance has so far played a minor role. For a more detailed analysis, the reader is referred to Caris *et al* (2013) and Winkler (2013).

The opinions expressed in this paper are personal and may not necessarily reflect the position and practices of the Inter-American Development Bank or the European Investment Bank.

- 1 Libor is the London Interbank Offered Rate.
- 2 Notwithstanding its legal interpretation, from an economic standpoint a repo is a borrowing collateralised by the underlying security, which is typically called “collateral”.
- 3 For illustrative purposes, no “haircut” is applied in the example.
- 4 Prior to its default, Lehman Brothers had unconventionally accounted a particular repo transaction, “Repo 105”, as a true sale, in order to reduce leverage in its balance sheet.
- 5 This is often done by “securities lending”, where repo activity on a portfolio of securities is essentially outsourced to a specialised agent.
- 6 The total profit and loss from this trade would be the change in the bond clean price, plus the bid–ask spread, plus the interests accrued on the bond coupon minus the cost of the repo (carry) for the holding period (Tuckman 2002).
- 7 Special rates could even go negative, should the related cost for the borrower of the security be lower than the cost of a fail on a delivery obligation (Fleming and Garbade 2004).
- 8 Due to their liquidity, on-the-run bonds are also often shorted by trading desks of big banks (eg, in order to hedge underwriting activities). This contributes to the tendency of such issues to trade “special”.
- 9 The main tri-party agents in Europe are Clearstream, Euroclear, JP Morgan Chase and Bank of New York Mellon. In the US, the only active tri-party agents are JP Morgan Chase and Bank of New York Mellon.
- 10 Optimisation algorithms scan through the security account of the borrower and allocate collateral across all its open trades at regular intervals during the day. The borrower can request specific collateral substitutions.
- 11 Eligibility and haircuts set as a direct function of equity historical or implied volatility could however prove impractical and are seldom implemented difficult (Choudhry 2007).
- 12 This was a factor in the Bear Stearns problems in 2007.
- 13 See the ICMA semi-annual survey of the European repo market (International Capital Market Association 2017).
- 14 The Global Master Repurchase Agreement (GMRA) is a master contract that two counterparties sign in order to discipline repo transactions between them. The GMRA is composed of a general standard part and a customisable Annex where the signatories can specify particular aspects (such as overcollateralisation).
- 15 The haircut method and the margin ratio method produce different results for the net exposure even when the haircut and the margin ratio are set at economically equivalent values (3% and 103.09279 in our example). While the actual difference could be negligible in most cases, in situations where haircut percentages are high (eg, in the case of long-dated repo transactions) the application of the

two methods could lead to substantial differences and reconciliation issues between the two counterparties.

- 16 This is similar to the potential future exposure approach for derivatives.
- 17 CCP-cleared repos can be traded bilaterally or on automated trading systems.
- 18 For some those CCPs that also clear other markets (eg, derivatives), the margining mechanism is applied across all markets.
- 19 A counter-intuitive feature of initial margin is that, despite its name, it can change during the life of one transaction, due to variations in volatility of the related risk factors and/or the counterparty rating. Initial margin is a “tail” measure particularly difficult to estimate for derivatives, as they are more volatile than repos and more complexly correlated.
- 20 Exposures on so called “qualifying CCPs” (ie, those that fulfil a series of criteria set out by the Bank for International Settlements and the IOSCO) attract a capital requirement of 2% according to Basel III. Exposures are a function of initial margin, variation margin and contribution to the default fund. At the time of writing, additional initiatives for CCP-cleared GC collateral were being pursued by LCH.Clearnet SA (France) and Cassa di Compensazione e Garanzia (Italy).
- 21 A notable volatility event and dislocation of the repo market occurred on the last three calendar days of 2016, with extreme repo rate movements. Specials rates on some government securities in high demand went as low as –15%. A detailed account can be found in Hill (2017b).
- 22 See <http://ecbc.hypo.org/>.
- 23 See <http://www.coveredbondlabel.com/>.
- 24 See <http://www.bis.org/bcbs/publ/d374.htm>.
- 25 See <http://bit.ly/2ASBegN>.

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Asset Encumbrance

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Asset encumbrance occurs when the bank's assets are used to secure creditors' claims or credit-enhance any transaction.

In Europe, asset encumbrance began to be discussed in 2012, after bank funding structures started to shift towards more secured funding compared with the previous decade, owing to the financial crisis that started in 2007.

During the financial crisis years, the interbank unsecured market fell, finally dropping dramatically due to higher counterparty risk concerns. Consequently, banks increased the use of their assets as collateral, having the need to resort to more secured market issuances, particularly covered bonds, and public-sector funding sources in addition to repurchase agreement (repo) funding. At the same time, greater collateralisation resulted from trading activities and related risk mitigation, such as for derivatives and wider use of central counterparty clearing houses (CCPs).

Assets that are encumbered are obviously not available for any other use.

The resulting regulatory concern was that it could be too risky (not only at individual bank level) should asset encumbrance reach excessive percentages due to the different potential types of associated risks, as we shall see later.

Because of that, in 2013, the European System Risk Board (ESRB), being responsible for the macroprudential oversight of the EU financial system,

promoted different quantitative and qualitative analyses aimed at providing support and recommendations for EU policymakers on this topic.

The ESRB's policy recommendations avoided requiring a specific regulatory new limit, choosing instead to support the development of guidelines on harmonised templates and definitions that could facilitate the monitoring of asset encumbrance (level, evolutions and types) by the supervisory authorities as part of their supervisory process. Furthermore, it was recommended that credit institutions put in place finalised internal policies to adequately identify, manage and control the different risks associated with collateral management and asset encumbrance. Following these ESRB recommendations, the European Banking Authority (EBA) published the implementing technical standard (ITS) on asset encumbrance reporting that became the final regulation in December 2014 (European Commission 2015b).

In this chapter, in addition to analysing in detail the different risks arising from asset encumbrance, we describe the definition of the so-called “asset encumbrance ratio” according to European Commission (2015b). Given the annual EBA report on asset encumbrance that follows the harmonised supervisory reporting framework based on the aforementioned EBA ITS, we shall also examine the evolution of the asset encumbrance ratio across European banks. Finally, in the concluding section we show that asset encumbrance is not necessarily “bad”; rather, it represents an important opportunity if properly managed. Towards that end, in this chapter we present a possible approach for the definition of a prudential internal limit that would support a sound asset and liability management (ALM) in banks.

RISKS FROM ASSET ENCUMBRANCE

Different consequences with connected risks can arise from an excessive level of asset encumbrance.

One of the main consequences of asset encumbrance is the subordination of unsecured creditors, since many more bank assets are used to guarantee other claims and fewer assets are available in the case of bank failure. In other words, asset encumbrance implies a shift of risks between investors, penalising unsecured bondholders and depositors, which become even more subordinated as the level of the bank's asset encumbrance increases.

The extent of the structural subordination of unsecured creditors depends on different factors, such as the level of protection, the type of other creditors

and the quality of the remaining unencumbered assets, as well as the bank's business model and the probability of default. The unintended negative consequence of this risk-shifting depends on the unsecured creditors' capacity to price and evaluate that risk: for example, existing unsecured creditors may not have the opportunity to modify their remuneration in the case of encumbrance changes. At the same time, they could have limited knowledge of a bank's asset encumbrance level, either because this phenomenon is complex to measure or because its disclosure can be poor. Moreover, considering that the structural subordination of depositors implies a higher level of risk of deposits (both sights or time deposits), the aforementioned risk-shifting could also involve the liabilities of deposit insurance funds, increasing their riskiness, with related concerns of EU member states that do not have special depositor's preference laws.

A second consequence of high levels of asset encumbrance involves a potential limitation to future access to unsecured markets because of the increased risk perception of unsecured investors and subsequent reduction in the variety of counterparties interested in investing in bank debt. This could imply greater concentration in the market and worsening of the functionality of the market itself, as well as an increase in the cost of unsecured funding. Since encumbrance can be difficult to price correctly, it could also imply an inefficient resource allocation, with undetermined effects on general financial stability.

A third important consequence is that encumbrance increases general funding and liquidity risks. In fact, high levels of assets encumbrance reduce the availability of unencumbered assets that can be transformed into liquidity and at the same time, increase the so called "contingent encumbrance". More specifically, a greater use of collateral for funding could imply a scarcity of unencumbered assets available to be transformed into liquidity via pledges in the private markets or through the main refinancing operations with central banks. A scarcity of central bank's available collateral could in turn limit its ability to transmit its monetary policy mechanisms, and reduce the possibility of providing liquidity assistance in crisis situations. That is to say, a high level of encumbrance may decrease the effectiveness of the central bank's actions, increasing the systemic risks in the banking sector and/or the credit risks of the central banks themselves should they decide to accept a wider range of collateral assets without the correct level of haircut.

Existing encumbrance also tends to increase in stressed or adverse situations, due to contractual obligations or increased counterparty risk perceptions, which lead to calls on further collateral as additional guarantees. This phenomenon, known as “contingent encumbrance”, results in potential outflows and higher liquidity risks depending on the type and magnitude of the adverse scenarios. In effect, contingent encumbrance could arise from different negative events, and does not have standard features. A typical adverse event that leads to further encumbrance is a significant increase in market volatility with negative effects on the quality of the collateral and a consequent margin call requiring additional haircuts. Another is when rating downgrades occur, since this can trigger contractual obligations with additional collateral needs.

A good example of how existing encumbrance could negatively affect liquidity risk is the case of covered bonds (CBs). A decrease in the value of the transferred loan book could require the issuer to reintegrate the cover pool, adding additional assets and/or more appropriate loans-to-value. In addition, adverse or stressed situations could trigger underlying contractual obligations and have negative consequences for liquidity, due to cash outflows and the additional need for collateral to fund them. As an example, covered bonds typically provide the outsourcing of specific activities when certain events happen, such as when downgrading occurs and low rating levels are reached. The need for outsourcing can concern the “Account Bank”, which is the bank where the vehicle company holds the liquidity arising from the creditor’s repayments, and is almost always the issuer itself. The implication is that if the Account Bank ceases to have the minimum required rating, it becomes necessary to move the collected amounts to another Account Bank, with resulting outflows and following negative impacts on liquidity, as mentioned above.

For all the reasons above, encumbrance tends to be procyclical, and thus can emphasise the procyclicality of the real economy: available collateral tends to grow in economic upturns and reduce in economic downturns, as in the case of increased needs for secured funding, additional haircuts and supplementary margin calls, triggers of contractual obligations, etc. A high level of encumbrance could also imply collateral scarcity and, ultimately, a high level of asset encumbrance could make the financial system riskier, since it becomes more sensitive to economic cycles.

In general, these risks are difficult to quantify, particularly because of the complexity of the phenomenon of encumbrance, which, in the absence of harmonised rules in the past, has also entailed a lack of public disclosure. Poor disclosure could in turn imply situations where encumbrance is not correctly priced by investors.

For all these reasons, the competent authorities deemed it important to define a uniform approach for measuring asset encumbrance through standard metrics across institutions; this helps the supervisory authorities to closely monitor the level, evolution and types of asset encumbrance and to support higher disclosure at the same time. In the next section, we shall analyse the definition of the “asset encumbrance ratio” as per the rules adopted by the EU.

ASSET ENCUMBRANCE RATIO

On July 24, 2014, the EBA published the final implementing technical standards (ITS) on asset encumbrance reporting under Article 100 of the Capital Requirements Regulation (EU) No 575/2013 (CRR), which mandated the EBA to develop reporting templates for all forms of asset encumbrance. At the end of 2014, the Commission Implementing Regulation (EU) 2015/79 (European Commission 2015b) entered into force based on the ITS submitted by the EBA.

The core metric defined in this Regulation is the asset encumbrance (AE) ratio, which measures the value of assets that are pledged as collateral relative to the total bank’s on- and off-balance-sheet assets. The formula for this ratio, which also considers the collateral received and reused by an institution, is defined as follows

$$\begin{aligned} &\text{AE ratio (\%)} \\ &= \frac{\text{total encumbered assets} + \text{total collateral received and reused}}{\text{total assets} + \text{total collateral received available for encumbrance}} \end{aligned}$$

In order to compile the numerator “an asset is considered encumbered if it has been pledged or if it is subject to any form of arrangement to secure, collateralize or credit enhance any transactions from which it cannot be freely withdrawn” (European Commission 2015b, Annex III, Paragraphs 9–11). This definition covers the following main types of transactions, even if it is not limited to them.

- Secured financing transactions (SFTs): these include repos, securities lending or other forms of secured lending where the lender receives collateral in the form of securities or cash to protect themselves against default by the seller.
- Collateralised deposits: secured lending where the lender receives collateral in forms other than securities or cash to protect themselves against default by the seller. For instance, funding from supranational counterparties can require as collateral the same loan that has been financed.
- Central bank facilities: all forms of secured funding (collateralised deposits or repo transactions) in which the counterparty is a central bank. Often, for these operations it is not possible to identify the specific underlying collateral, because normally it is managed through a “collateral pooling”. In this case Regulation (EU) 2015/79 requires the identification of the encumbered portion on a proportional basis, taking into account the overall composition of the pool of collateral.¹ It is important to note that assets already posted at the central bank are not encumbered until the credit institution uses them to fund some deals (excluding intraday credit).
- Derivatives (liabilities): the carrying amount of the collateralised derivatives that are financial liabilities, ie, with a negative value; this amount is primarily based on the replacement costs for derivative contracts (obtained by marking to market) that have a negative value and should be measured considering the “net replacement cost” if bilateral netting contracts are in place. Collateralised derivatives liabilities entail asset encumbrance, due to the corresponding collateral posted as variation margins, given the negative value of the contracts.
- Collateral placed in the clearing system or at central counterparties (CCPs): cash, bonds or other types of collateral (if any), allocated for access to the services of clearing systems and/or CCPs, including default funds and initial margins.
- Covered bonds: that is, all instruments defined in the first paragraph of Article 52(4) of Directive 2009/65/EU, whether they take the legal form of a security or not. The encumbrance corresponds to the cover pool backing the bonds that have been placed or pledged with third

counterparties. In the case of retained covered bonds (ie, of retention of an issuance, in whole or in part, by the same issuer), if they are not yet pledged, the underlying cover pool is considered to be unencumbered.

- **Securitisations:** the regulation in this category includes the debt securities originated in a securitisation transaction as defined in Article 4(61) of Regulation (EU) 575/2013. For securitisations that are not derecognised (ie, that remain in the bank's balance sheet), the encumbrance corresponds to the cover pool backing the securities that have been placed or pledged with third counterparties, and in the case of retained securitisations the same rules as for retained covered bonds apply.

Table 17.1 Encumbrance of instrument type by prevalent maturity

Instrument	Initial encumbrance
SFTs (repos)	1 day–3 months
Collateralised deposits (supranational funding)	5–10 years
Central bank funding (standard MRO)	3 months
Central bank funding (LTRO)	~5 years
Derivatives	2–5 years
Collateral placed in clearing system	>10 years
Covered bonds	5–10 years
Securitisations (weighted average life, including pre-payments)	~5 years

Notes: SFTs, secured financing transactions. MRO, main refinancing operations. LTRO, long-term refinancing operations.

Table 17.1 classifies the encumbrance of each previous type of instrument by its prevalent maturity: for instance, bonds underlying repos are normally encumbered with a very short-term maturity, while covered bond and/or asset-backed securities (ABSs) generally entail a medium- to long-term encumbrance.

For all these different types of transactions, the encumbrance must be represented using the carrying amount of assets held by the bank that are encumbered according to the definition in the regulation, as described above. The carrying amount means the amount that is reported on the asset side of the balance sheet and guarantees corresponding secured liabilities, called “matching liabilities”.

The amount of encumbrance (asset side) is normally greater than the amount of “matching liabilities”, for various reasons; the overcollateralisation requirements for secured issuances,² the level of haircuts applied or the different market values priced for assets pledged can affect the amount of matching liabilities differently. In general, higher-rated institutions are those less affected by the overcollateralisations, while lower-rated institutions are required to pledge a larger amount of collateral.

The following theoretical examples show both the impact of overcollateralisation on levels of encumbrance and the concept of the matched liabilities. We assume the following.

- Bank A holds its own as yet unpledged securitisation, whose nominal value is €100. The total carrying amount of the underlying assets is €110. At a certain point in time (T_1), Bank A uses the entire amount of this bond as collateral in a repo transaction with a central bank, which prices the securitisation at €90 and requires a haircut of 6%. Therefore, Bank A receives a central bank funding equal to €84.6.
- Bank B holds its own as yet unpledged securitisation, whose nominal value is €105. The total carrying amount of the underlying assets is €110, the same as in the previous case. At T_1 , ie, at the same time as Bank A, Bank B uses only a fraction (65%) of this bond as collateral in a repo transaction with a central bank, which prices the securitisation at €95 and requires a haircut of 4%. Therefore, Bank B receives a central bank funding equal to €62.4.

Given these assumptions we have the following.

- Bank A has encumbered assets for a total carrying amount of €110 compared with a matched liabilities value of €84.6. Assuming that the total assets of Bank A amount to €200, the asset encumbrance ratio is 55%.

- Bank B has encumbered assets for a total carrying amount of €97.5 compared with a matched liabilities value of €88.9. Assuming that the total assets of Bank B amount to €200, the asset encumbrance ratio is equal to 36%, much lower than that of Bank A, because Bank B has used its self-securitisation only partially. It should also be noted that the corresponding matching liabilities are proportionately better, because the haircut and overcollateralisation levels are lower and the price of the bond is higher. Assuming the utilisation of all Bank B's own securitisation and keeping the other hypotheses unchanged, the matched liabilities would be higher than for Bank A, reaching a total value of €95.76 despite the same asset encumbrance ratio (55%).

REGULATORY REPORTING AND PUBLIC DISCLOSURE

In order to closely monitor the AE ratio across banks in Europe, the supervisory reporting on asset encumbrance (based on the EBA ITS) came into force during 2014.

Through this new reporting, the supervisory authorities put in place a general monitoring framework to closely monitor the level, evolution and types of asset encumbrance in financial institutions. This new framework adopted uniform templates, ensuring a harmonised approach enabling competent authorities to better compare the reliance on secured funding and the degree of structural subordination of unsecured creditors across banks.

The reporting requirements on asset encumbrance are defined in European Commission (2015b). All institutions subject to prudential requirements at individual and consolidated level must comply with this new framework, and relevant reports must be submitted quarterly.

The EU regulatory reporting on asset encumbrance, besides measuring the value of assets that are pledged as collateral relative to the total bank's on- and off-balance-sheet assets, also aims to collect other granular information. For that purpose, it requires banks to fill in five sets of templates.

- **Part A – encumbrance overview:** the first set includes different templates that distinguish between positions that are encumbered at the balance sheet date and those that are available for potential funding needs on the same date. Towards this end, Part A requires a breakdown of the reporting institution's assets into encumbered and unencumbered

portions and between products (eg, collateralised deposits, repos, central bank funding, derivatives/liabilities, covered bonds, securitisations). The same breakdown into encumbered and unencumbered portions is required for received collateral, which should be shown in its own dedicated template. Other reports are required to represent the bank's own covered bonds and ABSs issued but not yet pledged and to provide information on the different sources of encumbrance, including those with no associated funding (eg, loan commitments).

- **Part B – maturity data:** this part requires the breakdown of encumbered assets and collateral received and reused by defined intervals (time buckets) corresponding to the matching liabilities' residual maturities.
- **Part C – contingent encumbrance:** this part requires information regarding the additional assets that the reporting institution may need to encumber in the case of adverse events, such as high market volatilities with a consequent decrease in the fair value of the existing encumbered assets. In this part, the institution should calculate the level of additional encumbrance under two specific scenarios:³
 1. a simulated decrease of 30% in the fair value of encumbered assets, with the subsequent need to add new collateral in order to maintain a constant collateral value, considering the previous levels of overcollateralisation;
 2. a hypothetical depreciation of 10% in significant currencies (ie, each currency for which the institution has aggregate liabilities amounting to 5% or more of the institution's total liabilities).⁴
- **Part D – covered bonds:** this part includes specific information on covered bonds issued by the reporting institution, comprising the nominal amount, the present value (swap), the asset-specific value (ie, the economic value of the cover pool assets), the carrying amount, the credit rating by agencies. Moreover, it also requires information on unencumbered assets that are eligible for cover pool and, towards this end, the eligibility for covered bonds should be evaluated on the basis of the relevant statutory covered bond regime.

- **Part E – advanced data:** this part includes different types of additional information about the reporting institution's assets and its received collateral. For instance, it requires an indication of all types of assets (specified as encumbered or unencumbered) and the amounts that are eligible for central banks. Another example concerns the additional information requested on the collateral received by the reporting institution, particularly with reference to the securitisations issued by other entities in the group.

As can be seen from the above, banks are required to produce a complex supervisory report that contains detailed information on all forms of asset encumbrance, including contingent encumbrance, which, being a substantial risk, is vitally important to better understand and analyse the liquidity and solvency profiles of the institution. The representation is structured in this way to allow the supervisory authorities a clear view of the relevant business concepts.

Alternatively, as far as the public information is concerned, in March 2017 the EBA published dedicated (draft) Regulatory Technical Standards (RTSs) on the disclosure of encumbered and unencumbered assets, while at the same time submitting its final report to the European Commission for the subsequent inclusion in the regulation by the competent European legislative bodies. Through these new RTSs, the EBA promoted transparent and harmonised information on asset encumbrance across banks in Europe, so as to enable market participants and investors (not just regulatory supervisors) to have a clear and consistent view of this phenomenon.

Considering the pre-existing supervisory reporting on asset encumbrance, in order to minimise the implementation costs for institutions, the EBA adopted the common definitions and format implemented for the above-mentioned supervisory report on asset encumbrance for public disclosure. Specifically, the EBA's technical standards on disclosure of encumbered and unencumbered assets require that institutions publish the following templates in their Pillar III reports, at least annually (see European Banking Authority 2017).

- **Template A:** encumbered and unencumbered assets in carrying amount and fair value by broad category of asset type, with the carrying amount of unencumbered assets broken down by asset quality.

- **Template B:** collateral received by an institution, by broad category of product type.
- **Template C:** carrying amount of encumbered assets and/or collateral received and associated liabilities.
- **Template D:** narrative information on the importance of asset encumbrance in the institution's funding model.

These templates enhance public information and cover a recognised gap, introducing requirements for additional pieces of information for both quantitative and qualitative templates. Quantitative templates require a highly granular breakdown of encumbered and unencumbered assets and off-balance-sheet items by asset class, introducing, for example, the reporting of encumbered and unencumbered assets by liquidity class, split using the same metrics as those adopted for liquidity risk monitoring (ie, extremely high-quality liquid assets (EHQLA) and high-quality liquid assets (HQLA)). Information is also required on self-securitisations, ie, on own-issued but retained ABSs or retained covered bonds.

Regarding the disclosure of qualitative information, recognising that asset encumbrance levels depend very much on the institution's business model, the EBA ITS focused on the need for clarification on, for example,

- the evolution of encumbrance over the period,
- the degree of overcollateralisation and its extent due to ABSs and/or covered bonds,
- the reasons behind changes in the level of encumbrance,
- the structure of encumbrance within a group,
- the potential presence of significant infragroup encumbrance.

However, these are only examples and, given the wide variety of possible business models, the EBA's standards offer some flexibility in their templates in order to enable institutions to disclose an appropriate set of information.

To conclude, all these standards, despite being somewhat complex and onerous to develop and implement for banks, represent a necessary step towards greater transparency in response to the market changes in banks'

funding and require credit institutions to put in place augmented procedures and controls to manage all the risks (including reputational risks) associated with collateral management and asset encumbrance. As a result of these regulatory changes, there is increased need for banks to have in place adequate risk management policies in order to define their approach to asset encumbrance and give better strategic guidance on the ALM choices.

THE LEVEL OF ASSET ENCUMBRANCE ACROSS BANKS IN EUROPE

Since 2015, the EBA has released annual reports specifically on asset encumbrance, as part of its ongoing commitment to monitor the composition of funding sources across European banking institutions. These reports resulted from the data on asset encumbrance that EBA began to receive in 2015, following the supervisory reporting (based on the EBA ITS) introduced by the EC (European Commission 2015b). The monitored institutions encompass all large banks at EU level and at least three banks in each country of the European Economic Area (EEA).

Within this area, at the end of 2016, the weighted average AE ratio (ie, the value of encumbered assets and collateral received in relation to the total unencumbered bank assets and collateral received) was 26.6%, a slight increase compared with those of previous years (25.4% for December 2015, and 25.1% for December 2014).

Even if such data is published in aggregate, the asset encumbrance ratios in the EU (analysed by EBA reports through the median, interquartile range and 5th and 95th percentiles) show a wide distribution across banks, with the highest levels of encumbrance reported by specialised mortgage institutions.

In more detail, we can observe from the AE ratios published by the EBA with reference to the end of 2016, that

- the distribution of total encumbered assets across countries ranges from less than 10% to more than 50%, with the highest levels of encumbered assets being shown by Denmark and Greece, whose average AE ratios are well above 50% and 40%, respectively, for different reasons (a large established covered bond market for Denmark and a high share of central bank funding for Greece),

- the distribution of encumbered assets by asset class shows that loans and debt securities were the most widely used products, with both making up more than 40% of the total encumbered assets and collateral.

At the end of 2016, the main source of asset encumbrance remained repurchase agreements, representing a large share of encumbrance, particularly in the UK (39%) and Italy (32%). The volume of covered bonds issued is likewise an important source of encumbrance (21% for the EU), with a total of covered bond issuances of €2.2 trillion in December 2016 compared with €2.0 trillion in December 2015 and €1.9 trillion in December 2014. A similar growth (in percentage terms) was seen for over-the-counter (OTC) derivatives, which amounted to 10% of the total sources of encumbrance in December 2016, increasing in volume up to a peak of €1.2 trillion in September 2015 compared with €1.0 trillion in December 2015. On the other hand, central bank funding is confirmed as an important source of encumbrance, albeit moderately in decline (8% of the source of encumbrance in December 2016, compared with 10% in December 2015).

Finally, the distribution of encumbrance by maturity shows that the largest share of assets and collateral are respectively encumbered and reused with an open maturity (on demand) or with very short-term maturities, generally up to two weeks. Long-term encumbrance (ie, for more than 10 years) was responsible for only about 11% of the total encumbered assets, which is consistent with the above-mentioned share of covered bonds as sources of encumbrance.

While the above data is not a cause for concern in EU banks' funding structures, it confirms the need for banks to rely on secured funding, given the constant difficulties in the unsecured financial market. In addition, the volume of OTC derivatives has further increased in most countries and this implies an additional demand for collateral, in particular for safe assets.

At the time of writing (a few years after the global financial crisis that started in 2007), the consequences of the changes in funding sources across Europe remained the same and the structural recourse for a greater use of collateral for secured funding had stabilised, even increasing the weighted average AE ratio, as described above. It follows that asset encumbrance has become a key element in the ALM of a bank, requiring new efforts to further improve institutions' management of liquidity and funding risks where encumbrance is involved.

CONNECTIONS WITH LIQUIDITY INDICATORS

Encumbrance is a key element in measurement and management of liquidity risk.

The “principles for sound liquidity risk management and supervision” defined by the Basel Committee on Banking Supervision (BCBS) in September 2008 state that “a bank should actively manage its collateral positions, differentiating between encumbered and unencumbered assets” (Basel Committee on Banking Supervision 2008, Principle 9).

The importance of BCBS Principle 9 is widely reflected in the international framework for liquidity risk measurement, which the BCBS promoted by developing two well-known regulatory indicators for liquidity (Basel Committee on Banking Supervision 2010) that were then implemented by Regulation (EU) 575/2013 and subsequent implementing documents: the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR).

In both indicators, the measurement of pledged assets relative to the amount of unencumbered bonds and other assets available to be pledged is a key measure for

- verifying the short-term resiliency of an institution’s liquidity risk profile by ensuring it has sufficient high-quality liquid resources to survive in an acute stress scenario lasting one month (LCR) and
- measuring the amount of longer-term, stable sources of funding it can employ relative to the liquidity profiles of the assets funded (NSFR).

In both cases, as we shall see immediately below, the projection over the time frame of encumbered assets and their related “matching liabilities” should be carefully evaluated and managed in order to respect the minimum liquidity requirements set by regulatory authorities.

Liquidity coverage ratio

This indicator aims to ensure that a bank maintains an adequate level of unencumbered HQLA that can be converted into cash to offset its liquidity needs for a 30-day time horizon under a significantly severe liquidity stress scenario. Towards this end, Regulation (EU) 2015/61 requires that the ratio of the HQLA to liquidity needs within the 30-day period must be at least 100%.⁵

The HQLA represent the numerator (ie, the liquidity buffer) of the LCR and should comply with the eligibility criteria for their classification as level 1 or level 2 assets in accordance with the regulatory definition (European Commission 2013, 2015a). At the same time, the HQLA also have to satisfy strict operational requirements and be unencumbered (ie, not subject to restrictions or guarantees) in order to be an available source of liquidity over a survival period of 30 days.

The denominator of the LCR is the total net cash outflow expected under the specified stressed scenario, as prescribed by Regulation (EU) 2015/61. Total expected cash outflows involve different inputs, including those associated with secured funding/lending maturities whose treatment depends on their underlying collaterals (ie, the corresponding encumbered assets).

Given these rules, it is clear how secured funding transactions simultaneously affect both the numerator and denominator of the LCR, making collateral management crucial for maintaining the ratio.

To better explain the mechanism, we should consider the fact that repo maturities are treated within the 30-day LCR window, depending on the quality of their underlying asset, since the type of collateral normally affects the bank's ability to roll over the repo.

In this regard, Regulation (EU) 2015/61 specifies that, if, for example, a repo is backed by level 1 assets, then there are no outflows associated with this transaction. Conversely, if a repo is backed by non-HQLA assets, there is a complete loss of funding, with a resultant outflow given in the denominator of the LCR, which accordingly reduces the liquidity buffer available. Other different run-off rates are set by the regulation for bonds that are considered sufficiently liquid and marketable to be included in the HQLA, but less liquid than level 1 assets: for instance, level 2A assets are subject to a 15% haircut before inclusion in the HQLA pool, while level 2B assets receive haircuts of 25% or 50% depending on type, to reflect both their lower liquidity and higher price volatility.

The same logic applies in the case of reverse repos or secured lending agreements, whose treatment considers that a bank normally decides to roll over, totally or partially, its maturing secured loan transactions, depending on the type of the underlying assets.

It follows that a bank should have the ability to project its encumbered assets, their matching liabilities and the collaterals received collectively, by analysing the maturities of secured funding/lending on the one hand, and the

underlying asset of that funding/lending on the other. Since Regulation (EU) 2015/61 sets caps in the LCR numerator to ensure asset diversification in banks' HQLA buffers, these caps increase the complexity of forecasting all these movements.

The bank's management of the HQLA is thus a fundamental part of the wider management of its collateral position and this, in turn, is strictly connected to the management of its encumbered and unencumbered assets.

In addition, the LCR also takes into consideration the potential increased liquidity needs related to the above-mentioned "contingent encumbrance", such as the outflows related to the potential for valuation changes on posted collateral securing derivatives transactions. This implies a bank should monitor the embedded trigger events that could give rise to the need to deliver additional collateral, by having a suitable information system able to report whether the bank has sufficient unencumbered assets of the right type and quality for such a contingency.

As we can see, the short-term liquidity coverage requirements confirm the importance of careful management of the asset encumbrance for banks.

Net stable funding ratio

The provisions on structural liquidity envisaged by the international reform (BCBS) undergoing formal implementation in the EU also placed a strong emphasis on the phenomenon of asset encumbrance in its definition of the net stable funding ratio.

The NSFR aims to promote greater use of stable funding so as to avoid medium and long-term operations giving rise to excessive imbalances to be financed in the short term. To this end, the NSFR establishes a minimum "acceptable" amount of funding above one year, in relation to the needs arising from the liquidity characteristics and the residual maturities of assets and off-balance-sheet exposures.

In fixing this minimum "acceptable" amount of stable funding, a key characteristic of assets that the regulation considers is the encumbrance.

In general, the amount of stable funding required by the NSFR depends on the liquidity of the bank's assets: positions that are more readily available to act as sources of extended liquidity require less stable funding (ie, smaller "required stable funding" factors) than those considered less liquid. Namely, the regulation intends that these required stable funding factors (RSFs) are calibrated by approximating the amount of an asset that could (or could not)

be monetised through sale or use as collateral in secured borrowing. When assets are already encumbered, the RSFs are therefore larger and their magnitude depends on the time remaining in the encumbrance period.

More comprehensively, as far as the treatment of encumbrance in the NSFR is concerned, the regulation requires banks first to consider the carrying value of encumbered assets, including their overcollateralisation, which means that if a bank had the need to overcollateralise a transaction (due to, for instance, the application of haircuts or to achieve a desired credit rating on an ABS issue), these excesses should be included as encumbered. The bank should then consider the residual maturity of encumbered assets by evaluating them with the remaining encumbrance periods, which are derived from the maturity of the matched liabilities. This is an important step for measuring the NSFR, and whenever the remainder of the encumbrance period is longer than the maturity of the asset the RSF is based on the former, assuming that the bank will be required to pledge new collateral in order to replace the previous collateral, which has expired in the meantime.

To better explain the cross-analysis between the residual maturity of encumbered assets and the remaining period of their encumbrance, we may imagine a residential mortgage (with credit risk qualified for 35% or lower weight) maturing in 10 years, which is used as an underlying cover pool in two different securitisations: one with a residual maturity of nine months and another with a residual maturity of two years.

- (i) In the former case the mortgage receives an RSF of 65%, because the remaining period of encumbrance is less than one year but longer than six months.
- (ii) In the latter case, the mortgage receives an RSF of 100%, because the remaining period of encumbrance is more than one year. In this case, it is therefore necessary to have more stable funding, since the utilisation of this asset is extended.

In the above example, the RSF would be lower than the percentages indicated above only if

- the residual maturity of the residential mortgage and its remaining period of encumbrance were less than one year, or

- the residential mortgage were unencumbered and with a residual maturity of less than one year.

Bearing in mind that this cross-analysis between assets and matched liabilities should be applied to an entire balance sheet, including all assets and liabilities and their maturities, it is clear how difficult this type of calculation can become. It requires detailed information and a sophisticated database in which the links between encumbered assets and matched liabilities are clearly identified. Where the encumbrance is allocated against a pool of assets, it is also necessary to implement a dedicated process to bring this allocation into line with the regulation.

The difficulty of such measures increases for banking groups made up of different legal entities, due to the need for different calculation processes at both a consolidated level and an individual level. For all these reasons, adequate and innovative engines to help the control and management of this encumbrance phenomenon and its development over time become necessary.

As a matter of fact, a detailed understanding of encumbrance across all products allows the appropriate management of the regulatory requirements and is crucial to meet obligations, while attempting to optimise the economic impact through an optimal asset allocation.

Table 17.2 Changes in LCR and NSFR if asset encumbrance increases following unsecured funding with a maturity of less than three months

Instrument	Maturity	LCR	NSFR*
Repo backed by level 1 assets	3M	No change	No change
Repo backed by non-HQLA	>30 days (3M)	Up	No change
Repo backed by non-HQLA	≤30 days	No change	No change
Covered bond funding backed by non-HQLA	>1Y	Up	Up
Covered bonds	>1Y	Up	Up
Securitisations	>1Y	Up	Up
More cash collateral placed for derivatives	>1Y	Down	Down
More cash collateral placed in clearing system	>1Y	Down	Down

Notes: * In accordance with the legislative package of amendments on NSFR submitted by the European Commission to the Parliament and the Council in November 2016. M, months. Y, years.

All the above clarifies how the appropriate management of the encumbrance has become an important dimension of integrated and efficient ALM.

Table 17.2 shows how LCR and NSFR could change if asset encumbrance increases following the use of the most common secured funding to replace short-term interbank deposits (ie, unsecured funding with a maturity of less than three months).⁶

RISK MANAGEMENT OF ASSET ENCUMBRANCE BY INSTITUTIONS

It is clear from the reasons discussed so far (ie, the potential risks embedded in excessive levels of encumbrance, the interconnections with liquidity management, regulatory reporting and increasing public disclosure) why it has become necessary for banks to put in place adequate risk management

policies to identify, monitor and control the phenomenon of asset encumbrance.

As previously mentioned, supervisory authorities have chosen to develop harmonised guidelines and have recommended that banks improve their internal policies, avoiding the requirement of a specific regulatory limit on asset encumbrance. Logically, this does not exclude the opportunity to fix an internal limit, but how can the correct target level be defined?

In defining its approach to asset encumbrance, each bank should take into account its own specificities, that is, its business model, the jurisdiction in which it operates, the characteristics of the current funding markets and the macroeconomic situation. Moreover, banks should consider possible strategies to address the need for additional encumbrance resulting from relevant stressed situations, such as is necessary when a bank designs its contingency plans.

This last point (incorporating the “contingent encumbrance”) is particularly important and requires accurate information on both the amount of additional encumbrance resulting from stress scenarios and the amount of the available unencumbered (but “encumberable”) assets.

In order to quantify the “encumberable” assets, a bank should assess the eligibility of each asset class to pledge it as collateral with central banks or with the major counterparties in secured funding markets (eg, repo and covered bond markets). This should be analysed by distinguishing availability between the group’s legal entities, by jurisdictions and by currency exposure. A bank should also be aware of the operational and timing requirements for making the asset usable as collateral; for instance, a new securitisation of mortgages may take several months, while the use of ready retained self-securitisations is immediate.

Therefore, the target level of asset encumbrance depends on different factors, which should be personalised according to the specificities of each institution and the macroeconomic scenario.

The stress test outcomes are very important for evaluating these specificities and supporting the definition of strategies and policies. While an institution manages its position under “normal” situations, it should be also prepared to manage additional disbursements under stressed conditions. This is why it is crucial to identify in advance the sources of potential liquidity strain, and to ensure the correct remedial actions in accordance with the bank’s established liquidity risk tolerance. In this regard, a prudent and

forward-looking recourse to asset encumbrance forms the basis for the necessary mitigation actions and effective contingency liquidity plans.

In the following example we present a possible approach to setting early warning thresholds, based on the objective of preserving unencumbered assets for stressed situations, while comparing the level of stable secured funding with relevant market benchmarks. In addition, different specific thresholds could also be calibrated on the basis of the maturity of encumbrance, given the different implications of long- and short-term encumbrance (such as for very short-term transactions, eg, repos).

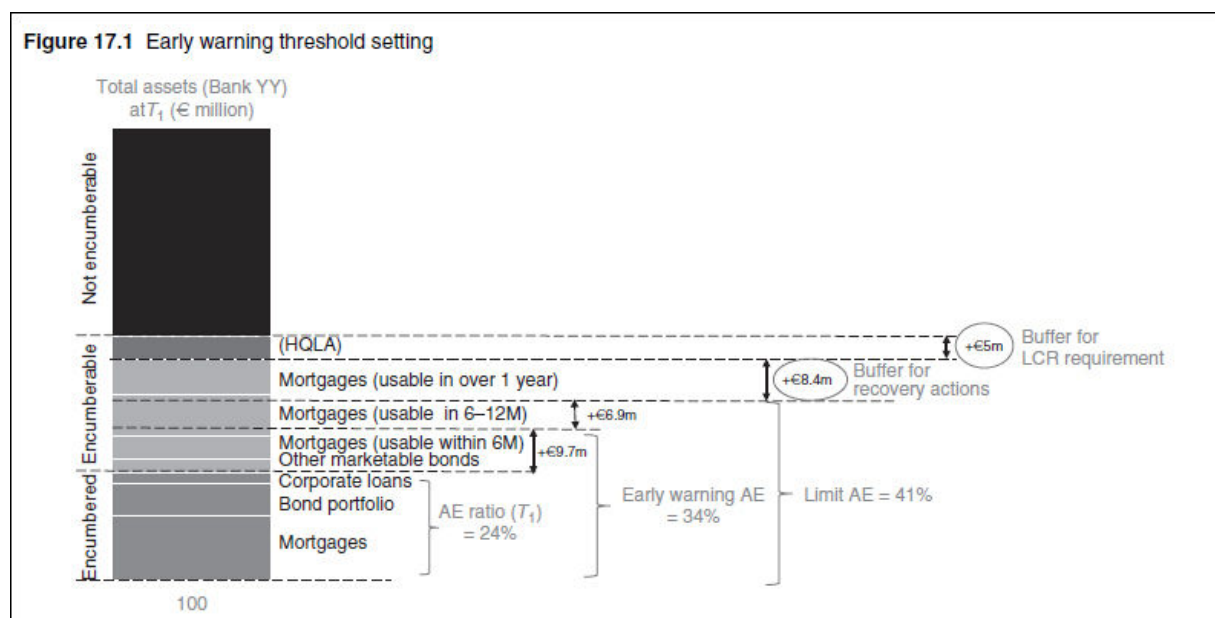
The simplified assumptions on which the example is based are the following.

- At a certain point T_1 , Bank YY shows an AE ratio of 24%, compared with an average AE ratio of 30% for its main peers. The 24% AE ratio of Bank YY is derived by a total of €24 million in encumbered assets relative to the bank's total assets of €100 million. For simplicity, this example does not consider the reuse of collateral received relative to the total collateral received available for encumbrance (both assumed to equal zero).
- The breakdown of the resulting unencumbered assets of Bank YY (ie, €76 million) shows:
 - €46 million of assets that are not “encumberable”;
 - €30 million of “encumberable” assets, of which €8 million are immediately available, including the €5 million needed (on average) to meet the minimum LCR requirement, while the other €22 million are usable over longer time frames, eg, €5 million within six months, €9 million in six to twelve months and another €17 million over one year.⁷
- The additional buffer required by the contingency funding plan (CFP) is €6 million for stressed cash outflows up to one year and €7 million for other outflows over one year; the corresponding increases in the encumbered assets are €6.9 million and €8.4 million, respectively.⁸

Under these assumptions, the internal “early warning threshold” on asset encumbrance of Bank YY could be set to an AE ratio of 34% (see [Figure](#)

17.1), with the following considerations:

- the possible increase in the AE ratio to 34% leaves sufficient available collateral to guarantee both that the LCR minimum requirement is met and an adequate cushion of unencumbered assets that can be sold or pledged should the CFP be activated;



- such a cushion of unencumbered assets, which should be reserved as insurance in addition to the minimum requirements of the LCR, is quantified according to the outcome of the periodic internal stress exercises;
- this cushion is based on a prudential estimation of the “encumberable” assets relative to the entire amount of unencumbered assets available in the balance sheet, also taking into account the timing requirements necessary for accessing (or preparing) the collateral;
- the possible increase in the AE ratio of Bank YY to 34% keeps it in line with the average AE ratio of the bank’s main peers, after adjusting it by adding a buffer based on the observed variability in the markets.

Alternatively, if we consider in this example an AE ratio of 41% (see “limit EA” in Figure 17.1), this higher level could be considered as a hard limit for Bank YY, ie, as a trigger for other policy measures. This is why this level

assumes the use of an additional €6.9 million in encumbered assets, leaving a limited amount of available collateral (ie, only €8.4 million in this example), which requires a long time before it is available to be used (see the breakdown of “encumberable” assets above). In other words, the 41% level could be considered quite critical, since Bank YY would have less room for manoeuvre in response to a shock, if this level were reached.

In general, the above example demonstrates how the absolute level of the AE ratio is not sufficient to understand the financial position of an institution, because it is very important to have a complete picture of the business model and the context in which the bank operates, taking into account the availability of the additional collateral that is already usable and/or potentially usable in the future.

In any case, the setting of an internal threshold on the AE ratio, when it is part of a wider risk control framework, represents a good instrument with which the relevant corporate bodies can address the bank’s effective recourse to asset encumbrance and to give guidance on its management with a comprehensive strategy.

CONCLUSIONS

We have seen that encumbrance is quite complex and potentially risky, but it also represents an opportunity for banks.

Asset encumbrance in effect represents an important way of reducing the cost of funding of a credit institution on the wholesale markets. At the time of writing, the differential between senior unsecured debt and covered bonds continues to be very interesting, even without considering the peaks reached during past crises. Also, with specific regard to the cost of interbank funding, unsecured transactions remain more costly than repo funding.

Moreover, we must not forget that secured funding has proved to be more resilient during past periods of stress, representing in many cases the sole source of funding available in the financial markets. For that reason, the use of secured funding may also be considered a sort of “credit stabiliser” during periods of crisis, because it can support business continuity and the bank’s role of credit intermediation that they regularly carry out.

Accordingly, in the light of all the pro and cons, asset encumbrance is not necessarily “bad” if it is properly managed.

In this chapter, we saw how the new regulatory framework promoted a greater awareness of this phenomenon and increased market discipline.

Credit institutions, in turn, had to adopt internal risk management policies, defining their approach to asset encumbrance, as well as procedures and controls for adequately measuring the asset encumbrance and the risks potentially arising from it. It is important that these policies, given their implications for the strategic setting of ALM in banks, are approved by the appropriate management bodies.

Ultimately, only adequate IT systems can support the integrated view of all balance-sheet items required for the appropriate management of the level, evolution and types of asset encumbrance (asset side) and their related sources of encumbrance (liability side). An advanced IT system is then the last requirement for a comprehensive governance framework aimed at optimising the funding structure and its costs using asset encumbrance, but this requirement is part of a broader need for an integrated ALM system, and we defer it to other dedicated analyses.

The opinions expressed in this chapter are personal and may not necessarily reflect the position and practices of the Intesa Sanpaolo.

- 1 This rule is different from the general requirement defined by the Commission Delegated Regulation (EU) N. 2015/61 of 10 October 2014 with regard to the liquidity coverage requirement (European Commission 2015a), which states in Article 7 that “Credit institutions shall assume that assets in the pool are encumbered in order of increasing liquidity on the basis of the liquidity classification..., starting with assets ineligible for the liquidity buffer”. Therefore, the supervisory regulation does not require a collateral breakdown on a proportional basis for measuring the liquidity coverage requirement, as alternatively required for the measurement of asset encumbrance, thus allowing the asset with better quality to be considered as unencumbered.
- 2 The level of overcollateralisation depends mainly on three factors: regulatory requirements; rating agencies’ requirements; and institutions’ strategic choices regarding the overcollateralisation buffer they wish to hold.
- 3 The general instructions define in detail the scenarios and specify that “the information reported shall be the institution’s reasonable estimate based on the available information” (see European Commission (2015b, Annex II).
- 4 As defined by the general instructions: “The calculation of a 10% depreciation shall take into account both changes on the asset and liability sides, ie, focus the asset–liability mismatches. For instance, a repo transaction in USD based on USD assets does not cause additional encumbrance, whereas a repo transaction in USD based on a EUR asset causes additional encumbrance” (see European Commission (2015b, Annex II).
- 5 The Commission Delegated Regulation (EU) 2015/61 (European Commission 2015a) defines the short-term liquidity requirements, including the LCR.
- 6 Obviously, the LCR and NSFR changes would be different if the asset encumbrance increases following the use of more secured funding instead of unsecured funding for the same maturities (the NSFR in particular could suffer).

- 7 This hypothesis assumes that there are timing requirements, when a bank activates the necessary steps to make the asset usable as collateral.
- 8 The estimated amount of new encumbered assets considers the effect of the overcollateralisation that is normally required, as previously described.

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Part IV

Balance-Sheet and Capital Management

Capital Management

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Capital management is a core activity of bank management. Its primary objective is to balance the supply of capital with the demand for it. In doing so, the interests and requirements of key stakeholders, most notably equity and debt investors, clients, analysts and the bank's supervisors and management, must be considered. The demand for capital arises as banks' business activities entail risks that need to be adequately covered to ensure that potential losses can be absorbed both in the ordinary course of business and under stress. Ultimately, the level of capital held needs to address all stakeholder requirements while permitting an adequate return on capital.

Capital management needs to operate as an advisor to bank management for day-to-day business decision-making and execution as well as planning and strategy formulation, ensuring that sufficient capital is available at all times, that it is invested wisely and that adequate returns are provided. Furthermore, from an asset and liability management (ALM) perspective, capital and capital instruments are valuable sources of long-term funding.

Multiple definitions of capital are used in bank management, ranging from the bank's internal definition of capital to shareholders' equity in financial accounting and regulatory capital as prescribed by banking regulation. In principle, the aim of such measures is to arrive at a well-defined articulation

of capital describing varying degrees of loss absorbency and hence quality of capital.

Equally, capital demand is measured and articulated in various ways. Economic capital is often used to quantify risks as perceived by bank management, typically through the application of statistical and internal models. Another important measure of capital demand is risk-weighted assets (RWA), the calculation of which is defined in specific banking regulation, related technical standards and supervisory guidance. Contrary to accounting values, economic capital and RWA aim to cover unexpected losses. Depending on the legal framework and supervisory practice, banks may be allowed (following an in-depth audit by supervisory authorities) to include some elements of their internal models in the calculation of RWA; in all other cases simplified standard measurement rules apply. RWA are designed to capture unexpected credit losses from loans or off-balance-sheet commitments, market risk losses emanating from trading positions and their value change in volatile or stressed market conditions, as well as operational risk losses, eg, those stemming from operational failures or fraud and related litigation.

Since the 2007–9 financial crisis, regulators have significantly heightened capital requirements for banks, principally by demanding more and higher quality capital, defining more conservative RWA measurement rules and implementing additional capital buffers. Further complexity was introduced by adding a backstop to the risk-based RWA measure, namely leverage, as a further regulatory constraint on capital. Additional bail-in liabilities were added to prepare banks for potential insolvency or resolution.¹ As banks pursue different business models ranging from ordinary deposit taking and lending to corporate banking, trading and asset management they need to understand the demand on capital that results from such activities. And as heightened regulatory requirements have become the binding constraint for bank management, it is even more critical for the capital management function to understand how the risks entailed in various business activities get measured.

Where businesses cannot provide adequate returns on such capital measures, either because bank-specific or market circumstances prevent it from making adequate returns or because the simplicity and conservatism in RWA (and leverage) measurement combined with heightened capital requirements leads to inflated capital needs not commensurate with

underlying risks and hence product pricing opportunities, business models need to be amended to remain viable and to continue to find investor support.

This chapter provides an understanding of the various regulatory capital definitions and related capital requirements imposed on banks by supervisors and banking regulation. It also describes key considerations regarding the desired mix of capital instruments, the balancing of capital supply and demand and related ALM aspects.

DEFINITION OF CAPITAL

In response to the financial crisis, in 2008 Group of Twenty (G20) leaders agreed to an ambitious and comprehensive strengthening of international bank regulatory standards (Group of Twenty 2008, Paragraph 8ff). Uncertainty around financial institutions and the interconnectedness of financial services proved to be a significant burden for bank customers and companies during the crisis, driving many economies into recession or (at a minimum) amplifying economic cycles. Capital levels at a large number of firms proved inadequate, and bank balance sheets required immediate repair, including state aid and taxpayer bail-outs in quite a number of cases. Against this backdrop, the Basel Committee on Banking Supervision (BCBS) undertook its key Basel III reform (Basel Committee on Banking Supervision 2011a).

In the following section we lay out the resulting regulatory capital definitions of the BCBS capital framework. Banking regulation in Europe (the Capital Requirements Regulation (CRR)) implements the corresponding three layers of capital.

Common Equity Tier 1 capital

Common Equity Tier 1 (CET1) capital consists of the nominal value of shares issued, any share premium, retained earnings, accumulated other comprehensive income (OCI) and other disclosed reserves and minority interest, minus specific regulatory adjustments.

It represents the highest quality of capital, ie, it is readily available to absorb losses, the most subordinated claim in liquidation, perpetual in nature and carries no obligation for distributions to be made.

As components, CET1 capital includes capital instruments (eg, shares) and capital items (eg, retained earnings). In its definition, it includes multiple

references to accounting concepts and values (eg, OCI). As a consequence, CET1 capital can also be derived by starting from shareholders' equity.

Regulatory adjustments include adjustments for several assets recognised on the balance sheet that might lose all, or a significant part of, their value in times of severe stress (or their value might be very difficult to realise at such a time). As such, they are deducted from shareholders' equity to arrive at CET1 capital. These adjustments include the following.

- **Goodwill and other intangible assets:** goodwill arises when the purchase price for a company or business exceeds the net present value of the assets and liabilities purchased. In stressed periods, this value may not be realisable, as the profit outlook and hence discounted cashflow of the purchased business may be impaired at such a time. Similarly, if a bank spends significant amounts of money to develop, eg, business-specific software,² the related intangible put on the balance sheet might become impaired under stress (eg, if the business it supports is shut down as a result of that stress). As a conservative precaution regulators thus require the deduction of such goodwill or intangibles.
- **Deferred tax assets (DTA):** there are two types of DTA. The first arises from temporary differences, ie, where a loss event is subject to earlier recognition in financial accounts compared with tax accounts (eg, when credit losses need to be booked in financial accounts but are not yet recognised for tax purposes). For these, only sizeable DTA above certain thresholds must be deducted from CET1 capital.³ A full deduction from capital is required for all other DTA, most notably DTA on net operating losses (NOL). DTA on NOL arise when a bank incurs a loss in a country where tax rules allow future earnings to be offset against prior period losses in the tax returns for those future periods. In this instance, and provided such future earnings are expected, accounting rules immediately allow the loss to be reduced in the current period by the applicable tax rate because a deferred tax asset exists via the future claim against the tax authorities. However, such DTA rely solely on future earnings and therefore are not considered CET1 capital by regulators.
- **Significant and insignificant investments in financial sector entities (FSEs):** in a systemic crisis, the interconnectedness between FSEs, be

these banks, insurance companies or other financial sector entities, is a particular regulatory concern. Notably, in a systemic crisis, the holdings a bank has in the equity instruments of failing FSEs increase the likelihood of the bank also failing (or becoming likely to fail) at the same time, given the write-downs to which the FSEs' holdings would be subject. Therefore, separate treatment of the respective aggregate investments in FSEs is considered in the regulation. Significant investments (ie, those above 10% of the respective FSE's share capital) are deducted from CET1 capital subject to certain thresholds identical to those for DTA arising from temporary differences.⁴ Insignificant investments in FSEs (below 10% share capital) are first aggregated and then follow a separate threshold test of 10% of a bank's capital. Significant FSE holdings that are below the threshold and hence not subject to deduction must be risk weighted at 250%; non-deducted insignificant investments in FSEs must be risk weighted according to the applicable rules.

- **Net pension assets:** banks that are exposed to pension obligations must consider such obligations prudently in regulatory capital. Defined-benefit pension fund liabilities reduce shareholders' equity and hence CET1 capital. If pension assets exceed the defined-benefit pension liabilities (eg, due to overfunding or the outperformance of the invested assets over the liabilities) the resulting net pension asset is also generally deducted from CET1 capital because transferring such "excess value" from the pension funds back into the bank is typically very difficult or impossible.

Several further deductions and prudential filters exist, notably for holdings in own shares, certain cashflow hedge reserves and securitisation-related gains on sale, gains and losses relating to own credit risk, and additional valuation adjustments as well as adjustments for the so-called expected loss shortfall.

Additional Tier 1 capital

Additional Tier 1 (AT1) capital can be described as a hybrid between equity and subordinated bonds. It combines bond-like features such as a repayment of 100% and the regular payment of a coupon or a predefined dividend with the characteristics of equity. The instruments are loss absorbing, as coupon

payments or dividend payments can be cancelled by the issuer at its sole discretion and the investor has no right to demand payment. Also, AT1 instruments can be written down or converted into equity at certain trigger levels. AT1 capital instruments are perpetual and the investor has no right to call or terminate the instrument. A right to call the instrument may exist after five years, but it may only reside with the issuer; supervisory approval will be required prior to any call being exercised. From an issuer perspective, AT1 capital has several advantages.

- It is more flexible than common equity and it is non-dilutive to existing shareholders.
- If issued in the form of a bond instrument, in some jurisdictions the coupon payments (or parts of them) are tax deductible.
- It can be denominated in any currency and provides a tool to economically manage foreign exchange (FX) volatility in capital ratios. However, depending on the accounting treatment and if recorded as an equity instrument, the foreign currency notional of an AT1 instrument is fixed in the accounts in the home currency at the rate at the issuance date.
- It provides a tool to manage the regulatory binding Tier 1 leverage ratio, particularly for banks with a low risk profile and hence a low average risk weight.

Since AT1 instruments are subordinated to depositors, general creditors and subordinated debt, and because non-payment of coupon does not constitute a default event, AT1 capital forms the second layer of capital.⁵

Tier 2 capital

Tier 2 (T2) capital is the third layer of regulatory capital; it is often referred to as gone-concern capital. T2 instruments are subordinated to claims from depositors, general creditors and other debt holders, and they provide loss absorption in bankruptcy and resolution. As with AT1 instruments, the issuer may have no right to call in the first five years after issuance. Under certain conditions, T2 capital may also include other items, notably the excess of

eligible provisions (particularly credit provisions) over expected losses for banks using internal models for RWA calculation.

After the financial crisis of 2007–9 the G20 agreed to establish a bank resolution regime, as the prevailing insolvency laws tended to be less suitable to unwind failed institutions without the risk of serious disruptions to financial markets and services, notably in the case of large bank failures. One of the consequences was to introduce a further layer of instruments that should increase the loss absorption amount a bank has in the case of resolution, so that a government bail-out becomes much less likely. The key feature of such bail-in instruments is subordination to all other senior creditors, ranging from derivative counterparts to corporate and retail depositors. Together with the regulatory capital components discussed above, this provides for a total loss-absorbing capacity (TLAC), which should prevent the need for state aid should a global systemically important bank fail. These bail-in liabilities are sometimes referred to as Tier 3.

CAPITAL REQUIREMENTS

Capital requirements prescribed in banking regulation are systematically articulated as a percentage of the respective regulatory measure of risk, most notably RWA or leverage exposure⁶

$$\text{capital ratio} = \frac{\text{capital supply measure}}{\text{capital demand measure}}$$

As discussed above, the RWA value is calculated for credit, market and operational risk based on various measurement techniques of different degrees of sophistication. Leverage is a much simpler measure of risk, even compared with the most basic standard measurements of RWA. It is mostly based on nominal exposure and it excludes the benefit of collateral.

Specifically, leverage uses the accounting value of exposure for all assets other than derivatives and securities financing transactions, for which a standard regulatory measurement must be used. Therefore, under leverage rules, €100 million of cash held at a central bank attracts the same amount of capital as the identical amount of money invested in a corporate loan or a high-yield bond.

A deeper analysis of the rules for calculating RWA or leverage exposure is beyond the scope of this chapter. However, it should be noted that the risk

weightings applicable under the standardised approach or internal model approach, and the measurement of leverage exposure as applicable to a bank's business and asset mix, are the principal drivers for the level of capital a bank must hold. As these measurements are continually being revised by regulators (through law) and supervisors (through evolving law interpretation and practice), the capital held by banks must also be continuously adjusted.

Against this backdrop, banking regulation defines minimum capital requirements for all forms of capital (CET1, T1 and total capital). It also prescribes buffers to be held in excess of the minimum requirements, and it provides supervisors with tools to request banks to hold even more capital in order to operate as a going concern.

Furthermore, in late 2015 the Financial Stability Board (FSB) issued minimum TLAC requirements for global systemically important banks (G-SIBs), which must be met from January 1, 2019 onwards. For smaller banks, similar but mostly less stringent requirements for minimum levels of bail-in liabilities have been formulated in national laws.

Solvency requirements

Minimum capital requirements and capital buffers

Solvency requirements are defined as CET1, T1 and total capital requirements relative to RWA.

The legal minimum CET1 capital requirement is 4.5% of RWA. The legal minima for Tier 1 capital and total capital ratios are 6% and 8%, respectively. This implies that up to 1.5% of RWA of AT1 capital can be recognised in Tier 1 capital, and up to 2% of RWA of Tier 2 capital can be recognised in total capital in order to satisfy the corresponding minimum requirements. Together, these requirements are the minimum own funds requirements.

In addition, the BCBS framework and implementing regulation entails three types of capital buffers to be held by banks in the form of CET1 capital, the so-called combined buffer requirements.

First, outside periods of stress, banks must maintain a capital conservation buffer of 2.5%, which acts as a buffer for losses that may be incurred, eg, in times of a bank-specific crisis.

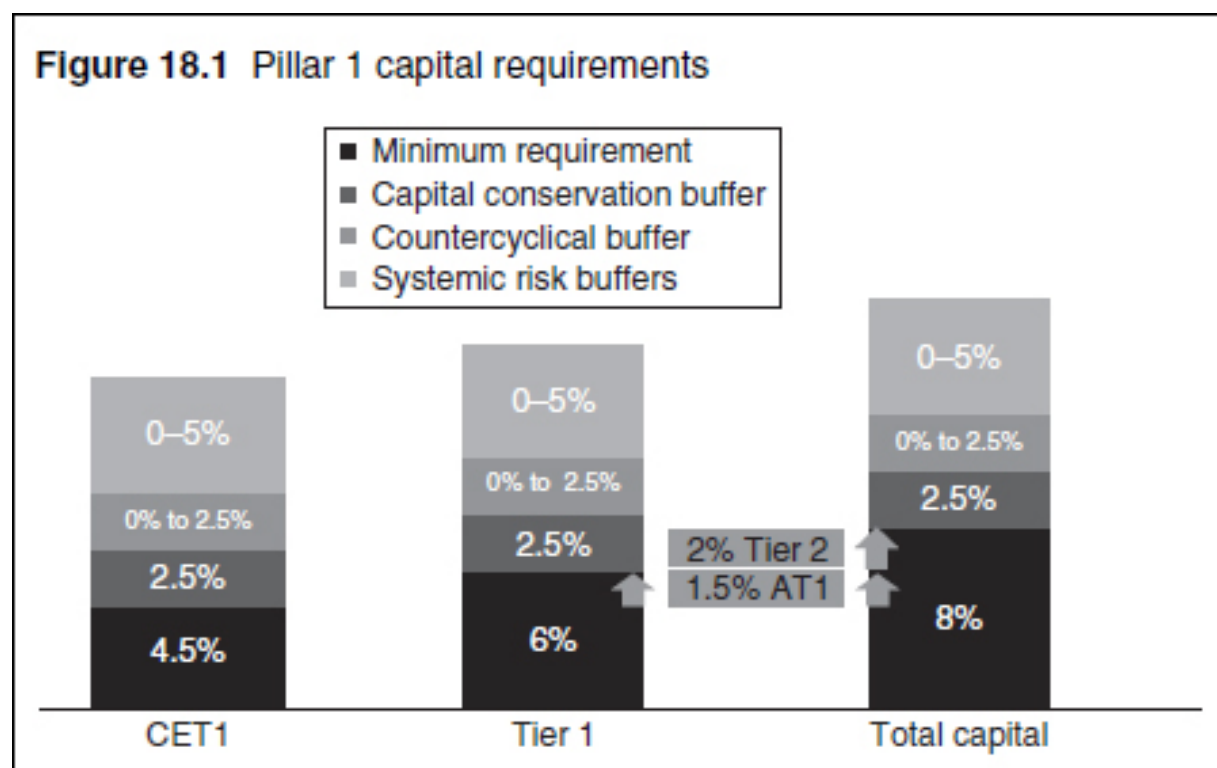
Second, a countercyclical buffer of between 0 and 2.5% has been introduced. This buffer aims to protect the banking system from

macroeconomic “overheating” if credit expansion is accelerating at unsustainable levels, increasing the risk of asset bubbles and subsequent future losses. Therefore, designated public authorities are asked to assess system-wide risk and the state of the economic cycle and to set adequate buffer levels applicable for assets extended to borrowers in their respective economies. This not only ensures that a buffer is built but also incentivises banks to slow down their credit growth. Each bank calculates its own buffer requirement based on the RWA related to its private sector credit exposures. The total countercyclical buffer requirement is the RWA-weighted sum of local countercyclical buffer requirements for each country in which the bank is exposed to credit risk.

Third, systemic risk buffers apply. At the height of the 2007–9 financial crisis it became apparent that systemically important financial institutions (SIFIs) could not be allowed to fail in a crisis given the externalities this would entail. Consequently, taxpayer money was required at the time in order to stabilise SIFIs likely to fail and to prevent an accelerated negative spiral. An institution-specific buffer requirement was thus agreed for systemically important banks. In a first step, the FSB developed an indicator-based measurement approach to identify G-SIBs (Basel Committee on Banking Supervision 2011b). The selected indicators reflect the size of the banks, their interconnectedness, the lack of readily available substitutes or financial institution infrastructure for the services they provide, their global (cross-jurisdictional) activity and their complexity (Basel Committee on Banking Supervision 2013a). Based on this assessment the FSB publishes a list of G-SIBs annually. In its 2017 list the FSB identified 30 institutions as global systemically important banks; it assigned them to four buckets (numbered 1–4) with corresponding G-SIB buffer requirements ranging from 1.0% to 2.5%. A fifth G-SIB bucket, which would attract 3.5% buffer requirements, remained empty. In parallel to the G-SIB assessment conducted by the FSB, national competent authorities are required to identify domestic systemically important banks (D-SIBs) in the context of their respective countries and their economies. For these, a corresponding D-SIB buffer is set. For banks that are a G-SIB and a D-SIB, the higher of the two requirements applies. Finally, competent authorities may set a general systemic risk buffer (SRB) for all systemic or macroprudential risks of a non-cyclical nature that are deemed not to be covered by any other provision. In the European Union

(EU), it is at the member state's discretion to set general systemic risk buffers.

When defining the bank's business strategy, including the geographical footprint and balance sheet structure, bank management must recognise the influence the bank's business strategy has on the outcome of the supervisors' assessment of the systemic importance of the institution, and hence its capital requirements. And it must justify these potentially higher requirements to shareholders by corresponding expectations of enhanced profitability.



As per the BCBS framework (and its implementation in EU legislation) the combined buffer requirements are subject to transitional arrangements and thus due to be phased-in between January 1, 2016 and year-end 2018, becoming fully effective on January 1, 2019. The applicable amount of the capital conservation and the SIFI buffer rises by 25% of its final value every year during the transition period. For the countercyclical buffer, the maximum buffer applied rises by 25% of the maximum 2019 final value of 2.5% every year. [Figure 18.1](#) summarises the Pillar 1 capital requirements that banks are required to comply with.

Additional capital requirements under Pillar 2

Three pillars exist under the BCBS framework. The first addresses minimum capital requirements and buffers as outlined above. The second deals with the need for banks to conduct their own internal capital adequacy assessment process (ICAAP) and with the obligation for bank supervisors to review a bank's ICAAP and to identify any potential need to add requirements over and above those specified in Pillar 1 to ensure that all risks are adequately covered. The third pillar outlines disclosure requirements for banks.⁷

One of the key outcomes of the supervisory review and evaluation process (SREP) is the determination of additional Pillar 2 capital requirements (P2R) to be held by banks in excess of the minimum own funds requirements outlined above. The European Banking Authority (EBA) has issued corresponding guidelines that provide a framework for how supervisors in Europe should determine additional Pillar 2 requirements and how these should be integrated into the Pillar 1 capital requirements discussed above (European Banking Authority 2014). Under this framework supervisors consider any weaknesses in business models, internal risk governance and control arrangements, risks to capital and risks to liquidity. Regarding the integration of Pillar 2 into Pillar 1 capital requirements, the EBA guidelines prescribe that the combined buffer requirements sit on top of the Pillar 2 capital requirements, which sit on top of the minimum own funds requirements.

Based on the EBA guidelines, the European Central Bank (ECB) first implemented its SREP in 2015, followed by a second SREP cycle in 2016, the core features of which are described in the SSM SREP methodology booklet (European Central Bank 2016).

With the 2016 process, the ECB articulated a further demand for banks to hold capital, through the introduction of Pillar 2 capital guidance (P2G), which banks are also expected to comply with.

The average P2R in 2016 was 2.0% and the average P2G was 2.1% of RWA: both were to be covered by CET1 capital. Pillar 2 capital guidance sits on top of the combined buffer requirements.

The application of Pillar 2 requirements differs in individual countries, with the UK following materially the same approach as the ECB, but with no comparable articulation of Pillar 2 capital requirements, for example, in the US.

Constraints on capital distributions

Banks distribute earnings, and hence capital, in a number of ways, most notably through share buy-backs or the payment of a dividend on common shares, a coupon on AT1 instruments or a variable compensation (bonus) to employees. Banking regulation places multiple constraints on such distributions in order to prevent them being made when capital levels are inadequate. By doing so, capital is preserved for use in potential resolution or insolvency proceedings until the overall amount of capital has risen to adequate levels.

In the EU, the mechanism to achieve this is enshrined in Article 142 CRD. In essence, banks are constrained in making distributions from capital when they fail to meet the combined buffer requirement. In such a case the calculation of the maximum distributable amount (MDA) is required. Where the shortfall in meeting the combined buffer requirement is less than 25% of that buffer, distributions are limited to a maximum of 60% of distributable post-tax income, and in any case are subject to supervisory approval. The bigger the gap towards the combined buffer requirement, the more the MDA is reduced, turning into a full prohibition of distributions when less than 25% of the combined buffer requirement is met.

Leverage ratio requirements

Following the financial crisis of 2007–9, excessive leverage was identified as one of the factors that undermined financial stability and exacerbated the downward spiral in asset prices (and credit contraction) when rapid deleveraging occurred in already stressed markets. In the light of this, the BCBS supplemented the risk-based capital requirements of the global capital framework with a 3% minimum leverage ratio constraint.

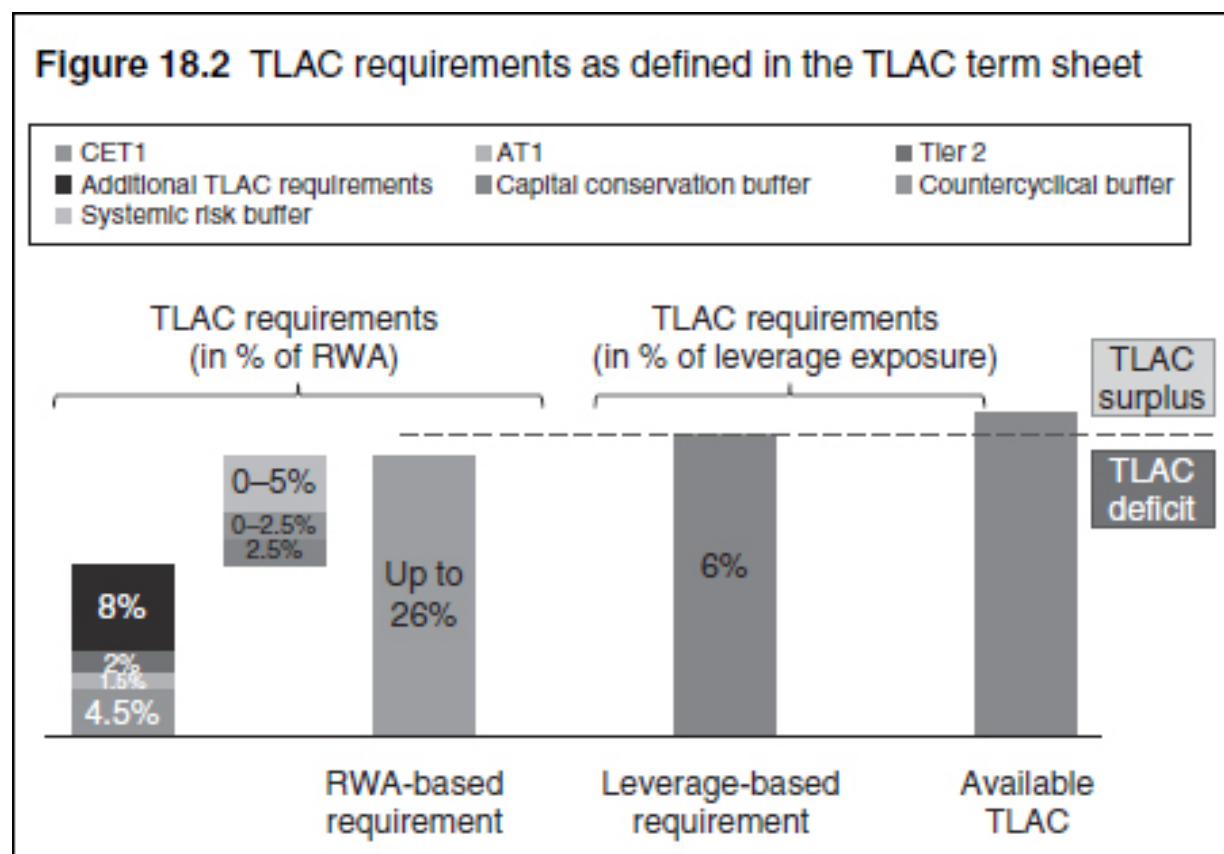
The application of leverage as a binding regulatory constraint differs across jurisdictions. In the US, previous leverage requirements were already well above the minimum 3% Tier 1 leverage ratio requirement articulated in Basel III, although on a somewhat different measurement basis. Meanwhile, at the time of writing, implementation of the BCBS minimum in most of Europe was expected to start from January 1, 2019.

As the ratio of RWA to leverage exposure (RWA density) differs depending on the business model and risk profile of a financial institution, the defined solvency and leverage requirements pose an optimisation challenge for capital management. In an optimised bank, the RWA density should equal the ratio of Tier 1 capital requirements (management targets)

under the solvency regulation and Tier 1 leverage requirements (management targets). As an example, a bank targeting 4% Tier 1 leverage and a 12% Tier 1 solvency ratio would aim at operating at approximately 33% RWA density.

TLAC requirements

As discussed above, effective recovery and resolution have been a political priority since the early days of the 2007–9 financial crisis. While the technical details of optimally setting up resolution regimes in financial institutions is beyond the scope of this chapter, we may at least note the additional requirements imposed on banks through the introduction of TLAC.



The FSB's TLAC term sheet (Financial Stability Board 2015) lays out the minimum requirements for TLAC to be met by G-SIBs (starting January 1, 2019) as the greater of 16% of RWA plus any applicable regulatory capital buffers or 6% of leverage exposure. These requirements should then rise to 18% and 6.75%, respectively, on January 1, 2022.

As for solvency requirements, competent authorities are required to assess the need for additional firm-specific TLAC requirements. In this assessment, due consideration must be given to the level of TLAC required for an orderly resolution to be implemented without recourse to taxpayer money. Also, the TLAC level should be set sufficiently high to ensure critical functions can continue to operate and provide services such that the impact on financial stability is minimised (European Central Bank 2016).

In Europe, requirements for loss absorbing capital (European Union 2014) were introduced for all banks in the Bank Recovery and Resolution Directive (BRRD), which has been translated into national law in member states. The BRRD aims to align the minimum requirement for eligible liabilities (MREL) with the TLAC term sheet in such a way that TLAC as defined for G-SIBs is fully included in MREL. That said, MREL allows the inclusion of some additional liabilities that do not qualify as TLAC.

At the time of writing, no final MREL requirements had been articulated to banks by the Single Resolution Board (SRB), the competent authority in Europe, but the finalisation of such requirements is expected in early 2018.

[Figure 18.2](#) illustrates the 2019 TLAC requirements as per FSB term sheet.

Global versus local requirements

The BCBS capital framework and FSB TLAC term sheet set minimum standards for large internationally active banks. The implementation into national law requires a transposition of the global framework, which at times requires adaptation to national specifics. To monitor the consistency of implementation across BCBS member states, the BCBS regularly reviews the status and quality of implementation of the Basel III rules and publicly reports progress in the adoption of the international standards.⁸ Despite the harmonised starting point and a high degree of consistency being achieved, some differences in national implementation remain, and may well continue in future.

Such differences may be justified and acceptable by definition, provided they do not put to question the soundness of banks and the material equivalence of outcomes.

First, differences in Pillar 1 capital measures and requirements exist as the result of national options and discretions explicitly entailed in the BCBS

framework. For example, in the case of retail and private sector entities, the generally used “90 days past due” default definition can be extended by supervisory authorities to up to 180 days if considered appropriate given local conditions. This discretion is used in the US and EU.⁹

Second, capital requirements as implemented at a national level may exceed the BCBS rules, which only set internationally agreed minimum standards. A good example of such “gold plating” is the leverage ratio requirement in the US that subjects US top-tier bank holding companies (BHCs) with more than US\$ 700 billion in total consolidated assets to a 5% minimum leverage ratio, an additional 2% above the minimum BCBS requirement of 3% (US Department of Treasury 2014). Failing to meet this enhanced requirement would trigger restrictions on capital distributions and discretionary bonus payments. Subsidiaries of such BHCs in the form of insured depository institutions are additionally required to operate at or above a minimum of 6% leverage to be considered well capitalised (US Department of Treasury 2014). Whether local standards are indeed set higher than the BCBS minimum standard becomes less clear when higher ratio requirements come in combination with a revised measurement basis. This can be seen from looking at the leverage requirement in the UK that allows banks to exclude central bank cash from leverage and simultaneously adds 25 basis points to the Basel baseline 3% leverage ratio requirement (Bank of England 2017).¹⁰ Evidently, for banks with significant liquidity buffers and related cash holdings, this leads to a relaxation of the requirements compared with the original Basel standard; and, in any case, reported figures become more difficult to compare across jurisdictions.

Another notable difference relates to the implementation of Pillar 2. As outlined above, European law and supervisory guidance and practice has given much weight to additional Pillar 2 capital requirements and guidance. In the US, such a cohesive and transparent framework leading to publicly disclosed additional requirements in similar breadth and size is not observed. Conversely, the US puts much more emphasis on regulatory stress testing, notably CCAR, to assess banks’ capital management and capital adequacy, deriving explicit conclusions on the banks’ ability to make distributions from capital. The regulatory stress test in Europe, which is designed by the EBA and conducted in conjunction with national competent authorities, most notably the ECB, leads to the transparent disclosure of

banks' risk profiles and their capital position under stress. However, it does not lead to a similarly direct conclusion on capital distributions.

Ultimately, national rules consider the specific banking system, business models and economic circumstances in the country of application. For example, credit losses recognised by US banks during the 2007–9 financial crisis (and also over much longer cycles) significantly exceed those experienced by most banks in continental European countries, such as those in Germany, France, the Netherlands or Switzerland. Consequently, RWA per loan unit, and hence RWA density, in the US must be higher than in Europe, which in turn means leverage for the same risk should be lower in the US. In other words, it is the difference in lending practice and business models that allows US banks to cope with higher leverage requirements more easily than European institutions,¹¹ and it is therefore adequate for US firms to be held to a higher leverage ratio standard.

Lastly, national implementations differ in that, in some countries (eg, the US), the BCBS framework is only applied to the narrow group of large internationally active banks for whom the rules were explicitly designed, while in others (eg, EU countries) the rules are applied to a much larger group of banks. An application to a wide range of banks, large or small, still allows higher capital requirements for large banks to be put in place where appropriate, eg, through the SIFI buffer and bank-specific Pillar 2 capital requirements and guidance, while maintaining more of a level playing field for banking services offered within a given jurisdiction. However, in such a framework, proportionality is typically applied to at least the regulatory reporting obligations and the intensity of supervisory oversight and audit activity.

With the near completion of the long-debated revisions to the Basel III rules (often also referred to as Basel IV given the wide-ranging amendments considered) at the time of writing, which will affect all RWA calculations from credit to market and operational risk, the path to full consistency in regulatory capital requirements remains difficult. For capital management this means banks would need to remain alert and flexible in planning capital requirements and capital structures for the foreseeable future. For ALM, a correspondingly holistic scenario- or sensitivity-driven approach must be taken.

CHARACTERISTICS OF CAPITAL INSTRUMENTS

Table 18.1 summarises the key characteristics of the various capital and bail-in instruments discussed above (Basel Committee on Banking Supervision 2011).

As a general condition none of the instruments may have, contractually or by creating an expectation:

- an incentive to redeem the instrument early (eg, a credit-sensitive element to distributions or step-up provisions),
- a guarantee by the issuer or a related entity to enhance the seniority of the instrument,

Table 18.1 Characteristics of capital instruments

	Common shares as part of CET 1 capital	Additional Tier 1 instruments	Tier 2 instruments	Additional bail-in liabilities
Subordination	Most subordinated	To depositors, general creditors, subordinated debt	Subordinated to depositors and general creditors	Unsecured senior debt subordinated to excluded liabilities
Maturity	Perpetual, only repaid in liquidation	Perpetual with issuer call option after five years	Minimum original maturity of five years	Minimum original maturity of one year
Repayment	Upon liquidation or through discretionary repurchase as legally allowed	Upon supervisory approval	Upon supervisory approval	Restricted if TLAC requirement is breached
Distributions	Fully discretionary out of distributable items	Coupon/dividend cancellable at full discretion; out of distributable items	Coupon not related to creditworthiness of the issuer	Coupon
Balance-sheet recognition	Equity (on the balance sheet)	Equity or debt	Debt	Debt
Default trigger	No	No	Yes	Yes
Loss absorption	First and proportionately greatest share	Conversion into common shares or write-down mechanism at prespecified CET 1 trigger point	Only in liquidation	Only in liquidation
Recognition as capital	Notional plus share premium	Paid-in amount	Amortised straight line in last five years of residual maturity	No; recognition as TLAC debt if residual maturity above one year
Issuance through a special purpose vehicle	No	Yes, only if proceeds are immediately available to an operating entity	Yes, only if proceeds are immediately available to an operating entity	Yes

- an investor closely related to the issuer, or
- a dividend pusher in the same instrument category (although payments on subordinated instruments may be restricted).

The BCBS criteria rule out shifts in the hierarchy of instruments or any structured features that increase the rights of the instrument holder at any point in the instrument's life cycle. As a result, the strict rules of subordination create a clear equity/liability hierarchy: the greater the coupon requirements, the greater the loss absorption capacity of the instrument. Consequently, the spread costs related to subordinated instruments become a significant factor when assessing the optimal funding mix of a bank.

MANAGING CAPITAL SUPPLY AND DEMAND

It is evident from the above that capital supply and demand must be balanced to ensure compliance with all applicable requirements and stakeholder expectations is achieved at the lowest possible cost, while business activity is supported through capital in order to deliver adequate returns.

Capital management will ensure that the required capital stack is put in place such that CET1 capital requirements are covered with CET1 capital, and AT1 capital is added to address the 1.5% higher T1 capital requirement (and more when there are AT1 capital requirements under Pillar 2, as in the UK). T2 capital comes on top in order to cover the additional 2.0% requirement for that category of capital followed by TLAC (MREL) liabilities to address corresponding needs. The composition of capital therefore principally reflects regulatory requirements and the relative cost of the instrument categories. Only if the respective instrument cannot be placed in the market (or if the price of lower quality capital in a disrupted market is above what must be paid for higher quality instruments) may higher quality capital be used to address lower capital quality requirements. A small buffer may also be added to each category of capital by banks in order to assure investors that, eg, a sudden increase in RWA may not put T2 capital below the 2% requirement, as this could trigger an earlier (total capital) MDA breach and hence constraints on AT1 coupon payments. Further complexities arise when the capital structure required for solvency reasons does not ensure that the T1 leverage requirement is met at the same time, notably where the RWA density is below the leverage-to-solvency ratio requirements as outlined above. In such a case, more T1 capital may be added to address the T1 leverage requirements, and hence less than 2.0% T2 capital may suffice for solvency purposes.¹²

In order to be able to manage a rational and optimised capital supply, capital management therefore often sets RWA and leverage limits for the bank's individual businesses such that a balanced capital structure in line with business needs and stakeholder requirements can be delivered continuously. The steering of capital demand, and hence business volumes, is critical for the capital management function. As an example, a bank may plan to expand in one area, eg, consumer lending, and shrink activities in another, eg, secured financing in the form of repo and reverse repo activities. In this case, the RWA density will increase, and adjustments to the capital composition (capital stack) may be needed. As such adjustments are not always deliverable in the short term (eg, due to issuance windows for capital instruments being constrained by compliance restrictions around earnings release dates, general market availability or market timing considerations), capital management must influence business demands for capital. Ultimately, a combination of holding some buffers over regulatory or other stakeholder requirements for capital and exercising control over business growth (or shrinkage) and related capital demand is needed.

To manage a bank's capital ratios, due consideration must also be given to the risk of changes in reported ratios as a result of movements in FX rates. Take a bank that holds all its CET1 capital in euro but 20% of its RWA in US dollars. In the case of a strengthening of the US dollar against the euro this bank will see a reduction in its reported CET1 capital ratio, as the US dollar component of its RWA will rise in euro terms. Therefore, capital management continuously monitors the currency composition of RWA (and leverage) and compares it against the currency composition of capital, with the aim to keeping ratio sensitivities within an agreed range of variability under a predefined rate movement scenario. To manage the currency composition of capital, capital injections in foreign branches may be adjusted up or down, or FX forwards may be entered into to align foreign-currency-denominated capital with RWA, leverage or TLAC requirements. Given the differences between the capital demand measures, a fully balanced capital demand and supply profile may not be achievable for all ratios at the same time. In this case, adjustments to the bank's business or geographical mix may be required, or relative preferences for one or the other ratio may need to be defined.

The FX example illustrates another important aspect of capital management: the interplay of capital requirements at a consolidated banking

group level versus those at a local branch or subsidiary level. At the local level entities usually need to comply with a set of regulations that is (even if materially aligned to BCBS recommendations) tailored to the respective country's specificities. Also, a different type of capital constraint (solvency, leverage or large lending requirement, etc) may be binding for the various individual entities at a local level. In this case, local capital requirements often differ from those required for the same activity from a consolidated group perspective, based on rules applicable to the group. This leads to another optimisation problem of balancing demand and supply, namely, how to distribute capital efficiently between entities, recognising that constraints at a local level differ from those at a group level.

INTEGRATING CAPITAL MANAGEMENT INTO ASSET AND LIABILITY MANAGEMENT

A bank's service of transforming short-term liabilities into long-term assets needs to be supplemented by

- longer term liabilities for funding,
- instruments with higher degrees of subordination for risk absorption and resolution,
- an adequate level of liquid assets for sudden spikes in short-term payment needs.

Only then a can bank convincingly demonstrate its safety and soundness to depositors and other counterparties with future claims on the bank. In this context, capital is one of the tools providing long-term funding and going-and-gone-concern loss-absorption capacity, making a bank attractive for depositors and others to deal with.

When making investment (placement) decisions, liability holders request interest payments considering, apart from term and liquidity premiums, an adequate compensation for their position in the creditor hierarchy and the probability of a bank becoming insolvent. It should be noted, however, that a bank's insolvency risk depends not only on its capital position in relation to the riskiness and liquidity of its assets and business model more broadly but also on the riskiness of the funding mix. The latter arises where funding mismatches exist, while assets cannot be liquidated in time to address

liability owners' requests for repayment and new funds cannot be sourced. In order for depositors and senior debt holders to bear the risk from a bank's business model that combines risky illiquid long-term assets and highly liquid short-term liabilities, an extra spread cost must be paid by the bank. To reduce this cost, the bank needs an adequate level of capital and subordinated debt in order to be able to attract cheap funding from depositors (and senior debt holders) who are willing to leave their money with the bank for a longer period of time. In aggregate, these relationships describe the optimisation problem that not only touches on capital management but is also at the heart of many other ALM-related questions addressed in this handbook.

Through the issuance of capital instruments, cash is generated that needs to be invested. Given the perpetual or very long-term nature of these instruments, capital can be used to fund long-term assets. Still, to segregate the management of the interest rate risk attached to long-term placements of cash, capital management typically tends to invest the proceeds from the issuance of capital instruments at short-term floating rates with an internal cash management pool. Long-term funding is then provided through this pool to the various businesses, whereby the pool manages the resulting interest rate risk mismatch. In such case, a bank's ALM function may manage the resulting rate profile by entering into swaps to stabilise returns for the bank. This process is often referred to as capital bucketing.

Another important link between capital management and ALM stems from the ALM goal to manage net interest income and economic value of equity. To do this, ALM generally makes use of interest rate derivatives. From this, the intricacies of accounting (in particular, the ineligibility of hedge accounting programmes) might lead to asymmetries. These asymmetries can result in CET1 capital sensitivity caused by temporary valuation mismatches. If left unmanaged, reported solvency and leverage ratios might not only be volatile but also deviate from economic outcomes over prolonged periods.

CONCLUSION

Capital management must balance capital demand and supply. With both sides of this equation being highly regulated, a deep understanding of regulation, notably requirements for capital and capital components, is critical. CET1 capital, as the highest quality capital, is readily available to absorb losses, the most subordinated claim in liquidation, perpetual in nature and carries no obligation for any distributions to be made. With other forms

of capital equally being perpetual or at least long term, capital is a very important source of long-term funding. As such, it is key to ALM. Optimising the capital structure and the distribution of capital is not only critical for capital management but also an important aspect of ALM. Hence, the review of a bank's business model and strategy from a capital management perspective needs to go hand in hand with its ALM optimisation. In the end, business activities must be managed such that adequate returns on capital can be delivered.

As Basel IV materially redefines all the existing regulatory measures of capital demand, a new wave of business model adjustments will be triggered, depending on the significance of the changes ahead. In any case, capital management always needs to be prepared, stay alert and be ready to act swiftly, as and when required.

The opinions expressed in this chapter are those of the author and do not necessarily reflect the Deutsche Bank position or practice.

- 1 Total loss-absorbing capacity (TLAC) introduces the most prominent form of such additional bail-in liability requirements; see Financial Stability Board (2015).
- 2 Accounting practices regarding the activation of software-related cost differ across accounting standards. While International Financial Reporting Standards filers usually recognise activated software cost explicitly as intangible assets (which then require deduction), US Generally Accepted Accounting Practices filers are found to activate it under property, plant and equipment or together with purchased hardware (which results in no corresponding deduction).
- 3 BCBS rules require deduction of DTA arising from temporary differences from CET1 capital if they exceed 10% of the capital amount before the deduction of such DTA and significant investments in FSEs. The DTA that remain non-deducted are risk weighted with 250% risk weight. In addition, the total amount of significant investments in FSEs and DTA arising from temporary differences that are not deducted needs to be less than 15% of the capital amount referred to above. Any excess would equally require deduction.
- 4 See Footnote 3. Note that Additional Tier 1 and Tier 2 instruments are subject to corresponding deduction rules.
- 5 Prior to the implementation of Basel III, instruments with less stringent requirements (often referred to as legacy or hybrid T1 instruments) were recognised as T1 capital. Such instruments may be recognised in T1 capital up to a certain amount until 2022.
- 6 The amount of capital required to support individual large lending exposures is based not on RWA or leverage exposure but on a derivation thereof.
- 7 The goal of the third pillar is to enforce market discipline on financial institutions. The prescribed disclosure requirements should ensure full transparency on a bank's capital requirements, demand and supply to the market on an ongoing basis, promoting comparability of banks' risk profiles within and across jurisdictions. It reduces information asymmetries and contributes to the safety and soundness of the financial system.

- 8 Under its Regulatory Consistency Assessment Programme (RCAP) the BCBS regularly evaluates the consistency and completeness of the adopted standards, including the significance of any deviations from the Basel III regulatory framework. These consistency assessments are carried out on a jurisdictional basis (see, for example, Basel Committee on Banking Supervision (2017a,b) for the most recent country summary report at the time of writing) and thematic basis (see Basel Committee on Banking Supervision (2013b) for credit risk, Basel Committee on Banking Supervision (2013a) for market risk and Basel Committee on Banking Supervision (2015) for counterparty credit risk).
- 9 For an overview of national discretions, see Basel Committee on Banking Supervision (2014).
- 10 Specifically, the Financial Policy Committee of the Bank of England recommends that leverage excludes claims on central banks, where they are matched by deposits denominated in the same currency and of identical or longer maturity.
- 11 Next to differences in lending practices, insolvency laws, general borrower culture and the relative size of the higher risk consumer credit lending, US banks sell most qualifying low-risk mortgages to government agencies (the Federal Home Loan Mortgage Corporation, Federal National Mortgage Association and Government National Mortgage Association), while European banks tend to hold significant volumes of low risk mortgages on their balance sheet.
- 12 Further complexity arises, eg, in the US, where for all banks a historical general leverage ratio requirement is set based on a simplified measure of leverage (excluding off-balance-sheet exposures), and for many banks a different supplementary leverage ratio requirement using the BCBS measure of leverage (including off-balance-sheet exposures) is applicable in parallel, requiring optimisation across the two measures.

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A Global Perspective on Stress Testing

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Since the 1990s, bank risk managers and regulators have become increasingly aware of the need to conduct stress tests on banks' balance sheets to assess the resilience of single banks (commonly referred to as "microprudential stress tests"), as well as the financial sector as a whole (widely known as "macroprudential stress tests"). In general, stress testing is a simulation technique to quantify the impact of (mostly) adverse market conditions on a financial portfolio. Likely outcomes are evaluated for historical and/or plausible but severe hypothetical stress scenarios. As such, these are easy to understand and communicate to board members and senior management, stakeholders and regulators.

In this chapter, we provide a short summary of stress testing history, discuss the details of major stress testing frameworks and give guidelines for a stress testing programme setup. For brevity, we restrict the scope of our analysis mainly to the US, eurozone and the UK. In addition, we outline likely future developments in stress testing design and methodology and their applications for internal bank steering.

Stress testing in the banking industry before the 2007–9 financial crisis

Microprudential stress testing

Until the early 1990s, stress tests were applied by single banks to complement their set of risk management approaches. Yet, it was not until 1996 that the application of stress testing to quantify market risk in trading books was a formal requirement: it was first formalised in the Basel Committee on Banking Supervision (BCBS) “Amendment to the Capital Accord to Incorporate Market Risks” (Basel Committee on Banking Supervision 1996). This stated that the application of banks’ own internal risk models required the implementation of bank-wide stress testing programmes for market risk. Stress testing quickly became common practice in bank risk management, and the amendment was further updated in 1998 (Basel Committee on Banking Supervision 1998).

In 1999, the BCBS proposed the adoption of stress testing to overcome major uncertainties in credit risk modelling, such as the estimation of default rates and joint-risk-factor distributions affecting the bank’s credit risk profile (Basel Committee on Banking Supervision 1999). Consequently, the requirement to apply stress testing via banks’ risk steering based on internal models was extended to credit risk with the 2004 Basel Accord (Basel II; Basel Committee on Banking Supervision 2004).

Macroprudential stress testing

Its broad acceptance as a proper risk management tool encouraged the application of stress testing in a macroprudential context too. The 1997 Asian financial crisis increased regulators’ demands to measure the resilience of the entire banking industry and to identify institutions with solvency concerns. Therefore, in 1999 the International Monetary Fund (IMF) and the World Bank launched the Financial Sector Assessment Program (FSAP), which included extensive stress testing exercises. Its scope was initially restricted to emerging market countries. As policymakers perceived possible stress contagion and rapid spillover effects of credit concerns into the economies of the developed world, the scope of the FSAP was extended to countries with systemically important financial sectors, starting with Japan in 2001, followed by the UK in 2002, Germany in 2003

and France in 2004 (International Monetary Fund 2014; Dent and Westwood 2016). At the time of writing, FSAP stress tests have been conducted in 145 countries worldwide.¹ Although their results hardly ever affected countries' financial policies, the use of macroprudential stress tests became common practice and frequently encouraged the implementation of macroprudential programmes of regular stress testing in the jurisdictions under investigation.

Stress testing after the 2007–9 financial crisis

In the wake of the global financial crisis, both regulators and practitioners in the financial industry rapidly faced up to the shortcomings of prevailing stress testing practices. In 2007, before the crisis got underway, the IMF had published a stress test of Lehman Brothers that measured potential systemic risks deriving from the growing US asset-backed securities market, which revealed that it “is not likely to pose a serious systemic threat” (International Monetary Fund 2007). The IMF report stated that equivalent stress tests of Bear Stearns and JP Morgan reached similar conclusions. The tests did not question the appropriateness of using investment-grade ratings for asset-backed securities issues and thus assumed practically zero probabilities of default. High-impact risk factors were assumed to be constant and/or the size of their potential changes was severely underestimated. The financial crisis unambiguously revealed that the omission of high-impact factors in banks' risk management can even result in systemic risk, especially if such negligence is common practice.

In addition, the crisis revealed that stress testing focused too much on solvency risk, while any funding and liquidity risk effects on banks' balance sheets were not monitored. The 2008 Lehman collapse resulted in a widespread simultaneous breakdown of interbank liquidity. Public attention turned to the highly pronounced funding maturity and currency mismatches in banks' balance sheets, amplifying solvency concerns. These vulnerabilities in the financial system had for the most part remained undetected prior to the crisis and emphasised the need for regular liquidity stress testing to complement regular solvency stress tests (Jobst *et al* 2017).

Microprudential stress testing

Microprudential stress tests are conducted by individual banks using their own set of risk management techniques. In addition to the development of the

Basel III Accord, the BCBS took a whole series of actions to develop the standards for bank liquidity and funding risk management. Focusing on stress testing, it released the “Principles for Sound Stress Testing Practices and Supervision” in 2009 (Basel Committee on Banking Supervision 2009). The framework in large part suggests banks’ stress testing design should integrate liquidity (Basel Committee on Banking Supervision 2013).

Macroprudential stress testing in the US

To measure the severity of the situation in the US banking system after the 2008 Lehman collapse and to pinpoint funding requirements in the financial system, the US Federal Reserve System (“Fed”) relied on a macroprudential stress test: in 2009, it launched the “Supervisory Capital Assessment Program” (SCAP), including 19 institutions regarded as systemically important. In contrast to previous stress tests, the Fed published the test results for the first time, in an attempt to calm down the dislocated interbank markets. Although the test included a large liquidity stress component to reflect funding risks in the interbank market, the main focus remained on solvency stress testing.

In subsequent annual stress tests, liquidity stress testing, which was still relatively under-developed with respect to the magnitude of its possible effects on bank balance sheets, was systematically extended. To shed light on future funding requirements and capital planning of financial institutions, the SCAP was converted into the annual Comprehensive Capital Analysis and Review (CCAR) in late 2010: systemically important banks are required to publish detailed financial plans for the following nine quarters. This allows financial institutions that have large future funding requirements and reliance on resilient capital markets to be publicly identified.

Macroprudential stress testing in the EU

In 2009, similarly to the Fed, the Committee of European Banking Supervisors (CEBS) conducted a Europe-wide stress test on 21 banks regarded to be of systemic importance. The scope of the test was extended to 91 European banks in 2010. In addition to the regular testing schedule, CEBS’s successor, the European Banking Authority (EBA), launched another stress test in 2011, as the 2010 test revealed many deficiencies in its design: results were based on the assumption that sovereign debt had zero probability of default. Even Cypriot banks, which filed for bankruptcy

shortly after the 2010 results had been published, passed the test's minimum requirements. In contrast, the 2011 stress test included scenarios with sovereign debt defaults, requiring 31 banks to increase their liquidity levels. The EBA has conducted frequent stress tests since 2011. The applied methodologies, scenarios and key assumptions were developed in collaboration with the European Systemic Risk Board (ESRB), the European Central Bank (ECB) and the European Commission (EC); the highly heterogeneous set of financial sectors in the different jurisdictions in the European Union (EU) spurred on the development of a highly standardised test approach. Extending the framework by emphasising liquidity and funding risk as in CCAR will certainly contribute towards a full picture of the risks. We shall discuss this in more detail below.

Macroprudential stress testing in the UK

In 2013, the Bank of England's Financial Policy Committee recommended the introduction of a regular stress testing programme for the UK banking system. The first test was conducted in 2014 as a variant of the EBA stress test with UK-specific stress scenarios: debt levels of both households and non-financial companies had significantly risen since 2013. House price growth had accelerated and became a nationwide phenomenon. Therefore, the Bank of England (BoE) introduced a country-specific housing market shock to the EBA scenario set, as a large proportion of the UK mortgage market was based on floating-rate mortgages, which may have rendered increasingly correlated and systemic mortgage filings. The UK stress test programme emphasises the requirement to explore threats of this kind to the resilience of the financial system. Every second year, the set of stress scenarios is therefore complemented by additional scenarios (the "biennial exploratory scenarios") to explore a wider range of risks of this kind. For future stress tests, the BoE has indicated its intention for a greater coordination with the international tests of, for example, the IMF and EBA. This approach can enhance the quality of the test design, increase the effectiveness in the supervision of large, cross-border institutions, and establish an environment in which to share expertise and information between regulators and supervisors. However, the BoE will retain the UK-specific elements and risks in which policymakers are particularly concerned (Bank of England 2015).

STRESS TESTING ENVIRONMENT IN THE US AND THE EU

Stress testing in the US

After the 2007–9 financial crisis, US regulators proposed huge changes to the US financial regulatory environment. One response to the crisis was the Dodd–Frank Wall Street Reform and Consumer Protection Act (Dodd–Frank Act), which was signed into federal law in 2010. As a key measure to enforce the safety and soundness of the US banking system, US regulators required large (and mid-sized) financial institutions to undergo periodic stress tests.

The Board of Governors of the Federal Reserve System (FRB) introduced coordinated, so-called “horizontal” reviews for large financial institutions (with more than US\$50 billion of US assets), a measure to maintain a comprehensive understanding and assessment of each firm and across firms. These reviews include, among others, the CCAR, the Dodd–Frank Act stress test (DFAST), the Comprehensive Liquidity Assessment and Review (CLAR) and the introduction of resolution plans.

As the most prominent and important stress testing exercise, the FRB introduced CCAR in 2011 to assess the largest banks’ capital adequacy under normal and stressed conditions, its robustness of internal risk management and capital planning processes and the feasibility of proposed capital actions.

CCAR’s primary objective is to enforce capital adequacy under normal and stressed conditions for large financial institutions, thereby considering planned capital distributions such as dividends and stock repurchases. The CCAR assessment is performed on both qualitative and quantitative bases.

As of January 2017, 39 financial institutions were under CCAR, of which 18 were “large and complex” (ie, large banks with non-bank assets of more than US\$75 billion) and the other 21 were “large and non-complex” banks. This list mainly consisted of large US banks, but also included major non-US banks with significant business in the US, such as Barclays, Credit Suisse, UBS and Deutsche Bank.

CCAR follows an annual cycle, with FRB releasing CCAR instructions and scenarios in January, banks submitting capital plans by April and the FRB announcing objections (or no objections) in June. Banks must demonstrate robust, forward-looking capital planning processes commensurate with their unique risks.

Since its inception in 2011, CCAR has evolved, with regulatory scrutiny of capital adequacy becoming increasingly sophisticated. The FRB consistently raises the bar every year, especially for large banks, eg, by publishing model risk management expectations (“SR 11-7”) in 2011,² capital planning best practices (their “seven principles”) in 2013 and capital planning process expectations (“SR 15-18”) in 2015. This ongoing rule-making and regulatory guidance renders compliance complex and cost intensive.

Failure to pass the CCAR can result in severe regulatory penalties, such as restriction of capital actions, eg, suspension of existing dividend payments, and supervisory intervention through direct enforcement actions. It may also lead to a reputational damage, destruction of shareholder value and increased costs to remediate CCAR issues and ensure regulatory compliance going forward. Therefore, CCAR has become an important strategic exercise with high relevance at board level.

CCAR requires a robust end-to-end capital planning process that typically consists of six major components (see [Figure 19.1](#)). An overview of the requirements for each component is described below.

Risk identification and scenario design

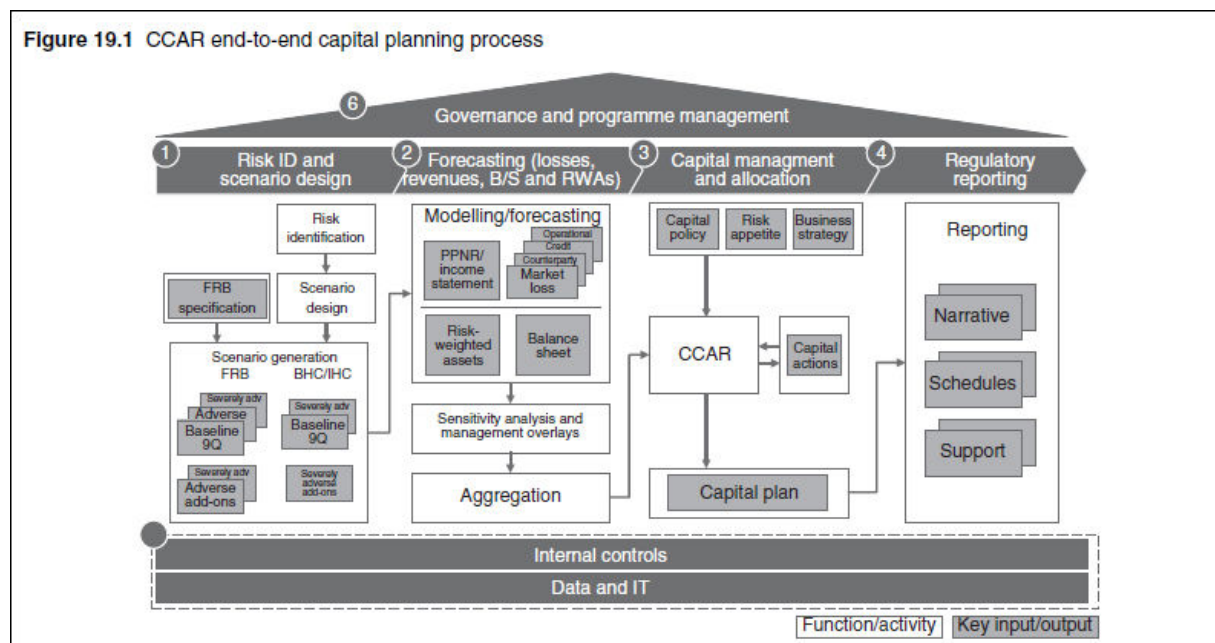
CCAR banks have to run both FRB-given and internal scenarios. Internal scenarios are often referred to as “BHC” (bank holding company) or “IHC” (intermediate holding company) scenarios.

The purpose of the FRB supervisory scenarios is to create a level playing field for all banks. Core components of the given scenarios are the nine-quarter projections for baseline, adverse and severely adverse (similar to the 2007–9 financial crisis) scenarios. These are prescribed through a narrative and nine-quarter projections for 28 domestic and international variables, such as GDP, unemployment, inflation, house prices, equity market indexes and volatilities, bond yields and major foreign exchange rates. The 2017 FRB’s severely adverse scenario is characterised by a severe global recession accompanied by a period of heightened stress in corporate loan and commercial real estate markets.

The largest and most complex financial institutions with significant trading positions are given additional global market shock (GMS) scenarios (adverse and severely adverse), which model instantaneous shocks to trading

books, private equity positions and counterparty exposures. The 2017 GMS was a set of approximately 20,000 risk factors across six main asset classes and was designed around sharp increases in general risk premiums and credit risk, as well as significant market illiquidity for non-agency securitised products, corporate debt and private equity.

Figure 19.1 CCAR end-to-end capital planning process



Large financial institutions with significant trading and custodial operations also have to model an instantaneous default of their largest counterparty across derivatives and securities financing activities.

In addition to the bank's calculations and results, the FRB runs the supervisory scenarios independently using its own internal models and banks' reported data.

The purpose of the internal, firm-specific BHC/IHC scenarios is to stress idiosyncratic vulnerabilities and risks. Banks must come up with their own baseline and severely adverse scenario for the upcoming nine quarters. The baseline scenario should typically be aligned to the firm's budget. The severely adverse scenario should be designed to stress idiosyncratic vulnerabilities and include idiosyncratic events, and should be at least as severe as the FRB severely adverse scenario in terms of net income and capital impact. Similar requirements apply to the internal scenarios and large counterparty default shocks.

The definition of reasonable internal BHC/IHC scenarios requires close integration of risk identification and scenario design and should engage a broad range of internal stakeholders. Banks have to identify and assess their risks first, which typically requires a risk taxonomy (ie, a sufficiently granular, standardised risk hierarchy and definitions across all business activities), a risk inventory with risk manifestation across all business units and a materiality assessment that assesses the identified risks based on business-as-usual reporting, aggregates across business units and identifies key firm-wide material risks.

The next step in creating the internal scenarios is for the bank to determine key scenario factors, define the core risk variable forecast and generate a full set of variables by using scenario expansion models. All scenarios must be supported by a clear narrative describing how economic conditions and idiosyncratic events, as well as target key firm-specific vulnerabilities, develop.

Forecasting of revenues, losses, balance-sheet components and risk-weighted assets

Under each defined scenario (baseline, adverse and severely adverse, both external and internal), CCAR stress testing requires forecasting of four key components: pre-provision net revenues (PPNR), stress losses, risk-weighted assets (RWAs) and balance-sheet components.

PPNR models are key for every CCAR submission. PPNR are typically decomposed into net interest income, non-interest income (fees) and non-interest expenses, and every component is modelled by using appropriate value drivers. Asset, deposit and funding balances, transaction volumes and margins typically drive income. To model these components, the business units (together with finance) use: historical internal and external data; regression approaches including external market data; assumptions on typical market sizes and shares; and expert judgement. Non-interest expenses such as compensation, occupancy and rent are often rule-based and centrally calculated by finance.

Stress losses are typically categorised as credit losses, trading losses, counterparty losses and operational losses. There are ample methodologies to model stress losses, eg, probability of default, exposure at default and loss given default models for credit losses or direct price shocks, and full revaluation and sensitivity-based approaches for trading losses. Operational

losses can be modelled via historical data simulation, regression or scenario analytics, but are often also forecasted using a judgment-based approach.

RWAs and balance-sheet changes are typically modelled centrally by applying Basel III models and regression-based approaches as well as assumptions on volumes and utilisation. All forecasted components are then aggregated to come up with a ratio forecast for each scenario.

Capital management and allocation

CCAR has explicit expectations for a firm's capital policy, capital adequacy assessment process and the capital plan.

The capital policy is a comprehensive document approved by the board, and should address major components of the capital planning process. It must adhere to the "seven principles" on capital planning best practices, issued by the FRB in August 2013. These serve as a qualitative evaluation metric as well as a guiding principle for a robust capital planning process. Key components are capital goals and (minimum) targets aligned with the firm's risk appetite, main factors that influence capital actions or trigger changes in capital actions and the definition of internal roles and responsibilities.

Capital adequacy is assessed based on four regulatory ratios over a nine-quarter horizon: Common Equity Tier 1 (CET1); Tier 1 capital; total capital; and Tier 1 leverage. These are calculated for each of the defined scenarios (internal and external), and planned capital actions have to be included in the calculations. Minimum thresholds for these ratios are given by the FRB and defined internally based on the bank's risk appetite and capital policy. The thresholds must not be exceeded throughout the planning horizon, ie, in every quarter for nine quarters. The results are disclosed to the public, based on the minimum value of the actual capital ratios throughout the planning horizon.

The capital plan should outline capital actions, as well as contingency actions and related triggers. In particular, it should include a detailed narrative of circumstances leading to a change in capital actions, eg, deterioration of economic environment or unfavourable market conditions. Corresponding trigger values have to be calculated for both baseline and adverse scenarios. Thresholds must reflect both point-in-time and forward-looking measures, provide early warning signs and trigger action. Depending on severity, they should be linked to escalation procedures for more immediate actions. The specifically defined contingency actions should be flexible with regard to potentially stressful situations, must be realistic, ie,

achievable during periods of stress, and prioritised, ie, ranked according to ease of execution and impact.

Regulatory reporting

The final CCAR submission is very extensive in terms of submission documents and data fields, and consists of the following three main components.

1. **Capital plan narrative:** a collection of documents providing information on the capital plan, capital policy, planned capital actions, capital adequacy process, risk identification programme, scenario design process, business plan changes, assumptions, limitation and weaknesses, etc.
2. **FR Y-14 schedules:** standard data templates on stress testing results across scenarios, time dimensions and business functions (FR Y-14A), detailed data on various asset classes, capital components and parts of PPNR (FR Y-14Q) and granular data from retail segment at loan/account/obligor level that is usable for stress testing (FR Y-14M).
3. **Supporting documents:** a collection of documents supplementing the narrative and the schedules, including policies and procedures, model design and validation, audit reports and contact lists.

Internal controls, data and IT

The reliability of CCAR results is critical for the FRB. Since the global financial crisis, most capital plans have been rejected for qualitative reasons rather than quantitative ones, the most common criticisms being: inadequate governance and weak controls around the end-to-end capital planning process; the inability to develop firm-specific scenarios that adequately reflect and stress the full range of business activities and exposures; weaknesses in loss estimation methodologies, assumptions and analyses.

To ensure reliable capital planning and effective review and challenge, the following controls and support functions should be involved in every step of the capital planning process:

- all models and processes involved in capital planning must be validated to determine their adequacy in terms of fit and performance and to

- uncover inherent weaknesses and limitations;
- internal audit conducts evaluation of models and processes involved in capital planning to ensure they function in accordance with supervisory expectations and internal policies;
 - policies define the capital planning process, including the roles of all parties involved and a formal process for policy exceptions, to ensure transparency and repeatability;
 - documentation capturing the capital planning process needs to be clear, comprehensive and detailed to allow effective review and challenge, to provide informed decisions and to inform outsiders;
 - data input and output must be correct, requiring reliable data transmission through the entire process, and, between models, data needs to be reconciled and manual adjustments should be transparent;
 - IT should enable the end-to-end process through provision, transmission and reporting of data, hosting models and process management tools, including tracking tools for issues/actions.

Governance and programme management

The overall CCAR governance should ensure that the board of directors and the senior management are informed and involved: the board has final approval of (and must approve) the CCAR submission, and the senior management is responsible for providing necessary and sufficient information.

Responsibilities include the effective review and challenge of the scenario and capital planning results, taking into consideration known weaknesses and limitations of the capital planning process. The board has the ultimate oversight and responsibility for the capital planning process, including steps taken to mitigate weaknesses, and should review and approve policies related to capital planning at least annually. The board is also responsible for all capital management decisions, and approves the capital plan as well as the CCAR filings and submission.

Stress testing in the eurozone

The stress test in the eurozone is conducted on a biennial basis and has become one of the most important regulatory exercises for banks. After the stress test in 2014 in association with the European Central Bank (ECB) in the course of the Comprehensive Assessment 2013–4,⁵ the European Banking Authority (EBA) became solely responsible for conducting stress tests in the eurozone.

The aim of this section is to give a brief overview of the “why, who, when, what and how” of EU stress testing; for technical details the reader should consult the original EBA guidelines (European Banking Authority 2017).

Why?

The purpose of the EBA stress test is to compare and assess the resilience of EU banks’ capital and balance sheets. Having a common stress test framework across different countries and legislations allows the stability of banks across the eurozone to be compared. The most significant outcome of the stress test is (still) the impact assessment on the capital ratio, indicated as losses in percentage points of CET1.⁶

Prior to the 2016 stress test, the regulators set a hurdle rate to “pass” the stress test. For the new exercise in 2018, the regulator will not set a capital hurdle rate (European Banking Authority 2017). However, stress test results serve as input to the Supervisory Review and Evaluation Process (SREP), a process for ongoing supervision of banks within the European Banking Union (European Central Bank 2016).

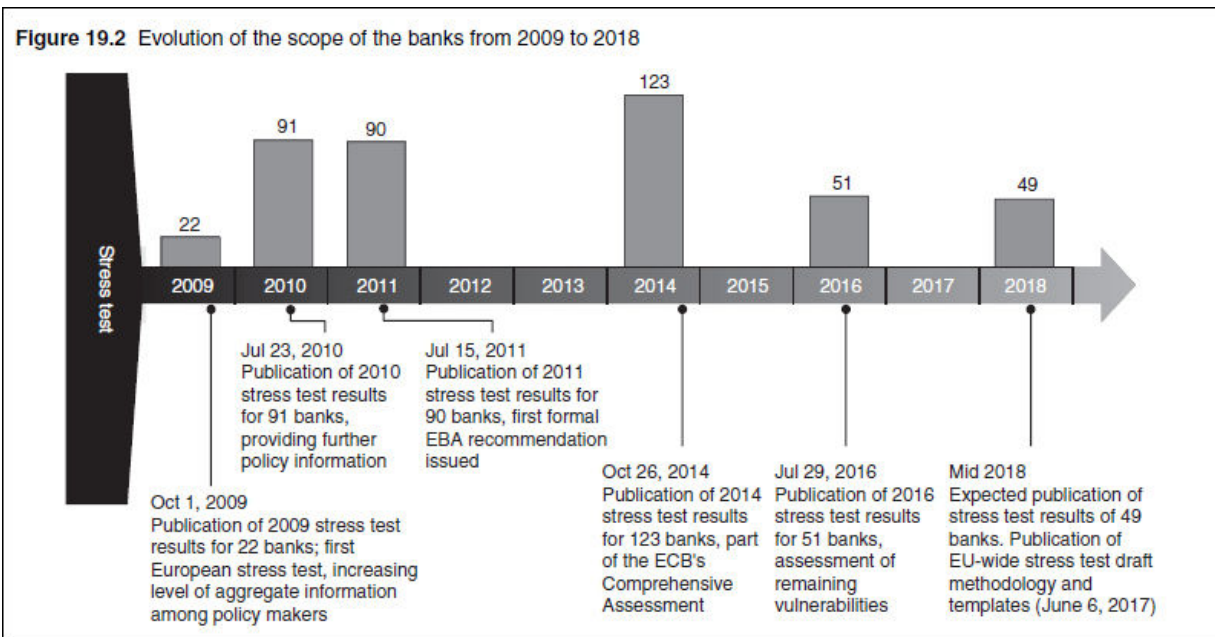
Who?

In 2018, the stress test will be carried out by a eurozone bank sample that covers 70% of total consolidated assets as of end 2016 and conducted on the highest “level of consolidation” (European Banking Authority 2017). To be included in this sample, a bank must have a minimum of €30 billion in assets (European Banking Authority 2017). [Figure 19.2](#) indicates the evolution of the bank sample over time from the first EU stress test in 2009. It shows first a strong increase and then a decrease in the number of participating banks. This effect is caused not only by a consolidation trend in the banking sector, but also by the changing definition of the scope. The complete list of participating banks can be found on the EBA’s website.⁷

When?

Since 2016 the stress test has always followed the same procedure, which could take up to 12 months.

1. The EBA publishes a “methodological draft” (European Banking Authority 2017) for the EU-wide stress test to outline amendments in scope, calculation methods, templates, etc.
2. During a consultation period, the draft is discussed, allowing the industry to submit questions or to push for changing the method in the event of a disproportionate high impact.
3. The stress test impact is calculated by the banks following the base and adverse scenarios for a given snapshot date. Up to this snapshot date, the bank is theoretically allowed to mitigate any potential stress test impact drivers. After the snapshot date, the regulator requires a so-called “static balance-sheet assumption” for the stress period of three years.



4. The results of the exercise are transmitted to the regulator via a set of predefined templates.
5. The regulator compares the results of the templates with peers and previous results, and examines the reliability in a question and answer

process with the bank.

6. The results of all banks are (typically) published by the EBA on its homepage.

What?

The stress test assesses a large variety of risks that are affected by the stress scenarios. The following risks are covered in this exercise.

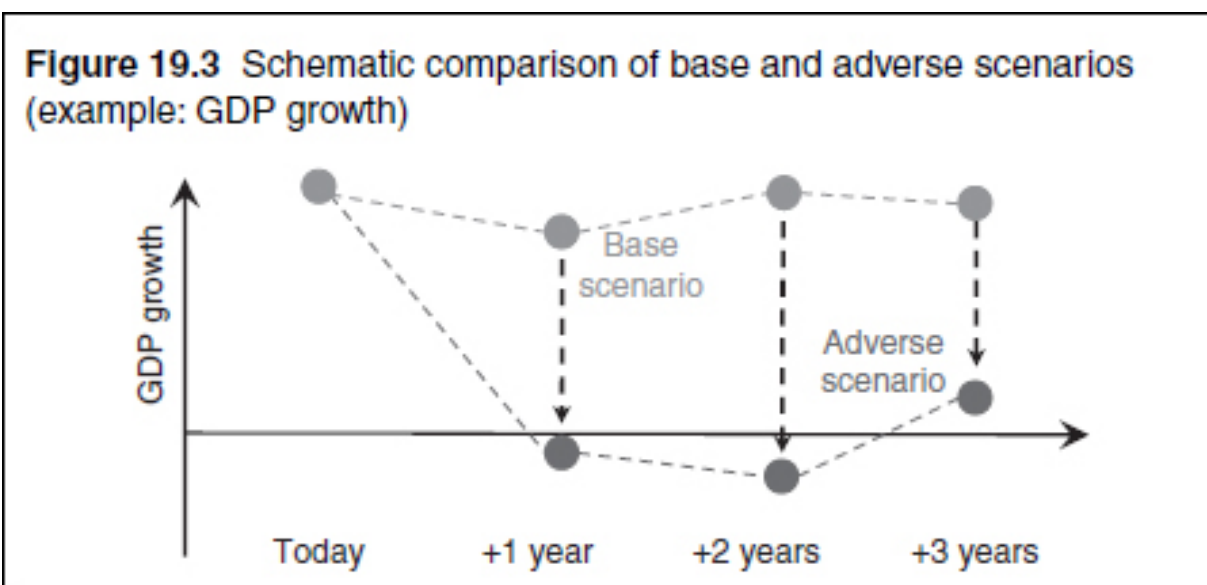
- Credit risk (including securitisation risk) covers the entire banking book with granularity by asset class, by country and by RWA approach. The securitisation, however, focuses on impairments for positions not held for trading, and on the mark-to-market treatment for positions at fair value.
- Market risk focuses on all positions included on the balance sheet at fair value, ie, held for trading, available for sale, fair value option. Note that risks arising from sovereign exposure are covered by credit and market risk.
- Operational risk (including conduct risk) incorporates the increase in capital/RWA because of stressing of future losses due to operational events such as fraud.
- The impact of stress scenarios on net interest income (NII), profit and loss (P&L) and capital items.

Since the 2018 stress test, a trend has been identified regarding modelling and predicting the stress impact on the entire balance sheet and the P&L, and not just on the capital ratio. Furthermore, the regulator has shown a growing reliance on internal models and avoided more punitive standard formulas (eg, conduct-risk-related operational losses). In addition, the granularity and forecasting requirements have increased from one exercise to the next.

How?

The EBA publishes a detailed description of how the stress period may look over the next three years. This gives a set of changing macroeconomic indicators (eg, a drop in GDP in specific regions). The stress scenario, the so-called “adverse scenario”, is compared with the “base scenario”, which

assumes a stable macroeconomic environment over the next three years, as shown in [Figure 19.3](#).



The two scenarios are applied to positions and risks within the scope of the stress test to forecast the P&L, balance sheet, capital ratios, etc, of the bank. The results of these forecasting exercises must be reported in predefined EBA templates, which serve as spreadsheets for calculation and validation as well as the final submission to the EBA and are to be assessed by the bank itself. [Figure 19.4](#) provides an overview of all templates and illustrates the two types of predefined template: 27 calculation support and validation data templates (input) and 10 transparency templates (output).

The results of all participating banks are (typically) published by the EBA on its homepage and used as input for the SREP by the competent authorities.⁸

Comparison of stress testing frameworks in the US and the EU

The EBA stress test in the EU and CCAR in the US have ideas and approaches on stress testing in common, but differ significantly in particular methodologies. The key differences are that CCAR requires internal scenarios and requests dynamic balance-sheet modelling, whereas EBA only requires external scenarios and a static balance-sheet approach (no business mix change). Furthermore, CCAR has a strong focus on qualitative elements,

with internal audit playing a key role in the CCAR submission. [Figure 19.5](#) gives an overview of the key differences.

It was unclear at the time of writing whether (and how) both stress testing approaches would converge. In the US, a review of the Dodd–Frank Act had been mandated to the US Treasury at the beginning of 2017, which could lead to a loosening of both US banking regulation and the stress testing environment. In contrast, there was a market view that the EBA stress test may converge towards a more comprehensive capital planning exercise. Moving towards US CCAR standards would require substantial additional effort for European banks. However, there are no official statements from regulators supporting this view and good reasons to maintain a more conservative approach.

Figure 19.4 Pre-defined templates by EBA

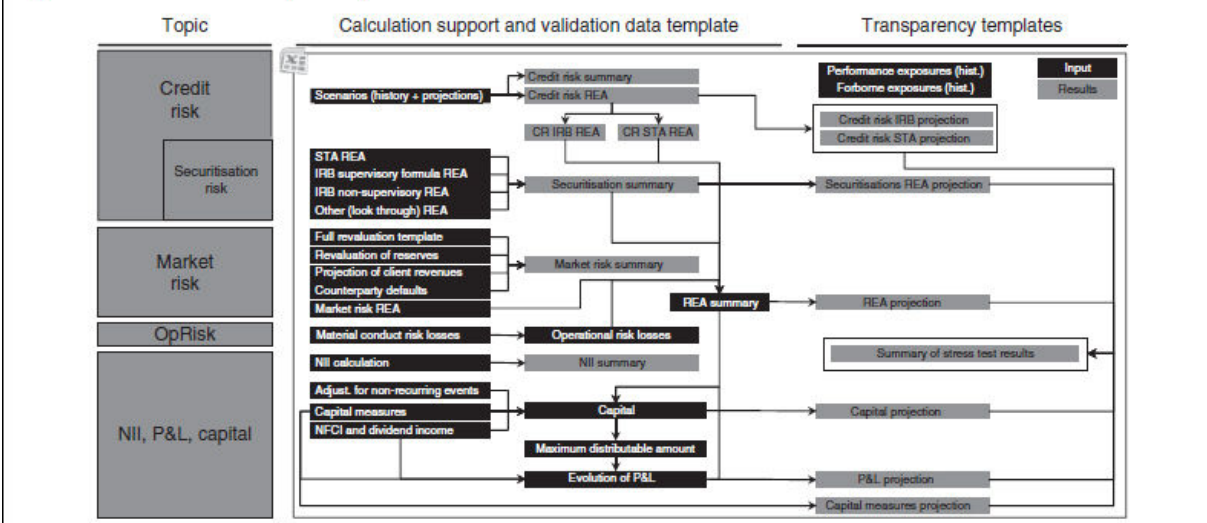


Figure 19.5 Key differences between EU and US stress testing approaches

	 CCAR	 EBA stress test	Impact assessment of potential adoption of US standards in EU
1 Risk identification and scenario design	• Three external and two internal (more severe) scenarios	• Only two external scenarios (no internal scenarios)	● Idiosyncratic scenarios require setup of robust risk identification and scenario design programme
2 Forecasting	• FRB calculates ratios via own independent models	• Internal models, EBA's own model "backtesting"	● Comparatively low additional effort but potentially high impact on results
3 Capital management and allocation	• Dynamic balance sheet	• Static balance sheet, ie, no business mix change	● Dynamic B/S requires additional modelling effort, but unlikely to be put in place (less conservative)
4 Regulatory reporting	• Large qualitative reporting component	• Focused on qualitative reporting via templates	● Large-scale qualitative reporting would require substantial additional documentation and resources
5 Internal controls, data and IT	• Internal audit plays a key role in submission	• No/less focus on internal audit function	● Build out internal audit to meet required coverage and skills
6 Governance and programme management	• Board sign-off required	• Sign-off by governing body accountable for programme	● More rigorous documentation and board education required

 High
  Medium
  Low

Despite potential future developments, banks should always be aware of the differences between these two jurisdictions, and consider learning from the different approaches, using select cutting-edge methodologies for their internal stress testing. Lessons learned from CCAR, eg, on scenario design or PPNR modelling, could help European banks to proactively adapt their stress testing and capital planning.

GUIDELINES FOR A STRESS TESTING PROGRAMME SETUP

Even though banks have acquired some experience in conducting stress tests (from, eg, EBA or the Fed), such exercises should not be underestimated. A comprehensive stress testing programme touches on various cross-functional topics across the organisation, and thus forces divisions to work together outside their known and business-as-usual procedures. As the EBA stress test guidelines incorporate new topics and require a natural fluctuation in personnel, parts of this exercise need to be tackled anew every two years. In this section we provide best practice methods for a proper setup of an effective and efficient exercise. Therefore, this section serves as a “handbook” both for readers new to stress testing programme management and for experienced readers wanting to enhance their approach and processes. To account for complexity, we focus on the EBA stress test as an

example, but the key components also apply to other stress testing frameworks.

The challenges of stress test programmes

Without the acquired knowledge and experience of previous stress test programmes and the continuity of previously involved employees, a biennial stress test programme can pose enormous challenges for banks. To set up a stress test programme, it is key to anticipate these challenges and undertake mitigation actions early. The following challenges can be differentiated.

Challenge 1: new methodology

As soon as the preliminary methodological guidance is published by EBA, a capability assessment of the new methodologies for the corresponding year's stress test reveals potential gaps that banks need to cover. Potential changes range from only minor adaptation, eg, minor template changes, to the integration of new regulations to be integrated for the first time, eg, IFRS 9 in the stress test in 2018.

Challenge 2: impact assessment and prediction

Stakeholders and banks' decision makers usually show an interest in the potential results of the stress test early on. Unexpected results, for example, that (by definition) indicate an instability of the bank in times of macroeconomic shocks, can lead to an unfavourable market position with clients and counterparties as well as the loss of a favourable reputation in the banking community. Therefore, the stress test programme management will often ask for predictions of stress test results during the early stages of the process. Furthermore, an assessment of the stress test capital impact even before the snapshot date allows the bank to derive potential mitigation measures to reduce the effects of the shock scenarios.

Challenge 3: consistent communication

As national competent authorities are (according to the EBA guidelines) officially responsible for data quality and bottom-up stress test calculation, consistent and timely communication throughout the whole stress test exercise is key in demonstrating a consistent data repository, adequate models and accurate stress test calculations. In cases of doubt, local

authorities are capable of running inspections in addition to the stress test exercise. Furthermore, banks (or their representatives) can participate in the consultation on the preliminary methodological stress test guidelines, which is recommended should the newly introduced methodologies have a disproportionately high impact.

Challenge 4: data and infrastructure

The most difficult point of the stress test is to link the bank's existing, somewhat complex infrastructure with an exercise that is mainly based on Excel spreadsheets and regulatory email correspondence. In addition, the required data for the scenario calculation, as well as the models for the calculation itself, is distributed across the whole organisation. Therefore, before handing in the official templates to the national competent authorities, data quality and consistency should be checked and rechecked.

Challenge 5: cross-functional management

The stress test exercise guidelines require organisations to collaborate with a wide variety of different functions as well as subsidiaries. As these processes are usually outside their known, customary day-to-day business procedures, the stress test exercise poses a management challenge. When there is a flood of new regulations directed to the same resources, each biennial stress test is essential in order for the programme management to successfully integrate data, infrastructure, risk management and accounting experts as well as an independent data reviewer. To reduce the considerable setup time and to ensure the continuity of data collection and impact calculations, these experts will ideally have worked in previous stress test teams.

Recommendations for setting up stress test programmes

In order to tackle the challenges discussed above, the following steps for setting up stress test programmes are recommended, and have already proven effective in practice.

Recommendation 1: clear programme timeline

The European stress test takes between six and twelve months from the reception of the preliminary guidelines to the disclosure of the results.

Figure 19.6 displays a typical 12-month timeline of a stress test exercise in Europe. Differentiating between these three steps is important in order to better steer the rare resources/experts during all phases of the programme.

Recommendation 2: pre-aligned project-structure and governance

Setting up a programme organigram, task description and programme governance (including steering committees) is generally not difficult and comes under “project management 101”. However, the specific stress test exercise, given its high methodological and organisational complexity, requires a predefined and prearranged project structure and governance that reflects the inherited processes and tools (see the next recommendation).

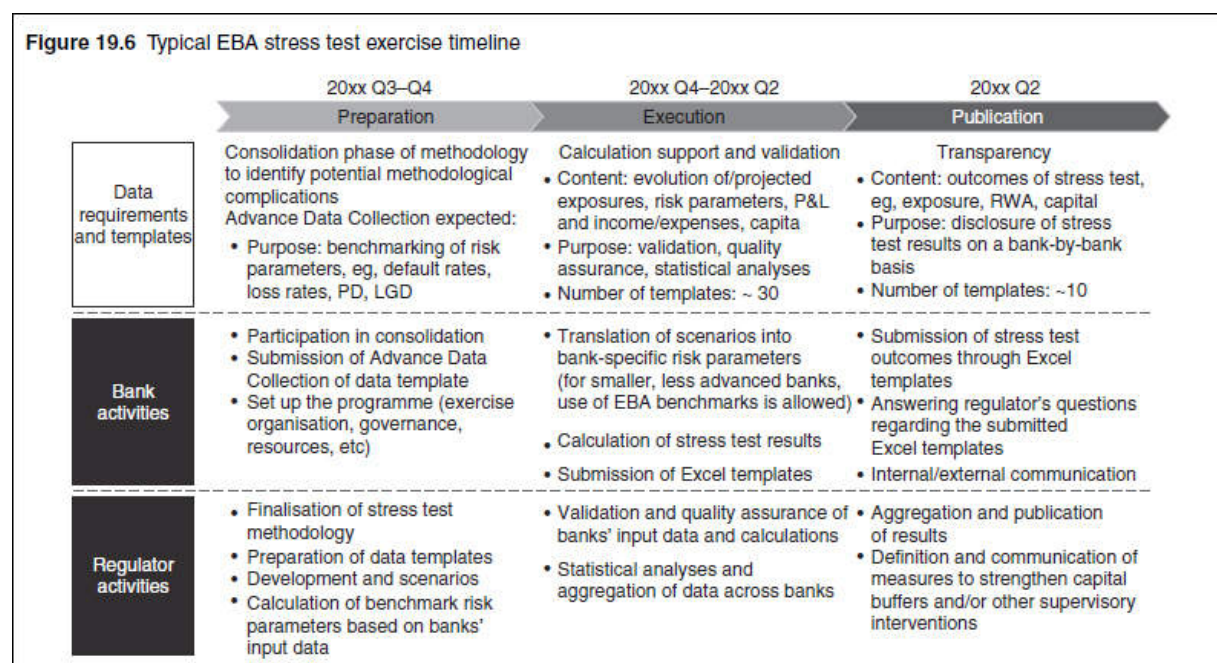


Figure 19.7 Typical stress test programme organisation, which can be adapted to specific organisational structures

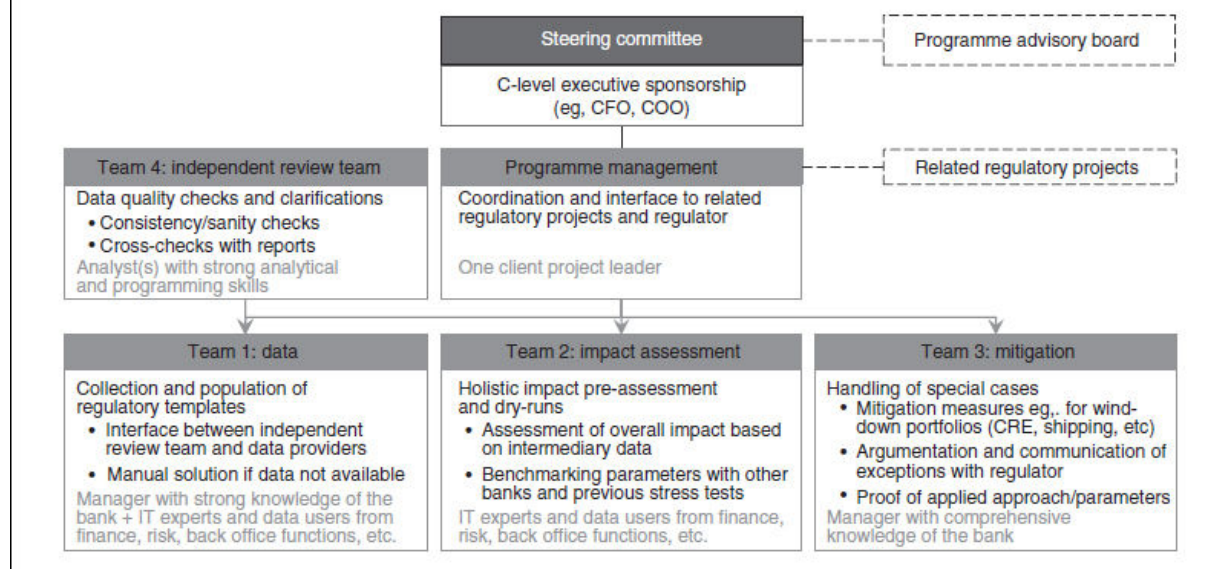


Figure 19.7 shows a typical project organigram, including the experts/managers necessary to lead the different teams. The consolidation team (team 1) plays a pivotal role. The impact and mitigation teams (teams 2 and 3) calculate the impact of the stress scenario and identify ways to mitigate it. Finally, the independent data review team (team 4) ensures the quality is deliverable.

Recommendation 3: predefined process and tools

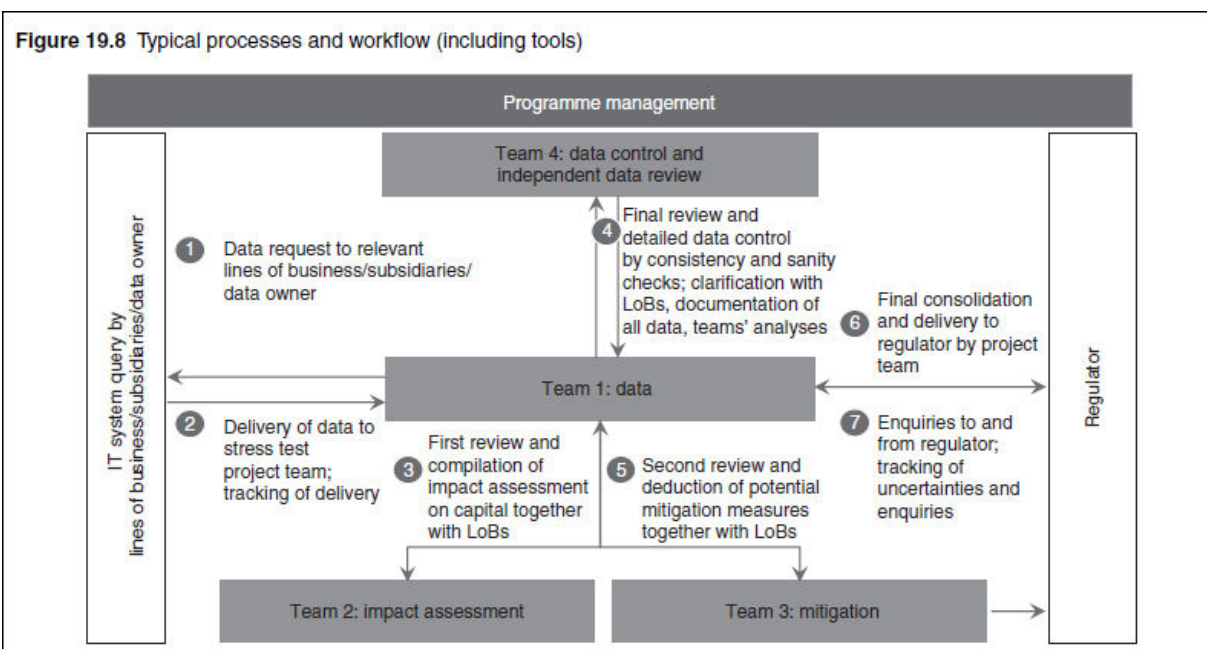
Together with the programme organisation and the task description, we also recommend setting up a process and workflow organigram to predetermine how (and in what order) the teams would work together. A clear definition of the workflow is particularly important, as several functions and business units across divisions would be working together on this exercise.

Figure 19.8 shows the typical workflow to align the four teams from the programme organigram in recommendation 2. At points 1 and 2, team 1 (data) requests and receives the data from lines of business (LoBs) subsidiaries and other data owners that consolidate “their” IT systems and data repositories. The tools used can be templates or properly programmed data queries (eg, SQL queries) that can be integrated directly into the stress test calculation spreadsheet, engine or software. At point 3 team 2 (impact assessment) reviews and calculates the likely impacts of the stress scenarios. Any striking abnormalities need to be listed for handover to team 4 (data

control and independent data review), which, at point 4, systemically conducts further consistency and sanity checks and conveys a list of necessary data remediation back to team 1. At point 5, team 3 (mitigation) further analyses the likely stress scenario impact and identifies tools to reduce it, ideally before the snapshot date (also called the static balance-sheet assumption by the EBA), when the data will, by definition, be frozen. In the last steps, at points 7 and 8 team 1 submits the final results to the regulator and thence functions as the “single point of contact” for the regulator.

Recommendation 4: ongoing result prediction

It is highly recommended that the potential impact is assessed throughout the exercise, even though this figure might only be a rough estimate during the first weeks. An ongoing impact assessment ensures proactive communication in the case of unexpected results and allows mitigation tools to be put in place before the snapshot date. However, the stress test forecast measured as the effect on capital ratio reduction (indicated in the EU in percentage points of CET1) should be stated as a range in order to account for unforeseen additional impacts/amendments that are not captured in the prediction.



Recommendation 5: “single point of contact” communication

Any information requests from inside the organisation (eg, board members) as well as from outside the organisation (eg, regulator or press) should be directed via a “single point of contact”. This person should be in close contact with the regulator and with the relevant units or the stress test programme in the bank. Ideally, they would be the head of the stress test programme or of the regulatory unit, if it exists. Such a single point of contact is necessary to ensure consistent communication of data and result predictions.

Recommendation 6: application of checklists for project

Before starting this project it is highly recommended that a checklist is formulated to ensure every relevant topic is included. The following questions may be seen as a checklist for managers setting up a successful stress test programme:

- what are the methodological changes to the previous stress test exercises?
- what were the main points of the last stress test exercises?
- what adaptations to the process and infrastructure are necessary compared with previous exercises?
- what other regulatory projects need to be considered during the course of this exercise?
- when does the bank expect to receive the preliminary methodological guidelines?
- should the bank participate in the consultation and, if so, how?
- when does the bank run the stress test programme and what are the phases?
- who are the relevant people and what are the relevant resources in the stress test programme?
- what additional external support for specific topics or management is required, if any?
- what is the appropriate programme organigram and structure?

- who is responsible for data continuity, run consistency and sanity checks?
- when and where will the programme teams gather to discuss open topics?
- how do the programme teams work together and integrate other subsidiaries?
- is it necessary to conduct a “dry run” data extraction before the official data request?
- what stress scenarios can be used to calculate the likely stress test result before the publication of the official scenarios?
- who is the single point of contact for internal and external communication?
- how does the programme communicate with the board and other internal stakeholders?
- how, when and where does the bank involve the national competent authorities, joint supervisory team and the supranational regulators (EBA, ECB, etc)?

DEVELOPMENTS AND EXTENSIONS OF STRESS TESTING

Integration of IRRBB and CCAR requirements into banks' risk management

In 2016, the Basel Committee on Banking Supervision published the interest rate risk in the banking book (IRRBB) standards (BCBS 368) (Basel Committee on Banking Supervision 2016). Initially planned as a standardised Pillar 1 requirement, the heterogeneity of banks and the accompanying difficulties in implementation led to the decision to implement this requirement as a Pillar 2 framework. Thus, at the time of writing banks are required to enhance their risk management for IRRBB. The BCBS 368 standard defines the minimum requirements for governance, roles and responsibilities, risk limitation and risk appetite, risk identification, monitoring, control and supervision, as well as requirements for risk data

and IT. Designed as a stress test for banking book positions, the framework balances solvency risk by quantifying the impact of stress scenarios on the economic value of equity (EVE) and future NII. Similar to the CCAR approach, banks are required to incorporate the results of the IRRBB stress test into the determination of economic capital amounts and bank-wide risk management and steering.

The stress test designs of the IRRBB and CCAR frameworks are largely similar. Therefore, US banks and internationally operating institutions with strong US branches have increasingly harmonised the implementation of both regulatory frameworks. Their goal is to increase consistency in the concept of risk management, as well as operational risk management, data and IT efficiency across the frameworks.

A simple comparison of both frameworks reveals four major elements for harmonisation.

1. **Governance and control:** CCAR and IRRBB influence a highly similar target governance infrastructure. The bank's governing body is responsible for the oversight and management of both approaches, with mandatory regular briefings of the board or delegated senior management. CCAR is more comprehensive, with a shorter minimum interval between board briefings (quarterly as opposed to semiannually for IRRBB) and a wider scope of reporting. Conformity of the governance structures can easily be achieved by defining a common target operating model, which is compliant with both frameworks.
2. **Risk management and capital policy:** CCAR and IRRBB stress test results must be considered for economic capital planning. The definition and implementation of an integrated risk measurement and management approach to quantify the bank's risk appetite and risk limits is crucial for avoiding contradictory implications of stress test results, and ultimately for the identification of capital adequacy. The IRRBB framework details the capital requirement for interest rate risk in the banking book, while CCAR covers capital adequacy for all market risks, not just interest rate risk. Integration is achieved by the definition and implementation of a common "risk appetite framework", which is compliant with both standards.

3. **Risk methodology:** the risk methodologies of CCAR and IRRBB standards are based on similar measurement and stress testing concepts, with CCAR including interest rate risk (not just interest rate risk in the banking book) and having stricter (dynamic) balance-sheet requirements. This calls for a harmonisation of stress test testing scenarios, as well as definitions of risk measure in the implementation of IRRBB and CCAR frameworks.
4. **IT and data management:** CCAR and IRRBB standards have a strong overlap and interdependence in data requirements, combining data fields from a very broad set of data types. This establishes the possibility of functional data harmonisation with effective data quality management. The development of a common, integrated data infrastructure compliant with BCBS 239 for both standards is a target for industry best practice.

All the elements above can be further outlined in greater detail. As an example, the remainder of this section discusses the major elements to be harmonised in CCAR and IRRBB “risk methodology”.

- **Definition of the risk horizon:** CCAR has the mandatory requirement to disclose projected capital plans for the a risk horizon of nine post-stress quarters. This is stricter than the BCBS approach, which states that disclosed changes in economic equity values (ΔEVE) should be standardised with a risk horizon over the remaining life of the balance sheet. Disclosed values of net interest income (ΔNII) should have a horizon of 12 months.
- **Balance-sheet assumptions:** CCAR risk modelling is based on a dynamic balance-sheet assumption; the required projections explicitly consider originations, defaults and customer behaviour in line with dynamic risk factors. In contrast, disclosed ΔEVE values in BCBS 368 only require consideration of a run-off assumption and disclosed ΔNII values require a constant balance sheet.
- **Behavioural assumptions:** CCAR establishes an explicit modelling requirement for scenario-dependent prepayments, as well as borrower and depositor behaviour. In BCBS 368, behavioural assumptions should

also be factor dependent, yet are more detailed than in CCAR, eg, in the “standardised framework”.

- **Risk measures:** CCAR is based on loss- and earnings-based measures; it defines losses, including fair value losses (EVE). In addition to loss forecasting, it establishes the requirement to define pre-provision net revenues, including NII. In contrast, BCBS 368 gives more detailed definitions of disclosed EVE and NII measures. Thus, CCAR has less detailed definitions, yet broader risk measures.
- **Scenario definitions:** as well as the standard scenarios applicable to all institutions, CCAR has requires use of multiple additional bank-specific scenarios across material risks (not necessarily interest rate risk). In contrast, BCBS considers additional internal scenarios, addressing the bank’s IRRBB profile with a focus on identified risk concentrations.

Integrated balance-sheet steering

Based on continuously enhanced stress testing frameworks and on the corresponding increased scenario analytics and forecasting capabilities, banks should rethink their internal budgeting and planning, and move towards integrated balance-sheet steering. First, it is important that a bank’s internal business planning is consistent with the baseline in a stress testing and capital planning exercise. Second, banks should use insights from their stress testing exercise to enhance their business planning and include different future scenarios. In a target state, banks should have an integrated framework and central tool to model both their business planning and their stress testing across different jurisdictions.

An integrated balance-sheet view can support the joint optimisation of capital, liquidity, funding and leverage. In contrast to most budgeting and planning processes, the impact on steering processes should be considered under both normal and adverse/stressed scenarios. It is critical to assess optimal balance-sheet and P&L modelling choices, even beyond the focus on stress testing.

Developing integrated balance-sheet management requires three building blocks:

1. a methodological approach and modelling choices;

2. implementation with a focus on integrated tools, governance and workflows; and
3. development of strategy and management actions, ie, optimisation tools based on steering metrics and quantification of the corresponding impact.

By applying this integrated view, stress testing will cease to be just an expensive regulatory exercise, and become a bank's central and effective steering instrument, forming the best foundation for management decision-making.

- 1 See <https://www.imf.org/external/np/fsap/faq/index.htm#q1>.
- 2 See <http://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm>.
- 3 See <http://bit.ly/2zkZoxQ>.
- 4 See <http://www.federalreserve.gov/supervisionreg/srletters/sr1518.htm>.
- 5 See European Central Bank (2014a,b). The Comprehensive Assessment was the initial assessment by the ECB before it took over the supervision of the largest banks in the European Union under the framework of the newly legally established Single Supervisory Mechanism (Pillar I of the European Banking Union).
- 6 The Common Equity Tier 1 ratio is based on Basel III integration of the Capital Requirements Regulation (CRR) in the European Union; see European Commission (2012).
- 7 See <http://www.eba.europa.eu>.
- 8 Note that the regulator does not always publish all results or templates for stability reasons, as sometimes full transparency of bad stability results of certain stress test participants can trigger a self-fulfilling prophecy due to a pullback by counterparties or clients.

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*Reverse Stress Testing: Linking Risks,
Earnings, Capital and Liquidity – A Process-
Orientated Framework and Its Application to
Asset–Liability Management*

Michael Eichhorn; Philippe Mangold

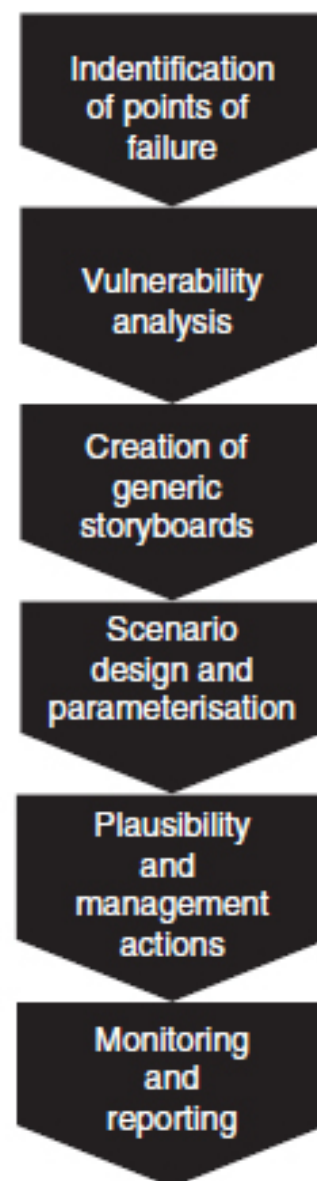
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Reverse stress testing (RST) is commonly understood to be the identification of adverse scenarios that render the business model unviable. Prior to the 2007–9 global financial crisis, few, if any, banks worked on RST. After the financial crisis, regulators and de facto regulatory bodies introduced different requirements and recommendations for banks to perform RST. EY (2013), in a survey of major international financial institutions, noted that the importance of RST as a tool for risk measurement and risk management strongly increased. However, there is still neither detailed regulatory guidance nor an industry standard or “best practice” on how to implement RST in a meaningful way.

While both traditional stress testing and RST involve the analysis of adverse scenarios and their respective impacts, they differ in two key aspects.

1. **Direction:** in traditional stress tests, banks start by defining a scenario specifying adverse macroeconomic or financial conditions (or a combination thereof). Banks then assess the impact on their business, typically in terms of earnings, and capital and liquidity adequacy, over a specific period of time. Conversely, RST starts by defining the outcome then reverse-engineering scenarios that, should they unfold, would lead to the specified result.

Figure 20.1 A six-step generic approach to reverse stress testing



2. **Severity of the stress:** RST goes further into the tail end of the probability distribution than traditional stress tests or other risk measures such as value-at-risk or economic risk capital, since RST scenarios are usually, by construction, designed to be so severe that they “break the bank”.

The primary challenge in RST is usually not a lack of imagination by individual stakeholders within any bank. Often, it is rather the process of collecting and aggregating the data and knowledge across the bank in a consistent and coherent manner. This chapter outlines a process that is both sufficiently generic and adaptive, which aims to engage the key stakeholders across an institution in one-off exercises as well as regular reviews of the respective scenarios and their likelihood of unfolding. Asset–liability management (ALM) experts are usually key contributors to this process, due to their macro view across business lines and their key role in managing structural balance-sheet risks and financial resources. Thus, they are key contributors not only to the inputs (eg, the scenarios), but also to the evaluation of the impacts and, if required, the design and implementation of mitigating actions.

The following sections propose a framework for RST that supports deep and deliberate thinking around highly adverse scenarios. Since there are usually a large number of scenarios that could lead to a specific outcome, we advocate beginning at a generic level, and moving on to a deeper examination of specific scenarios. Finally, if a given scenario is deemed highly likely, the management needs to derive actions to make the analysis meaningful in practice. Based on this framework, we next discuss how ALM experts can apply and contribute to RST. The penultimate section provides further guidance on what users should consider when applying RST. Finally, we give a brief summary of the chapter.

A GENERIC FRAMEWORK FOR REVERSE STRESS TESTING

Our proposed process for collecting and aggregating the data, information and knowledge across the bank is composed of the following steps:

1. identification of points of failure,

2. vulnerability analysis and creation of a risk inventory,
3. creation of generic storyboards,
4. scenario design and parameterisation,
5. plausibility checks and management actions,
6. monitoring and reporting.

These steps, shown in [Figure 20.1](#), should form the fundamental basis for RST discussions across the bank and involve the respective functional representatives, including ALM experts.

Step 1: identification of failure points

The first step is the identification of so called “failure points”, ie, points at which a bank’s business model becomes unviable. We highlight failure points from an earnings perspective, from a capital perspective and from a liquidity perspective. While there may be interconnections, it is important to consider these perspectives separately, as this will help us to devise a sufficiently broad range of RST scenarios.

Earnings failures involve sustained (ie, over a number of quarters or years) low or deteriorating pre-tax income (PTI) for a cluster, business area or an entire bank. Such failures could call into question whether a bank should continue to operate from a commercial point of view. The earnings failure point could be reached well before a bank’s capital or liquidity resources are exhausted. For example, regulators have stressed that reputational risk may break a bank even if it has sufficient capital and liquidity.

From a capital perspective, the bank could become insolvent. Prior to reaching insolvency, however, the capital figures may breach internal thresholds. The latter may, therefore, also be used as failure points in the context of RST. For example, capital losses that exceed the risk appetite may trigger actions to restore the capital base, while a further decline in capital levels could trigger mandatory conversions of convertible instruments into equity or result in breaches of regulatory thresholds. Accordingly, failure points could be linked to the recovery and resolution plan. However, the business model can also become unviable without a reduced capital base. This may be the case if the capital requirements increase, eg, due to a loss of

an internal model waiver or to a new regulation that results in higher capital requirements. In this case RST may be cross-linked to other plans, such as trigger points of the solvent wind-down plan. Finally, the bank may become illiquid. This may occur, for example, at the point when the high-quality liquid asset buffer drops below a predefined level. Again this point may potentially be linked to either risk appetite or recovery action triggers. Alternatively, liquidity failure points may be linked to a minimum day-count survival horizon.

Step 2: vulnerability analysis and creation of a risk inventory

The purpose of the vulnerability analysis is to identify the primary material risks to which the bank is exposed through its business model and client base.

A vulnerability analysis can be performed top-down or bottom-up¹ and combine various qualitative or quantitative techniques.² Whichever techniques are used, the output should be a comprehensive list of vulnerabilities that are material to the bank under consideration. This list may include ALM risks, eg, structural interest rate and foreign exchange (FX) risks. It should feed into a bank-wide vulnerabilities inventory. Ideally, specific risk inventories should exist for different business units, departments (including the treasury) or risk types, which are then aggregated into a bank-wide risk inventory, typically maintained by an enterprise-wide risk function.

In one example, a private bank, we used qualitative expert interviews. With the support of two members of the senior management, in a series of semi-structured dialogues we asked 15 business and risk executives what could cause the business model to become unviable from their perspective.

In the next step we analysed the interview responses in detail,³ clustered the responses by risk type and established a two-dimensional heat map. The analysis led to valuable insights. For example, it became clear that the framework needed to differentiate between purely reputational risk incidents and those arising from a domino effect of operational or conduct risk incidents. Most of the executives particularly emphasised this domino effect. Another insight was the importance of considering certain types of operational risk (OpRisk) that may give rise to conduct risk.⁴ We also noted sequential patterns, eg, liquidity risks were repeatedly seen as a consequence of the materialisation of other risks.

Depending on the frequency with which the different risk types were mentioned during the interviews, and the perceived severity of the events, they were assigned to the red, amber or green zone of the heat map. Accordingly, we distinguished between “key vulnerabilities” and “other vulnerabilities” of the business model.

We also examined patterns. For example, the vulnerability analysis may suggest that most key risks in the red zone of the heat map are idiosyncratic, whereas risks in the other zones of the heat map are more systemic, or vice versa.

Another interesting finding resulted from the question of the moment at which the business model becomes unviable. We distinguished between incidents where this point is reached after a period of days or months (“a sudden fatal punch”) and incidents where it is reached over a period of years (“a slow bleeding out”).

Table 20.1 Classification of scenarios by type of failure and time horizon

Time horizon of scenario	Point of failure			
	Liquidity failure	Capital failure	Capital-induced liquidity failure	Earnings failure
Instantaneous (“sudden fatal punch”)	Case 1a Severe risk event results in liquidity failure (impact on capital is minor)	Case 2a Sudden exhaustion of capital basis	Case 3a Sudden exhaustion of capital basis raises substantial doubt about future viability of business, leading to liquidity failure	Case 4a Sudden changes resulting in negative PTI impact for the foreseeable future, making business model unviable
“Slow bleeding out”	Case 1b Series of risk events result in liquidity failure (impact on capital is minor)	Case 2b Gradual exhaustion of capital basis; emergency liquidity facility still available	Case 3b Continued deterioration of capital basis raises substantial doubt about future viability of business, leading to liquidity failure	Case 4b Continuing changes resulting in negative PTI impact, making business model unviable

The output of step 2 was not simply a list of relevant incidents, but a risk inventory and a corresponding classification. This is summarised in [Table 20.1](#), where the rows differentiate the various time horizons over which the failure point is reached and the columns display the different dimensions of failure. We added a fourth dimension, called “capital-induced liquidity failure”, which will be explained later.

This classification permits some degree of completeness, given the wide range of possible scenarios. It will also be useful when storyboards need to

be created, as discussed in step 3.

Step 3: creation of generic storyboards

The storyboard shows the development of scenario narratives (from status quo to the point where the business model becomes unviable). One obvious, and to some extent trivial, way is to derive scenarios by scaling single or multiple risk factors to extreme levels; however, this seems overly simplistic in that it restricts RST to a sensitivity analysis.⁵

Regulatory requirements and recommendations, eg, those issued by the Prudential Regulatory Authority, Federal Reserve System, the Swiss Financial Market Supervisory Authority and Bank for International Settlements, provide high-level guidance on storyboards. Regulators expect banks and other financial institutions to review a range of scenarios, including the failure of major counterparties and coinciding idiosyncratic and macroeconomic events.

We also spoke to different consultancies. They recommended developing 6–12 well-defined storyboards. Even this number may prove challenging in practice, while potentially still being insufficient: as discussed at the outset, there may be hundreds or thousands of scenarios for each business model. Considering all of these is likely to prove impractical and difficult to handle computationally.

Therefore, our framework proposes to obtain a “big picture” perspective first (so as not to get lost in the detail of many possible storyboards). By drawing from the classifications displayed in [Table 20.1](#), we differentiate eight groups of storyboards. Specifying, say, one storyboard for each group results in a manageable number of scenarios while providing some comfort around the degree of completeness. The objective of the next step is to develop scenarios for each of the eight groups. However, instead of jumping straight to one specific scenario and calibrating the full set of variables, we propose to develop so-called “generic scenario storyboards”.

Figure 20.2 Illustrative example of a generic storyboard

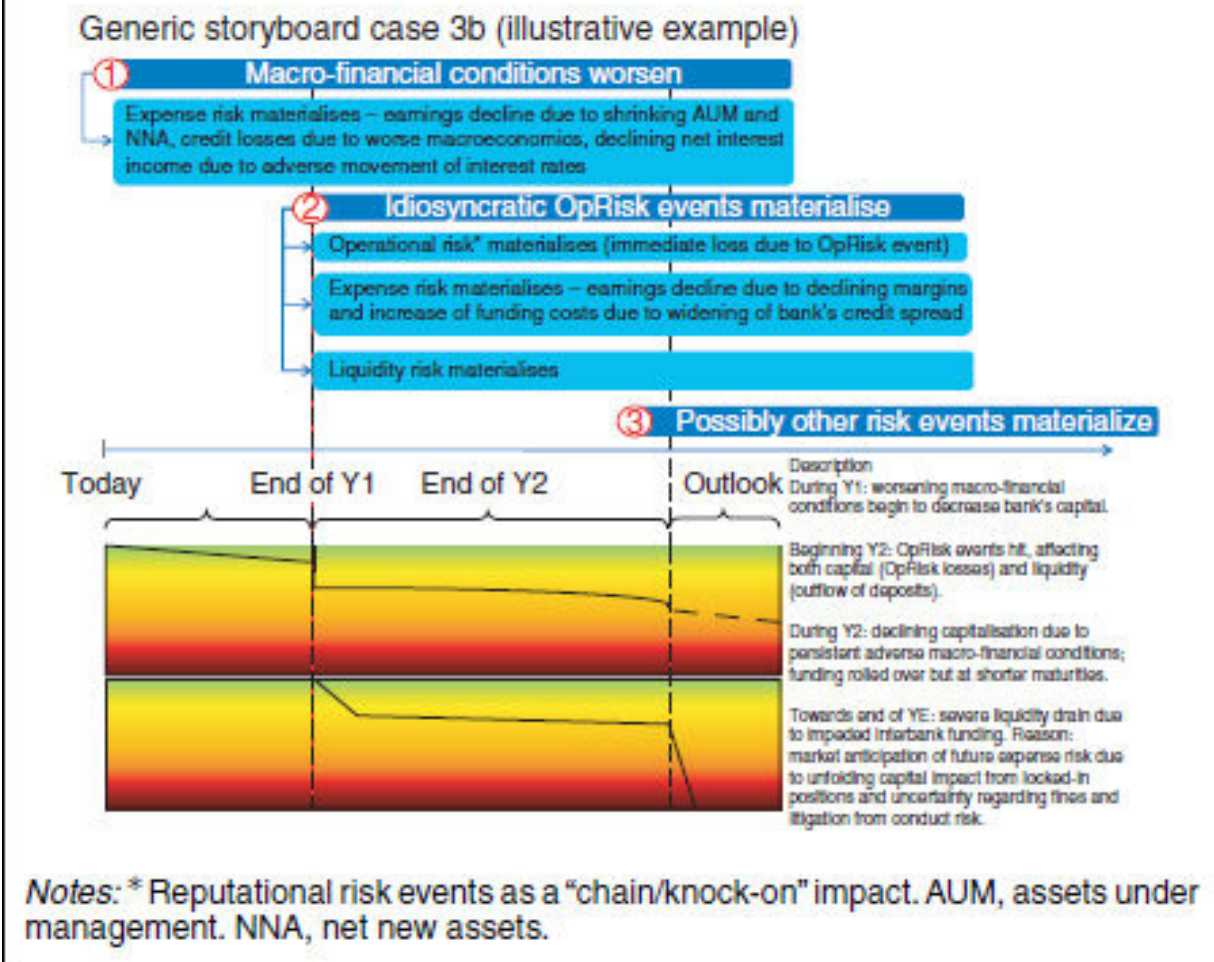


Figure 20.2 shows an example of a generic storyboard for the case 3b group from Table 20.1, which is a capital-induced liquidity failure unfolding over time, ie, a “bleeding out” type of scenario. The timeline in the middle divides the figure into an upper and lower part (the length of the arrow illustrates the time horizon as per the vulnerability analysis). The upper part of the figure shows generic event types, eg, a macroeconomic crisis and an idiosyncratic operational risk event. The graphs in the lower part show how the capital and liquidity resources, respectively, develop over time and use the traffic light zones, red, amber, and green. The solid black lines show the projected behaviour of the capital and liquidity resources over time as the storyboard specified in the upper part of the figure unfolds.

In this example the storyboard is as follows: in year 1 the worsening macroeconomic conditions negatively affect the capital situation of the bank.

However, the capital resource is still in “green” territory: as can be inferred from the lower plot, there is no significant impact on the high-quality liquid asset buffer during year 1. At the beginning of year 2, an idiosyncratic operational risk event results in a loss that induces a drop in capital. With a time lag, this also results in a reduction in the liquidity buffer, which can be seen in the lower plot, which shows the liquidity perspective. During year 2, the persistent adverse macroeconomic conditions cause a further gradual decline in capital, leading to both capital and liquidity ending up in the amber zone at the end of the year. At this point, however, the market starts to anticipate the materialisation of future losses, which would lead to insolvency. This may be associated with other risk events materialising (eg, anticipation of further operational risk incidents that may also trigger conduct and reputational risks) or with losses from locked-in positions that cannot be unwound. The dashed line in the capital plot illustrates this anticipation of a future deterioration of capital. However, before this effect can materialise, the bank suffers a severe liquidity drain and hits a liquidity failure point. The reason is that, within this scenario, counterparties are not willing to roll over existing funding or provide new funding because of the depressed capitalisation associated with the anticipated future losses. This is what we therefore label a “capital-induced liquidity failure”.

Step 4: scenario design and parameterisation

To give more meaning and content to the storyboards,⁶ the next step in the framework is to specify the building blocks of the scenario, ie, to provide the specific calibration of the scenario. This step gives further insight into the range of risk factors contributing to the scenario, their financial impacts and their change under stress. The calibration may (but is not required to) rely on guidance derived from standard scenario analysis, such as historically observed or regulator-imposed scenarios. It may include expert judgement as well as formal quantitative methods, such as principal component and maximum likelihood approaches.

Step 5: plausibility checks and management actions⁷

With regard to plausibility checks, the calibration should be carefully sense-checked and, where possible, backtested against existing stress events. If the likelihood of an RST scenario, and hence risk of business failure, is deemed

to be high, the bank must take mitigating action. In fact, without determining tangible action, RST may be perceived as a purely academic exercise and lack credibility with stakeholders, including regulators.

The open-ended nature of RST means that management's mitigating actions very much depend on the type of scenario identified, and its constituent material risk drivers. In our view, banks should think broadly and deeply at this stage. The proposed framework differentiates between actions that avoid or hedge the risks, and actions that primarily address the effect of a risk materialisation. For example, instead of only asking whether the bank should restructure the business, management may also consider the following questions.⁸

- Can the bank postpone decisions until a time when a major uncertainty has been resolved (eg, following the outcome of a pending regulation or litigation)?
- Can the bank create tests to probe and reduce uncertainties surrounding decisions?
- Can the bank create real options that give the right but not the obligation to take certain actions?

With regard to the effects of RST, banks may, among other things, consider changes to their capital planning, liquidity planning, contingency planning and living will.

Furthermore, management should use the knowledge gained from RST to improve the existing risk management. For example, a bank may decide to review netting agreements with counterparties or initiate systems and process enhancements to reduce the identified operational risks.

Management actions should take into account the likelihood of the scenario. Mitigating actions should be taken immediately for scenarios close to materialisation. Given RST focuses on “very severe but plausible” scenarios, it would be very rare for a scenario to appear in this “high likelihood” category. Lower likelihood scenarios may not be escalated, but should be monitored so that any change in scenario impact or likelihood is captured.

On identifying a new scenario, the first action should be to check whether the risk management framework provides sufficient coverage. Then a

decision should be taken on whether the scenario should be included in the ongoing monitoring and reporting (the final step in the framework).

Step 6: monitoring and reporting

RST scenarios and vulnerabilities should be monitored regularly to trigger management discussions and actions. Over time users will observe changes in likelihood levels. If, for a given RST scenario, the overall scenario likelihood increases, eg, to above a predefined threshold, managers should propose formal mitigating actions and escalate these to the relevant governing body.

Scenario monitoring may be conducted primarily through indicators linked to their constituent risk factors. If a risk is particularly unquantifiable or difficult to monitor empirically (eg, reputational risk), a specific risk management function may be required to assess its status. Ideally, the bank should establish dashboards and combine them with top-down judgement and qualitative reports (eg, from functional area heads) to monitor whether the real world is moving away from or towards a generated scenario.

As well as scenarios, the bank should directly monitor the vulnerabilities identified in step 2. For example, the operational risk function should constantly monitor their top 10 operational risks.

Finally, the overall RST owner, usually a working group or a committee, should be responsible for consolidating the RST results for presentation to the governing bodies. Reports should be produced at least annually to generate awareness of the insight gained from the findings as well as from running the process itself.

REVERSE STRESS TESTING AND ASSET-LIABILITY MANAGEMENT

Bank-wide RST working groups often focus on cross-risk scenarios, similar to the example of the liquidity-induced capital failure illustrated in [Figure 20.2](#). The applications cover all types of risks and are usually run company-wide. As such, they are often led by a function looking holistically across different risk types, such as enterprise risk management.

In these applications, the materialisation of risks usually forms part of a chain of events that ultimately causes an earnings, liquidity or capital failure. This may include ALM risks, such as structural interest rate risks. For

example, Grundke and Pliszka (2015, p. 29) found that, for a particular bank in question,⁹

negative realisation of the latent credit risk factor, a slight to medium downturn of the economy as well as decreased risk-free interest rates for short-term maturities and increased risk-free interest rates for long-term maturities represent the most probable reverse stress test scenario.

In other banks other risks (eg, reputational and conduct risk) may have a greater impact than ALM risks, depending on, eg, the scenario, business model and other factors. For instance, banks may deem certain ALM risks as less material, particularly if risk mitigation strategies and mature ALM processes are in place.

Whatever the case, ALM experts can contribute to RST in at least two capacities, including reviewing the feasibility of hedging strategies in an RST scenario and assessing the scenario impact on the capital and liquidity resources, eg, by providing an assessment of the liquidity outflows or the triggering of capital conversion features. ALM experts can also apply RST and its way of thinking with a narrower scope to find answers on specific ALM questions and the susceptibility of the management of ALM risks to severe stress scenarios. Using the framework introduced earlier, [Tables 20.2–20.4](#) illustrate examples of a wide spectrum of potential applications.

PRACTICAL APPLICATION TIPS

If ALM experts contribute to a bank-wide working group or use RST with a narrower focus, they are likely to appreciate practical application tips.

[Table 20.5](#) summarises lessons learnt by risk managers and ALM experts who have used RST in practice.¹⁰ While their relevance is likely to differ from bank to bank, success may also be a question of how users apply RST in practice. For example, despite the theoretically infinite number of scenarios, new users may find it challenging to reach a failure threshold. This may be interpreted as a positive affirmation that the risk management processes are working. However, it does not mean that RST becomes less relevant. For instance, in his post mortem of the financial crisis, Haldane (2009) considers “disaster myopia” and the poor understanding of network externalities as major reasons why “banks failed the stress test” and explicitly warns that risk managers may succumb to cognitive biases.¹¹ However, even if it is interpreted as an affirmation, RST users can choose a

different operationalisation to increase the relevance of RST for day-to-day management.

Table 20.2 Illustrative example: earnings failure

RST phase	
Definition of points of failure	Loss of earnings of, for example, US\$1 million.
Vulnerability analysis	Larger than expected shifts from modelled non-maturing deposits into maturing deposits, leading to breakage costs of interest rate swaps as the existing ones need to be unwound.
Generic storyboard	What size of unexpected outflows of non-maturing deposits could result in a loss of US\$1 million (failure point), eg, due to the need to unwind hedges that are no longer matched but become open positions?
Scenario design and parameterisation	Instead of using outflows as input to determine the loss under stress, the loss is predefined at, for example, US\$1 million. Then the outflows are stressed until they hit the US\$1 million loss threshold. For illustration, this may be computed similarly to the “goal seek” function in Microsoft Excel.
Plausibility check and management actions	Question whether the probability that the resulting size of outflows will materialise is sufficiently low to feel comfortable with the existing modelling assumptions, hedge policies and positions. If not, management actions may trigger changes in one or more of these aspects.
Monitoring and reporting	The analysis may be re-run on a regular basis (eg, semi-annually) and discussed in executive forums.

- Users may reduce their level of conservatism and severity by adjusting the failure points and identifying and discussing less extreme scenarios. For example, it is often easier to think of scenarios that cause an earnings failure in a particular area than it is to think of a severe capital failure for the entire bank. Even though this does not correspond to the primary purpose of RST, it may nevertheless prove to be a useful application, particularly for the discussion of mitigation of risks that are adverse but not extreme.

Table 20.3 Illustrative example: liquidity failure

RST phase	
Definition of points of failure	Reduction of liquidity buffer, for example, by US\$25 billion.
Vulnerability analysis	Haircuts to mark-to-market value of assets held in liquidity buffer and/or sizeable net outflows by liquidity risk factors (wholesale, retail, derivatives, intraday, franchise, trapped liquidity etc).
Generic storyboard	Idiosyncratic and/or market-wide scenario entails risk factor shifts that are generally larger than those assumed as part of internal stress tests and regulatory liquidity adequacy assessments. Which liquidity risk factors need to change, and by how much, to reduce the liquidity buffer below a given threshold/amount (failure point)?
Scenario design and parameterisation	Experts review which risk factors would change the most in the scenario in question. Parameterisation to determine which combinations of adverse changes in risk factor could result in an outflow of US\$25 billion. Practical parameterisation may verify by how much certain existing stress outflow assumptions need to be scaled to reach failure point.
Plausibility check and management actions	Question whether probability of scenarios with these enlarged/scaled outflow assumptions is sufficiently low. If not, management actions can include changes in liquidity outflow profile and/or increases in the liquid asset buffer.
Monitoring and reporting	The analysis may be re-run on a regular basis (eg, quarterly) and discussed in executive forums.

- Users may consider a combination of multiple adverse events that individually may not be not very severe but which coincide so that

failure points are reached. The adverse scenarios do not need to occur contemporaneously but could occur in quick succession, such that the consequences of the events build up. This can help to not only illustrate that the bank is breakable but also indicate how much it would take to bring the bank's business model to its knees. It also shows the extent to which the bank is projected to survive specific stress scenarios (which is another helpful piece of information). [Figure 20.2](#) provides a practical example.

Table 20.4 Illustrative example: capital failure

RST phase	
Definition of points of failure	Reduction of Common Equity Tier T1 (CET1) ratio, for example, by 25 basis points.
Vulnerability analysis	Mismatch between risk-weighted assets (RWA) by currency and CET1 composition by currency, eg, euro may represent 50% of the RWA (denominator) but less than 50% of the CET1 (numerator).
Generic storyboard	Sizeable currency shock of one or multiple currencies that show mismatch causes reduction of CET1 ratio leading to a breach of risk appetite and/or regulatory requirement.
Scenario design and parameterisation	Application of shocks (appreciation/depreciation of different sizes for major currencies) until the failure point is reached. If options are used as currency hedges, the volatility may also be stressed in discrete steps (eg, increase by 5%, by 10% etc).
Plausibility check and management actions	Question whether probability of scenarios with these enlarged shifts is sufficiently low. If not, application of (additional) hedges, actions to change composition of RWA (eg, fewer mortgage offers in euro) and/or CET1. Where relevant, the bank may also review trade-off between the immunisation of CET1 ratio and changes in the shareholder equity, eg, by adjusting the respective risk appetite.
Monitoring and reporting	The analysis may be re-run on a regular basis (eg, monthly) and reported as part of the standard management information.

- Users may decide to narrow the scope, to focus on specific failure points, ie, to run RST for individual business fields, products, types of

risk and time horizons, with failure points for the intended level of severity. The examples in [Tables 20.2–20.4](#) provide practical illustrations.

Table 20.5 Practical takeaways and examples

Step in RST	Theme	Practical takeaway and example
Definition of points of failure	It may be challenging to identify scenarios that reach failure thresholds	Explain <i>ex ante</i> that capital and liquidity failures often require large exogenous shocks and challenging assumptions. Example: every position in the bank is within the defined risk limits; these limits are derived from the overall capital and liquidity buffers, which are already calibrated to cover extreme losses.
Vulnerability analysis	Positions that provide large gains	Challenge positions that provide large gains to stress scenarios. Investigate the assumptions that give rise to those gains. Example: an RST working group discusses whether these positions should be considered when designing shocks and pathways that push through the failure points.
Vulnerability analysis	Structural hedges of tail risks	Challenge the reliability of structural hedges for the overall portfolio. Example: an RST working group investigates the behaviour of structural hedges in a suite of stress scenarios; the group explores the consequences of the failure of those structural hedges, discusses any magnifying mechanisms for systemic risks and possible management actions.
Vulnerability analysis	Second-order losses	Consider losses even if the bank may not be legally liable. Example: a bank's special purpose vehicles (SPVs) are off balance sheet and a long way from bankruptcy. Failure of these SPVs should normally not affect the capital position but lack of support may drive regulatory and reputational risk.
Vulnerability analysis	Second-order losses	The RST working group considers (credit) contingency risks and network effects. Example: concentration of collateral from margin finance business; failure of a counterparty can lead to material exposure to the underlying securities.
Storyboard and scenario design	Simplistic storyboards	Avoid simplistic storyboards. Example: an ALM risk expert informed their chief risk officer (CRO) that the loss of an internal model waiver may equate to a capital failure. The CRO answered that they were well aware of this without RST and that this one-off "capital hit" would not represent a storyboard that lent itself to deeper RST discussions. By contrast, another RST user considered the loss of the internal model waiver alongside changes in the RWA quantification under the standard approach, capital reductions through defaults, further losses resulting from the reduced monetisation value of collateral and adverse FX moves.
Storyboard and scenario design	Number of scenarios	Develop multiple scenarios, potentially between six and twelve, including a shortlist of a few key scenarios, to consider a sufficient number of vulnerabilities. Example: an RST user received very positive feedback from their board. They presented multiple scenarios to discuss with the board, explaining which of the scenarios required the most management attention and why.
Plausibility check	Underestimation of risks	Check whether parallels can be drawn, eg, from existing businesses, and whether chain effects need to be considered. Example: for some banks, regulatory fines have been significantly higher than expected, and operational incidents triggered conduct and reputational risks.
Plausibility check	Overestimation of risks	Give points of comparison to explain the plausibility of the applied shocks. Example: In one bank, the RST scenario assumed that spreads for government bonds widen to levels many times higher than observed during the financial crisis in 2007–9. The board challenged this assumption and later dismissed the entire RST work.
Management actions	Narrow focus and poor definition of management actions	Consider management literature on, eg, scenario planning (see, for example, Schoemaker 2002) and other tools to facilitate broader discussions on management actions. Example: RST discussions only focused on the reactive measures following the risk materialisation in the assumed crisis; how these risks could be avoided or hedged, eg, by building optionality and flexibility into strategic plans, was not discussed.
Management actions	Execution risks	Use range estimates for management actions instead of the spurious accuracy of (subjective) point estimates. Example: an RST user had intense debates with regulators and auditors around the credibility of the management actions. As management actions are subject to different sources of uncertainty, (eg, feasibility, implementation time, governance approvals) a range may better reflect execution risks.
Monitoring and reporting	Dashboards and ownership	Regularly report whether the "real world" is moving towards or away from the "scenario world". Example: a working group assigned ownership at different levels (eg, scenarios, vulnerabilities, risk factors) and continuously asked each owner for regular updates to get a granular and differentiated picture of where the "real world" is moving.

The other points listed in [Table 20.5](#) illustrate potential “quick wins” and typical pitfalls.

In our view, users should consider RST primarily as a thinking tool. Users should consider it not only in isolation but also as a tool with which the organisation can collaboratively think about risk management. They should apply RST in a structured way. However, it should be filled with problem-specific, agile and at times “unimaginable” thoughts. This may include the integration of various other tools. For example, strategy tools¹² and creativity techniques¹³ may help to identify vulnerabilities, storyboards and management actions.

SUMMARY

RST can be a powerful tool to improve our understanding of exposures of the bank. As the starting point is the result rather than the scenario, RST averts the danger of scenario creep.¹⁴ In particular, it can help to identify the so-called hidden risks that are not considered when running traditional stress tests. Furthermore, RST can facilitate a better understanding of the dynamic interplay of risks over time (eg, between operational, conduct and reputational risk or macroeconomic, credit and structural interest rate risks) and their impact on capital and liquidity. In this sense, it can complement existing tools.

From our experience, we believe, when it comes to RST, that the process is actually more important than the exact figures or detailed scenarios that are being derived. The process should lead senior executives and risk managers through the relevant thought process and thus spread risk awareness across the organisation.

Against this background, we introduced a generic process-orientated framework, which should support a structured process and facilitate discussions on a deeper level. The generic setup means that the framework can remain stable over time, while its content is continually evolving. For example, banks may accommodate new vulnerabilities and different scenarios with the potential to break the bank.

However, we appreciate that there is “no one size fits all framework”.¹⁵ Banks should modify and extend the framework discussed above to suit their specific needs. For example, the proportionality argument also seems relevant in the context of RST:

For smaller, simpler firms, reverse stress testing may primarily be an exercise in senior management judgment focused on scenario selection. For very small firms the submission may be a short written explanation of these factors, which would simply need to be periodically refreshed. It does not necessarily involve detailed modelling. For larger, more complex firms, a more structured and comprehensive approach to reverse stress testing is expected.¹⁶

We propose to experiment with our approach and to consider the individual steps as “adjustment screws”, which should be tailored to this purpose.

We provided different examples of how ALM experts can use RST and the proposed framework to find answers to typical ALM questions and gave an example how of risk managers and ALM experts can work together on RST to consider the impact of scenarios on the liquidity and capital resources.

This was rounded off with practical takeaways, including typical pitfalls. In our experience, it is rare that RST does not result in small tangible measures (eg, changes in legal netting agreements) at least. Where RST does not add value, its application may lack structure and diligence. At the same time, we believe users need to be much more creative in the way they use RST to evaluate the strengths and deficiencies of their business models.

We believe ALM experts and other RST users may also find it useful to link their work with initiatives aimed to improve our understanding of the systemic risks inherent in financial markets. This groundbreaking research (for example, on endogenous risks, amplifying mechanisms, regulatory responses¹⁷ and financial terrorism¹⁸) may help to identify new sets of scenarios.

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- 1 Bottom-up means the vulnerability analysis is first performed in subsectors (eg, business-unit-by-business-unit or risk-class-by-risk-class). A subsequent aggregation step identifies themes and (inter)dependencies.
- 2 Techniques may range from qualitative expert interviews to quantitative risk factor modelling using value-at-risk-type (Monte Carlo) simulations, maximum loss analytics, error trees and impact chains with a critical path. Likewise, a bank may think about an executive off-site workshop that uses strategic management tools such as strengths–weaknesses–opportunities–threats (SWOT) analysis, Porter’s five forces analysis, political–economic–social–technological–environmental–legal (PESTEL) analysis or scenario planning, possibly supported by creativity techniques such as the Delphi method.
- 3 In parallel we reviewed historical cases of banks that failed (eg, Northern Rock and US banks) and incidents that brought financial companies close to failure (eg, “balance-sheet holes” identified by UK financial firms). We also conducted an extensive literature review. For example, triggered by the

interviews, we reviewed empirical studies on the reputational damage caused by certain types of operational incidents.

- 4 In subsequent discussions with risk management representatives, we found a detailed scheme, which mapped operational risk incidents to conduct risk incidents over the life cycle of a client relationship.
- 5 If a single risk factor is identified to have a sufficiently high impact that it breaks the bank by itself, it may become a scenario in its own right. However, multi-factor cross-risk scenarios should still complement it.
- 6 Generic storyboards should support discussions rather than restricting them. Scenarios developed beyond the scope of the generic storyboards are equally valid, and should trigger the same monitoring and escalation action discussed later.
- 7 Steps 5 and 6 of the proposed framework are based on the scenario planning work by Schoemaker (2002).
- 8 There are various further measures (eg, staging of commitments, switching of uses or customer groups or abandoning).
- 9 This paper also gives details on the calibration approach.
- 10 We thank Shan S. Wong for her input on [Table 20.5](#).
- 11 In line with his thinking, RST users may check, among other things, whether they can stretch the sample data set of macroeconomic and financial variables back further (eg, a century instead of a decade).
- 12 For example, the Ansoff matrix, Porter's five forces, PESTEL analysis, SWOT analysis.
- 13 For example, 6-3-5 method, force fitting method, mind mapping, lateral thinking.
- 14 Analysing scenarios that are not bound by historical precedence is a good complement to stress testing, which often relies (implicitly or explicitly) on historically observed scenarios.
- 15 Katalysys Ltd, "Reverse stress testing: tool to improve business planning and risk management" (see <http://www.katalysys.com/reverse-stress-testing.html>).
- 16 See Endnote 15.
- 17 For relevant literature see, for example, <http://www.systemicrisk.ac.uk/>.
- 18 See, for example, Belleron's, "How to Kill a Bank" vision paper (available at <https://www.belleron.net/>).

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XVAs and the Holistic Management of Financial Resources

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Faced with the seemingly unending spawning of XVAs – the valuation adjustments to the fair value of derivatives – since the 2007–9 financial crisis, anyone who used to trade interest rate derivatives prior to the crisis could be forgiven for thinking, as William of Occam allegedly said, *entia non sunt multiplicanda praeter necessitatem*, ie, entities should not be multiplied without necessity. Those were the days when a swap was a swap and you knew what you were trading. Then came the realisation that a swap was actually the carrier of multiple diseases, hidden in the small print of the contracts, and the world has not been the same since.

In the aftermath of the crisis, market players had to come to terms with the reality of so-called “second-order” risks, in particular counterparty risk and funding risk, impinging on their positions. As their counterparties could no longer be counted on to honour their commitments, and as liquidity became a scarce and costly resource, banks had to go through their derivatives portfolios thoroughly and identify all the asymmetries in what looked like matched books from a market risk perspective. First, the banks had to tell apart collateralised and uncollateralised transactions, to ascertain their

exposure to counterparty default; this meant also seeing whether or not their International Swaps and Derivatives Association (ISDA) master agreements allowed netting of payables and receivables.¹ Then, for collateralised counterparties, they had to pore over the features of their credit support annexes (CSAs) and map the frequency of margining, the kind of collateral that could be posted and other technicalities (thresholds, material transfer agreements (MTAs), etc): all features that had been negotiated in a pre-crisis world when they really made no difference to risk perception and pricing.

In short, what looked like a matched book of linear interest rate swaps turned out to be a bundle of credit and funding risks contingent on the market-to-market of underlying derivatives, or “netting sets”, when netting was allowed: what used to be a swap had actually become a complex multi-asset derivative, embedding hybrid options, with payoffs depending on the interaction of the underlying interest rate (and exchange rate, for cross-currency swaps), the credit spread of the counterparty and the bank’s own credit risk or funding spread.

This sparked a veritable revolution in pricing, risk management, market practice and organisation, which can be summarised by two major trends.

On the one hand, there was a huge push within the industry for the collateralisation and standardisation of derivative contracts. This has also been forcefully encouraged by the regulator: at the time of writing, a standard interest rate swap is not just collateralised, but also cleared with a central counterparty (CCP). Thanks to the strict margining rules and other risk mitigation measures enforced by the CCP (initial margins, default fund, etc), centrally cleared swaps come closest to the “riskless” derivative of yore, even though, as we shall see, the risk mitigation measures can themselves entail funding costs that must be taken into account in pricing and risk management.

On the other hand, banks have had to develop the capability to price and manage the second-order risks embedded in swaps that differ from the standard, whether they are old trades sitting on banks’ legacy books or new trades that have been negotiated with counterparties that (not being required by regulation) are either unwilling or unable to clear or even collateralise their transactions. These include primarily the uncollateralised derivative transactions with corporate customers, and then all the bilaterally collateralised transactions that are not cleared, are “weakly collateralised”

in terms of frequency of margining and/or quality of the collateral and have embedded “cheapest-to-deliver” options (eg, under multi-currency CSAs).

The pricing and risk management of these transactions pose multiple challenges.

- First of all, in the fair value of the derivative banks have to include the valuation adjustments (XVAs) representing the underlying risks. This process started with the credit valuation adjustment (CVA), which was already an accounting item before the crisis, but was not normally priced or hedged, and then moved on to the debt valuation adjustment (DVA) for own credit risk, the funding valuation adjustment (FVA), and so on and so forth, down to discussions about new items for initial margin valuation adjustment (MVA) and capital valuation adjustment (KVA); as the evidence from market pricing builds up, the XVAs find their way into accounting practice.
- Pricing and revaluation of the XVAs is computationally highly complex: apart from the simplest payoffs, there are no closed formulas for pricing, and Monte Carlo simulation methods are required to project the expected exposure over the life of the transaction; the calculation of the sensitivity of XVAs to risk factors is even harder: smart maths (eg, adjoint differentiation) and computing power are key resources.
- Provided that the ISDA master agreement with the counterparty permits netting, the computations must be performed not for the single transaction, but for the whole netting set: what must be computed is the marginal contribution of the transaction to the XVAs of the netting set; this is a completely different logic from the traditional stand-alone pricing of derivatives and one that does not fit the (commercial) front-office platforms.
- Hedging XVAs requires a dynamic multi-asset hedge portfolio to be set up: the XVAs are subject not just to the level of the underlying, counterparty credit spread and own credit spread, but also to their volatilities and correlations; for most of these risks no traded instrument is available, which means that the hedges (which are often proxy hedges) must be periodically rebalanced, and that the position is subject to potentially adverse correlations (so-called “wrong-way risk”).

These complexities prompted a double response from the banks.

- First, there has been heavy investment in financial engineering and technology in order to come up with not just pricing formulas and numerical methods but also proprietary platforms that can handle Monte Carlo simulations on multi-asset, multi-currency netting sets (sometimes including hundreds of transactions) and provide pricing and revaluation of XVAs, and compute risk sensitivities in a time frame consistent with the requirements of pricing new transactions and portfolio hedging.
- There has been an organisational change in the front office, with the creation of dedicated units that take care of pricing and hedging XVAs (the “XVA desk”).
 - In the pre-deal analysis phase of a non-standardised derivative, the XVA desk provides the pricing of all the valuation adjustments to the sales and market-making desk; these will be the building blocks out of which the final price will be quoted to the customer or counterparty.
 - Once the derivative is traded, second-order risks are transferred from the market-making desk to the XVA desk against the payment of a fee: given that managing these risks typically exceeds the expertise and mandate of the market-maker, and in view of the fact that the various cross-correlations between risk factors make for economies of scale in hedging, it makes sense to concentrate them in a single place.
 - While the XVA desk can be found in different places in the banks’ organisational charts, hedging “second-order risks” dovetails with two other areas of “bank resource management” (BRM): collateral and liquidity management. The idiosyncratic features of CSAs determine the residual counterparty and funding risk of collateralised netting sets (eg, collateral mismatches, jump to default risk, downgrade triggers), while the collateral balance represents one (possibly the most dynamic) of the components determining the funding needs of the bank and, according to the regulatory treatment in the calculation of liquidity ratios (liquidity coverage ratio (LCR) and net stable funding ratio (NSFR)), the maturity structure of the funding

strategy. In addition, to the extent that variation and initial margins can be covered with securities, the bank can engage in “collateral optimisation”, in order to minimise the cost of meeting its margin calls.

- Ultimately, for investment banks and corporate and investment banking divisions of commercial banks, where the derivatives business absorbs a relevant share of risk and balance sheet, BRM units (usually including treasury, counterparty risk and capital management) have widened the scope of traditional ALM units, of which in a sense they are the more dynamic heir, and have developed a holistic approach to transfer pricing and resource optimisation across the banking book and the trading book. This requires an effort to bring together the “deterministic” and the “contingent” components of liquidity and credit risk, and to apply a consistent pricing methodology across different products, in order to achieve a bank-wide efficient allocation of scarce resources.

To sum up, the challenges of the new derivatives market call for a holistic approach to the management of financial resources on multiple levels: a unified pricing framework, a shared technological platform, an integrated risk management framework and a consistent organisational structure. In what follows we go through the different XVAs in greater detail, focusing on their interactions, as well as on the synergies between XVA management and the wider management of “scarce resources”.

CAPITAL STRUCTURE OF DERIVATIVE REPLICATION

The textbook approach to derivatives pricing is based on the fundamental theorem of asset pricing. In particular, the possibility of replicating a contingent claim in a complete market leads to the “law of one price” via no-arbitrage strategies. This theory was developed in the second half of the twentieth century and led to the valuation paradigm of expectation under the risk-neutral measure associated with the bank account numéraire. It also led to the amazing growth of the derivatives market, which came to a halt, however, when the 2007–9 crisis showed that some of its assumptions were only valid as a first approximation.

In fact, it could be argued that the 2007–9 crisis coincided with a systemic crisis in the classic assumptions for pricing contingent claims, which included:

- market completeness, which implies the availability of replicating strategies;
- no-arbitrage, which implies the law of one price;
- information relating to the two parties to the transaction is irrelevant;
- bank accounts carry negligible risk of default;
- replicating strategies can be built at the individual deal level (no portfolio effects).

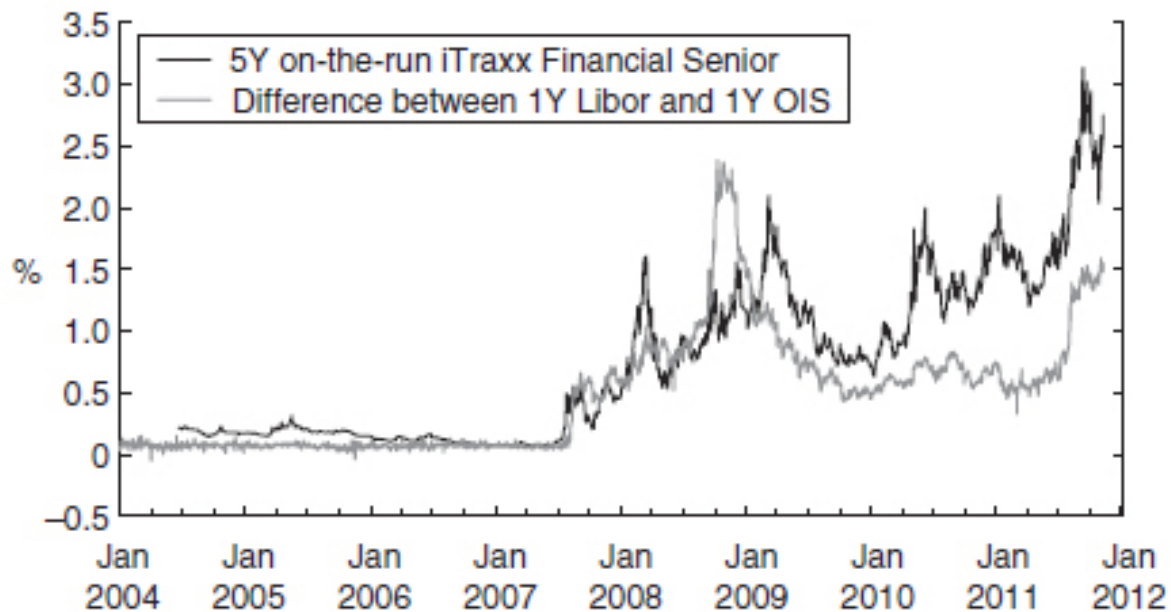
The emergence of CVA risk provided a privileged perspective on this historical development. With regard to the first two points above, as we shall discuss below (see pp. 539ff), CVA is a hybrid payoff, which often cannot be fully hedged in the market. Moreover, unlike the third assumption, information relating to the two parties is a key ingredient of bilateral CVA. Portfolio effects (the final assumption) due to nonlinearity and netting make CVA hard to evaluate; traditional front-office systems based on additive sensitivities cannot cope, so new systems must be built to manage CVA risk.

Before discussing CVA hedging, we look in more detail at a key assumption that was challenged: the risk-free nature of the bank account.

Pricing with a risky bank account

The bank account numéraire sits at the heart of the risk-neutral pricing approach. The fact that it grows at the risk-free rate is testimony to an era when derivatives trading was far removed from treasury operations. The 2007–9 crisis changed the perspective of the market on the remoteness of bank default events. Banks are really defaultable entities that finance their operations at a positive spread over the risk-free rate. This was soon reflected in the price of default protection, and in the London Interbank Offered Rate (Libor) basis ([Figure 21.1](#)). A first attempt to deal with the problem was simply to increase the bank account rate by the funding spread (Piterbarg 2010). However, this approach created consistency issues with the modelling of own default in DVA, as we discuss later (see pp. 542ff).

Figure 21.1 Banks lost their default-free status during the 2007–9 crisis, as evidenced by the selected market data



Source: Banca IMI.

The theory of XVA is essentially a theory of how to adjust the textbook approach to pricing by reflecting secondary risks and costs incurred in the production of derivatives. It therefore requires a theory of derivatives production, or “replication” as it is known in the field, including a consistent model of its capital structure.

Some considerations on hedging CVA

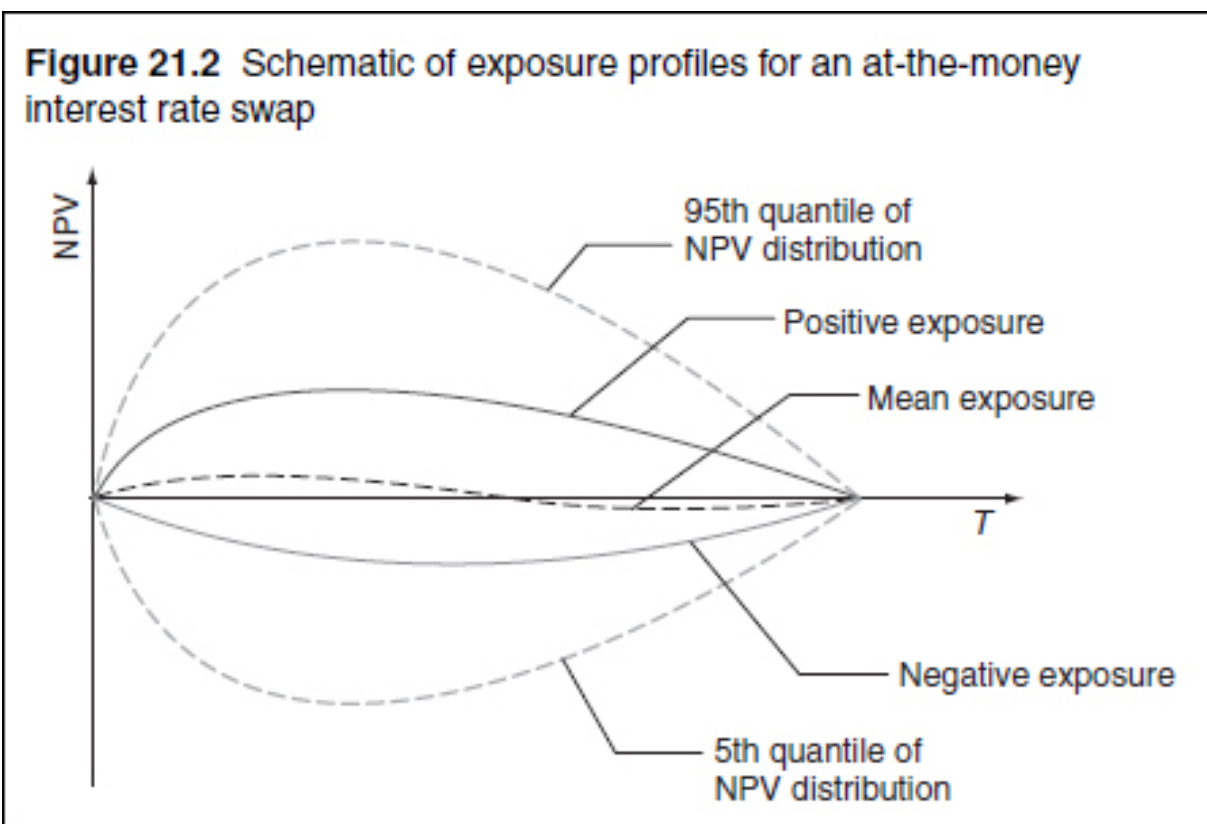
CVA is the expected discounted value of the loss due to counterparty risk

$$\begin{aligned} \text{CVA}(t) &= \mathbb{E}_t[D(t, \tau_C) \text{loss}(\tau_C)] \\ &= -\mathbb{E}_t[D(t, \tau_C) 1_{\{t < \tau_C \leq T\}} 1_{\{\tau_C < \tau_I\}} \underbrace{\text{LGD}_C(V_0(\tau_C))^+}_{\text{EAD}}] \quad (21.1) \end{aligned}$$

where $D(t, \cdot)$ is the risk-free discount, τ_C is the default time of the counterparty, LGD_C its loss given default, V_0 is the close-out value of the derivatives position, EAD denotes exposure at default, and the indicator functions $1_{\{t < \tau_C \leq T\}}$ and $1_{\{\tau_C < \tau_I\}}$ activate the payoff only when the

counterparty defaults before the maturity of the deal T and before the bank's default time τ_D , respectively.

Netting is a major mitigant of counterparty risk. However, when it is applicable V_0 must represent the value of the entire derivative netting set: in this case, the nonlinearity of the positive part operator $(\cdot)^+$ means that it is not possible to split the CVA calculation according to the asset class of the underlying, thus creating portfolio effects.



At the same time, CVA can be seen as the expected cost of hedging counterparty risk. CVA risk has two main drivers.

1. **Credit spread** contains the information on counterparty's creditworthiness; it determines the distribution of τ_C .
2. **Derivative exposure** represents the notional of the (contingent) credit risk.

Hedging CVA means hedging credit spread and exposure drivers. The positive part operator in exposure at default (EAD) introduces nonlinear

dependence on exposure drivers. Moreover, the fact that V_0 is evaluated at default time τ_C creates a functional relationship between credit and exposure risk: if exposure drivers move, CVA's credit spread sensitivity will change. Similarly, a movement in the credit spread of the counterparty will change CVA's sensitivity to underlying exposure drivers. This interplay is captured by cross-gammas (second-order mixed sensitivities).

CVA is a hybrid risk, as it joins risk factors characterised by a broad range of market liquidity. Assuming that CVA can be hedged is often a bold statement: in practical terms, for some risk factors (eg, the credit spread of smaller corporate clients, or long-dated inflation volatility), a hedging market may not exist. This has implications in terms of the economic capital associated with unexpected losses of CVA net of hedges. On the other hand, the Basel III capital regulation takes a prudential view of basis risks that may arise when standard credit default swap (CDS) protection is used to mitigate counterparty risk; often for this reason CVA hedges cannot be fully recognised as mitigants of regulatory capital requirements. In general, regulation was relatively slow to acknowledge the emergence of the hybrid market risk nature of XVAs.

The role of capital

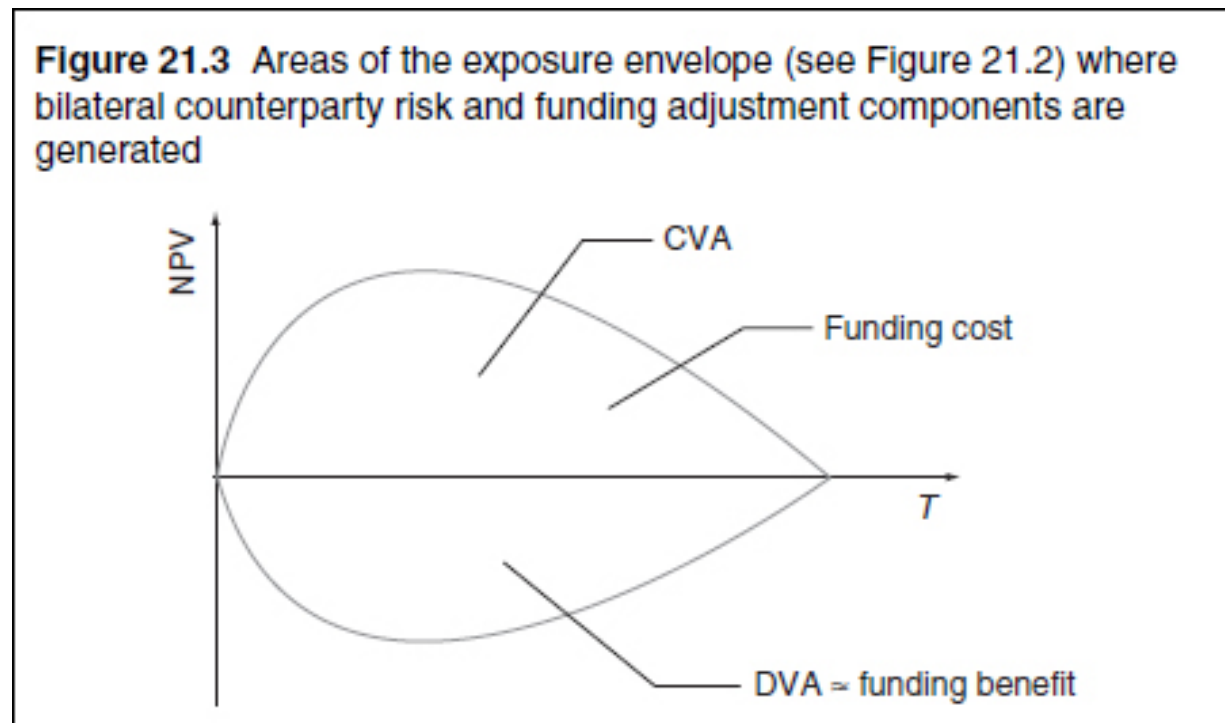
KVA is an established acronym of the valuation adjustment for the cost of capital. The interaction with CVA is quite strong, as discussed in Prampolini and Morini (2017). From an economic perspective, capitalisation is essentially a strategy that complements replication; it may be the only strategy available when hedging is not possible, and to account for model risk. In a way, protection from risk by CVA is like buying insurance, while protection using economic capital is like acting as an insurance company.

On the other hand, hedging CVA mitigates economic capital for counterparty risk, again pointing to the need for a holistic XVA approach. Regulatory requirements, however, may recognise hedging only partially, such as in the mitigation of counterparty (counterparty credit risk) jump-to-default risk with credit default swaps, thereby forcing banks to hold capital against risks that are substantially mitigated by CVA hedges.

In any case, the value of KVA cannot be calculated by using a replication approach, precisely because the capitalisation of risk emerges where hedging is ineffective. This again is indicative of the crisis in textbook pricing theory,

based on market completeness and no-arbitrage assumptions. On the other hand, return-on-capital considerations are becoming key measures of the profitability of investment banks, as profit must be evaluated against the capital structure that can sustain it.

In fact, “profit” is a better characterisation of capital remuneration: capital has a “cost” only in the sense that the net profit it attracts is expected to meet some target. This shows how hybrid XVA is: some of its parts are included in accounting fair value and lend themselves to hedging and replication; others are not really hedgeable and require a different optimisation approach.



The value of DVA

Interactions between the XVAs are not limited to CVA–KVA. In this section we explore the funding implications of DVA, which represents the obverse of counterparty risk, in the sense that my DVA coincides with the CVA of my counterparty

$$DVA(t) = \mathbb{E}_t[D(t, \tau_I)1_{\{t < \tau_I \leq T\}}1_{\{\tau_I < \tau_C\}}LGD_I(V_0(\tau_I))^+] \quad (21.2)$$

As discussed in Morini and Prampolini (2011), DVA essentially coincides with the benefit part of FVA (apart from differences in the spread applied for measuring the two adjustments); see [Figure 21.3](#). Acknowledging this

overlap is essential to avoid double-counting the benefit, and it also has implications in terms of managing DVA risk.

Hedging DVA is a notoriously controversial proposition: DVA takes the form of a positive payoff associated with default on own obligations. However, DVA is a (positive) component of fair value with a negative carry; if left unhedged it becomes a cost and expires, worthless, at the maturity of the derivative position. Common strategies for realising the value of DVA include exposing the bank to correlated credit risk, and using contingent liquidity generated by derivative liabilities to mitigate the cost of required funding.

DVA hedging has obvious synergies with managing the risk of CVA exposure drivers. With regard to own credit spread risk, particularly for related contingent funding allocation, coordination with the bank's treasury department is essential.

Pricing the capital structure consistently

A particular interaction exists between KVA and FVA: capital allocated to a trade should reduce its funding requirement. But does capital effectively provide funding resources to the business? In principle, with the issuance of equity liabilities, cash resources should become available in the form of free capital; these cash resources do not attract a cost of funding, but rather attract the excess margin generated by the firm.

A first consideration is that pricing KVA entails a capital allocation discipline, whereby free capital is made available to the business to offset contingent financing requirements for the production of derivative instruments. If KVA is introduced without this discipline, derivative production would be less efficient, as it would have to pay the inflated funding costs of a balance sheet where the liability side is composed entirely of borrowed money obligations (excess margin would still remunerate capital as measured by KVA).

If the discipline is implemented, the introduction of KVA should be reflected in the FVA formula by taking into account a trade's lifetime capital profile.

CONTINGENT LIQUIDITY

Funding value adjustment (FVA) is the adjustment to the derivative risk-free valuation that accounts for its funding cost/benefit calculated in a manner consistent with the other XVAs (mainly CVA and DVA) (Castagna and Fede 2013). It is usually calculated by taking into account the incremental effect of the insertion of a new deal into the current portfolio, by assuming for convenience that market risk is hedged by a perfect mirror deal traded with a market counterparty, according to the prevalent market standard (cleared through a CCP or simply margined under a golden CSA, depending on the typology of the derivative).

FVA is complex: finding and properly pricing the asymmetry that can alter the funding needs of your derivative portfolio is the name of the game. Derivatives with uncollateralised counterparties (ie, corporates) are the tip of the iceberg, but by scratching the ice you can easily verify that FVA is also actually required for derivatives with weak or asymmetrical collateralisation, basically for all those trades margined with features inconsistent with a golden CSA.

Regulatory requirements, such as leverage ratio (LR), NSFR and LCR play a significant role in defining the liquidity profile of the derivative portfolio and how liquidity needs can be managed. For example, LR and NSFR do not allow the positive present value (PV) of a specific netting set to be offset with the collateral received if margining is scheduled on a weekly basis; according to the NSFR framework you are requested to sum

$$[(PV^+ - \text{collateral received}) - (PV^- - \text{collateral posted})] \quad (21.3)$$

If this quantity is positive, the liquidity shortage has to be funded by available stable funding for more than one year; if it is negative, the liquidity surplus can be invested only on maturities of less than six months.

This asymmetry poses a big challenge in pricing a fair FVA compliant with regulatory requirements, because, for a very balanced portfolio, every change of sign of the expected exposure calls for the application of a wide bid–offer spread. Since the pricing of a FVA is based on the evaluation of contingent liquidity profiles, this approach would be detrimental to business by generating significant drawbacks. A more pragmatic solution is to set your price according to the average expected portfolio profile in the long run: if you expect to get a large contribution from uncollateralised receivables, you should price FVA by using the whole term structure of your funding; otherwise, you should use the cost of funding on short-term tenors only.

Unlike CVA and DVA, hedging the FVA means managing the profit and loss (P&L) volatility by gradually adjusting the multi-year funding plan and the mix of funding sources. Let us consider the key sensitivities of a portfolio of uncollateralised interest rate receivable derivatives, hedged by mirrored collateralised transactions. This portfolio is negatively correlated to an interest rate reduction and/or an increase in the funding cost. At first sight these sensitivities are similar to those of the CVA/DVA. Thus, you could manage your portfolio by a strip of interest rate linear instruments and a proxy hedge of your funding spread, such as a CDS on the iTraxx Financial Senior index or on a basket of peers (buying your own CDS by posting the nominal amount as independent amount is not a viable solution).

This approach generates a negative carry, since you still have to fund the liquidity shortage over the lifetime of the portfolio: then it is essential to use a funding source that is sensitive to the market changes of your own credit spread in order to fund incoming margin calls and simultaneously offset daily P&L.

Let us move on and analyse the case of collateralised derivatives. FVA is still required in many cases, whenever we are margining the derivative portfolio not according to a golden CSA. The main features of a golden CSA are: daily calculation and bilateral posting of both variation margin (VM) and initial margin (IM); zero threshold; MTA < 500k, euro and US dollars as eligible currencies; cash-only margining. It is easy to verify that this constraint is not respected in the case of CSA with weekly margining, one-way CSA (ie, with supranational entities), CSA with special purpose vehicles or pension funds that can post only bonds or CSA that requires posting collateral to a third-party account if the rating of the receiving party goes below a predefined threshold.

One of the trickier cases is represented by the bilateral bond-only CSA. First, it requires that all derivative cashflows are discounted by the repurchase agreement (repo) curve of the “cheapest to deliver” collateral up to the latest maturity of the portfolio. Actually, this is a challenging task since derivatives are likely to have maturities longer than 10 years, while the repo market is not liquid beyond a one-year tenor. Furthermore, with the implementation of the European Central Bank quantitative easing, the core-Europe general collateral (GC) itself is now many basis points far away from the euro overnight index average (Eonia) fixing and it is no longer

viable to use the overnight indexed swap curve as a good proxy of the “cheapest to deliver” GC curve.

Second, LR and NSFR set some regulatory constraints that are likely to reduce the chance to recourse to the market in order to restore the net balance of securities received or to be posted under CSA through repos and reverse repos. In order to tackle and manage these issues holistically, an effective collateral optimisation is required that is designed to cover both IM and VM posting requirements on bilateral CSA and CCPs and the needs of the LCR liquidity buffer, by selecting the optimal mix of cash and bonds that reduces liquidity costs and capital absorption.

Derivatives traders are familiar with the concept of FVA, but asset and liability managers, and treasurers, should also approach this topic with a holistic view. Whenever assets or liabilities are swapped from fixed to floating or vice versa in order to manage the interest rate risk, these derivatives are likely to produce changes in the collateral account. These deals, cleared through the CCP or margined under bilateral CSA, produce an MVA and affect contingent liquidity needs in the same way as the hedging deals of uncollateralised derivatives. At first sight you could compare this case to that of the asset swap; actually, if the bond purchased can be repoed in the market on overnight tenor, every delta on interest rates will produce margin calls and changes in the repo equivalent value with opposite sign. On the other hand, securities or assets such as mortgages, which cannot be effectively self-financed through market channel with a daily market reset mechanism (eg, the daily margining required by the global master repurchase agreement), produce the same effects as the uncollateralised derivatives portfolio. It is easy to verify that this situation almost always occurs for swapped liabilities, because it is quite impossible to get funding collateralised by your own paper.

Therefore, practitioners should consider extending FVA coverage to the whole balance sheet, with a holistic view, in order to optimise the contingent liquidity risk across trading and banking books, even if the existing regulatory framework does not recognise this kind of interplay and ultimately does not support all-inclusive management of the daily P&L of the contingent liquidity risk.

Initial margins, MVA and collateral optimisation

As the bulk of over-the-counter derivatives moves to central clearing, market players have to factor in further costs coming from the IM requirement and a raft of similar charges that are levied by the CCPs. IMs are posted over and above VMs in order to mitigate the closeout risk, ie, the risk of adverse changes in the market value of a defaulted member's position during the so-called "margin period of risk", between the day they default and stop posting VMs and the day the close-out amount is calculated: should the latter exceed the VMs, IMs provide a protection for the CCP. According to the risk profile of a member's position, the CCPs may ask them to post various add-ons to IMs; in addition the member has to contribute to the default fund, which is calibrated to cover the default of the CCP's two largest members. Whereas VMs can be posted only in cash, members can meet their IM requirements either in cash or by posting securities, according to eligibility criteria and haircuts peculiar to each CCP. While IMs and default funds are a pillar of the CCP's risk management framework (which contributes to the CCP's low credit risk weighting), they represent a significant cost for the members, especially as they attract an 85% required stable funding factor under NSFR and hence must be funded for tenors of more than one year. Since the IM requirement is computed through a (stressed) value-at-risk-like (VaR-like) simulation of each member's position, it depends on the volatility of the underlying instrument and the sensitivity of the overall position: this means that players with one-sided flows are prone to large margin requirements.

When a bank enters into a new transaction that entails a change in the sensitivity of its netting set with a CCP (eg, an uncollateralised interest rate swap with a corporate customer, hedged with a CCP-cleared swap with a market counterparty) it should factor-in the price the cost (benefit) of an increase (decrease) in its IM requirement. To the extent that the position is expected to stay on the books until maturity (which may well be the case for a corporate customer), the bank should compute the expected cost over the whole life of the transaction: this requires an estimate of the profile of the incremental IM requirement over time, which in turn involves a complex simulation of the transaction's marginal contribution to the VaR of the netting set with the CCP. Based on this profile, the funding costs are computed using the term structure of the bank's own credit spread and discounted back to today. This expected funding cost is computed, and possibly charged, at the time of trading, but it has not yet graduated to a proper value agreement: no market player has yet adopted the MVA as a component of fair value in its

accounts. However, this is a widely expected development, especially since, from September 2016, major derivatives dealers have also been posting IMs on non-cleared derivatives (eg, swaptions and cross-currency swaps under bilateral CSAs), and the regulation should extend progressively to the whole banking industry.

Managing the cost of IMs is a multi-faceted exercise that a bank can approach (in principle) along three dimensions.

1. Minimising the risk profile of the netting set with a CCP: this can be achieved by sourcing two-way business from customers and/or squaring positions with counterparties with the opposite exposure, for instance, “backloading” legacy transactions from bilateral CSAs to the CCP.
2. Routing their business to a CCP where the IM requirement is less costly because of different computation methods, collateral eligibility criteria, haircuts or fee structure: clearly this is possible only when a product is cleared by more than one CCP (eg, interest rate swaps are cleared by LCH.Clearnet, the Chicago Mercantile Exchange and Eurex) and when the counterparty to a trade is willing (and able) to clear through the same CCP; different CCP rules and dealers’ positioning have even evolved a basis market for moving transactions from one CCP to another.
3. Optimising the allocation of collateral between IM calls: since CCPs accept securities as well as cash for IMs, a member will try to post the “cheapest to deliver” asset to each CCP, in order to minimise the funding cost. When a bank has multiple IM calls, it can engage in a “collateral optimisation” exercise and allocate eligible securities in its inventory to the various CCPs with a view to maximising their “collateral value”, which is a function of the repo rates and of the haircuts and fees set by each CCP; a bank could even source cheap collateral it does not own from the market and post it to a CCP. In this optimisation exercise a bank should consider the LCR liquidity buffer alongside the IM calls; in this respect an organisation setup where collateral management, repo financing and liquidity management sit in the same team is clearly a competitive advantage.

CONCLUSIONS

The developments in regulation and market practice that followed the financial crisis upset the derivative landscape and have definitely changed the economics of the business. To compete in the derivative arena, banks need to be able to tackle the complexity of second-order risks and more generally to reap the gains of an efficient management of scarce resources. A holistic approach to XVA management is a key component of this integrated effort, usually entrusted to BRM units that aim to widen the scope of traditional ALM, bringing together the banking book and the trading book and the “deterministic” and the “contingent” components of liquidity and credit risk. In doing so they apply a consistent transfer pricing methodology across different products and enforce a bank-wide efficient allocation of scarce resources, reaping the benefits of synergies along the whole value chain.

As we argued, to be successful this approach needs three basic ingredients.

1. **Human talent:** you need the right mix of quants and traders to stay on top of a field of finance that is still evolving and to successfully manage a complex multi-asset book with hybrid exposures and often unpleasant cross-gamma effects.
2. **Technology:** in order to adapt to a fast-changing environment, a flexible platform is needed that allows you to quickly deploy the new functions; significant investment in computing power and techniques is needed to deal with the sheer computational complexity of XVA values and sensitivities, which means that developing your own proprietary system gives you a key competitive advantage, given the relative lack of established commercial vendors.
3. **Organisation:** XVA management is at the heart of an efficient deployment of the bank’s scarce resources, and the ability to accurately measure the second-order risks of your derivative book is vital in order to price new derivative transactions properly and to optimise liquidity and capital consumption. To this effect, having the XVA desk within a BRM unit that includes treasury and possibly RWA management is a distinct advantage, allowing you to have an integrated view on how the derivative business fits within the wider liquidity and capital management of the bank and reap the benefits of synergies along the whole value chain.

This chapter represents the views of its authors alone, and not the views of Banca IMI.

- 1 Banks got a foretaste of the impact of credit risk on derivatives pricing with the move to a multi-curve environment according to the tenor of the underlying Libor index; see [Figure 21.1](#).

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Optimal Funding Tenors

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Barclays

Since the financial crisis of 2008, funding matters have received considerable attention from financial institutions. This concern can be easily understood by observing that spreads for unsecured long-term borrowing have dramatically increased,¹ even for the leading global financial firms, having a large impact on the accrual cost of their unsecured balance sheets.

Also, as banks painfully realised during the crisis of 2008, the infinite liquidity assumption – the ability to always raise money – does not hold for any financial institution, and the business model of lending long term (mortgages), borrowing short term (deposits) and earning the carry implies a considerable liquidity risk.

This chapter focuses on funding requirements for uncollateralised derivatives. For such trades, the main paradigm is that the mark-to-market (MTM)² needs to be funded at all times, and all funds need to be raised using the firm's unsecured funding rate (see the section starting on page 557). To understand the main issues, consider the simple case of a European call option. Using the paradigm, at trade initiation day, the bank needs to fund the option's MTM, that is, borrow that amount for a certain time. This borrowing will result in accrual costs at a rate that depends on the chosen funding tenor. On the next business day, assume that the option price has increased. Using the paradigm again, the bank needs to fund the new MTM. The amount it

needs to raise will depend on the MTM move and on how much of the previously raised funding has expired, which in turn depends on the funding tenor chosen on day zero. In general, this amount will be non-zero, and the bank is exposed to two kinds of risk: the availability of funding (liquidity risk) and changes in funding cost (funding risk), which is determined by the prevailing funding rate.

In this chapter we introduce a framework that makes it possible to analyse different components of funding cost. These components include costs associated with borrowing at a reference rate,³ costs associated with spreads of the firm's unsecured funding rate over the reference rate and costs associated with friction and regulatory requirements (see the section starting on page 559). These costs are computed as an adjustment to the trade's MTM available in systems live at the time of writing using a framework whose methodology closely follows the actual business logic, that is, it issues bonds of various tenors when raising funding (see the section starting on page 557).

The correct choice of funding tenors depends on risk preference and trade details. For example, consider a trade with only static cashflows. Because the tenors and the size of the cashflows are known at trade initiation, we can expect to eliminate liquidity risk by term funding, that is, by issuing funding instruments such that their payoffs and tenors match the static cashflows. However, this is clearly not possible for derivatives with variable payoff, eg, for European call options. For these trades, we could choose a set of tenors and fund to the expected cashflows. However, in addition to not eliminating liquidity risk, we would overfund in about 50% of the market scenarios,⁴ thus introducing friction cost when lending the excess money back to the market. The choice of funding tenors and corresponding funding amounts defines the funding strategy.

Our framework allows different funding strategies to be discriminated quantitatively in terms of costs and risks. Explicit results comparing costs and risks between a long-term strategy and a short-term strategy on an interest rate (IR) portfolio are presented (see p. 566). Readers who do not have a deep interest in scenario specifics should consider jumping straight to the fourth section (see pp. 566ff) to get an overview of derivative funding strategies. Advanced, real-world funding strategies are considered in the fifth section (see pp. 571ff). We conclude the chapter by relating our methodology to theoretical developments (Piterbarg 2010; Burgard and

Kjaer 2011a,b) in the risk-neutral framework, including a discussion on the impact of different choices for the reference rate (Libor versus OIS).

SIMULATION FRAMEWORK

Our simulation framework is based on a Monte Carlo engine where each Monte Carlo path can be considered as a possible evolution of today's market. As we shall show, this approach makes it easy to compute various risk measures, eg, percentiles, which will be used to determine funding risks. It also allows us to implement a funding methodology closely following the actual business logic, that is, issuing bonds when raising funding, and to decouple the simulation model from funding methodology.

This section contains three subsections. The first introduces the Monte Carlo engine and gives an overview of the implementation of our funding methodology. Then the details of the simulation model are presented, and we explain on simple examples the graphical representation of the results returned from our framework.

Monte Carlo engine

Within the Monte Carlo engine, a large number of Monte Carlo paths on a given set of stopping dates $\{T_0, \dots, T_N\}$ are simulated. Here T_0 represents the start date of the simulation, henceforth called day zero. Each path should be considered as a possible evolution of day zero's market. At each stopping date and path, a market is constructed which can include foreign exchange (FX) rates, IR yield curves, funding curves, volatility surfaces and other market objects. Various cross-asset portfolios specified at day zero can be evolved along these paths, and the MTM values of their life-cycled trades can be computed at each stopping date and path using the generated markets. At each stopping date and path, trades can be added and removed from these portfolios, reflecting, for example, refinancing transactions.

To apply this engine to our funding problem we begin by specifying a portfolio representing the trading desk's book on day zero of the simulation. Then we simulate a set of future market evolutions. Using asset-class-specific pricing models we compute the MTM of the portfolio at future stopping dates, and, in particular, determine its future funding needs. Note that these MTM values are path dependent. Depending on the funding needs at each future stopping date, we build IR bonds, equivalent to issuing funding

instruments, and add them to a second portfolio. The coupon payments of these funding instruments represent the funding cost (paid on accrual basis), and we collect them in dedicated accounts, thus obtaining pathwise results on the funding cost during the simulation. More details on this methodology will be given in the third section.

Simulation models

In general, day zero markets are live markets or closed markets stored in some database. To generate future markets we have to use a model. An important observation is that banks are interested not only in the expected funding cost, but in the associated funding risk. Since this risk cannot be hedged, it is described correctly by the distribution of the funding cost (eg, 95th percentile) in a real-world measure and not a pricing measure. In addition, our simulation model allows us to model funding rates such that forward rates are not necessarily realised.

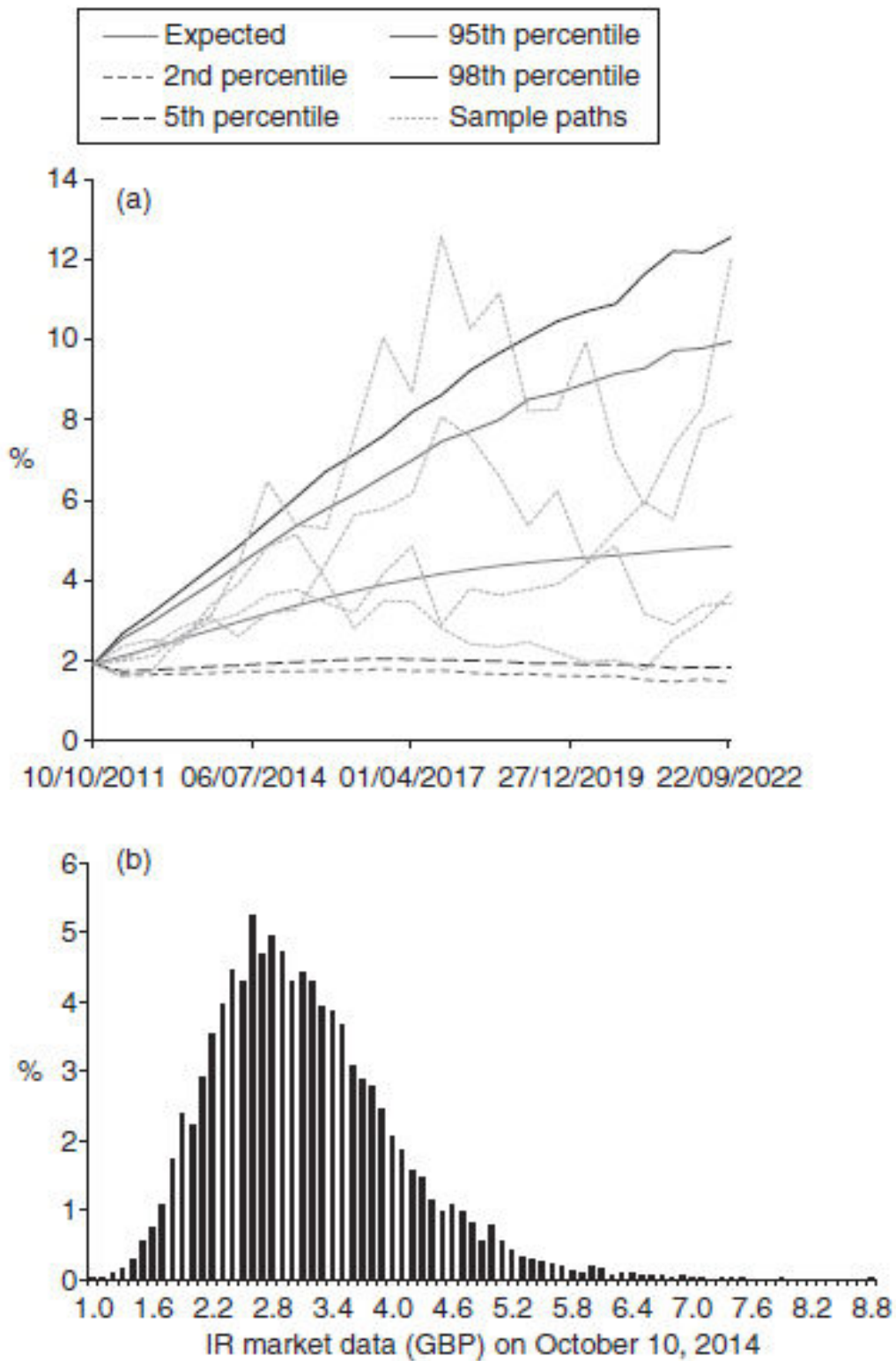
We use a cross-currency model with support for additional asset classes. In this chapter we focus on modelling the IRs. To begin with, we model a set of swap rates of constant maturity. Each rate is driven by its own lognormal, mean-reverting stochastic process with time-dependent coefficients. Using these simulated rates, a reference yield curve is built for each currency at each future stopping date via bootstrapping. Unsecured funding rates are modelled via funding spreads over the reference yield curves. These spreads are modelled as a set of corporate bond constant-maturity spreads of various tenors. Each spread is driven by its own normal or lognormal mean-reverting process with time-dependent coefficients. Using these spreads and the corresponding reference yield curves, uncollateralised funding curves are built at future stopping dates. To calibrate these processes at day zero, a term structure of trends and volatilities for each process, a mean reversion speed parameter and a correlation matrix between all stochastic processes are specified. These parameters can be read from a historical database, can be implied from the current market or manually specified.⁵

Although we have focused on a specific model in the real-world measure, its worth mentioning that the Monte Carlo engine is agnostic to the simulation model, allowing us to use simulation models of any dynamics, including pricing models.

Cones for market data and trades

Cones are used to display most of our quantitative results within this chapter, including the evolution of market data, trade MTM values and funding costs. A cone is a diagram displaying the time evolution of the expected values and various percentiles of a modelled quantity. In particular, a cone contains information on the marginal distribution at each stopping date.

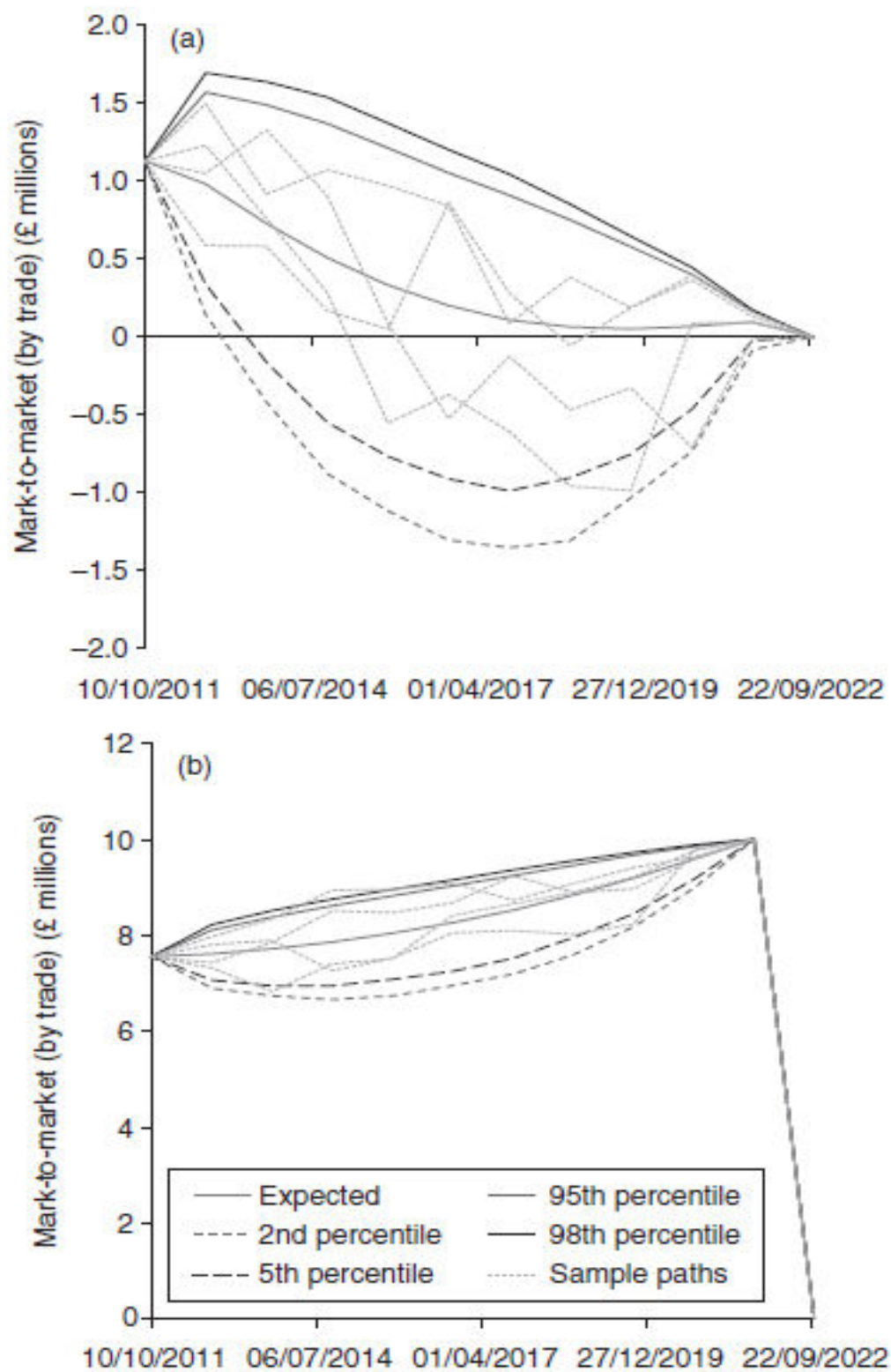
Figure 22.1 5Y swap rate



(a) Cone; (b) distribution.

To illustrate the concept, we consider the results of a 10Y semiannual simulation which includes modelling Libor swap rates. The day zero market implies a 5Y swap rate of 1.9%, while the parameters of our model imply an expected swap rate of 4.8% in 10Y. By running 1,000 paths of our simulation, future markets are generated and queried for the 5Y swap rate.

Figure 22.2 MTM cones



(a) 10Y swap; (b) zero-coupon bond.

To construct the cone of the 5Y swap rate (Figure 22.1(a)), the market at each path and stopping date is queried for the simulated yield curve and its implied 5Y swap rate, and (for each stopping date) the expected value and various percentiles are computed. In addition to the cone, the marginal distribution at each stopping date can be reported, eg, the distribution for the 5Y swap rate at after 3Y of simulation (Figure 22.1(b)). Note that the distribution confirms our lognormal assumption for the swap rates.

Using the simulated markets, trades can be priced on each stopping date and path. We consider two examples which will be important later in the chapter, a 10Y swap (see p. 566) and a zero-coupon bond (see p. 562).

The cone for 10Y Swap MTM (Figure 22.2(a)) displays the MTM evolution of a 10Y receiver swap with £10 million notional, 4% fixed rate. The cone for the zero-coupon bond (Figure 22.2(b)) displays the MTM evolution of a bond with £10 million notional expiring in 10Y.

FUNDING METHODOLOGY

This section introduces the funding methodology used in our simulation framework to fund uncollateralised derivatives. It consists in four subsections. In the first, the main assumptions are stated and motivated. Then, in the second, the different components of the funding cost which will be considered in this chapter are defined. The funding methodology is introduced in the third subsection. It mirrors the actual behaviour of the trading desks when raising funding for their derivative books from Treasury. This section should be read in close conjunction with the final subsection, where the methodology is illustrated on the simple example of funding a zero-coupon bond.

Assumptions

The two main assumptions are as follows.

- (i) MTM values are computed via discounting using a reference rate.
- (ii) The full MTM value of our portfolio has to be funded at all times. In particular, if the MTM is positive, we have to borrow the MTM from the treasury at an unsecured funding rate, while a negative MTM can be lent to the treasury at an unsecured funding rate.

We shall show (see p. 575) that under suitable constraints these assumptions guarantee that our funding cost agree with the theoretical funding adjustment derived in Piterbarg (2010) and Burgard and Kjaer (2011a). We shall now give some heuristic motivation justifying our assumptions.

The first assumption reflects the fact that the available MTM values in the existing live banking systems are computed via a reference rate, eg, OIS or Libor.⁶ Specifically, these MTM values reflect market prices and do not depend on the bank's funding cost. The funding cost should be computed as an adjustment to these MTM values.

To explain the second assumption we can follow two lines of reasoning.⁷

1. Consider an uncollateralised derivative with positive MTM. To buy this trade, the desk must raise the money at the unsecured funding rate. Assume that the MTM has increased by the next day. The desk could either sell the trade immediately and post the MTM change as profit to the shareholders (dividends) or keep the trade. When the desk keeps the trade, the MTM change should still be considered as profit, and be made immediately available to the shareholders (dividends⁸). Since the trade cannot be used as collateral, the MTM change has to be raised using the unsecured funding rate; hence, the full MTM must be funded. Now consider the case of a trade with negative MTM. Assuming that there are other trades with positive MTM, this trade will reduce the amount of required funding on the total portfolio by its MTM. Assuming the MTM of the bank's derivative portfolio is positive, all trades with negative MTM can be considered as reducing the funding cost.
2. Again consider the case of the desk holding an uncollateralised trade with positive MTM. The desk will be required to hedge the associated market risks. This could be done by executing the opposite trade with another big financial institution, which is usually done on collateralised basis. Hence, the hedging trade has no funding adjustment.⁹ Since the hedging trade has negative MTM, the desk needs to post collateral. Assuming full collateral posting, the desk must raise the trade's MTM at all times (at the unsecured funding rate, since the original trade cannot be used as collateral). Since the desk earns the reference rate on the collateral, funding costs are caused by the spread of the unsecured funding rate over the reference rate. When holding an uncollateralised

trade with negative MTM, its corresponding hedging trade will generate cash from collateral postings, which in turn can be given to the treasury to earn an unsecured funding rate.

Funding costs

In this section we introduce a set of different costs that are associated with funding.

- The cost occurring when borrowing at the rate used in live banking systems to compute the derivative's MTM. Historically, this reference rate was the Libor rate; hence, we shall call its associated cost "Libor charge".¹⁰ In the classical Black–Scholes context this cost offsets (on an expected level) the MTM evolution of the trade, eg, for a zero-coupon bond (discounted with the reference rate), this cost describes the difference in today's MTM and the bond's notional payment at expiry.
- The spread cost associated with borrowing at the firm's unsecured funding rate instead of borrowing at the reference rate. In the risk-neutral funding framework (see p. 575), this cost describes the funding adjustment to the MTM computed using the reference rate. We call it the "cost of funds".
- The bid–offer cost represents friction cost: it is introduced into the framework by a bid–offer spread over the unsecured funding rate that is applied when borrowing and lending money to match the funding requirements. The bid–offer spread is expected to be around 10bp.
- Spread cost associated to regulatory requirements: the regulator has introduced various requirements to reduce liquidity risk associated with short-term funding. In general, these requirements translate into a tenor based cost which is represented in our framework by curve (eg, a 6M rate of 30bp and a 5Y rate of 10bp). We call this curve regulatory spread curve and add its rate as a spread to the unsecured funding rate. We call the associated cost "regulatory liquidity cost".

Methodology

Our methodology to compute funding costs follows closely the actual business logic, which can be summarised as follows: assume the MTM of the

portfolio of a trading desk at a given business day is positive. This implies that the desk needs to raise funding from the treasury. Assume the desk decides to raise funding for one year. Then it would issue a bond with a notional of the current MTM.¹¹ The coupons of this bond reflect the funding costs to the desk, and hence must include Libor cost, cost of funds, bid–offer cost and regulatory liquidity cost. Assume that at the next business day the MTM has increased. Hence, the desk needs to raise additional funding, so it issues another bond. Actually, since the original bond has not yet expired, it needs to raise only the increment in the MTM (minus any unpaid accrual from the original bond). An account which collects these coupon payments will reflect the true funding costs to the desk. Negative portfolio MTM is associated with lending money, that is, with buying the corresponding bonds.

In order to differentiate between various components of the funding costs, in our framework we issue a set of bonds, as detailed below, and collect the charges in different accounts. Note that the tenor of the bonds corresponds to the period over which the funding is raised and that the costs are charged via accrual.

We describe below in detail the different types of bonds and accounts. The reader is advised to read these subsections in close conjunction with the example presented on page 562.

Libor charge

To capture the Libor charge, at each stopping date T_i a set of Libor floating bonds $B^L(T_i, M)$ are built with quarterly coupons and tenor M . For funding shorter than 3M the coupon is paid at expiry. The floating leg ensures that the bonds price at par every quarter and no additional IR risk is introduced. The reference rate¹² will be picked up from the generated markets during the simulation. The tenor M of the bonds corresponds to the period over which the funding is raised and the sum of the notional of all issued bonds agrees with the amount to fund, $AMF(T_i)$, at stopping date T_i . Coupons representing the Libor charges are paid into the Libor account, where they are rolled at the reference rate.

Cost of fund charge

To capture the cost of fund charge, at each stopping date T_i , a set of fixed bonds $B^S(T_i, M)$ are built with the same coupon dates, notional and tenors as issued Libor bonds (see above). The fixed coupon rate corresponds to the stochastically simulated funding spread $s(T_i, M)$ and is picked up from the simulated markets. Coupons representing the cost of fund charges are paid into the cost of fund account, where they are rolled at the unsecured funding rate. Note that charges will result in a cost when borrowing and an income when lending money.

Bid–offer charge

To capture the bid–offer charge, at each stopping date T_i a set of fixed bonds $B^{BO}(T_i, M)$ with the same coupon dates, notional and tenors as issued Libor bonds is built (see the earlier paragraph on Libor charge). The fixed coupon rate corresponds to the bid–offer spread and is picked up from a rolled deterministic bid or offer curve within the simulated markets representing the bid and offer spreads for different tenors. A bond with positive notional implies that we need to borrow (hence, we use the offer curve), while a bond with negative notional implies that we lend money (hence we use the bid curve). Coupons representing bid–offer charges are paid into the bid–offer account, where they accrue over time. Note that charges will result in a cost when borrowing and when lending money; hence, they do not offset each other.

Regulatory liquidity charge

To capture the regulatory liquidity charge, at each stopping date T_i a set of fixed bonds $B^{RLC}(T_i, M)$ with the same coupon dates, notional and tenors as the issued Libor bonds is built. The fixed coupon rate corresponds to tenor-dependent spread, representing costs associated to regulatory liquidity requirements and is picked up from a rolled deterministic curve within the simulated markets. Coupons representing regulatory liquidity charges are paid into the regulatory liquidity cost account, where they accrue over time. Note that a set of netting rules depending on the tenors is applied, eg, borrowing £2 million for 5Y and depositing £1 million for 2Y results in a charge of only £1 million.

Amount to fund

To finalise our methodology we must determine the amount to fund, $ATF(T_i)$, at each path and stopping date T_i . It follows from our assumptions that at day zero

$$AMT(T_0) = MTM(T_0)$$

On later stopping dates, bonds already issued but not yet expired must be taken into account. (They correspond to funding raised at previous stopping dates.) We denote by $M_B(T_i)$ all bonds issued but not expired by T_i , and by $M_B^L(T_i)$ all such Libor bonds. In addition, for each bond B_l we denote by N_l its notional and by $A_l(T_i)$ its unpaid accrual at T_i . Then, the amount to fund at stopping date T_i is given by

$$AMF(T_i) = MTM(T_i) - \sum_{l \in M_B^L(T_i)} N_l - \sum_{l \in M_B(T_i)} A_l(T_i) \quad \text{for all } i \quad (22.1)$$

Example

To clarify the definitions given above, we illustrate our funding methodology on the simple example of the zero-coupon bond (see p. 554). Our strategy is executed semi-annually, starting at the bond issuing date and ending at the bond expiry date. All funding instruments are build to the next execution day and a 6M funding spread of 50bp, a flat bid–offer rate of 10bp and a 6M regulatory spread of 30bp are assumed. We use the Libor rate as reference rate.

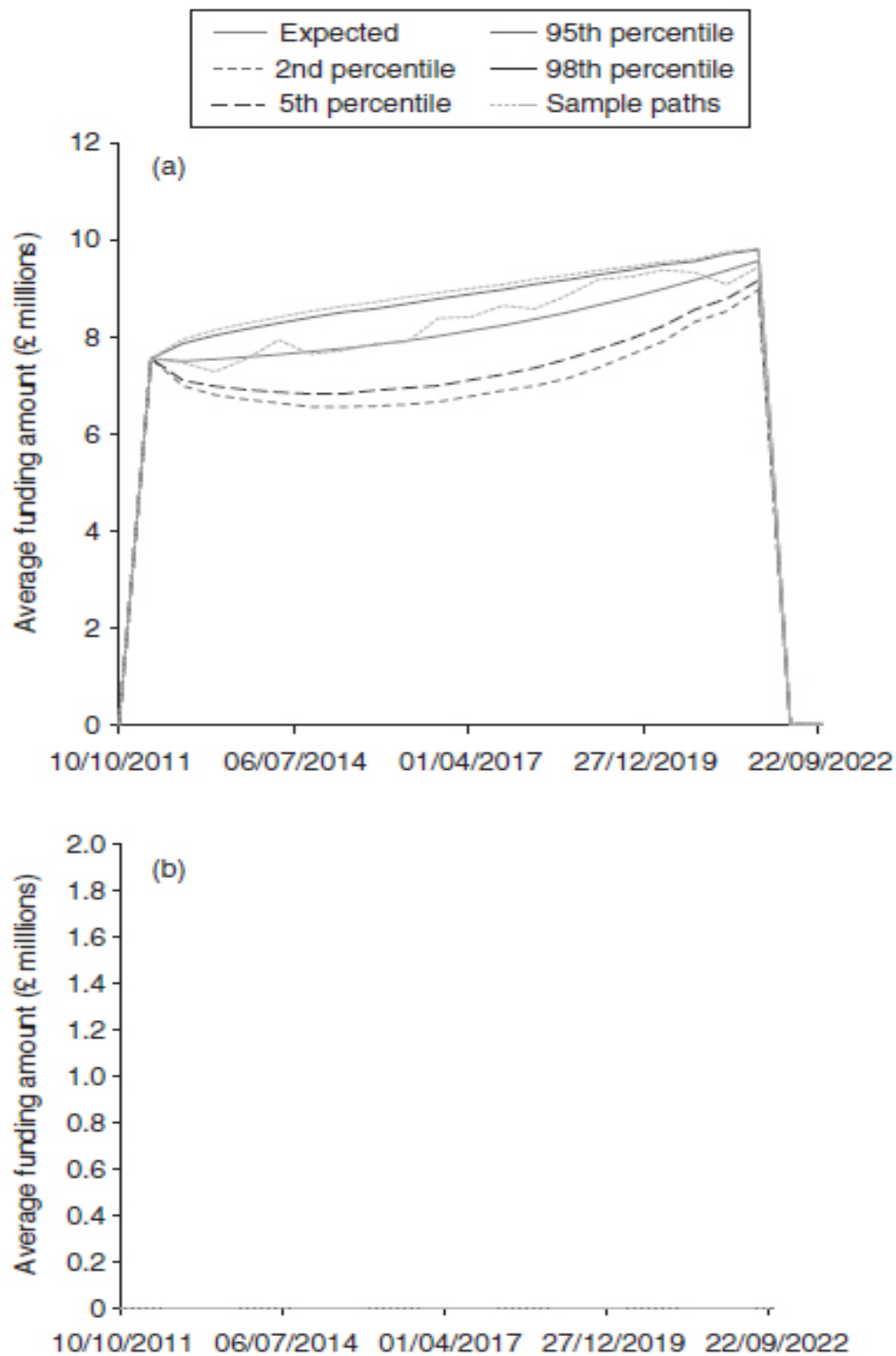
To understand the different funding charges, it will be useful to consider first the evolution of the borrowed and lent amount.

The amount to borrow (Figure 22.3(a)) describes the amount which has to be raised to fund the trade, while the amount to lend (Figure 22.3(b)) describes the overborrowed amount from previous stopping dates. Hence, in this strategy we only borrow, but never lend money. This can be understood by recalling that at the simulation start date we need to fund the full MTM, which (for a zero-coupon bond) is positive; hence, we borrow. On all future stopping dates, the funding instruments from the previous dates have expired. Thus, we always need to fund the full MTM, which for all paths is positive;

hence, we borrow. In particular, the evolution of the borrowing amount follows the MTM evolution of the zero-coupon bond for each path and we never lend money back.

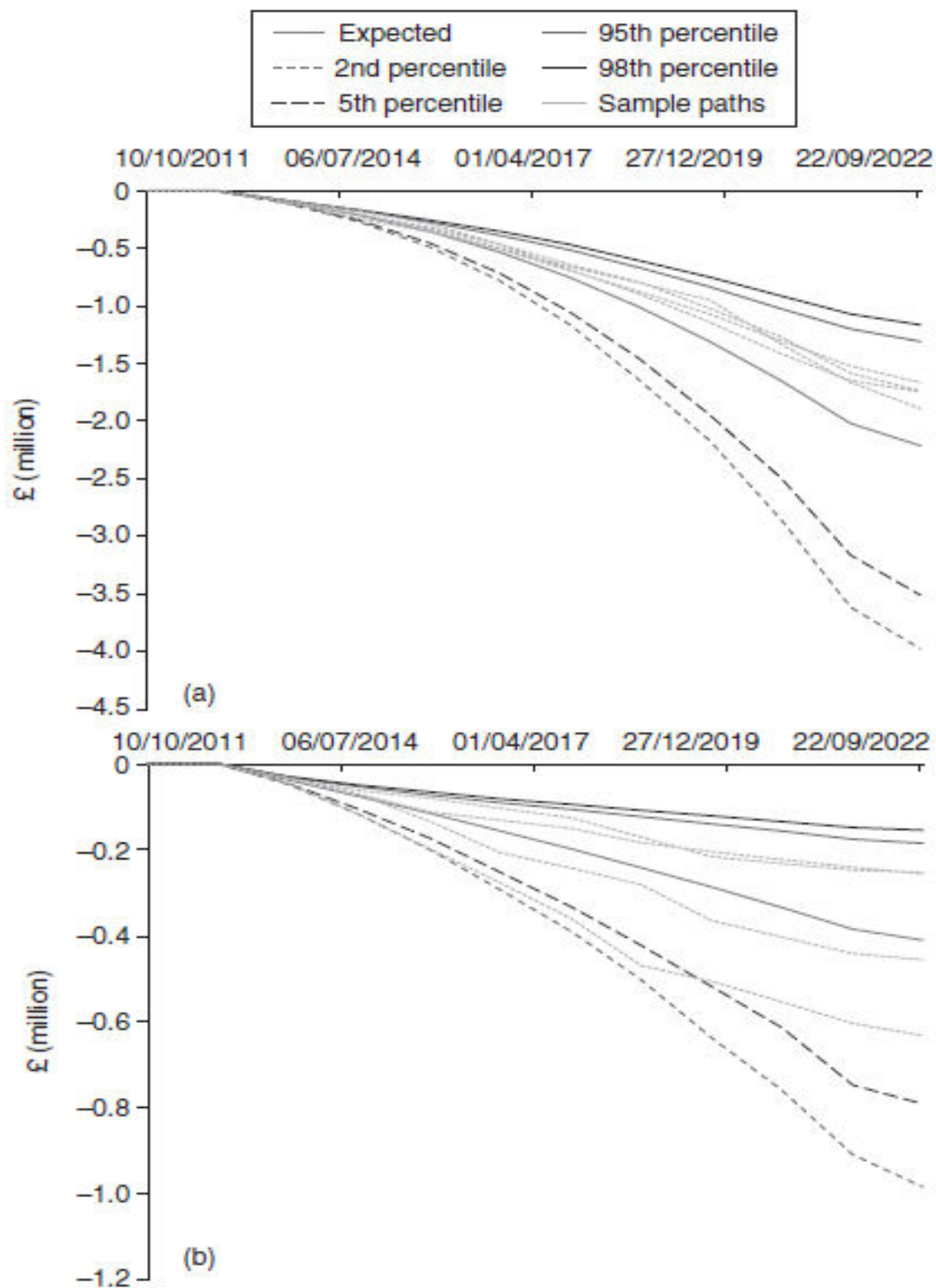
Using the Libor curve as the discount curve, we find an MTM at simulation start date of £7.7 million. This agrees with the expected Libor charge (Figure 22.4(a)) of –£2.3 million plus the £10 million notional. This reflects the fact that the Libor costs (on the expected level) are offset by the MTM evolution of the trade. For the average cost of fund charge we find £0.4 million (Figure 22.4(b)). This can be understood by assuming an average funding need of £8 million with a funding spread of 50bp for 10Y.

Figure 22.3 Amounts



(a) Borrowed; (b) lent.

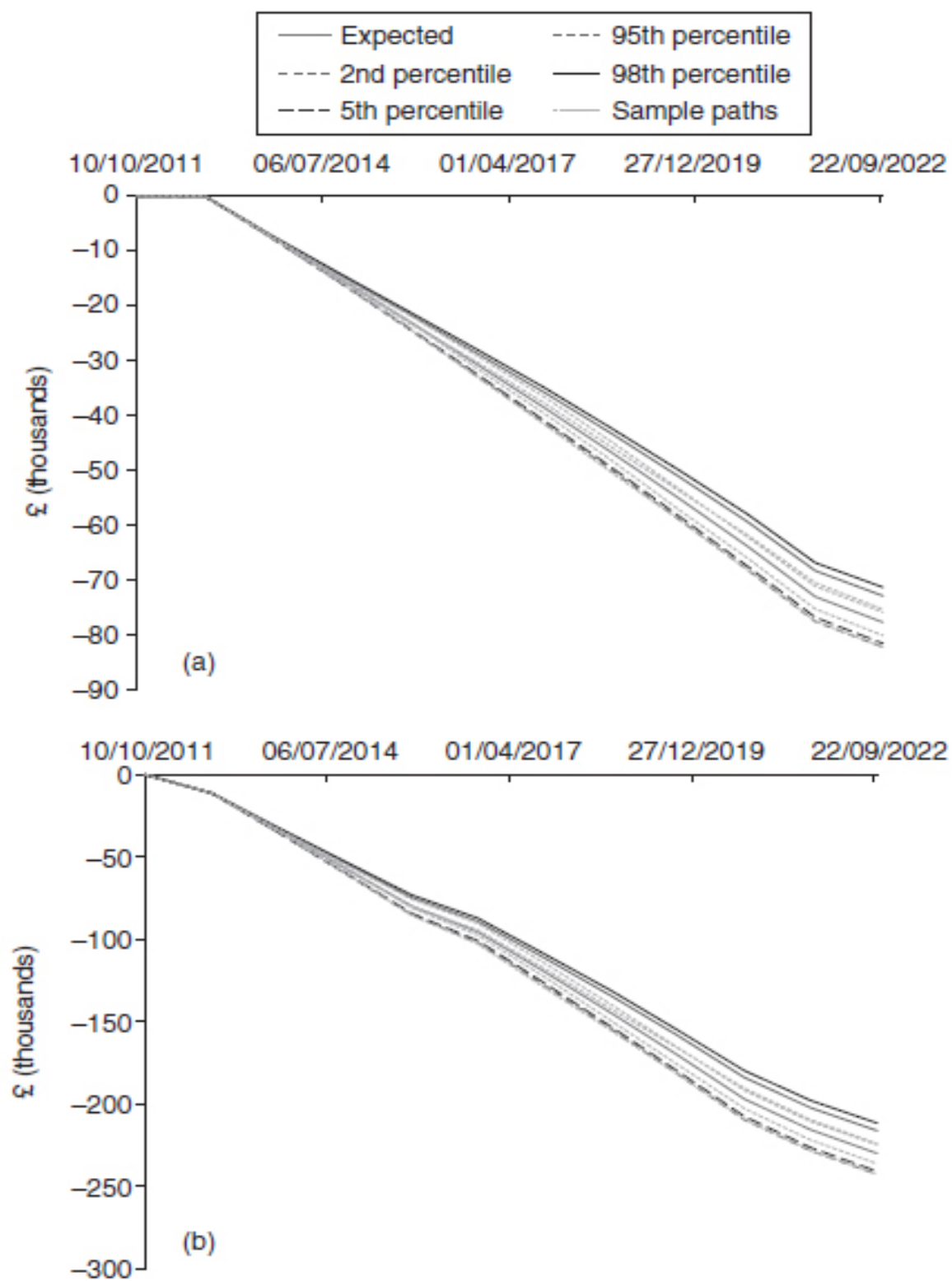
Figure 22.4 Charges



(a) Libor; (b) cost of fund.

For the average bid–offer charge ([Figure 22.5\(a\)](#)) we find £80,000, which can be explained by again assuming an average funding need of £8 million for 10Y and a flat bid–offer spread of 10bp. A similar argument explains the regulatory liquidity charge ([Figure 22.5\(b\)](#)) of £240,000 from the assumed regulatory spread of 30bp. In both cases, the observed volatility in the cones comes solely from the volatility in the MTM of the bond.

Figure 22.5 Charges



(a) Bid-offer; (b) regulatory liquidity.

The strategy described in this section represents the limit of funding as being as short as possible (funding instruments are built to the next strategy execution date). In general, such a strategy leads to high liquidity risk.

LONG-TERM VERSUS SHORT-TERM STRATEGY

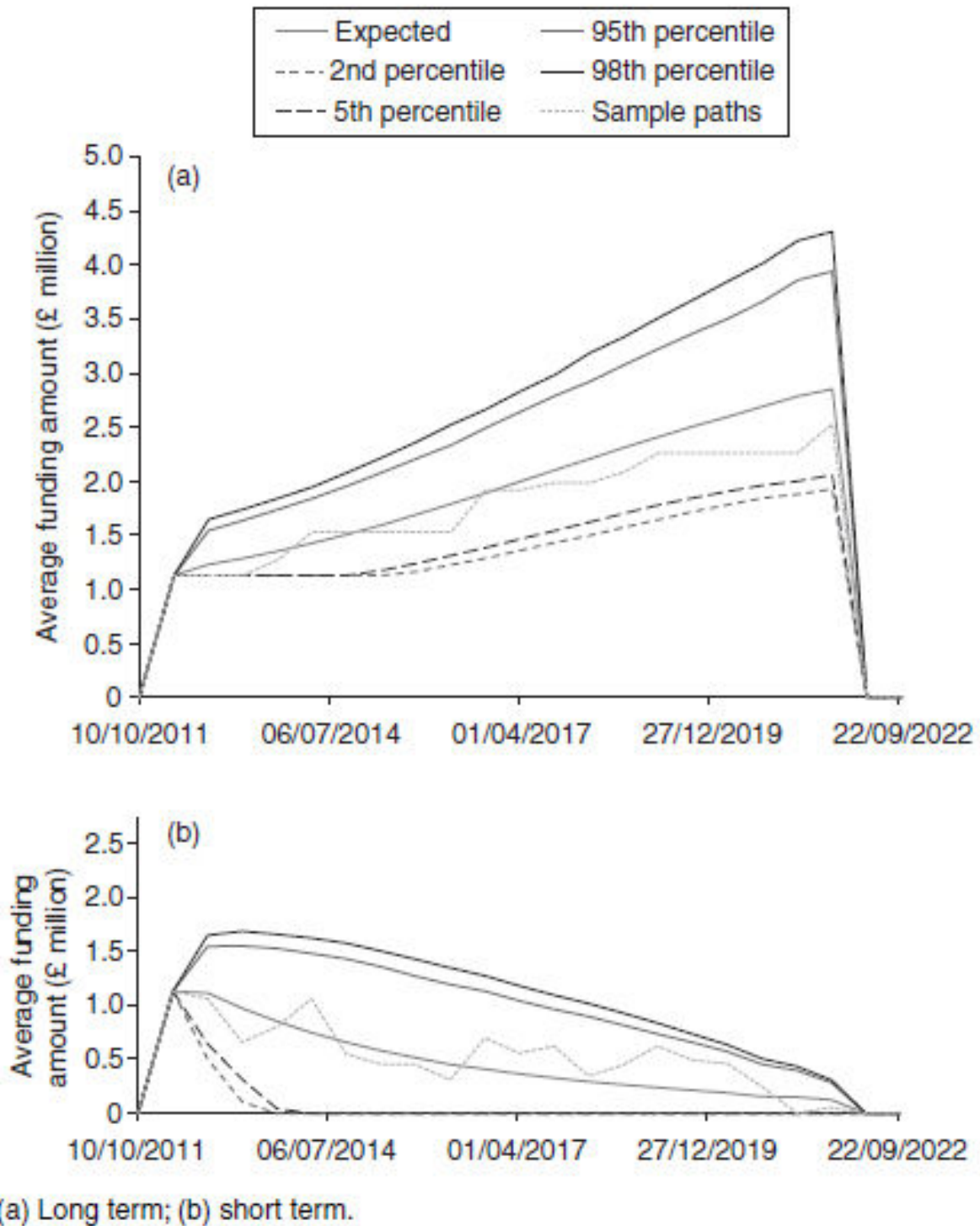
In this section we use the different components of funding costs (see pp. 559ff) to quantitatively discriminate between a short-term strategy and a long-term strategy on a portfolio of a fixed-income exotic book. In general, short-term strategies are naturally incentivised by the upward sloping funding curve to minimise the cost-of-funds charge. Long-term strategies are exploited to reduce liquidity risk. Both strategies are executed semi-annually for 10Y, during the lifetime of the portfolio. Within the short-term strategy all instruments are built to the next execution date (we fund as short as possible), while for the long-term strategy all instruments are built to the portfolio expiry date (we fund as long as possible).

Our results will be displayed using cones (see p. 554) which show the evolution of expected cost and its percentiles, and hence give a measure of risk on these costs. It is important to note that we consider cost and risk associated to funding rates and liquidity as synonyms; in particular, we assume funding at high rates implies funding with high liquidity risk.

We represent the portfolio by the IR receiver swap introduced on page 554. The expected 6M and 5Y funding spreads are 50bp and 250bp, respectively, the bid–offer rate is 10bp and 6M and 5Y regulatory spreads are 30bp and 10bp. Analysing the borrowed amount in both strategies will aid our understanding of the funding charges.

In the short-term strategy, the borrowing cone ([Figure 22.6\(b\)](#)) corresponds with the positive MTM cone. This can be explained by the fact that all issued funding instruments expire at the next stopping date; hence, at each stopping date we must borrow the positive MTM of the portfolio. In the long-term strategy, however, it can be observed ([Figure 22.6\(a\)](#)) that after 10Y the average borrowing amount is about 250% of day zero's MTM, while the 95th percentile is more than 400%. To understand this, recall that no funding instrument, including the lending instruments, expires before the swap expiry date. Hence, there are many market scenarios where we need to borrow more than the actual MTM. A dramatic effect of this overborrowing can be seen on the bid–offer costs.

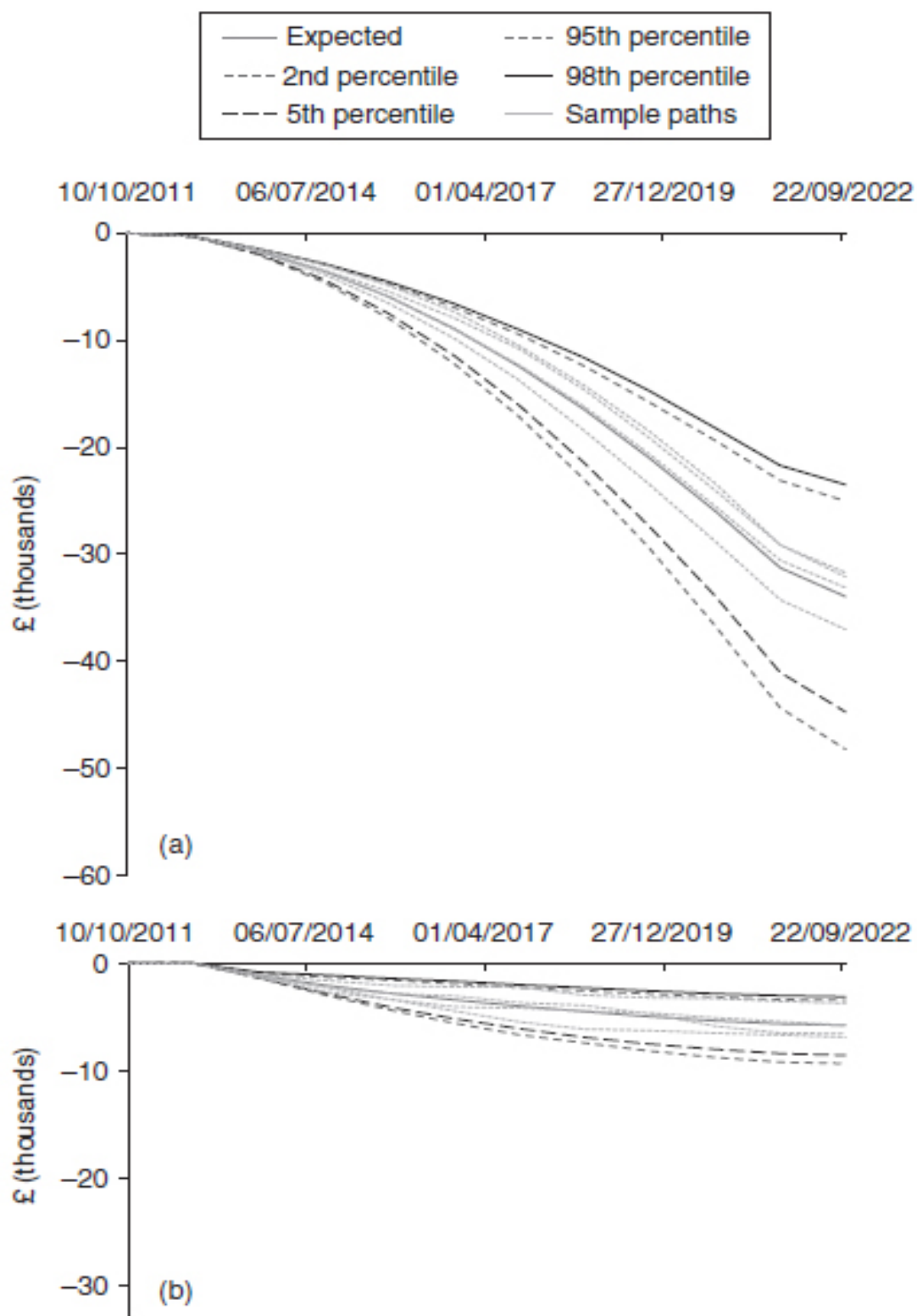
Figure 22.6 Borrowed amount



After 10 years, the expected bid–offer cost in the short-term strategy (Figure 22.7(b)) is about £5,000. This can be explained by a bid–offer

spread of 10bp, and an average borrowing amount of £0.5 million for 10Y. In the long-term strategy we overborrow; hence, the average borrowing amount increases to £1.5 million. In addition, we pay the bid–offer on lending, and hence expect a charge of $£2 \times 10\text{bp} \times 1,500,000 \times 10\text{Y}$ matching approximately the cost, of £32,000 ([Figure 22.7\(a\)](#)).

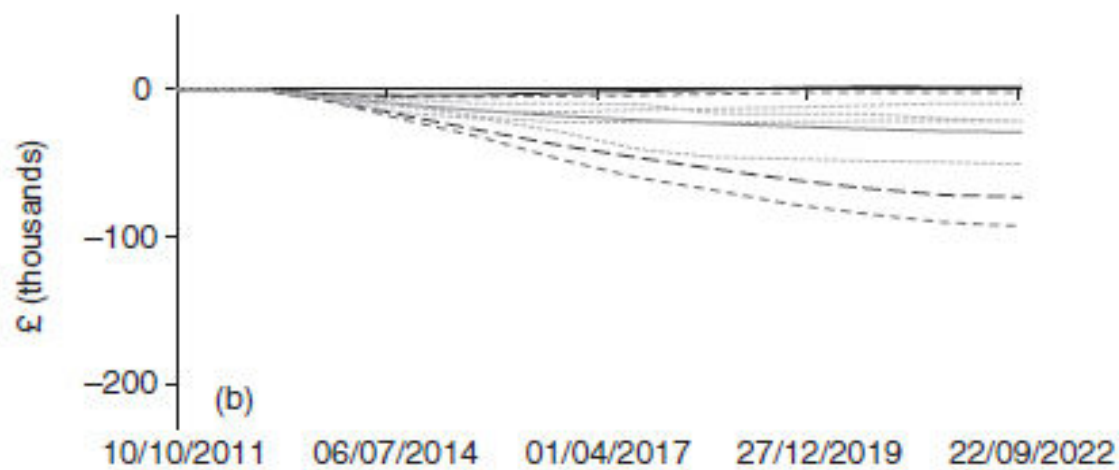
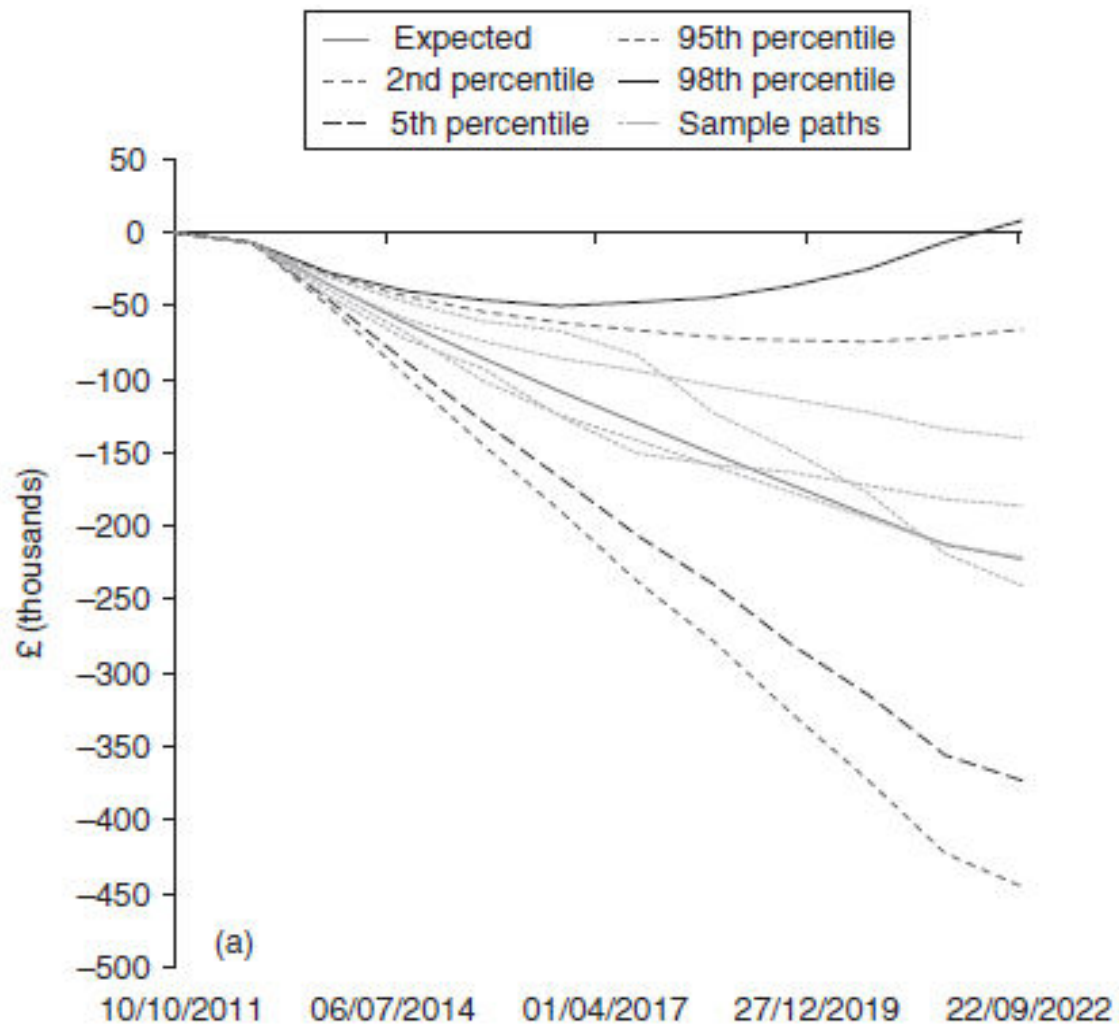
Figure 22.7 Bid–offer charge



(a) Long term; (b) short term.

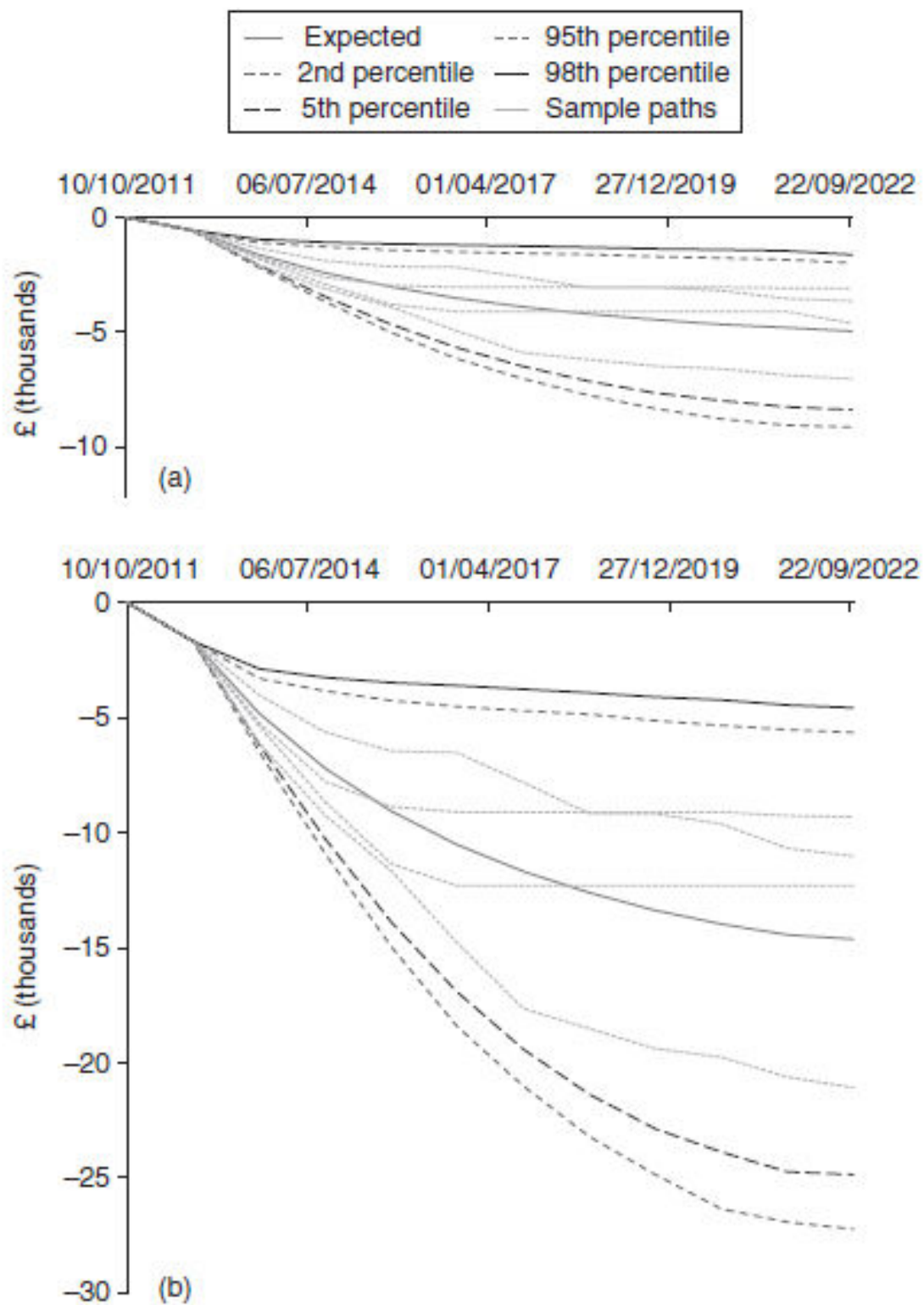
In the short-term strategy all funding instruments are built with a tenor of 6M. Hence, a 6M funding spread of 50bp and an average funding need of £0.5 million for 10Y leads to a cost of £25,000 after 10 years ([Figure 22.8\(b\)](#)). Following the same argument for the long-term strategy, recalling a 5Y spread of 250bp, we expect an average funding cost of £100,000 ([Figure 22.8\(a\)](#)). Recall that the funding cost allows netting between borrowing and lending, and hence our overborrowing should not affect this cost. However, the simulation tool reports a funding cost of £220,000. The difference can be explained by the imposed correlation of -70% between Libor rates and funding spread. Declining Libor rates result in an increased MTM of the swap, and hence require additional funding, which has to be bought at a higher funding spread.

Figure 22.8 Cost of fund charge



(a) Long term; (b) short term.

Figure 22.9 Regulatory liquidity charge



(a) Long term; (b) short term.

The regulatory liquidity charge for the short-term funding ([Figure 22.9\(b\)](#)) can be explained by recalling a 6M regulatory spread of 30bp applied to an average funding amount of £0.5 million for 10Y.

Table 22.1 Comparison summary

	Long-term strategy	Short-term strategy
Cost of funds charge	High (EV) Low volatility (RV ~ 2)	Low (EV) Large volatility (RV ~ 3)
Bid–offer charge	High (EV)	Low (EV)
Regulatory liquidity charge	Low (EV)	High (EV)

EV denotes expected value and RV denotes second percentile/EV.

A similar argument gives a regulatory liquidity charge of £5,000 ([Figure 22.9\(a\)](#)) for the long-term strategy. Note that overborrowing and corresponding lending do not increment the regulatory liquidity charge, since we are allowed to apply netting.

[Table 22.1](#) summarises our findings for the various components of the funding cost.

In the short-term strategy we find a low cost-of-funds charge, a low bid–offer charge and a high regulatory liquidity charge. In the long-term strategy we find a large cost-of-funds charge, a high bid–offer charge and a low regulatory liquidity charge. Although these quantitative results depend on the specific numerical assumptions made in our simulation model (see p. 554), they agree with our intuitive expectations. In addition, in the short-term strategy we find higher relative volatility in the cost-of-funds charge (high liquidity risk), while the long-term strategy results in a lower relative volatility (reduced liquidity risk).

ADVANCED STRATEGIES

So far we have discussed a short-term strategy where all funding instruments expire at the next stopping date, and a long-term strategy where all funding instruments expire at the portfolio’s expiration date (see the previous

section). In this section we introduce a wider set of advanced, real-world strategies available in our framework which allow us to choose more customised funding tenor profiles.

In general, after running a sufficiently large set of different funding strategies, we should be able to construct an efficient frontier plot between funding cost and risk and choose the optimal strategy matching the cost–risk appetite.

Funding to expected cashflows

When funding to expected cashflows, also called matched funding, the funding instruments are chosen such that their principal payoffs match the expected cashflows of the portfolio. This strategy is suitable for sticky payoffs, eg, IR bonds, which do not exhibit large MTM fluctuations.

In our implementation, a set of tenors are chosen; the corresponding expected cashflows of the portfolio within the tenor buckets are computed using tenor discount risk, and the funding instruments are issued. If the portfolio contains only static cashflows, the discount risk is independent of the path and stopping date and this strategy will eliminate most liquidity risk. However, for a generic portfolio, the discount risk is dependent on the path and stopping date, and hence it must be recomputed at each path and stopping date, and the funding instruments adjusted. Hence, for non-sticky portfolios, the strategy does not eliminate liquidity risk, but suffers the drawbacks of a large cost of fund charge and bid–offer charge, similar to the long-term strategy considered in the previous section.

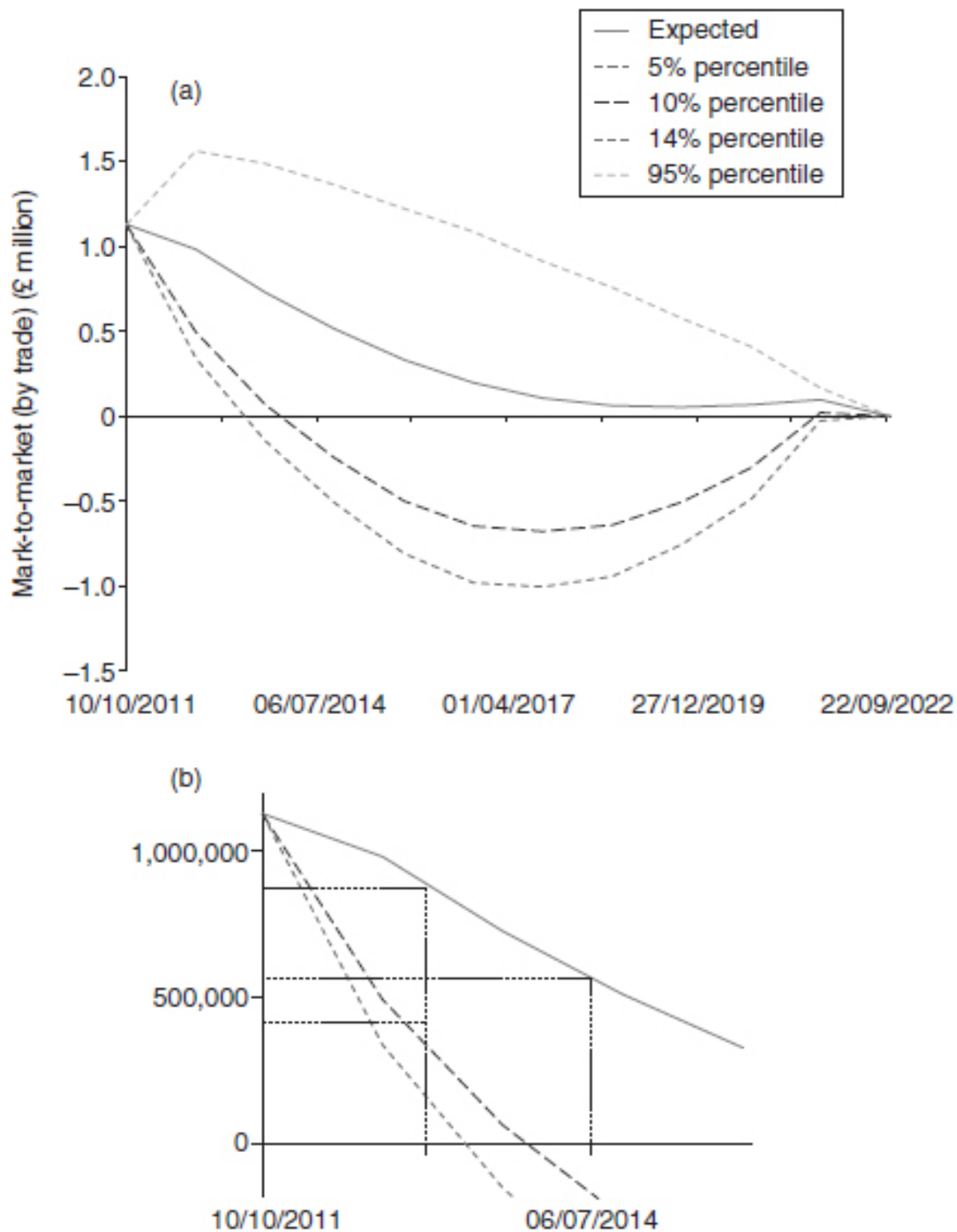
Term strategy

The term strategy can be considered as an interpolation between the short- and long-term strategies considered in the previous section. It can be applied when the funding profile, in particular the weighted average life (WAL), should be kept constant during the life time of the portfolio. Enforcing a large weighted average life reduces the liquidity risk associated with short-term funding.

In our implementation, the input consists of a set of percentages and tenors determining which percentage of the MTM is funded to which tenor. For example, the percentages (50%, 50%) and the tenors (1Y, 2Y) imply that, at each stopping date, funding instruments are built such that

- 50% (of notional) expire within the next year;
- 50% expire between one and two years.

Figure 22.10 MTM evolution of IR swap



(a) Regular; (b) zoomed.

If the percentages add up to 100%, the full MTM is funded. Note that this algorithm takes non-expired funding instruments from previous stopping dates into account.

Executing the specific term strategy above leads to an interesting observation. Assume a strategy with annual stopping dates on a portfolio with flat MTM evolution is executed. At day zero, funding instruments for 1Y and 2Y are issued, each accounting for 50% of the portfolio's MTM. On the next stopping date the 1Y instruments have expired and the 2Y instruments have a remaining lifetime of 1Y. Hence, only 2Y funding instruments must be issued. This pattern will apply to all future stopping dates. Although we always raise 2Y funding (except on day zero), the WAL is only 1.5 years.

Forward volatility cone strategy

A forward volatility cone strategy can also be considered as an interpolation between the short- and long-term strategies considered in the previous section. However, instead of keeping the funding profile constant, it uses the expected MTM profile and its envelope (the volatility cone) to determine the funding tenors.

Consider the example of the MTM evolution of the a receiver swap. To fund day zero MTM we could choose a set of tenors and match fund, that is, fund to expected cashflows. Assume we have chosen tenors of 14M, 33M and 10Y. The corresponding match funding amount can be read off from the zoomed MTM evolution ([Figure 22.10\(b\)](#)) and amounts to £0.2 million, £0.3 million and £0.6 million. On the other hand we could use the 16th percentile of the MTM profile,¹³ that is, fund £0.6 million for 14M and £0.5 million to 33M and nothing for 10Y. Funding to the 16th percentile implies that we fund shorter and overfund in only 16% of the cases and not in 50% (match funding). Shortened funding tenors imply a reduction of the expected cost of funds, while reduced overfunding reduces bid–offer cost. Note that the volatility cone is path dependent. It depends on the market data and on the state of portfolio, and hence has to be recomputed at each path and stopping date. We use an analytic approximation which is fast compared with Monte Carlo within Monte Carlo.

Limit strategy

In general, match funding should only be applied for trades with sticky payoffs. However, diverting from match funding will introduce a mismatch between the chosen funding profile and the expected cashflow profile, thus potentially increasing liquidity risk. In [Table 22.2](#) we determine this mismatch for the forward volatility cone strategy described in the previous subsection.

The introduction of limits on the mismatch gives senior management a tool to control the corresponding risk. Hence, the limit strategy is a conservative strategy which ensures that the funding mismatch at all paths and stopping dates is within preset limits. The strategy still allows for shortening of the average funding tenor, thus reducing bid–offer costs and the cost-of-funds charge. The input for the strategy is a set of tenors and a set of limits, one for each tenor. Note that the limit structure provides a cone that looks quite similar to the volatility cone.

Table 22.2 Funding mismatch

	Expected MTM profile	Expected cashflow	Fund to expected cashflow	Forward volatility cone	Mismatch
Day 0	1.1				
14M	0.9	0.2	1.1	0.6	0.5
33M	0.6	0.3	0.9	0.5	0.4
10Y	0	0.6	0.6	0	0.6

Buffer strategy

The buffer strategy allows us to reduce the regulatory liquidity charge (see p. 561) while simultaneously restricting the bid–offer cost and cost-of-funds charge. It works most effectively for portfolios whose MTM values show small fluctuations around a flat profile. Purely short-term funding would accumulate a high regulatory liquidity charge, while borrowing only in the long term results in a high cost of funds and high bid–offer cost (driven by many offsetting funding instruments required by the MTM fluctuations). The buffer strategy resolves this by overborrowing a fixed amount (the buffer) long term. The buffer should be sufficiently large to capture small MTM fluctuations. The amount funded in excess to the MTM is lent on a short-term

basis. The result of this strategy avoids overborrowing (reduced bid–offer cost) and a reduction in short-term borrowing (reduced regulatory liquidity charge).

RISK-NEUTRAL FUNDING ADJUSTMENTS

In this section we relate our funding methodology (see p. 557) to existing theoretical models. We shall focus on two models (Piterbarg 2010; Burgard and Kjaer 2011a,b), both using Black–Scholes-like hedging arguments to derive the funding adjusted price. In the first subsection we review the main assumptions and the relevant results of these models. In particular, the concept of a funding adjustment is introduced, representing a correction to the trade’s MTM computed in a world without funding costs. In the second subsection the relation between our methodology and the theoretical results is discussed, and it is shown that under specific assumptions the funding adjustment agrees with the funding cost computed within our framework. In the final subsection we discuss the impact of choosing Libor or OIS as the reference discount rate.

Review of theoretical models

Piterbarg (2010) considers a funding adjusted derivative price in a world with general collateral postings. We consider his results in the limit of uncollateralised derivatives. Burgard and Kjaer (2011a,b) consider derivative pricing while taking funding adjustments and counterparty risk into account. We consider their results in the limit of vanishing collateral and vanishing counterparty risk. The relevant assumptions for our discussion of Piterbarg (2010) and Burgard and Kjaer (2011a,b) can be summarised as follows:

- the existence of a risk-free lending curve $P_C(t, T)$ with a corresponding (stochastic) overnight rate $r_C(t)$ paid on cash collateral;
- the existence of an asset, $S(t)$, driven by a lognormal stochastic process and short rate, $r_R(t)$, paid on funding secured by the asset $S(t)$;
- a short rate, $r_F(t)$, for unsecured funding and the definition of a corresponding funding spread $s_F(t) \equiv r_F(t) - r_C(t)$.¹⁴

Under these assumptions,¹⁵ both models derive the same partial differential equation for the funding adjusted price V^ω for any derivative on $S(t)$

$$\partial_t \hat{V} + A_t \hat{V} - r_F \hat{V} = 0$$

Here A_t denotes the standard second-order differential operator for derivatives on $S(t)$ in the Black–Scholes world. Applying the Feynman–Kac formula, and defining the discount factor

$$D_{r_F}(t, T) = \exp - \left(\int_t^T r_F(u) du \right)$$

the solution for a derivative with payoff $V(T)$ at T can be written as

$$\hat{V}(t) = E_t[D_{r_F}(t, T) V(T)] \quad (22.2)$$

This equation confirms the usual assumption that the funding adjusted price for uncollateralised derivatives can be computed by discounting with the unsecured funding rate. Burgard and Kjaer (2011a) define a funding adjustment $U(t)$ by

$$\hat{V}(t) = V(t) + U(t) \quad (22.3)$$

where $V(t)$ solves the standard Black–Scholes equation using the cash collateral rate $r_C(t)$ as discount rate. The Feynman–Kac solution for the partial differential equation for $U(t)$ is given by

$$U(t, T) = - \int_t^T s_F(u) D_{r_F}(t, u) E_t[V(u)] du \quad (22.4)$$

It follows immediately from this equation that the funding adjustment is negative for derivatives with positive MTM and positive for derivatives with negative MTM. This agrees with the assumptions of funding costs and benefits in our funding methodology (see p. 557). In the next section we show how $-U(t)$ can be identified with the “costs of funds” in our framework.

Comparison

In this section we shall assume that the funding costs in our framework are computed under the following restrictive assumptions:

- vanishing regulatory liquidity rate, which implies a vanishing regulatory liquidity charge;
- vanishing bid–offer cost;
- use of a pricing model with a dynamics representing the assumptions of the section on the review of theoretical models (see p. 576) as simulation model;
- we identify the risk-free lending curve $P_C(t, T)$ with our reference curve; for simplicity we restrict to deterministic rates;
- restrict to deterministic funding spreads and approximate $s_F(t)$ with the overnight funding spread.

Discretisation of Equation 22.4 along the stopping dates $\{T_0, \dots, T_N\}$ with $t = T_0$, $T = T_N$ and $\Delta T_k = T_k - T_{k-1}$ and multiplying by $D_{r_F}(t, T)^{-1}$ leads to

$$D_{r_F}(t, T)^{-1}U(t, T) = -E_t \sum_{k=0}^{k=N-1} s_F(T_k)D_{r_F}(T_k, T)^{-1}V(T_k)\Delta T_{k+1} \quad (22.5)$$

It follows from our methodology that

$$\sum_{k=0}^{k=N-1} s_F(T_k)D_{r_F}(T_k, T)^{-1}V(T_k)\Delta T_{k+1}$$

agrees with the pathwise cost-of-funds charge computed using an overnight strategy in our framework. Hence, under the above assumptions, our framework computes the (negative) undiscounted funding adjustment

$$-D_{r_F}(t, T)^{-1} \cdot U(t, T)$$

as the cost-of-funds charge.

Funding cost and funding tenor

We have shown in this chapter that funding cost in our framework depends in general on the chosen funding tenors. This feature, which relies on the fact that we can simulate markets using real-world measures such that funding

rates are not necessarily realised, allows us to optimise funding cost against risk.

On the other hand, in a completely risk-neutral model, the funding charge should be independent of the chosen funding tenor as long as the forwards are realised.

This section shows that under the restrictions of page 577 the funding charge in our framework is also independent of the chosen funding tenor. We define the forward spreads by

$$f_s(T_1, T_2) = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} s_F(u) du$$

and assume that the funding spreads follow their forwards. We use Equation 22.5 and compare the funding adjustments for two strategies for a simple two period funding model with three stopping dates $\{T_0, T_1, T_2\}$. In the first strategy all instruments are built to the next stopping date, while in the second strategy all instruments are built to the last stopping date. We find the discounted funding cost

$$\begin{aligned} C_1(T_0, T_3) &= V(T_0)\tau f_s(T_0, T_1)D_{rf}(T_0, T_0) \\ &\quad + V(T_1)\tau f_s(T_1, T_2)D_{rf}(T_0, T_1) \\ C_2(T_0, T_3) &= 2V(T_0)\tau f_s(T_0, T_2)D_{rf}(T_0, T_0) \\ &\quad + (V(T_1) - m(t)V(T_0))f_s(T_1, T_2)D_{rf}(T_0, T_1) \end{aligned}$$

Choosing

$$m(t) = D(T_0, T_1)^{-1}$$

corresponds to our methodology introduced in the third section of this chapter (pp. 557ff), namely rolling the unexpired funding amount from T_0 to T_1 . With this choice of $m(t)$, the funding costs $C_1(T_0, T_2)$ and $C_2(T_0, T_2)$ agree. This argument can be extended to a multiperiod model showing that under the assumption of page 577 our funding charge is indeed independent of the chosen funding tenor.

Libor versus OIS

In this section we discuss the impact of choosing Libor or OIS as the reference rate. In general, the reference rate is related to secured funding, or risk-free rate. While, historically, the Libor rate was considered the best proxy for risk-free rate, it became clear after the crisis of 2008 that the Libor rate includes a risk premium, and OIS is now viewed as the best proxy for a risk-free rate.

In the context of the assumptions above it follows that the funding-adjusted MTM can be computed either by using the unsecured funding rate as discount rate (Equation 22.2) or by adding a funding adjustment (Equation 22.3) to the MTM computed using the reference rate. That implies that splitting the unsecured funding rate into an OIS rate and a corresponding OIS spread or into Libor rate and a corresponding Libor spread leads to two different decompositions of the funding-adjusted MTM, but does not affect the value of the funding-adjusted MTM itself.

In this chapter we have focused on the distribution of the funding costs as a measure of liquidity risk. In this context it is important to be aware that different choices of the reference rate will lead to different results on these distributions, and thus on the associated liquidity risk.

CONCLUSION

Within this chapter we introduced a framework that allows quantitative analysis of the funding cost on uncollateralised derivative portfolios using different dynamic funding strategies.

Our funding strategies include simple short- and long-term strategies and advanced, purpose built strategies. Their application to different portfolios and client requests are discussed, and explicit results comparing funding costs and risks between long-term and short-term strategies on an IR portfolio are presented.

We used a funding methodology that closely follows actual business logic (issue bonds when raising funds) and described its relation to theoretical developments.

The author thanks Gerson Riddy, Barry Mcquaid, Tom Hulme, John Taylor and Vladimir Piterbarg for valuable discussion and feedback on the chapter. In addition, the author thanks Mats Kjaer and Nicolas Millot for valuable discussions on funding adjustments in the risk-neutral context.

- 1 This increase is reflected, for example, in the 5Y iTraxx for senior financials, which increased from 40 basis points (bp) prior to 2008 up to 360bp.
- 2 The MTM is computed with respect to a specific reference rate.
- 3 The reference rate is related to secured funding, eg, overnight indexed swap (OIS) or London Interbank Offered Rate (Libor). We discuss the impact of its specific choice in the section starting on page 575.
- 4 This assumes that the nonparametric skew in the MTM distributions vanishes. Then 50% of the possible market scenarios imply realised cashflows above the expected values, and 50% below.
- 5 Note that this set-up allows, for example, the correlation effect between swap rates and funding spreads on funding cost and risk to be quantified by running the same simulation several times with a different set of correlation parameters.
- 6 The impact of this choice is discussed in the section on page 575.
- 7 The author thanks Gerson Riddy and Barry Mcquaid for explanations.
- 8 Ignoring tax, etc.
- 9 This assumes that the reference rate corresponds to the secured funding rate
- 10 The impact of choosing a different rate, eg, OIS, is discussed in the sixth section.
- 11 Although the desk does not literally issue a bond to treasury, the mechanics of its fundraising and the associated cost can be described as a “bond issue”.
- 12 Libor or OIS; see the discussion on pp. 575ff.
- 13 The 16th percentile corresponds approximately to the 1σ deviation in a standard distribution.
- 14 Burgard and Kjaer (2011a) assume $r_C(t)$ and $r_F(t)$ to be deterministic.
- 15 In the limit of vanishing collateral and vanishing counterparty risk

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Funds Transfer Pricing in the New Normal

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Funds transfer pricing (FTP) is a term used to describe the sum of policies and methodologies a bank applies in its internal steering systems to charge for the use (and credit the generation of) funding and liquidity. Commonly, FTP is embedded in economic value added (EVA) or return on equity (ROE), return on risk-adjusted capital (RORAC) performance measures of single transaction or a business line's risk-adjusted profitability or performance as a complement to capital consumption. It focuses on distributing the bank's funding and liquidity costs to the beneficiaries of these scarce and valuable resources. As such, FTP is an extremely powerful tool, which is deeply rooted in any divisional profit and loss (P&L) or profit centre calculation. In fact, FTP is the means by which a bank's overall net interest income can be split into originating units and subunits, thus enabling the bank's management to perform an effective planning, monitoring and control cycle. As a consequence of the 2007–9 global financial crisis these costs are material and need to be allocated. They can turn a business line's performance or product's profitability from positive to negative and vice versa. Thus, FTP is a deeply strategic steering instrument and needs to be carefully and thoroughly designed according to the bank's business model.

This chapter is organised as follows. In the next section we explain why banks need an effective FTP scheme. Then we give our point of view on how

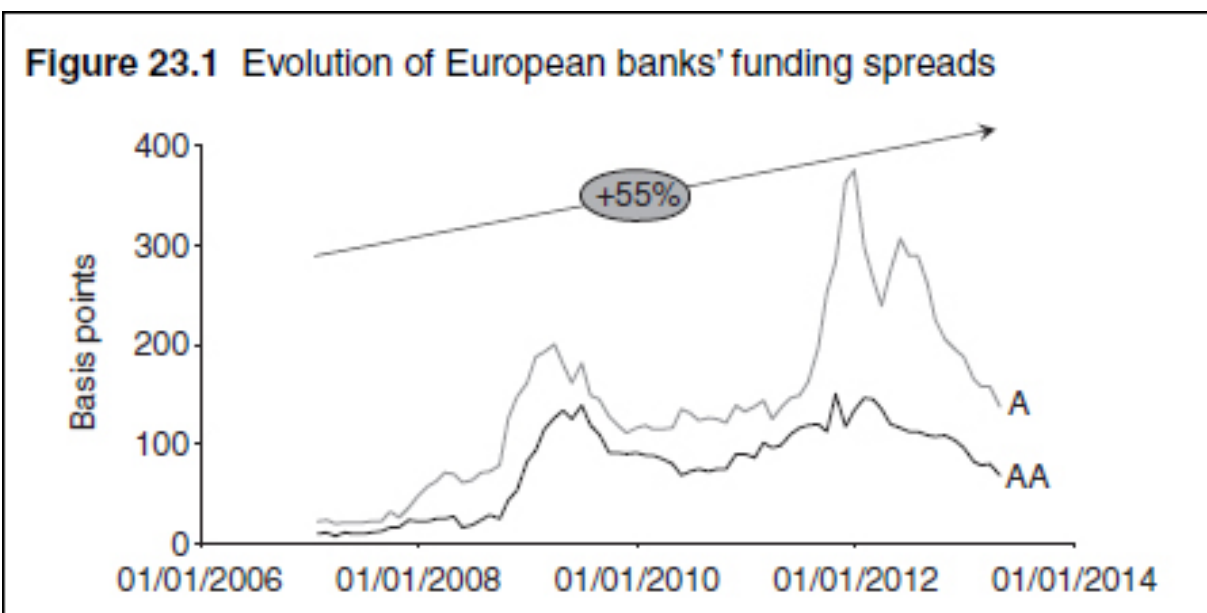
a best practice FTP scheme should be built. Next, we aim to take a more strategic viewpoint on FTP as a tool for effective balance-sheet management. We conclude with six key take-home messages.

BANKS NEED AN EFFECTIVE FUNDS TRANSFER PRICING SCHEME

Reflecting our client work from the early 2000s onwards, we were surprised to see how frequently the impact of the FTP scheme was underestimated. Based on years of experience and habit, managers tend to focus on a bank's (or, for that matter, business unit's) net interest income as an aggregate metric to measure the top line of an interest-generating business. In reality, the disaggregation into interest income and interest expense shows that a significant portion of the net interest income is usually primarily driven not by client margins but by internal models that value the cost of funding provided from or to the bank's treasury department. Any change in such models – such as adjusting the behavioural maturity assumption for a corporate sight deposit portfolio, or a different interest rate applied to calculate capital benefit – will distort the net interest income significantly without the slightest change to the bank's actual business relationships to the outside world.

The importance of FTP schemes correlates directly with the cost of funding and liquidity in the money and capital markets. It is not uncommon to observe that funding spreads have doubled or tripled (depending on the size, rating and geographical location of the bank) over the course of a few years of financial crisis ([Figure 23.1](#) shows a compound annual growth rate of 55% in the period January 31, 2007, to April 30, 2013). The higher the funding spreads, the more precisely FTP schemes need to be designed and parameterised, in order to achieve the desired optimum allocation of funds. Flaws in the system will result in inefficient allocation of a valuable resource. Take as an example the contingent liquidity facility a bank provides to a corporate customer (or to portfolio of such customers). The bank's commitment to provide liquidity on demand will require the bank to hold a substantial amount of liquidity itself (eg, in the form of unencumbered securities that can be used for repo business to generate liquidity upon demand, but cannot be used for long-term secured funding, thus increasing the bank's overall funding cost). Not charging this cost to the business unit will

generate a disincentive, driving up liquidity risk without adequate revenues: take the example of the Fixed Income Clearing Corporation trading desk, measured solely by their trading result, without taking into consideration the volume and time horizon of tied-up funds.



Regulators are increasing the pressure on banks to revamp their FTP systems accordingly. For example, Grant (2011) warns that banks “with poor [liquidity transfer pricing] practices are more likely to accrue larger amounts of long-term illiquid assets, contingent commitments and shorter-dated volatile liabilities, substantially increasing their vulnerability to funding shortfalls”. Grant concludes with recommendations on liquidity transfer pricing best practices, such as clear governance processes or recognition of both on-balance-sheet funding liquidity risk and off-balance-sheet contingent liquidity risk. National legislators, such as Germany’s BaFin, have passed dedicated FTP requirements. In their 2012 update of their “MaRisk” (Minimum Requirements for Risk Management) principles, BaFin demanded that banks should set up an “adequate pricing system for internal charging of cost, benefits and risks of liquidity” (BaFin 2012). The Basel III regulations, with requirements for balance funding structures and minimum liquidity cushions, should also force banks towards revamping their FTP schemes to create the right incentives. New data and reporting requirements will disclose shortcomings, and additional balance-sheet requirements (eg, the

minimum requirement for eligible liabilities (MREL)) will constantly push institutions to react by adapting the FTP methodology.

Governance considerations should play an important role in the design of an FTP scheme. With the internal transfer of fees comes the transfer of risks, and thus of responsibilities. For example, a bank may have the capital markets' swap desk receive internally the swap rate for a fixed rate loan from a business unit, and in exchange deal with the interest rate risk of that position, while there may be a liquidity management desk in the treasury that receives the bank's liquidity spread in exchange for dealing with liquidity risk. There needs to be alignment between risk taking and clearly defined responsibility, or, in other words, a separation of net interest income components arising from client business margins and interest rate or liquidity maturity transformation.

On the following pages, we lay out some best practices in setting up a successful FTP scheme and try to illustrate the strategic perspective: how FTP helps to achieve the larger goal of balance-sheet structure management. Our observations are based on client work with large international banks' treasury departments, as well as benchmark surveys and interviews.

ELEMENTS OF A BEST PRACTICE FUNDS TRANSFER PRICING SCHEME

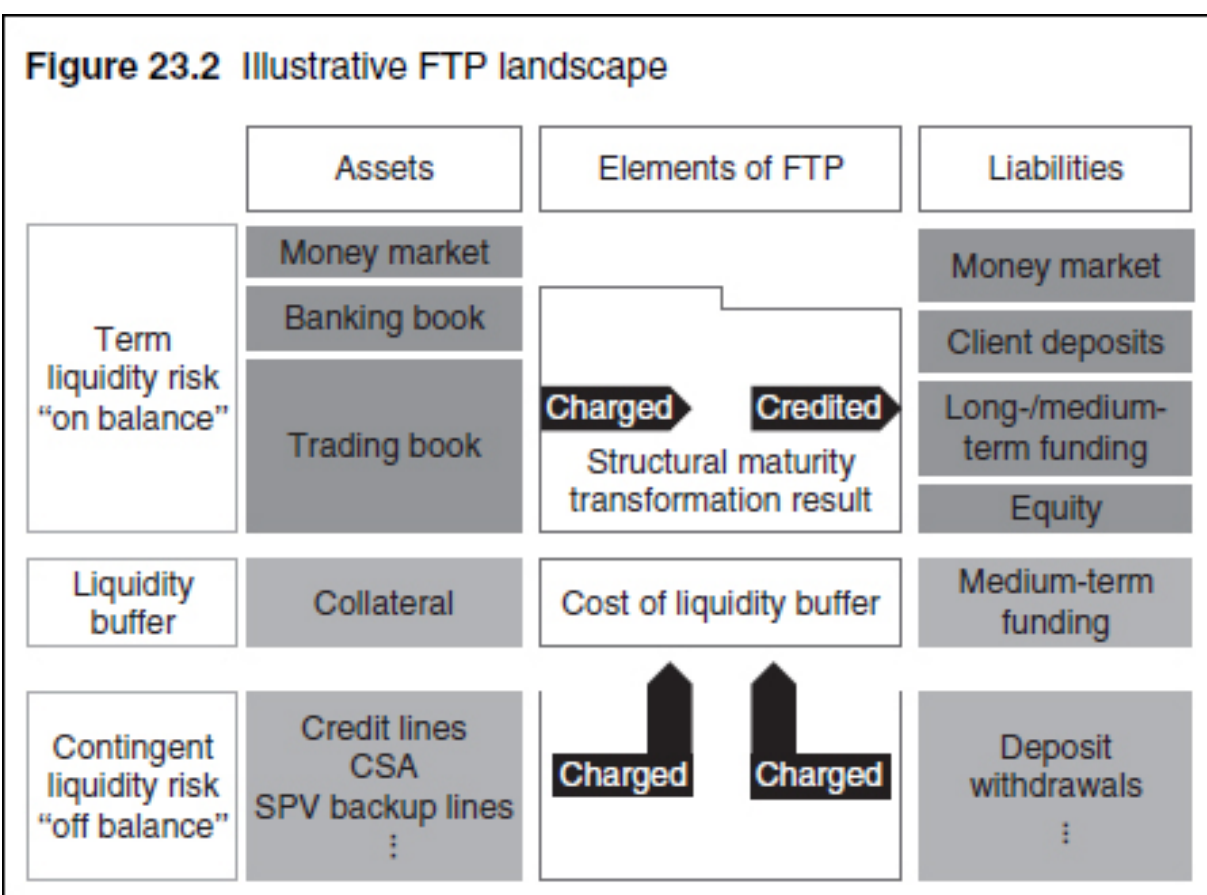
From our point of view, a best practice FTP scheme consists of four elements. First, we need to define the elements of the bank's balance and off-balance-sheet position that need to be included in a holistic FTP setting: we call this the "FTP landscape". In fact, this landscape should entail all sources of liquidity cost, ie, both for term liquidity (based on deterministic cashflows) and for contingent liquidity arising from stochastic cashflows. Second, suitable methodologies have to be defined to calculate transfer prices. Third, the "right" curve for the calculation has to be selected. Finally, FTP can only be effective if the treasury operating model ensures that the results from FTP can be measured and monitored to allow an effective steering process.

The FTP landscape

The natural idea of FTP is that assets get charged their cost-of-funds, which is credited to the bank's funding units.

FTP for the banking book (eg, loan books)

Neu *et al* (2012) show that most European banks apply a marginal cost-of-funds concept to charge their loan books and to credit their issuances and deposit units. However, it is less common to also apply a full transfer pricing scheme to the trading book or for sources of contingent off-balance-sheet liquidity risk.



From our point of view, both the trading portfolios and off-balance-sheet commitments should be considered in a holistic FTP approach.

- **Trading book:** security positions (eg, investment portfolios or cash-equivalent securities) are shown in the balance sheet according to their current market or book value. For liquidity risk this information is, however, not sufficient. What is really needed is the future collateral value or the cash value when pledging or selling these positions, as only the net difference between the initial funding volume and the liquidity (cash) equivalent needs to be funded by the bank. Thus, haircuts for

future market value volatility and liquidation discounts must be considered when funding these positions, and hence need to be reflected in a fair funds transfer price.

- **Off-balance-sheet commitments:** these also need to be considered in a holistic FTP approach, as they bear a significant liquidity risk. Examples are: unexpected drawdowns on committed credit lines; collateral agreements with swap counterparts; corporate customer swaps without credit swap agreements (CSAs), which in turn will cause liquidity implications when closing the position with a banking counterpart, where CSAs are common (many large banks react by setting up a liquidity management desk for derivatives, similar to a CVA desk); commercial paper backup lines to special purpose vehicles or any other bank-specific guarantees or triggers. To hedge against the contingent liquidity risk arising from these positions, banks hold a liquidity buffer of highly liquid assets (such as government bonds or cash), which is usually funded for a medium term of 6–12 months. The cost of holding this collateral, ie, the negative carry of this portfolio, needs to be allocated as shown in [Figure 23.4](#).
- **Money market:** assets and liabilities are usually traded in the same business unit, and hence the charging and crediting is already incorporated in the P&L of the money market and does not need to be reflected in the FTP. In other words, there is no need to design internal cost-of-funding schemes when the actual cost of funding will be determined each and every day by the interaction with the external market (the “street”).
- **Equity:** this will usually be included via an equity model book whose results will be allocated business lines, which will correct for the usually applied “100% wholesale funding” within the FTP for the fact that the bank needs to run on equity, and most assets will require some underlying equity anyway. Since this equity has not only a solvability component, but also a liquidity implication, it is fair to account for this fact, which is done via equity model books whose asset side might be “invested” into the own loan book. Most frequently, we have observed banks to credit the business side with a risk-free rate of varying tenors, ranging from as short as two weeks until as long as ten or more years.

A quick guide to FTP methodology

Defining the methodology for FTP is a wide remit. Thus, in this chapter we only give a short guide to four selected topics in the FTP landscape defined in the previous section. We shall give a short overview of how FTP works for loans (including revolving loans and Basel III adjustments), tradeable products and deposits and how to define the size and cost allocation of a suitable liquidity buffer.

FTP for loans

This is largely implemented by almost all European banks. It is standard within the market to calculate the liquidity spread. This is usually derived from the difference in the internal rate of return (IRR) or par rate with respect to the swap curve and the bank's funding curve (swap curve shifted by the bank's term funding spreads). The IRR for the swap curve fixes the fair transfer price paid by loan origination to asset and liability management (ALM) as the transfer price for interest rate risk in the loan (see, for example, Neu and Matz 2007)

$$\sum_{t=1}^T (N_{t-1} \times \text{IRR}_{\text{ALM}} + A_t) \times \exp(-r_{\text{swap}}^t \times t) = N_0 \Rightarrow \text{IRR}_{\text{ALM}}$$

where N_t denotes the outstanding notional at time t , A_t denotes the amortisation amount in period $t-1$ to t and r_t denotes the swap rate at time t .

Analogously, the IRR on the bank's term funding curve is the transfer price for interest rate and mismatch liquidity risk, where BLS_t denotes the bank's term funding spread above the swap curve, and LM stands for "liquidity management"

$$\sum_{t=1}^T (N_{t-1} \times \text{IRR}_{\text{LM}} + A_t) \times \exp(-(r_{\text{swap}}^t + \text{BLS}^t) \times t) = N_0 \Rightarrow \text{IRR}_{\text{LM}}$$

The transfer price for liquidity mismatch risk, relative to the term funding level of the bank and the amortisation schedule of the loan, is given by the difference between these two IRRs

$$\text{liquidity cost} = \text{IRR}_{\text{LM}} - \text{IRR}_{\text{ALM}}$$

The liquidity transfer price of a revolving loan facility contains three components: mismatch liquidity costs for the current and expected usage and contingency liquidity costs for the open limit. The current usage of a revolving loan facility is characterised by the drawn amount and the tenor of the drawdown. For the bank, only the current usage, the current open line and the current tenor of a drawdown are known. When hedging the mismatch liquidity risk of individual draws under the loan facility, eg, by buying maturity-matching term funding, the treasury faces the challenge of generating a large number of funding tickets that have to be adjusted upon each change of the drawn amount. Furthermore, the treasury would neglect the fact that many short-term draws under a loan facility will be rolled over, so that a core amount will be drawn permanently until the final maturity of the facility. Instead of hedging mismatch liquidity risk of each individual draw, it is more appropriate in practice to hedge it on a portfolio level. For this purpose the current usage is separated into a core usage, which will remain drawn for the entire lifetime of the facility, and a volatile usage, which fluctuates in the short term. The volatile usage is funded in the short term, whereas the core usage is funded until the end of the facility. For the uncertainty of the volatile part, a liquidity buffer needs to be maintained, for which the costs should be allocated to the facility.

At the time of writing, all banks have implemented the net stable funding ratio (NSFR) and the liquidity coverage ratio (LCR). However, for many banks, how to integrate the cost to comply with these ratios into FTP remains an open problem. It is clear that matched funding will not be enough to reach an NSFR ratio of 100%. As an example, consider a three-year bullet loan of US\$100 million with a matching three-year term funding. Of course, in the first two years the resulting NSFR is equal to 1 (100% required stable funding (RSF) match with 100% available stable funding (ASF)). However, due to an incongruent change in ASF and RSF factors in late 2012, the NSFR is equal to 0. One possibility of adapting the FTP scheme in the example under Basel III would be to split the funding between three years' funding and four years' funding. Then, again, in year 3 the NSFR would be equal to 1. However, this would lead to a higher term funding cost for the three-year loan, which would have to pay the average of the bank's three-year and four-year term funding spread.

FTP for tradeable products

This is a three-step process ([Figure 23.3](#)). In the first step, assets are clustered into liquidity classes according to their market liquidity based on clearly documented criteria, which have to be reviewed on a regular basis and need to be coded within the front office system. Typical criteria are asset class, central bank eligibility and credit quality. In a second step, the economic liquidation profiles are modelled under a “going concern” assumption (ie, considering the ability to use alternative funding sources, eg, repo, central bank, and the ability to liquidate assets in the market at market-sensitive prices).

The definition of the liquidity profiles is an art in itself, and based, for example, on volume in relation to daily turnovers, central bank eligibility and policy, market depth, economic and bank-specific conditions and, of course, some expert judgement. The third step is the derivation of the liquidity cost, charging the amount per time bucket given by the liquidation profile by the respective term funding spread and rebating the respective buckets if assets are self-funded (eg, in the repo market) and summing over all term charges.

Figure 23.3 FTP for tradeable products

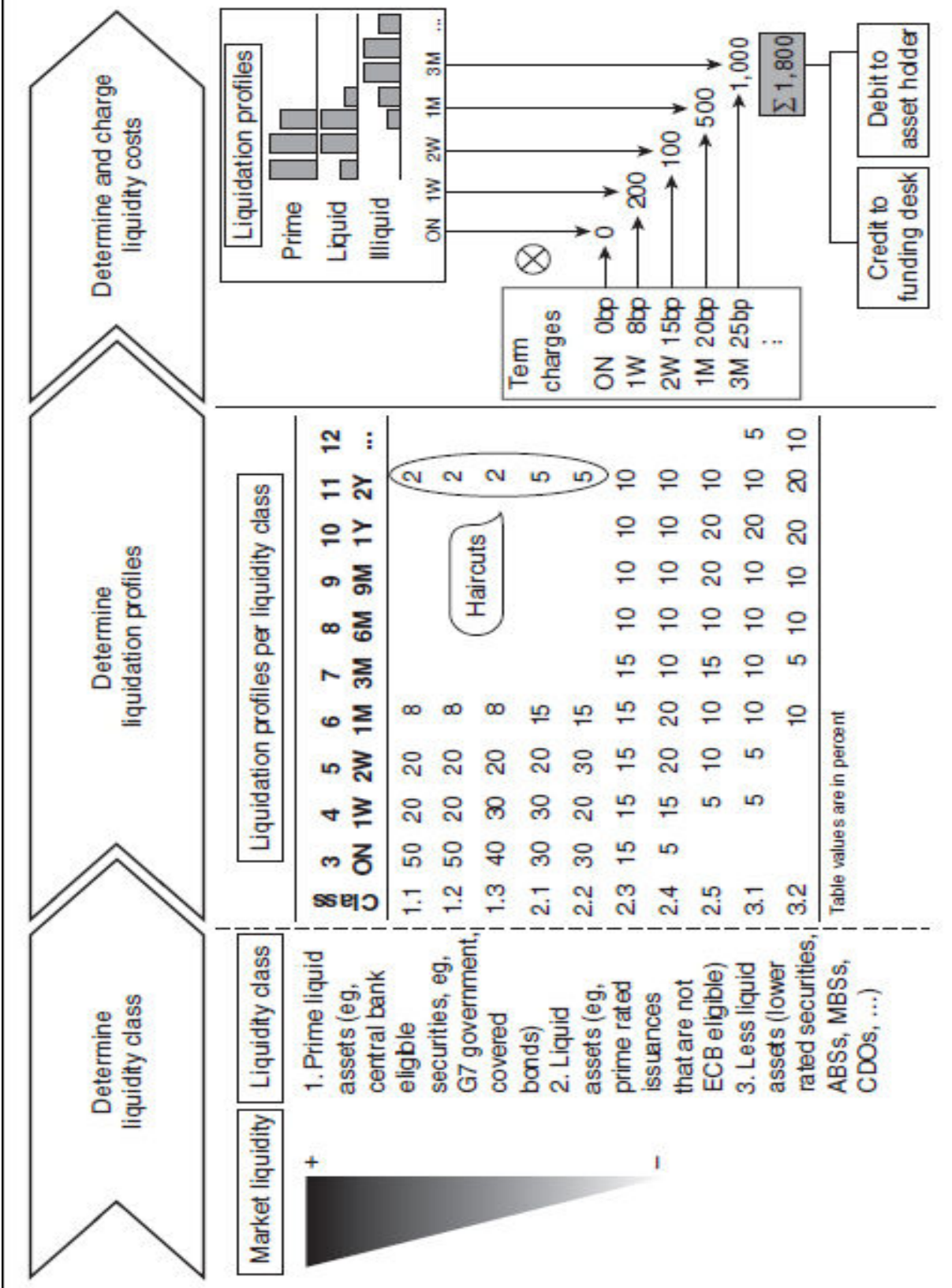


Table values are in percent

Table 23.1 Illustrative example for core deposit modelling

Position	Model funding matrix (%)						
	1D	1D–1M	1–6M	6–12M	1–5Y	5–10Y	10Y+
Deposits from banks	90	0	0	10	0	0	0
Deposits from corporates	15	15	20	20	10	10	10
Deposits from retail customers	10	0	0	20	20	20	30
Other deposits	10	0	10	40	40	0	0

The last three columns show crediting of deposits to business units using moving averages of (unsecured) liquidity costs of the bank for respective maturities.

The last three columns show crediting of deposits to business units using moving averages of (unsecured) liquidity costs of the bank for respective maturities.

FTP for deposits

Since the maturity of deposits is not predictable, but it is plausible that deposits are not all withdrawn at their legal maturity (eg, sight deposits), they are modelled in the liquidity gap profile with the use of assumptions on the deposit base. Deposit-base assumptions depend on the counterparty/customer class (eg, retail or corporate deposits) and need to be estimated using historical data. Of course, a regular model validation is necessary. It is clear that the “stable” part of the deposits (eg, tenors longer than 1Y or longer than 2Y) should be credited, as only this part can replace long-term funding, while the short-term fraction should be treated like money market funding with correspondingly lower credit. [Table 23.1](#) gives an illustrative example of models for three different deposit classes (banks, retails and corporate).

Clearly, to credit stable deposits with the bank’s unsecured funding spread in the respective tenor incentivises the deposit business for growth. A growing deposit base raises the question of whether the modelling assumptions should be applied to the full deposit volume. We believe that an upper bound (as an absolute number of euro) for each deposit class should be

defined to make sure the overall deposit volume remains in line with the bank's overall balance-sheet strategy.

FTP for contingent liquidity risk

This consists of two elements. First, a liquidity buffer of suitable size to hedge against contingent liquidity risk has to be defined. Second, the cost of this portfolio, resulting from a negative carry by funding highly liquid assets over a medium term of, eg, 6–12 months, needs to be allocated to respective businesses units, which are a source of the contingent liquidity risk.

In determining the size of the liquid-asset buffer, a stress scenario will be applied to the economic liquidity gap model, a discretionary management buffer or additional regulatory stipulation will be added and the liquidity requirement in the resulting model will be measured ([Figure 23.4](#)).

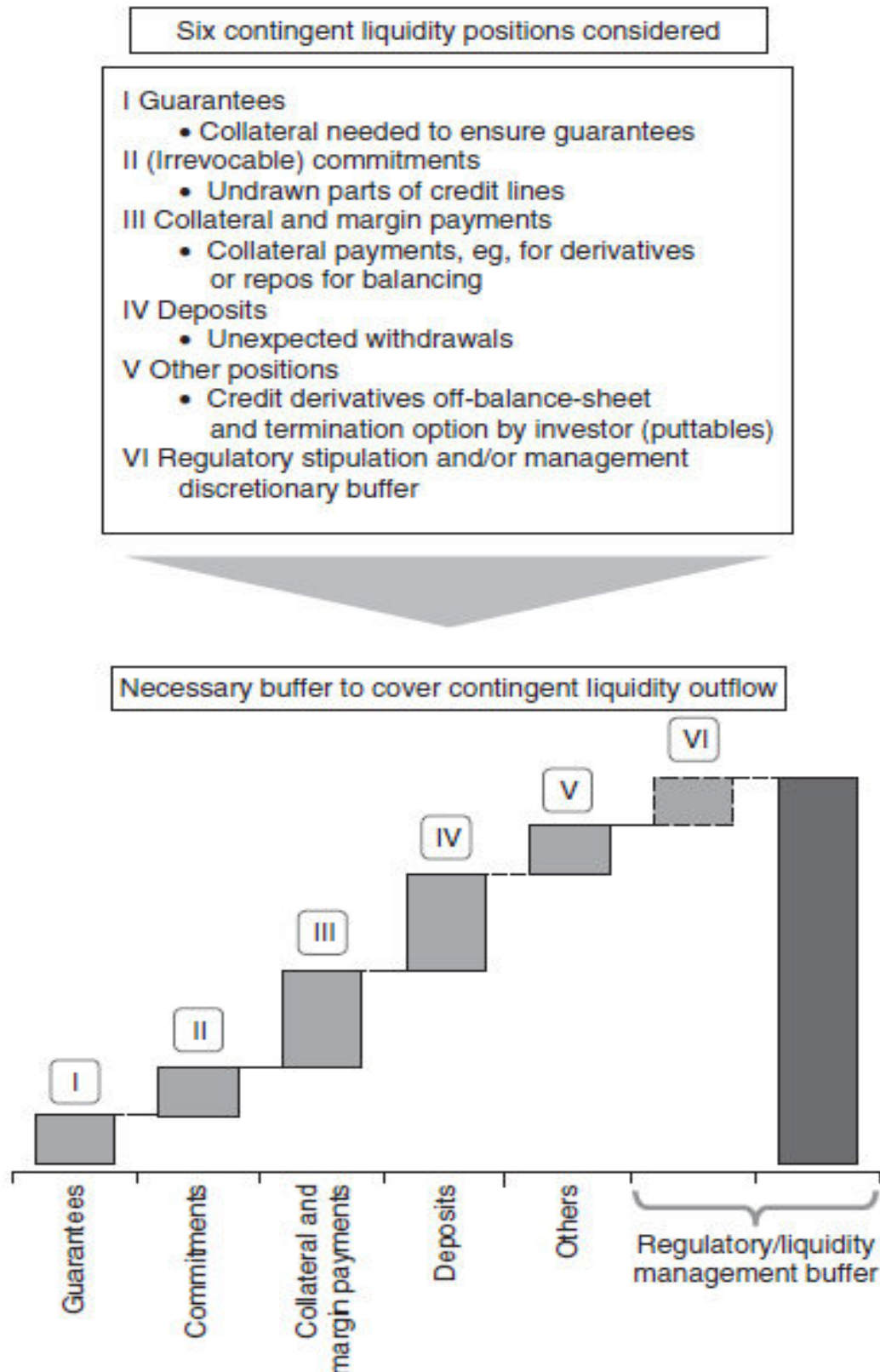
The overall cost of holding the liquid-asset buffer is usually calculated by the treasury (for example, as a negative carry on the portfolio) and broken down to the business units, revealing the underlying liquidity needs of each unit under the stress scenario.

Finally, we believe two additional best practice elements are required to make such schemes successful.

1. **Ring-fenced versus synthetic buffer:** either a treasury owns a ring-fenced liquidity portfolio or it has robust processes in place (internal liquidity agreements; internal liquidity and collateral market) to have a sufficient synthetic liquidity buffer in a liquidity crisis. A ring-fenced portfolio most often is less complex and feels safer than a synthetic buffer. Nevertheless, it is expensive to ring-fence valuable assets, and the increased flexibility in a synthetic buffer most often lowers the costs of the buffer. Modern liquidity buffer management is bedded into the overall collateral management of an institution.
2. **Central collateral management:** whether the liquidity buffer is ring-fenced or synthetic, it is very important that centrally within the institute there is transparency about all unencumbered assets, which can be used to generate liquidity.
3. ***Ex post* versus *ex ante* pricing:** the cost of the liquidity buffer can be allocated to the business units via *ex ante* estimation of the costs and via *ex post* allocation. *Ex ante* estimation has the advantage that

businesses can forward the costs to the client and are not surprised by the costs afterwards. Still, we often see that the *ex ante* pricing of liquidity buffer costs is a complex mechanism for insignificant costs, so a simple *ex post* allocation is better suited for some product types. The institution must find the right balance depending on product type.

Figure 23.4 Illustrative determination of a suitable liquidity buffer



Setting the “right” curve

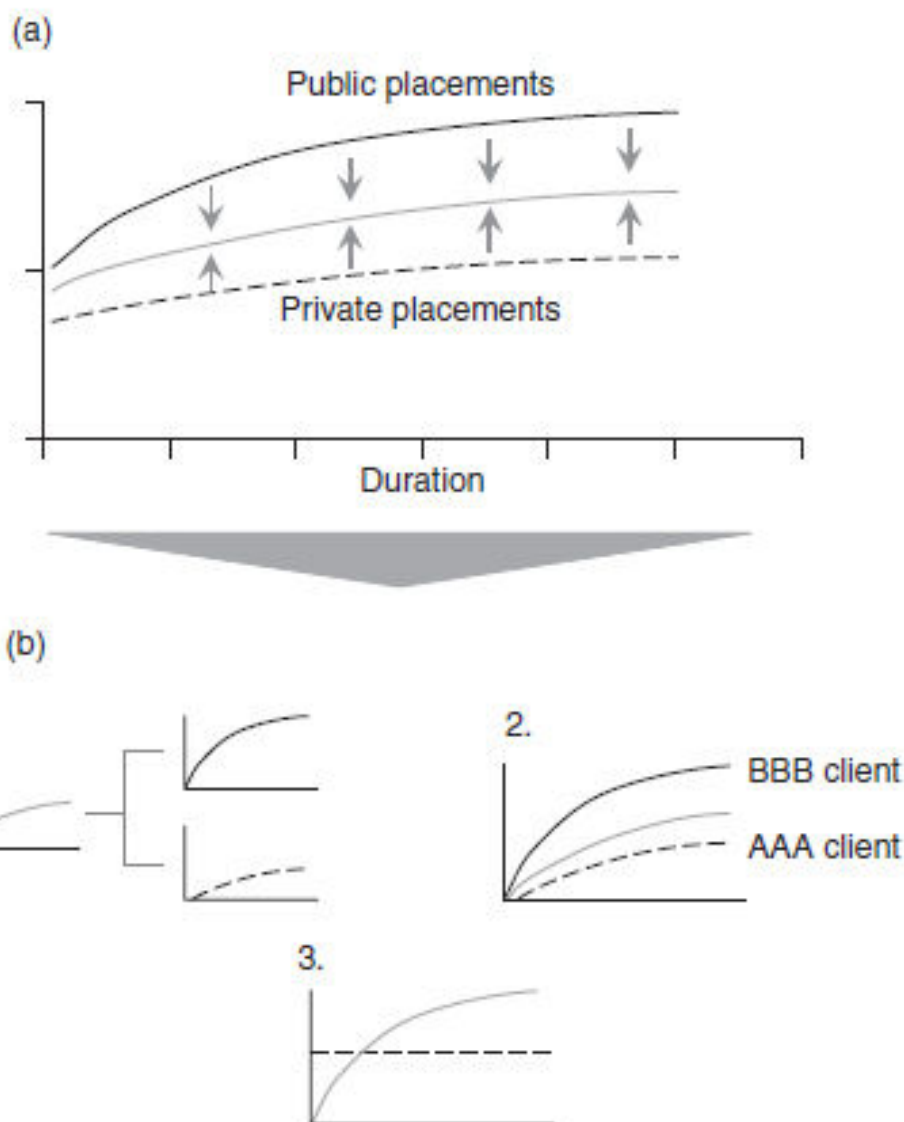
In the “classical” set-up, the most common FTP methodology is a gross marginal cost approach which is based on a simple paradigm: what is the opportunity cost to fund one unit of new asset volume in the wholesale market? In consequence, loans pay the term-matching wholesale funding spread which is paid out to the funding desk and also granted to stable deposits. This approach is widely implemented among European banks. Neu *et al* (2012) showed that approximately 92% of the 25 large European banks based their FTP approach on this paradigm. However, at the time this approach was developed and implemented at most banks, balance sheets were growing and the growth was funded by additional capital markets issuances. In consequence, FTP based on the marginal cost approach incentivised profitable growth.

Banks will usually use different funding curves for different asset classes in the first place, to account for the self-funding characteristics some assets may have. For example, assets that can be used to refill the collateral pools of covered bond funding or Pfandbrief funding will systematically receive a lower cost-of-funding curve, as the bank will be able to refinance at a lower cost using these secured funding instruments. It should be noted, however, that the full asset will not receive this favourable treatment, as haircuts need to be taken into consideration. This is usually already factored in to the applied curve, which in essence is a weighted average cost-of-funding curve for the asset class.

In the “new normal” after the 2007–9 crisis, things are different: banks shrink their balance sheets and try to reduce their dependence on wholesale funding. In addition, treasurers will be more and more restricted in issuing specific tenures on capital markets. Strong balance sheets will have a strong, stable deposit base with a clear competitive advantage for loan pricing. In particular, banks with a loan-to-deposit ratio less than 1 may use a risk-free investment curve as the basis for FTP instead of the classical wholesale funding curve. Of course, this would be a complete change in the underlying paradigm: the opportunity cost of funding one unit of new asset in the wholesale market would be replaced by the opportunity benefit of investing one unit of new deposit. On the other hand, investment unit segments need to be funded “at arm’s length” given the discussion around the institutional separation of commercial and investment banking functions. We believe that this new balance-sheet paradigm needs to be reflected in the banks’ FTP

approach. We currently see three trends in dynamic curve setting ([Figure 23.5](#)).

Figure 23.5 Dynamic approaches to funding curve setting



(a) Basic approach: generation of a benchmark curve from public and private bond placements. Curve generation is based on public and private bond placements. Differentiation is made between senior unsecured and covered bond-eligible funding (different curves applied). (b) Alternative approaches: “de-averaging” of benchmark curve over business segments and clients. 1. Segment-specific curves. 2. Higher consideration of client specifics in large individual transactions (eg, rating, deposits). 3. Duration-independent (flat) funding curves.

- **Segment-specific curves:** for universal banks, the significant deviation in the “stand-alone” funding cost of segments versus the blended funding cost of bank needs to be considered, eg, core bank versus wind-down unit or stable (eg, retail) versus volatile (eg, investment banking) segments leading to segment-specific funding curves. In practice, it is not trivial to calculate these curves: usually stand-alone spreads are derived from peer benchmarking. In particular, segment-specific curves are needed to set the right incentives to grow in strategic business areas for banks with a large amount of non-core assets, but also account for the fact that the funding position of the bank as a whole is in fact determined by the sum of its parts, which may drive the bank’s overall cost of funding up or down.
- **Greater consideration of client specifics in single large transactions (eg, rating, deposits):** some banks have even started to use rating-related funding costs for portfolio-relevant client loan business, eg, AAA versus BBB long-term project finance. Another example we have seen in the market is the introduction of client-specific credits when pricing a loan for a client with a strong deposit base. This, of course, is an example of stimulating a strategic view by subsidising via the FTP, which we shall discuss more extensively in the next section.
- **Duration-independent (flat) funding curves:** treasurers are facing limited funding opportunities in certain maturity buckets. Hence, some banks use duration-independent flat funding curves, or at least step functions averaging for several time buckets, leading to an average marginal cost-of-funds approach.

Since the financial crisis, several balance sheet requirements have made the setting of the “right” curve even more complex:

- the asset encumbrance ratio forces institutions to find the right balance between encumbered and unencumbered assets and refinancing;
- MREL and total loss-absorbing capacity (TLAC) have led to the introduction of a new liability class between Tier 2 capital and senior unsecured capital, which affects the curves for preferred and non-preferred senior unsecured capital;

- the leverage ratio increases the balance-sheet costs that can be included in the curve setting.

We believe that there is no best practice or single approach to curve setting for FTP. Indeed, as the examples above indicate, setting the right curve is a very bank-specific and individual task.

Measuring the results from FTP

Of course, an FTP scheme is only “complete” if the results from FTP can be measured in a clear and transparent way. We believe that treasury operating models need to ensure that the treasury result for the banking book can be split into at least the following components:

- the result of interest rate positioning and interest rate maturity transformation;
- the result from FTP, ie, liquidity maturity transformation;
- the result of taking the credit spread risk in strategic investment book management by treasury.

To achieve this transparency, long-term interest rate risk and long-term liquidity risk need to be separated into different portfolios within treasury, eg, an ALM portfolio for the interest rate risk management and a LM portfolio for the strategic liquidity management. Assume, for example, the bank underwriting a five-year loan with 4.0% client rate, assume the five-year swap rate to be 2.5% and the banks five-year funding spread to be 0.5%. Then this fixed rate loan should be split internally by treasury into a five-year fixed rate payer swap at 2.5%, transferring the interest rate risk to ALM, and into a five-year floating deposit at Libor + 0.5%, transferring the structural liquidity risk and FTP to the LM desk. Similarly to the loan, any fixed rate liability should be split into its interest and liquidity components. Hence, the bank has a fully transparent view on the interest (ALM desk) and liquidity maturity transformation (LM desk).

THE STRATEGIC PERSPECTIVE: BALANCE-SHEET STRUCTURE MANAGEMENT VIA FTP

Active balance-sheet structure management has become a key priority for banks, and frequently it is the bank's treasurer who has the overall responsibility. There are many reasons why active and conscious balance-sheet structure management has increased in importance: scarcity of (both capital and funding) resources requires banks to allocate these very precisely to their businesses. New (and existing) regulatory requirements (eg, leverage ratio) control the leverage a bank can take and set minimum levels for funding structure and liquidity buffers. Recovery and resolution requirements (MREL/TLAC) led to the introduction of a new funding class between Tier 2 and senior unsecured capital, and will improve the clarity on the liability recovery cascade. Maturity transformation, which used to be an important revenue source as well as one of banks' key functions, was under regulatory scrutiny. Since banks were increasingly moving into secured funding instruments, driving up their balance sheets' asset encumbrance (with severe side effects on the remaining unsecured funding positions) the asset encumbrance ratio was introduced to steer the right balance between encumbered and unencumbered assets and refinancing.

As a result, banks have enhanced their planning processes significantly, paying more attention to funding and liquidity aspects. As part of their periodic Treasury Benchmark Survey in 2012, the authors showed that more than 80% of banks had redesigned their financial planning processes to account for liquidity and funding aspects, more than 75% had introduced new structural liquidity risk ratios and more than 66.6% of banks had actively reduced the amount of maturity transformation they performed. These changes were partly due to the increased effort to design a "target balance-sheet structure", linking asset and funding strategy and setting overall growth ambitions.

However, defining a target structure is only the first step; arriving in the target state usually takes years, as existing portfolios need time to run off, and it is the structure of incoming new business that gradually reshapes the balance sheet. Now, how does the bank make sure that these new business and new funding flows help the bank to reach their designed target state? In certain conditions, setting strict limits will help, eg, maximum volumes in certain asset classes; loan-to-deposit thresholds or other self-funding requirements; restricted overall funding volumes for asset-financing businesses. In particular, in situations where the bank as a whole has to obey to growth limits (for example, as a result of state aid restructuring

proceedings or a deleveraging strategy), it may be adequate to impose volume guidelines on business segments or individual desks. However, such strict limits usually run the risk of allocating resources suboptimally, as a central planner would need to consider all costs and benefits and typically lacks sufficient information to do so. Therefore, we recommend using the bank's FTP scheme as the tool of choice, incentivising those behaviours that help to achieve the bank's goals, and penalising all others.

In doing so, we recommend the following five key elements for success be considered.

1. The bank's asset strategy needs to be aligned to the funding strategy. A bank that uses its balance-sheet strength to finance high-volume, long-term assets while seeking to keep liquidity transformation risk low will inevitably need to define an appropriate long-term funding strategy. If the funding mix relies on a significant portion of covered funding or securitisations, the assets need to be eligible for use in such vehicles. The bank's FTP scheme should reflect these requirements, by setting differentiated funding curves for cover pool-eligible assets. Furthermore, as increased asset encumbrance will drive up the cost of unsecured funding, any assets relying on unsecured funding need to be priced according to a more expensive funding curve.
2. We should be alert to potential "virtuous circles", positive self-fulfilling prophecies, as we observed in the direct banking market: banks need to make assumptions on the stickiness of their retail deposits. The stickier the deposits, the higher their value (ie, upward-sloping liquidity spread curve). The higher their internal value, the more the bank's business segments can profitably pay to their customers. And the more the bank pays, the greater the likelihood that the deposits actually will stay for longer. While we may not make the mistake of hoping that each progressive assumption will in the end become proven, we need to accept that the modelling of fair internal transfer prices should be based on the assumed mid-term target state, and not necessarily the starting position. On the other hand, if we believe that the designed target state is achievable in the medium term, we should be prepared to act accordingly in the transition phase.

3. Banks still have a long way to go in the fair pricing of “hidden” liquidity costs. These are frequently, but not exclusively, contingent liquidity costs. Examples include the liquidity tied up in trading portfolios for longer periods of time (frequently priced overnight or not priced at all), contingent liquidity requirements arising from off-balance-sheet items such as guarantees or current account limits, or the funding of cash collaterals posted for out-of-the-money derivatives. In our project experience, we observe a natural resistance from the business segment managers to the allocation of such hidden liquidity costs to the businesses; they frequently argue that doing so would “destroy the business’s profitability”. In fact, these costs may indeed be significant, which emphasises the strategic importance of FTP in the first place. As a matter of fact, sometimes these businesses actually turn out to be unprofitable, having benefited in the past from implicit subsidies received in the form of cheap or free liquidity. Naturally, treasurers seeking to spread their structural and contingent liquidity costs will face opposition of the business unit, which will invariably result in a rigorous discussion of the methods and models used to arrive at the “fair” liquidity price allocated to the businesses. To succeed in these discussions, treasurers will need to make sure their calculations withstand such increased scrutiny, and will need to build on their credibility, making sure they cannot be accused of optimising methodology to their own (ie, the profit centre’s) benefit.
4. The time factor needs to be taken into account. Deleveraging a bank will eventually result in a lower risk profile and thus reduced funding cost. The target funding curve will match the target balance-sheet structure. To arrive at the target state, the bank will need to set aside a budget to fund the journey, being prepared to set internal prices below the current funding costs, and allowing time for the required portfolio structure to change accordingly.
5. Banks must have a true and fair view. We believe that the setting of strategic goals benefits greatly from banks being incentivised by the FTP scheme, but such incentives or subsidies need to be made very transparent and come on top of a “true and fair view” of the bank’s liquidity cost. Consumption (or generation) of liquidity should initially always be priced in a way that reflects the overall bank’s true cost or

benefit. On top of this economic view, the bank may define strategic initiatives that should be supported by internal pricing: for example, market share growth, strategic product initiatives or diversification of funding channels (build-up of new, more expensive, but diversified funding sources). As a best practice, we recommend specific “subsidy pools” be defined, against which liquidity can be booked, allowing maximum transparency and budgetary control of investments and benefits.

Introducing a balance-sheet structure management process via the FTP scheme can be a very complex process. Liquidity charges and credits need to be calculated on a product-by-product basis, taking into consideration self-funding aspects, contingent liquidity implications, behavioural tenors and, potentially, altogether different funding curves. Leaving plain vanilla products aside, at the time of writing there is still much conceptual discussion going about the “right” way to price certain asset classes. For example,

- how should cash collaterals for cross-currency swaps be funded and priced, and how would this affect the funding cost of foreign currency funded via these swaps?
- how should run-down portfolios with long maturities but a clear mission to sell off assets as quickly as possible be priced?
- what is the “right” stress scenario to apply when simulating contingent liquidity outflows?

We believe that ultimately banks need to find the right balance in the trade-off between methodological precision and academic detail on the one hand, and a certain level of pragmatism and transparency on the other. After all, the purpose of setting and charging the “right” price for consumption of liquidity is not rooted in some sort of justice or other ideological consideration. Instead, it is all a question of finding the optimum allocation of resources: making sure that resources get used in the most efficient way. However, in order to make the pricing mechanism work (high liquidity costs discourage business activities that tie up much liquidity for small returns and vice versa), it is a crucial prerequisite that all involved stakeholders, primarily market-facing business segments, have a high level of transparency on the mechanism and can anticipate how certain behaviours will affect their P&L.

If, for example, the bank's goal is to avoid long-term asset financing, the big impact comes from making it clear to business units that long tenors will need to pay significantly higher spreads than shorter tenors, thus incentivising sales teams to build in cancellation/renegotiation clauses into their contracts. If a bank's funding strategy relies on covered bond funding, there needs to be a noticeable spread between assets that will be eligible for the desired funding scheme and those that are not. However, pricing schemes of course need to be not only transparent, but also sufficiently consistent that they can withstand challenges by business units (which they always will be, as most banks' business units we have seen claim that their treasury is using all sorts of flawed models, while the competitors' treasury departments use much fairer models, allowing competitors to offer better prices on the market, etc.

CONCLUSION

We conclude with six important points.

1. FTP is a deeply strategic steering instrument and needs to be carefully and thoroughly designed to meet the bank's business model.
2. Banks need an effective funds transfer pricing to incentivise profitable growths.
3. The FTP landscape of a bank should include all relevant balance sheet and off-balance-sheet positions that need to be considered from a liquidity risk perspective.
4. Banks' treasury operating models need to ensure transparency on the results from FTP (liquidity maturity transformation).
5. There is no best practice to derive the "right" curve for FTP. Instead, setting the curve is an individual and bank-specific task.
6. FTP is a tool to assist strategic balance-sheet management. However, incentives and subsidies need to be made very transparent and come on top of a "true and fair view" of the bank's liquidity cost.

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Balance-Sheet Management with Regulatory Constraints

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This chapter will update the reader on the variety of relevant constraints for treasury professionals within banks, and how these constraints have modified the traditional approaches to asset and liability management (ALM). It is no longer sufficient to consider balance sheet management just through a normal business environment; extreme and highly stressed environments should also be considered. The tools needed to achieve this are evolving, and will vary by institution, but here we describe the key considerations for the reader in designing the relevant framework.

With the finalisation of Basel III, three new metrics for balance-sheet risk management will be binding:

1. the liquidity coverage ratio (LCR) defines a minimum for the liquidity buffer relative to potential outflows in a stress scenario;
2. the net stable funding ratio (NSFR) defines a minimum for stable sources of funding to the term funding requirements on the asset side;
3. the leverage ratio sets a minimum for capital as a percentage of total balance-sheet size.

These ratios will supplement the already implemented and updated Basel II solvency ratio, which sets a minimum level of capital relative to risk-weighted assets from credit risk, market risk and operational risk.

Two additional regulatory concepts concerning the issuance of debt instruments eligible for bail-in, which should support the absorption of losses and facilitate potential recapitalisation in the case of failure and resolution were introduced in the aftermath of the global financial crisis: the “total loss-absorbing capital” (TLAC) concept, introduced by the international Financial Stability Board (FSB) will be applicable for globally systemic banks; this will be mirrored by the “minimum requirement for eligible liabilities” (MREL), a similar concept for European Union (EU) banks. We aim to explain the relationship between the various regulatory and market liquidity constraints that must be considered in managing a bank’s balance sheet, and its asset and liability position.

In addition to the ratios introduced by the Basel Committee on Banking Supervision (BCBS), the FSB and the EU, a further set of ratios are applied to measure and contain the risks in a bank’s balance sheet. Of these, the loan-to-deposit ratio (LDR), which relates the volume of loans to the volume of deposits on the balance sheet, is probably the most widely applied, eg, in China, where hard limits are set by regulators.¹

A further metric that has merited increasing attention is the asset encumbrance ratio, which measures the percentage of assets used for secured funding transactions such as covered bonds, securitisations or repurchase agreements (repos).

Such constraints around liquidity and capital will be outlined in the following section, which is followed by a description of how these constraints affect key balance-sheet positions and how they can feed into an optimisation framework. This is followed by a section discussing implementation and steering of the balance sheet, before a summary concluding the chapter.

EXPLANATION OF LIQUIDITY BUFFERS

The liquidity of banks in the context of funding can be defined as their ability to meet their liabilities, and unwind or settle their positions as they become due (Basel Committee on Banking Supervision 2008). Liquidity risk captures

the inability of a financial institution to service their liabilities as they fall due. Definitions of funding liquidity risk usually involve a time horizon.

Typically, funding liquidity risk depends on the availability of various liquidity sources, such as disposal of assets, loan syndications, client deposits, secured and unsecured issuances, interbank funds, repos and direct funding from central banks. Liquidity risk materialises where a financial institution is unable to meet its obligations due to insufficient liquid resources. One of the regulators' preferred methods of measuring liquidity risk is stress testing. The applied stresses can be specific to the financial institution, market wide or a combination of the two. In order to mitigate measured liquidity risk, banks have built up liquidity buffers significantly in past years. The introduction of the LCR fostered this development by explicitly quantifying requirements for a stock of high-quality liquid assets as a function of outflow arising from a stress scenario.

The liquidity buffer should be composed of cash and a core of assets that are both central bank eligible and highly liquid in private markets. For the longer end of the buffer, a broader set of liquid assets might be appropriate, subject to the bank demonstrating the ability to generate liquidity under stress from them within the specified period of time. The liquidity buffer has to be managed to ensure that, to the maximum extent possible, assets will be available in times of stress.

It is important to assess whether the benefits of holding incremental buffer assets outweigh the costs. From an individual institution's perspective, the ideal size and quality of the buffer are obtained when the marginal benefits of holding it are equal to the marginal costs.²

The estimation results of Bordeleau and Graham (2010) provide some evidence that the relationship between liquid assets and profitability depends on the bank's business model and the risk of funding market difficulties. Adopting a more traditional, ie, deposit- and loan-based, business model allows a bank to optimise profits with a lower level of liquid assets. This will also depend on the cost of raising deposits compared with alternative sources of funding.

From a policy perspective, the empirical results of Bordeleau and Graham suggest that policymakers and bank managers should bear in mind the trade-off between resilience to liquidity shocks and the cost of holding lower-yielding liquid assets. While holding liquid assets will make banks more resilient to liquidity shocks, thus reducing the negative externalities they

might impose on other economic agents, holding too many may impose a significant cost in terms of reduced profitability. Indeed, as retained earnings are the primary means of organic capital generation, low profits may prevent banks from expanding and extending additional credit to the real economy. These benefits and costs are equally applicable for both individual institutions and the financial system as a whole. When local regulations permit it, the liquidity buffer itself can be used to implement maturity transformation, thus mitigating the costs of holding the buffer to some extent.³ Integration of the respective risks into the overall asset–liability risk management framework is advisable.

REASONS FOR TERM FUNDING, AND THE IMPACT ON ASSET AND LIABILITY MANAGEMENT

One of the key reasons banks encountered liquidity stress during the global financial crisis was an overreliance on short-term funding, particularly short-term wholesale funding. As a consequence, regulators now expect banks to hold sufficient term funding going forward. A key measure for the supply of term funding is the NSFR, which was introduced by the BCBS alongside the LCR, and will be binding from 2018. This measures the stable funding available to a bank as a percentage of the required long-term funding, and has a target of at least 100%. Stable funding is made up of equity, preferred stock, wholesale funding with tenors longer than one year and certain types of deposits, such as those from retail and small business customers. To a lesser degree, funding from non-financial corporates, sovereigns, central banks, multilateral development banks and public sector entities is also eligible as stable funding. Requirements for stable funding originate from certain on-balance-sheet and off-balance-sheet exposure. The lower the marketability or eligibility as central bank collateral, the greater the amount of required stable funding for assets.⁴

The higher standards for term funding have a significant impact on the flexibility of ALM. The restrictions explicitly limit the capacity for maturity transformation of liquidity. While maturity transformation beyond one year for retail deposits is limited only to some degree, it is virtually impossible for short-term wholesale funding from financial institutions. Some maturity transformation is still possible for wholesale funding from non-financial institutions.

The different sources of stable funding can be associated with varying costs. Those funding sources which carry higher costs may provide additional value for the bank in the case of eligibility as capital instruments or eligibility for bail-in. The cheaper sources of stable funding may have disadvantages: deposits may be subject to ring-fencing, while covered bonds increase the asset encumbrance of a firm.⁵

ASSET ENCUMBRANCE

Asset encumbrance expresses the degree of funding obtained by assigning or pledging existing assets on a bank balance sheet. Assets that are already clearly assigned to certain funding sources are not available for secured funding in the future, and may therefore limit refinancing opportunities in times of stress. Asset encumbrance also measures the degree of structural subordination on a bank's balance sheet. The main sources of secured funding are the following.

- **Repos:** these are economically similar to a secured loan, with the buyer (effectively the lender or investor) receiving securities as collateral to protect them against default by the seller. The party who initially sells the securities is effectively the borrower. Unlike a secured loan, however, the legal title to the securities passes from the seller to the buyer. Coupons (interest payable to the owner of the securities) falling due while the repo buyer owns the securities are, in fact, usually passed on directly to the repo seller. A key aspect of repos is that they are legally recognised as a single transaction (important in the event of counterparty insolvency) and not as a disposal and a repurchase for tax purposes.
- **Covered bonds:** these are issued by credit institutions to fund certain loans that are secured by real estate clients, ship mortgages, aircraft mortgages and claims against public-sector bodies. These covered assets are reported in the credit institution's balance sheet and are registered and monitored separately. The issuer undertakes to pay the covered bond bearers the promised interest and, at maturity, to repay the principal amount of the covered bond. In the event of the issuer's insolvency, the covered bond bearers have a preferential claim in respect of the assets entered in the cover registers. The cover pools and

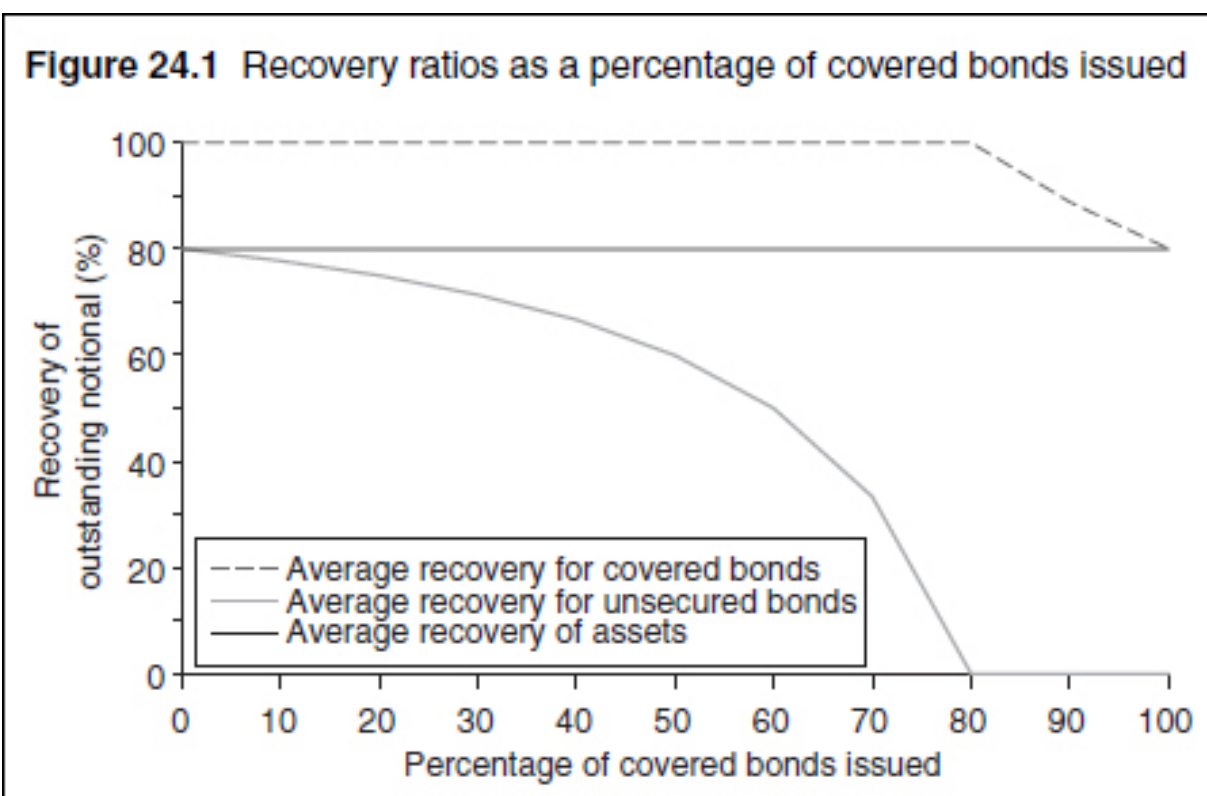
the covered bonds are not included in the insolvency proceedings under the insolvency administrator, but are managed separately by the cover pool administrator. Securitisations (asset-backed securities, residential mortgage-backed securities, etc) provide similar preference to the underlying security, but with no support from the issuer; that is, the underlying loans also service the debt issued.

- **Derivative contracts:** these are typically designed so that counterparties have no claims against each other at inception. But, as time passes, one of the counterparties typically accumulates claims against the other. To safeguard these claims, the debtor must post collateral, leading to asset encumbrance.⁶ Asset encumbrance may change over time due to deliberate changes in a bank's funding strategy or due to factors beyond their immediate control, such as margin calls due to changes in related market parameters and perceived asset liquidity.

One of the effects of asset encumbrance is that it tends to shift risks to unsecured creditors, a process called structural subordination. Structural subordination takes place via two channels. First, the issuance of secured funding is usually overcollateralised. This means that unsecured claimants finance some part of pledged assets or collateral. Second, even if the issuance of secured funding has no overcollateralisation, structural subordination usually takes place. The covered bond, securitisation and derivatives collateralisation structures are typically selective in the suitability and quality of collateral. The higher quality collateral on the balance sheet is therefore pledged into the structure, leaving the unsecured creditors with recourse to the lower quality collateral. For both these reasons, unsecured claim holders tend to be worse off after asset encumbrance increases (Juks 2012, p. 8).

The impact of increasing asset encumbrance on holders of covered and unsecured bonds is depicted in [Figure 24.1](#). The graph shows the recovery ratio measured as a percentage of the aggregate outstanding notional of covered bonds and unsecured bonds as a function of the encumbrance level, which is itself a function of the covered bonds issued. The example assumes an average recovery rate of 80% for all assets. With very low encumbrance, recovery ratios for covered bonds will be 100%, while recovery rates for unsecured bonds will be slightly below 80%. The greater the proportion of

secured bonds, the larger the drop in the average recovery ratio for unsecured bonds. When the share of covered bonds surpasses a certain level, the average recovery ratio for unsecured bonds will drop to zero, and the average recovery ratio for covered bonds will approach the average recovery ratio of the overall asset pool. Capital and subordinated debt can mitigate this to some extent.



A higher level of asset encumbrance thus tends to weaken the position of unsecured funding providers; this is relevant not only for holders of regular senior notes but also for depositors when neglecting the additional protection of deposit insurances. In fact, depositors are able to withdraw their funds within a short period of time and therefore destabilise the funded financial institution, while holders of unsecured debt may only be able sell their claims in the market. The introduction of depositor preference has a similar effect to encumbrance, as the “uncovered” lenders are further subordinated.

However, it should be noted that the introduction of secured funding enables funding from a variety of sources, which may reduce the probability of default for unsecured creditors, thereby offsetting subordination from asset encumbrance. Further, as illustrated above, unsecured recoveries are only

significantly affected at very high levels of asset encumbrance. Most banks operate in the range of 10–30% asset encumbrance, and at these levels the impact of subordination from rising encumbrance levels is less pronounced.

Asset encumbrance does, however, affect resilience to funding stresses. Encumbrance levels tend to increase during periods of funding stress, both as a result of deteriorating terms on existing collateralised obligations and because unsecured issuance tends to be replaced with secured issuance, as secured funding is generally less credit sensitive than unsecured wholesale funding. Where encumbrance levels are already high, there tends to be less unencumbered collateral of a suitable quality available to meet contingent encumbrance requirements in a period of stress.

In order to limit the risk from increased encumbrance to unsecured funding providers, banks should consider establishing limits on the level of asset encumbrance. The amount and quality of unencumbered collateral necessary to meet contingent encumbrance needs are attracting increasing attention.

CAPITAL INSTRUMENTS

Solvency can be defined as the degree to which the current assets of an individual or entity exceed their current liabilities. It can also be described as the ability of a corporation to meet its long-term fixed expenses and to accomplish long-term expansion and growth. One key instrument to ensure solvency is the sufficient supply of capital, which can serve to absorb challenges to the long-term ability to meet financial obligations.

Basel III rules increased the requirements for capital held by banks, with specific rules to be applied for systemically relevant banks.⁷ The increased requirements mean that capital instruments are a significantly more important long-term funding source for banks, and careful consideration needs to be given to the duration assumptions: Tier 1 capital is perpetual and generally can be seen as a source of long-term funding with the ability to cover short and medium term losses. The NSFR recognises capital as well as preferred stock as stable funding (with an implicit funding tenor of at least one year) to the full extent. Additional Tier 1 and Tier 2 issuances can be regarded as (initial) term funding for at least five years owing to their call tenors of at least five years. Hence, with the increased requirements, capital instruments will be able to absorb some of the stable funding requirements that would otherwise be covered by senior unsecured and secured term issuances on the

one hand or stable deposits on the other. When determining the optimal funding mix between Common Equity Tier 1, additional Tier 1 and Tier 2 notes, cost considerations should be taken into account. While common equity has to be regarded as the most expensive source of capital, dividends can be set according to business development. Additional Tier 1 and Tier 2 issuances (contingent capital notes) increase issuance costs and need to be factored into the treasury budget in a going-concern business environment.

BAIL-IN INSTRUMENTS

The bank recovery and resolution regime introduced in the EU at the beginning of 2015 was designed to ensure the orderly resolvability of even systemically important institutions. A key element of the European regime is the bail-in tool, which makes it possible, for the first time, for holders of non-subordinated debt instruments to be exposed to bank losses outside insolvency proceedings, alongside the institution's shareholders and subordinated creditors. While it would be possible in principle to bail-in all of a bank's liabilities, some exemptions are permitted to ensure that the resolution objectives can be achieved. Officially, the minimum requirement for total loss-absorbing capacity (TLAC) will come into force in 2019 for global systemically important banks (G-SIBs).

Generally, the following financial instruments count as instruments eligible for bail-in:

- common equity;
- additional Tier 1 capital;
- Tier 2 capital;
- unsecured debt issuances (unless further qualified by national laws);
- term notes;
- bank deposits not covered by a qualifying deposit insurance scheme;

while the following are typically ineligible:

- deposits, sight deposits and short-term deposits (deposits with an original maturity of less than one year) that are covered by a qualifying deposit insurance scheme;

- liabilities arising from derivatives and debt instruments with derivative-linked features, such as structured notes;
- liabilities arising other than through a contract (such as tax liabilities);
- liabilities that are preferred to senior unsecured creditors under the relevant insolvency law (eg, eligible deposits of natural persons and small and medium-sized enterprises in the EU);
- liabilities that are legally excluded from bail-in or cannot be written down or converted into equity without legal risk.

In the following analysis, we assume that TLAC requirements are an approximation of MREL requirements and therefore only consider the final TLAC rules in our analysis. Minimum TLAC will be introduced in two steps. According to this standard, from 2019, G-SIBs will be required to maintain TLAC amounting to at least 16% of their risk-weighted assets (RWAs) and to 6% of the Basel III leverage ratio denominator, whichever is higher. From 2022, the requirement will increase to 18% of RWAs or 6.75% of the leverage ratio.

CONSIDERATION OF LEVERAGE

Excessive leverage was identified as one of the factors behind the financial crisis. As a consequence, the 2011 Basel III framework included rules on the leverage ratio which state that the average of the monthly Tier 1 capital as a share of the leverage ratio exposure (LRE) measure shall be 3% over a quarter. The definition of the Basel LRE has been subject to some revisions (see Basel Committee on Banking Supervision 2011, 2014, 2016; JP Morgan 2014; Hartmann-Wendels 2014) and comprises the items in [Table 24.1](#).

While the Basel leverage ratio is scheduled to be implemented globally from beginning of 2018 on a global basis, the US bank regulators (the Federal Reserve, Federal Deposit Insurance Corporation and the Office of the Comptroller of the Currency) implemented their interpretation as the supplementary leverage ratio (SLR) in 2013 as part of the revised Basel III capital rules. A 3% SLR applies to the largest banking organisations with US\$250 billion or more in total consolidated assets or US\$10 billion or more in on-balance-sheet foreign exposure. In April 2014, the US bank regulators finalised an “enhanced” SLR on the eight US global systemically

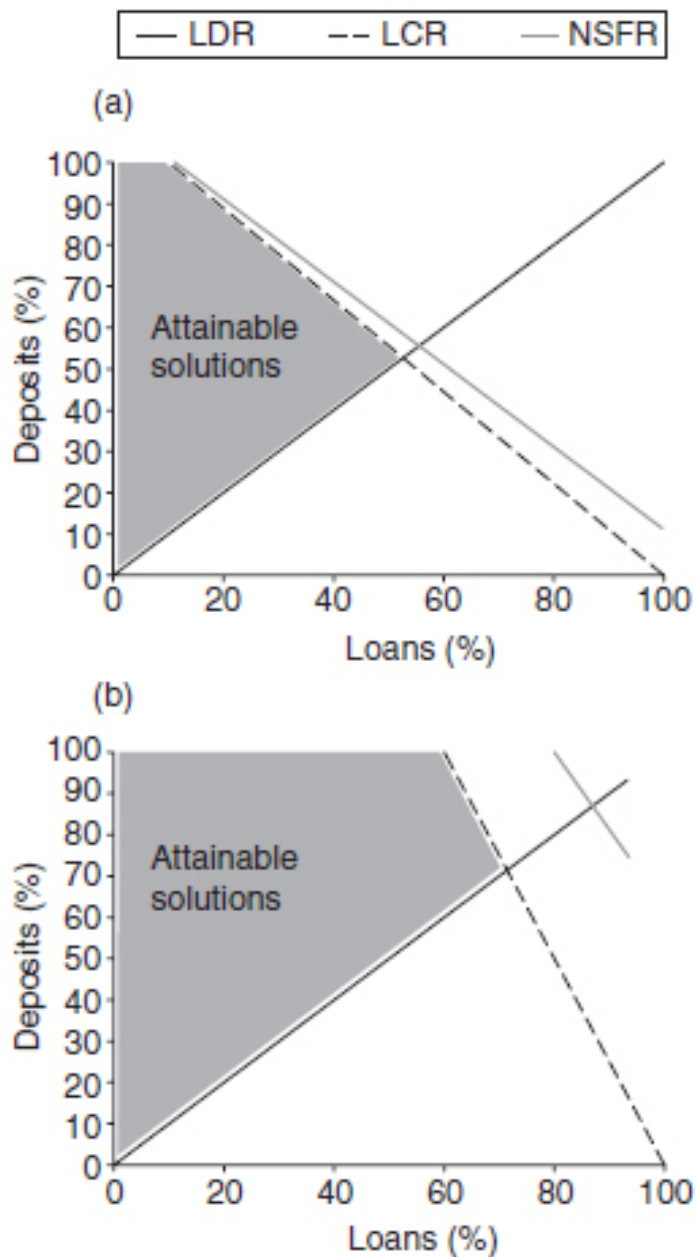
important banks (G-SIBs) and their insured depository institutions. This heightened standard is US-specific and not required by the BCBS. As part of the “enhanced” SLR, the eight US G-SIBs must meet a 5% SLR at the holding company level and a 6% SLR at the bank level. Banks also must hold a cushion above this 6% to account for volatility.

Table 24.1 Key elements of the exposure measure of the Basel III leverage ratio

Item	Comment
On-balance-sheet assets	• Assets deducted directly from Common Equity Tier 1 (eg, deferred tax assets, goodwill) as well as inter-group exposure are deducted • Accumulated other comprehensive income and other disclosed reserves • Common shares issued by consolidated subsidiaries of the bank and held by third parties (ie, minority interest)
Off-balance-sheet derivatives exposure	• Derivative exposure not subject to a netting or collateral agreement are captured at replacement costs plus potential future exposure • Netting effects from netting agreements are partially deducted from the exposure value • Variation margins from collateral agreements can be deducted • Client exposure from third-party clearing services if the bank guarantees the performance of the clearing house • Written credit derivatives
Other off-balance sheet exposure	• Commitments including liquidity facilities, direct credit substitutes, acceptances, standby letters of credit, trade letters of credit, failed transactions and unsettled securities with credit conversion factors between 20% and 100% • If commitments are unconditionally cancellable, a credit conversion factor of 10% is applied

Leverage limits introduce conflicts in the management of a bank's balance sheet and liquidity. In order to manage within both constraints, banks need to be selective about the volume of short-term cash they accept from depositors that is absorbed in the liquidity buffer. Also, where leverage is the more binding constraint, the economic incentives favour taking more risky assets than lower risk, highly liquid assets.

Figure 24.2 LDR, LCR and NSFR liquidity ratios

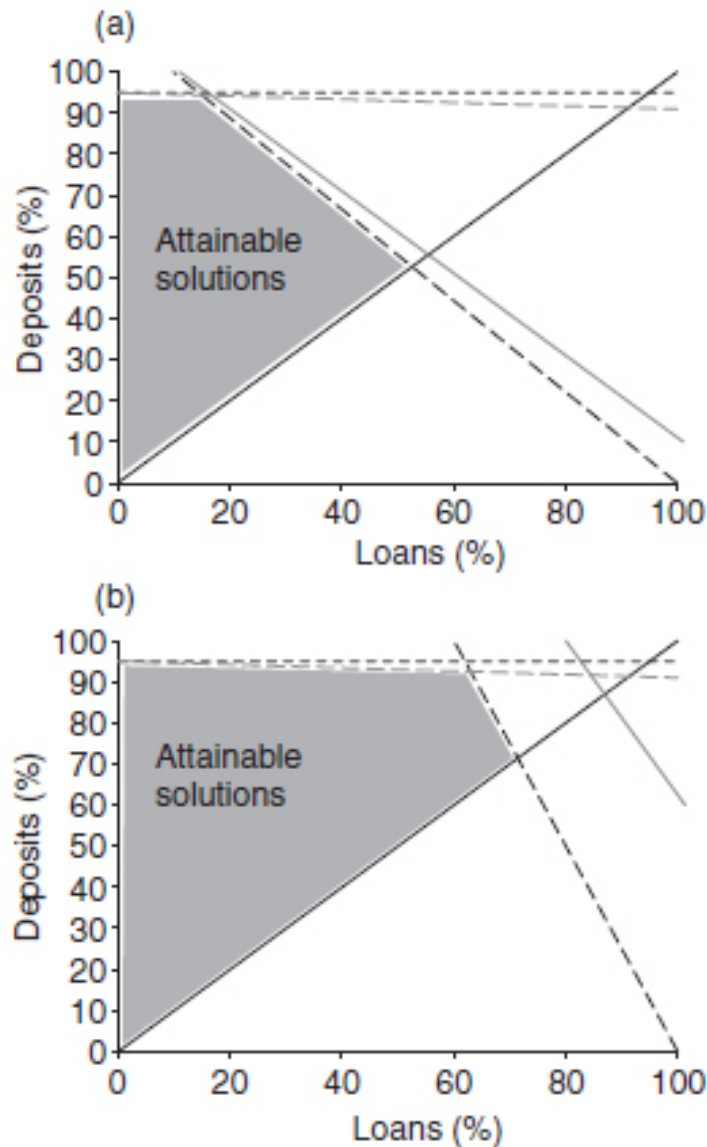


(a) Investment banking model; (b) universal banking model.

This section explores the combined repercussions of the LCR, NSFR and LDR on a simplified bank balance-sheet structure and, by extension, the ALM challenges. [Figure 24.2](#) depicts these ratios as constraints of a linear optimisation problem.

A bank is considered whose asset side consists of only two types: loans to customers and securities held as liquidity reserve against potential outflows of deposits. The abscissa of [Figure 24.2](#) depicts the structure of the asset side as the share of loans as a percentage of total assets. This number is equivalent to 1 minus the liquid asset ratio. The liability side also contains only two liabilities: equity and customer deposits. The ordinate of [Figure 24.2](#) shows the percentage of deposits of total liabilities.

Figure 24.3 LDR, LCR and NSFR leverage and capital



(a) Investment banking model; (b) universal banking model.

Taking returns in isolation, maximising the share of deposits lowers the cost of funds and maximising the share of loans (minimising the liquidity reserve) maximises return on assets. However, due to the liquidity coverage ratio, the volume of loans granted to clients is constrained by the amount of reserve assets (high-quality liquid assets) to be held. The size of the reserve

buffer in turn depends on, among other things, the type of deposits gathered and the client type. Generally, the LCR induces a negative relationship between the share of loans on the asset side and the share of deposits on the liability side. The more risky the type of deposits seen in a stress period, the higher the buffer and the lower the share of loans. This relationship is depicted by the downward-sloping dashed lines in [Figures 24.2 and 24.3](#).

The constraints provided by the NSFR are depicted by the grey dashed line in [Figures 24.2 and 24.3](#). This line implies a negative relationship between deposits and loans as loans will always require some proportion of stable funding, but deposits do not qualify to 100% as stable funding. Consequently, in the absence of securities as liabilities, a greater proportion of equity will be necessary to accommodate a higher share of loans on the balance sheet. The impact of this restriction is significantly dependent on the business model. For retail banks, the restriction is rather limited due to the higher degree of recognition of retail deposits as available stable funding. On the other hand, the restriction is rather significant for corporate and investment banks, as there is less recognition as stable funding in the NSFR.

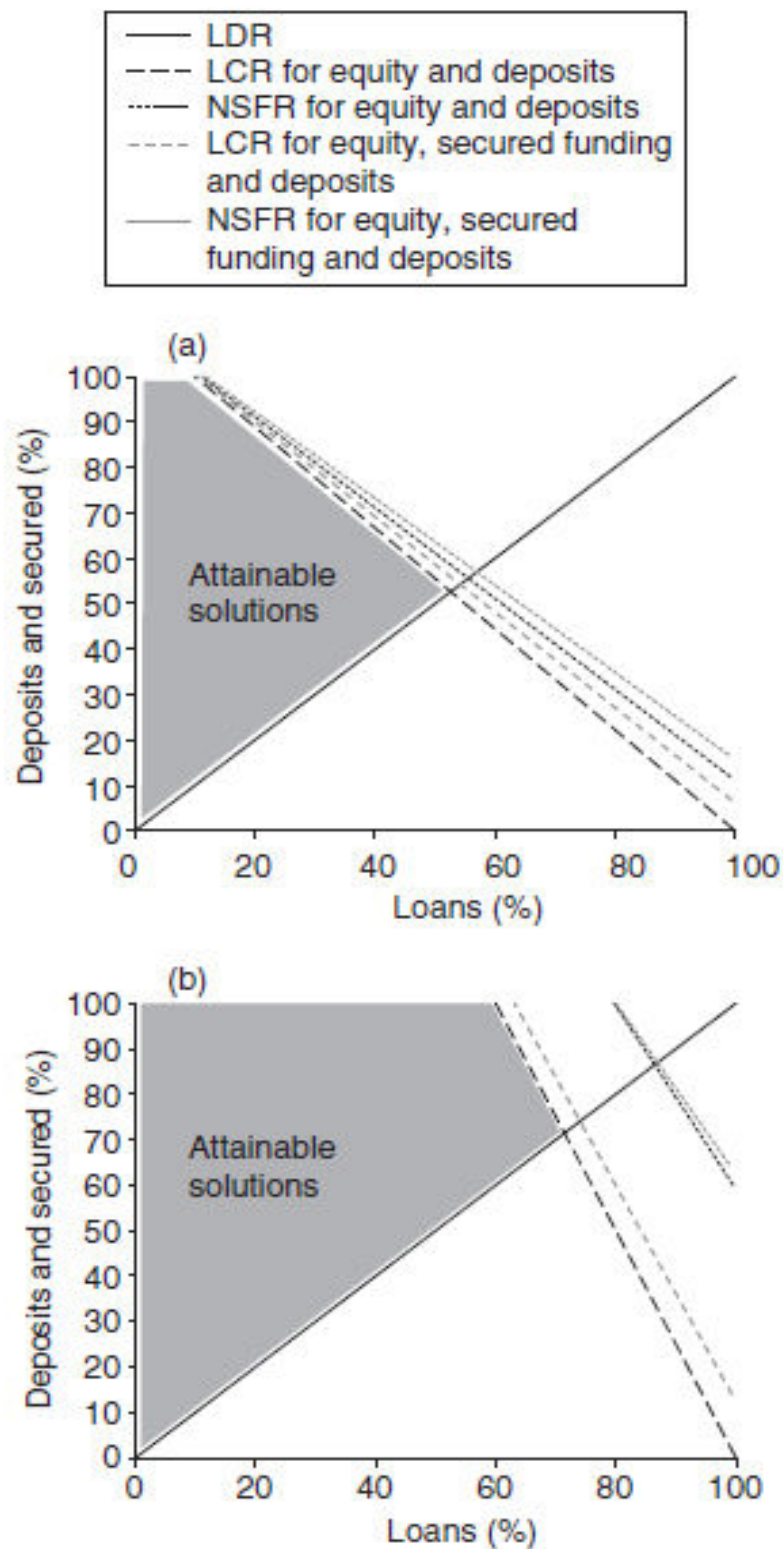
The attainable deposit–loan combinations are shown for two banking models: the investment bank model in part (a) is characterised by a high demand for cash reserves to mitigate risk from the less stable deposits under the LCR and the need to limit lending to stable funding sources. The universal banking model, which has a greater proportion of business and residential customers, benefits on the other hand from lower outflows in deposits under the LCR and greater stability of the deposits under the NSFR (Schmaltz and Pokutta 2012). The solid line in the graphs reflects the loan-to-deposit ratio. A loan-to-deposit ratio of 100% implies a positive relationship between loans and deposits and therefore an upward sloping line. A loan-to-deposit ratio of less than 100% implies that the field to the left is attainable for the bank.

The graphs also show the loan-to-deposit ratio as a constraint on the attainable combination. A loan-to-deposit ratio of 100% or less excludes combinations where loans are financed by non-deposits. In practice, many banks will have an appetite for a loan-to-deposit ratio greater than 100%, market conditions permitting, so the loan-to-deposit ratio represents more of a prudential guide than a fixed constraint.

[Figure 24.3](#) depicts the linear optimisation problem with the inclusion of constraints by the Basel III leverage ratio and the Basel II solvency ratio. The

leverage ratio will require a minimum equity buffer to be held as a percentage of the total balance sheet. Leaving off-balance-sheet exposure aside, the leverage ratio constraint will impose an upper ceiling on the deposits to be raised as a minimum level of equity (3–5%) must be held. The constraints of the leverage ratio are represented by the respective dashed black lines in [Figure 24.3](#).

Figure 24.4 LDR, LCR and NSFR, leverage and capital



(a) Investment banking model; (b) universal banking model.

The simplified model for a retail bank is further complicated by access to other sources of funding, particularly secured funding and unsecured wholesale funding. These extend the range of options, but are best represented by combining with deposits to reflect total sources of funding. Additionally, the need to incorporate the different liquidity characteristics of deposits from divergent sources needs to be reflected in the optimisation constraints.

Figure 24.4 shows the effect of introducing term secured financing backed by loans on the asset side of the balance sheet. Term secured funding typically has a lower cost of funds than either equity or other term unsecured funding, enabling a reduction in cost of funds compared with the model where only equity and deposits are available. Issuance of term secured funding such as securitisations and covered bonds mitigates a portion of the liquidity reserve requirement against short-term deposits, created by the LCR, thereby allowing for a greater share of loans on the asset side.

The NSFR is also affected by the introduction of term secured financing, because funding for more than one year attracts a 100% stable funding factor. This reduces the requirement for equity or other term unsecured funding for a given proportion of loans. Secured funding also self-finances a portion of the loan portfolio, thereby creating capacity for a greater loan-to-deposit ratio.

In a universal banking model there is some potential to optimise across a diversified loan portfolio and sustain higher levels of encumbrance given the greater share of traditionally securitisable loan types. For an investment banking model, a smaller portion of assets may be suitable for raising secured funding. However, asset securitisation has a greater impact on the range of obtainable loan and deposit combinations, given the greater constraints imposed by the LCR and NSFR on this funding model.

The benefits of secured funding on the attainable combinations and optimisation of cost of funds are constrained by the impact of subordination of unsecured creditors on the cost of unsecured funds and the lower resilience to stress as a result of less unencumbered collateral to raise replacement funds in situations of funding stress. In some jurisdictions banks may also be subject to regulatory encumbrance ratio constraints.

While the introduction of other funding sources supplements the picture of attainable combinations, the end result is an optimisation with many factors, which is difficult to interpret in a graphical representation. In order to obtain

more meaningful results, the optimisation is better achieved by quantitative simulations that incorporate⁸

- the return on equity from asset options,
- the return on funding from assets, given both equity and funding constraint,
- access to secured and unsecured funding markets,
- costs of different sources of funding.

Additionally, the starting position of the bank's balance sheet is the most meaningful constraint, as asset decisions are often multi-year and contractual bank funding is also locked-in. Consequently, simulations should include multi-year options to ensure one year's best outcome does not create problems in the future.

The results of the quantitative optimisation can lead to conclusions with respect to the optimal balance-sheet allocation which may call for significant changes in the strategy of a business. As an example, repo businesses traditionally deliver low returns on notional volumes and thus are becoming less attractive, given the leverage ratio constraint. Furthermore, plain derivatives with low margins are also becoming less attractive, given the leverage constraint, such that trade netting and bilateral settlement need to be enforced in order to reduce leverage exposure. Deposits with high LCR risk weights are affected by both costs for holding a liquidity buffer with potential negative carry and the higher equity requirements due to the leverage ratio. Also, loan books with low risk-weighted assets may become less attractive if margins do not compensate for the relatively higher leverage weights.

The optimisation framework can also help to assess reactions of banks in a crisis situation. For example, a study by Halaj (2013) comes to the conclusion that in order to optimise their profitability, given a set of balance-sheet constraints, banks may reduce loan books in times of crisis, due to the lower profitability, and thus may undermine the monetary transmission impulse from lower interest rates.

STEERING TOWARDS AN OPTIMAL BALANCE SHEET

Limits and transfer pricing are the most prominent tools with respect to steering towards an optimal balance sheet. Limits on balance sheet use may be more effective in the short term, but can leave businesses with a lack of clarity concerning the profitability of individual products and external pricing.

Transfer pricing is the preferred tool for providing the right incentives to the originators of assets and liabilities. In this respect it can be seen as the central axis linking the liquidity and ALM management within the constrained balance sheet. Transfer prices should price the key scarce resources such as capital, term funding, short-term funding, liquidity buffer and balance-sheet usage in such a way that businesses can price resources to external clients so that the most efficient businesses grow, while less efficient businesses are reduced or changed to become more resource efficient.

The transfer pricing mechanism needs to be unified to provide clear incentives and recovery of costs, but needs to reflect both the expected behaviour of assets and liabilities in a normal business environment and the expected behaviour within a stressed liquidity environment. The need to include different market environments (from a liquidity perspective) means that the same assets and liabilities have to be modelled across a wide range of outcomes, with the normal business environment measured by some proxy of the median, and the stressed liquidity conditions measured in the tails.

For the pricing of borrowing and lending to the business, which is achieved through a unified transfer pricing mechanism, both outcomes need to be reflected in the rates paid. This is best expressed in floating rate spreads, to provide clear separation between the liquidity value (or liquidity term premium) and the market value (reflecting the interest rate hedge rate for the corresponding term).

In order to determine the all-inclusive transfer price, several components are needed:

- the normal business conditions behavioural term;
- the idiosyncratic stress behavioural term;
- the return on liquidity buffer investments;
- the market rate for external funding, on a floating rate basis;

- the overlay adjustments to the above market rate for the targeted term liquidity mismatch.

For assets the market rate should also be adjusted to reflect the ability to fund the assets in the secured market. The market rate should reflect the observable rate for funding the assets (with a mix of secured and unsecured funding). Where assets have contingent outflows (such as undrawn components), this contingent risk needs to be added to the funding cost (on a probability adjusted basis). A different term structure will generally be required for the contingent outflows.

For liabilities, the same principles can be applied, although typically the contingent outflow component is greater. This would require the term structure of the deposits to be split between the short-term (stressed market) expected outflows and the normal business condition term component.

This approach allows a comprehensive picture of the liquidity term structure of the balance sheet to be created and priced to the originating businesses. The market hedges then need to be layered onto the pricing to support a complete product pricing proposition. The term structure would in most cases be consistent with the liquidity pricing term, but hedges may be constructed with some mismatch to meet other objectives, such as earnings stability.

When leverage becomes a binding constraint for a bank it will need to be included in the transfer pricing scheme. This is relevant not only for assets but also for liabilities. Consequently, it is important that both the optimisation framework and the transfer-pricing framework permit a comprehensive modelling on all relevant restrictions. The optimisation framework is somewhat more complicated, as it needs to include not only internal transfer pricing but also the external pricing and costs for non-financial resources.

CONCLUSION

The rapid changes in the liquidity management requirements and banks' management practices have profoundly altered the necessary approaches to asset and liability management.

Tools need to be developed to allow the contingencies of liquidity stress events to be captured within ALM modelling. Further, modelling needs to be able to incorporate the constraints on the balance sheet imposed by encumbrance considerations, and more generally the need to be resolvable.

The constraints imposed by balance sheet management requirements mean that the matching of assets and liabilities in a traditional ALM model are no longer valid. The modelling needs to incorporate the additional assets, and the contingent requirements. Consequently, the following need to be considered.

- Separation between cash instruments' liquidity duration and interest rate duration: this increases the need for and efficacy of using derivatives to provide the appropriate interest rate profile.
- Inclusion of liquidity buffer assets in the overall ALM risk framework.
- Extension of behaviour modelling to include the stresses created in liquidity events, enabling some duration dimension to be included for the assets supporting the contingent outflows.
- There needs to be a very clear distinction between behavioural profiles for different products and client groups.
- The development of appropriate models for the liquidity tenor and interest tenor of capital products, given their greater importance as funding sources.
- Increased term funding and the resulting impact on maturity transformation and bank profitability should be reflected in the balance sheet.
- More general recognition of multiple scenarios in assessing the ALM risks on the balance sheet. For example, incorporation of recovery or resolution levels of stress events in the modelling of risk and potential outcomes, in order to assess the ALM profile.

These complications will increase further with the regulatory requirements of organising banks into multiple units that are manageable on a stand-alone basis should another part of the banking organisation fail.⁹ This leads to capital and funding plans being defined for, and restricted to, subunits in addition to the group level. The analysis and conclusion of this chapter will need to be adjusted to reflect the impediments to the free flow of capital and liquidity.

- 1 The loan-to-direct-funding ratio aims to broaden the loan-to-deposit ratio by considering the extent to which (the majority of) the institution's business is funded by medium- and long-term debt. The liquid-assets-to-total-deposits ratio can be seen as a simpler version of the LCR, as it measures the extent to which deposits are covered by liquid assets. The liquid-assets-to-total-assets ratio provides information about liquid assets as a share of the overall balance sheet.
- 2 See Committee of European Banking Supervisors (2009). The statement ignores the externalities associated with liquidity distress (eg, spillover effects and the systemic impact of banking failures) as well as moral hazard issues (institutions might not bear the full consequences of the risks they are bearing due to possible state aid).
- 3 For example, in order that banks cover the first week of outflows, the German regulator requires them to hold cash and high-yielding securities that can be liquidated without significant losses. In order to cover the subsequent period (up to one month), the bank can hold other assets that can be sold without significant losses (BaFin 2012).
- 4 See Basel Committee on Banking Supervision (2010, pp. 25ff) for a detailed description of the NSFR and respective weights.
- 5 The concept of bail-in was introduced by a legislative proposal of the UK in HM Treasury (2012). It requires that, in addition to capital, certain wholesale funding issuances can also be seen as loss-absorbing capacity in the case of a bank failure.
- 6 See Juks (2012). In addition, insurance contracts can also be collateralised and thus lead to asset encumbrance.
- 7 See Basel Committee on Banking Supervision (2011) for details.
- 8 See Puts (2012) for a quantitative linear optimisation approach given Basel II and Basel III restrictions.
- 9 Examples are ring-fencing certain banking activities according to the Banking Reform Bill in the UK, or the US IHC requirement for foreign banks, as well as the proposed EU equivalent in CRD5/CRR2.

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